APPROVAL FORMAT

APPROVED BY : ______________ **DATE** 23 Mar 2026

VP FLIGHT OPERATIONS

APPROVED BY: ______________ **DATE** 26-03-2026

Captain Assefa Birru
Director Air Operators
Certification and Survillance
ECAA

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FLIGHT OPERATIONS MANUAL

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FLIGHT OPERATIONS MANUAL

REVISION 25B
30-MAR-2026

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FLIGHT OPERATIONS MANUAL

Manual control no:_________________________

Assigned to :_____________________________

Date:________________________________

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CHAPTER 0 ADMINISTRATION AND CONTROL OF MANUAL

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CHAPTER 0 ADMINISTRATION AND CONTROL OF MANUAL

0.0. FOREWORD

This Flight Operation Manual (FOM) defines operational policies, procedures, instructions and guidance necessary for company operations personnel, to perform their duties in standard manner.

The manual is approved by Ethiopian Civil Aviation Authority (ECAA) to be compliant with ECARAS and applicable regulations. Compliance with this manual will also ensure compliance with FAA, EASA, IOSA, CAAC and other regulatory bodies requirements as applicable.

The FOM is issued to all crew members and all other persons and officials involved in Ethiopian operations per the distribution list in section 0.4.7.

All affected persons shall be familiar with policies, rules, procedures and the contents of this FOM relevant to the performance of their duties. However, nothing in this manual shall prevent personnel from exercising their own best judgment during emergency or during irregularity whereby no provision is given by this manual. This exercise must be reported in writing as soon as the flight is completed.

Proper use of this manual requires thorough knowledge of airplane system and company policies. Operational staff of the Division is expected to refer to and use the policies and procedures outlined in this manual as applicable.

This FOM is prepared in six Chapters with related Sections:

Chapter 0 Manual Control

Chapter 1 Administration

Chapter 2 Operational Policy

Chapter 3 Emergency and Irregularities

Chapter 4 Weight and Balance

Chapter 5 Training

Director Flying Operations shall continually review and update accordingly the content and currency of this manual from time to time to ensure compliance with operational and regulatory requirements.

CHAPTER 0 ADMINISTRATION AND CONTROL OF MANUAL

Strict adherence by the cockpit crew to the prescribed procedures in this manual will enhance the effort for coordinated performance in the endeavor to establish professionalism.

Vice President Flight Operations is the primary owner of this manual and his approval is mandatory.

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CHAPTER 0 ADMINISTRATION AND CONTROL OF MANUAL

0.1. INTRODUCTION

The legal basis for **Ethiopian** to provide and use an Operations Manual is defined in the ECARAS Part 2, part 7, part 8 and part 9.

The content of the Operations Manual complies with:

a) The applicable regulations of the Authority;

b) Applicable ICAO standards and recommended practices;

c) The required subparts of the Federal Aviation Regulation (FAR), EASA;

d) Additional rules and regulations applied by States within which **Ethiopian** is authorized to conduct flight operations; and

e) The provisions of **Ethiopian's** Operator Certificate (AOC) and Operations Specifications (OPS SPECS), as issued and approved by the Authority.

0.2 SUPPLEMENT MANUAL

The Flight Operations Manual (FOM) shall ensure familiarity and contains or references the policies procedures, checklists, limitations and/or restrictions pertinent to the performance of duties in areas and conditions where operations are conducted and other guidance or information necessary for compliance with applicable regulations, laws, rules and Ethiopian standards.

The following manuals and documents supplement this manual as applicable:

- Approved Flight Manual (AFM)
- Aircraft Operating Manual (AOM) including performance data, weight and balance data/manuals, checklists;
- Flight Crew Operations Manual (FCOM);
- Jeppesen Airway Manual;
- Flight Crew Training Manual (FCTM);
- Flight Crew Techniques Manual (FCTM);
- Standard Operating Procedures Manual (SOP);
- IATA Dangerous Goods Regulation (DGR);
- Emergency Response Guide for A/C Incidents involving Dangerous goods;
- Security manual;
- Flight Dispatch & Navigation Manual (FDNM);

- Flight Operations Training Program;
- Cabin Crew Operations Manual (CCOM);
- Emergency Response Manual;
- Route Manual;
- Aviation Safety Program Manual;
- MEL/CDL;
- Airport analysis/OPT;
- PBN pilot training handbook for Ethiopian airlines flight crew;
- Flight Data Monitoring Manual
- Ground Handling Manual (For all cargo airplanes)
- North Atlantic Operations and Airspace Manual
- Airport Handling Manual
- PBN Training and Procedure Manual
- Flight Operation Quality Assurance Manual
- Airworthiness Directives (ADs)
- As applicable, Aeronautical Information Publications (AIP) and NOTAMS;

0.3 PERSONS AFFECTED

Latest revised documents relevant to the flight crew is to be distributed in part or in their entirety electronically in the class I EFB (IPAD or the latest WINDOWS). All contents and revisions are controlled by the Manger FOPS Performance and Technical Support and Team Leader Documentation. Flight crew will comply with the revision requirement upon the synchronization of their class I EFB prior to each flight and as a minimum every 7 days. The updating process will be monitored by Manager FOPS Performance and Technical Support and the flight crew will be contacted through personal email or telephone if they fail to update their devices accordingly.

All Flight Operations must comply with the prescribed methods of operation, detailed procedures and operating limitations established by ECAA (e.g. AOC, ECARAS etc.), equipment manufacturer (Approved Flight Manual (AFM) and Flight Crew Operations Manual) for each aircraft type used in operations.

The Flight Operations Manual (FOM) has priority over all other documents unless it is otherwise stated in the manual.

This manual applies to both female and male operations personnel although some references may refer to one gender only. In all cases, reference to one gender shall also be deemed equally applicable to both genders. Any question about the content or use of this manual can be directed to Dir. Flying Operations.

0.4. AMENDMENT AND REVISION

0.4.1 MANUAL CONTROL

This manual requires the approval of Vice President Flight Operations. The revision number and effective date shall be annotated in the top-right corner of the page header. All manual users shall be provided with the latest revision on or before the effective date.

The normal revision cycle for this manual is once at the end of each year to verify its legibility, currency and easy identification. Any required change prior to this time will be inserted in the bulletins section after having passed through the standard manual approval process. Director Flying Operations upon receiving any amendment proposal using the form provided below will present it to the review committee. The revised manual shall be sent to approval upon completion of the review and update process.

The approval process will be in accordance with the directives in the Management Policy Manual (MPM).

Revision pages will be annotated to show the date of issue (and date of effect if different), the amendment list number and the portion of the text which have been revised, as indicated by the vertical margin lines adjacent to the changes.

Each amendment will be accompanied by a revised list of effective pages, with their date of issue and by certificate of receipt/incorporation. An amendment list record will be maintained at the front of each manual.

Master copy of this manual shall be retained by Mgr. FOPS Performance & Technical Support.

0.4.2. RECORD OF REVISION

This manual shall be amended or revised as necessary to ensure that the information contained herein is kept up to date. All such amendments or revisions shall be issued to all personnel that are required to use this manual.

This revision reflects the most current information available to Flight Operations through the subject revision date. The following revision highlights explain changes in this revision.

Pages containing revised technical material have revision bars associated with the changed text or illustration. Editorial revisions (for example, spelling corrections) may have revision bars with no associated highlight.

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Retain this record in manual. Up on receipt of revision, insert revised pages in manual and enter the revision number, revision date and filed date in the appropriate block.

Rev. No	Revision Date	Insertion date	Inserted by	Initial
0	05-06-75			
1	26-04-92			
2	01-04-97			
3	31-08-05			
4	11-09-06			
5	26-02-07			
6	17-12-07			
7	25-03-08			
8	03-10-08			
9	01-12-08			
10	11-05-09			
11	22-12-09			
12	15 -7 -11			
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18B	30-11-18			
19	10-09-19			
20	06-01-20			
21	01-10-20			
22	15-12-21			

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22B	15-03-23			
23	10-01-24			
24	16-06-25			
25	15-10-25			
25A	03-11-25			
25B	16-02-26			

0.4.3. REVISION HIGHLIGHTS

- This manual is a whole revision of REV. 25A of the FOM.
- CRM Instructor selection requirement
- This revision is required to comply with IOSA and ECARAS requirements.

0.4.4. REVISION TRANSMITTAL SHEET

Transmittal Number	Chapter / Section	Effective Date	Cancels, supersedes (Insert old effective date if any)	Date inserted	Entered by (Name & Initial)
24	16-06-25				
25	15-10-25				
25A	03-11-25	01-12-25			
25B	16-02-26	30-03-26			

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0.4.5. REVISION TRANSMISSION

The Flight Operations Manual must: (ECARAS 9.2.2.4)

a) Include instructions and information necessary to allow the personnel concerned to perform their duties and responsibilities with a high degree of safety.
b) Be in a form that is easy to revise and contain a system which allows personnel to determine the current revision status of each manual.
c) Have a date of the last revision on each page.
d) Not be contrary to any applicable Ethiopian regulations and the AOC holder's operation specifications.
e) Include a reference to appropriate civil aviation proclamation, rules and standards.

Dir. Flying Operations shall:

a) Maintain an up to date list of manuals together with the Manual Control Numbers and locations or nominated individuals or crewmembers that will be required to amend particular numbered copy(s).

b) Transmit revisions to all manual holders. Acknowledgment of receiving and insertion of revision by all Flight Operation Manual holders shall also be followed up and a record is kept by Director Flying Operations.

c) Temporary revision - Temporary revisions shall be issued pending final revisions of the Flight Operation Manual per the distribution list to all manual holders. A Temporary revision record shall be filed between the list of effective pages and contents of the affected chapter.

d) Emergency revision - This method is used for revisions that are urgent and need to be immediately circulated pending revision or incorporation in the relevant section of the Flight Operation Manual. Emergency revisions shall be transmitted to all manual holders electronically, by fax or by any other means of fast communication.

e) Identification and location of revisions-The temporary revision record shall be filed between the list of effective pages and contents of the affected chapter. Current revisions and copy revisions can be obtained from Documentation office.

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0.4.6. AMENDMENT PROPOSAL FORM

Proposed amendment: (Continue on separate sheet if necessary)
Reason for Amendment
Signature of Proposer: Position:
Authorized by: ______________________________________ Dir. Flying Operations ____ GUIDELINES ON THE COMPLETION OF THIS FORM 1. Proposed amendments to the Flight Operations Manual will only be considered if submitted on this form. 2. This form shall be completed, signed and handed to Dir. Flying Operations for comment/approval. Amendments to specific parts shall then be forwarded as follows: 3. The final decision on amendments to the Flight Operations Manual rests with Director Flying Operations. The company reserves the right to amend any proposal.

0.4.7 DISTRIBUTION

After receipt of the approved manual, Flight Operations Documentation Section prints enough copies for distribution. Dir. Flying Operations shall ensure the manual is properly reviewed / revised and duly approved prior to distribution.

This manual shall be distributed to all operational control personnel and external service providers (if there will be) on the effective date. It shall be made available to all in the distribution list below through the Flight Operations Documentation Section. Those recipients shall as well get the revised portion (s) of the manual.

The primary method of distributing the FOM is by electronic means. Alternatively, the FOM is available on the Flight Operations page of the Ethiopian portal intranet. ETG IT department shall provide scheduled backup for FOM as per ITPM 01.06.

Distribution of the Flight Operations Manual can be accessed through ET-PORTAL &

Individual Logipad or COD shall be available per the distribution list established below.

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Manual Cont. No	Issued to
1	VP Flight Operations
2	Dir. Flying Operations
3	Dir. Flight Training and Standard
4	Dir. Corporate QA, SMS, EMS, Industrial Safety & ERP
5	Chief Operations Officer
6	Chief Commercial Officer
7	VP Maintenance Repair Organization
8	VP Corporate HRM
9	VP Customer Services
10	VP Internal Audit and Compliance
11	VP Legal Counsel & Corporate Secretariat
12	All Chief Pilots
13	Manager Boeing fleet training
14	Manager Airbus and Q400 fleet training
15	Manager MPL and cadet
16	Training coordinators and other supporting staff
17	Dir. IOCC
18	Dir. Airport Operations
19	Dir. In Flight Services
20	Ethiopian Civil Aviation Authority
21	All Area Managers
22	Manager Airport Services
23	Flight Operations Documentation
24	Mgr. Flight Dispatch and Movement Control
25	Mgr. FOPs Performance and Technical Support
26	Mgr. Crew Scheduling
27	All Aircraft Library
28	Training coordinators and other supporting stuff

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0.4.8 RETENTION AND DISPOSAL OF OBSOLETE DOCUMENTS AND RECORDS

Mgr. FOPS Performance & Technical Support is responsible for retention and disposal of Flt Ops documents and records.

Despite expiry of retention periods, Flt Ops managers and team leaders are responsible for the extension of retention period of documents and records required for ongoing or pending audits, investigations or lawsuits (including reasonably anticipated lawsuits) without damage and alteration until the matter is resolved. It is also their responsibility to permanently retain master copies of manuals they own with their source, reference and revision documents as historic records.

Head IT Services Operation is responsible for deletion of obsolete or expired electronic documents and records from ETG IT systems when notified.

Mgr. FOPS Performance & Technical Support is responsible to regularly check for timely deletion of obsolete or expired softcopy documents and records from ETG IT systems.

0.4.8.1 RETENTION OF FOM

FOM shall have a retention period of 5 years after its. It shall be kept safe and secured expiry in Flt Ops Documentation library with controlled access to a minimum period. Both soft and hard master copies of each document shall be safely retained in Mgr. FOPS Performance & Technical Support office as backups and historic records with their source, reference and revision documents.

Mgr. FOPS Performance & Technical Support shall take scheduled inventories of documents and records under their custody to sort expired ones for disposal.

If the document retention period has expired and the document is no more required, it shall be processed for disposal or deletion as appropriate.

0.4.8.2 RETENTION OF RECORDS (ECARAS 9.2.2.5)

The following documents will be kept in hard copy or soft copy (on the IT SERVER) until the specified minimum period mentioned below is met.

a) The following records will be kept for the period specified below:

I. Flight, duty and rest time records for 18 months.
II. Flight crew records for 2 Years.
III. Other AOC holder personnel for which a training program is required for 12 months.

b) The following flight documents will be retained for a period of 90 days:

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- Dispatch releases;
- Load sheet;
- Operational flight plans;
- Flight weather briefings;
- Original copy of all load and trim sheets;
- Passenger manifests;
- Fuel and oil records;
- Aircraft registration, date and flight number;
- Flight crew names and duty assignment;
- Fuel onboard at departure, en-route and arrival;
- ATD, ATA and flight time;
- Incidents/observations;
- Dangerous Goods Notification;
- A copy of all ATC flight plans;
- NOTAMS

c) Aircraft technical logbook for 2 years, including:

- journey records section
- maintenance records section

d) Flight recorder records for 60 days.
e) Quality system records for 5 years.
f) Dangerous Goods transport document for 6 months.
g) Dangerous Goods acceptance checklist for 6 months.
h) Records on cosmic and solar radiation dosage for 12 months.
i) Other records as may be required by ECAA.

j) Ethiopian Airlines shall maintain:

I. Current records which detail qualifications and training of all its employees and contact employees, involved in operational control, flight operations, ground operations and maintenance of the air operator.

II. Records for those employees performing crewmember or flight operations officer duties in sufficient detail to determine whether the employee meets the experience and qualification for duties in commercial air transport operations.

k) Flight Operations shall maintain records in a form and format that ensures readability, security and integrity of the records at all times and in a manner acceptable to ECAA.

l) After being retained for a minimum period mentioned above, the documents may be destroyed.

m) Flight Operations shall ensure the above operational records as subjected to standardized processes for:

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I. Identification
II. Legibility
III. Maintenance
IV. Retention and Retrieval
V. Protection, integrity and Security
VI. Disposal, Deletion (electronic records) and archiving.

All documents shall be easily identifiable, legible, easily maintainable. It is also required to be retainable, well protected and the deletion process has to be followed as per the Ethiopian Airlines IT Procedure Manual.

0.4.8.3 DISPOSAL OF FOM AND RECORDS

a) Flt Ops manuals and records shall be disposed in compliance with all relevant laws, regulations, and ETG standards in an environmentally friendly way.
b) Hardcopy documents shall be shredded down or recycled and it shall never be thrown to the trash.
c) Electronic documents and records shall be deleted in collaboration with the IT department. Documents that are held in electronic media may be overwritten or physically destroyed, but not placed in the trash.
d) Disposal of Dangerous Goods materials shall be as per the State Regulations and DG disposal instructions.
e) Sensitive documents and records containing confidential or personal information shall be disposed of securely in accordance with government regulations and ETG requirements
f) Respective managers and team leaders shall keep records of document and record disposal activities that includes the following information:
   - the document / record type
   - the disposal date
   - the disposal method
   - the person disposed the document
   - the authorizing personnel
   - the witness for the disposal

0.4.9 FLIGHT CREW RECORDS

a) The following flight crew certifications, qualifications, training and currency requirements shall be recorded and retained at the training department in hard and soft copy (Q-Pulse system). The crew scheduling shall ensure that each flight crew member prior to being assigned to duty, are qualified and current in accordance with the below listed items, and the flight crew members shall not accept a flight if not qualified for duty in accordance with the requirements as listed below.

I. Licenses/certification including eligibility to exercise privileges of pilot license/certificate in international operations (entire period of employment);

II. Specific pilot license/certification limitations;

III. Specific qualification (LVP, RVSM, EDTO etc.);

IV. Equipment qualification (TCAS/ACAS, GPWS/EGPWS, PBN, PBCS);

V. Recency-of-experience;

VI. Medical status, including Medical Certificate;

VII. Initial training, route training, route check, recurrent training, Proficiency check, line check and checking results;

VIII. Right seat qualification (as applicable);

IX. Type(s) of qualification (entire period of employment);

X. Airport and Route competence including special airports.

XI. Instructor/Examiner/Line Check airman qualification (as applicable);

XII. CRM/Human Factor training;

XIII. Dangerous Goods training;

XIV. Security training;

XV. Accrued flight time, duty time, duty periods and rest periods

b) All the training records will be kept in hard copies. The "Q-Pulse" digitalized system shall be used as a backup digital file.

c) It is the responsibility of Training Department and Manager Crew Scheduling to maintain and control records of:

I. Type(s) of qualification

II. Recency of experience

III. Medical status

IV. Right seat qualification

V. Airport and route competence

VI. Instructor/Examiner/ Check Airman qualification

VII. Fight time, duty time and rest time

d) Manager Crew Scheduling shall ensure that there is a due date alert system to be utilized by each Scheduler before scheduling any crew member for a flight duty especially when dual qualification applies for a particular flight crew member. He shall also, in close cooperation with each respective Chief Pilot, plan the renewal of all re-currency requirements.

e) Flight Operations shall ensure that at least the following records are kept for 18 months for each flight crew member: (ECARAS 8.12.1.15)

I. The start, duration and end of each flight duty period

II. Rest periods

III. Flight time

The data in the Crew Management System (CMS) as well as the backup digital file in the Information Service department shall serve as the overall back up for these files.

0.4.10 DISPATCHERS RECORDS

Mgr. Flight Dispatch shall keep all Flight Dispatchers' records this includes:

a) Initial training

b) Recurrent training

c) Familiarization flight records for the current year

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0.5 TERMS AND DEFINITIONS

When the following terms are used in this manual, they have the following meanings.

"Shall" means that the procedure is mandatory and must be carried out.

"Should" means that the procedure is recommended.

"May" means the procedure is optional.

"Note" is used when an operating procedure, technique etc. is considered essential to be emphasized.

"Caution" is used when an operating procedure, technique etc. may result in damage to equipment if not carefully followed.

"Warning" is used when an operating procedure, technique etc. may result in personal injury or loss of life if not carefully followed.

Accelerate Stop Distance Available (ASDA) – The length of the takeoff run available plus the length of stop way if provided.

Active Frost – is a condition when frost is forming. It occurs when airplane surface temperature is at or below 0°C (32°F) and at or below dew point

Aerial Work– Operation in which an aircraft is used in specialized services such as agriculture, construction, photography, surveying, observation and patrol, search and rescue, aerial advertisement etc.

Aerodrome – A defined area on land or water (including any buildings, installations and equipment) intended to be used either wholly or in part for the arrival, departure or movement of aircraft.

Aerodrome Operating Minima – The limits of usability of an aerodrome either for takeoff or landing, usually expressed in terms of visibility or runway visual range, decision altitude/height (DA/DH) or minimum descent altitude/height (MDA/H) and cloud conditions.

Airline – Any air transport enterprise offering or operating international scheduled air service.

Aircraft – Any machine that can derive support in the atmosphere from the reactions of the air other than the reactions of the air against the earth's surface.

Aircraft Flight Manual (AFM) – A manual associated with the certificate of airworthiness, containing limitations within which the aircraft is considered to be airworthy, instructions and information necessary to the flight crewmembers for the safe operations of the aircraft. (Flight Crew Operations Manual, Approved Flight Manual, MEL/CDL, etc.)

Air Operator Certificate (AOC) – A certificate authorizing an operator to carryout specified commercial air transport operation.

Anti-icing Fluid – a) mixture of water and Type I fluid, b) Premix Type I fluid, c) Type II fluid, Type III fluid, or Type IV fluid and d) mixture of water and Type II fluid, Type III fluid, or Type IV fluid.

> **NOTE**: Fluids mentioned in a) and b) must be heated to ensure a temperature of 60°C (140°F) minimum at the nozzle.

Authority – Is the Ethiopian or any other Civil Aviation Authority.

Commercial Air Transport Operation – An aircraft operation involving the transport of passengers, cargo or mail for remuneration or hire.

Contamination Check –check of airplane surfaces for contamination to establish the need for de-icing.

Crewmember – A person assigned by an operator for duty on an aircraft during flight.

Cruising Level – A level maintained during a significant portion of a flight.

Day Off – Periods available for leisure and relaxation free from all duties.

Deadhead Transportation – The period from the time a crewmember reports for the purpose of a positioning flight until the positioning flight is complete.

Dangerous Goods – Articles or substances, which are capable of posing significant risk to health, safety or property when transported by air.

Decision Altitude/Height (DA/DH) – A specified altitude/height in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

NOTE: 1) Decision altitude (DA) is referenced to mean sea level (MSL) and decision height (DH) is referenced to the threshold elevation.

2) The required visual reference means that the section of the visual aids or the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of positions in relation to the desired flight path.

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Destination Alternate – an alternate aerodrome to which an aircraft may proceed should it become impossible or inadvisable to land at the aerodrome of intended landing.

NOTE: The aerodrome from which an aircraft departs may also be an en-route or destination alternate for that flight.

Duty': Any task that a crew member is required to perform by the operator, including flight duty, administrative work, training, positioning, and standby.

'Duty period: A period which starts when a crew member is required by an operator to report for or to commence a duty and ends when that person is free from all duties.

Enroute Alternate – An aerodrome at which an aircraft would be able to land after experiencing an abnormal or emergency condition while en-route.

Examiner (Aeroplane): A person who is qualified, and permitted, to conduct an evaluation in an aeroplane, in a flight simulator training device for a particular type of aeroplane, for a particular AOC holder.

Examiner (Simulator): A person who is qualified to conduct an evaluation, but only in a flight simulation training device for a particular type of aircraft, for a particular AOC holder

Flight Crewmember – a licensed crewmember charged with duties essential to the operation of an aircraft during flight.

Flight Duty Period (FDP)': A period which commences when a crew member is required to report for duty, which may include a flight or a series of flights and finishes when the aircraft finally comes to rest and the engines are shut down, at the end of the last flight on which he/she acts as a crew member.

Flight Plan – Specified information provided to air traffic services units, relative to an intended flight or portion of a flight.

Flight Recorder – Any type of recorder installed in the aircraft for the purpose of complementing accident/incident investigation.

Flight Time – The total time from the moment an aircraft first moves under its own power for the purpose of taking off until the moment it comes to rest at the end of the flight.

NOTE: Flight as here defined is synonymous with the term "block to block" or "chock to chock" time in general usage which is measured from the time the aircraft moves from the loading point until it stops at the unloading point.

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General Aviation Operation – An operation other than commercial air transport or aerial work operation.

Instrument Meteorological Conditions (IMC) – Meteorological condition expressed in terms of visibility, distance from cloud and ceiling less than the minimum specified for visual meteorological condition.

Isolated Aerodromes – aerodromes are typically considered isolated, if the fuel required (alternate + final reserve fuel) to the nearest alternate aerodrome is more than fuel to fly for 2 hours at normal cruise consumption above the destination aerodrome, including final reserve fuel.

Landing Distance Available (LDA) – The length of the runway, which is declared available and suitable for the ground run of an airplane landing.

Local Day – A 24-hour period commencing at 00:00 local time.

Local Night - A period of 8 hours falling between 22:00 hours and 08:00 hours local time.

Master Minimum Equipment List (MMEL) – A list established for a particular aircraft type by the manufacturer with the approval of the State of manufacturer containing items one or more of which is permitted to be unserviceable at the commencement of the flight. The MMEL may be associated with special operating conditions, limitations or procedures.

Maximum Mass – Is maximum certificate takeoff mass.

Minimum Descent Altitude/Height (MDA/H) – A specified altitude /height in a non-precision or circling approach, below which descent may not be made without visual reference.

Minimum Equipment List (MEL) – A list that provides for the operation of an aircraft, subject to specified conditions, with particular equipment inoperative, prepared by the operator and approved by the state of registry in conformity with, or more restrictive than the MMEL established for the aircraft type.

Moderate and Heavy Freezing Rain -

Night – The hours between the end of evening civil twilight and the beginning of morning civil twilight or such other period, between sunset and sunrise, as may be prescribed by the appropriate authority.

Night Duty: A flight duty period encroaching on any portion of the period between 02:00 and 04:59 hours in the time zone to which the crew is acclimatized.

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Obstacle Clearance Altitude/Height (OCA/H) – The lowest altitude/height above the elevation of the relevant runway threshold or above the aerodrome elevation as applicable used in establishing compliance with appropriate obstacle clearance criteria.

Operational Control – The exercise of authority over the initiation, continuation, diversion or termination of a flight in the interest of safety regulatory and efficiency of the flight.

Operational Flight Plan – The operator's plan for the safe conduct of the flight based on considerations of aircraft performance, other operating limitations and relevant expected conditions on the route to be followed and the aerodrome concerned.

Operator - A person, organization or enterprise engaged in or offering to engage in an aircraft operation.

Pilot-In-Command (Captain)- The pilot responsible for the operation and safety of the aircraft during flight time.

Pressure-Altitude - An atmospheric pressure expressed in terms of altitude, which corresponds to that pressure in the Standard Atmosphere.

Rating - An authorization entered on or associated with a license and forming a part thereof, stating special conditions, privilege or limitations pertaining to such license.

'Rest Facility': A bunk, seat, room, or other accommodation that provides a crew member with a sleep opportunity:

'Class 1 Rest Facility': A bunk or other surface that allows for a flat sleeping position and is located separately from both the flight deck and the passengers cabin in an area that is temperature controlled, allows the crew member to control light, and provides isolation from noise and disturbance;

'Class 2 Rest Facility': A seat in an aircraft cabin that allows for a flat or near flat sleeping position, which is separated from passengers at least by a curtain to provide darkness and some sound mitigation, and is reasonably free from disturbance by passengers or crew members;

'Class 3 Rest Facility': A seat in an aircraft cabin or flight deck that reclines at least 40 degrees, provides leg and foot support and is separated from passengers by at least a curtain to provide darkness and some sound mitigation, and is not adjacent to any seat occupied by passengers.

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'Rest Period': A continuous and defined period of time, subsequent to and/or prior to duty, during which a crew member is free of all duties

Regular Aerodrome - An aerodrome which may be listed in the flight plan as an aerodrome of intended landing.

Runway Visual Range - The range over which the pilot of an aircraft on the center line of runway can see the runway surface markings or the lights delineating the runway or identifying its center line.

Safety Harness - Includes shoulder strap and a seat belt that may be used independently.

Standby': A defined period of time during which a crew member is required by the operator to be available to receive an assignment for a flight, positioning or other duty without an intervening rest period.

State of the Operator - The state in which the operator's principal place of business is located or, if there is no such place of business, the operator's permanent residence.

State of Registry - The State on whose register the aircraft is entered.

Synthetic Flight Trainer - Any one of the following three types of apparatus in which the flight conditions are simulated on the ground.

A Flight Simulator - which provides an accurate representation of the flight deck of a particular aircraft type to the extent that the mechanical, electrical, electronic, etc., aircraft systems control functions, the normal environment of flight crew members, and the performance and flight characteristics of that type of aircraft are realistically simulated.

A Flight Procedures Trainer - which provides a realistic flight deck environment, and which simulates instrument responses, simple control function of mechanical, electrical, electronic, etc., aircraft systems, and the performance and flight characteristics of aircraft of a particular class.

A Basic Instrument Flight Trainer - which is equipped with appropriate instrument, and which simulates the flight deck instrument of an aircraft in instrument flight conditions.

Takeoff Alternate – An alternate aerodrome at which an airplane can land should this become necessary shortly after takeoff and it is not possible to use the aerodrome of departure.

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Take-Off Run Available (TORA) - The length of the runway which is declared available and suitable for the ground run of an airplane taking off. This in most cases corresponds to the physical length of the runway.

Visual Meteorological Conditions (VMC) - Meteorological conditions expressed in terms of visibility, distance from c loud and ceiling, equal to or better than specified minima.

0.6 ABBREVIATIONS

("A")	Lead Cabin Crew
("B")	Cabin Crew in charge of forward galley
("C")	Cabin Crew in charge of First Class
("C2")	Cabin Crew in charge of Business Class
("D1")	Cabin crew in charge of Economy Class and Duty free sales
("D2")	Cabin Crew in charge of Economy Class and duty free sales
("E1"/"E")	Cabin Crew in charge of Economy Class Galley
("E2")	Cabin Crew in charge of Economy Class Galley
("F")	Senior Cabin Crew in charge of Business & Economy Class
A&P	Airframe & Power Plant
A/C	Aircraft
Act.	Actual
Altn	Alternate
AMSL	Above mean Sea Level
AOM	Aerodrome Operated Minima
App	Approach
Arr	Arrival
ATA	Actual Time of Arrival
ATC	Air Traffic Control
ATD	Actual Time of Departure
ATIRF	Air Traffic Incident Reporting Form
ATO	Actual Take-off
ATS	Air Traffic Services
AVO	Avionics

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AVT	Audio Visual Trainer
B/C	Business Class
BLK	Block
C/C	Cabin Crew
CAPT	Captain
CBT	Computer Based Trainer
CEO	Chief Executive Officer
CFIT	Controlled Flight Into Terrain
Cont.	Contingency
CRM	Crew Resource Management
D/S	Distance/Speed
Decl.	Declaration
Dist.	Distance
DME	Distance Measuring Equipment
DPM	Department Manager
DPT	Department
DVM	Division Manager
E/C	Economy Class
Eng	Engine
Est.	Estimated
ECAA	Ethiopian Civil Aviation Authority
ETA	Estimated Time of Arrival
ETAF	Ethiopian Air Force
ETD	Estimated Time of Departure
ETO	Estimated Take-off
EO	Executive Officer Operations
FAA	Federal Aviation Administration
F/A	Flight Attendant
F/E	Flight Engineer
F/O	First Officer
F/P	Flight Purser

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FDRT	Flight Dispatcher Recurrent Training
FIR	Flight Information Region
FL	Flight Level
FLT. OPS	Flight Operations
FMC	Flight Management Computer
FMR	Flight Movement Report
FMS	Flight Management System
FOM	Flight Operations Policy Manual
FR/C	First Class
Freq.	Frequency
FRT	Freight
Ft	Fleet
FWD	Forward
Gen	General
GMT	Greenwich Mean Time
G/S	Glide Slope
GS	Ground Speed
GTOW	Gross takeoff weight
HF	High Frequency
HRS (Hrs.)	Hours
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
Inst.	Instrument
IOCC	Integrated Operation Control Center
KGS (Kgs)	Kilograms
L/C/C	Lead Cabin Crew
LAT	Latitude
LAV	Lavatory
LDG (Ldg.)	Landing
LOC	Localizer
Long.	Longitude

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LW	Landing Weight
Max	Maximum
MBS (Mbs.)	Millibars
MEL	Minimum Equipment List
MGT	Management
Min	Minimum
MMEL	Master Minimum Equipment List
Mnvr	Maneuver
NM	Nautical Mile
NO.	Number
NOTAM	Notice to Airmen
OPT	Onboard Performance Tool
ORL	Original
P3	Employee Termination Form
PA	Passenger
PCM	Passenger Count Memo
PIC	Pilot-In-Command
PLD	Payload
POB	Persons On Board
PRM	Persons with Reduced Mobility
Proc	Procedure
PTS	Pilot Training School
RA	Resolution Advisor
Reg.	Registration
RTE (Rte.)	Route
RWY	Runway
SAT(S)	Satisfactory
Sched.	Scheduled
SDT	Set-up Departure Time
SIM (SIMUL)	Simulator
SUPV.	Supervisor

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SVCS	Services
T/O	Take-off
TA	Traffic Advisor
TAS	True Air Speed
TBD	To Be Defined
TLCS	Team Leader Cabin Services
TQM	Total Quality Management
TRK	Track
TRT	Technical Recurrent Training
TTL	Total
TU/C	Tourist Class (Economy class)
UNSAT(U)	Unsatisfactory
UTC	Co-ordinate Universal Time
VFR	Visual Flight Rule
VHF	Very High Frequency
VMC	Visual Meteorological Condition
VOR	VHF Omni directional Range
VP	Vice President
VTR	Video Transmitter Recorder
WT(wt)	Weight
XTRA	Extra
ZFW	Zero Fuel Weight

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SECTION 1.1. ORGANIZATIONAL CHARTS

SECTION 1.1. ORGANIZATIONAL CHARTS

1.0 PURPOSE

The purpose of this organizational chart procedure is to establish guidelines for displaying a divisional or departmental structure for publishing announcements/processes of organizational changes.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

All employees under Flight Operations Division are affected.

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1. Organizational Charts shall be provided in a timely manner to identify and communicate the departmental structures within the Division as they happen.

4.2. Organizational Charts are reviewed and approved by Human Resource Management Division.

4.3. Organizational Charts are maintained, coordinated, published and distributed to all concerned personnel within the Enterprise.

4.4. Ethiopian delegates the authority and assign responsibility for the management and supervision of specific areas of the organization relevant to the flight operations management system.

5.0 DEFINITION

5.1. Organizational Chart is a visual illustration of a divisional structure which includes function name, the assigned title, professional positions, etc.

6.0 RESPONSIBILITY

6.1. The VP Flight Operations shall ensure that:

6.1.1. Accurate and current organizational chart(s) is/are maintained.

6.1.2. The paperwork for employee changes is quickly processed and that the current organizational charts accompany the paperwork to the Human Resource Management (HRM) Division for review and approval.

6.1.3. The corrected or redrawn organizational chart(s) is/are forwarded to HRM for review and publication

7.0 PROCEDURE

7.1. When a new person(s) is/are hired, transferred, or other major changes are being made, a new organizational chart is required.

7.2. Each time there is a change involving title, addition, deletion, or change in personnel, revision of the organizational chart is required.

7.3. The head of the position shall standardize the organizational chart prior to final review, approval and publication.

7.4. The approved chart shall be published and distributed per the Distribution List.

7.5. Temporary delegation of authority will be granted to direct subordinates only when the delegating supervisor is to be away from his normal place of assignment. Such temporary authority may be delegated to only one individual at a time and becomes automatic when the subordinate is appointed "Acting (title)".

7.6. Flight Operations follow the below procedure for delegation of duties when the operational manager, director or Vice President is unable to be on their normal place of assignment;

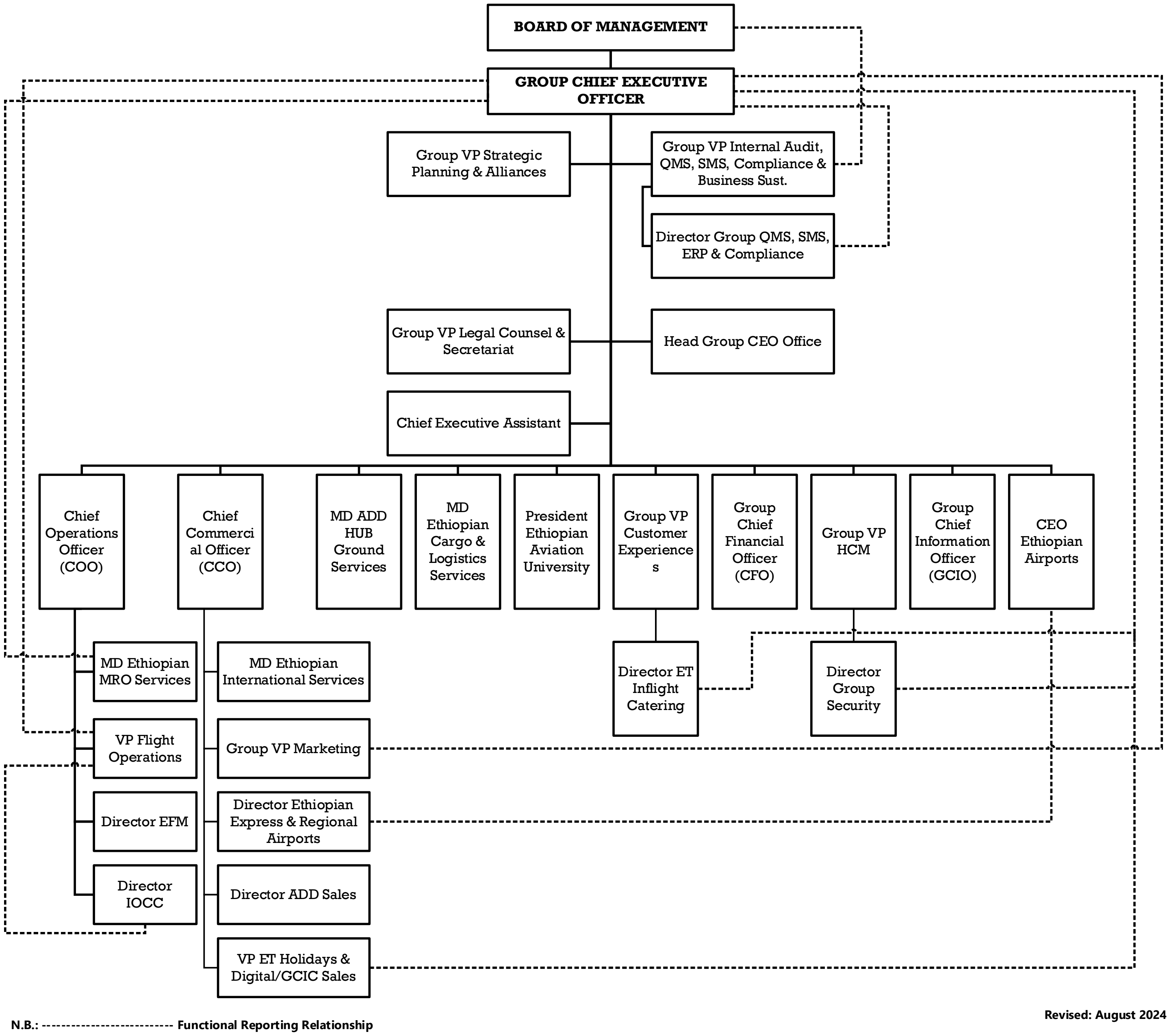
a) The delegating supervisor will communicate and brief his subordinate if there is any pending issue to be completed, and the basic functions that he/she has to perform during his/her absence.

b) The delegating supervisor shall make his/her email or work-related contacts to auto reply for the sender to contact his/her subordinate on his/her behalf.

c) Then the delegating supervisor will fill the Temporary Delegation of Authority format and the delegating supervisor and the VP Flight Operations will sign on the format and will be handed over to the acting personnel.

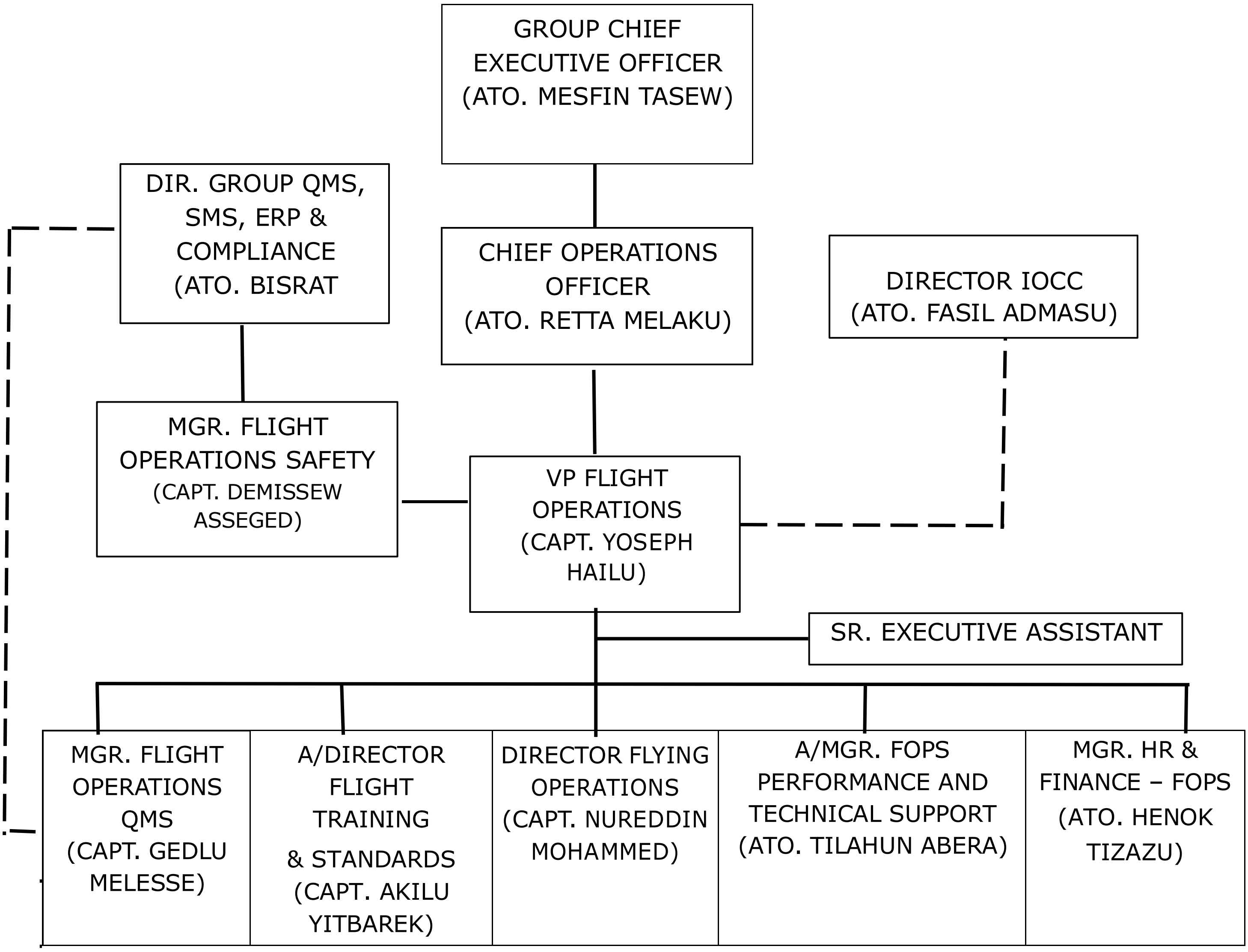
SECTION 1.1. ORGANIZATIONAL CHARTS

7.7 MANAGEMENT (UP TO DIRECTOR LEVEL) STRUCTURAL CHART

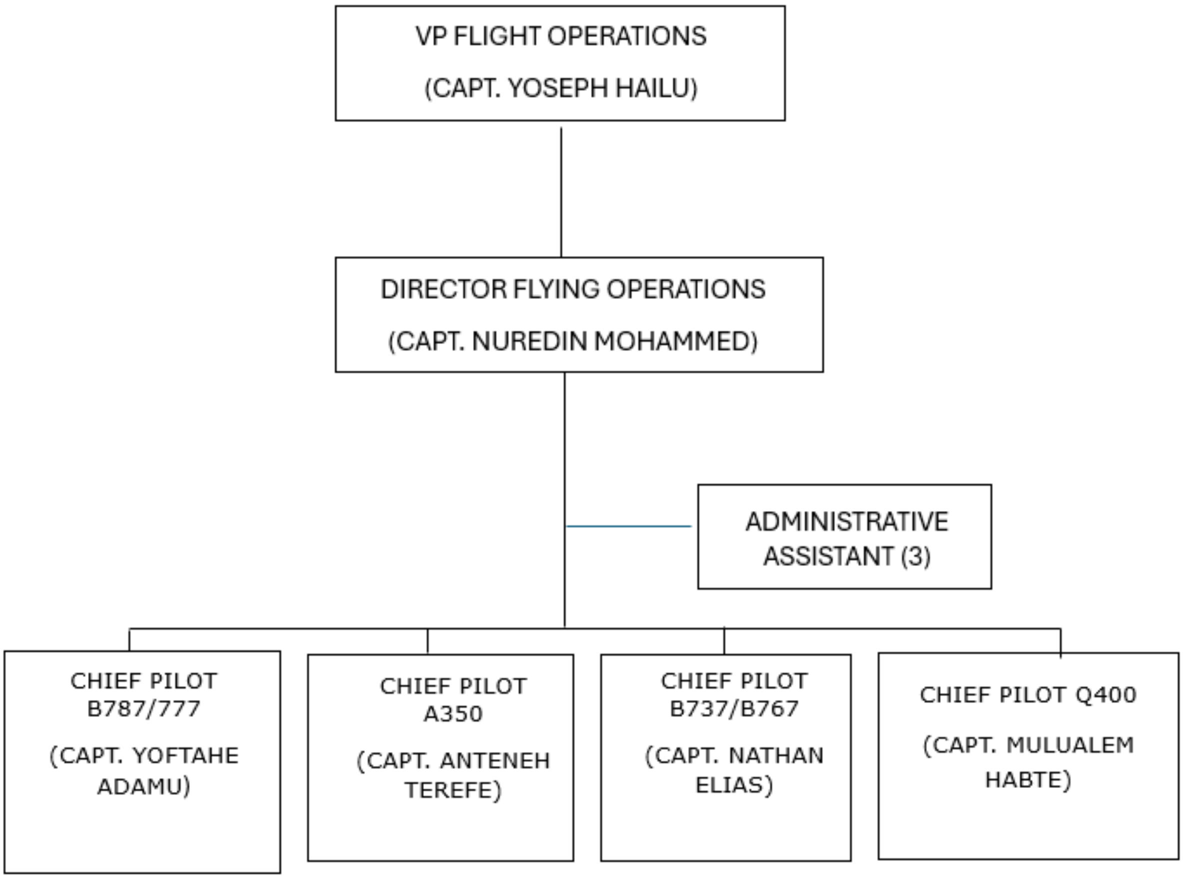


SECTION 1.1. ORGANIZATIONAL CHARTS

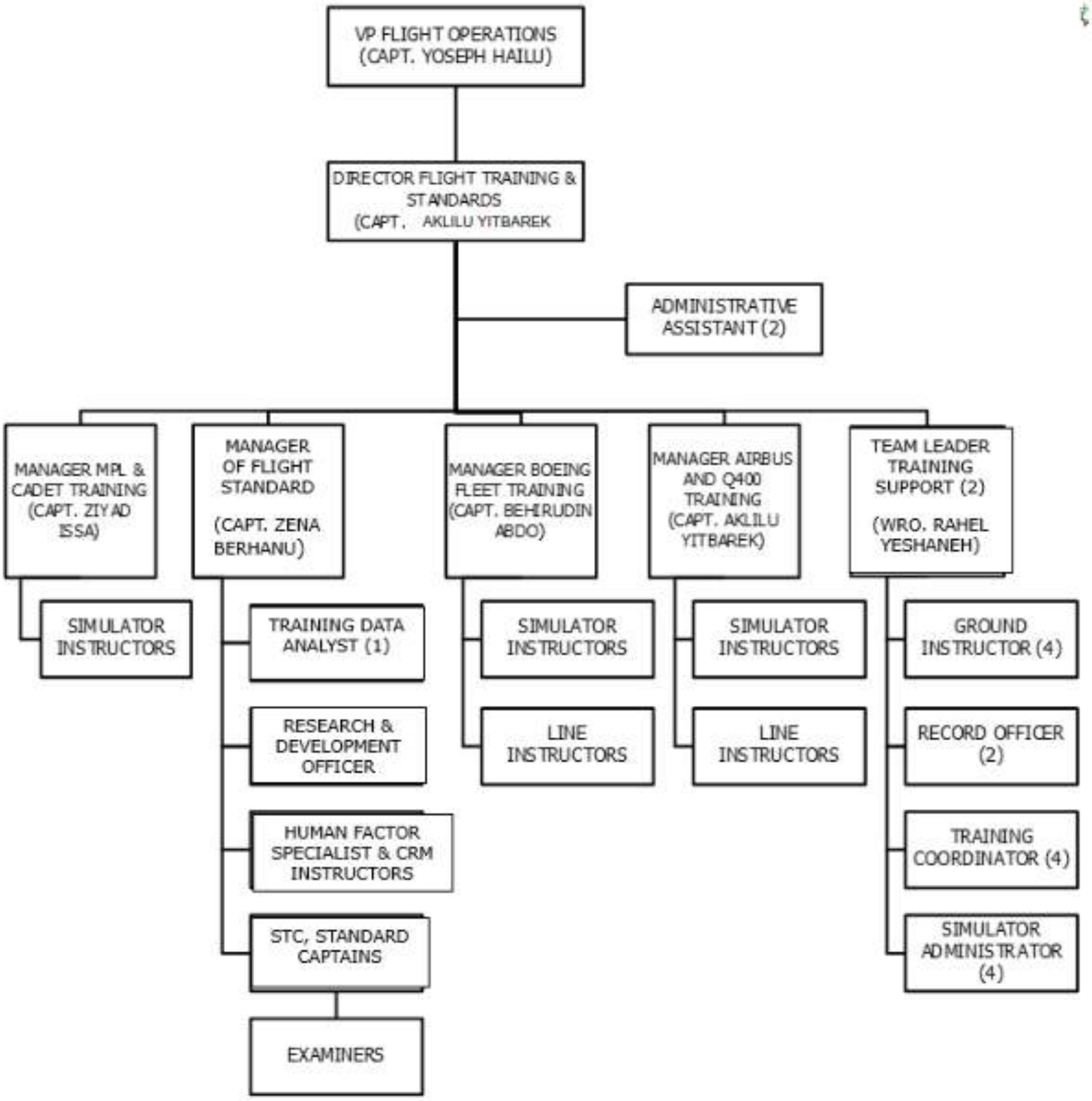
7.7.1 VP FLIGHT OPERATIONS STRUCTURE



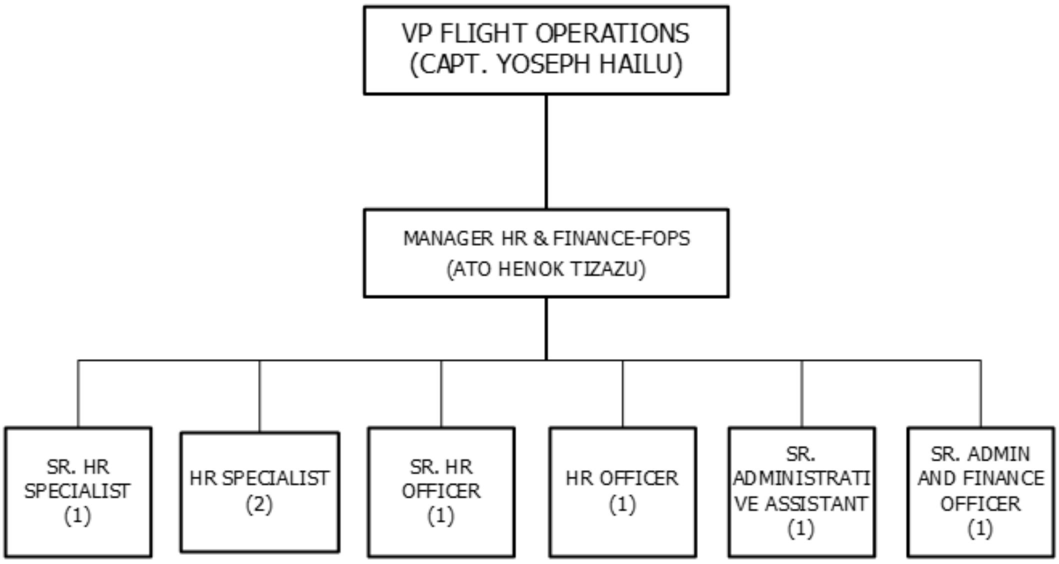
7.7.2 DIRECTOR FLYING OPERATIONS STRUCTURE



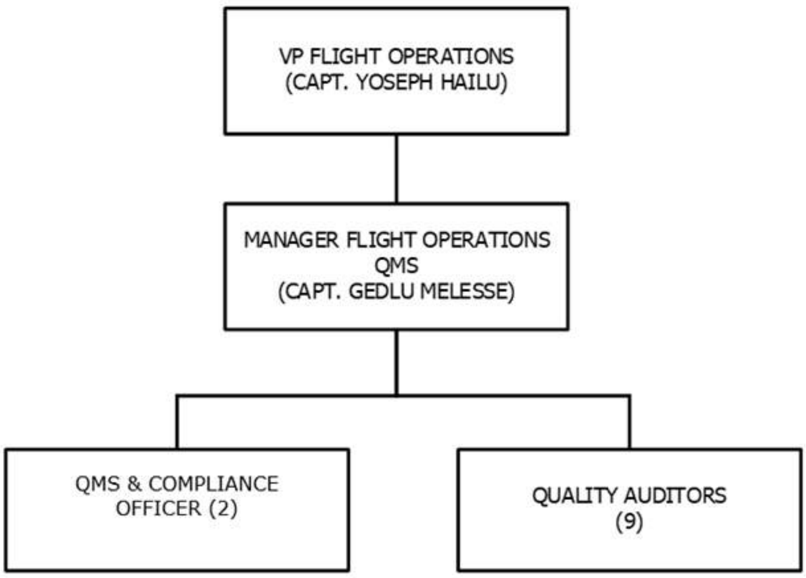
7.7.3 DIRECTOR FLIGHT TRAINING AND STANDARD'S STRUCTURE



7.7.4 MANAGER HR & FINANCE – FLIGHT OPERATIONS STRUCTURE



7.7.5 MANAGER FLIGHT OPERATIONS QMS STRUCTURE



SECTION 1.2. DUTIES AND RESPONSIBILITIES

SECTION 1.2. DUTIES AND RESPONSIBLITIES

1.0 PURPOSE

The purpose of this duties and responsibilities procedure is to outline the duties and responsibilities of management personnel in Flight Operations Division and to provide a summary of the duties and responsibilities assigned to each divisional, departmental or sectional head as outlined in the Organizational Chart.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Personnel involved in the respective division, department or section

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1 All personnel under flight operations division have to complete the company indoctrination curriculum approved by ECAA, appropriate to that person's duties and responsibilities. (ECARAS 9.2.2.9)

4.2 The indoctrination curriculum shall include training in knowledge and skills related to human performance, including coordination with other AOC personnel. (ECARAS9.2.2.9)

4.3 Flight Operations Management that are required for the performance of functions relevant to the safety and security of Aircraft operations and Flight Crew Members are required to have the appropriate level of knowledge, skills, training and experience for their specific position.

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

6.1 CHIEF EXECUTIVE OFFICER (CEO)

NOTE: The CEO reports to the Board of Management

6.1.1. Serves as the Company's Chief Executive Officer responsible for directing the business in a way which will produce maximum profit and return on invested capital consistent with the best interest of Ethiopia, passengers, employees and general public.

6.1.2. Establishes current and long-range objectives, plans and policies, interprets and applies policies of the Board of Management.

6.1.3. Co-ordinates and maintains the various functions and operations of the company.

6.1.4. Establishes policies that ensure safety, quality & security management systems are sufficiently resourced for smooth & safe operation of the company.

6.1.5. The CEO has an overall accountability including the authority and control of resources necessary to finance, implement and enforce policies and procedures within the operation.

6.2 CHIEF OPERATING OFFICER (COO)

NOTE: The COO reports to the Chief Executive Officer

6.2.1 Chief Operating Officer is responsible for defining future directions and strategies of ETHIOPIAN Operations and customer services.

6.2.2 Provide services to Enterprise through the development and implementation of objectives, plans and policies covering the activities of the A/C Maintenance, Overhaul and related Quality control services, Flight Operations and Customer Service activities.

6.2.3 Establishes medium and long-range objectives for his or her area of assignment, reviews and plans for their implementation, reviews policies of all Divisions under his/her supervision and ensures conformance with overall company objectives, plans and policies.

6.2.4 Develops, implements and maintains applicable processes and procedures, establishes policies that ensure safety, quality & security and management systems which are sufficiently resourced for smooth & safe operation of the company.

6.3 VICE PRESIDENT FLIGHT OPERATIONS (VP FLT OPS)

NOTE: The VP FLT OPS reports to the Chief Operating Officer

BASIC FUNCTIONS

6.3.1. Provides service to the Airline through the development of flight operations policies and procedures designed to provide safe and economic operation of aircraft.

6.3.2. Determine staff requirements to obtain maximum utilization of flight operations personnel through the system.

6.3.3. Participates in the evaluation of flight equipment proposed for use by the airline.

6.3.4. Directs the training standard of cockpit crew.

6.3.5. Directs the overall activities of flying operations.

6.3.6. Directs the activities of operational planning, flight dispatching flight movement control and crew scheduling.

6.3.7. Implements procedures which will safeguard Ethiopian customers and employees' safety and property, protect Ethiopian property, asset and equipment and ensure its efficient utilization for the purpose intended.

6.3.8. Establishes procedures, manages and control methods necessary for the protection, safety and security outcomes of Ethiopian flight operations.

6.3.9. Establishes Safety and quality as core value in all Ethiopian operations and give due attention and immediate action to conditions that may affect safety and quality.

6.3.10. Approves crew hotel accommodation contracts on behalf of Ethiopian Airlines.

6.3.11. Ensures ongoing compliance with conditions and restrictions of the Air Operator Certificate (AOC) as outlined hereunder.

a) Ethiopian shall hold an AOC for the terms and conditions being conducted. (ECARAS9.1.1.4)

b) Ethiopian shall at all times continue in compliance with the AOC terms, conditions of issuance and maintenance requirements in order to hold the certificate. Failure to comply with may result in the revocation or suspension of the AOC.

c) AOC will consist of two documents: (ECARAS 9.1.1.7)

I. A one-page certificate for public display signed by the AOC, and

II. Operations specifications containing the terms and conditions applicable to the AOC holder's certificate.

d) The Authority (ECAA) will issue an AOC which will contain: (ECARAS 9.1.1.7)

I. The name and location of the AOC holder

II. The date of issue and period of validity for each page issued

III. Description of the type of operations authorized

IV. The type(s) of aircraft authorized for use

V. The authorized areas of operations or routes

VI. Exemptions, deviations and waivers (listed by name);

VII. Other special authorizations, approvals and limitations issued by the Authority, to include, as applicable:

(a) Low visibility operations (LVO);

(b) CAT II and/or III approaches;

(c) Automatic landing, head-up display (HUD) or equivalent displays, vision systems operations and associated operational credit(s) granted (if such systems are used to gain operational benefit);

(d) Use of GPS to conduct any approach;

(e) ETOPS/EDTO, as applicable, includes the applicable threshold/maximum diversion times established for each particular aircraft and engine combination;

(f) RVSM operations;

(g) MNPS/NAT HLA operations;

(h) Area of Magnetic Unreliability (AMU);

(i) Basic RNAV/RNP operations;

(j) AR navigation specifications for PBN operations;

(k) Performance-Based Communication and Surveillance (PBCS) operations;

(l) Transport of dangerous goods as cargo;

(m) Electronic Flight Bag (EFB) operations.

e) Concerning duration of an AOC issued by the Authority, it is effective for 12 months unless the Authority amends, suspends, revokes or otherwise terminates the certificate, or the AOC holder surrenders it to the Authority, or the AOC holder suspends operations for more than 60 days. (ECARAS 9.1.1.8)

f) Ethiopian shall make application for renewal of an AOC at least 30 days before the end of the existing period of validity. (ECARAS 9.1.1.8)

g) Ensures compliance with:

I. Conditions and restrictions of AOC,
II. Applicable regulatory requirements and
III. Standards established by Ethiopian.

h) Ensures that the Air Operator Certificate (AOC) issued by Ethiopian Civil Aviation Authority is valid and current. Ascertains the AOC and all its associated documents shall define the scope of the current operations and identify senior managers responsible for AOC function.

6.3.12. PERSONALLY PERFORMED RESPONSIBILITIES

a) Approves policies governing flight dispatch (Operational Planning) communication and flight crew activities and plans for the efficient utilization of personnel in these functions.

b) Evaluates and recommends new aircraft and flight equipment for use on present or proposed aircraft, to ensure that the airline keeps pace with industry development and avoids economic or safety penalties.

c) Directs daily operations of the division and ensures quality requirements are met and maintained in accordance with the regulatory authorities' such as, ICAO, EASA, FAR, IATA and etc. regulation and the good standard practices dictated by aviation industry and internal standards of ETHIOPIAN.

d) Has the authority, responsibility & accountability for the management and supervision of all flight operations activities.

e) Reviews and approves all flying procedures, manuals and emergency procedures.

f) Reviews and approves all communications procedures covering ground and flying duties.

g) Ensures the operations' risk management policy is implemented in his respective areas of responsibility and risk management is incorporated in the planning & management processes.

h) Ensures that the risk register (database) is developed in his respective operations which includes prioritized risks, appropriate risk response, defined risk mitigation processes and measures that monitor effectiveness

i) Liaisons with the ECAA, other regulatory authorities, Original Equipment Manufacturers, and other external entities.

j) Reviews and approves all Airports in regard to facility, equipment and condition of field for a safe operation.

k) Oversees/delegates the investigation of accidents whenever flight equipment is involved in an accident causing damage to equipment or injury to passengers or crew.

l) Coordinates charter and non- routine flights with Marketing division and the Chief Executive Officer's Office.

m) Serves as chairperson of "flight delay meeting" and reports monthly on time performance to management committee meeting.

n) Approves payments, requests for expenditures, expense reports per limits of authority.

o) Approves new employees' pay authorization, authority to terminate employees, acceptance of resignation and acceptance of leave without pay letters of the division's employees.

p) Reviews and approves operating & capital budget, compiled goals and objectives, compiled periodic achievement reports on goals and objectives, compiled cost reduction plan and its periodic achievement reports and compiled monthly activity report of the division before directed to the appropriate organ of Ethiopian.

q) Handles grievances of employees of the division and directs remedial actions to minimize the negative effects.

r) Approves trip evaluation requests of the division before approval by the Chief Operating Officer.

s) Participates in regular and AD HOC COO and staff meetings.

t) Ensures that Flight Operations' employee shall comply with all policies, procedures and regulations specified for the position in Ethiopian Airlines Security Manual Chapter-One under the topic of "Assignment of responsibilities on Aviation Security."

u) Monitors the overall safety and security activities and reports deviations with action taken to Safety and QA and Corporate Security.

v) Ensures procedures and develops governing flight deck security and personnel are properly trained to execute flight deck security procedures.

w) Performs other duties from time to time as may be assigned by the Chief Operating Officer.

x) Has the authority, responsibility and is accountable to perform functions relevant to the safety and security of A/C operations in areas of the flight operations.

6.4 DIRECTOR FLYING OPERATIONS

NOTE: Director Flying Operations reports to VP Flight Operations

BASIC FUNCTIONS

6.4.1 Directs jet and turbo prop aircraft operations.

6.4.2 Directs the planning development and continued maintenance of the overall pilots with operational requirements of the airline.

6.4.3 Maintains company standards related to appearance, personnel conducts and the technical performance of cockpit crew personnel as well as staff and reliable operation of flights.

6.4.4 Is responsible for accident investigation/prevention and fight safety programs.

6.4.5 Approves requests of operational expenditures related to scheduled and non-scheduled flights.

6.4.6 Develops updates and maintains, operations manuals and programs and policies and procedures of the division.

6.4.7 Maintains close relationship with aircraft operators, manufacturers and keep abreast the latest technological innovations.

6.4.8 Monitors the implementation of fuel conservation, directs the fuel monitoring program and information management system of the department.

6.4.9 Responsible for the dissemination of safety Critical Operations information to appropriate personnel including as a minimum flight crew Bulletins or Directives etc.

6.4.10 Reviews and recommends all flying operations procedures, manuals, power charts and emergency procedures.

6.4.11 Guides and advises subordinates and flight crew members of their responsibilities for overall conduct of a flight including the maintenance of customer and cockpit relations, understanding of operating and administrative policies and procedures, and need for adherence to established standards.

6.4.12 Monitors daily flight Captain's Reports and their compilation, debriefing reports and line report; directs required investigation by subordinates into facts relating to delays, damages, incidents and other non-routine operations and follows-up for the implementations of recommended actions.

6.4.13 Addresses cockpit crew on technical, policy, management and administrative matters as deemed necessary.

6.4.14 Ensures the consistent implementation of the airline safety & Security policy and procedures by reporting deviations to safety or security offices as appropriate; initiating and taking preventive and corrective actions to avoid the recurrence of the deviation; arranging appropriate safety and security awareness/trainings and updating records of employees to ensure they are current and qualified; and ensuring the serviceability of equipment's to prevent safety risks.

6.5 DIRECTOR FLIGHT TRAINING AND STANDARDS

NOTE: The Director Flight Training & Standards reports to the VP Flight Operations

BASIC FUNCTIONS

6.5.1 Directs both intermediate and advanced (3rd and 4th training phase) multi crew pilot license (MPL) and flight crew training.

6.5.2 Monitors and observes the delivery of training development and research for the advancement of training.

6.5.3 Directs the planning, development and continued training of the entire pilot consistent with the operational requirements of the airline.

6.5.4 Monitors cockpit crew proficiency, line training and final checkout of pilots on new equipment.

6.5.5 Manages the training for cockpit crew consistent with the operational requirements of the Airline.

6.5.6 Monitors the functioning of the AB-INITIO pilot training school and provides recommendations where applicable.

6.5.7 Develops, updates and maintains training manuals, Operation Manuals, programs, policies and procedures for the department.

6.6 CHIEF PILOT A350

NOTE: Chief Pilot A350 reports to the Director Flying Operations.

BASIC FUNCTIONS

6.6.1 Plans and monitors the activities of A350 cockpit crew personnel all phases of their designated duties, the safe and reliable conduct of flights and maintenance of company standards related to appearance and personal conduct.

6.6.2 Plans training in consultation with Manager Recurrent & Transition Training to upgrade, maintain and improve proficiency levels of A350 cockpit crew personnel.

6.6.3 Coordinates with Team Leader Crew Scheduling in Flight Control Section to assign student- Captains, First Officers to various flights; follows-up their progress and participates in final check-out.

6.6.4 Coordinates with Manager Recurrent & Transition Training in establishing guiding principles for CPT/Simulator and Flight Training.

6.6.5 Coordinates with other departments on operational, scheduling, airport (airfield) and equipment matters for efficient operation.

6.6.6 Ensure that Air Operator Certificate (AOC) issued by Ethiopian Civil Aviation Authority is valid and current. Ascertains the AOC and all its associated documents shall define the scope of the current operations, and the compliance is thereof.

6.6.7 Adhere consistently to the airline’s safety and security policy and procedures through identification and mitigation of operational hazards and related security risks and implements all precautionary action/measures/processes etc.

6.6.8 Determines the suitability of airports and equipment for efficient and smooth operations.

6.6.9 Performs periodic audit of training organizations including contracted training provisions.

6.6.10 Performs periodic audit of pilot records.

6.6.11 Ensures the currency and qualification of cross qualified pilots as appropriate.

6.6.12 Assists in all investigations of aircraft incidents.

6.6.13 Performs other duties as may be assigned from time to time by the Director of Flying Operations.

6.6.14 Develop and recommend policies and procedures related to A350 aircraft operational activities.

6.7 CHIEF PILOT B777/787

NOTE: Chief Pilot B777/787 reports to the Director Flying Operations.

BASIC FUNCTIONS

6.7.1 Plans and monitors the activities of B777/787 cockpit crew personnel all phases of their designated duties, the safe and reliable conduct of flights and maintenance of company standards related to appearance and personal conduct.

6.7.2 Plans training in consultation with Manager Recurrent & Transition Training to upgrade, maintain and improve proficiency levels of B777/787 cockpit crew personnel.

6.7.3 Coordinates with Team Leader Crew Scheduling in Flight Control Section to assign student- Captains, First Officers to various flights; follows-up their progress and participates in final check-out.

6.7.4 Coordinates with Manager Recurrent & Transition Training in establishing guiding principles for CPT/Simulator and Flight Training.

6.7.5 Coordinates with other departments on operational, scheduling, airport (airfield) and equipment matters for efficient operation.

6.7.6 Ensure that Air Operator Certificate (AOC) issued by Ethiopia Civil Aviation Authority is valid and current. Ascertains the AOC and all its associated documents shall define the scope of the current operations, and the compliance is thereof.

6.7.7 Adhere consistently to the airline's safety and security policy and procedures through identification and mitigation of operational hazards and related security risks and implements all precautionary action/measures/processes etc.

6.7.8 Determines the suitability of airports and equipment for efficient and smooth operations.

6.7.9 Performs periodic audit of training organizations including contracted training provisions.

6.7.10 Performs periodic audit of pilot records.

6.7.11 Ensures the currency and qualification of cross qualified pilots as appropriate.

6.7.12 Assists in all investigations of aircraft incidents.

6.7.13 Performs other duties as may be assigned from time to time by the Director Flying Operations.

6.7.14 Develops and recommends policies and procedures related to B777/787 aircraft operational activities.

6.8 CHIEF PILOT B737/767

NOTE: Chief Pilot B737/767 reports to the Director Flying Operations.

BASIC FUNCTIONS

6.8.1 Plans and monitors the activities of B767/737 cockpit crew personnel all phases of their designated duties, the safe and reliable conduct of flights and maintenance of company standards related to appearance and personal conduct.

6.8.2 Plans training in consultation with Manager Recurrent & Transition Training to upgrade, maintain and improve proficiency levels of B767/737 cockpit crew personnel.

6.8.3 Coordinates with supervisor Crew Scheduling in Flight Control Section to assign student-captains, First Officers to various flights; follows-up their progress and participates in final check-out.

6.8.4 Coordinates with Manager Recurrent & Transition Training in establishing guiding principles for CPT/Simulator and Flight Training.

6.8.5 Coordinates with other departments on operational, scheduling, airport (airfield) and equipment matters for efficient operation.

6.8.6 Ensure that Air Operator Certificate (AOC) issued by Ethiopia Civil Aviation Authority is valid and current. Ascertains the AOC and all its associated documents shall define the scope of the current operations and the compliance is thereof.

6.8.7 Adhere consistently to the airlines safety and security policy and procedures through identification and mitigation of operational hazards and related security risks and implements all precautionary action/measures/processes etc.

6.8.8 Determines the suitability of airports and equipment for efficient operations.

6.8.9 Performs periodic audit of training organizations including contracted training provisions.

6.8.10 Performs periodic audit of pilot records.

6.8.11 Ensures the currency and qualification of cross qualified pilots as appropriate.

6.8.12 Assists in all investigations of aircraft incidents.

6.8.13 Performs other duties as may be assigned from time to time by the Director Flying Operations.

6.8.14 Develops and recommends policies and procedures related to B767/737 aircraft operational activities.

6.9 CHIEF PILOT Q400

NOTE: Chief Pilot Q400 reports to the Director Flying Operations.

BASIC FUNCTIONS

6.9.1 Plans and monitors the activities of Q400 cockpit crew personnel in all phases of their designated duties, the safe and reliable conduct of flight and maintenance of company standards related to appearance and personal conduct.

6.9.2 Plans for training in consultation with Manager Recurrent & Transition Training to upgrade, maintain and improve proficiency levels of Q400 aircraft cockpit crew personnel.

6.9.3 Coordinates with Team Leader Crew Scheduling in Flight Control Section to assign student-captains, First Officers to various flights; follows-up their progress and participates in final check-out.

6.9.4 Coordinates with Manager Recurrent & Transition Training in establishing guiding principles for CPT/Simulator and Flight Training.

6.9.5 Coordinates with other departments on operational, scheduling, airport (airfield) and equipment matters for efficient operation.

6.9.6 Ensure that Air Operator Certificate (AOC) issued by Ethiopia Civil Aviation Authority is valid and current. Ascertains the AOC and all its associated documents shall define the scope of the current operations, and the compliance is thereof.

6.9.7 Adhere consistently to the airline's safety and security policy and procedures through identification and mitigation of operational hazards and related security risks and implements all precautionary action/measures/processes etc.

6.9.8 Determines the suitability of airfields and equipment for efficient and smooth operations.

6.9.9 Performs periodic audit of training organizations including contracted training provisions.

6.9.10 Performs periodic audit of pilot records.

6.9.11 Ensures the currency and qualification of cross qualified pilots as appropriate.

6.9.12 Assists in all investigations of aircraft incidents.

6.9.13 Performs other duties as may be assigned from time to time by the Director of Flying Operations.

6.9.14 Develops and recommends policies and procedures related to Q400 aircraft operational activities.

6.10 MANAGER BOEING FLEET TRAINING

NOTE: Manager Boeing Fleet Training reports to Director Flight Training & Standards.

BASIC FUNCTIONS

6.10.1 Manages the development and continued maintenance of Boeing flight Crew training consistent with the existing regulations and operational requirements of the airline.

6.10.2 Is responsible for upgrading and maintaining proficiency of all flight crew in consultation with the Mgr. Airbus and Q400 Training and concerned chief pilots.

6.10.3 Develop training manuals and programs, policies and procedures to maintain standards.

6.10.4 Initiates improvement ideas to AB-INITIO pilots' training school functions.

6.10.5 Maintains relationship with aircraft manufacturers, operators and training institutions.

6.11 MANAGER MPL & CADET TRAINING

NOTE: Manager MPL & Cadet Training reports to Director Flight Training & Standards.

There is a functional relationship between the Manager, MPL & Cadet Training and the Head, Pilot Training School under Ethiopian Aviation University for the conduct of MPL and Cadet pilot training programs.

BASIC FUNCTIONS

6.11.1 Oversees the conduct of Phase I through Stage 5 of MPL training in coordination with Ethiopian Aviation University.

6.11.2 Manages the implementation of Stage 6 through completion of Line Training of the MPL and cadet program.

6.11.3 Coordinates with Corporate HRM for the hiring and placement of trainees upon successful completion of the MPL training program.

6.11.4 Monitors the delivery of advanced MPL training, including ground school, simulator, and line training sessions, ensuring alignment with ECAA and applicable regulatory standards.

6.12 MANAGER AIRBUS AND Q400 TRAINING

NOTE: The Manager Airbus and Q400 Training reports to the Dir. Flight Trainings & Standards.

BASIC FUNCTIONS

6.12.1 Manages the development and continued maintenance of Airbus and Q400 flight Crew training consistent with the existing regulations and operational requirements of the airline.

6.12.2 Is responsible for upgrading and maintaining proficiency of all flight crew in consultation with Mgr. Boeing Fleet Training and concerned chief pilots.

6.12.3 Develop training manuals and programs, policies and procedures to maintain standards.

6.12.4 Initiates improvement ideas to AB-INITIO pilots' training school functions.

6.12.5 Maintains relationship with aircraft manufacturers, operators and training institutions.

6.13 MANAGER FLIGHT TRAINING STANDARD

NOTE: Manager Flight Training Standard reports to Director Flight Training & Standard. All examiners and standards captain reports to manager Flight Training and Standard.

BASIC FUNCTIONS

6.13.1 CBTA Program Oversight

- Design and implement Competency-Based Training & Assessment (CBTA) and Evidence-Based Training (EBT) programs in line with regulatory standards.
- Develop scenario-based curricula addressing core pilot competencies under real-world conditions.

6.13.2 Instructor Standardization, Supervision and continual improvement

- Ensure instructors use established assessment models.
- Conduct Train-the-Trainer programs, instructor evaluations, and standardization audits.
- Distributes standardization meeting minutes and collects feedback from instructors and flight crews to support the continual improvement of the training program.

6.13.3 Compliance & Quality Assurance

- Monitor training program compliance with regulatory standards.
- Track performance indicators and implement corrective actions.

6.13.4 Assessment & Data-Driven Performance Monitoring

- Tracking pilot development in technical and non-technical skills.
- Use data analytics to enhance assessment reliability and training effectiveness.

6.13.5 Innovation in Simulator & Flight Training

- Integrate CBTA into simulator sessions using adaptive learning, AI tools, and VR techniques.

6.13.6 Safety & Just Culture Integration

- Apply SMS principles throughout training design and delivery.
- Promote Just Culture to encourage open error reporting and continuous improvement.

6.13.7 Stakeholder Coordination & Representation

- Liaise with Flight Operations, Safety, and Regulatory Affairs to align training with operational needs.
- Represent the training department at regulatory meetings and CBTA/EBT industry forums.

6.14 MANAGER FLIGHT OPERATIONS PERFORMANCE & TECHNICAL SUPPORT

NOTE: The Manager Flight Operations Performance & Technical Support reports to the VP Flight Operations.

BASIC FUNCTIONS

6.14.1. Serves as Technical advisor to the Flight Operations divisions.

6.14.2. Ensures the existence of a high technical standard through the maintenance of a functional relationship with planning & engineering and the flight crew.

6.14.3. Evaluates old and new aircraft and recommends actions.

6.14.4. Is responsible for all matters pertaining to aviation fuel and technical publications in the Flight Operations divisions.

6.14.5. Provides the Flight Operations offices and flight crews with the necessary documentation, such as Operations Manual, Dispatch Deviation Manual, Flight Training Manual, Performance Engineer Manuals and software systems to the user departments.

6.14.6. Provides system-wide services in the distribution, upkeep of Policy and Technical Manuals, books, standard forms and all material published by Flight Operations division.

6.14.7. Provides specialized assistance in information management as needed.

6.14.8. Examine the organization's procedures and needs to identify potential areas that could benefit from mechanization.

6.14.9. Ensures that the manufacturer's manual of each aircraft has enroute performance data to include engine-out service ceiling and drift down data.

6.14.10. Provides all engine takeoff climb gradient information or ensures that the manufacturer manual of each aircraft contains the all-engine takeoff climb gradient information.

6.14.11. Is responsible to advise Director Flying Operations, whenever there is a safety issue regarding the use of aircraft FMC/GPS navigation database.

6.14.12. Monitors the implementation of best fuel and environmental efficiency practices of the industry.

6.14.13. Maintains statistics of fuel consumptions of all aircraft operated by Ethiopian and conducts necessary analysis and recommendation on required improvement on to aviation fuel usage.

6.15 MANAGER HR & FINANCE FLIGHT-OPERATIONS

NOTE: The Manager HR & Finance – Flight Operations reports to the VP Flight Operations.

BASIC FUNCTIONS

6.15.1 Monitors and ensures the proper human power requirement planning, financial procedures and management of the division.

6.15.2 Assists the Vice President Flight Operations and all management organs of Flight Operations divisions in all non-technical functions of the division.

6.15.3 Make sure that Coaching, Mentoring, Training and leadership development programs are implemented in the division.

6.15.4 Manages Crew Scheduling administrative function with a view to maintaining an efficient cost-effective workable system that can address all variables.

6.15.5 Manages the administration of expatriate pilots in co-ordination with corporate HRM offices.

6.15.6 Is responsible for effective administrative co-ordination between Flight Operations and other divisions of the airlines.

6.16 MANAGER FLIGHT OPERATIONS QMS

NOTE: The Manager Flight Operations QMS reports to VP Flight Operations.

BASIC FUNCTIONS

6.16.1 Ensure audits are conducted per plan in their respective areas.

6.16.2 Provide feedback to VP FOPs on critical quality issues.

6.16.3 Ensure enforcement of recommendations are applied in time.

6.16.4 Submit periodic quality assurance reports to Dir. Group QMS, SMS, ERP & Compliance.

6.16.5 Develop audit scopes and ensure provision of the required resources to audit activities.

6.16.6 The overall implementation of the LOSA program.

- Forming LOSA steering committee.
- Proper selection and training of observers.
- Schedule and conduct of the audit.
- Data analysis and retention.
- Organize a secure site for the data and oversee the availability and collection of the observation forms.
- Protects the identity of the observers and the observed and ensures complete confidentiality.
- Preparation of report and feedback.
- Performing safety action (recommendations) mitigation follow-up.

6.17 TEAM LEADER TRAINING SUPPORT

NOTE: Team Leader crew training planning and coordination reports to the Manager Recurrent & Transition Training.

BASIC FUNCTIONS

Prepares and supervise short and long-term cockpit crew training requirements on all equipment to meet the needs of ETHIOPIAN and optimize the crew utilization and to upgrade, maintain and improve proficiency level of all cockpit crew members. Keeps informed on new training requirements of regulator body and industry standards; Generates report of training status of flight crew annually, quarterly and monthly; Develops FAA, CAA, and Collective Agreement compliant fair cockpit Crew flight distribution by scheduling technique.

6.18 STANDARDS CAPTAIN

NOTE: Standards captain reports to the Manager Flight Training standards and Facilities.

BASIC FUNCTIONS

Supervise the development and continued maintenance of the standard of all trainings consistent with the existing regulations and operational requirements of the airline; participates in the development of training manuals and programs, policies and procedures to maintain Line Operations standards; Maintains relationship with aircraft manufacturers, operators and training institutions to keep up with the latest changes and technological advances. Standard Captains will witness simulator evaluation sessions conducted by qualified examiners to ensure that training and assessment standards are consistently met.

6.19 RESEARCH & DEVELOPMENT OFFICER

Ensure continuous development and innovation of flight training programs aligned with international aviation standards and organizational goals.

BASIC FUNCTIONS

6.19.1. Training Design Innovation Develop and enhance CBTA/EBT-based curricula using emerging technologies (VR/AR, data analytics).

6.19.2. Research & Benchmarking Identify global best practices and evaluate advanced training tools and methodologies.

6.19.3. Compliance & Quality Ensure alignment with regulatory requirements and continuous improvement.

6.19.4. Program Evaluation Analyse training data to improve effectiveness and close performance gaps.
6.19.5. Project Management Lead R&D projects ensuring timely, cost-effective, and quality delivery.
6.19.6. Stakeholder Engagement Collaborate with internal teams and external partners; promote knowledge sharing and innovation.

6.20 TRAINING DATA ANALYST

Analyse training data—including flight, simulator, and ground course performance to enhance crew proficiency, training effectiveness, and regulatory compliance.

BASIC FUNCTIONS

6.20.1. Collect and assess data from training records, simulator sessions, ground courses, and flight performance.
6.20.2. Identify trends, performance gaps, and training needs.
6.20.3. Recommend corrective actions and strategic plans for flight and ground training improvement.
6.20.4. Collaborate with training, operations, and IT teams for data-driven enhancements.
6.20.5. Ensure all training data processes meet aviation safety and regulatory standards.
6.20.6. Produce dashboards, reports, and visual summaries for management.
6.20.7. Stay current on training analytics trends to drive continuous improvement.

6.21 TEAM LEADER DOCUMENTATION

NOTE: The Team Leader Documentation reports to the Manager Flight Operations Performance & Technical Support

BASIC FUNCTIONS

6.21.1. Supervises, controls and ensures various manufacturer’s manuals, company policy and procedure manuals and Jeppesen airway manuals are kept in the central library, on board the aircraft and necessary locations and are up to date and current.
6.21.2. Ensures the proper distribution of operational documents, manuals, aircraft certificates and other publications.
6.21.3. Flight operation documentation section shall ensure that all manuals including associated revisions have the following;

a) Contains information that is legible and accurately represented;
b) Is written in an English language which is understood by flight operations personnel;
c) Is presented in a useable format that meets the needs of flight operations personnel;

d) Is accepted or approved by the Authority.

6.22 GROUND INSTRUCTORS

Deliver and manage ground training and digital learning systems to support pilot competency development, regulatory compliance, and operational excellence.

BASIC FUNCTIONS

6.22.1. Conduct classroom, CBT, and systems integration sessions per Training Course Outline.

6.22.2. Prepare, review, and update lesson plans, syllabi, and training schedules in compliance with regulatory requirements.

6.22.3. Monitor trainee ground course performance and progress, report delays or deficiencies promptly.

6.22.4. Collect trainee feedback and compile statistical reports for management.

6.22.5. Oversee online training content, eLearning platforms (e.g., Logipad), and digital training materials, ensuring alignment with SMS, GDR, LPRO, CRM, and Safety & Security training standards.

6.22.6. Collaborate with content developers to design, update, and manage multimedia learning resources.

6.22.7. Provide user support through tutorials, training sessions, and troubleshooting.

6.22.8. Analyse engagement and performance metrics to improve training outcomes.

6.22.9. Support the development and implementation of new digital training platforms.

6.22.10. Maintain a portfolio of customized eLearning courses for operational needs.

6.22.11. Conduct trainee evaluations objectively at all times, and if they encounter any internal or external interference, they must immediately report it to the head of Ethics and complete the interference reporting form.

6.23 LINE INSTRUCTORS (Q-400, B737, B767, B787/777 & A350)

NOTE: Line Instructors report to Manager Recurrent & Transition Training

BASIC FUNCTIONS

6.23.1. Serves as company Flight Examiner and flight instructor as applicable for cockpit crew members both on-line and route proficiency check of their respective equipment;

6.23.2. Assists in monitoring all aspects of Flight Operations pertaining the equipment;

6.23.3. Serves as a regular Captain on the line of respective equipment when not functioning in the capacity of check pilot.

6.23.4. Conduct trainee evaluations objectively at all times, and if they encounter any internal or external interference, they must immediately report it to the head of Ethics and complete the interference reporting form.

6.24 SIMULATOR INSTRUCTORS (Q-400, B737, B767, B787/777 & A350)

NOTE: The Simulator Instructor reports to Manager Recurrent & Transition Training and MPL simulator instructors reports to manager MPL and cadet trainings.

BASIC FUNCTIONS

6.24.1. Serves as simulator instructor for cockpit crew members per the respective equipment.
6.24.2. Assists in monitoring all aspects of Flight Operations pertaining to the equipment he is qualified in as he develops the pilot's ability to perform the tasks of a line pilot in a safe and reliable manner.
6.24.3. Conducts proficiency Checks on company and customer pilots.
6.24.4. Recommends and develops Standard Operating Procedures training materials, etc.
6.24.5. Implements policies and procedures regarding the development of cockpit crew personnel through Technical Training Programs.
6.24.6. Ensures Computer Based Training (CBT) devices are functioning appropriately to be used by cockpit crew trainings.
6.24.7. Prepares, maintains and administers questionnaires for testing system Knowledge of cockpit crew training.
6.24.8. Conduct trainee evaluations objectively at all times, and if they encounter any internal or external interference, they must immediately report it to the head of Ethics and complete the interference reporting form.

6.25 FLIGHT DATA MONITORING OFFICER

NOTE: The Flight Data Monitoring Officer reports to the Manager Flight Operations Quality Assurance.

BASIC FUNCTIONS

6.25.1. Is responsible for the overall management, security, administration, and maintenance of the Flight Data Monitoring Program.
6.25.2. FDM officer is responsible for monitoring, reviewing & interpreting of the detail analysis.
6.25.3. FDM Officer is responsible to monitor, review & interpret the GDRAS-related reports which will be approved by Manager FOQA and Safety.

7.0 PROCEDURE

7.1 QUALITY AUDITORS

NOTE: The Quality Auditors report to the Manger Flight Operations Quality Assurance

BASIC FUNCTIONS

When assigned by Manager Flight Operation Quality Assurance, Quality Auditors will be released from their flying duties for the period of the audit and audit report submission.

7.2 FLIGHT OPERATIONS REVIEW COMMITTEE

7.2.1 The Review Committee (RC) is the highest body in Flight Operations delegated and is empowered to assess and enact changes in policies, procedures, rules and instructions, all matters concerning introduction of new aircraft, system and fleet modification/upgrade, changes to manufacturers' checklists and standard operations. The RC is charged with ensuring compliance to safety regulations, company policies and procedures, human factor principles etc. It is the responsibility of the ground instructor to make sure that all the above listed elements are properly addressed to all flight crew members via company e-mail, training in a classroom or any available means.

7.2.2 In the event a modification(s) to either a procedure(s) or checklist or items there in, is required, the committee will notify the Original Equipment Manufacturer(s) of the proposed modification for approval or acceptance. This process may be repeated as necessary to reach a consensus; however, safety critical changes shall be given highest priority and may be implemented as is for a retroactive review. The final approved procedure, checklist or any change will be communicated to all users through the revision process before the due dates.

7.2.3 In addressing training system reviews and making improvements and or changes, feedback from the output of the training and assessments system, trend analysis and audit activities will be utilized.

7.3 COMPOSITION OF THE REVIEW COMMITTEE

7.3.1. The committee will be comprised of the following personnel:

a) VP Flight Operations
b) Dir. Flying Operation
c) Dir. Flight Training & Standard
d) Mgr. HR & Finance-FOPS
e) Mgr. Boeing Fleet Training
f) Mgr. Airbus and Q-400 Training
g) Mgr MPL and Cadet Training
h) Chief pilots
i) Instructors / Examiners (As required)
j) Others as appropriate

NOTE: For training review committee refer to FOM 5.5.

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SECTION 1.3. CORPORATE SAFETY PROCEDURES

SECTION 1.3. CORPORATE SAFETY PROCEDURES

1.0. PURPOSE

The purpose of this corporate safety procedure is to provide guidelines on how all concerned personnel shall strictly adhere to the safety regulations to prevent/avoid accidents and/or injuries while on duty.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Chief Executive Officer
- Accountable Manager
- Each division Vice President (VP)

4.0. P0LICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1. Ethiopian supports all the legal requirements and the best aviation practices considering safety as a core value. It is Ethiopian's safety goal to have zero accidents, mitigate incidents and eliminate the risk of injury to customers, personnel and damage to equipment and property.

4.2. Ethiopian Safety Management System shall include all aspects of flight, maintenance, ground, industrial and environmental safety including emergency response planning and management.

4.3. Ethiopian should have a program that ensures personnel throughout the flight operations organization are trained and competent to perform SMS duties. The scope of such training should be appropriate to each individual's involvement in the SMS.

5.0. DEFINITION

5.1 **Accountable Manager** is a person who has corporate authority for ensuring that all maintenance/operation required by the enterprise/customer can be financed and carried out to the standard required by the National Aviation Authority (Ethiopian Civil Aviation Authority).

5.2 **SMS**–means Safety Management System

SECTION 1.3. CORPORATE SAFETY PROCEDURES

6.0. RESPONSIBILITY

6.1. The Chief Executive Officer (CEO) is responsible for the overall safety of Ethiopian Airlines.

6.2. The Accountable Manager (the Chief Operating Officer – COO) is responsible for implementing safety standards and resolving operational safety issues. This responsibility includes the authority to establish support for the program in each operational unit.

6.3. Responsibility for further implementing the Safety Program in all divisions rests with respective vice Presidents and their designate.

6.4. The Accountable Manager responsible for assuring implementation of Ethiopian Safety Management System is the Vice President Internal Audit and Integrated Management Systems Compliance (VPIA & IMSC) and/or his designate. This responsibility includes the authority to make safety recommendations, revise the system, devise appropriate data analysis, hazard identification, safety reporting, safety communication, awareness and orientation, conduct investigations, operational Audits and safety risk assessments.

6.5. Each division directors and managers will be held accountable to ensure that all reasonable steps are taken to prevent incidents and accidents to identify all laws and regulations that affect their area of responsibility and to develop systems for ensuring compliance. Each operational unit shall develop, implement and integrate elements of Ethiopian Aviation Safety program within their operating procedures.

6.6. Employees at all levels are required to focus on safety, assume individual responsibility to be proactive and take preventive safety actions by immediately informing their Team leaders if they are aware of any immediate threat to the safety of flight, aircraft, equipment, property, themselves or others.

7.0. PROCEDURE

7.1. A non-punitive reporting system is in place within the organization to encourage employees to report all hazards, errors, incidents and operational deficiencies, in a blame free environment, in order to promote a safe working environment. Information leading to the identity of any employee who reports an error, accident, incident, hazard etc… under this policy shall never be disclosed unless agreed to by the employee or required by law.

7.2. No reprisal shall be taken against any employee who acts to prevent an accident by reporting hazard, incident or error unless it rises from actions of illegal activity, willful misconduct, gross negligence, deliberate sabotage and substance abuse.

7.3. Individuals are encouraged to report hazards and safety deficiencies to management and will be praised for providing essential safety related information. Ethiopian believes sincere input of all employees to be highly desired and vital to safe operations.

7.4. All employees are expected to participate in the program and take an active role in the identification, prevention, reduction, and elimination of human error and hazards. In addition, all employees are required to adopt the standards and procedures set forth in various company manuals.

* * * *

SECTION 1.4. COORDINATION, COMMUNICATION AND MEETINGS

SECTION 1.4. COORDINATION, COMMUNICATION AND MEETINGS

1.0. PURPOSE

The purpose of this coordination, communication and meetings procedure is to establish procedure to provide guidelines on how the concerned personnel perform coordination, communication and meeting procedures.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Operation Management Personnel

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

5.0. DEFINITION

N/A

6.0. RESPONSIBILITY

6.1. Mgr. Flight Operations Performance & Technical Support

a) Is responsible for advising the Director Flying Operations and Dir. Flight Training & Standard whenever there is a safety issue on applicable aircraft performance requirements prior to implementation.

EXAMPLE: Performance degradation, engine upgrade or downgrade, use of new performance appendix forms AFM, etc.

b) provides consultations and communicates information to enable flight planning, dispatch and flight crew to determine that the airports intended use are adequate, to include:

c) Ensure that flight planning systems on use indicate Minimum Safe Altitude (MSA) in the flight plan for all phases of the flight as established by the responsible state.

d) Is responsible for advising Director Flying Operations, whenever there is a safety issue regarding the aircraft FMC/FMS/GPS navigation database.

6.2. Flight crew members are required to determine if airports of intended use meet the operational requirements. The following is a guide for a flight crew;

SECTION 1.4. COORDINATION, COMMUNICATION AND MEETINGS

I. Runway characteristics: Flight crew has to use Jeppesen charts & manual, NOTAMS, etc.…

II. Applicable performance requirements: Flight crew has to use OPT, fly smart, Airport analysis & A/C manufacturer manuals to determine performance requirements

III. Air traffic services & associated communications, Flight crew has to use Jeppesen manual's different sections, AIP, NOTAM, etc.... for proper usage of the services

IV. Navigation aids and lighting; Flight crew has to use Jeppesen charts & manual, NOTAMS, etc....

V. Weather reporting: Flight crew has to use Jeppesen charts & manual, NOTAMS, etc....

VI. Emergency services, including temporary periods of reduced rescue & fire Fighting (RFF); Flight crew has to use Jeppesen manual, NOTAMS, etc....

7.0. PROCEDURE

7.1. COORDINATION

7.1.1. Flight Operations have a process (as depicted on SMSM manual) to ensure issues that affect operational safety and security are coordinated among personnel with expertise in the appropriate areas within the flight operations organization and relevant areas outside of flight 0perations to include:

a) accident prevention and flight safety;

b) cabin operations;

c) engineering and maintenance;

d) operations engineering;

e) operational control/flight dispatch;

f) human resources;

g) ground handling, cargo operations and dangerous goods;

h) manufacturers (AFM/AOM, operational and safety communication);

i) regulatory agencies or authorities;

7.1.2. FOPs Performance and Technical Department shall coordinate with Ethiopian MRO, Original Equipment Manufacturers and Airframe Manufacturers to ensure that all aircraft operated by Ethiopian have the following.

a) Instrumentation and/or avionics readily visible to the intended pilot flight crew member, necessary to conduct operations and meet applicable flight parameters, maneuvers and limitations.

b) Equipment necessary to satisfy applicable operational communication requirements including emergency communication.

c) Avionics equipment and/or components necessary to satisfy applicable Navigation requirements to provide necessary redundancies as applicable authorized by the State for use in RNP, MNPS/NAT HLA, RVSM and/or PBCS operations.

d) Avionics, instrumentation and/or radio equipment necessary to satisfy applicable approach and landing requirements.

e) Other components and /or equipment necessary to conduct operations under applicable flight conditions, including instrument meteorological conditions.

7.2. COMMUNICATION

7.2.1 To enable and ensure effective exchange of operationally relevant information throughout the management system and among operational personnel in all areas where operations are conducted. The system and procedures elaborated on the SMS manual chapter 12 will be used. In addition, the following communication systems are used.

7.2.2 Methods of communication:

a) Regular meetings (e.g. management, Instructors, Crew, Safety committee, and other operationally relevant departments).

b) Reports (Captains' reports, Safety/Security reports, etc....)

c) Review/Ad hoc committee

d) Bulletins

e) E-mails

f) Telephone

g) Radio

7.2.3 Accountable Flight Operations Managers will communicate within and external to Flight Operations, including the top management (VP flight operations).

7.3. MEETINGS

7.3.1 Regular communication and meetings shall be maintained between the following to achieve continuous improvement of ground, simulator and aircraft training and line operations.

a) Management twice a month

b) Instructors/ Examiners quarterly

c) Flight crew quarterly

7.3.2 Minutes of the meetings will be distributed to all concerned.

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SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

1.0 PURPOSE

The purpose of this specific duties procedure is to provide specific duties of certain key personnel listed here in.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1 In addition to the preceding material given under Responsibility in Section 1.2, it is necessary to provide detailed listings of the specific duties of certain key personnel. This paragraph provides the information.

Subordination when on-duty and off-duty will be as follows:

4.2 OFF-DUTY

a) B777/787 pilots report to Chief Pilot B777/787.
b) A350 pilots report to Chief Pilot A-350
c) B767/757 pilots report to Chief Pilot B767/757.
d) B737/Q400 Pilots report to chief pilot B737/Q400.

4.3 ON-DUTY

a) All Captains report to their respective Chief Pilots.
b) All First officers’ report to the captain of the aircraft they are assigned to operate.

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

The Captain and the First Officer shall be responsible for the assigned duties.

The flight crew has authority, responsibility and is accountable to perform functions relevant to the safety and security of A/C operations.

6.1. AUTHORITY OF PILOT-IN-COMMAND

The captain is directly responsible for and is the final authority as to the operation of his airplane.

6.2. THE CAPTAIN IS RESPONSIBLE FOR:

a) Conducting safe, efficient and secured flights in compliance with appropriate ATC and government rules and regulations and the policies and procedures specified in the company Flight Operations Manual System.

b) The safety of all passengers, crewmembers, mail and cargo onboard when the doors are closed.

c) The operation and safety of the airplane, its proper service, and the maintenance of airworthiness from the moment the aircraft is ready to move for the purpose of taking off until the moment it finally comes to rest at the end of the flight and the engine(s) are shut down.

d) Cooperating with the flight dispatcher in accordance with policies and procedures specified in the flight operations manual. And should have final authority for decisions regarding airworthiness of the airplane and flight planning.

e) Delegating his responsibility for the safety & security of the airplane and its cargo as well as the passengers while on the ground to the company's representatives or to specific crew members.

f) Not permitting any person to be carried in the airplane who is obviously under the influence of intoxicating liquors or drugs except for medical patients under proper care or in case of emergency.

g) Ensuring that standard operating and emergency procedures and regulations are known and adhered to by all crewmembers in the air and on the ground, including adherence to the prescribed cockpit checklist.

h) Ensuring that in all cargo operations, the availability, accessibility and serviceability of restraint systems, emergency equipment in the cargo compartment & shall also ensure that the 9G Barrier/Net is secured before each take off is checked by load master.

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

i) Coordinating the duties of the respective crew members in flight and on the ground to ensure that the provisions outlined herein are complied with.

j) Familiarizing himself with the records of First Officers and trainees and for giving them the fullest possible benefit of his experience to improve their proficiency and bring them up to a standard within their category. This shall include giving permission to first officers to perform takeoffs and landings considering operational capabilities and requirements. Whenever a First Officer performs takeoffs or landings, the captain shall be prepared to take control immediately should such an action becomes necessary. The PIC shall not allow the First Officer to taxi the Aircraft during ground maneuver.

k) Acting within regulations in the best interest of Ethiopian at all times and taking into account all known factors.

l) Maintaining proper liaison with supervisory staff at all stations in order to ensure an efficient and punctual operation.

m) Monitor, check and verify the navigational performance, maintaining a particular RNP and accuracy of the present position of the aircraft during all Phases of flight, after prolonged in-flight operations and before commencing approaches, by using the FMS RNP/ANP alerts or a VOR/DME distance and bearing against FMS FIX distance/bearing of the same station.

n) Upholding at all times and in all circumstances the prestige of ETHIOPIAN by setting a high standard of personal behavior and appearance.

o) Ensuring the discipline of his flight and cabin crew at all times, in the air and on the ground, and for overseeing that, when in uniform, crew members present a good personal appearance, and uniforms are worn smartly and correctly.

p) PIC shall ensure that all crew members & passengers are prohibited from smoking onboard or within the vicinity of the aircraft.

q) Submitting evaluation reports on crew members as required.

r) Cockpit voice recorder recordings may not be used for purposes other than for the investigation of an accident or incident subject to mandatory reporting except with the consent of all crew members concerned.

NOTE: It is prohibited to intentionally switch off or erase the CVR and FDR data during flight time. CVR and FDR are only switched off by the flight crew after a flight when required to preserve data in the event of an accident or serious incident that would otherwise be lost.

s) The PIC ensures the continuous operation of the aforementioned recorders such that: for the flight recorder; from when the aircraft begins its takeoff roll until it has completed the landing roll and for the cockpit voice recorder; from the initiation of the pre-start checklist until the end of the securing aircraft checklist. (ECARAS8.5.1.24)

NOTE: Flight crews are required to do the proper operational tests before every flight.

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

t) Decide whether to accept an airplane with defects allowed by the CDL or MEL.

u) Ensure that the pre-flight inspection has been completed.

v) Ensure the proper implementation and application of the manufacturer provided procedures and checklists approved for use by Ethiopian.

w) Ensure that the aircraft carries sufficient required fuel, oil and additional fuel if operationally required. (FOM 2.12)

x) The pilot to whom conduct of the flight has been delegated shall, in an emergency that requires immediate decision and action, take any action he considers necessary under the circumstances. In such cases he may deviate from rules, operational procedures and methods in the interest of safety refer to FOM 1.5 (6.5) for more information.

y) For submitting confidential reports, when necessary, on members of his flight and cabin crew. Where such a report contains criticism of an air crew member this shall be discussed with the individual before submission.

z) The captain is responsible for the safety of his aircraft and its contents throughout the time he is in command.

aa) At all times and in all circumstances to behave correctly and shall not deliberately violate company operational standards.

bb) To set a high standard of behavior and for the discipline of his flight and cabin crew at all times both in the air and on the ground.

cc) To require another Captain to be called upon to exercise authority over the crew members of other Captains, in which case he will do so, while making every effort to contact the Captain concerned.

dd) For ensuring that his orders are promptly obeyed and do his utmost to develop a high level of esprit de corps, both in his crew and with other Ethiopian divisions with which his duties bring him into contact.

ee) In all cases where a Captain decides to suspend a member of his crew from duty on the grounds of being unfit for service, he shall submit a report to Dir. Flying Operations upon completion of the flight.

ff) To ensure that there is no doubt whatsoever as to which pilot is officially in charge of an aircraft, every pilot required to be responsible for an aircraft must sign the aircraft papers as the captain of the aircraft, regardless of his rank or seniority. (Signing the aircraft papers officially confers the overriding responsibility and authority with respect to the aircraft).

gg) For the safety of his aircraft and its contents throughout the time he is in command.

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

6.3. THE FIRST OFFICER SHALL BE RESPONSIBLE FOR:

a) Attending pre-departure briefings, be fully aware of the intended flight planned route, contents of dispatch releases sheet, meteorology forecast conditions, NOTAMS and any other evaluation data.

b) Is responsible for the completion of the required forms like load sheet, trim sheet and fuel sheet etc... During unassisted revenue departure.

c) Performing checks and drills, normal and emergency, in the manner prescribed by current Flight Operations Manual.

d) Monitoring the execution of Emergency Checklist actions whenever possible.

e) Conducting Radio communications, recording flight progress and important communications.

f) Keeping the Captain informed of Navaid and communication frequency changes, ensuring that all facilities are properly identified.

g) Monitoring the flight at all times, checking that correct procedures and techniques are being followed. In particular he must advise the captain clearly and concisely if and when, the aircraft departs significantly from its intended flight path e.g. climbing through a clear altitude on departure, or going below Glide Slope on ILS approach, or whenever he considers that a hazardous situation is developing.

h) Complete a periodic crosscheck of instrument indications of both Captain and First officer instrument panels.

i) Monitor, check and verify the navigational performance, maintaining a particular RNP and accuracy of present position of the aircraft during all Phases of Flight, after prolonged in-flight operations and before commencing approaches, by using the FMS RNP/ANP alerts or a VOR/DME distance and bearing against FMS FIX distance/bearing of the same station.

j) Monitor destination, destination alternate and enroute alternate(s) weather information while enroute.

k) Carry out any other duties required by the captain.

l) Record any system or component malfunction in the aircraft Maintenance Logbook (AML) when instructed by the captain.

m) Maintain competency at all times to carry out basic navigation procedures using conventional methods.

n) Assume command of the aircraft while in flight in the event of incapacitation of the captain and take over the responsibilities implied.

o) First Officers are not allowed to taxi an Aircraft under any circumstances.

p) Carry out his duties as defined in this manual and /or other relevant manuals and instructions and in conformity with legal requirements.

q) Act within regulations, in the best interests of the Company, at all times taking into account all known factors.

r) Uphold at all times and in all circumstances the prestige of Ethiopian by setting a high standard of personal behavior and discipline.

s) Check and compute the take-off and landing weights and pass the correct V Speeds.

t) Assist the Captain in conducting safe and efficient operation of the aircraft.

u) Maintain a special relationship with:

I. Captains and all other crew members, including cabin staff, to ensure the smooth and efficient running of the flight.

II. Ground technicians for the establishment of goodwill and co-operation as well as close liaison on technical delays.

III. Refueling staff for dealing with refueling, particularly on stations where there is no engineering back-up.

6.4. AUTHORITY OF THE CAPTAIN

6.4.1. The captain has full control and authority over the aircraft and its operation without limitation. This includes:

a) Full authority regarding operational conduct and preparation of the flight, acting through his joint authority with the Dispatcher.

b) Control over the aircraft, its passengers and its cargo upon its release by the marketing and/or maintenance department and/or flight control department.

c) Full authority over other crew members and their duties from the time of their reporting for duty until their release from duty. This authority extends to him whether he holds valid licenses authorizing him to perform the duties of those crewmembers.

d) Full authority and responsibility for preflight planning and the operation of the flight in compliance with national and international law, regulations, company policy and procedures.

6.5. EMERGENCY AUTHORITY OF THE CAPTAIN

6.5.1. The captain is permitted to deviate from prescribed rules, regulations, minimums, company policy and procedures as required for safety of flight considerations in emergencies. An aircraft in distress has the right-of-way over all other air traffic. Air traffic control should be kept informed of deviations from clearances of flight plans and will give priority to an aircraft that has declared an emergency.

6.5.2. Dumping fuel to meet LGW limitations is an option of the Captain’s emergency authority. The Captain may exercise his emergency authority to exceed the maximum LGW if the landing is necessary as the result of an incident or a mechanical irregularity and he determines that such a landing would be safer than dumping fuel.

6.5.3. In an emergency arising during flight time the Captain shall keep the appropriate air traffic control or services facility and flight control fully informed of the progress of the flight.

6.5.4. Whenever emergency authority is exercised, the Captain shall submit a written report of any deviation to the VP Flight Operations within 2 days after the flight is completed.

6.6. OPERATING RESTRICTIONS FLIGHT CREW

6.6.1. No person may take off or continue normal operation of an aircraft:

- Unless thoroughly familiar with reported and forecast weather conditions the route to be flown including terminal and alternate weather forecasts.
- When, in the opinion of the captain, existing or anticipated icing conditions might adversely affect the safety of the flight.
- That has frost, snow or ice adhering to the wings and control surfaces of the aircraft.
- Unless a valid flight plan has been prepared as outlined in the flight preparation section of this manual.
- Unless it is airworthy and is equipped as outlined in the flight preparation section of this manual.
- Unless the aircraft's weight and center of gravity have been officially determined to be within the limits imposed by the approved aircraft flight manual.
- Unless the aircraft, in revenue service, is operated along and within approved routes and areas to and from designated airports specified in the route manual.
- Unless before and during a flight, the captain obtains all available current reports or information on airport conditions, irregularities of navigation facilities and irregularities of service that may affect the safety of a flight.
- Unless accountability has been made regarding the takeoff, enroute and landing aircraft performance limitations.

7.0 PROCEDURE

7.1 CARRIAGE OF SUPERNUMERARIES

NOTE: Supernumeraries report to the Pilot-In-Command

7.1.1 If supernumeraries are carried onboard, the pilot-in-command shall:

I. If applicable, define and describe the duties or responsibilities assigned to such personnel that are related to safety.

II. Ensure supernumeraries do not impede flight crew members in performance of their duties.

III. If a cabin crew is used, ensure supernumeraries do not impede cabin crew members in performance of their duties.

7.1.2 During carriage of supernumeraries on all passenger and cargo aircraft, supernumeraries shall be briefed by flight crew or cabin crew to address on compliance with normal, abnormal, turbulent weather conditions & emergency situation procedures related to safety.

7.1.3 Supernumeraries shall be notified by flight crew or cabin Crew in the cabin or supernumerary compartment prior to T/off and landing to provide notification to supernumeraries by either the flight crew or cabin crew to prepare for takeoff, when in descent phase of flight & to prepare for landing. If a cabin crew is not used, the flying crew has to verify at least the following: supernumerary seatbelts fastened, tray tables and seat backs in a stowed and upright position, cabin baggage and other carry-on baggage secured in designated areas, galleys and associated equipment stowed and restrained, and as applicable, inflight entertainment system viewing screens off and stowed.

7.1.4 If cabin crew or load master are not used, to ensure safety during normal door operation, upon appropriate signal for door opening indicated from the ground personnel that is, 'two knocks' by hand on the outside of the door and an appropriate signal (e.g. 'thumbs-up' signal through the door observation window) that the door may be opened, flight crew shall arm the doors once they are closed for the flight and disarm at the end of the flight and prior to opening for supernumeraries and/or crew deplaning. Supernumeraries shall be advised not to touch it without the flying crew command. For additional safety guidelines please refer to CCOM 4.32.2.

7.1.5 For all cargo A/C, the flight crew has to confirm that smoke barrier is closed and secured for taxi, takeoff and landing. Flight crew has to make sure and brief supernumeraries that they should not open the smoke barrier during such periods.

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

7.1.6 If cabin crew members are not used, preparation of the cabin prior to take off and landing would normally require the flight crew to verify certain conditions are in effect. Items checked by the flight crew typically include:

- Passenger seat belts fastened.
- Tray tables and seat backs in a stowed and upright position
- Cabin baggage and other carry-on items are secure in designated areas.
- As applicable, in-flight entertainment system viewing screens off and stowed.
- Galleys and associated equipment stowed or restrained.

7.1.7 If no cabin crew or load master is used in carriage of supernumeraries on all passenger and cargo aircraft flight crew shall brief.

a) Prior to departure of a flight, the supernumeraries to be familiar with the location & usage of safety equipment to include the following:

I. The dissemination of supernumerary safety information as per FOM 2.5 (7.7)

- seat belts (usage how to fasten and unfasten)
- brief that smoking is prohibited at all times.
- emergency exit and evacuation procedures (to wait for the flight crew instruction) and show them how to open the doors, evacuate & where to proceed after.
- Life jackets (life vests) (when and how to use it if flown over water).
- Life rafts (where to access if applicable and how to use it)
- Oxygen masks (when and how to use it)
- Emergency equipment (when and how to use it as per the flight crew instruction).

II. Cabin or supernumerary compartment readiness prior to first aircraft movement, takeoff and landing.

III. As applicable the arming or disarming of door slides.

IV. Preparation for and encounter with Turbulence through seat belt sign or verbal command from the flying crew.

V. Medical situations (when and how to use it).

VI. Abnormal situation (how and when to report it to the flying crew and receive command from them)

VII. Verification that baggage is stowed (when to say cabin is secured as long as the baggage is stowed properly).

SECTION 1.5. SPECIFIC DUTIES OF FLIGHT CREW PERSONNEL

b) The flight crew is responsible for checking all emergency equipment, securing the cabin and arming & disarming of door slides as applicable.

c) The Flight crew shall make sure that all emergency equipment & emergency oxygen are readily accessible to the supernumeraries. The flight crew will allow the supernumerary to be seated to the nearby emergency equipment so that they can access it easily.

d) The flight crew shall ensure that supernumeraries are seated with their seat belts fastened through communication or checking in person.

I. During the taxi phase of a flight
II. During the takeoff and landing phase of a flight
III. Prior to and/or during turbulence
IV. During an emergency situation, if considered necessary.

e) For all Ethiopian fleets during Emergency Evacuation the Flight crew shall leave their seat after completing their own procedures & assist supernumeraries/passengers to evacuate the airplane per the following procedures;

I. If evacuation is not already started, initiate evacuation by evaluating outside condition for example Fire, obstruction, etc....
II. Open the door in armed position, if slides do not inflate pull the manual handle.
III. Re-direct supernumeraries/passengers to alternate exit due to unusable exit or changing condition of the cabin.
IV. Assemble supernumeraries/passengers in a safe area upwind and away from the aircraft (at least 200 meters).

NOTE 1: Ethiopian does not transport passengers without a cabin crew.

* * * *

SECTION 1.6. FLIGHT CREW REGULATION

SECTION 1.6. FLIGHT CREW REGULATION

1.0. PURPOSE

The purpose of this flight crew regulation procedure is to establish guidelines on flight crew regulations for compliance.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Crewmembers
- Crew Members

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1. FLIGHT OPERATIONS & IOCC

4.1.1 Flight operations shall have a program of checking and standardization of crew members approved by the Authority. It shall also check pilots proficiency on those maneuvers and procedures, including emergency procedures and instrument flights, as applicable, that are prescribed by the Authority.

4.1.2 IOCC shall schedule, and the PIC shall ensure that the minimum number of required cabin crew members is on board passenger-carrying flights. (ECARAS 9.3.1.7)

4.2. PROHIBITION OF USAGE OF ELECTRONIC DEVICES ON THE FLIGHT DECK

4.2.1. Flight crew members are prohibited to use wireless communications (WIFI) while they are in the flight deck.

4.2.2. Flight crew members are prohibited from using their personal laptops, tablets, mobile phones or company provided devices to watch movies or listen to music while the Aircraft is being operated.

4.2.3. Prohibition doesn’t apply to the use of their class I EFB for a purpose directly related to operation of the Aircraft, for emergencies, or safety related tasks. However, only one pilot at a time shall be used after critical phase of flight to ensure that one pilot is always able to focus on the flight deck displays, is able to maintain situational awareness and have full access to flight controls.

4.2.4. **Use of Non-Standard Adapters and Portable Devices**

Only company-approved electrical or electronic devices and accessories may be used in the flight deck to ensure operational safety and equipment integrity.

4.3. SCHEDULING POLICY AND PHYSICAL CONDITION

4.3.1 All crew members are responsible for maintaining themselves in good physical condition.

4.3.2 A flight crew member shall not accept flight duty when he is aware that his physical condition does not meet the requirements of his current medical certificate.

4.3.3 In determining fitness for duty, the following factors shall be considered:

a) Alcohol and psychoactive substance use;

b) Pregnancy;

c) Illness or use of medication(s);

d) Surgery;

e) Deep diving;

f) Fatigue whether occurring in one flight, successive flights or accumulated over a period of time.

4.4. PREGNANCY

4.4.1. A pregnant flight crew member must be suspended from flight duties in case of confirmed and reported pregnancy to the respected chief pilot.

4.4.2. She shall produce certificate of her pregnancy within 24 hours of such notification.

4.5. DIVING PRIOR TO FLYING

Flight crewmembers shall not engage in underwater activities involving the use of breathing apparatus (scuba diving) within the 48-hour period preceding a flight. (The combined effects of water compression and in-flight cabin altitude could result in incapacitation). There is no restriction on normal shallow diving and swimming before flight duty.

4.6. USE OF DRUGS

Flight crewmembers shall not use sleeping pills within 10 hours preceding departure time.

4.7. BLOOD DONATIONS

Flight crewmembers shall not donate blood except with permission of a company-approved physician. Time limit before flight will generally be 48 hours.

4.8. CORRECTIVE LENSES

Flight crews are required to put on their physician ordered corrective lenses as stated on medical certificate whenever they are flying, and they shall also have a spare set readily available.

4.9. VACCINATIONS AND IMMUNIZATIONS CERTIFICATE

Crewmembers shall maintain vaccinations and immunizations in accordance with health conditions and regulations applicable to countries entered. Common vaccinations are needed or recommended, and certificate validity periods are as follows:

CONDITION	CERTIFICATE VALIDITY
Yellow Fever	120 Months beginning 10 days after vaccination
Cholera	6 Months beginning 7 days after Vaccination
Malaria	Tablets as prescribed

Time limit before flight duty for vaccinations is as prescribed by the nurse or the doctor, except for malaria which has no time limit.

4.10. USE OF INTOXICATING LIQUOR, NARCOTICS OR DRUGS

No person shall pilot an airplane or act as flight crewmember of an airplane while under the influence of intoxicating liquor or any narcotic or drug or psychoactive substance by reasons of which his capacity to perform his duties is impaired, and under no circumstances shall be permitted to take any such intoxicants during his duty periods.

4.11. ALCOHOL & DRUGS TEST

Ethiopian will perform control substance/drug & alcohol test for flight crew, cabin crew & dispatchers. A team of Occupational Health Officer and Laboratory Technician from Medical Department & Security person from corporate security department will perform this test.

The test will be performed per the following conditions.

a) Random screening testing

b) Post-accident testing

c) Testing at outstations in collaborations with local airport officials

g) Before initial employment in EAL.

An employee who refuses to submit to any required drug/alcohol testing will be treated in the same manner as an employee who tested positive.

4.12. SICKNESS/INJURY/MEDICAL CONDTION REPORTING

a) When reporting sick, crewmembers must advise the Crew Scheduling Officer (out of office hours – the duty Scheduler) immediately and shall see Ethiopian Airlines medical unit as soon as possible for medical checkup. A Flight crew shall get a medical release from company medical doctor when fit for duty before accepting any flight.

b) Ethiopian is responsible for the costs involved when flying staff become ill or disabled while away from Home base. To meet this obligation, Ethiopian station staff at all stations where there is crew layover have been instructed to pay all relevant medical costs.

4.13. Fatigue Whether occurring in one flight, successive flights or accumulated over a period

To ensure pilot safety and operational integrity by allowing timely reporting of fatigue, whether acute or cumulative. A pilot needs to report that he is fatigued to scheduling. At the same time, he needs to notify the respective chief pilots. The report needs to include any relevant duty logs or rest history if requested. The Chief Pilot may follow up for

a) Further assessment
b) Recommendations (e.g., rest period, medical check, roster adjustment)
c) Support resources

4.14. collision Avoidance Policy

Flight crew needs to maintain vigilance for conflicting visual traffic to avoid a collision.

5.0. DEFINITION

N/A

6.0. RESPONSIBILITY

All flying crew shall be responsible to abide by the regulation in this section.

SECTION 1.6. FLIGHT CREW REGULATION

7.0. PROCEDURE

7.1. GENERAL CONDUCT

7.1.1 Crewmembers serving with Ethiopian Airlines are subject to company rules and regulations. Noncompliance will result in disciplinary action such as a warning, temporary removal from duty without pay or termination of employment.

7.1.2 Crew members should be courteous and tactful in all dealings and conversations with fellow employees and passengers. Company business shall not be discussed in the presence of non-company personnel.

7.1.3 Discussions between crew members, other employees or contract personnel must be kept on a quiet level in the presence of passengers and the public. Any disagreements conveying an impression of heated argument must be avoided. Physical violence is not allowed under any circumstances.

7.2. AGE RULE

7.2.1 In all Ethiopian flights a person who has attained the age of 60 may act as a pilot in a command or a co-pilot on an actual flight up to the age of 65, only if the other crew member assigned on the same flight is below the age of 60.

NOTE: Scheduling is responsible not to assign two flight crew members above the age of 60 to fly together during augmented crew operation.

7.2.2 The maximum age a person can operate a flight as a minimum flight crew is 65.

7.3. UNIFORMS

7.3.1 **General Policy**

The appearance of flight crew members reflects the professional image and reputation of the airline. All crew shall maintain a high standard of personal grooming and uniform presentation at all times.

It is the captain’s responsibility to ensure that all crew members are properly attired and present a consistent and professional image.

Crew uniforms shall be of the style and design approved by the Company. Each crewmember is responsible for maintaining the uniform in a clean, neatly pressed, and serviceable condition. Shoes must be polished, and all accessories kept tidy and professional.

Uniforms shall be worn at all times while on flight duty, including deadheading, and may be worn when traveling to and from duty or in restaurants where alcoholic beverages are not served.

NOTE: Consumption of any alcoholic beverages while in uniform is strictly forbidden.
NOTE: All operating crew shall wear a white shirt with a tie, appropriate shoulder tab, and company-issued hat when visiting the passenger cabin for operational or safety-related reasons that require their presence.

SECTION 1.6. FLIGHT CREW REGULATION

To avoid confusion or unnecessary concern among passengers, flight crew members who are deadheading, changing position, or leaving the flight deck for rest must remove their shoulder tabs and tie before entering or joining the passenger cabin.

7.3.2 **Uniform Components**

a) The Suit
- The standard Ethiopian green uniform coat and trousers shall be worn while on flight duty.
- Dresses are not permitted.
- Under conditions of high temperature or humidity, coats may be removed but must be folded properly and carried on the arm.
- All pilots operating together must be identically attired.
- The hemline of trousers shall sit just below the ankle and must not touch the ground.
- Trousers must not be skinny fit.
- Bulky or protruding items such as cell phones, wallets, or other personal objects must not be carried in front trouser pockets while in uniform, as they distort the professional appearance.
- Jackets must be well-fitted, neither too tight nor shorter than standard, maintaining a professional tailored appearance.

b) Topcoat
- The standard Ethiopian Airlines Flight Operations topcoat shall be worn when required or desired.
- A plain black or green muffler/scarf may be worn with the topcoat.

c) Shirts
- Only the official white Ethiopian Airlines shirt is authorized for uniform wear.
- A tie and appropriate rank shoulder tabs shall be worn at all times.
- Shirts must be clean, pressed, fully buttoned, and tucked in.
- A white undershirt or vest may be worn.
- All flight crew shall wear a white shirt with tie and rank tabs whenever in the cockpit.

d) Tie
- The standard green Ethiopian Airlines tie is approved.
- The official plain gold tie clip shall be worn with the tie at all times.

e) Shoes
- Black dress shoes with smooth upper surfaces and laces are standard.
- Sneakers, lazy-man, or ornamented shoes are not permitted.
- Female pilots may wear square-heeled black shoes with heel height not exceeding 3 cm.
- High-heeled shoes may be worn enroute but must be changed before operating flight duties.
- Shoes must be clean, polished, and in good condition.

f) Socks
- Plain black or plain green socks are permitted.

g) Belt
- Black belt with plain gold or silver buckle must be worn at all times.
- The design must be simple and dress type.

h) Raincoat
   - The nylon green alligator-style raincoat is the recommended uniform outerwear.

i) Sweater
   - A green or black sweater of compatible style may be worn as needed.

j) Recognition Pins
   - A maximum of one recognition pin may be worn, centered ¾ inch below the globe of the coat wing.

k) Cap
   - The standard company-issued flight crew hat with black shell cordovan visor shall be worn at all times in uniform, except in the cockpit.
   - The cap must be clean, undamaged, and properly maintained.

l) Ornamentation
   - Wing insignia, stripes, and decorated caps shall be worn per company rank designation.
   - Captains shall wear four stripes.

7.3.3 Grooming and Personal Appearance

A. Hands and Nails
- Hands and nails must be clean, neat, and well maintained.
- Nail length must not exceed the fingertips and should be square in shape.
- Female pilots may use clear or natural shades or a French manicure; bare nails are acceptable.
- Bright colors, glitter, gems, metallic finishes, or decorative nail art are not permitted.

B. Hair

B1. General Requirements
- Hair must always appear clean, tidy, and professionally styled.
- Hair products may be used but must not leave a greasy or wet look.

B2. Male Pilots
- Hair must be short, well-trimmed, and must not touch the shirt collar.
- Acceptable styles are conservative and natural in color (black or dark brown).
- Facial hair is limited to a clean shave with a trimmed mustache.
- Sideburns must not extend below the mid-ear

B3. Female Pilots
- Oxygen Mask Seal: In an emergency, a pilot must be able to "don" an oxygen mask in under 5 seconds. Bulky hairstyles or large clips can prevent the mask from sealing against the face, which is a major safety violation.
- The "Hat" Rule: The pilot hat is a mandatory part of the uniform in all public areas. Hairstyles must be flat enough on top and low enough at the back to allow the hat to sit perfectly level.
- Color Consistency: Hair must be single, natural-looking shade. High-contrast highlights or "fashion colors" are not allowed. Roots must be touch up regularly; visible regrowth is not permitted.
- Security: Hair must be "secured away from the eyes." Loose strands or "wispy" pieces that fall into the face are not permitted.
- Bangs: Bangs are allowed but must be trimmed so they do not touch the eyebrows or interfere with vision.

- Subtle Accessories: Hair ties, pins, and clips must be "plain and subtle." They should ideally match your hair color or be black/brown. Large bows, jeweled clips, decorative beads, and metal cuffs are strictly prohibited as they constitute a Foreign Object Debris (FOD) hazard in the cockpit.
- Zero Flyaway Policy: Female pilots are expected to use "grooming products" (hairspray, wax, or gel) to ensure a perfectly smooth finish. Loose strands or "wispy" bits are considered a grooming violation.
- Length: If hair is longer than shoulder length, it is often required to be pulled back in a ponytail, braid, or bun to maintain a crisp appearance.

Recommended Styles

- The Low Bun (Chignon): This is the most highly recommended style. It must be a neat, round bun positioned at the nape of the neck.
  - It must be a neat, no 'flyaway", frizz or stray hairs.
  - Bun should be round and positioned at the nape of the neck.
  - Bun must be secured with a hairnet that matches the hair color exactly.
  - For thick or very long hair, braid the tail tightly, and then wrap the braid into a bun.
- Wet Low Bun: The hair should be neat, slicked flat, with no 'flyaway", frizz or stray hairs.
  - The hair could look damp, not dripping water.
  - Secure the hair into a low ponytail at the nape of your neck. Ensure there are no "bubbles" or loose sections at the back of your head.
  - Twist the ponytail tightly into a rope and wrap it into a flat, circular bun.
  - Wrap a fine, invisible hairnet that matches your hair color over the bun.
- Natural Low Bun/Ponytail: hair must maintain a polished, authoritative image while accommodating natural hair textures and protective styles.
  - Edges should be slicked back with zero visible flakes or dry product residue.
  - Braids must travel in a direct line toward the back or nape of the neck. No "swooping" across the forehead or "extreme" asymmetrical parting is permitted.
  - Braids must be styled flat to the scalp. No stray hairs or "fuzz" should be visible between the braid and the hairline.
  - Braids must be high-tension to prevent flyaway. "Boho" or loose braids are considered non-compliant.
  - Braids should be secured into the bun so it won't move during a long duty day.
- The French Twist: A classic alternative for medium-to-long hair that is considered highly professional and keeps hair entirely contained.
  - It must be tucked in completely; no loose ends should be visible at the top of the roll.
- Single Braid or French Braid: A hair style for medium-to-long hair that is often preferred for long-haul flights because it stays secure for many hours. the braid shall start at the very base of your neck. It shall be tight, high-tension braid.
- The Ponytail: For medium-length hair, a low ponytail is permitted.
  - It must be secured with an undecorated black/brown scrunchie, and the tail itself generally cannot exceed 10 inches in length. If it exceeds, it has to be put in a bun.
- Short Hair (The Bob and Pixie Cuts): pilots who prefer shorter hair, bobs and pixie cuts are recommended.
  - Hair can be worn down if it does not touch or fall below the top of the shirt collar, provided it is neatly styled and tucked behind the ears (with bobby pins) so it doesn't fall forward during critical phases of flight (takeoff and landing). But if the hair touches the collar, it must be tied back.

- Micro-Braids or Box Braids: Hair must be either "Micro" or "Small" box braids which look much more professional and flexible. "Jumbo" Braids are not permitted.
  - Length should be 10 inches, if longer should be secured in a bun.
  - The bun should not be bulky. Bulky braids make it nearly impossible to wear the uniform hat correctly and can interfere with the seal of an oxygen mask. The braids must not create gaps around the temples or jawline that would let air leak out.
  - The braids should look sleek from the forehead back. Boho Braids are not allowed.
  - If the braids start to look "fuzzy" due to friction or the roots have grown out significantly (usually past 2 weeks), the style should be refreshed. Visible "new growth" that looks untidy is a grooming violation.
  - the braiding hair must match your natural hair color exactly. "Ombré" or "highlights" are not permitted.
  - No Decorative Beads/Jewelry are allowed on the braids.

C. Makeup (Female Pilots)
- Makeup is optional.
- If worn, it must be subtle, matching natural skin tone.
- Excessive or dramatic styles are not acceptable.

D. Jewelry
- Jewelry must complement the uniform and maintain a conservative, professional appearance.
- One ring per hand (not on the index finger or thumb) is permitted; moderate in size.
- Visible piercings and earrings are not allowed for male crew.
- Visible tattoos are prohibited.
- Watches must be conservative with metallic or leather straps and faces no larger than the wearer’s wrist.
- Bracelets or any wristband is not allowed.

For Female Pilots:
- One matched pair of earrings (stud or small hoop) may be worn on the earlobe only.
- Dangling, tribal, or oversized earrings are not permitted.
- One slim bracelet or bangle matching the uniform may be worn.
- Other facial jewelry is prohibited.

7.3.4 Uniform Accessories

a) Gloves
- Plain black leather or fabric gloves are authorized.
- Must be clean and in good condition.

b) Umbrella
- It must be short, foldable, and plain black.
- Must be kept clean and serviceable. Use is not permitted on the air side.

c) In-flight Bag and Luggage
- A Company-issued or equivalent black professional flight bag must be used.
- Backpacks or casual rucksacks are not permitted.
- Luggage must be clean, undamaged, and free from non-company stickers or decorations.
- All crew are responsible for loading their own luggage onto company transport; the captain may delegate verification duties.

d) Sunglasses / Spectacles
- Must have a professional, conservative design.
- Frame colors: black, brown, silver, or gold.

- Frameless glasses are permitted.
- Frames must be free of ornaments or decorative elements.
- Sunglasses must not be worn indoors, on top of the head, or attached to the uniform.

7.3.5 Conduct While in Uniform

Only light, subtle fragrances may be used.
Crew must be on the hotel bus at least five (5) minutes before pickup time, with personal tasks completed. Professional conduct and demeanor must be always maintained while in uniform.

The First Officer is responsible for verifying that all crew members are listed on the hotel accommodation list upon arrival.
If any crew member's name is missing, the First Officer shall record it from the General Declaration (GD) and ensure that the correct wake-up time is clearly communicated to the hotel front desk.

7.3.6 Crew Rest Attire (Pajamas / Sleepwear)

General Principles:
- Crew rest attire should prioritize comfort, hygiene, and safety.
- Attire must be always clean and in good condition.

Male Crew:
- Sleepwear may include plain T-shirts and shorts or trousers.
- Avoid clothing with offensive logos, graphics, or text.
- No sleeveless undershirts in shared crew areas.

Female Crew:
- Sleepwear may include plain T-shirts, tops, or nightwear and shorts/trousers.
- Avoid clothing that is overly revealing in shared crew areas.
- Brightly colored or patterned sleepwear is acceptable if not offensive or inappropriate.

Additional Notes:
- Crews should change into appropriate uniform or casual attire before leaving the hotel room for public areas.
- Footwear such as flip-flops or slippers may be used in private or crew rest areas only.
- All attire must not interfere with the safe use of safety equipment during any in-flight rest.

7.3.7 Handing Over of Crew Hotel Room Keys

Crew hotel room keys are company property during layovers and must be handled with care and accountability.

Procedure:

- Hotel room keys are **strictly personal and non-transferable**.
- **Under no circumstances** shall a crew member hand over or give access to their hotel room key to **any third party**, including other crew members, hotel staff, or external visitors.
- In the event of an emergency or lockout, crew shall contact the **hotel front desk**; replacement or access must be arranged through official channels only.

- Violation of this policy may be treated as a **breach of company security protocol** and could lead to disciplinary action..

7.3.8 Baggage Handling at Check-in and Layover Hotels

Crew members are expected to adhere to baggage weight limits and maintain professionalism at airports and hotels.

Procedure:

- Baggage allowance shall follow company policy (2 pieces "where allowed", max 23 kg each).
- **Unnecessary opening or rearrangement of heavy bags** at check-in counters, terminal areas, or hotel lobbies is strictly prohibited.
- Crews shall prepare and weigh baggage at home/hotel room or before reporting to the airport.
- Any overweight or excess luggage must be handled personally and shall not delay check-in or crew processing.

7.4 ALCOHOL AND PSYCHOACTIVE SUBSTANCES

7.4.1 Crewmembers shall not consume alcoholic beverages while on duty or within eight hours of flight departure time. Crewmembers shall not commence flight duty if under the influence of alcohol or psychoactive substances, including drugs prescribed by a physician. In addition to this, no heavy drinking will be permitted 24 hours prior to scheduled departure.

7.4.2 People occupying observer's seats shall not be served alcoholic beverages. Persons who have been consuming alcoholic beverages shall not occupy observer seat and no person shall consume alcoholic beverages while in uniform.

7.4.3 Deadheading crew members traveling as passengers shall not be served alcoholic beverages.

7.4.4 Narcotics shall not be used at any time except under medical direction and when the crew member is not on flight duty.

7.4.5 The problematic use of any psychoactive substances by any flight crew member is prohibited. Any flight crew member that is identified as engaging in any kind of problematic use of alcohol or psychoactive substances will be removed from flight duty and may be dismissed immediately.

7.4.6 In addition, flight crew members are subject to the regulations set by ECAA regarding the use of such substances.

7.5 PERSONAL DOCUMENTS

7.5.1 Passports, visas, licenses, pilot Logbooks, vaccination cards, and crew identification cards, etc… must be carried by all crewmembers. The scheduling department shall normally maintain records of such documents and remind each crewmember of such documents when expiration dates are imminent. However, it is mainly the crew member's responsibility to watch expiration dates himself and maintain valid documents.

7.5.2 If any required documents are lost, the responsible flight crewmember shall notify the Flight operations immediately.

7.5.3 **Renewal of Personal Documents**

All flight crew members are required to ensure that any renewed personal documents (license, medical certificate, passport, Visa, etc.) are submitted promptly to the Director of Flying Administration.

Procedure:

- Renewed copies shall be **sent via email** to **Director, Flying Administration** **DirFlyingAdmin@ethiopianairlines.com**
- **Physically delivered** in person during office hours to the Flying Administration office.
  Failure to provide updated documentation may result in administrative hold or suspension from active duty until compliance is achieved.

Responsibility:
Each crew member is personally responsible for maintaining current documentation and ensuring timely submission.

7.6. BEFORE FLIGHT THE CAPTAIN SHALL:

7.6.1 Obtain the necessary information regarding hazardous areas, serviceability of airport, and aids to navigation for the route to be flown by examining the briefing sheet (NOTAM applicable to the enroute phase of flight and departure, destination and alternate airport). In addition to this, ensure all essential information pertinent to the flight is on board the airplane.

7.6.2 Obtain the necessary meteorological data and ensure information relevant to the flight, including departure, departure alternate (if applicable), enroute, destination and destination alternate weathers are available.

Ensure that Operational flight plan (OFP) is received. If hard copy is prepared, Operational flight plan will be printed in duplicate by fight dispatch and the front page of the OFP will be prepared as well to be signed by the captain. The signed OFP shall be retained in file.

7.6.3 Under normal circumstances, all the above documents will be retrieved online through approved CLASS I EFB on any compatible own device. Whenever the Flight Crew are unable to access the documents online, paper backup has to be given to the Flight Crew. PIC is responsible for checking the briefing package/s before departure for the final OFP and confirm by completing Flight Release Form in the e-Form section of the Logipad and signing electronically.

7.6.4 The captain for which a Captain is detailed extends from the time he signs the aircraft dispatch release papers until the aircraft is handed over to Ground technician.

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7.6.5 Ensuring that ATC flight plans are duly filed, checked, and signed by him or his designate. File the flight plan with ATC except at station where the company's representative performs this.

7.6.6 Perform flight planning (if not performed by a flight dispatcher or another designated person) and determine quantity of fuel, oil, weight and balance information etc., as required.

7.6.7 If the flight is to be delayed or canceled, he should confer with the operational planner and/or the aircraft dispatcher on duty and ensure that the appropriate actions are taken, and delay notices are issued and that the passengers are informed of the delay or cancellation. At stations where there is no Ethiopian station staff, it is the captain's duty to initiate the appropriate correspondence with the appropriate aircraft dispatcher or scheduler.

7.6.8 On long delays the captain should be mindful of his passengers, but the offer of free alcoholic drinks should not be made without exercising the greatest discretion. An infringement of IATA regulations in this respect could result in the airline being fined a considerable sum of money. Furthermore, in Muslim countries all alcohol must be securely stowed and locked. Failure to do this could result in severe penalties being imposed by the authorities.

7.6.9 Ensure that all members of the crew are on board the aircraft one hour before the scheduled departure or revised departure time, so that departure can be made without delay immediately after the passengers are comfortably seated.

7.6.10 Ensure that the prescribed emergency equipment is on board and in its proper location prior to departure.

7.6.11 Ensure for each flight a description of known or suspected defects that affect the operation of the aircraft is recorded in the ATL.

7.6.12 Ensure by reference to the maintenance log that all previous discrepancies have been cleared or processed in accordance with the MEL/CDL and the certificate of airworthiness is valid before departure.

7.6.13 Ensure that Navigation Kit is on board the airplane and contains the prescribed items.

7.6.14 Check all documents well in advance of the expected flight departure and ensure that they are complete. If a document is incomplete or missing, get a replacement from Flight Operations Documentation in good time to avoid delay.

7.6.15 Ensure that the preflight check (exterior inspection) as outlined in the Aircraft Flight Manual has been satisfactorily accomplished by each assigned flight crewmember. The exterior inspection shall focus on safety in critical areas of the aircraft and as a minimum ensure:

a) Pitot and static ports, MFPs (multi-function probes), AOA (angle of attack) are not damaged or obstructed.

b) Flight controls are not locked or disabled (as applicable, depending on aircraft type);

c) Frost, snow or ice is not present on critical surfaces;

d) Aircraft structure or structural components are not damaged.

7.6.16 Ensure that he and his crew possess and have with them valid licenses, passport and health certificates in connection with the duties for which they have been detailed.

7.6.17 Ensure pre-flight check (exterior inspection) of the airplane and equipment as outlined in the Aircraft Flight Manual has been satisfactorily accomplished by each assigned flight crewmember. The PIC shall not delegate the accomplishment of the exterior aircraft inspection (walkaround) to other individuals except to the qualified Fight crew member. The exterior inspection shall focus on safety in critical areas of the aircraft and as a minimum ensure:

a) Blocked or damaged pitot/static ports;
b) Locked or disabled flight controls (as applicable depending on aircraft type);
c) Locked or disabled steering and/or landing gear systems;
d) Landing gear strut compression;
e) Tire pressure, wear and/or damage;
f) Fluid leaks;
g) Unlatched/open doors and access panels;
h) Presence of frost, snow or ice on critical surfaces;
i) Aircraft structural integrity (damage);
j) Flight crew notification procedures;
k) Any other safety-critical and/or aircraft type-specific items as defined by the aircraft manufacturer or operator.

7.6.18 Check all radio equipment and ensure that all cockpit clocks are set at the correct time (UTC).

7.6.19 Check his oxygen equipment to ensure that the oxygen mask is functioning properly, and that the oxygen supply and pressure are adequate for use as described in the airplane Flight Manual before each flight.

7.6.20 Check proper amount of fuel is on board in accordance with OFP.

7.6.21 Ensure repair or replacement of any instrument or equipment found to be malfunctioning or not in compliance with MEL/CDL if the aircraft is to be dispatched with inoperative equipment.

7.6.22 Ensure that on the first flight for the flight crew on an aircraft during duty period after an aircraft has been left unattended by a flight crew for any period of time, flight crew members shall perform a preflight inspection of flight deck systems and emergency equipment for their availability, accessibility and serviceability.

7.6.23 Ensure that when no cabin crew is used, a new flight crew has assumed control of the aircraft, the flight crew shall perform a preflight inspection of emergency systems and equipment in the cabin and supernumerary compartment for their availability, accessibility and serviceability.

7.6.24 Accept the airplane from the responsible technician and thereafter be responsible for the airplane until the fight is complete.

7.6.25 Ensure that the Maintenance Inspection form and A/C logbook are complete, signed and valid for the intended duration of the flight.

7.6.26 Ensuring that the aircraft is securely loaded as prescribed in official instructions. As it is the normal duty of station staff to supervise the actual loading and complete the load and trim documentation, the captain may accept the accuracy of the data on such documentation. Refer FOM 4.2 (7.1,7.2 & 7.3)

7.6.27 Shall personally assume control of special diplomatic, army and/or police mail, precious cargo, and supervise its stowage aboard the aircraft. The responsibility for the security of these items will be his from the time he signs for them and all doors closed until they are handed over at their destination.

7.6.28 Ensure compliance with regulations concerning transportation of restricted or dangerous goods.

7.6.29 In the case of overbooking, where seats are available, but the aircraft is apparently overweight according to standard passenger weights, he shall consider taking individual weights. Captains should liaise with the dispatchers on duty whenever this situation occurs. On no account should standard weights be mixed with individual weights. One or the other must be used for all passenger weights.

7.6.30 Check with cabin staff that the correct number of passengers are on board, including dead heads and any accommodated on the flight deck; that they are briefed on the position and method of use of cabin emergency exists and equipment and are given a practical demonstration of method of use of life jackets for flights of 30 minutes or more over the sea, and in the use of oxygen masks in the event of pressurization failure. A passenger in the flight deck for take-off or landing should be briefed on the position of the flight deck emergency and the action required of him in an emergency.

7.6.31 Approve the distribution and amount of the load by checking and signing the load sheet, general declaration, passenger and NOTOC as necessary.

7.6.32 Ensuring that aircraft performance is determined by taking into account weather and runway condition and technical status of the aircraft.

7.6.33 Ensure that all passengers are given briefings on location and use of emergency equipment.

7.6.34 Assigns Pilot flying (PF) and Pilot monitoring (PM).

7.6.35 Briefs or delegates the first officer to conduct the briefing on the intended flight.

7.6.36 Ensure that crewmembers and passengers are properly secured in their seat by seatbelts and/or shoulder harness for take-off and landing, and at other times when turbulence or emergency conditions make it necessary.

7.7. DURING FLIGHT THE CAPTAIN SHALL:

7.7.1 Be responsible for all required radio communication, proper position reporting, adherence to ATC clearances, and routing.

7.7.2 The captain is responsible for monitoring and verifying the navigational performance and all essential instruments at regular intervals.

7.7.3 Ensure that current forecasted weather information is received Checked and monitored during the enroute phase of flight to include as a minimum destination, destination alternate and for any enroute alternate airports.

7.7.4 The captain shall maintain an effective lookout, during all phases and conditions of flight, traffic in order to avoid collisions.

7.7.5 Shall record and monitor time over way points and fuel status on OFP at least every 30minutes provided that cruise time is an hour or more, for the purpose of identifying trends and for comparison to the OFP.

7.7.6 The captain is responsible for the care and comfort of his passengers and the wellbeing of his crew.

7.7.7 The captain is expected to make regular use of the aircraft PA system or through information documents to communicate with his passengers after having ensured through consultation with the team leader cabin services that they are awake. Complaints are often received from passengers that they rarely hear from the Captain on PA or receive flight progress reports. The captain can improve passenger relations by keeping the passengers informed of flight progress, reasons for delays etc. On short sectors and night sectors on which passengers are asleep, it is not possible or desirable to make an announcement or to distribute flight progress reports, but on all suitable sectors the passengers should be addressed in one form or another or be given a brief on the route before pushback/taxi whichever is earlier.

7.7.8 Be responsible in providing early warnings through company radio (HF or VHF) or ACARS to communicate any aircraft condition that may affect subsequent dispatch. At all subsequent stops, ensure that Aircraft Technical Log (ATL) entries are updated with all known or suspected defects, clearly reflecting the current status of the aircraft in a legible and non-erasable manner. All defects and operational remarks must be recorded in the ATL, regardless of whether they are repetitive or previously reported, to maintain continuous and accurate tracking of aircraft condition. In case of errors, corrections must be made in a clear and identifiable manner and initialed. All ATL entries must be complete and up to date at all times.

NOTE:

a) In the case of PA, the most suitable time to make an announcement is when the aircraft is set in climb for the final level. It is suggested that passengers are welcomed abroad for a short reason and an apology for delay if any be given and ETA destination, weather, points of interest enroute etc.... be included.

b) The use of PA can be overdone, but it is usually possible to make an announcement in most sectors after checking with cabin staff that passengers are not asleep.

c) Attention shall be paid to the use of PA and passenger relations during qualification route checks. First officers when flying are expected to make PA announcements and/or provide flight progress reports.

d) The captain shall pay careful attention to the convenience of passengers at each stop. Passengers must be informed as soon as possible if any change to the schedule of the flight is contemplated. If an airfield has to be over-flown due to weather or a diversion is decided upon, it shall be the captain’s duty to apologize and give an explanation to the passengers whose care and comfort will remain his responsibility until handed over to the station staff at the ultimate destination.

7.7.8 Be responsible in giving early warnings through company radio (either HF or VHF) or ACARS to maintain the status of the aircraft that adversely affect the next dispatch. At all subsequent stops they are responsible for updating Aircraft Technical Log (ATL) entries with known or suspected defects reflecting the present status of the aircraft in a legible and no erasable manner. In case errors were committed corrections should be made in a legible and identifiable manner and initialed. All ATL entries need to be up to date.

7.7.9 Immediately inform the station/Authorities concerned of any irregularities (e.g. technical troubles, diversions etc...) that may affect planning and handling activities on the ground.

7.7.10 Report the inadequacy of any facilities to the concerned authority without delay.

7.7.11 Be responsible that the following information recorded for each flight:

a) Aircraft registration;
b) Date;
c) Flight number;
d) Flight crew names and duty assignment;
e) Departure and arrival airports;
f) ATD, ATA & flight time.

7.7.12 Be ultimately responsible for cabin service standards and is requested to submit constructive criticism and comments on his Captain's Report form regarding the standard attire and punctuality of cabin staff.

7.7.13 Not deliberately violate the company's operational standards.

7.8. AFTER FLIGHT THE CAPTAIN SHALL:

7.8.1 Sign and ensure completion of all logs and records maintained during flight & supervise the placement of NAV charts in the proper Jeppesen folders if hard copy is used.

7.8.2 Make a report by using Logipad e-Form section when appropriate, to the Dir. Flying Operations for any irregularity for which the company could be held liable.

7.8.3 Send a report on any irregularity.

7.8.4 Be responsible for the proper safeguarding of the airplane, passengers' personal belongings and cargo at airports where no company personnel are present to whom this duty is assigned.

7.8.5 After each landing, a flight crew shall complete the post flight inspection. The crew members will be inspecting the A/C for general condition and look for any abnormalities which may have occurred during the flight and log in the maintenance logbook.

7.8.6 The captain is responsible for entering into the aircraft technical logbook all known and/or suspected defects and malfunctions affecting the aircraft.

7.8.7 The captain is responsible for completing and signing the flight logbook.

7.8.8 After every flight debrief the crew on overall flight occurrences.

7.8.9 Complete the necessary report forms.

7.8.10 Brief the Flight dispatcher and/or Operational planner all the relevant facts concerning the flight like inadequacy of facilities, etc...

7.9. PRIOR TO FLIGHT THE FIRST OFFICER SHALL

7.9.1 Obtain the necessary information regarding hazardous areas, serviceability of airport, and aids to navigation for the route to be flown by examining the briefing sheet (NOTAM). In addition to this, ensure all essential information pertinent to the flight is on board the airplane.

7.9.2 Obtain the necessary meteorological data and ensure information relevant to the flight, including departure, departure alternate (if applicable), en-route, destination and destination alternate weathers are available.

7.9.3 Ensure that Operational flight plan (OFP) is received. If hard copy is prepared, Operational flight plan will be printed in duplicate by fight dispatch and the front page of the OFP will be prepared as well to be signed by the captain or by his designated First Officer, this serves as the acceptance of the OFP by the captain. The signed OFP shall be retained in file.

7.9.4. Under normal circumstances, all the above documents will be retrieved online through approved CLASS I EFB on any compatible own device. Whenever the Flight Crew are unable to access the documents online, paper backup has to be given to the Flight Crew. PIC is responsible for checking the briefing package/s before departure for the final OFP and confirm by completing Flight Release Form in the e-Form section of the Logipad.

7.9.5. Ensuring that ATC flight plans are duly filed, checked, and signed by him when designated by the captain. File the flight plan with ATC except at station where the company's representative performs this.

7.9.6. Perform flight planning (if not performed by a flight dispatcher or another designated person) and determine quantity of fuel, oil, weight and balance information etc., as required.

7.9.7. Be at the airplane at least one hour before departure and observe cabin crews are also onboard; if not, coordinate with crew scheduling to avoid delay.

7.9.8. Ensure that the prescribed emergency equipment is on board and in its proper location prior to departure.

7.9.9. Ensure by reference to the maintenance log that all previous discrepancies have been cleared or processed in accordance with the MEL/CDL and the certificate of airworthiness is valid before departure.

7.9.10. Ensure that Navigation Kit is on board the airplane and contains the prescribed items.

7.9.11. Check all documents well in advance of the expected flight departure and ensure that they are complete. If a document is incomplete of missing, get a replacement from Flight Operations Documentation in good time to avoid delay.

7.9.12. Perform the preflight check (exterior inspection) as outlined in the Aircraft Flight Manual. The exterior inspection shall focus on safety in critical areas of the aircraft and as a minimum ensure:

a) Pitot and static ports, MFPs (multi-function probes), AOA (angle of attack) are not damaged or obstructed;

b) Flight controls are not locked or disabled (as applicable, depending on aircraft type);

c) Frost, snow or ice is not present on critical surfaces;

d) Aircraft structure or structural components are not damaged.

7.9.13. Perform pre-flight inspection of the airplane and equipment on the ground.

7.9.14. Check all radio equipment and set all cockpit clocks at the correct time (UTC).

7.9.15. Check his oxygen equipment to ensure that the oxygen mask is functioning properly and that the oxygen supply and pressure are adequate for use as described in the airplane Flight Manual before each flight.

7.9.16. Check proper amount of fuel is on board in accordance with OFP.

7.9.17. Ensure repair or replacement of any instrument or equipment found to be malfunctioning or not in compliance with MEL/CDL if the aircraft is to be dispatched with inoperative equipment.

7.9.18. Assist the captain that the aircraft is securely loaded as prescribed in official instructions. As it is the normal duty of station staff to supervise the actual loading and complete the load and trim documentation, the captain may accept the accuracy of the data on such documentation.

7.9.19. Assist the Captain in compliance with regulations concerning transportation of restricted or dangerous goods.

7.9.20. Ensuring that aircraft performance is determined by taking into account weather and runway condition and technical status of the aircraft.

7.10. DURING FLIGHT THE FIRST OFFICER SHALL:

7.10.1. Be responsible for all required radio communication, proper position reporting, adherence to ATC clearances and conduct communication in the prescribed manner & monitor ATC clearances compliance thereto.

7.10.2. Notify Holloway of the ATD, ETA, at company reporting points, en- route flight movement and/or significant deviation from the operational flight plan for flight following purposes.

7.10.3. Monitor and verify the navigational performance and all essential instruments at regular intervals.

7.10.4. Be familiar with the available navigation and communication facilities, ATC procedures, takeoff and approach procedures along the route and for the airport concerned.

7.10.5. Advise the Captain of any conflicting traffic & obstruction which could be hazardous.

7.10.6. Obtain Check and monitor current and forecasted weather information during the en-route phase of flight to include as a minimum destination, destination alternate and any en-route alternate airports. Observe changes of wind, ice formation, and other conditions, which may influence the normal progress of the flight and advise the captain.

7.10.7. Maintain an effective lookout during all phases of flight for conflicting traffic.

7.10.8. Check and record time over waypoints and fuel status on the operational flight plan, for flights less than one-hour, verbal discussion with the PIC suffices inflight fuel monitoring. Monitor flight time and fuel burn to identify trends and for comparison to the OFP.

7.10.9. 7.10.9 Be responsible in giving early warnings through company radio (either HF or VHF) or ACARS to maintenance on the status of the aircraft that adversely affects the next dispatch. At all subsequent stops they are responsible for updating Aircraft Technical Log (ATL) entries with known or suspected defects reflecting the present status of the aircraft in a legible and no erasable manner. In case errors were committed corrections should be made in a legible and identifiable manner and initialed as necessary.

7.10.10. Record the following information for each flight:

a) Aircraft registration;

b) Date;

c) Flight number;

d) Flight crew names and duty assignment;

e) Departure and arrival airports;

f) ATD, ATA & flight time.

7.10.11. Inform the Captain if and when any airplane or navigational abnormality exists.

7.10.12. Take orders directly from the captain and assist him in all phases of the flight.

7.10.13. Not deliberately violate the company's operational standards.

7.11. AFTER FLIGHT THE FIRST OFFICER SHALL:

7.11.1 Place NAV charts in the proper Jeppesen folders if paper chart is used.

7.11.2 Enter all known and/or suspected defects and malfunctions affecting the aircraft in the aircraft technical logbook when instructed by the captain.

7.11.3 Assist the Captain in completing the flight documents.

7.11.4 After each landing, a flight crew shall complete the post flight inspection. The crew members will be inspecting the A/C for general condition and look for any abnormalities which may have occurred during the flight and advice the PIC.

7.11.5 Provide the Flight Dispatcher/ station representatives with all information required for the arrival message through Holloway Radio.

7.11.6 Submit all completed documents to the Flight Dispatch Office if hard copy is used.

7.12. SAFETY OF COMPANY AIRCRAFT ON GROUND

7.12.1 The captain is responsible for adhering to the schedule as closely as circumstances permit. He should be satisfied when making up lost time (e.g. reducing time on the ground) that the crew will not be excessively fatigued.

7.12.2 After the aircraft is safely positioned for unloading on ground, the safety of the aircraft is the captain's responsibility partly to the First Officer, until a Local Agent or his Representative takeover as follows:

a) At normal night stop stations where there are adequate station staff and good facilities for ensuring the safety of the aircraft.

b) At unscheduled stops or stations where the staff is insufficient to accept responsibility, the captain shall take all possible precautions to ensure the safety of the aircraft, if necessary detailing members of his crew to guard it.

c) The captain shall take such steps as practicable to ensure that no attempt is made by any member of the crew to evade Customs or Censorship regulations, or have any unauthorized load carried on the aircraft.

7.13. GENERALLY, CABIN CREW MEMBERS SHALL:

7.13.1 Report to the PIC any safety related situation for timely action.

7.13.2 Not deliberately violate the company's operational standards.

7.13.3 Review company policies and procedures and keep assigned manuals current.

7.13.4 Cooperate with other crew members to ensure that the highest possible standards of flight safety and efficiency are maintained.

7.13.5 Know and comply with airport procedures for flight crews including requirements to carry passports, certificates, license, and vaccination cards required for flight duty and follow customs, immigration and currency control procedures as prescribed.

7.13.6 Be familiar with the handling of the airplane equipment for normal, abnormal and emergency operations.

7.13.7 Possess adequate and reliable personal equipment necessary to perform assigned duties.

7.13.8 Perform the duties and procedures prescribed in the appropriate Airplane Operations Manual and Company bulletins.

7.13.9 Maintain proficiency and qualifications in the airplane.

7.13.10 Report procedures in the Flight Operations Manual System that are impractical, inconsistent with specified standards or detrimental to safety. Make suggestions for improvement.

7.13.11 Report for duty at the prescribed time and location. When on layovers away from domicile, keep the Captain and the Team Leader Cabin Services informed of whereabouts if they leave the assigned hotel.

7.13.12 Brief crew members coming on duty concerning routing and status of the airplane and its equipment.

7.13.13 Pick up the mail at assigned company mailboxes regularly, maintain familiarity with information posted on the bulletin board, changes since last duty assignment and other data essential to job performance.

7.13.14 Maintain a clean and smart appearance and adhere to uniform regulations.

7.13.15 Cabin attendants shall cooperate with the Team Leader Cabin Services and with the cockpit crew to ensure the efficient conduct of a flight, providing passengers with safety, comfort and service.

7.13.16 From the time of the briefing prior to departure from home base, until return to the home base, cabin attendants are subordinate to the captain.

7.14. PRIOR TO FLIGHT CABIN ATTENDANTS SHALL:

7.14.1 Arrive punctually, well rested and prepared at the time specified for the cabin briefing.

7.14.2 Attend the cabin briefing.

7.14.3 Thereafter attend crew briefing of the captain during which the details of the flight are discussed. This is best done at a time when all crew are conveniently together before boarding passengers.

7.14.4 Ensure that on the first flight for the cabin crew or after a crew change or after an aircraft has been left unattended by a flight crew or cabin crew for any period of time cabin crew members shall perform a preflight inspection of cabin emergency systems, emergency equipment and (supernumerary compartment on cargo aircraft as applicable) for their availability, accessibility and serviceability.

7.14.5 After boarding the aircraft; check cabin equipment, catering and emergency equipment and report to the team leader cabin services any inadequacies as appropriate.

NOTE: The Team Leader cabin services in turn shall inform the captain or his deputy whether the cabin will or will not be ready for boarding in time.

7.14.6 Take up the prescribed positions in the aircraft for welcoming the passengers.

7.15. DURING FLIGHT THE CABIN ATTENDANTS SHALL

7.15.1 Ensure the efficient conduct of the prescribed services.

7.15.2 At transit stations, ensure that the exchange of dishes, the cleaning of the cabin, the loading of food and beverages are carried out as prescribed and that the passengers are adequately provided for.

7.15.3 This includes accompanying the passengers to the transit room and handling them to the new cabin team, when appropriate.

7.15.4 Ensure that the necessary documents are received, maintained and processed.

7.15.5 Report any incidents or irregularities that occur during the flight to the captain.

7.16. AFTER FLIGHT CABIN ATTENDANTS SHALL

7.16.1 Ensure the orderly handover of the aircraft to the new cabin attendants, when appropriate or to the catering personnel.

7.16.2 Report any incidents or irregularities that occurred during flight to the captain and the appropriate department verbally, followed by a written report.

7.17. To enhance flight safety through proactive visual scanning and situational awareness, especially in environments with potential traffic conflicts flight crew need to maintain vigilance for conflicting visual traffic.

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SECTION 1.7. FLIGHT CREW MANAGEMENT

SECTION 1.7. FLIGHT CREW MANAGEMENT

1.0 PURPOSE

The purpose of this procedure is to establish guidelines on how Flight crew management entails the planning of crew requirements, rostering, tracking and assuring qualification and re-currencies per type of aircraft and the regulation in force.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Crew Scheduler
- Duty Planner

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1 GENERAL

4.1.1 Manager crew scheduling and each flight crew member shall ensure that all crew duty assignments including relief and augmented operations that the flight crew will operate in compliance with the following:

a) Flight crew qualifications and recency requirements as listed in Chapter 0 SECTION 0.4.9 of this manual;

b) Crew composition requirements;

c) Flight/Duty and time limitations;

d) Minimum rest time requirements.

e) Inexperienced flight crew pairing limitations;

f) Fitness for duty (valid 1st Class medical certificate) and;

g) As stipulated in the ECAA regulatory directives and the collective bargaining agreement articles.

4.2 FLIGHT CREW MEMBERS FLIGHT, DUTY AND REST TIME LIMITATION

4.2.1 Ethiopian shall consider the following as duty time for the purposes of determining required rest periods and calculating duty time limitations for each operating flight crew member:

b) Entire duration of the flight;

c) Pre-operating deadhead time;

d) Training periods prior to a flight;

e) Office time prior to a flight for management flight crew members.

4.2.2 All flight crewmembers are obliged to forward all flight activities outside of Ethiopian, if any, to Crew Scheduling office and the latter shall monitor flight and duty time limitations.

NOTE: No Ethiopian flight crew member shall engage in commercial operation outside of company operations without the explicit approval of VP Flight Operations.

4.2.3 Flight deck crewmembers are required to keep a record of their total time in a logbook. The company will keep records of flight crew and cabin crew flight time, duty time and days worked.

4.2.4 The flight deck crew records the beginning and end of block and flight times to the nearest minute. The F/O or flight deck crew occupying the F/O seat records these flight times.

5.0 DEFINITION

'Acclimatized': a crew member is considered to be acclimatized to the WOCL of the time zone where he/she is in when he/she has spent at least 36 consecutive hours free of duty or 72 hours conducting duties in an area within 3 hours of time. Until that time the crew member remains acclimatized to his/her previously acclimatized time zone.

NOTE: acclimatized means the physiological and mental state of a crew member whose biorhythms and bodily functions are considered aligned with local time and applies only when 3 time zones or more is crossed.

'Augmented flight crew': A flight crew which comprises more than the minimum number required to operate the aircraft allowing each flight crew member to leave their assigned post and be replaced by another appropriately qualified flight crew member for the purpose of inflight rest.

'Block time' starts when the airplane starts to move. In case of pushback, it starts when pushback is initiated. In case of no pushback, it starts when the airplane starts to move under its own power. Block time stops when the airplane stops at its parking place at the end of the flight. Block time is used for recording flight time of crew members and flight-scheduled time.

Brake release before receiving explicit pushback clearance from Ground Control and confirmation from the ground crew is strictly prohibited, except in cases of immediate safety concerns.

Duty: Any task that a crew member is required to perform by the operator, including flight duty, administrative work, training, positioning, and standby.

NOTE: Administrative work, training, positioning & standby are not Flight Duty Period as long as they are not followed by a Flight Duty per FOM 1.7(7.16.3).

'**Duty period':** A period which starts when a crew member is required by an operator to report for or to commence a duty and ends when that person is free from all duties. In case of flight, duty starts at the reporting time and ends 30 minutes after block in.

'Flight Time': The total time from the moment the aircraft first moves from its parking place for the purpose of taking off until the moment it finally comes to rest on the designated parking position at the end of the flight and all engines or propellers are stopped.

'**Flight Duty Period (FDP)'**: A period which commences when a crew member is required to report for duty, which may include a flight or a series of flights and finishes when the aircraft finally comes to rest and the engines are shut down, at the end of the last flight on which he/she acts as a crew member.

'Home Base': The location nominated by the operator to the crew member from where the crew member normally starts and ends a duty period or a series of duty periods and where, under normal circumstances, the operator is not responsible for the accommodation of the crew member concerned.

'**Night Duty'**: A flight duty period encroaching on any portion of the period between 02:00 and 04:59 hours in the time zone to which the crew is acclimatized.

'Operating Crew Member': A crew member carrying out his/her duties in an aircraft during a flight.

'Positioning': The transferring of a non-operating crew member from one place to another, at the request of the operator, excluding the time from home to the designated reporting place at home base and vice versa, as well as the time for local transfer from a place of rest to the commencement of duty and vice versa.

'Rest Facility': A bunk, seat, room, or other accommodation that provides a crew member with a sleep opportunity:

- '**Class 1 Rest Facility'**: A bunk or other surface that allows for a flat sleeping position and is located separately from both the flight deck and the passengers cabin in an area that is temperature controlled, allows the crew member to control light, and provides isolation from noise and disturbance;

- '**Class 2 Rest Facility'**: A seat in an aircraft cabin that allows for a flat or near flat sleeping position, which is separated from passengers at least by a curtain to provide darkness and some sound mitigation, and is reasonably free from disturbance by passengers or crew members;
- '**Class 3 Rest Facility'**: A seat in an aircraft cabin or flight deck that reclines at least 40 degrees, provides leg and foot support and is separated from passengers by at least a curtain to provide darkness and some sound mitigation, and is not adjacent to any seat occupied by passengers.

'Rest Period': A continuous and defined period of time, subsequent to and/or prior to duty, during which a crew member is free of all duties.

Standby': A defined period of time during which a crew member is required by the operator to be available to receive an assignment for a flight, positioning or other duty without an intervening rest period.

WOCL: window of circadian low.

6.0 RESPONSIBILITY

6.1. The Vice President Flight Operations is primarily responsible to oversee compliance with the activities (rules and regulations) listed in this section.

6.2. The concerned department director(s) and the affected section manager(s) are jointly or individually responsible to ensure compliance with the related issues addressed in this section.

6.3. Flight crew management entails the planning of crew requirements, rostering, tracking and assuring qualification and re-currencies per type of aircraft and the regulation in force. Director Flying Operations in conjunction with Manager Crew Scheduling and team leader Crew Planning are the overall responsible parties for this purpose.

6.4. Manager Crew Scheduling and Duty Planner are responsible for assuring that all conditions as set in these instructions are carried out by the individual crew schedulers.

6.5. Flight crewmembers are responsible for not exceeding flight deck time limitations.

6.6. Flight Crew Members are responsible to report emergency, paternity & mourning leave to crew scheduling by any available means, & subsequently the respective chief pilot and/or Dir Flying Operations shall be notified through e-mail or text messages.

7.0 PROCEDURE

7.1. NOTIFICAITON OF FLIGHT ASSIGNMENT

7.1.1 Notification of all assignments shall normally be made by the monthly crew roster. It shall be the individual crewmember's responsibility to notify the appropriate section if his or her roster has not been received prior to the start of the new month.

7.1.2 Flight Crew members are required to check their Roster upon arrival Addis to see if there is any alteration to their monthly roster.

7.1.3 In times of operational irregularities, Director Flying Operations and manager Crew Scheduling may authorize such changes in short notice in as much as the operation requires.

a) Notification on the change of the roster will be made through telephone, electronic message or any other available means to make the crew aware of the change.

b) Receipt for changes to the roster shall be acknowledged by the crewmember concerned as soon as possible and a written record of the date, time, and method of such acknowledgement shall be kept by Crew Scheduling personnel.

c) A minimum advise time for an off-duty crew member before duty assignment shall not be less than 10 hours

d) Flight Operation shall have the authority to assign any pilot on any route if the office deems it necessary. Dir. Flying Operations, Dir. Flight Training or respective chief pilots shall authorize this assignment at any time if the need arises.

e) A Flight Crew member on duty can continue a flight or take additional flights without a rest period as long as the FDP is within limit.

7.2. FIRST-IN, FIRST-OUT SCHEDULING

7.2.1 During periods of extremely high non-scheduled flight activity, critical shortages of any or all categories of flight crews, or national emergency, normal scheduling processes may be temporarily suspended and the crew members in the aircraft type concerned may be scheduled on a day-to-day "First-in, first-out basis". During the period of time that this type of scheduling is in effect crewmembers will be assigned duties based on who first becomes legal for a particular flight after his or her return from a previous duty assignment. In order for this type of scheduling to work efficiently, an accurate and complete record of each individual crewmember's accumulated flight and duty time, rest time computations must be made several times each day to determine crew legality and position on the crew availability list regarding future crew assignments.

7.2.2 As soon as possible after the termination of the condition that precipitated the use of the "first-in, first–out" method of scheduling, normal monthly block scheduling will be resumed.

7.2.3 During the time period that "first-in, first-out" scheduling is in effect, notification of future duty assignments shall be made by telephone, electronic message service or any other means. Normal acknowledgement procedures shall apply.

7.3. LISTING NAMES WITH RESERVATION

It is the responsibility of Crew Scheduling at Addis Ababa or Station Manager at outstations, to notify the local reservations office of intent to travel including the names, route and destination. Such notification must be made at least 24 hours before departure. The same applies for the return segment of travel.

7.4. ROUTING ARRANGEMENT BY CREW SHCEDULING

7.4.1 When it is necessary to deadhead flight crew personnel, the routing will be determined by Crew Scheduling. Once established, the Team Leader of Crew Scheduling has the authority to change the routing.

7.4.2 Emergency changes in routing must be communicated immediately to the Crew Scheduling department in Addis Ababa.

7.4.3 Flight crew personnel may request changes in deadhead routing, and such requests must be made at least two weeks in advance.

7.4.4 In the event it is necessary to deadhead a complete crew (or part of crew) all crew members will travel together whenever this is practical.

7.5. TRADE OF FLIGHT ASSIGNMENTS

Due to many factors, this may have adverse effects on operations. However, the trading of flight assignment with prior approval of manager crew scheduling in coordination with respective chief pilots can be considered as far as it does not affect normal operations.

7.6. STANDBY ASSIGNMENTS

7.6.1 Flight personnel can be assigned for a maximum 12-hour standby period in a 24-hour time to cover all flights operating out of home base during such period. They may also expect to be called for special flights, test flights, ferry flights, or other duty assignments. When on standby assignment, flight personnel shall normally be available by telephone contact and shall be in a position to ensure reporting for flight duty within one hour following notification. When a flight crewmember on standby duty cannot be contacted by telephone, it shall be the responsibility of the individual crewmember to contact the Crew Scheduling office/dispatch at crew briefing time prescribed for all flights concerned and at least once each hour to ensure that all flights are covered.

7.6.2 When a standby crewmember is used, he/she shall be returned to his/her original rotation as soon as possible and another crewmember of the same rank and equipment category shall immediately be assigned the standby duty assignment of the utilized crewmember. Any flight crew member who had to be replaced or had a flight canceled shall immediately and automatically become the first standby for that day and shall be assigned before the regular standby crewmember unless such assignment will adversely affect future scheduling

7.6.3 Following the elapse of the minimum crew rest at layover stations, all flight crew are considered to be on standby and should avail themselves for duty. All efforts will be made to alert crew of schedule changes in good time by duty manager IOCC.

7.6.4 AIRPORT STANDBY

A crew member shall be considered on standby from reporting at the reporting point until the end of the notified standby period.

If standby is immediately followed by an FDP, the following shall apply:

a. If no accommodation is provided to the crew, the FDP shall count in full of the start of the standby;

b. If accommodation is provided to the crew the FDP shall count from the start of the FDP. If the time spent on standby is 6 hours or more the maximum FDP shall be reduced by the amount of standby time exceeding 6 hours. (ECARAS 8.12.1.9)

7.6.5. OTHER STANDBY

a) Standby other than airport standby shall be at home or in suitable accommodation.

b) If the time spent on standby is 8 hours or more the maximum FDP shall be reduced by 25% of the amount of standby time exceeding 8 hours.

c) The maximum number of standby hours shall not exceed 72 hours in any 7 consecutive Days. (ECARAS 8.12.1.10)

7.7. INABILITY TO REPORT FOR A FLIGHT OR ACCOMPLISH A SCHEDULED DUTY

When for any reason, a crewmember discovers that he/she will not be able to report for a flight or accomplish a scheduled duty assignment; the crewmember concerned shall immediately inform the Crew Scheduling office. Whenever possible, this information should be received by the Crew Scheduling office sufficiently in advance to prevent a delay due to the replacement of the crew member. Failing to report at the normal report time for a duty assignment will not be considered as adequate and sufficient notice and may earn a disciplinary measure.

7.8. BASIC MINIMUM REST (ECARAS 8.12.1.12)

7.8.1. MINIMUM REST PERIOD AT HOME BASE

The minimum rest period provided before undertaking an FDP starting at home base shall be at least as long as the preceding duty period, or 12 hours, whichever is the greater.

7.8.2. MINIMUM REST PERIOD AWAY FROM HOME BASE

The minimum rest period provided before undertaking a flight duty period starting away from home base shall be at least as long as the preceding duty period, or 10 hours, whichever is the greater. The minimum rest period away from home base shall include an 8-hour sleep opportunity taking account of travelling and physiological needs.

7.8.2.1. Every effort should be made to ensure that the rest time will not be interrupted by messages or telephone calls from the airport or company representatives. Line station personnel, without the express approval of the Dir. Flying Operations, shall make no assignment of flight crew members other than that which is in accordance with established policies and procedures.

7.8.3. RECURRENT EXTENDED RECOVERY REST PERIODS

The minimum recurrent extended recovery rest period to compensate for cumulative fatigue is 24 hours, such that there shall never be more than 168 hours between the end of one recurrent extended recovery rest period and the start of the next.

7.8.3.1. A cockpit crew that has flown 20 hours or more in any 48 consecutive hours or 24 hours or more in any 72 consecutive hours shall be given a rest period of not less than 18 hours before being assigned to any duty with the operator.

7.8.4. TIME ZONE DIFFERENCES (ECARAS 8.12.1.13)

Time zone differences shall be compensated by additional rest, as follows:

a) At home base, if one sector encompasses 4 to 5 time zones, the minimum rest shall be 36 hours if the sector encompasses 6 or more time zones, the minimum rest shall be 48 hours.

b) Away from home base, if one sector encompasses 4 time zones or more, the minimum rest shall be at least as long as the preceding duty period, or 14 hours, whichever is the greater (ECARAS 8.12.1.13)

7.9. FLIGHT CREW MEMBERS FLIGHT, DUTY AND REST TIME LIMITATION

7.9.1. FLIGHT TIME LIMITATIONS

Flight crew members shall not be scheduled, and no flight crew member may accept an assignment or continue an assigned flight duty period if the total flight time:

a) Will exceed 9:30 hours, if the operation is conducted with a 2-pilots flight crew.

b) Will exceed 13 hours if the operation is conducted with a 3-pilots flight crew

c) Will exceed 17 hours if the operation is conducted with a 4-pilot flight crew.

7.9.2 If unforeseen operational circumstances arise after takeoff that is beyond control, a flight crew member may exceed the maximum flight time specified and the cumulative flight time limits specified in section 7.9.1 & 7.14to the extent necessary to safely land the aircraft at the next destination airport or alternate, as appropriate. (ECARAS 8.12.1.4)

7.9.3 Flight time that exceeded the maximum flight time limits permitted by this section. must be reported to the Authority within 10 days. (ECARAS 8.12.1.4)

The report must contain the following:

a) A description of the extended flight time limitation and the circumstances Surrounding the need for the extension;

b) If the circumstances giving rise to the extension were within the certificate holder's control, the corrective action(s) that are intended to be taken to minimize the need for future extensions.

c) Corrective action must be implemented within 30 days from the date of the extended flight time limitation.

7.10. PERMITTED EXCEPTIONS TO LIMITATIONS

7.10.1. Limitations may be exceeded up to the limitations set by the law of the State due to the following reasons:

a) To protect the safety of flight.

b) For search and rescue operations.

c) For test flights at stations other than Addis Ababa.

d) To complete unavoidable operational delays (the monthly flight time ceiling shall not limit completion of the flight.

e) Seasonal variation in traffic load.

f) Charters or short leases, if the undertaking is unable to operate within flight time limitations.

g) In an emergency.

7.11. LIMITATIONS WHERE THE FLIGHT CREW CONSISTS OF TWO PILOTS (UNAUGMENTED OPERATIONS) (ECARAS8.12.1.5)

7.11.1 No flight crew member may accept an assignment for an unaugment flight operation if the scheduled flight duty period exceeds the limits in Table 1 below.

7.11.2 If the flight crew member is not acclimated the maximum flight duty period in Table 1 below is reduced by 30 minutes.

TABLE 1 Maximum Flight Duty Period Limits for Two Pilot Operations based on number of Flight Segments

Scheduled Time of Start (Acclimated Time) (LT)	Flight Segments					
	1	2	3	4	5	6	7 or more
0000-0559	12	12	10:30	10	9	9	9
0600-0659	13	13	12	12	11.5	11	10.5
0700-1259	14	14	13	13	12.5	12	11.5
1300-1659	13	13	12	12	11.5	11	10.5
1700-2359	12:30	12	11	11	10	9	9

7.12. LIMITATIONS FOR AUGMENTED FLIGHT CREW (ECARAS 8.12.1.7)

7.12.1 For Flight operations conducted with an acclimated augmented flight crew, flight crew may not be assigned, and no flight crewmember may accept an assignment if the scheduled flight duty period will exceed the limits specified in **TABLE 2**.

7.12.2 If the flight crew member is not acclimated the maximum flight duty period is reduced by 30 minutes.

7.12.3 Flight crew may not be assigned, and no flight crew member may accept an assignment involving more than three flight segments under this section.

TABLE 2 Maximum Flight Duty Period Limits for Augmented Flight Crew

Scheduled Time of Start (Acclimated Time) (LT)	Maximum Flight Duty Period (hours) Based on Rest Facility					
	Class 1 Rest Facility		Class 2 Rest Facility		Class 3 Rest Facility	
	3 Pilots	4 Pilots	3 Pilots	4 Pilots	3 Pilots	4 Pilots
0000-0559	16	18	15	16.5	14	14.5
0600-0659	17	19.5	16	17.5	15	15.5
0700-1259	18	20	17.5	19	16	16.5
1300-1659	17	19.5	16	17.5	15	15.5
1700-2359	16	18	15	16.5	14	14.5

7.13. FLIGHT TIME CUMULATIVE LIMITATIONS (ECARAS 8.12.1.11)

A flight crew member shall not fly and shall not be scheduled to fly if the flight crew members' total flight time will exceed the following;

a) 32 hours in any 7 consecutive calendar day period
b) 110 hours in any 30 consecutive calendar day period
c) 300 hours in any period of 90 consecutive calendar day period
d) 1000 hours in any 365 consecutive calendar day period
e)

7.14. DUTYTIME CUMULATIVE LIMITATIONS

If the flight crew member's total flight duty period will exceed:

a) 60 duty hours in any 7 consecutive days; and
b) 190 duty hours in any 30 consecutive days, spread as evenly as practicable throughout this period.

7.15. CONSECUTIVE NIGHTTIME OPERATIONS (ECARAS 8.12.1.14)

Not more than 3 consecutive flight duty periods that infringes on the window of circadian law can be undertaken, nor may there be more than 4 such duties in any 7 consecutive days.

7.16. MIXED DUTIES

7.16.1. MIXED OFFICE AND FLYING

When a crewmember is required to report for duty in advance of the stipulated report time for a scheduled flight, to carry out a task at the best of the company, the time spent on that task shall be part of the subsequent flight duty period.

7.16.2. MIXED TRAINING AND FLYING

When a flight crewmember flies in the simulator, either on a check or training flight, or as a training pilot or instructor, and then within the same duty period flies as a flight crewmember on a scheduled flight, all the time spent in the simulator/training is counted in full towards the subsequent flight duty period. The simulator session shall not be counted as sector, but the flight duty period allowable is calculated from the reporting time of the simulator.

7.16.3. MIXED GROUND ACTIVITY AND FLYING

When a crewmember is required to report for a flight duty after any ground activity assigned by the operator like (physical examination, training, administrative work, etc …), the time spent on that task shall be part of the subsequent flight duty period.

NOTE: The minimum advise time for duty assignment after these activities shall not be less than 10 hours.

7.17. COORDINATION WITH THE CREW SCHEDULING OFFICE

7.17.1 When monthly scheduling is in effect, each concerned section shall submit its request for the monthly schedule to the Crew Scheduling office by the 15th of each month. When “first-in, first-out” scheduling is in effect, each concerned section shall submit its request as soon as possible after they are established and, whenever possible, shall give the Crew Scheduling office at least 24 hours advance notice of these requirements.

7.17.2 In both cases, all concerned sections shall maintain a close liaison with the Crew Scheduling office and shall keep that office currently informed of any changes in personnel status or other factors affecting the efficient scheduling and notification of flight personnel.

7.18. CREW REPORTING TIME

7.18.1 All crew members shall report for duty to the dispatch office by checking in on the CMS on the designated computers one hour and thirty minutes before the scheduled departure time out of home base. At outstations duty starts one hour before the scheduled departure time. If special circumstances require an earlier reporting time, the crew members concerned will be so informed.

NOTE: For VIP flight report to the dispatch office three hours and thirty minutes before the scheduled departure time.

7.18.2 Deadheading crew members shall report in uniform for flight at least one hour before departure.

7.18.3 If a particular crewmember has not reported on time, the duty Scheduler shall accomplish the following: -

a) Make a reasonable effort to determine whether the crew member is present and has merely forgotten to report or if information regarding his or her delay in reporting has been received.

b) Make a reasonable effort to contact the crewmember by telephone and determine the situation.

7.18.4 If, after the above steps have been taken, the information available does not indicate that the crewmember will report at least an hour before scheduled departure time, to prevent further delays the standby crewmember shall be called. The regular crewmember is automatically removed from the flight, and the flight is assigned to the standby crewmember regardless of the fact that the regular crewmember may subsequently report prior to scheduled departure time. The only exception to this rule is that no standby crewmember is available or waiting for the standby crewmember to report for the flight will result in extreme delay.

7.19. NOTIFICATION OF IRREGULARITIES

If for any reason, other than caused by a flight crew member, a flight is delayed, IOCC shall inform the appropriate flight crew the departure delay in good time.

7.20. CREWMEMEBRS ON EXTENDED ABSENCE FROM DUTY

A crew member that is absent from duty for an extended time period due to illness, reassignment of primary duties, leave of absence or other cause will be removed from scheduled duty assignments and his responsibilities will be reassigned to another crewmember. When the crewmember concerned returns to duty, he shall be assigned standby duty until the start of a new scheduling period at which time normal scheduling practices will be resumed. The same as above is applicable when any crew member cancels his leave.

7.21. CREW TRANSPORTATION

Crew transportation will be provided by the company to and from the crewmember's home and the airport at the home base and to and from the crew hotel and the airport at a layover station. Crew members will be picked up between thirty minutes and one hour before the reporting time for their flight and are expected to leave as soon as this transportation arrives. Should the crew transportation fail to arrive by thirty minutes prior to the reporting time, a reasonable effort should be made to contact the company's representative at the layover station and make appropriate arrangements. If this effort is unsuccessful, satisfactory arrangements cannot be made; the use of a taxi at company expense will be authorized.

7.22. EQUALIZATION OF FLIGHT TIME

Crew Scheduling should try it's very best to balance flight time among crewmembers.

7.23. CABIN CREW FLIGHT DUTY AND REST PERIODS (ECARAS 8.12.1.16

The standard provisions on Cabin Crew flight duty and rest periods set up in this section apply to all Cabin Crew operating a flight and not only to the minimum cabin crew complement carried on the aircraft to meet provisions of the ECAA.

7.23.1 FLIGHT DUTY PERIOD (FDP)

a) Ethiopian may assign a duty period to a cabin crew only when the applicable flight duty period (FDP) limitations of this paragraph are met.

b) Unless the conditions stated in "C" below are met, Ethiopian may not assign a Cabin Crew a FDP of more than:

I. 14 hrs - 0600 - 1759 hrs

II. 13 hrs - 1800 - 0559 hrs

c) Ethiopian may assign a Cabin Crew for a FDP up to 20 hours provided the following conditions are met.

I. Horizontal rest facilities are provided.

II. The divisions of duty and rest are fairly distributed among all Cabin Crew members on a flight.

III. A minimum in-flight rest period of 3 hours must be provided for a FDP up to 18 hrs. and 4 hours must be provided for a FDP up to 20 hrs.

7.23.2 REST PERIOD AFTER A FDP

The minimum rest period for Cabin Crew immediately after a FDP shall be:

a) At home base as long as the preceding duty time or 12 hrs whichever is longer.

b) At outstation, shall be as long as the preceding duty period or 10 hrs whichever is longer.

7.23.3 DAY OFF

a) A maximum of 7 days off in a month shall be provided to a Cabin Crew.

b) Shall not be on duty for more than 7 consecutive days between day off.

7.23.4 DUTY HOURS LIMITATIONS

a) 60 hours in any 7 consecutive days (however, in the event of unforeseen delays after the Commencement of a rostered duty period covering a serious of duty periods, this limit may be increased to 65 hours).

b) 210 hrs. in any consecutive 30 days.

* * * * *

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

1.0 PURPOSE

The purpose of this flight crew composition is to establish guidelines on flight crew composition compliance.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Cabin Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1. No Ethiopian flight shall be operated unless at least the minimum flight crew in **TABLE1** in this section is provided.

4.2. No Ethiopian flight shall be operated unless at least the minimum safety cabin crew is provided as in TABLE 2 or as prescribed in 7.5.6 of this section.

5.0 DEFINITIONS

N/A

6.0 RESPONSIBILITY

All captains are responsible to make sure that the requirements in this section are met.

7.0 PROCEDURES

7.1 DESIGNATION OF PILOT-IN-COMMAND

7.1.1 Ethiopian shall designate one pilot as Pilot-In-Command (PIC). (ECARAS 9.3.1.6).

7.1.2 Except as provided below, a Captain is assigned on each flight as Pilot-In- Command and will occupy the left seat.

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

7.1.3 During upgrading or a transition training flight, the Check Captain shall be designated as the Pilot-In-Command and shall occupy the right pilot seat. On the final check the qualifying Captain shall be designated as pilot-in- command and the check Captain may occupy the observer's seat. The check Captain shall only assume command if the situation warrant, in which case he shall complete the flight as Pilot-In-Command.

7.1.4 During routine line checks, the qualifying Captain shall be designated as Pilot- In-Command and shall occupy the left pilot seat. The check Captain may occupy the observer's seat and shall only take command if he deems it necessary.

7.1.5 A change of Pilot-In-Command shall be noted in the airplane log as well as the journey log. Should the first officer assume command due to the captain's incapacitation, the event shall be noted in the same manner. IOCC shall also be notified via company VHF/HF, ACARS, SATCOM, (if applicable); such notification is forwarded to the company via enroute communication facilities.

7.2 CAPTAIN ASSIGNED AS PILOT-IN-COMMAND

7.2.1 The captain assigned as Pilot-In-Command is the commanding officer of the airplane and has full command over all crewmembers during flight and when away from home base and over all passengers as long as they are on board the airplane.

7.2.2 The Pilot-in-Command ensures the airworthiness of the aircraft and the serviceability of both operational and emergency equipment by: (ECARAS 9.3.2.2)

- Assuring the accomplishment of preflight inspections.
- Assuring the correction of any defect and/or damage affecting safe operation of an aircraft to an approved standard, taking into account the MEL and CDL of the aircraft type;
- Assuring the accomplishment of any operational directive, airworthiness directive and any other continued airworthiness requirement made mandatory by the Authority.

7.3 TWO CAPTAINS ASSIGNED TOGETHER

7.3.1 On all flights other than check or training flights, where two Captains are assigned together, the PIC shall always occupy the left seat.

7.3.2 For the duration of each flight Ethiopian designate only one pilot as a PIC and he/she will occupy the left seat. The PIC of the flight shall be the one to sign the ATL.

7.3.3 Normally, When the flight is originating from home base, the senior Captain will have the priority to choose which leg to fly after which the sequence will be by alternation. The company can identify the PIC of the flight by checking the hard or soft copy of the ATL.

7.3.4 Where the operation begins outside home base,

a) The captain who arrived at the station first and who has a longer layover will be the Pilot-In-Command of the first leg of the augmented crew operation.

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

b) When the Captains arrive at the station together or when their layover time is the same after arriving on different flights, the senior Captain has priority to choose which leg to fly.

c) The captain who served as a relief pilot or who was a DH crew (as applicable) will take the command responsibility on the subsequent leg

7.3.5 For safety reasons in augmented crew operations, the additional flight crew members(s) of the relief crew shall occupy the cockpit observer seat(s) for departure and arrival or in critical times where applicable. For departure, it shall be from engine start until the seat belt sign is off and for arrival from top of descent to block in; so that they shall attend the departure and arrival briefing together with the operating crew. This does not infringe the final authority and responsibility of the Pilot-In-Command.

7.3.6 However, with understanding among the operating crew, a suitable pattern can be arranged without affecting the pattern issued by crew scheduling.

7.3.7 If any abnormality or emergency occurs during the rest time of the Pilot-In-Command, the relief Captain will accomplish the emergency procedure or non- normal procedures and once the aircraft is under control, the Pilot-In-Command will be called to the flight deck and assume control of the aircraft. (Consideration should be given to the length of time the Pilot-In-Command had been resting before calling him to the flight deck.)

7.3.8 The crew shall discuss any decision and/or diversion for best result. The Pilot-In-Command shall make the final decision.

7.3.9 When a case arises during the layover time that requires coordination and decision of the Pilot-In-Command, the senior Captain will have the final authority and inform Flight Operations of his decision. Captain's report has to be filed to Dir. Flying Operations.

7.3.10 During cabin announcement, the names of Captains, F/O., and cabin Team Leaders of the operating crew shall be mentioned.

7.4 SUCCESSION OF COMMAND

7.4.1 The PIC is always in command except in the event of incapacitation in flight. In such situations the First Officer would assume the functions of the captain. FOM 3.6 (7.3).

7.4.2 The following chain of command shall be respected at all times.

a) Assigned Captain (PIC)
b) Relief Captain (if applicable)
c) First Officer
d) Relief First Officer (if applicable)
e) Team Leader Cabin Services
f) Senior Cabin Crew II
g) Senior Cabin Crew I

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h) Cabin crew

i) Junior cabin crew

7.5 CREW COMPOSITION

7.5.1 The number and composition of the flight crew may not be less than that specified in the flight manual or other documents associated with the airworthiness certificate. (ECARAS 8.4.1.1)

7.5.2 **SINGLE CREW: -** All company aircraft shall be operated by a minimum of two pilots, one of whom shall be designated as PIC and the other as first officer on the aircraft type concerned with the exception of training flights, where a student pilot may be substituted as first officer. On flights requiring two qualified Captains, one shall be designated Pilot-In-Command and have full authority over the conduct of the flight.

7.5.3 **MULTIPLE CREW:** - Two captains and one first officer.

7.5.4 **DOUBLE CREW: -** Two captains and two first officers.

NOTE: Throughout this manual, the Pilot-In-Command is referred to as "the captain."

7.5.5 Minimum Flight Crew: - No company flight shall be operated unless at least the minimum flight crew in **TABLE 1** is provided.

NOTE: Crew Scheduling Department shall be responsible for ensuring that two inexperienced flight crews are not assigned to operate together as per FOM 1.10(4.1.7).

TABLE 1

AIRCRAFT TYPE	NUMBER OF PILOTS
A350	2
B777/787	2
B767	2
B737	2
Q-400	2

7.5.6 Cabin crew composition: - On company revenue and non-revenue flights carrying passengers, the safety minimum cabin crew number shall be as follows:

a) For aircraft having a seating capacity of more than 9 but less than 50 passengers one Flight Attendant.

b) For aircraft having a seating capacity of more than 50 but less than 101 passengers, two Flight Attendants.

c) For aircraft having a seating capacity of more than 100 passengers, two Flight Attendants plus one additional Flight Attendant for each unit (or part of a unit) of 50 passenger seats above a seat capacity of 100 passengers.

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

d) On aircraft with nine passenger seats or less, a cabin attendant need not be carried provided a member of the flight crew can perform the safety functions of a cabin attendant or by competent person designated by the captain.

e) Additional cabin crew members may be assigned to a flight to provide the quality of in-flight service that the company desires.

f) On ferry and company business flights, the number of cabin attendants may be reduced by the substitution of qualified flight crew members that are not members of the assigned crew. If no passengers are carried, no cabin attendants need be assigned.

g) During takeoff and landing cabin attendants required in this section shall be located as near as practicable to the required floor level exits and shall be uniformly distributed throughout the aircraft to conduct emergency evacuation duties. During taxi, Flight Attendants must remain at their stations with safety belts and shoulder harnesses fastened.

7.5.7 **TABLE 2** below provides safety minimum cabin crew numbers by Aircraft Type & configuration.

7.5.8 In the event of cabin crew incapacitation, illness, non-availability or an unforeseen change of aircraft type, the cabin crew composition may be reduced using the following conditions and procedures from Out Stations.

a) Cabin crew composition below the safety minimum shall be with the authority of VP Flight Operations.

b) For each cabin crew reduced below the safety minimum cabin crew composition the PIC has to ensure:

I. 50 passenger seats in the vicinity of the vacant cabin crew seat shall be blocked from being utilized.

II. The team leader cabin services shall reassign areas of responsibility (including arming & disarming doors) of reduced cabin crew as required.

III. Re-seating of passengers with regard to exits & other applicable aircraft type limitations & the level of commercial service provided by the remaining cabin crew shall be at the discretion of the captain in consultation with the Team Leader Cabin Services.

IV. Able Bodied Persons (ABP) shall be selected to occupy the vacant cabin crew seats for Takeoff & Landing and shall be briefed.

V. Upon completion of the flight the captain shall complete an Air Safety Report which shall be duly submitted to the ECAA.

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

TABLE 2

A/C Type	Total Configuration	Safety minimum Cabin Crew
Q400	67-78	2
B737-700	118	3
B737-800	154	4
B737-8MAX	160	4
B767-300	214-270	5/6
B777-200LR	315	7
B777-300ER	393	8
B787-800	270	6
B787-900	315	7
A350-900	343-355	8
A350-1000	394	11
All cargo	-	0

NOTE: All cargo flights shall be operated with cabin crew if there are no Load Masters.

7.6 RELIEF PILOTS POLICY

7.6.1. Purpose

The purpose of this policy is to define the roles, responsibilities, and operational procedures, applicable during multiple crew operations when the relief captain seats on the right seat to ensure safety, compliance with regulations, and efficient crew management.

7.6.2. Scope

This policy applies to all flights

- Operational requirements for relief captain to Manage abnormal and emergency issues when seating on the right seat without the captain’s presence in the cockpit which ensures continuous alertness and safety.

7.6.3. Definitions

- **Relief pilot:** A pilot (Captain) assigned to monitor the aircraft and relieve the operating pilots during the flight on multiple crew operation (2 Captains and One First officer). The relief Captain will occupy the right seat to relief the first officer
- **Operating Pilot:** The Pilot in command or First Officer responsible for takeoff, landing, and other critical flight phases.
- **Multiple crew operation:** An operation conducted by 2 Captains and One First officer.

SECTION 1.8. FLIGHT AND CABIN CREW COMPOSITION

7.6.4. Eligibility

A relief pilot must be qualified Captain on the respective Aircraft.

7.6.5. Duties and Responsibilities

The relief pilot is responsible to perform all duties and responsibilities of F/O when assuming the F/O seat.

Limitations:

- The relief pilot shall **not perform takeoff, landing, or taxi** from the right seat unless he is a qualified examiner and current.

7.6.6. Rest and Duty Periods

a. Relief pilot duty periods shall comply with regulatory flight and duty time limitations.

b. Relief pilots may be assigned to long-haul flights in accordance with FOM Limitations

7.6.7. Training and Qualification

a. There is no specific training requirement other than the normal training of Aircraft systems, SOPs, and CRM practices, recurrent training .

b. Depressurization and engine failure in cruises has to be discussed /Practiced during Recurrent Simulator training based on the 3 year Matrix.

7.6.8. Safety and Compliance

a. Relief pilots must continuously monitor flight instruments, systems, and crew actions to ensure safety.

b. Relief pilots are required to comply with all regulatory and company safety directives.

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SECTION 1.9 FLIGHT CREW EMPLOYMENT, SENIORITY AND TERMINATION

SECTION 1.9 FLIGHT CREW EMPLOYMENT, SENIORITY AND TERMINATION

~~SECTION~~ **1.9 FLIGHT CREW EMPLOYMENT AND TERMINATION**

1.0 PURPOSE

This section provides guidelines for establishment of employment, termination, re-employment.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

The Vice President Flight Operations is primary responsible to oversee compliance with the company established policy and procedure.

7.0 PROCEDURES

7.1. EMPLOYMENT

7.1.1. The following process as illustrated on annex II of this chapter must be completed for all flight crew employment.

a) Credentials and Licenses

b) Flight Operations and/or Human Resource Management will assess psychological analysis or interpersonal skills.

c) Security background check.

SECTION 1.9 FLIGHT CREW EMPLOYMENT, SENIORITY AND TERMINATION

d) Medical examination.

e) Sufficient fluency in the English language for ATC communication, for flight crew and cabin crew communication and for adequate understanding of company manuals and documentation.

f) Aviation experience checks (if applicable).

g) A competency-based interview to check the applicant's language and technical competence.

h) Check the technical skill capability of the applicant through SIM evaluation.

7.1.2. An Individual may join Ethiopian's Flight Operations Division as a pilot either by graduating from Ethiopian aviation academy Pilot Training School, Ethiopian Air Force or any credible aviation facility. A graduate from Ethiopian aviation academy Pilot Training School or Ethiopian Air Force training school automatically join the airline as a First Officer. A person applying for employment from other than these institutes must take and pass a flight test, physical check, written and oral examinations. The initial job grades on joining Flight Operations for the above mentioned, up on employment, will be First Officer on the smallest of the Ethiopian fleets.

7.1.3. All new pilots will remain on a probationary status for a period of one (1) year after beginning on the line as flight crewmembers. Flight Operations may at any time and solely at its own discretion terminate a pilot during this probationary period. There will be no recourse from such termination.

NOTE: A commercial license with instrument rating and first-class medical certificate is required to join Ethiopian Flight Operations Division as a First Officer.

7.2. DIRECT ENTRY PILOTS

In addition to the requirement in (7.1) above, direct hire for the position of PIC shall as a minimum require line operation experience of 500hrs as PIC on type.

7.3. FLIGHT CREW SENIORITY

7.3.1 New entrants from Ethiopian Aviation academy will seniority placing per their ranking as provided from Manager MPL & CADET Training Department.

7.3.2 Pilots who failed transition training will lose seniority automatically and will be dealt with as per FOM5.2 (7.29).

7.3.3 Pilots who failed assessment interview, who couldn't succeed in their simulator or line training during upgrade training will lose their seniority. Second chance for assessment interview will be given after three months if the office believes that the candidate has improved on his previous weakness. Failure of simulator or line training will be dealt with FOM 5.2 (7.30)

7.3.4 A pilot who refuses to sit for upgrade assessment interview will lose his/her seniority and needs to present his/her justification in writing. He/she will be given another chance after a request for readiness in writing is submitted by him/her. The case will then be evaluated by the office for reconsideration.

7.3.5 Pilots who refuse to take transition training will lose their seniority.

7.3.6 Penalty based demotion should be treated in such a way that the seniority of the pilot will be revoked during his/her lower equipment operation due to the demotion. After the pilot is reinstated to his previous position, his seniority will be on top of the pilots who have less experience from his previous type of experience.

NOTE: If the demotion takes more than six months the pilot is required to accumulate at least 1000 hours of a type after being reinstated before being considered for promotion of higher equipment.

7.3.7 Disciplinary measures will not have an impact on pilots' seniority.

7.4. TERMINATION

7.4.1 If a new hire trainee does not successfully pass all written examinations, he may be given the opportunity to retake examinations in the subjects he failed in within six months. Failure to pass all examinations after such retakes shall justify termination.

7.4.2 A student that does not meet prescribed standards during their Flight Training period shall be terminated before being accepted as an employee of Ethiopian.

7.4.3 First officer's transition training for higher equipment will be handled according to FOM 5.2(7.29).

7.4.4 Student Captain failing to fulfill the prescribed ground and flight requirements, for the captain position will be handled according to FOM5.2 (7.30).

7.4.5 If Flight Training isn't successful and warrants termination from Ethiopian, as stated above, shall be given the opportunity to fill other available non-flying positions within Ethiopian, provided they meet the requirements for the positions available, and provided that the individuals have proven to be employees in excellent standing, their grounding was solely their ability to meet the company's standards for flying positions.

7.5. POLICY ON RE-EMPLOYMENT OF COCKPIT CREW

Flight Operations will not consider rehiring any cockpit crewmember that has resigned or been terminated as a result of demotion, limited ability, professional quality or low moral standards. This policy will apply to all Pilots irrespective of the remarks on the P-3 form.

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SECTION 1.10. MINIMUM FLIGHT CREW EXPERIENCE

SECTION 1.10. MINIMUM FLIGHT CREW EXPERIENCE

SECTION 1.10. MINIMUM FLIGHT CREW EXPERIENCE

1.0 PURPOSE

This section establishes guidelines for Ethiopian Airlines to carry minimum flight crew that can provide adequate safety operations.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1 PILOTS

4.1.1. All Ethiopian aircraft shall carry not less than two pilots as members of the flight crew. These pilots should be qualified in type and provide adequate safety for the operations involved.

4.1.2. Any Captain and/or First Officer deliberately infringing on these instructions may be considered to be guilty of disobedience or conduct prejudicial to the interests of the company.

4.1.3. The captain is authorized to allow a First Officer to carry out takeoffs and landings when the Captain has accumulated 300 hours' command time on type.

4.1.4. A Captain may allow a student Captain to make takeoffs, landings from the left seat provided he himself is qualified to do so, on type.

4.1.5. If the First Officer has less than 200 hours on type, and the captain is not an appropriately qualified examiner/flight instructor pilot, the Captain must make all takeoffs and landings. This limitation does not apply for SFO pilot positions.

4.1.6. The Captain shall be PF when takeoff and land under the following situations.

SECTION 1.10. MINIMUM FLIGHT CREW EXPERIENCE

a) At special airports (category C)

b) During low visibility operations

c) During heavy weight

d) Adverse weather conditions (contaminated runway, Strong cross wind, tail wind etc.).

e) Emergency or abnormal situations

4.1.7. Inexperienced Flight Crew (Captain who has less than 300 hours and F/O who has less than 200 hours on type after the completion of initial training/qualification) shall not be scheduled together. Such pairings do not apply to pilots when they are recruited with a higher-flying hours on a specific fleet or SFO positions.

NOTE: Flight crew scheduling office must ensure that such pairings of inexperienced flight crew members shall not be assigned together, and flight crew shall not accept such assignments.

4.1.8. Experience of PIC & FO for low visibility approach is as stated in FOM 2.20 (7.24.2 & 7.24.3)

5.0 DEFINITION

In relation to an aircraft "flight crew", means those members of the crew of the aircraft who undertake to act as Captain and/or First Officer of the aircraft. The composition of the operational flight crew should include qualified crew members as are necessary for the safety of the operations involved.

6.0 RESPONSIBILITY

N/A

7.0 PROCEDURE

N/A

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SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

1.0. PURPOSE

This section provides general regulations to be followed by all flight crew members and authorized inspector.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Crewmembers
- Authorized Inspector
- Air Traffic Controller

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.01

4.1. DUMPING OF FUEL

4.1.1. Fuel may be dumped in an emergency when the Captain considers such action is required.

4.1.2. Fuel dump could be initiated without any forewarning and at most any altitude, as might be the case should an engine failure, in combination with another performance limiting malfunction, occur on takeoff. Whilst it is likely that the pilot would declare an emergency in this situation, it is equally unlikely that they would have the time immediately available to detail any specific actions (including dumping fuel) that they might be taking.

4.1.3. Fuel Dumping (jettison) should be considered when situations dictate landing at high gross weights and adequate time is available to perform the jettison. When fuel jettison is to be accomplished, the following precautions shall normally be taken;

a) Air Traffic Control (ATC) should be informed.

b) Where possible, the aircraft shall be kept pressurized to keep out fumes.

c) Ensure adequate weather minimums exist at airport of intended landing.

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

d) Dumping shall not be carried out when there is evidence of precipitation, statistics or probability of lightning discharge if the aircraft is in close proximity to the ground or water.

e) Fuel jettisons should be done 5,000 to 6000 feet AGL to ensure complete fuel evaporation.

f) Airspeed shall be controlled within the limits dictated in the respective flight manuals.

g) Avoid jettisoning fuel in a holding pattern with other airplanes below.

h) Only electrical circuits essential to the operation shall be used during dumping.

i) No HF transmissions shall be made during dumping. A listening watch may be maintained. Other radio and radar transmissions shall be restricted to the minimum necessary in the circumstances.

NOTE: For more information on dumping of fuel procedures refer to FOM 2.7 (7.2)

4.1.4. The following separation minima to be used in respect to other known traffic in case of fuel dumping.

At least 10 NM horizontally, but not behind the A/C dumping fuel.

a) Vertical separation if behind the A/C dumping fuel within 15 minutes flying time or a distance of 50 NM by at least 1000 feet if above the A/C dumping fuel & at least 3000 feet below the A/C dumping fuel.

b) The horizontal boundaries of the area within which other traffic requires appropriate vertical separation extend for 10 NM either side of the track flown by the A/C which is dumping fuel, from 10NM ahead, to 50NM or 15 minutes along track behind it (including turns).

4.2. SMOKING IN THE AIRCRAFT

SMOKING IS BANNED ON ALL ETHIOPIAN FLIGHTS.

4.3. USE OF CABIN JUMP SEAT

4.3.1 The jump seats may be used for extra members of the crew whenever necessary. The captain has the authority to permit the use of the jump seat for a member of Ethiopian Staff or Staff family. It is entirely within the prerogative of the PIC to permit or refuse.

4.3.2 Flight Operations inspectors to Ethiopian Civil Aviation Authority travel on a firm booking. However, if the service on which an inspector is traveling becomes overbooked at any point, the Station will ask the captain to permit the Inspector to use the jump seat or a cabin crew seat in order to avoid off- loading a revenue passenger. Although Flight Operations Inspectors will normally have a firm booking for a passenger seat, it should be noted that under the terms of our Air Operation Certificate, an Inspector on duty has the authority to travel in the cockpit of the aircraft should he require for the purpose of observing the conduct of any part of the operation. The captain may decline such a request only if he considers that the Inspector's presence is prejudicial to flight safety.

4.3.3 Existing procedures are designed to ensure that problems of accommodating an Inspector on the cockpit are not caused by a flight inspection coinciding with the presence of additional crew members on duty. (I.e. during route checks, line checks, supervisory flight, navigation training, etc. …)

4.3.4 If such problems should arise for unforeseen reasons (e.g. unexpected change of plan or crew, etc.,) Captains should co-operate in meeting the Inspector's requirements, if possible, without disrupting the check/supervisory/training program. Although these problems will normally be satisfactorily resolved by co-operation between the Captain and Flight Operations, attention is drawn to the authority held by flight Operations should the inspector insist on traveling in the cockpit.

4.4. PORTABLE ELECTORONIC DEVICES (ECARAS 8.5.1.28)

Passengers should be advised regarding the restriction of the use of personal electronic device (PEDS). No PIC may permit any person to use, nor may any person use a portable electronic device on board an aircraft that may adversely affect the performance of aircraft systems and equipment. For commercial air transport operations, Ethiopian makes a determination of acceptable devices and publishes that information in FOM 2.8(7.3) for the crewmembers use and the PIC delegates the team leader cabin services to inform passengers of the permitted use.

4.5. DOCUMENTS TO BE CARRIED ONBOARD

All manuals, checklists and procedures developed and provided by the manufacturer shall be used unless processed for modification as stated in ETHIOPIAN AIRLINES QUALITY MANUAL APP. II. Such process shall ensure:

(i) Human factors principles are observed in the design of the OM, checklists and associated procedures;

(ii) The specific parts of the OM relevant to flight crew are clearly identified and defined;

(iii) If applicable, any differences from procedures and checklists provided by the manufacturer(s) are based on operational considerations.

4.6. AIRCRAFT LIBRARY

4.6.1. The following documents shall be carried on board each flight and stowed in the appropriate location either electronically or in hard copy in a manner that provides ready access by the flight crew during flight preparation and inflight. These include all the necessary documents for the route served by the particular fleet.

a) Flight Operations Manual (FOM);

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

b) Aircraft Flight Manual (AFM);

c) Aircraft Operations Manual (AOM) or Flight Crew Operations Manual (FCOM) and QRH as a minimum;

d) Normal and emergency checklists for each operating flight crew member;

e) Dispatch Deviation Guide (DDG);

f) Configuration deviation list (CDL):

g) Minimum Equipment List (MEL);

h) Master Minimum Equipment List (MMEL);

i) Fault Reporting Manual;

j) Airport Analysis for destination and alternate airports with special engine out departure procedures;

k) A/C specific Weight/Mass and Balance instructions/DATA including Load Sheet.

l) Company Route Manual containing Route and airport instructions and information for each flight to include departure airport, destination airport, En-route alternate airport & emergency airports. The escape routes used in case of decompression or engine failure in an area of high terrain.

m) The applicable departure, navigation and approach charts for use by each operating flight crew members;

n) For each operating crew a set of departure, destination, en-route, en- route emergency procedures and alternate Jeppesen air way manual containing route and airport instructions, as a minimum & Jeppesen airway manual.

o) Airport charts including the method for determining airport operating minima;

p) En-route charts;

q) Cabin crew Operations Manual;

r) Airline Security Program Manual;

s) IATA Dangerous goods Manual;

t) Flight plans (OFP and ATS) for each flight;

u) Trim sheet;

v) Ground Handling Manual (Ground Services Procedure manual (GSM), Ground Operation Procedure Manual (GOM) and Cargo Services Procedure Manual (CSM)) For all cargo airplanes;

w) ICAO Emergency Response Guidance for aircraft incidents involving Dangerous Goods;

x) Live Animals Regulations, Flight Operation Pool Manual;

y) Emergency Response Plan Manual;

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

z) European RVSM Flight Planning Requirements, North Atlantic operations & Airspace Manual.

aa) United States of America Automated operations specification;

bb) Guidance materials on the implementation of a 300m (1000ft) Vertical Separation Minimum in the MNPSA of the North Atlantic Region;

cc) Ethiopian Operation specification containing certificates;

dd) Conversion chart Feet to Meter;

ee) Physical characteristics of ETOPS Alternate-Airport;

ff) Specific performance calculation & all operational relevant information (like take off performance deviation e.g. due to inoperative equipment or abnormal situation) will be available on (class I, II & III EFBs) as applicable. For Q-400 use speed card, performance handbook & (MEL/CDL for takeoff deviations).

gg) Aircraft Journey Log

hh) Copy of the release to service

4.7. LICENSES/CERTIFICATES

- Noise Certificate
- Certificate of Dangerous Goods
- Radio License
- Air Operator Certificate
- Certificate of Airworthiness
- Insurance Certificate
- Certificate of Registration
- Fuel Carnet

4.7.1 Onboard documentation availing will be the responsibility of the flight crew. On no account will any crewmember remove or change any item from the aircraft. Any missing or damaged items must be reported on arrival in order to effect replacement.

4.7.2 Flight Operations Documentation section is responsible to keep current and provide the applicable documents, including revisions to the State (ECAA) for review, acceptance and/or approval.

4.7.3 All manuals that are used by Flight Operations shall be kept at Documentation section for reference.

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

4.8. DOCUMENTS TO BE CARRIED BY THE PILOT

4.8.1 Each pilot who is assigned for flight duty or simulator shall carry or have immediate access to at least the following manuals & documents electronically (class I EFB) or in hard copy.

- Relevant licenses and medical certificate
- Valid passport and vaccination card
- The appropriate Flight Crew Operations Manuals and QRH
- The appropriate Standard Operating Procedures Manual
- The appropriate Flight Crew Training Manual
- Operational flashlight

4.8.2 It is the responsibility of each pilot to revise and keep current manuals and documents in his possession. Each pilot shall check that all the required data on the Crew Management System (roster) are current at dispatch before undertaking any flight assignments.

4.8.3 It is the responsibility of each pilot to update his/her documents contained in his/her class I EFB (any compatible own device) tool before each departure from ADD or as a minimum every 7 day. And he/she must have access to the most current manuals at all times.

4.9. MEALS ONBOARD

4.9.1 Onboard catering is prepared considering caloric intake and culinary test. Three thousand calories per day is considered a normal daily intake. The proportion between main components of meals should be allowed. Protein, carbohydrate, fats will be approximately as follows:

Protein	15 %,
Carbohydrate	60 %,
Fats	25 %*
*Preferably of vegetable origin	

4.9.2 To avoid contracting dysentery, food poisoning and gastro-enteritis along the routes in warm climate, certain precautions should be taken by all flying staff, whether on the aircraft or on the ground.

4.9.3 Fresh salads and uncooked vegetables should not be eaten, unless they are known to have been hygienically prepared. Fruits with thin skins are potentially dangerous, as also are watermelons because the water in which they are grown can be polluted and the germs in the skin contaminate the melon when it is cut.

4.9.4 Over-ripe fruit or that which has been damaged or left cut should not be eaten. Fish, unless cooked shortly after being caught, is liable to undergo rapid decomposition in warm climates, leading to presence of powerful poisons that may not affect the taste.

Shellfish of all kinds should be particularly avoided. Ice and ice cream are often infected with germs unless hygienically made and handled. Generally, drinks should be refrigerated or put on ice, but ice (unless safe) should not be put in drinks. Drafts of cold drinks can be harmful.

4.10. SAFETY OF PASSENGERS IN FLIGHT

4.10.1 The captain must ensure that passenger is familiar with the location and use of seat belts, emergency exits, life jackets, and oxygen dispensing equipment provided for individual and collective use.

4.10.2 Passengers should be briefed on such emergency equipment as appropriate to the circumstances in the event of an emergency during flight.

4.10.3 The captain should ensure that during takeoff, landing and whenever precautions may be considered necessary by reason of turbulence or any emergency occurring during flight that all passengers should be secured in their seats by means of the seat belts or harness as applicable.

4.11. HOTEL ACCOMMODATION

4.11.1 It should be ensured that Ethiopian flight crews are accommodated at a convenient and well-secured first-class standard hotel while on duty at both local and foreign locations where crew layover is required.

4.11.2 Flight Operations Division shall negotiate and sign crew hotel contracts on behalf of Ethiopian.

4.11.3 The Area Manager may assist, on request, in the negotiation, obtainment of hotel rates and other related information that facilitates the decisions by Flight Operations division on the selection of appropriate hotels.

4.11.4 Hotel contracts and current status of contract expiration dates shall be kept by the Senior Staff, Flight Operations.

4.11.5 Flight Operations Division will conduct a survey of the general conditions and proposals of the available first-class standard hotels as soon as it is informed of the need for crew layover. In the event that the location happens to be abroad, Flight Operations may delegate the survey task and report to the Area Manager.

4.11.6 Renewal negotiations with available hotels at outstation locations will be handled two months before the expiration date of the contract. If the negotiations are handled by the Area Manger, the results of the negotiations shall immediately be transmitted to Flight Operations along with his recommendation and pertinent documents.

4.11.7 A decision by Flight Operations has to be made within five days of receipt of relevant data. While making a decision, security, cost, convenience and other related factors need to be considered.

4.11.8 The contract proposal has to be initiated by Flight Operation and submitted to the Legal Department using "Contracts Requisition and Summary Form" for the preparation of a contract, which shall be completed within five days of receipt of the requisition.

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

4.11.9 The signing of the contract must be finalized at least a week before the expiration of the existing contract or the commencement of new operation.

4.12. STANDARD REFERENCE FOR OUTGOING LETTERS

4.12.1 In order to establish reference numbers for correspondence originating from the Flight Operations Division, a system of uniform indexing shall be used which will provide systematic procedure for outgoing correspondence and maintenance of files.

4.12.2 Each Flight Operations“ Division/Department/Section Managers will prepare outgoing letters under centralized reference number bearing index ‘FOP’ followed by number showing a sequence of the letter in a given year.

a) Conventional System
A conventional numbering system with a dash and slash-numeric breakdown which provides a means to identify the Division, the sequence for outgoing letters, and the year in which the letter was written shall be used.

b) Originating Division
Codes have been assigned for the identification of originating Division by three letters, and the year in which the letter was written shall be used.

c) Serial Number
The secondary level of the reference number will be the serial number, which will show the sequence number of the outgoing letters in the current year that shall be logged in a Record Book.

d) Year
The tertiary level of the reference number is the year in which the letter was written.

4.13. OPERATIONAL LANGUAGE

4.13.1 The official working language (common language) between flight crew, ATC, cabin crew and Dispatchers is English. It shall be used by all flight crew during line operation in the flight deck and between flight crew and cabin crew.

4.13.2 All operational documentations are published in English language.

4.13.3 Trainings and evaluation activities are conducted in the English language.

4.13.4 All Crewmembers and aircraft dispatchers must be able to read, write, and understand the English language.

4.13.5 All flight crewmembers including instructors and examiners shall demonstrate a level of proficiency in the English language that will permit them to understand the information in the Flight Operations Manuals, effectively communicate during the performance of operational duties.

SECTION 1.11. GENERAL REGULATIONS BASED ON ADMINISTRATION

4.13.6 The language proficiency of all pilots who demonstrate proficiency below the Expert Level (Level 6) shall be formally evaluated at intervals in accordance with an individual's demonstrated proficiency level, as follows:

a) Those demonstrating language proficiency at the Operational Level (Level 4) should be evaluated at least once every three years; and

b) Those demonstrating language proficiency at the Extended Level (Level 5) should be evaluated at least once every six years.

4.13.7 Pilots who do not meet the English Language Proficiency of level 4 shall not be scheduled.

4.14. COMMUNICATION DUTIES

4.14.1 A member of the operational flight crew shall be qualified to operate the radio equipment to be used.

4.14.2 All communications shall be in the English language using Standard Phraseologies.

4.15. DISCIPLINE

No Flight Operations personnel shall deliberately violate Flight Operation policies and procedures. If any flight operations personnel make a willful and deliberate violation, it will be dealt with GHCMPM manual (corporate matters section 7.13).

5.0. DEFINITION

N/A

6.0. RESPONSIBILITY

6.1. The Vice President Flight Operations is primary responsible to ensure maintenance of the general rules and regulations stated in this section.

6.2. Flight Operations Division shall be responsible for negotiating and signing crew hotel contracts on behalf of Ethiopian.

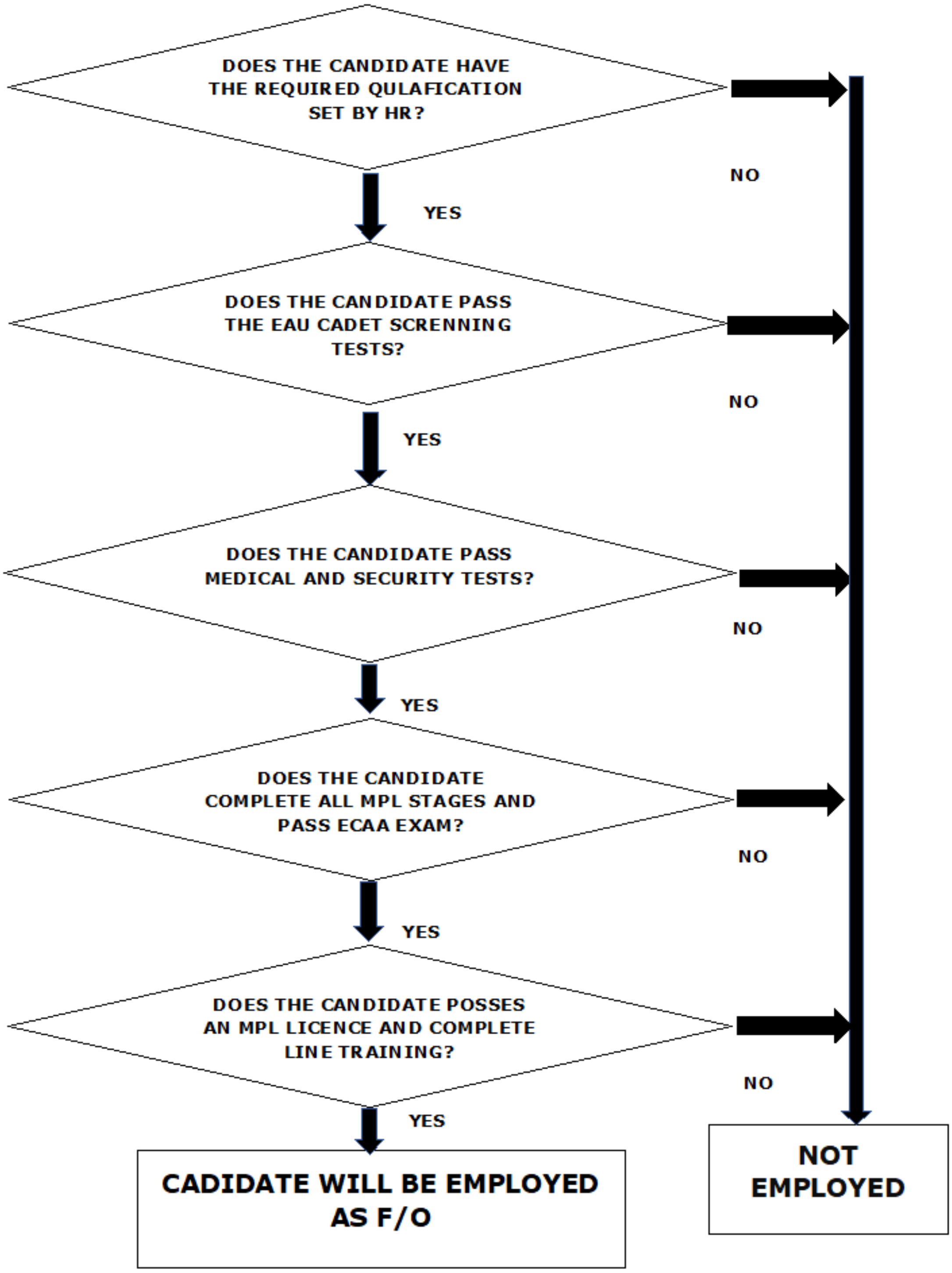
6.3. Each Flight Operations Vice President, Department Director and/or Section Managers are responsible for using systematic reference procedures for outgoing correspondence and filing all operational data and correspondence pertaining to the Division, Department or Section, per the established policy and procedure.

7.0. PROCEDURE

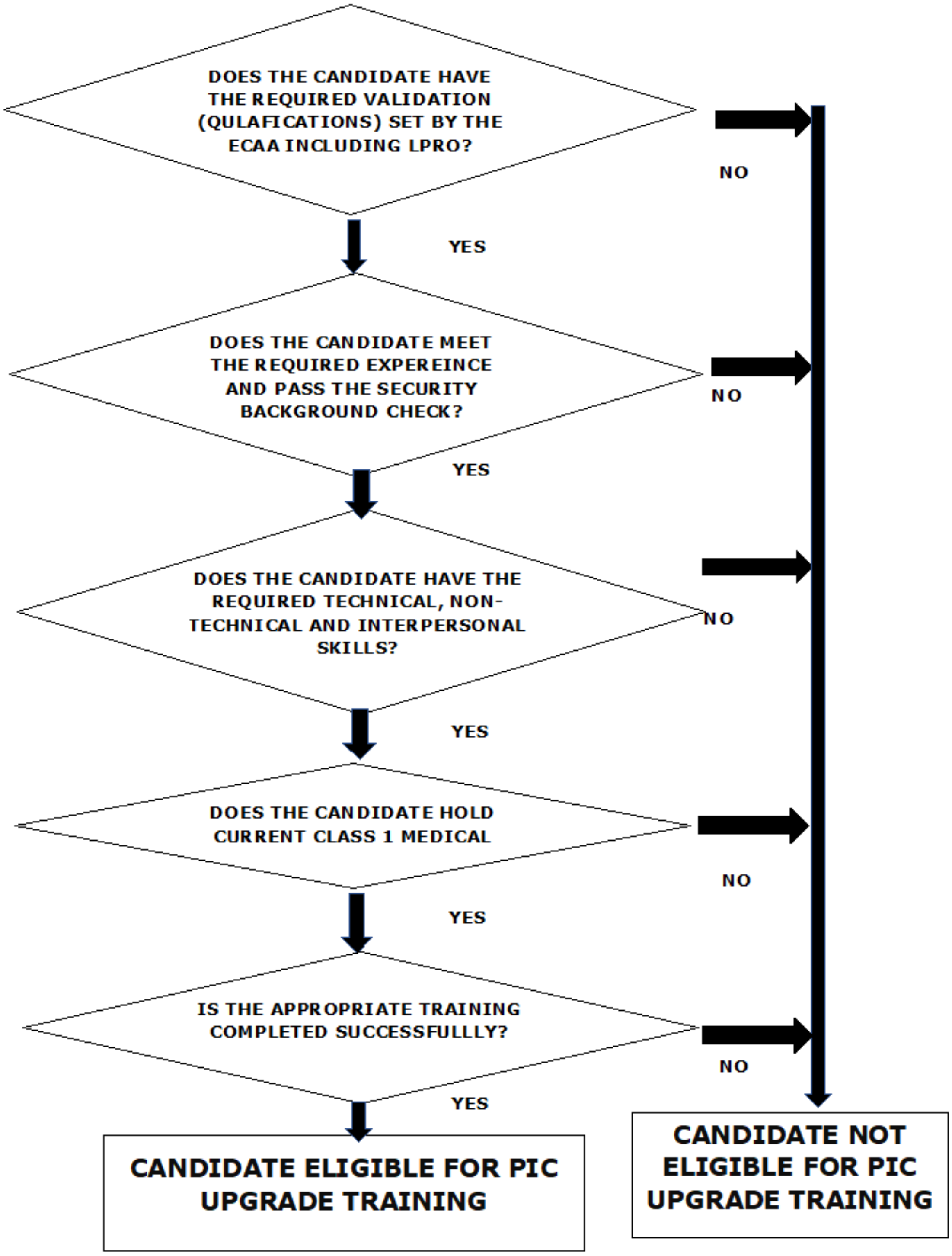
N/A

* * * * *

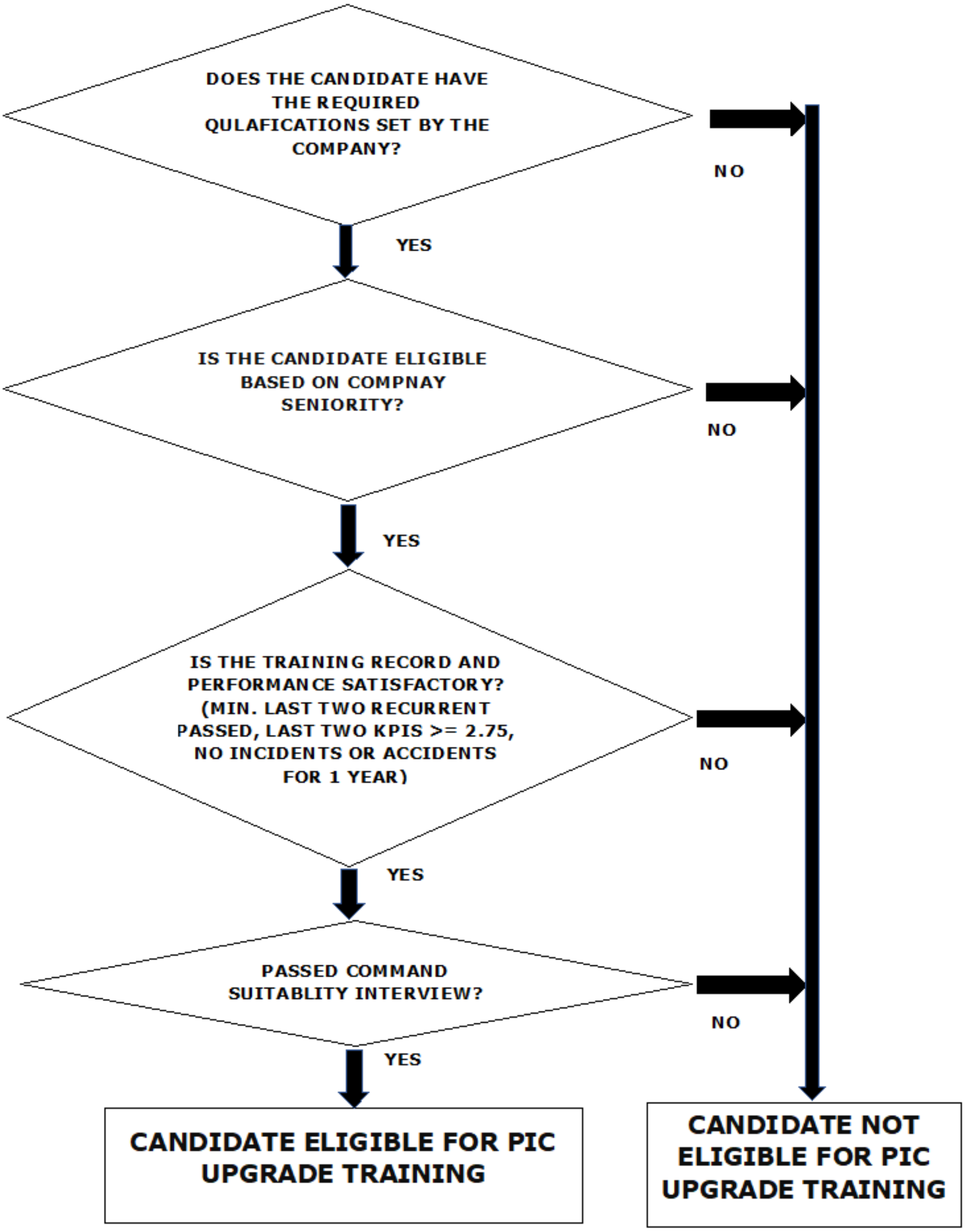
ANNEX I: MPL CADET EMPLOYEMENT PROCESS



<u>ANNEX II</u>: DIRECT ENTRY FLIGHT CREW EMPLOYEMENT PROCESS



ANNEX III: PIC UPGRADE PROCESS



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SECTION 2.1 FLIGHT OPERATIONS STANDARDS AND OPERATIONAL CONTROL

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SECTION 2.1 FLIGHT OPERATIONS STANDARDS AND OPERATIONAL CONTROL

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SECTION 2.1 FLIGHT OPERATIONS STANDARDS AND OPERATIONAL CONTROL

1.0 PURPOSE

This section provides established operational standards to be incorporated within the Flight Operations Division.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Personnel under Flight Operations

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. BASIC PRINCIPLES

4.1.1. Ethiopian shall not operate an aircraft unless that aircraft has an appropriate current airworthiness certificate; is in an airworthy condition; and meets the applicable airworthiness requirements for the intended operations. (ECARAS 9.2.3.1)

4.1.2. All flights shall be planned and operated in accordance with company regulations and AOC. These company regulations are intended to conform to rules and regulations established by ECAA. However, should an inadvertent discrepancy become apparent, ECAA rules and regulations shall be complied to and the discrepancy shall immediately be reported to the Flight Operations Division for correction.

4.1.3. Ethiopian shall not operate an aircraft unless it has both maintenance release, if maintenance has been performed prior to the flight, and a valid airworthiness release. The PIC shall not operate an aircraft unless he is in possession of a valid airworthiness release to indicate that any maintenance, preventive maintenance or inspections performed on the aircraft have been satisfactorily performed and appropriately documented. (ECARAS 9.3.2.10)

4.1.4. The following basic principles, listed in their order of importance, shall be used as guides in planning and conducting all company flight operations;

SECTION 2.2 OPERATIONAL SPECIFICATIONS

a) Safety
b) Economy
c) Standardization
d) Passenger comfort
e) Schedule

5.0 DEFINITION

Operational Control - is the exercise of the company's authority over the initiation, conduct, and termination of a flight.

6.0 RESPONSIBILITY

6.1. The Vice President Flight Operations has the primary responsibility to monitor the implementation of rules and regulations established by Ethiopian Civil Aviation Authority (ECAA).

6.2. The Captain and the dispatcher are jointly responsible for the initiation, continuation, diversion, cancellation and termination of flight.

6.3. The Captain and the dispatcher are responsible for assuring that each flight is planned, conducted and monitored with respect to:

a) Adherence to company policy and government regulations.
b) Issuance of necessary advisory information and instructions to the Captain to ensure the safety of the flight.
c) Preflight preparation.
d) Adherence to published schedules.

PROCEDURE

7.1. OPERATIONAL CONTROL

7.2. OPERATIONS FLIGHT WATCH / FOLLOWING

7.2.1 The Captain and the Flight Dispatcher shall use a common set of documents for a planned flight.

7.2.2 Any operational issues such as:

- Documentation
- Route change

- MEL issues
- Weather
- Delay etc. …

shall be communicated between the Dispatcher and flight crew using company VHF, HF radio, ACARS or SATCOM as appropriate

7.2.3 Assertion by all reasonable means that conditions and ground facilities required for the flight is adequate for the type of operation.

7.2.4 If in the opinion of the Dispatcher or the Captain there is a reasonable doubt that a flight cannot continue to be operated; cancellation, delay or diversion must be initiated immediately.

7.2.5 The flight dispatcher shall be available for inflight assistance through all phases of the flight establish communication with flight crew and maintain continuous tracking of all Ethiopian flights to ensure flight follows planned vertical and horizontal profile without deviations. He shall proactively monitor weather phenomena, SNOWTAMS and NOTAMS affecting the flight and advise the flight crew through available means.

7.2.6 The flight watch team should collect any relevant information for the current flight operation including update of weather forecasts and segment reports for ETOPS enroute alternates, update of enroute weather forecasts at cruise altitude but also at lower altitude including FL100, NOTAMs and SNOWTAMs etc.

7.2.7 The flight watch should also be ready to assist the crew if a diversion is required following a failure re-routing and fuel status reassessment.

7.2.8 The flight dispatcher shall keep record of flight departure and arrival times.

7.2.9 The flight dispatcher shall advise the pilot in command not to continue, and the pilot in command shall not continue, towards the airport of intended landing or alternate airport, if the following situation indicates that a safe landing cannot be made at the time of arrival:

a) Weather information, both enroute (SIGMET) and at the airport of intended landing (such as thunderstorm, turbulence, icing and restrictions to visibility)
b) Fuel monitoring where re-calculation of the OFP with respect to fuel on board determines that fuel on board is not sufficient to continue beyond the ETOPS, RE-DISPARCH or PNR points or any time upon advice by the PIC where aircraft burns all the contingency fuel.
c) Filed conditions, such as runway condition and availability, and navigation aid status.
d) En-route navigation system and facilities where possible failures may occur which could affect the safe continuation of completion of the flight.

SECTION 2.2 OPERATIONAL SPECIFICATIONS

7.2.10 For aircraft engaged in EDTO operations:

a) A flight will not proceed beyond the threshold time unless the identified en route alternate airports are re-evaluated for availability and the most up-to-date information indicated that, during the estimated time of use conditions at those airports will be at or above the established airport operating minimal for the operation;

b) If any conditions are identified that would preclude a safe approach and landing at and identified enroute alternate airport during the estimated use, an alternative course of action has been determined.

7.2.11 Within two hours prior to the flight's arrival at any designated re-dispatch point, and prior to executing the re-dispatch, flight watch office shall provide the PIC with the additional information concerning Meteorological conditions, ground facilities and services at the destination and alternate aerodromes specified in the re-dispatch. If the route of flight to be used to the new destination aerodrome is different from the planned route, the new route of flight should be specified and advised to the PIC.

7.2.12 The flight dispatcher shall monitor all flights for positions, OPS normal report and shall communicate with and assist the flight crew as required while en-route. The main voice communication will be VHF, HF and surveillance will use WSI fusion (ACARS, RADARS, flight plan data and ADS-B position reports).

7.2.13 The flight dispatcher shall address operational inefficiency observed during the flight non-economic level and in adequacy of NAVAIDS to appropriate authority and organizations like IATA without undue delay.

7.2.14 The flight dispatcher shall receive arrival time and ensure flight is terminated.

7.2.15 Ethiopian shall have an adequate system approved by ECAA for proper dispatch, flight supervision and monitoring of the flight progress. The dispatch and monitoring system shall have enough dispatch centers, adequate for the operations to be conducted, located at points necessary to ensure adequate flight preparation, dispatch and in-flight contact with the flight operations. Qualified flight operation officers at each dispatch center are required to ensure proper operational control of each flight. (ECARAS 9.3.1.23)

7.2.16 Fuel supply, including actual enroute consumption compared to planned consumption, as well as the impact of any changes of alternate airport or additional enroute delays.

7.2.17 Any item of aircraft equipment becomes inoperative that will result in increased fuel consumption or a performance or operational decrement; this must be considered and planned for so that the aircraft will be able to make a safe landing at an approved airport.

a) Air Traffic Management issues, such as reroutes, altitude or speed restrictions and facilities or system failures or delays.

b) Security issues, which could affect the routing of the flight or its airport of intended landing.

7.3 OPERATIONAL CONTROL HANDOVER PROCEDURE

7.3.1. The company policy with regard to the control and responsibility for the operation, security, and safeguarding of a company aircraft and its constituents while on the ground and/or in the air are as follows: -

a) The Maintenance department shall have control and responsibility for the aircraft while on the ground until such time as:

I. The aircraft have been fully prepared for flights.
II. All required mechanical work has been completed.
III. The airworthiness release has been signed by an authorized person and
IV. The captain of the flight is on board the aircraft and has assumed control and responsibility by signing the required forms.

b) At this point, there is a joint responsibility between the captain and local station agent as to the security of the aircraft, the conduct of the passengers on board the aircraft and the actual departure time of the aircraft.

c) The captain does not assume complete control and responsibility until the doors have been closed, the passenger steps removed or stowed, and an engine is started. Until this time the local Station Manager has the right to assume control of the security of the aircraft, board or remove passengers or cargo, and delay the departure of the flight. All Stations Managers' actions pertaining to crewmembers shall be coordinated with the captain.

d) Once the Captain has assumed complete control and responsibility for the flight as outlined previously, he will retain it throughout the entire flight. He will relinquish control and responsibility at the completion of the flight when the aircraft has been parked, the engines shut down, the exit doors open, all engine shutdown procedures have been accomplished, and all passengers disembarked.

e) Currently joint responsibility is shared between Maintenance and the local Station Manager. Normally, this joint responsibility lasts only for the period required for the Station Manager to complete unloading of cargo and accomplish other required post-flight duties.

f) Maintenance personnel will then assume complete control and responsibility for the aircraft.

g) At intermediate stations where crew changes occur with the passengers still on board, the joint responsibility continues with Captain, Station Manager and Maintenance. During crew handovers and whenever passengers are still on board, the flight deck must be attended by at least one pilot or by qualified personnel to handle any emergency.

h) At off-line stations lacking ground personnel qualified to assume control and responsibility over the aircraft, the captain will retain such control and responsibility. He will complete all necessary arrangements for the security of the aircraft, the reloading and/or unloading of passengers and cargo, required mechanical servicing, and other duties that might be required.

7.4 AIRCRAFT SCHEDULING AND ROUTING

7.4.1 The company shall establish flight operations schedules. These schedules shall allow sufficient time for proper servicing of the aircraft at intermediate stops and shall consider the prevailing winds en-route and the cruising speed of the type of aircraft used. This cruising speed must be based on the anticipated aircraft weight, temperature, and altitude capabilities at the time the flight is to be conducted. The cruising speed must be obtained from the appropriate aircraft operations manual.

7.5 USE OF AIRPORTS

7.5.1 Those airports, which are authorized by the company, as listed in the Nav Kit will be used for company flight operations. Other airports shall not be used except with the approval of the VP Flight Operations. Such approval shall be issued by means of flight disposition message or operational bulletin. In an emergency condition during flight, the captain has the authority to use any appropriate airport.

7.5.2 Flight operations into and out of airports not listed in the Nav Kit shall not be authorized as indicated above unless it is determined that all aircraft performance requirements and limitations have been met. All special conditions such as terrain, availability of navigational facilities, meteorological reports and local restrictions, must be taken into consideration. A determination must be made that the intended operation can be completed safely and in compliance with all limitations and restrictions. The pilot-in- command must check and accept all aspects of conformity before taking a flight.

7.6 NOISE ABATEMENT CURFEWS

7.6.1 When company flights are operated in accordance with published schedules, compliance with the various takeoff and/or landing curfew regulations is automatically ensured. Whenever operational irregularities occur which might require specific exemption from a curfew regulation, Flight Control shall be responsible for initiating and coordinating the actions required for an alternate disposition in the event the exemption is not approved.

It cannot be overemphasized that permission to initiate and continue into certain airfields is dependent upon the degree of compliance by the operator with the mandatory noise abatement procedures. Disregard to procedures laid down will result in the withdrawal of permission to operate or the refusal by the Airport Authority concerned to extend the period during which operations are permitted.

7.6.2 Noise abatement procedures are secondary to obstacle clearance requirements, Air Traffic Control requirements and any aircraft emergency, including engine failure at or above V_1 followed by continued take-off.

7.6.3 A fundamental requirement in determining noise abatement procedures is that the aircraft is at no time below the flight path calculated to ensure obstacle clearance with one engine inoperative. Whenever exemption from a takeoff curfew is required at a remote station, the concerned Area Manager, who will have the responsibility for coordinating the request, will communicate the decision of the appropriate authority to the captain.

7.6.4 When executing noise abatement procedure, the following should be considered:

a) The captain is responsible for conducting the flight in accordance with relevant regulations.
b) The procedure must not result in degradation of flight safety.
c) Execution of the procedure must not require above average piloting skill or result in unnecessary pilot workload.
d) Terrain clearance and adverse weather, which are likely to jeopardize safety, must be identified and considered.

7.7 ROUTE AND FLIGHT DOCUMENTS

7.7.1 Only such charts, maps, route manuals, and instrument approach charts as have been reviewed, approved, and distributed by the company may be used on company flights. However, if unforeseen circumstances necessitate the use of other material, such material, may only be used provided:

a) No other option is available.
b) The captain has carefully examined the material and is satisfied as to its accuracy and completeness.
c) The greatest caution is exercised while using this material.
d) The material has been approved by the state provider authority for operations conducted in the same aircraft category and under the same conditions.

7.8 RESTRICTION OR SUSPENSION OF OPERATION

When the company or the captain knows of conditions including en-route, airport, or runway conditions that are hazardous to safe flight operation, actions shall be taken by either or both to restrict or suspend operations until those conditions are corrected.

7.9 WEATHER REPORTING FACILITIES

7.9.1 Neither the company, nor the captain, or any flight operations officer may use any weather report to control flight unless it was prepared by an agency approved and certified to provide aeronautical meteorological services under a controlling state authority.

7.9.2 The company must show that enough weather reporting facilities are available along each route or route segment to ensure that timely reports and forecasts are available for the flight operation. The controlling governmental authority must approve the weather reporting facility, or the facility and personnel utilized must be adequately equipped and properly trained to make weather observations.

7.9.3 Ethiopian shall use sources approved by ECAA for weather reports and forecasts used for decisions regarding flight preparation, route and terminal operations. For passenger carrying operations, the AOC holder shall have an approved system for obtaining forecasts and reports of adverse weather phenomena that may affect safety of flight on each route to be flown and airport to be used. (ECARAS 9.3.1.21)

7.10 METEOROLOGICAL INFORMATION

7.10.1 All flight crews are required to develop and maintain a sound working knowledge of the system used for reporting aerodrome actual and forecast weather conditions and for the codes associated with it.

7.10.2 The Jeppesen Airways Manual Supplementary Text - Meteorology – provides comprehensive explanatory material concerning the availability of meteorological information and its interpretation.

7.10.3 The following is a summary of the routine information provided to flight crews for flight planning purposes, pre-flight and in-flight.

a) The significant weather charts should cover the following layers:

I. The layer between FL 250 and FL 450
II. The layer between FL 100 and FL 250 for limited geographical areas, as determined by regional air navigation agreement.

b) Aerodrome Meteorological Data

c) METARs and TAFs are produced by aerodrome meteorological offices and used by Pilots in Command to decide whether the actual/forecast conditions would allow a safe takeoff and landing within the permitted aerodrome operating minima.

d) METARs (Aviation routine weather reports), are compiled half-hourly or hourly and contain the following information:

- Type of report (METAR or SPECl/special report)
- ICAO station identifier

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- Time of observation
- Wind direction and speed (usually in degrees true and kts, but some counties vary)
- G (gusts)
- Horizontal Visibility
- RVR - if visibility is less than 1500 m
- Weather phenomena - if any
- Clouds in six characters groups
- The first three characters indicate the cloud amount
- Sky coverage is expressed as in the table below

FEW	1 to 2 oktas
SCT (scattered)	3 to 4 oktas
BKN (broken)	5 to 7 oktas
OVC (overcast)	8 oktas
SKC	Sky clear

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NOTE: In some countries the cloud amount may still be given in oktas instead of FEW, SCT, BKN OR OVC.

METAR - TREND FORECAST OF WEATHER CONDITIONS	
BECMG (BECOMING)	Indicates an expected permanent change in the meteorological conditions occurring at a regular or irregular rate
TEMPO	Indicates expected frequent or infrequent temporary fluctuations in the meteorological condition lasting for a period of less than one hour and, in the aggregate, cover less than one- half of the period during which the fluctuations are forecast to occur
AT	At
FM	From
TL	Until
CAVOK	Shall be used when the following conditions prevail simultaneously at the time of observation: • Visibility is 10 kilometers or more • No clouds below 5.000 ft. or below the highest minimum sector altitude, whichever is greater, and no cumulonimbus • No precipitation, thunderstorm, sandstorm, dust storm, shallow fog or low drifting sand, dust or snow.
SKC	Sky Clear - should be used if there are no clouds and the term CAVOK is not applied.
NSC	No Significant Cloud - should be used when: • No cumulonimbus and • No cloud below 5.000 ft. or below the highest minimum sector altitude whichever is greater are forecast and CAVOK or SKC are not appropriate.
NSW	No Significant Weather

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TAF (Terminal Aerodrome Forecast)

NOTE 1: TAF is The period of validity of routine TAF should not be less than 6 hours nor more than 30 hours this period of validity should be determined by regional air navigation agreement. Routine TAF valid for less than 12 hours should be issued every 3 hours and those valid for 12 to 30 hours should be issued every 6 hours.

NOTE 2: The structure is similar to the METAR but lists forecast changes to weather conditions.

NOTE 3: Conditional forecast elements need not be considered, except that **PROB40 or TEMPO** condition below the lowest applicable operating minima must be taken into account.

DETERIORATION will be taken into account if below applicable operating minima, but **IMPROVEMENT** is to be disregarded.

7.11 NON-ROUTINE AERONAUTICAL INFORMATION

7.11.1 The following "non-routine" meteorological information is provided when applicable:

a) As a SPEC, a special report amending METAR.
b) Amended TAFs.
c) SIGMET (significant meteorological reports) when significant weather occurs.
d) Aerodrome warnings, such as microburst or wind shear.

NOTE: Full details of weather reports and meteorological data presentation are available in the Jeppesen Airway Manual supplementary text - meteorology. The preceding information shall be considered as an excerpt.

7.12 FLIGHT CONTROL FACILITIES

7.12.1 The company must show that it has sufficient flight control centers for the operations to be conducted, and these must be located at points necessary to ensure proper operational control.

7.12.2 The use of contract flight control facilities acceptable to the controlling governmental aeronautical authority is permitted, whenever such facilities are established in future.

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SECTION 2.2 OPERATIONAL SPECIFICATIONS

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SECTION 2.2 OPERATIONS SPECIFICATION

1.0 PURPOSE

This purpose of this section is to provide guidelines on how all operational personnel, flight crew, dispatchers etc., shall comply with the laws, regulations and procedures of those states in which operations are conducted.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- All operational personnel
- Flight crewmembers
- Dispatchers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. Jeppesen/state published minimums for all types of approach are applicable to all of Ethiopian's transport equipment serving on both domestic and international operations.

5.0 DEFINITION

- **ASR**- Airport Surveillance Radar
- **PAR**- Precision approach Radar

6.0 RESPONSIBILITY

Flight crewmembers are responsible to strictly follow the established policy and procedure.

7.0 PROCEDURE

7.1. METHODS FOR ESTABLISHING AERODROME OPERATING MINIMA (AOM)

Ethiopian's public transport operations are to be carried out in accordance with IFR. Pilot-in-command does not have approval to operate a public transport flight under Visual Flight

Rules without special authorization.

7.2. CONCEPT OF MINIMA

7.2.1. The term Minima refers to the aerodrome weather conditions and the applicable navaid available and defines the minimum visibility (horizontal and vertical) prescribed for the taking off and landing of Ethiopian's aircraft.

7.2.2. The factors affecting minima are:

a) Aircraft capability given in the Aircraft Flight Manual, defines the lowest minima for which the aircraft has been certified.

b) State operating minima published on the AIP aerodrome chart, established by the state in which an aerodrome is situated.

c) Operator's minima are established by ECAA through the issuance of approved Operations Specifications. These minima may be the same as, but not lower than, the minima established in above.

7.2.3. Crew minima represent the minima that a Pilot-in-command is authorized to operate to and is based upon the qualifications and experience of both pilots.

7.3. PORTRAYAL OF AOM

7.3.1 The approved AOM for take-off and commencement and continuation of approach and landing at a particular aerodrome is represented by:

a) The minima published under AIR-OPS on the relevant Jeppesen aerodrome or instrument approach chart: or

b) The minima published collectively on a separate Jeppesen JAR 10-9X or 20-9X minimums page placed in front of the first instrument approach chart for a particular aerodrome; or

C) For aerodromes where no AIR-OPS AOM is published, Pilots in command are authorized to use the Jeppesen published AOM under FAR-121.

7.4. WEATHER MINIMA

7.4.1. Ethiopian shall not commence an IFR flight unless the available information indicates that the weather conditions at the aerodrome of intended landing will be the minimum ceiling and visibility values for the standard instrument approach procedure to be used or minimum operating altitude, if no instrument approach procedure is to be used, which would allow a VMC descent to the aerodrome. (ECARAS 8.6.2.5)

7.4.2. The company is authorized to utilize the following types of facilities in executing instrument approach procedures: ILS, VOR, VOR/DME, PAR/ASR, NDB/ADF, GPS (Check AOC)

7.4.3. The weather minima specified in the applicable instrument approach procedure and this section apply when the navigation facilities, related airborne equipment, and visual aids upon which such weather minima are predicated are in satisfactory operating condition. No instrument approach will be commenced using a navaid unless the airborne equipment is provided with an approved device to indicate automatically failure or malfunction of the ground facilities or related airborne equipment.

7.4.4. The weather minima specified in the Jeppesen airway manual are the lowest authorized for scheduled and alternate airports specified. Flight crew shall not commence an approach beyond the initial approach fix unless the available information indicates that the weather conditions at the aerodrome of intended landing will be the minimum ceiling and visibility values for the standard instrument approach procedure to be used, if no instrument approach procedure is to be used, which would allow a VMC descent to the aerodrome. RVR reporting must be available when the landing visibility minimum is below 800 meters to conduct approach and landing operations in all categories.

7.4.5. On all non-precision approaches, when applicable, it is Ethiopian standard policy that the PF stays on MAP display mode while the PM monitors VOR, LOC, or NDB raw data as required. Refer to the relevant FCOM, FCTM or SOP.

7.4.6. Missed approach must be initiated prior to or at the DA/DH or MDA/MDH when the weather minima, the navaid (the approach is based on) or the airborne equipment is degraded to the extent that a safe approach and landing cannot be executed upon reaching the minimum.

7.5. TAKE-OFF OPERATING MINIMA

7.5.1. Unless restriction is shown on approach charts, all runways are authorized for take-off provided gross weight take-off data is available.

7.5.2. Takeoff minimums listed are applicable to all runways without specific reference to runway number, unless otherwise noted.

7.5.3. A take-off must not be commenced unless:

a) The visibility or RVR is equal to or better than the published AOM; and

The weather conditions at the departure aerodrome are equal to or better than the applicable AOM for landing at that aerodrome unless a suitable and nominated take-off alternate is available.

b) When the meteorological visibility is below that required for take-off and the

RVR is not reported (or no meteorological VIS or RVR is not report is available) a take-off may only be commenced if the Pilot-in-command can determine that the actual visibility along the take-off runway is equal to or better than the required minimum.

c) At aerodromes where no published take-off AOM is available, the following is approved for use to establish a take-off AOM not lower than the value given in the table below.

7.5.4. RVR/VISIBILITY FOR TAKEOFF

TABLE 1

FACILITIES	RVR / VISIBILITY (m)	
	CAT A, B & C	CAT D
Nil (day only)	550	
RL and/or CL5	250	300
RL & CL	200	250
RL CL and multiple RVR information4	150	200
Approval required	150	150

a) For RVR/visibility below 400m, LVP must be in forced.

b) The reported RVR/visibility of the initial part of the runway can be replaced with pilot assessment.

c) For night operations, at least RL and runway end lights are required.

d) The required RVR values must be achieved for all relevant RVR reporting points except the initial part of takeoff run.

e) Ethiopian is approved for takeoff minima 150m for B737, B757/767, B777/787 & A-350 aircraft provided:

I. Low Visibility Procedures are in forced.

II. High intensity runway centerline lights spaced 15m or less and high intensity edge lights spaced 60m or less are in operations.

III. Flight crew members have satisfactorily completed training in a simulator approved for this procedure.

IV. A 90m visual segment is available from the cockpit at the start of the takeoff run. (Please refer to 2.20 (7.21).

V. The required RVR value has been achieved for all of the relevant RVR reporting points.

NOTE: Q-400 is not approved for low visibility takeoff and approach.

7.6. BASIC MINIMA FOR ALL INSTRUMENT APPROACHES

7.6.1. Pilot in Command is authorized to operate to the AOM published on the appropriate Jeppesen aerodrome instrument approach chart and pertinent AOC approval.

7.6.2. Jeppesen published instrument approach chart minima are the lowest authorized.

7.7. BASIC MINIMA FOR CIRCLE TO LAND

Jeppesen published instrument approach chart minima are the lowest authorized.

7.8. INEXPERIENCED PIC LANDING MINIMA

Until a PIC has 15 sectors performing PIC duties in the aircraft type (which include 5 approaches to landing using CAT I or CAT II procedures), he/she may not plan for or initiate an instrument approach when the DH or MDA is less than 100m (300 feet) or the visibility is less than 1.5km. (ECARAS 8.10.1.31)

NOTE: The PIC can fly CAT I or II operations after he has flown 15 sectors as a PIC on type. However, he can fly CAT I up to his 15 sectors flight with the above limitations.

7.9. AIRCRAFT APPROACH CATEGORIES

TABLE 2

CATEGORY	AIRCRAFT	VAT RANGE
C	A-350/B777-200LR/B757/B737/Q400	121-140 Kts
D	B763/B787/B777-300/B738/B737-8MAX	141-165 Kts

7.10. TABLE FOR CONVERTING RVR

TABLE 3

VISIBILITY (STATUTE MILE)	FEET	RVR (YARDS)	METERS	KMS.
¼	2000	440	400	0.4
½	2400	880	800	0.8
¾	4000	1320	1200	1.2
1mile	5000	1760	1600	1.6
1 ¼	6000	2000	2000	2.0

7.11. VISUAL APPROACH

7.11.1. Visual Approaches - A visual approach takes place when either part or all of an instrument approach is not completed and the approach is executed by visual reference to the terrain. Visual approaches are permitted with ATC approval. A visual approach shall not be conducted when the visibility is less than 5000 meters and ceiling less than the appropriate minimum safe altitude or as published by the authority whichever is higher.

7.11.2. Visual Maneuvering- Visual maneuvering in the vicinity of an aerodrome is permitted with ATC approval. The reported cloud ceiling must be above the Minimum Safe Altitude and the reported visibility not less than 5000 m. Once descent from the final fix on an instrument approach has commenced, no deviation from the approach track for vertical profile adjustment is permitted. Where an approach cannot safely be continued, due to profile errors, the published missed approach procedure must be executed.

7.11.3. A flight crew may accept a clearance for a visual approach or conduct a visual approach if the following conditions are carefully balanced and checked.

a) If the prevailing weather condition is above the required VFR minima (ceiling and visibility conditions).

b) No fog or smoke which impairs visibility

c) For inexperienced flight crew they should be familiar with the airport and airport environment

d) VASI or PAPI should be available

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7.12. NIGHT VISUAL APPROACH

7.12.1 Whilst man-made obstacles in the vicinity of an airport such as building, or towers are normally lit during the hours of darkness, natural obstacles such as hills or trees are not. As a consequence, unless there is exceptional illumination such as full moon on new snow, natural obstacles will be largely invisible to the pilot during night visual approach Black hole effect is another inherent risk of night visual approaches. Black hole conditions exist on dark nights (usually with no moon or star light) when there are no ground lights.

7.13. CONTACT APPROACHES

7.13.1. A contact approach is an approach wherein an airplane on an IFR plan operating clear of the clouds with at least one-mile flight visibility and having received a ATC authorization, may deviate from the prescribed instrument approach procedure and proceed to an airport by visual reference to the surface.

7.13.2. Contact approaches will be conducted in accordance with the following conditions and limitations:

a) When a circling approach to the runway of intended landing is required, the reported ceiling must be equal to or better than the IFR minima prescribed in the applicable instrument approach procedure for circling approaches.

b) When a straight-in approach to the runway of intended landing is contemplated, the reported ceiling must be equal to or better than the MDA prescribed in the applicable instrument approach procedures for a non-precious (VOR, NDB, ASR. etc.,) straight-in approach to such runway. The reported visibility must be at least one mile and in no case less than such higher IFR visibility minimum prescribed in the applicable instrument approach procedure for a non-precision straight-in approach to the runway of intended landing.

c) Descent below MDA is not authorized unless the runway threshold of intended landing is in view at all times.

7.14. IFR LANDING MINIMA FOR LOCAL CONDITIONS

Unless prohibited in the applicable approach procedure, a landing may be made at an airport when the local visibility is reduced to not less than one half mile or RVR 2400 by purely surface weather conditions such as smoke, haze, dust, ground fog, or blowing snow or sand, provided the ceiling is not less than 1000 feet, the airplane is aligned with the runway of intended landing before entering the local surface visibility conditions, and the runway of intended landing is plainly visible, allowing the pilot to have adequate visual reference to the line or forward motion at all times during final approach and landing.

7.15. SPECIAL PROVISIONS

7.15.1. Except as provided below, during the enroute phase of flight, large airplanes will be operated within the navigable airspace of the country in accordance with instrument flight rules. However, during the terminal phase of flight, operations may be conducted under visual flight rules provided:

a) The airplane is being given radar vectors by ATC.
b) At controlled airports, the captain is in direct communication with the control tower, approach control, or departure control servicing the airport of arrival/departure.
c) At uncontrolled airports, the captain is in direct communication with an air/ground radio communications facility capable of providing airport traffic advisory service.

7.16. VFR WEATHER MINIMA FOR TAKEOFF AND LANDING

7.16.1 No person may land or take off an aircraft under VFR from an aerodrome located within a control zone, or enter the aerodrome traffic zone or traffic pattern airspace unless the:

a) Reported ceiling is at least 450m (1,500ft); and
b) Reported ground visibility is at least 5km; or except when clearance is obtained from ATC.

7.16.2. No person may land or takeoff an aircraft or enter the traffic pattern under VFR from an aerodrome located outside a control zone, unless VMC conditions are at or above those indicated in part. (ECARAS 8.8.3.1.)

7.16.3. The only exception to the required weather minimums of this subsection is during a special VFR operation.

7.17. DESCENT BELOW AUTHORIZED IFR LANDING MINIMA

7.17.1. Operation to or from an aerodrome using instrument approach shall not be conducted lower than operating minima those which may be established for the aerodrome by the State unless the State specifically approves that operation. This is referred to as Instrument Approach Minima. (ECARAS 8.8.1.7)

7.17.2. When conducting an instrument approach, no person may operate an airplane below:

a) The prescribed MDA,
b) The sum of the ceiling minimum and the airport elevation, for a non-precision approach,
c) The decision height (in the case of continuing a precision approach below the decision height),
d) The sum of the ceiling minimum and touch-down zone (TDZ) elevation unless:

I. The airplane is in a position from which a normal approach to the runway of intended landing can be made and,

II. The approach threshold of that runway, or approach lights or other markings on or associated with the approach end that runway, are clearly visible to the pilot.

7.17.3. If upon arrival at the missed approach point or DH, or at any time thereafter, any of the above requirements are not met, the pilot shall immediately execute the appropriate missed approach procedure.

7.17.4. If the TDZ elevation for a particular runway is not available where it is necessary to calculate a DH by adding the ceiling minimum and TDZ elevations, threshold elevations shall be used. If threshold elevation is not available, airport elevation shall be used.

7.18. VISUAL CUES AT DA/H OR MDA/H

7.18.1. A pilot may not continue an approach below the applicable DH or MDH unless the aircraft has arrived at a position from which a normal approach profile can be followed to the runway in use and at least one of the following visual references for the intended runway is distinctly visible and identifiable to the pilot.

7.18.2. For non-precision and CAT I Operations elements of the approach light system;

- The threshold;
- The threshold lights or markings
- The threshold identification lights;
- The visual glide slope indicator;
- The touchdown zone or touchdown zone markings;
- The touchdown zone light; or
- Runway edge lights;

7.18.3. For CAT II Operations a segment of at least three consecutive lights being:

- The centerline
- The approach light
- Touches down zone lights
- Runway center line lights
- Runway edge lights
- A combination of these is attained and can be maintained.

The above visual reference must include a lateral element of the ground pattern, i.e.:

- An approach lighting cross bar
- The landing threshold
- A barrette of the touchdown zone lighting.

7.18.4. For CAT IIIA Operations at least 3 consecutive lights being:

- The centerline of the approach lights
- The touchdown zone lights
- The runway center line lights
- The runway edge lights
- A combination of these is attained and can be maintained.

7.18.5. For CAT IIIB operations with fail-operational flight control systems using a decision height (DH), a pilot may not continue an approach bellow the Decision Height (DH) unless a visual reference containing at least one centerline light is attained and can be maintained.

7.18.6. For CAT III operations with no DH there is no requirement for visual contact with the runway prior to touchdown. The permitted RVR is dependent on the level of airplane equipment.

7.19. APPROACH BAN (COMMENCEMENT AND CONTINUATION OF APPROACH)

7.19.1. After passing the IAF, the flight crew may commence an approach regardless of the reported controlling RVR/visibility but if the reported RVR/VIS is less than the applicable minimum the approach shall not be continued:

a) Beyond the outer marker or equivalent position (see NOTE below),
b) Into the final approach segment in the case where the DA/H or MDA/H is more than 1000 feet above the aerodrome.

NOTE: The equivalent position referred to above can be established by means of DME distance, a suitably located NDB or VOR or marker beacon or any other suitable fix that independently establishes the position of the aircraft. Where no outer marker or equivalent position exists, the pilot in command shall make the decision to continue or abandon an approach before descending below 1000 feet above the aerodrome on the final approach segment.

7.19.2. Where RVR is not available, the pilot-in-command may derive a RVR value by converting the reported visibility for non-precision and CAT I approaches only (see table below). FOR FLIGHTS INTO the US PLEASE REFER TO THE USA OPERATION SPECIFICATION DOCUMENT.

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7.19.3. If, after passing the outer marker or equivalent position in accordance with (7.18.1) above, the reported RVR/visibility falls below the applicable minimum, the pilot-in-command may continue the approach to DA/H or MDA/H.

7.19.4. A pilot may continue the approach below DA/H or MDA/H, and the landing may be completed provided that the required visual reference is established at DA/H or MDA/H and is maintained.

7.20. CONVERSION OF MET VISIBILITY TO RVR

7.20.1. AOM is generally expressed in RVR. If RVR reports are not available, and only meteorological visibility is reported, then for straight in instrument approaches only, the reported visibility may be converted according to the table below. The converted visibility (the equivalent RVR) may then be compared with the charted RVR values to determine whether the approach may be commenced or continued.

TABLE 4

CONVERSION TABLE	RVR = REPORTED MET VISIBILITY MULTIPLIED BY (FACTOR)	
LIGHTING ELEMENTS IN OPERATION	DAY	NIGHT
High intensity approach lights and High intensity RWY lights	1.5	2.0
Any type of lighting installation other than above	1.0	1.5
No lighting	1.0	Not applicable

7.20.2. The table above shall not be used when calculating takeoff or Category II / III minima, when a reported RVR is available, or if state minima are published. (Refer to the ATC section in the Jeppesen Airway Manual to determine whether state minima are published, if no AIR OPS/JAA minima charts are available).

a) If RVR and VIS are charted together (RVR/VIS values are preceded "R" respectively "V"), the RVR value is compulsory. If no RVR is reported the VIS shall be used without conversion.

b) No prefix will be charted if RVR and VIS are identical. The RVR report is compulsory. If no RVR is reported the VIS shall be used without conversion.

c) If only VIS is charted the visibility value is preceded by a "V" and shall be used without conversion.

7.20.3. This means that conversion may only be considered when minima are depicted as RVR (i.e. preceded by an "R" alone).

7.21. MONITORING OF RADIO AIDS

7.21.1. The pilot monitoring (PM) must identify and monitor the serviceability of the radio aid utilized for the approach before the commencement of the approach and there after till the approach is completed.

7.21.2. Descent when using ILS glide slope information as the sole means of vertical guidance must not be made below the glide slope intercept altitude until established on localizer and at or within the glide slope point.

NOTE: Presence of an ILS station identifier does not confirm the integrity of the glide slope signal.

7.21.3. The station identifier and sign must be reidentified if:

a) ILS

I. Whenever warning flags have appeared and cleared,

II. Whenever indications are in doubt.

b) VOR

I. Whenever warning flags have appeared and cleared including passing overhead the VOR,

II. Whenever indications are in doubt.

c) NDB

I. The station identifier is to be monitored by pilot monitoring (PM) throughout the approach and missed approach when relevant.

7.22. MISSED APPROACH

7.22.1. The flight crew shall always discontinue or go around from an approach or landing (prior to the selection of reverse thrust) when a safe landing cannot be assured (e.g. aircraft not stabilized in accordance with the criteria established) or a go-around is otherwise required (e.g. when instructed by ATC). Go-around is considered as a normal procedure, and the flight crew shall always be prepared for a go-around. The PM shall make a call for a go-around at any time during approach and landing until the selection of reverse thrust. The crew should also consider the risk of the go-around maneuver itself for the decision making. In any case, the decision to make a go-around or land shall not affect the PIC's emergency authority in the event of (impending) abnormal or emergency situations.

7.22.2. It does not also inhibit the flight crew from reporting go-around related events.

7.22.3. An approach **must be discontinued** if visual reference has not been attained or cannot be maintained at or below the minima. In addition, the following conditions warrant go around.

a) If an approach becomes un stabilized at or below 1000 feet AFE or touch down point is expected to be beyond the touch down zone.
b) Warning flags indicate failure;
c) The station identifier of the NAVAID ceases;
d) Indications are in doubt;
e) The aircraft is displaced vertically and/or laterally beyond predetermined limits;
f) On ASR or PAR approach if communications cease.

7.22.4. Except in an emergency or when there has been a significant change in reported weather conditions, no more than two successive approaches to an aerodrome shall be carried out where both approaches have resulted in a go-around.

7.22.5. Refer to SOP for more information.

NOTE: If there is a deliberate violation of the criteria laid down above or a failure to go around from an approach that does not meet the criteria the crew will be required to explain their actions.

7.23. AIRCRAFT STABILIZED APPROACH AND CFIT AVOIDANCE

7.23.1. In order to ensure a safe final approach and landing it is essential to maintain appropriate speed, thrust, vertical speed limitations and establish the required configuration (flaps & gear) to stabilize the aircraft in a timely manner with speed break retracted.

7.23.2. During any instrument approach without visual reference and when conducting approaches with visual reference to the ground the aircraft shall be fully stabilized in the required configuration in accordance with the AOM at least 1000 feet above field elevation (AFE).

NOTE: Runway alignment shall be accomplished not later than 1000 feet above field elevation (AFE). However, where certain types of approaches (e.g. circling), necessitate alignment turns below 1000 AFE it is essential that specified attention is being given to bank angle.

7.23.3. If it is determined that a stabilized approach requirement mentioned above is not achievable a go around must be executed at no less than 1000 AFE in IMC & VMC. Do not attempt to land from unstable approach.

7.23.4. Touchdown shall be within the touchdown zone of the runway i.e. do not attempt to extend the flare by increasing the pitch attitude in an attempt to achieve a perfectly smooth touchdown. Follow specific type of A/C proper landing technique to land on touchdown zone.

7.23.5. In order to reduce exposure to crew errors and CFIT accidents, all flight crew shall use a stabilized constant descent profile during the final segment of all approaches.

NOTE: As guidance for non-ILS approach the flight crew shall use VNAV, IAN, FLS, FPA or other methods that provide a stabilized constant path angle for the final segment of a non-ILS approach. The procedures on how to use these approach types are included in the SOP.

7.23.6. Rate of descent should not be more than 1000 feet/minute in the vicinity of high terrain and when aircraft height above terrain is 2000 feet or less at all times.

7.23.7. Flight Crew shall limit the vertical speed of an A/C to not more than 1000 feet per minute for the last 1000 feet climbing or descending to an assigned altitude or flight level when the flight crew are aware of another A/C at or approaching an adjacent altitude or Flight level unless otherwise instructed by ATC. Therefore, the flight crew shall ascertain that the V/S of the A/C is reduced to 1000ft/min or less to avoid unnecessary Resolution Advisory.

7.23.8. During a real or spurious GPWS or other terrain avoidance alert provided by onboard equipment, the flight crew shall accomplish the maneuvers described on the specific fleet's QRH and SOP. And the flight crew shall report the incident to the flight operations after the completion of the flight.

7.23.9. If all the stabilization criteria are not established, or cannot be maintained utilizing minimal control inputs from the applicable gates until touchdown, a mandatory go around maneuver shall be executed unless such maneuver cannot be safely accomplished under existing conditions e.g., traffic conflict, severe weather, obstacle clearance, aircraft performance, etc. Momentary airspeed deviations made necessary by atmospheric conditions are acceptable if such are immediately corrected. Frequent or sustained deviations are considered unstable, and execution of a go-around is mandatory.
The decision to execute a go-around from an un-stabilized approach shall be a shared responsibility between the Pilot-in-Command (PIC), Second in Command (SIC), or other pilot on the flight deck (ECAA Inspector, Flight Instructor/Evaluator, Observer, Safety Pilot, etc.) who is qualified on the aircraft type being operated. Regardless of the reason or source of the go-around call, the Captain / Pilot Flying (PF) must honor and execute the go around without question or undue delay. If the Pilot Flying (PF) is advised of an un-stabilized condition at or below the applicable approach gate more than two (2) times, and does not react or initiate appropriate corrective actions, the PM shall announce "I have control" and take positive action to return the aircraft to a safe condition of flight. No attempt shall be made to reverse a go-around decision once it is initiated, and no punitive actions shall be taken against any pilot (regardless of status or position) who initiates or calls for a go-around maneuver in accordance with this Stabilized Approach Policy.

Note: Willful or deliberate continuation of an approach or landing under any conditions below the stabilized approach gates will be subject to investigation in accordance with SMS Just Culture criteria.

7.23.10. **Landing Criteria:** Following a stabilized approach, a missed approach/balked landing must be executed if any of the following conditions exist (unless unsafe due to traffic, weather, obstacle clearance, or performance constraints):

- IAS is below Vref at runway threshold crossing.
- Target airspeed is excessive (>Vref+20) approaching the flare.
- Touchdown cannot be achieved within the designated touchdown zone.
- Touchdown occurs with both main landing gears on one side of the runway centerline.
- Once reverse thrust is initiated, a full-stop landing must be made.

7.23.11. **Flight Crew Roles & Responsibilities**

- **Captain / Pilot Flying (PF):** Execute stabilized approach and landing, respond immediately to all callouts, and initiate go- around when criteria not met.
- **First Officer / Pilot Monitoring (PM):** Continuously monitor the approach; issue mandatory "Unstabilized, Go-Around" callouts when criteria not met beyond **applicable gate(s)**
- **Escalation Protocol:**
  - o If PF does not respond to the first call-out → PM repeats call-out.
  - o If PF still does not respond → PM must state "I have control," assume control, initiate a go-around, return aircraft to safe flight condition
- **Irreversibility:** Once initiated, a go-around decision must not be reversed.
- **Accountability:** The execution of a go-around from an unstabilized approach is a shared responsibility between the Pilot-in-Command (PIC), Second in Command (SIC), or other pilot on the flight deck qualified on the aircraft type being operated.

  Regardless of the reason or source of the go-around call, the Captain/ Pilot Flying (PF) is obligated to honor and execute the go-around maneuver without question or undue delay.

  No action shall be undertaken by ET or any of its management personnel as a result of any execution of a mandatory go-around maneuver or appropriate transfer of control.

  Any willful or deliberate continuation of an unstabilized approach or landing below the applicable gate(s) will be investigated, and if necessary, dealt with under SMS Just Culture criteria.

7.24. MAXIMUM PERMISSIBLE RATE OF DESCENT

7.24.1. During descent until 10000 ft. above AGL it is recommended to descend at rates not higher than 6000 ft. /min.

7.24.2. During descent below 10000 ft. AGL the following rates must not be exceeded to reduce terrain closure rate and increase recognition/response time in the event of unintentional conflicts with terrain:

- 5000 ft. /min. down to an altitude 5000 ft. AGL

- 4000 ft. /min. down to an altitude 4000 ft. AGL
- 3000 ft. /min. down to an altitude 3000 ft. AGL
- 2000 ft. /min. down to an altitude 2000 ft. AGL

7.25. TURNS AFTER TAKE-OFF

7.25.1. Minor heading changes (up to 10 degrees bank) are not considered to be a turn.

7.25.2. Turns up to a bank angle of 15 degrees may be executed at or above a height of 300 feet (AGL).

7.25.3. Normal bank (25 degrees) may be executed at or above a height of 400 feet (AGL).

7.25.4. Turns up to a bank angle of 30 degrees may be executed at or above a height of 500 feet (AGL).

NOTES: For the above maneuvers the minimum speeds as per FCOM shall be observed.

- The maximum bank angle for normal operations shall not exceed 30 degrees.

7.26. IMMEDIATE TURN AFTER TAKEOFF – ALL ENGINES

7.26.1. Obstacle clearance, noise abatement or departure procedures may require an immediate turn after takeoff. The minimum height to commence a turn in such cases shall not be below 400 feet AGL with takeoff flaps and appropriate speed.

7.26.2. In the event of engine failure on takeoff, runway heading must be maintained regardless of the SID and ATC clearance unless a special engine inoperative procedure is called for in the Airport analysis/OPT or Logipad onboard performance calculator or compatible own device.

7.26.3. A maximum bank angle of 25° shall be used during all phases of flight unless stated otherwise in some non-normal conditions. The pilot-in-command may deviate from this requirement during emergency situations.

* * * * *

SECTION 2.3 NEW ROUTES OR NEW AIRPORTS FOR SCHEDULED OPERATION

SECTION 2.3 NEW ROUTES OR NEW AIRPORTS FOR SCHEDULED OPERATION

1.0 PURPOSE

The purpose of this procedure is to provide guidelines how to operate over routes or route segments approved by the controlling government authority.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1 NEW ROUTES OR NEW AIRPORTS

4.1.1 Prior to operating on any new route or to any new airport, Manager Flight Operations Performance & Technical Support in coordination with flight control shall complete a route and aerodrome analysis, including:

- Obstacle clearance for all phases of flight (minimum safe altitudes);
- Runways, Taxi ways, Ramp Areas (width, length, and pavement loading);
- NAVAIDs and lighting;
- Field conditions
- Curfews
- ARFF (Airport Rescue and Firefighting)
- PPR (Prior Permission Required)
- Applicable operating Minima
- Weather reporting;
- Emergency services;

- Fuel burn calculations;
- Fuel freeze considerations;
- ETOPS requirements if applicable;
- Air Traffic Services;
- Critical engine inoperative operations;
- Depressurization over critical areas per the company route manual;
- Special airport classification.

4.1.2 Mgr. Flight Operations Performance & Technical Support in coordination with Flight Dispatch shall ensure a flight will not be commenced unless it has been ascertained, to the extent possible, that conditions and ground facilities required for the flight are adequate for the type of operation per current Jeppesen manual or AIP.

4.1.3 For ETOPS route or operations over remote or sparsely populated areas Mgr. Flight Operations Performance & Technical Support in coordination with Flight Control should provide a summary of information on airports for flight crews to use in an emergency situation.

4.1.4. Ethiopian may conduct operations only along such routes and within such areas for which: (ECARAS 9.3.1.26)

a) Ground facilities and services, including meteorological services, are provided which are adequate for the planned operation.
b) The performance of the aircraft intended to be used is adequate to comply with minimum flight altitude requirements.
c) The equipment of the aircraft intended to be used meets the minimum requirements for the planned operation.
d) Appropriate and current maps and charts are available.
e) If two-engine aircraft are used, adequate airports are available within the time/distance limitations.
f) If single-engine aircraft are used, surfaces are available which permit a safe forced landing to be executed.

4.2 ROUTE APPROVAL

4.2.1 For each proposed route or area, the navigational systems and facilities shall be capable of navigating the aircraft:

a) Within the degree of accuracy required for ATC and;
b) To the airports in the operational flight plan within the degree of accuracy necessary for the operation involved.

4.2.2 In situations without adequate navigation systems reference, day VFR operations can be conducted safely by pilotage, only with approval from ECAA, because of the characteristics of the terrain.

4.2.3 An actual flight over a proposed route or route segment is not required if, by analysis, the company shows that such a flight is not essential to safety, considering the availability and adequacy of facilities and services.

4.2.4. Ethiopian may not operate over any route or route segment unless approved by ECAA. When seeking a route approval, Ethiopian must show ECAA that: -

a) It is able to satisfactorily conduct operations between each airport over that route or route segment.
b) Facilities and services required by this section are available and adequate for the proposed operation.

4.3 SPECIAL ROUTE

4.3.1 The training department shall ensure that, a pilot has obtained adequate knowledge of the route to be flown prior to being assigned for a flight during his initial training or prior to a flight crew member used as a PIC. This can be achieved through pilot self-briefing by using pictorial representation which is slide video show and being briefed by a qualified line instructor or examiner in addition to his simulator or line training.

4.3.2 Depending on the complexity of the area and route, the following methods of familiarization shall be used:

a) For non-complex routes, familiarization by self-briefing with area and route documentation, or by means of programmed instruction.
b) For complex routes, such as those involving through airspace with special navigational requirements (e.g. metric altimetry, escape routes), then in addition to sub-paragraph a. above, in flight familiarization during line flying under supervision or familiarization in a simulator will be conducted.

NOTE: some of the "escape procedures" over high terrain routes are complex and include multiple step-downs and track reversals/strict track adherence and/or dog-leg procedures.

EXAMPLE: Flying over Himalayas and flight through China involves (metrics flight levels)

4.3.3. Any one of the above will establish qualification on routes and at airports that are unrestricted and hence have no additional qualification requirements. Validity of complex route is 12 calendar months in addition to the remainder of the month of qualification or

12 months from the latest operation on the route or to the airport. If route or airport competence is revalidated within final 2 calendar months, the period of validity extends from the date of revalidation until 12 calendar months from the expiry date of that previous route and airport competence qualification.

4.4. ROUTE WIDTH

Approved routes over foreign airways or advisory routes have a width equal to the width designated and published by the applicable aeronautical authority controlling the airway or advisory route.

5.0 DEFINITION

Air Navigation Facility: Any facility used in, available for use in, or designed for use in aid of air navigation, including aerodromes, landing areas, lights, any apparatus or equipment for disseminating weather information, for signaling, for radio directional finding, or for radio or other electrical communication, and any other structure or mechanism having a similar purpose for guiding or controlling flight in the air or the landing and take-off of aircraft.

Controlled Flight: Any flight which is subject to an ATC clearance.

6.0 RESPONSIBILITY

The Director Flying Operations is primary responsible for this section.

7.0 PROCEDURE

7.1 CRITERIA FOR DETERMINING THE USABILITY OF AERODROMES

7.1.1. Before an aerodrome is first utilized for company operations as a destination, alternate or en-route alternate, it shall be approved as adequate. In general, an operation to or from an aerodrome will only be permitted provided normal operating procedures can be used. Such procedures must apply not only to the approach/landing and takeoff phase but shall also cover all forms of ground handling and operation.

7.1.2. In approving an aerodrome as adequate for company operations, the following aspects shall be considered:

a) Aerodrome dimensions with regard to performance requirements;
b) Runway, taxiway and apron dimensions with regard to surface maneuvering requirements of the Company’s aircraft types;
c) The obstacle situation in the approach, missed approach and departure sectors;
d) Instrument approach and departure facilities;

e) Lighting and communication facilities;

f) Local conditions such as special weather situations, night flying restrictions or even political aspects that may affect operations;

g) Rescue and firefighting capabilities availability;

h) Ground service facilities for fueling, loading, maintenance, de-/anti-icing, catering, general handling and the availability of immigration and customs authorities.

NOTE: For adequate ETOPS en-route aerodromes see ETOPS POLICY.

7.1.3. Aerodromes so approved as adequate, are listed in the Route Manual to be used as enroute alternate. Only those aerodromes shall be used which are listed therein. Aerodromes not listed therein, may only be used when specifically authorized, or in the event of an in-flight emergency taking into consideration the adequacy of the airport and weather condition.

7.1.4 When an aircraft has landed at an aerodrome not listed, as a result of the Pilot-in- command having exercised his emergency authority, the subsequent takeoff requires approval from Dir. Flying Operations. The Pilot-in-command shall ensure that he has sufficient performance data available to allow him to conduct a safe take-off and departure from the aerodrome of landing.

7.2. AERODROME RESCUE AND FIRE FIGHTING CATEGORIES (RFF)

7.2.1 Aerodrome rescue and firefighting categories have been developed and recommended for use by ICAO for the purpose of providing information concerning the availability of rescue and firefighting services at aerodromes. These categories are based on the overall length of the longest airplane, normally using the airport and its maximum fuselage width and are expressed by figures from 1 to 9 for increasing levels of protection.

NOTE: The airport directory section of the Jeppesen Airway Manual lists the respective aerodrome’s RFF category.

7.2.2 The use of an aerodrome with reduced rescue and firefighting capacity shall be governed by the policies outlined below. These policies are based on the philosophy that a fire in its initial stage, e.g. brake or engine fire, is controllable even with less than the minimum firefighting capacity and in the event of an extensive fire, passengers are evacuated, and the aircraft is abandoned.

7.3 REDUCED RFF - PRE-FLIGHT NOTIFICATION

7.3.1 If a reduction of the RFF capacity of an aerodrome is officially notified, the following shall apply:

a) No protection available (RFF capacity 0) = NOT USEABLE.

b) RFF capacity reduced by one category below the minimum level of protection, the aerodrome may be used after due consideration of all relevant aspects, such as:

I. Cargo/Dangerous Goods on board;

II. Aerodrome condition;

III. Weather; and

IV. Aircraft status.

7.3.2 The company may not conduct any flight operation over a route or route segment unless adequate ground aids are:

a) Available over the route or route segment within the degree of accuracy required for ATC.

b) Located to allow navigation to any airport of destination within the degree of accuracy necessary for the operation involved.

7.3.3 Ground aids are not required for:

a) Day VFR operations that can be conducted safely by pilotage because of the characteristics of the terrain.

b) Route segments where the use of RNAV or other specialized means of navigation is approved by the controlling aeronautical authority of the state.

7.4 COMMUNICATIONS FACILITIES

7.4.1 The company may not conduct any flight operation over a route or route segment unless a two-way air/ground communication is maintained which will ensure reliable radio contact, under normal condition, between each aircraft and the flight-following center responsible for flight watch. This communication system must be independent of any government system.

7.4.2 Ethiopian shall not operate an aircraft unless it is equipped with radio communication equipment required for the kind of operation being conducted. (ECARAS 7.3.1.1)

7.4.3 For all operations Ethiopian shall be able to have rapid and reliable radio communications with all flights over the entire route structure under normal operating conditions. This radio communication system should be independent from the ATC system. (ECARAS9.3.1.25)

7.4.4 International air navigation shall have at all times available communication facilities for immediate communication to rescue coordinating centers, information on the emergency and survival equipment carried on board any of the airplane, including: (ECARAS9.3.1.25)

i. The number, color and types of life rafts and pyrotechnics.

ii. Details of emergency water and medical supplies.

iii. The type and frequencies of the emergency portable radio equipment.

7.4.5 The flight crew is responsible in giving early warnings through company radio (either HF, VHF or ACARS) to maintenance on the status of the aircraft that adversely affect the next dispatch for appropriate action by maintenance and Flight Dispatcher.

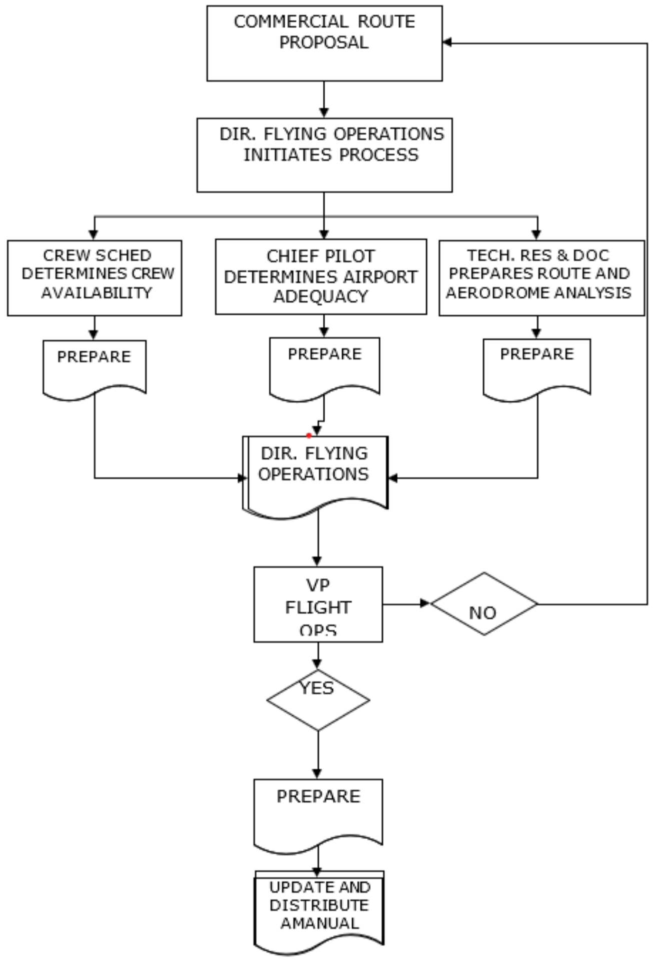
7.5 AIRCRAFT LIMITATIONS

7.5.1 Type of route – unless otherwise authorized by the controlling aeronautical authority, the company may not operate a two–engine aircraft over a route or route segment that contains a point farther than one hour flying time from an adequate airport in still air at normal cruising speed with one engine inoperative. In granting variation from this requirement, the controlling aeronautical authority considers the character of terrain, the kind of operation, and the performance of the aircraft.

(See ETOPS POLICY FOM 2.15)

7.5.2 The company may not operate an aircraft over a route or route segment unless it shows compliance with the performance requirements outlined in aircraft's manual.

7.6. ROUTE APPROVAL PROCESS CHART



7.7. OPERATION TO SPECIAL PROCEDURES AIRPORTS

7.7.1. Aerodromes within the Ethiopian Area of Operations are categorized according to their operational complexity, based upon an assessment of their terrain characteristics, minimum safe altitudes, approach aids, approach procedures, seasonal weather conditions, performance limitations, and any other unusual characteristics. Airports designated as Category B & C shall be communicated to crew scheduling department for the purpose of complying with the requirements specified below using the process illustrated at the end of this section. Certification that the qualification requirements for a Category B or C aerodrome have been completed is to be accomplished before the flight commences.

7.7.2. It is imperative that briefings be read in conjunction with current Jeppesen documentation and that where any conflict exists, Jeppesen takes priority.

SECTION 2.3 NEW ROUTES OR NEW AIRPORTS FOR SCHEDULED OPERATION

a) Category A

Category A aerodromes are considered not to pose any special operational problems for approach, landing or takeoff.

Category A aerodromes will have:

I. An approved instrument approach procedure,

II. At least one runway with no performance limiting procedure for take-off or landing,

III. Night operations capability.

NOTE: Ethiopian pilots are authorized to operate into any Category A aerodrome, provided their Annual Line Check is valid, and provided they conduct a self-brief for the aerodrome using the flight documentation provided in the Jeppesen Airway Manual and Route Manual or the aerodrome concerned is adjacent to another aerodrome at which the captain is currently qualified to land.

b) Category B

Category B aerodromes are considered to pose certain challenges to the approach and/or landing and/or takeoff.

An aerodrome may be classified as Category B for any or all of the following:

I. Non-standard approach aids or approach patterns.

II. Unusual weather conditions.

III. Unusual characteristics or performance limitations.

IV. Any other relevant consideration including obstructions, physical layout, lighting, etc.

NOTE 1: Prior to operating to a Category B aerodrome or aerodromes, a captain is to be briefed or should self-brief by means of programmed instruction or other material (Jeppesen etc.) on the relevant characteristics of aerodrome(s) of concern. Where possible, the Briefing should be supplemented by a discussion with the Fleet Chief Pilot or with pilots who have recently operated to the same aerodrome. The Briefing may be studied during the period of pre-flight preparation, or may be studied at any convenient earlier time, prior to the applicable expiry date.

NOTE 2: Specific Aerodrome Briefings are in the Route Manual. The training requirements applicable to each Category B aerodrome are specified in Flight Operations Training Manual.

NOTE 3: The captain shall certify that the briefing or self-briefing has been completed by recording and signing the fact on the appropriate CAT B competence qualification form.

NOTE 4: A Category B Aerodrome Competence Qualification Form shall be completed and signed by the captain to certify that the briefing requirements for a particular aerodrome have been carried out. These forms shall be available in-Flight Control. Completed forms shall be placed in the Flight Folder and returned to Flight Control.

NOTE 5: Aerodrome competence qualification shall be re-validated by operating to the aerodrome, within the period of validity prescribed above. In the case of a Category B aerodrome, re-validation may also be achieved by self-study of the applicable briefing material, and completion of the appropriate Category B Aerodrome Competence Qualification form.

NOTE 6: Validity is for 12 months.

c) Category C

Category C aerodromes are considered to pose special problems in the approach, landing and/or take-off that require pilot awareness and training additional to that applicable to Category B aerodromes.

Prior to operating to any Category C aerodrome, the captain is required to:

I. Undergo a briefing on the aerodrome and visit the aerodrome as an observer or;
II. Make an actual approach accompanied by a pilot observer or an instructor pilot who is qualified for the airport or;
III. Undertake instructions on operations at the aerodrome, in an approved flight simulator for this purpose, acting as an operating pilot or;
IV. To have observed or have actually made an approach and landing (or take off) during initial training or prior to a flight crew member being used as a PIC, at the aerodrome concerned, within the previous twelve (12) months.

NOTE 1: For Category C aerodromes, the Crew Scheduling Office will arrange the necessary visits and/or training to be carried out, in coordination with the applicable Fleet Chief Pilot.

NOTE 2: The specific aerodrome qualification training requirements applicable for Category C aerodrome are summarized in Flight Operations Training Manual. Such training requirements may also be extended to flight crew members other than the captain at certain aerodromes. Specific Aerodrome Briefings are produced for each of these aerodromes and included in the Route Manual.

NOTE 3: A record of the above qualifications shall be communicated to supervisor crew scheduling and maintained by the Training Department.

NOTE 4: The Aerodrome Qualification Form for a Category C aerodrome shall be completed by the instructor and/or Examiner assigned to conduct the briefing, training and/or visit, upon successful completion of each of the required elements. These certificates, once completed, shall be countersigned by the Fleet Chief Pilot and the completion date recorded in the pilot's training record.

7.8 UNLISTED AIRFIELDS

7.8.1 Whenever a flight is to be operated to an aerodrome not contained in the Aerodrome Listing, Director Flying Operations or the applicable fleet Chief Pilot will determine the Category of the respective aerodrome and/or if a verbal/written aerodrome brief is required.

7.8.2 If a special briefing is required, Flight Operations will forward the relevant details to the Pilot-in-Command to enable him to complete the appropriate briefing requirements.

7.8.3 If special training is required, the flight will not take place until such training is completed.

7.9 OPERATION ON SPECIAL ROUTES

7.9.1 Routes in the Ethiopian Area of Operations are categorized according to their operational complexity, based upon an assessment of their terrain characteristics, minimum safe altitudes, performance limitations, and any other unusual characteristics. Routes designated as Complex shall be communicated to crew scheduling department to comply with the requirements specified below using the process illustrated at the end of this section. Certification that the qualification requirements for a Complex route have been completed is to be accomplished before the flight commences.

a) Non-Complex routes

Non-complex routes are considered not to pose any special operational.

NOTE: Ethiopian pilots are authorized to operate in any non-complex route, provided their Annual Line Check is valid, and provided they conduct a self-brief for the route using the flight documentation provided in the Jeppesen Airway Manual and Route Manual.

b) Complex routes

Complex routes are considered to pose special problems for the operation that require pilot awareness and training additional to that applicable to non-complex routes. Airspaces with special navigational requirements or special "escape procedures" over high terrain which include multiple step-downs and track reversals/strict track adherence and/or dog-leg procedures are considered complex.

Prior to operating on any Complex route, the captain is required to:

I. Undergo a briefing on the route and fly over the route as an observer or;

II. Fly on the route accompanied by a pilot observer or an instructor pilot who is qualified for the route or;

III. Undertake instruction on route operation, in an approved flight simulator for this purpose, acting as an operating pilot or;

IV. To have observed or have actually made a flight over the route during initial training or prior to a flight crew member being used as a PIC, on the route concerned, within the previous twelve (12) months.

NOTE 1: For Complex routes, the Crew Scheduling Office will arrange for the necessary visits and/or training to be carried out, in coordination with the applicable Fleet Chief Pilot.

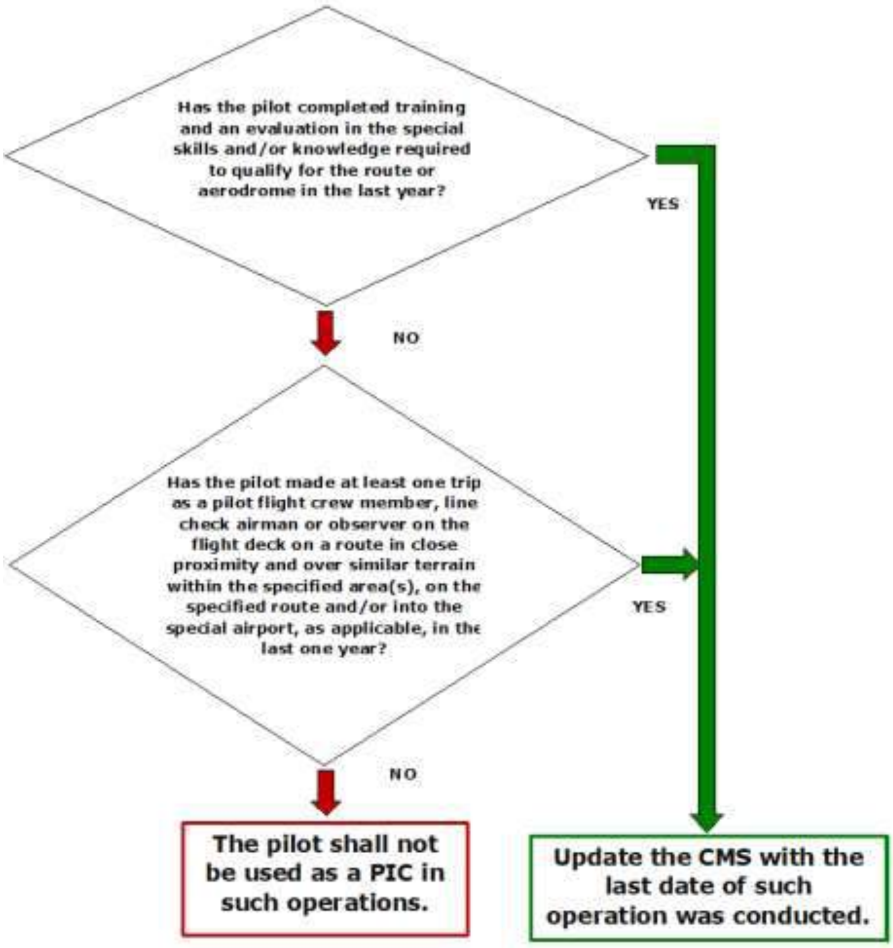
NOTE 2: The specific route qualification training requirements applicable for Complex route are summarized in Flight Operations Training Manual. Such training requirements may also be extended to flight crew members other than the captain on certain routes. Specific Route Briefings are produced for each of these routes and included in the Route Manual.

NOTE 3: A record of the above qualifications shall be communicated to supervisor crew scheduling and maintained by the Training Department.

NOTE 4: The Route Qualification Form for a Complex route shall be completed by the instructor and/or Examiner assigned to conduct the briefing, training and/or visit, upon successful completion of each of the required elements. These certificates, once completed, shall be countersigned by the Fleet Chief Pilot and the completion date recorded in the pilot's training record.

SECTION 2.3 NEW ROUTES OR NEW AIRPORTS FOR SCHEDULED OPERATION

7.10. PROCESS FOR ASSIGNING PIC TO SPECIAL AERODROME OR ROUTE COMPETENCY



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SECTION 2.4 FLIGHT PREPARATION

SECTION 2.4 FLIGHT PREPARATION

SECTION 2.4 FLIGHT PREPARATION

1. PURPOSE

The purpose of this flight preparation procedure is to establish guidelines on how flight crew members prepare flight database before the commencement of each flight.

2. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3. PERSONS AFFECTED

- Flight Crewmembers

4. POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. All flight crew are required to update their Class I EFB before commencement of each flight and check if there is anything new (bulletins, notices etc..) and review the following in addition to items listed on FOM1.5 & 1.6.

- Aircraft airworthiness determined from ATL, MEL/CDL status;
- The operational flight plan;
- Current maps and charts;
- Weather at departure, destination and alternate airports;
- NOTAMs applicable to the enroute phase of flight and to departure, destination and alternate airport
- Aircraft performance, weight and balance;
- Verify that the most current FMS database applicable to the date of operation is available.

4.2 Crew must verify and select the latest Navigation database prior to first flight on the effective date of the database. If the database has to be used beyond the date of validity the MEL relief must be consulted prior to departure.

4.3 Landing Performance Margin Policy

I) General

Ethiopian Airlines shall determine landing performance using both regulatory requirements and Company-defined operational safety margins. Certified landing distances are obtained under ideal test conditions; therefore, supplementary safety margins are required to accommodate operational variability, runway condition uncertainty, and human factors.

II) Dispatch (Pre-Flight) Landing Performance Requirements

a. For all dispatch-related landing assessments, a regulatory landing factor of 1.67 shall be applied to the certified dry landing distance, unless otherwise mandated by the regulatory authority.

b. A destination or alternate aerodrome shall be considered acceptable at dispatch only when the factored landing distance does not exceed the Landing Distance Available (LDA).

c. The dispatch factor accounts for:

- Variability in pilot technique and operational reaction time
- Environmental uncertainties (temperature, pressure altitude, wind variation)
- Runway surface condition differences
- Aircraft configuration influences and performance degradation

III) Time-of-Landing (TOL) Performance Assessment

a. At the time of landing, performance shall be assessed using the Actual Landing Distance (ALD) for the prevailing aircraft configuration and runway condition.

b. Ethiopian Airlines applies the following operator-defined margins:

- Dry Runway: ALD plus 15% safety margin.
- Wet Runway: ALD plus the regulatory wet-runway performance increment in accordance with manufacturer and authority guidance.
- Contaminated Runway: Use manufacturer-published contaminated-runway performance data only, with no additional factors applied.

c. Landing is permitted only when the required landing distance (including the applicable margin) does not exceed the LDA.

IV) Additional Margin Requirements

Additional adjustments shall be applied when operational conditions reduce braking performance or increase landing distance. These include, but are not limited to:

a. MEL items affecting braking, reversers, antiskid, spoilers, or aerodynamic drag.

b. Tailwind component, whether forecast or actual.

c. Short or performance-critical runways requiring enhanced stopping assurance.

d. High-elevation aerodromes, where density altitude significantly influences landing distance.

e. Autobrake requirements as specified in fleet-specific SOPs or performance manuals.

These adjustments maintain an appropriate safety buffer during demanding or degraded operational scenarios.

V) Contaminated Runway Performance

a. No Company-defined margin or factor shall be applied to contaminated-runway landing performance calculations.

b. Required landing distance must be derived solely from approved manufacturer contaminated-runway performance data.

c. Flight crew shall ensure the required landing distance for the applicable runway contaminant category remains within the LDA.

VI) Operational Responsibility

a. The flight crew shall:

- Confirm landing performance at dispatch.
- Recalculate landing performance in flight when conditions change.
- Ensure required landing distance (including applicable margin) remains within the LDA before committing to land.

The flight crew and IOCC shall coordinate whenever destination or alternate conditions differ from dispatch assumptions to maintain compliance with landing performance requirements.

5. DEFINITION

ATL- Aircraft Technical Log
CDL- Configuration Deviation List
MEL-Minimum Equipment List
MARGINAL WEATHER- Ceiling 1000 feet and visibility 3 miles inclusive

6. RESPONSIBILITY

The Pilot-in-command is responsible for flight preparation with regard to airworthiness, operation and installation of instruments and equipment, weight and center of gravity and check of operational limits.

SECTION 2.4 FLIGHT PREPARATION

7. PROCEDURE

7.1. PRELIMINARY ARRANGEMETNS

7.1.1. The procedures outlined in this section shall be followed during the conduct of all company flight operations. The final responsibility for flight preparation rests with the Captain. By signing the dispatch release, load sheet and aircraft acceptance.

7.1.2. The captain certifies that: -

a) He is satisfied that the aircraft is airworthy.
b) The instruments and equipment required for the given operation are installed and operative or deferred as appropriate.
c) The weight and center of gravity of the aircraft are within limits and that cargo loads are properly secured.
d) A check has been completed indicating that the operational limits of the airports listed in the company dispatch release and those of the aircraft can be complied with for the flight to be undertaken.
e) The flight has been planned and meets the requirements regarding prevailing weather conditions as well as required fuel and oil supply.
f) He accepts control of the aircraft and the requirements of intended operation and that the flight may be conducted safely.

7.2. PRE-FLIGHT DOCUMENTATION

All required pre-flight documents should be completed prior to the departure of flight. At stations providing flight control services, the flight crew is relieved of certain pre-flight paperwork duties, but not of the responsibility for its accuracy and timely completion. At stations where pre-flight assistance is not available, the flight crew shall complete all the required pre-flight documents including the load and trim sheet if necessary. The PIC shall normally delegate these tasks to other crew members, but he must always determine that they have been satisfactorily completed prior to the departure of the flight.

7.2.1. FLIGHT FOLDERS

a) Pre-flight folder containing the following information will be provided by dispatch and shall be accessible to the flight crew during flight preparation and in flight in hard or soft copy.

I. Air traffic services flight plan;
II. Operational flight plan;
III. En-route and terminal weather;

IV. Appropriate class I and II NOTAMS;

V. Permit numbers; (if required)

VI. Flight Control dispatch release form. (as applicable)

b) The following documentation will be included in the folder for the purpose of the reporting of various issues and shall be accessible to the flight crew in flight.

I. Captain's report form
II. First Officers grade form
III. Flight attendants grade form
IV. Auto land report
V. ICAO Bird strike/incident report form
VI. FMC Report
VII. Engine In-Flight shutdown form
VIII. Faulty radio communication, visual & non-visual navigation aid report form
IX. TCAS RA incident form
X. Aircraft accident/ incident report form
XI. Air traffic incident report form
XII. IATA Air Traffic Services Occurrence information
XIII. Air safety report form
XIV. CAT B & C airport/route competence qualification form
XV. Guideline for unruly, rage(misconduct), flight disturbance
XVI. Violation of smoking, unacceptable behavior form
XVII. Aircraft search checklist (specific threat)

7.2.2. Additionally, Marketing department personnel, Maintenance personnel and Fueling Company will provide the following material:

I. General declaration
II. Passenger manifest
III. Load and trim sheet
IV. Cargo manifest
V. Fuel load and servicing record form
VI. Fuel receipt
VII. NOTOC (Captain's information for dangerous goods, live animals, perishable etc.).

7.3. CREW BRIEFING

7.3.1. Flight crewmembers shall familiarize themselves prior to the commencement of any flight, with all available information appropriate to the intended operation. Prior to each takeoff and landing, the PIC or the pilot flying shall brief the other pilot and shall review the normal and non-normal departure and approach procedures and considerations. The maintenance status of the airplane and the acceptable deferred defects, if any, should be briefed and their effects must be discussed. In addition, the flight crewmembers are required to conduct flight deck jump seat briefing as described on FOM 2.5 (7.13).

7.3.2. It is imperative that the required engine inoperative flight path is reviewed and briefed prior to each takeoff.

NOTE: All operations to any airport require the availability and use of approved approach plates and charts for the associated types of approaches. This may be substituted by a visual approach in day VMC conditions from the initial approach altitude.

7.3.3. Each flight crewmember shall use a separate set of charts and approach plates for increased situational awareness and cross checking.

7.4. FLIGHT DECK TO CABIN CREW BRIEFING

7.4.1. The purpose of the cabin crew briefing is to develop a team concept between the flight deck and cabin crew. An ideal developed team must share knowledge relating to flight operations, review individual responsibilities, share personal concerns, and have a clear understanding of expectations.

7.4.2. Upon flight origination or whenever a crew change occurs, the captain will conduct a verbal briefing with all the cabin crews present. However, preflight duties, passenger boarding, rescheduling, etc. may make it impractical to brief the entire cabin crew complement. Regardless of time constraints, the captain must at least brief the Team Leader Cabin Services.

7.4.3. The briefing should cover the following items:

a) Brief introduction between flight and cabin crew in a professional and friendly manner to help create the team spirit.
b) Check that each crew member holds valid documents. (This can be delegated to the Team Leader Cabin Services).
c) Logbook discrepancies that may affect cabin crew responsibilities or passenger comfort (e.g., inoperative intercom, failure of the automatic pressurization, etc.)
d) Weather affecting the flight (e.g., turbulence – including appropriate code levels, thunderstorms, weather near minimums, etc.). Provide the time when the weather may be encountered. (e.g., "Turbulence can be expected approximately one hour

after takeoff.”)

e) Responsibility for PA announcements when the Fasten Seat Belt sign is turned on during emergency or any other conditions appropriate to the flight.

f) Delays, unusual operations, non-routine operations (e.g., maintenance delays, ATC delays, re-routes, etc.)

g) Shorter than normal taxi time for preflight announcements.

h) Total flight time for cabin service planning and passenger announcement.

Any other items that may affect the flight operation or in-flight service such as catering, fuel stops, armed guards (escorts), etc.

i) A review of the sterile flight deck policy,

j) Flight deck entry protocol shall be respected by stating name and established entry code. Confidentiality must be respected by all.

7.4.4. During the briefing, the captain should solicit feedback for operational concerns (e.g., does each person understand the communication process when there is a threat on board?). The captain should also solicit feedback for information which may affect expected team roles. Empower each crew member to take a leadership role in ensuring all crewmembers are made aware of any item that might affect the successful completion of the flight.

7.4.5. The Team Leader Cabin Services will inform the captain of any inoperative equipment and the number of cabin crew on board.

7.4.6. The captain will inform the Team Leader Cabin Services when there are significant changes to the operation of the flight after the briefing has been conducted.

7.5. METEOROLOGICAL BRIEFING

7.5.1. This generally is the first item to be checked. Such things as routing, fuel required for alternate airports, and many others are based on the meteorological information obtained in this briefing. The meteorological briefing will normally be conducted by the meteorological office personnel of the station concerned at their facility except where company flight operations officers or contract flight control services personnel are qualified to do so and possess the required facilities.

7.5.2. This meteorological briefing shall include at least the following:

a) Route and area forecasts as appropriate

b) Actual and forecast weather reports for the destination and alternate airports

c) The latest available synoptic, surface, and upper air charts

d) Pilots report (if available)

e) Wind aloft reports

f) Other significant meteorological observations

7.5.3. Of the above, at least the en-route and airport forecasts shall be in a written form, which will become a part of the flight records.

7.5.4. A complete meteorological folder will be obtained so that the greater amount of meteorological information is available on the flight deck during flight and can be used as a basis for in-flight decisions.

7.6. NOTAMS AND AIP

7.6.1. The dispatch office is responsible for the management and control of NOTAMS and AIPs.

7.6.2. The dispatcher will provide the required NOTAM for the operating Flight Crew for their review.

7.6.3. The dispatchers are responsible for reviewing the applicable AIP and briefing the Flying crew accordingly. If the operation is conducted in enroute remote airspace from which strategic lateral offset procedures (SLOP) are published in the relevant AIP, the dispatcher is responsible for providing such information to the Flight Crew.

7.6.4. Other safety critical operational information to include

I. Airworthiness directives

II. Manufacturer bulletins

III. Flight crew bulletins or directives will be disseminated using the following process on appendix 2.1

7.7. SPECIAL COMPANY REQUIREMENTS

7.7.1. The flight crew will be provided or shall obtain a detailed briefing regarding special company requirements or procedures for the intended operation.

a) Information regarding VIP's, couriers, prisoners, medical patients, unaccompanied children, wheelchair dependent passengers.

b) Change in aircraft, crew, or published flight schedule.

c) Special arrangements for overnight accommodation crew transportation, passenger handling, maintenance, landing, and over flight permit Number, servicing and handling complete details shall be included for non-scheduled operations.

d) Dangerous or restricted cargo.

e) Registered mail, security shipments, diplomatic pouches, live animal, etc.

7.8. MAINTENANCE STATUS OF THE AIRCRAFT

7.8.1. The flight crew will obtain a preliminary briefing regarding the maintenance status of the aircraft from flight control personnel to determine any significant effect upon the intended operations. Should this be the case, the appropriate limitation will be observed, and the operation planned accordingly.

7.8.2. When the flight crew boards the aircraft, a final and more complete maintenance status briefing will be conducted by the duty Foreman or his delegate technician. During preflight and inflight the ATL will be accessible to the flight crew.

7.8.3. Additionally, the flight crew shall:

a) Review the previous maintenance discrepancies and corrective actions in the technical logbook, which contains a journey records section and an aircraft maintenance record section (ECARAS 9.3.2.9), together with the current list of differed deficiencies and MEL/CDL requirements to determine the airworthiness of the aircraft.

b) Review all items entered in the Aircraft Technical Log (ATL) and addressed for proper sign off. Review a reasonable number of days to familiarize and discuss with maintenance on repeated items.

c) Review the Acceptable Deferred Defect (ADD) page and check if all items are in accordance with MEL/CDL and that those items have not expired beyond the category limit as shown on the system definitions section of MEL.

d) In the event there is a Time Limit Overrun (TLO) on any item, ensure that a properly and duly signed document is inserted in the ATL.

e) A maintenance discrepancy that requires airworthiness release should be communicated with Production Department of the Technical Service. Flight Control should facilitate the required communication. Under no circumstances should any aircraft depart with defects affecting airworthiness that have not been processed according to this procedure.

f) The provisions of MEL/CDL at both home base and out stations are applicable until the aircraft starts to move under its own power for the purpose of takeoff. Any decisions to continue a flight following a failure or un-serviceable item after brake release for taxi must be subject to the good judgment and airman ship of the PIC. The following guidelines may help the captain to better assess the situation for the current and next flight sectors.

1. If possible, confirm the availability of the maintenance support at destination to prevent operational disturbances
2. Consider the performance penalties as informed in the MEL (i.e. takeoff performance, fuel consumption increase, etc....)

3. Consider operational restrictions as informed in the MEL (e.g. degradation of approach, landing, RVSM/RNP or ETOPS capability, etc.)
4. Consider the environmental restrictions as stated in the MEL (e.g. OAT limitations, icing conditions limitations)
5. Consider the possible inter-relationship between all deferred MEL items (cumulative dispatch conditions requirements, etc.)

g) The PIC has the authority to reject an aircraft if, prior to each flight, he is dissatisfied with any aspect of airworthiness and or maintenance status of the aircraft.

7.8.4. MINIMUM EQUIPMENT LIST (MEL) AND CONFIGURATION DEVIATION LIST (CDL) (ECARAS 9.3.1.12)

a) MEL (approved by ECAA) shall be provided for the use of the flight crew members and maintenance personnel assigned to operational control functions during the performance of their duties. The MEL shall be specific to the aircraft type and variant which contains the circumstances, limitations and procedures for release or continuance of flight of the aircraft with inoperative components, equipment or instruments.

b) CDL (provided and approved by the State of Design) shall be provided for the use of flight crew members and maintenance personnel assigned on operational control functions during performance of their duties. The CDL shall be specific to the aircraft type. The AOC holder operations manual shall contain those procedures accessible to the Authority for operations in accordance with the CDL requirements.

c) An aircraft is not permitted to depart with a defect that has not been processed in accordance with the MEL/CDL.

7.9. TEST FLIGHT GENERAL

7.9.1. A test flight should be carried out following maintenance or repairs, which could affect the flight performance or handling characteristics of an aircraft. Also, as it is impracticable to prove the serviceability of all components by ground testing, it will be necessary to carry out test flights whenever it is impossible to ensure airworthiness without doing so.

7.9.2. Normal maintenance, component changes, disconnection of flight control, or any reported defect does not in itself warrant a test flight since the deciding factors are whether or not:

a) Flight characteristics could be affected and/or

b) Airworthiness can be established without a test flight.

7.9.3. Test flights are not required if ground tests or inspections or both show conclusively that repair or alternations have not appreciably changed the flight characteristics or substantially affected the flight operation of the aircraft.

7.9.4. Test flights must take place during daylight.

7.9.5. Permission from Director Flying Operations or from the appropriate chief pilot must be granted to perform the test flight.

7.10 FERRY FLIGHT

7.9.6. A certified aircraft shall not be operated except for the purpose of flight testing if malfunctions are observed during its operation that are impairing or likely to impair the airworthiness.

7.9.7. Permission from Director Flying Operations or from the appropriate Chief pilot must be granted to ferry aircraft in exceptional cases with impaired airworthiness (below MEL requirements). Such ferry flight may only be conducted if:

a) The required repairs cannot be performed locally and
b) The safe conduct of the ferry flight is assured.

7.9.8. For the conduct of ferry flights within the scope of the above permission the following minimum requirements shall be adhered to:

a) The certification and licensing authority have not withdrawn the airworthiness certificate of the aircraft.
b) Neither an airworthiness directive nor a mandatory inspection preludes the ferry flight.
c) The causes of the malfunctions and their possible effects on other components or systems are known.
d) During the ferry flight the only people allowed on board shall be those required for the operation handling and if necessary, inspection of the aircraft.
e) The operating weight of the aircraft shall be kept as low as possible.
f) The flight shall be conducted in compliance with relevant procedures and regulations of the Flight Operations Manual.

7.11. FLIGHT PLAN

7.11.1. SUBMISSION OF A FLIGHT PLAN

a) It is the duty of the Flight Dispatcher to file the ATC flight plan, and the PIC must ensure that ATC flight plans have been filed for all flights.
b) No company aircraft will take off unless an ATS flight plan is filed with the nearest ATS facility. If communications are not readily available, the flight plan

may be filed as soon as practicable by radio after takeoff. ATS flight plans for regularly operated flights (Scheduled flights) may be submitted in the form of repetitive (Stored) flight plan. At stations not providing flight control services the flight plan shall be completed by the flight crew.

c) Information relating to an intended flight or portion of a flight shall be submitted in the form of a flight plan to Air Traffic Service (ATS) UNIT prior to operating:

I. Any flight or portion thereof to be provided with air traffic control services.

II. Any IFR flight within advisory airspace or within other areas as may be required by ATS.

III. Any flight within or into designated areas or along designated routes when required by ATS to facilitate alerting and search-and-rescue activities.

IV. Any flight across international borders.

NOTE: The term "**Flight Plan**" is used to mean variously: full information on all items in the flight plan description, the entire planned route of a flight, or limited information required to obtain a clearance for a minor portion of a flight such as crossing an airway or taking off from or landing at a controlled airport.

d) Unless otherwise required by the appropriate authority, a flight plan for a flight to be provided to ATC or air traffic advisory service shall be submitted at least 30 minutes before departure, or if submitted during flight, at a time which will ensure its receipt by the appropriate ATC unit at least 10 minutes before the airplane is expected to reach the intended point of entry into a controlled airspace control or advisory area, or the point of crossing an airway or advisory route.

7.11.2. CONTENTS OF A FLIGHT PLAN

a) A flight plan shall cover any of the following items considered relevant by the appropriate ATS authority:

I. Airplane identification

II. Flight rules

III. Flight Status

IV. Type of airplane

V. Communications equipment

VI. Navigation & approach aids

VII. Secondary surveillance radar

VIII. Airport of departure
IX. Time of departure
X. Estimated times at flight information region boundaries
XI. Cruising speeds
XII. Cruising levels
XIII. Route to be followed
XIV. Airport of intended landing and ETA
XV. Alternate airport

b) For flight plans submitted during flight, the information provided concerning departure airport would be an indication of the location from which supplementary information concerning the flight may be obtained, if required.

c) For flight plans submitted during flight, the information to be provided concerning time of departure will be the time over the first point of the route which defines the route to be flown and should also include the following;

I. Fuel endurance.
II. Total number of people on board.
III. Emergency and survival equipment.
IV. Other information as requested.

7.11.3. COMPLETION OF A FLIGHT PLAN

a) Unless otherwise prescribed by the appropriate ATS authority for IFR flights, a flight plan shall contain applicable information on relevant items from the foregoing list of contents up to and including “Alternate Airports” regarding the whole route or the portion there for which the flight plan is submitted.

b) The flight plan shall also contain information on all other listed items when submitted or prior to departure for an IFR flight, for facilitation.

7.11.4. AMENDMENT OF ORIGINAL FLIGHT PLAN

a) The flight plan may be amended enroute to include any alternate airport that is within the fuel range of the aircraft as specified by the FUEL POLICY.

b) Any person may specify any scheduled alternate airport authorized for the type of aircraft as a destination in the original flight plan.

c) No person may change an original destination or alternate airport that is specified in the original flight plan to another airport while the aircraft is enroute unless;

I. The airport is authorized in the route manual for the type and the requirements for fuel, performance limiting weights, and meteorological conditions may be confirmed at the time of amendment of the original flight plan.

II. Each person who amends an original flight plan enroute shall record such amendment on the original flight plan. Pilots accomplishing such amendments shall notify Flight Control of the action to be taken.

7.11.5. CHANGES TO FLIGHT PLAN

a) All changes to a flight plan submitted for an IFR flight or a controlled VFR flight and significant changes to a flight plan submitted for an uncontrolled VFR flight shall be reported as soon as practicable to the appropriate ATS unit.

NOTE: Information submitted prior to departure regarding fuel endurance or total number of persons carried on board, if incorrect at time of departure, constitutes a significant change to the flight plan and, as such, must be reported.

b) Type of change information to be requested

I. Change of Cruising level

- Airplane Identification
- Requested new cruising level and cruising speed at this level.
- Revised estimated time (when applicable) at subsequent flight information region boundaries.

II. Change of Route

- Airplane Identification
- Flight Rules
- Description of new route of flight including related flight plan data beginning with time and position from which requested change to route is to commence.
- Revised estimated time at destination.
- Other pertinent information as requested ATS.

III. Change of Route & Destination

- Airplane Identification
- Flight rules

- Description of new route of flight to new destination including related flight plan data beginning with the time and Position from which requested change or route is to commence.
- Estimated time of arrival at new destination
- Alternate airport
- Other pertinent information as requested by ATS

7.11.6. CLOSING A FLIGHT PLAN

a) Unless otherwise instructed by ATS, the PIC shall report the arrival either in person or by radio at the earliest possible opportunity after landing, to the appropriate ATS unit at the airport if a flight plan has been submitted covering the entire flight or the remaining portion thereof to destination.

b) When a flight plan has been submitted for only a portion of a flight (other than the remaining portion of a flight destination), it shall when required, be closed by report to the appropriate ATS unit.

c) When communication facilities at the airport of arrival are known to be inadequate and alternate arrangements for the handling of arrival reports on the ground are not available, a message comparable to an arrival report shall be transmitted to ATS by radio immediately prior to landing. Normally, this transmission shall be made to the aeronautical station serving the ATS unit in charge of the flight information region in which the airplane is operating. Arrival reports made by airplanes shall contain the following:

I. Airplane identification
II. Airport of departure
III. Time of arrival
IV. Airport arrival

NOTE: Failure to provide an arrival report, when required, may cause serious disruption in the air traffic services and may incur expenses in unnecessary search and rescue operations.

7.11.7. FLIGHT PLAN CANCELLATION

Cancellation of an IFR flight plan enroute is not permitted except to amend the original flight plan. This requirement does not prevent the PIC from the use of a VMC approach, climb or descent.

7.12. SLOT TIME

7.12.1. At airports where slot times are assigned by the appropriate flow management units or ATS, the arrangement for departures shall ensure that the flight will be ready for takeoff at the assigned slot time or in case of slot period at the beginning of the time window.

7.12.2. Pilots will be informed of an assigned slot time by station or by flight dispatch center. In case pilots are advised by ATS of an assigned or changed slot, dispatch or station shall be informed immediately. When it becomes apparent that the ground handling process cannot be completed in time to meet the scheduled departure time for an unforeseen reason a revised deadline shall be determined. The station manager and the PIC will then determine if the new assigned slot can still be met.

7.13. OPERATIONAL FLIGHT PLAN (OFP)

7.13.1. Operational flight plan (OFP) must be prepared for all flights except test and training flights.

7.13.2. OFP shall be available during flight preparation & be accessible to the flight crew during flight.

7.13.3. The operational flight plan provides detailed fuel, routing, and technical information for in flight use by the flight crew. The operating flight crew are responsible to check the information's on the OFP must be consistent with the filled ATS flight plan. If the ATS flight plan is different from the OFP, the flight crew are required to communicate with dispatch to change the OFP or to file ATS as per the OFP if accepted by the controllers.

7.13.4. After both flight crew checked the ATS flight plan and OFP have the same flight planning, the flight crew will enter the data to the FMS. After completion of the FMS data entry, both flight crew are required to check the OFP is consistent with the data programmed in to the navigation system.

7.14 USE OF HISTORIC OPERATIONAL FLIGHT PLAN

7.14.1 Historic OFP shall be planned only if the system is down and unable to produce updated OFP.

7.14.2 The following conditions shall apply on the preparation of Historic OFP;

a) The OFP that will be prepared shall be within the last 10 days of the same fleet
b) A/C registration can be different, but the "k" factor used on the OFP has to be higher than the proposed A/C to be flown.
c) If the current weight is higher, fuel will be added according to the required additional fuel burn per ton (1000KGS) on the Historic OFP.
d) Contingency fuel will be calculated at 5%.
e) The worst wind condition (the least tail wind or the highest head wind) and other limitations from the previous 10 days has to be considered.
f) The Historic OFP shall not be the one that was used by other flight crew in the

past (nothing should be written on the historic OFP). It shall be a clean copy.

g) Flying with historic flight plans is prohibited to or from Europe, South and North America.

7.15. CONTENTS OF OPERATIONAL FLIGHT PLAN

7.15.1. Either company or contract provided operational flight plans are authorized, provided they are based upon approved data, airways and airspace structure, and fuel consumption data for aircraft type.

7.15.2. The operational flight plan shall include at least:

a) Aircraft registration;
b) Aircraft type and variant;
c) Date of flight and flight identification;
d) Departure airport, STD, STA, destination airport;
e) Types of operation (ETOPS, IFR, etc.);
f) Route and route segments with check points/waypoints, distances, time and tracks;
g) Planned cruising speed and flying times between waypoints/check points;
h) Planned altitude and flight levels;
i) Minimum safe altitude (MSA) for all phases of the flight as established by the responsible state;
j) Fuel calculations;
k) Fuel on board when starting engines;
l) Alternate(s) for destination and, when applicable, take off and en-route including information required in item (f) to (i) above;
m) Relevant meteorological information.

7.15.3. Detailed description of OFP is found in route manual.

7.16. ROUTE REQUIREMETNS

7.16.1. The route manual lists approved routes and areas of operations approved by the Civil Aviation Authority.

7.16.2. Route selection during flight preparation shall be based upon approved routes or route segments. Deviation from standard routing for the purpose of sight seeing is not authorized.

7.16.3. Whenever an approved area of operation does not specify particular routes or, for any reason, it is necessary to deviate from an approved route, the following factors shall be considered: -

a) Availability of enroute, terminal and alternate navigation aids.
b) Availability of meteorological services.
c) Diplomatic clearance: verify if traffic rights over politically sensitive areas, or special national security requirements exists.
d) Availability of ground handling and servicing facilities.
e) Terrain to be over-flown and aircraft performance limitations.
f) Fuel availability.
g) Reliability of available charts and maps.

7.16.4. Whenever possible, company flights shall be conducted within controlled airspace or, if not practical within airspace, providing air traffic advisory or similar service.

7.16.5. Within the restraints of route requirements, the route of flight selected shall be that which will result in a minimum time operation consistent with ATC restrictions, meteorological conditions and passenger comfort.

7.17. SELECTION OF CRUISING LEVELS

7.17.1. Cruising levels will be selected based on the following factors.

a) Distance between departure airport and destination.
b) Height of terrain over which the flight is to operate.
c) Fuel consumption at a given flight altitude.
d) Wind conditions and resulting wind components at a given flight altitude or level.
e) Other meteorological conditions such as turbulence, icing or thunderstorm activity.
f) Aircraft or equipment performance capability and limitations etc.…

7.17.2. Ideally, a cruising level would be selected which would satisfy all flight requirements and result in optimum aircraft performance with regard to the above factors. However, this is not always possible.

7.17.3. A single factor such as terrain elevation, wind component or aircraft performance limitations may on a given flight determine the selection of cruising level, but all factors must be considered.

7.18. CRUISING LEVEL WITHIN CONTROLLED AIRSPACE

7.18.1. IFR flights shall obtain an air traffic control clearance when operating in controlled airspace. The cruising levels to be used by IFR flight for operation in controlled airspace shall be selected from:

a) The table of cruising levels from the Jeppesen Airway Manual,

b) A modified cruising level (**TABLE-3 & 4**), when so prescribed for flight above Flight Level 290.

NOTE: Tracks prescribed in table of cruising levels do not apply when otherwise indicated in ATC clearances.

7.19. OUTSIDE CONTROLLED AIRSPACE

7.19.1. An IFR flight operating in level cruising flight outside of controlled airspace shall be flown at a cruising level appropriate to its track as specified in:

a) The tables of cruising levels except when otherwise specified by ATC for flight at or below 3000 feet (900 m) above mean sea level.

b) A modified table of cruising levels, when so prescribed, forfeit above flight level 290.

7.19.2. VFR cruising altitude or flight level, except while holding in a holding pattern of 2 minutes or less, or while turning, each person operating an aircraft under VFR in level cruising flight more than 3,000 feet above the surface shall maintain the appropriate altitude or Flight Level prescribed below, unless otherwise authorized by ATC: (Unless authorized by the appropriate ATC authority, Ethiopian may not operate in VFR flight above FL 200 or at transonic and supersonic speeds). (ECARAS 8.8.3.6)

a) When operating below transition altitude:

I. On a magnetic course of zero degrees through 179 degrees, any odd thousand feet MSL altitude+ 500 feet (such as 3500, 5500, or 7500):

II. On a magnetic course of 180 degrees through 359 degrees, any even thousand feet MSL altitude + 500 feet (such as 4500, 6500, or 8500)

b) When operating above transition altitude up to flight level 290 inclusive:

I. On a magnetic course of zero degrees through 179 degrees, any odd Flight Level + 500 feet (such as 195, 215, or 235): or

II. On a magnetic course of 180 degrees through 359 degrees, any even

Flight Level + 500 feet (such as 185, 205, or 225)

c) A modified table of cruising levels, when so prescribed for flight above Flight Level 290.

NOTE 1: Authorization for VFR flights to operate above FL 290 shall not be granted in areas where a vertical separation minima of 300meters (1000 feet) is applied above FL 290. (REFER **TABLE -7 & 8**).

NOTE 2: Tracks prescribed in table of cruising levels do not apply when otherwise specified in ATC clearances.

TABLE 1

VFR Flights (RVSM AIRSPACE)

From 000 degrees to 179 degrees	
LEVEL	
FL	Feet
035	3500
055	5500
075	7500
095	9500
115	11500
135	13500
155	15500
175	17500
195	19500
215	21500
235	23500
255	25500
275	27500

TABLE 2

VFR Flights (RVSM AIRSPACE)

From 180 degrees to 359 degrees	
LEVEL	
FL	Feet
045	1350
065	2000
085	2600
105	3200
125	3800
145	4400
165	5050
185	5650
205	6250
225	6850
245	7450
265	8100
285	8700

SECTION 2.4 FLIGHT PREPARATION

TABLE 3

IFR Flights (RVSM AIRSPACE)

From 000 degrees to 179 degrees	
LEVEL	
FL	Feet
010	100
030	3000
050	5000
070	7000
090	9000
110	11000
130	13000
150	15000
170	17000
190	19000
210	21000
230	23000
250	25000
270	27000
290	29000
310	31000
330	33000
350	35000
370	37000
390	39000
410	41000
450	45000

TABLE 4

IFR Flights (RVSM AIRSPACE)

From 180 degrees to 359 degrees	
LEVEL	
FL	Feet
020	2000
040	4000
060	6000
080	8000
100	10000
120	12000
140	14000
160	16000
180	18000
200	20000
220	22000
240	24000
260	26000
280	28000
300	30000
320	32000
340	34000
360	36000
380	38000
400	40000
430	43000

SECTION 2.4 FLIGHT PREPARATION

TABLE 5

IFR Flights (Non-RVSM AIRSPACE)

From 000 degrees to 179 degrees	
LEVEL	
FL	Feet
010	100
030	3000
050	5000
070	7000
090	9000
110	11000
130	13000
150	15000
170	17000
190	19000
210	21000
230	23000
250	25000
270	27000
290	29000
330	33000
370	37000
410	41000

TABLE 6

IFR Flights (Non-RVSM AIRSPACE)

From 180 degrees to 359 degrees	
LEVEL	
FL	Feet
020	2000
040	4000
060	6000
080	8000
100	10000
120	12000
140	14000
160	16000
180	18000
200	20000
220	22000
240	24000
260	26000
280	28000
310	31000
350	35000
390	39000
430	43000

TABLE 7

VFR Flights (Non-RVSM AIRSPACE)

From 000 degrees to 179 degrees	
LEVEL	
FL	Feet
300	30000
340	34000
380	38000
420	42000
460	46000
500	50000

TABLE 8

VFR Flights (Non-RVSM AIRSPACE)

From 000 degrees to 179 degrees	
LEVEL	
FL	Feet
320	32000
360	36000
400	40000
440	44000
480	48000
520	52000

7.20. REDUCED VERTICAL SEPARATION MINIMUM (RVSM) CRUISING FLIGHT LEVELS

7.20.1. RVSM shall be applicable in that volume of airspace between FL290 and FL410 inclusive in the flight information regions (FIR/UIR) along specified routes outside controlled airspace shall maintain a listening watch on the appropriate radio frequency and establish two-way communication as necessary with the ATS unit providing flight information service.

7.20.2. RVSM CRUISING FLIGHT LEVEL

TABLE 9

WEST BOUND	EAST BOUND
FL 400	FL 410
FL 380	FL 390
FL 360	FL 370
FL 340	FL 350
FL 320	FL 330
FL 300	FL 310
	FL 290

7.21. FLIGHT ALTITUDE RULES

7.21.1. Neither the company nor the captain may operate an aircraft below the minimum altitudes as indicated in ICAO Annex 2 Chapter 4 for VFR and Chapter 5 for IFR. Ethiopian operation is conducted as set forth below.

7.21.2. Except when necessary for take-off or landing & when specifically authorized by the appropriate authority, an IFR flight shall be flown at a level which is not below the minimum flight altitude established by the state whose territory is overflown or, where no such minimum flight altitude has been established.

a) Over high terrain or in mountainous areas, at a level which is at least 600m (2000ft) above the highest obstacle located within 8 km of the estimated position of the aircraft.

b) Elsewhere than as specified in a), at a level which is at least 300m (1000ft) above the highest obstacle located within 8 km of the estimated position of the aircraft.

NOTE: The estimated position of the aircraft will take account of the navigational accuracy which can be achieved on the relevant route segment, having regard to the navigational facilities available on the ground and in the aircraft.

7.22. MINIMUM EN-ROUTE FLIGHT ALTITUDES

7.22.1. The following IFR minimum altitudes are published in the Jeppesen enroute, area and terminal charts:

a) Minimum En-route Altitude (MEA)
b) Minimum Obstacle Clearance Altitude (MOCA)
c) Route Minimum Off Route Altitude (MORA)
d) Grid Minimum Off Route Altitude (Grid MORA)
e) Minimum Safe Altitude (MSA)
f) Minimum Vectoring Altitude (MVA)
g) Minimum Descent Altitude/Height (MDA/H)

7.22.2 The above terms are defined in the INTRODUCTION section of the Jeppesen Airway Manual. These published minimum altitudes will be applied as follows:

7.23. TERMINAL AREA

7.23.1. During IFR, normal and emergency operations, descending below the published MSA is not allowed unless the flight crew is fully VFR or following the standard instrument Approach Procedure, being radar vectored to MVA, following SID or STAR.

7.23.2. The Minimum Vectoring Altitudes may be utilized only upon the radar controller’s determination that an adequate radar return is being received from the aircraft being controlled. Unless on published IFR procedure, any clearance below the published MVA shall only be accepted if visual conditions allow the pilot to maintain positive visual separation from terrain.

7.24. MINIMUM ALTITUDE FOR NORMAL OPERATIONS

7.24.1. In normal operations enroute IFR flight levels or altitudes shall be higher than the published Minimum En-route IFR Altitude (MEA) indicated on enroute charts.

7.24.2. The minimum enroute altitude shall be the higher of the Minimum Off-Route Altitude (MORA) and the published Minimum Obstruction Clearance Altitude (MOCA). Both minimum altitudes are indicated on enroute charts when they exist.

7.24.3. In case of incomplete or lack of safety altitude information, obstacles and reference points have to be located on Operational Navigation Charts (ONC) or topographic maps. The minimum safe enroute altitude must clear all obstacles within 5 NM (9.3 km) of the route centerline by 1000 ft. (300m) if the reference point is not higher than 5000 ft. (1500m) MSL or 2000 ft. (600m) if reference point is higher than 5000 ft. MSL. If available and not limiting, the grid MORA may be used as minimum flight altitude. These minimum altitudes must be respected along the track with all engines operative unless a procedure has been approved to cope with depressurization.

7.24.4. Descending below MEA, MOCA, MORA or GRID MORA is allowed only if the flight is conducted under VFR or there is a specific escape route which clears all obstacles around.

7.25. MINIMUM ALTITUDE FOR NON-NORMAL OPERATIONS

7.25.1. In case of one engine inoperative, the following minimum altitude requirements apply:

a) The net flight path must have a positive gradient at 1500 feet (450m) above the aerodrome where the landing is assumed to be made after engine failure.

b) The gradient of the net flight path must have a positive gradient at an altitude of at least 1000 feet (300m) above all terrain and obstructions along the route within 5 NM (9.3 km) on either side of intended track.

Or

c) From the cruising altitude to the aerodrome where the landing can be made, the net flight path must clear vertically with a negative gradient by at least 2000 feet (600m) all terrain and obstructions along the route within 5 NM (9.3 km) on either side of the intended track.

NOTE: Fuel jettisoning (if applicable) is permitted to an extent consistent with reaching the aerodrome with the required fuel reserves.

7.25.2. Whilst maintaining the centerline of an airway or ATS route, during normal and emergency flight operations, the minimum flight altitude will be the lowest of Minimum En route Altitude (MEA), Minimum Obstruction Clearance Altitude (MOCA), or Route Minimum Off-route Altitude (Route MORA)

7.25.3. When deviating from the centerline of an airway or ATS route during normal flight operations, the applicable minimum flight altitude shall be the Grid MORA containing the aircraft's position. Deviation from ATS routes is not permitted during normal flight operations where the GRID MORA is not published, unless a specific minimum flight altitude has been issued for that route and flight.

7.25.4. Descending below MEA, MOCA, MORA or GRID MORA is allowed only if the flight is conducted under VFR or there is a specific escape route which clears all obstacles around.

7.24.5. If deviating from an airway or ATS route for emergency reasons (i.e. decompression or drift down), the minimum flight altitude shall be the Grid MORA, unless a lower mandatory altitude is published on a specific Escape Route Chart produced for the route being flown (see Route Manual).

7.26. ALTIMETER SETTING

7.26.1. All Ethiopian flights shall be operated on QNH altimeter setting reference only during takeoff, approach and landing. Should the flight crew receive altimeter setting information in QFE, they shall request a QNH setting from the ATC service. (ECARAS8.8.1.4.)

7.26.2. Altimeters shall be checked during the preflight phase as stipulated in the Aircraft Operations Manual.

TABLE 10

FLIGHT PHASE	No 1 (LEFT)	No 2 (RIGHT)	REMARK
Before T/off	QNH	QNH	
Climb & cruise	QNH	QNH	If remaining below Transition altitude
Climb	1013.2	1013.2	At Transition altitude
En route	1013.2	1013.2	
Descent	1013.2	1013.2	When cleared to intermediate Flight level
Descent	QNH	QNH	When cleared to an altitude and no further Flight level reports are required by ATC
Initial approach	QNH	QNH	
Final approach	QNH	QNH	
Missed approach	QNH	QNH	

7.27 ALTIMETER METER/FEET CONVERSION

Altimeter meter/feet conversion table is provided below, and the flight crews are required to use it on those flight sectors, which require its use.

7.27.1. METERS TO FEET CONVERSION CHART

TABLE 11

WESTBOUND (180° - 359°)		EASTBOUND (000° - 179°)	
Flight Levels		Flight levels	
Meter	Feet	Meter	Feet
15500	50900	14900	48900
14300	46900	13700	44900
13100	43000	12500	41100
12200	40100	11900	39100
11600	38100	11300	37100
11000	36100	10700	35100
10400	34100	10100	33100
9800	32100	9500	31100
9200	30100	8900	29100
8400	27600	8100	26600
7800	25600	7500	24600
7200	23600	6900	22600
6600	21700	6300	20700
6000	19700	5700	18700
5400	17700	5100	16700
4800	15700	4500	14800
4200	13800	3900	12800
3600	11800	3300	10800
3000	9800	2700	8900
2400	7900	2100	6900
1800	5900	1500	4900
1200	3900	900	3000
600	2000	300	1000

NOTE 1: When communicating with ATC in metric altimetry airspace, all references to FL/Altitude/Height will be in meters. However, all inter-cockpit standard calls for climb and descent shall be referenced to feet (e.g. 1000 to go etc.) where applicable, the metric Altimeter function should only be used as a cross-reference.

NOTE 2: If a landing is to be made at an aerodrome in a metric altimetry region, DH/DA/MDA are depicted in feet on the Jeppesen chart.

7.27.2. CONVERSION CHART FOR ANNONCEMENT

TABLE 12

FEET	METER	FEET	METER
28000	8550	29000	8850
30000	9150	31000	9450
32000	9750	33000	10050
34000	10350	35000	10650
36000	10950	37000	11300
38000	11600	39000	11900
40000	12200	41000	12500

7.28. CORRECTIONS TO MINIMUM FLIGHT ALTITUDES AND FLIGHT LEVELS

All minimum flight altitudes and flight levels shall be corrected for the effects of pressure and temperature variations from standard, and for the effects of high wind speed in areas of high terrain as follows:

TABLE 13

CORRECTIONS	MINIMUM FLIGHT ALTITUDE	MINIMUM FLIGHT LEVEL
Wind	X	X
Temperature	X	X
Pressure	N/A	X

7.29. WIND CORRECTION-HIGH TERRAIN

When flights are conducted within 20nm of terrain that rises over 2000ft the minimum flight altitudes shall be corrected for the wind effect in accordance with the following table:

TABLE 14

ELEVATION	0 – 30 KTS	31 - 50 KTS	51 - 70 KTS	MORE THAN 70KTS
2000ft to 8000ft	Add 500ft	Add 1000ft	Add 1500ft	Add 2000ft
Above 8000ft	Add 1000ft	Add 1500ft	Add 2000ft	Add 2500ft

7.30. TEMPERATURE CORRECTION

7.30.1. Pressure altimeters are calibrated to indicate true altitude under ISA conditions. Any deviation from ISA will result in an erroneous reading on the altimeter. In a case when the temperature is higher than ISA, the true altitude will be higher than the figure indicated by the altimeter, and the true altitude will be lower when the temperatures are lower than ISA.

7.30.2. The altimeter error may be significant and become extremely important when considering obstacle clearances in very cold temperatures. When ambient temperature is lower than ISA, the following additions to minimum flight altitudes shall be made:

TABLE 15

ISA DEVIATION	CORRECTION
Lower than ISA	NOT less than 5%
Lower than ISA MINUS 15oC	NOT less than 10%
Lower than ISA MINUS 30oC	NOT less than 20%
Lower than ISA MINUS 50oC	NOT less than 25%

7.31. PRESSURE CORRECTION

7.31.1. All published Jeppesen safety altitudes relate to height above mean sea level and are measured with reference to QNH. To ensure true altitude does not compromise minimum altitude clearance requirements, the following corrections shall be added to the minimum flight altitudes to determine the minimum usable flight level.

TABLE 16

QNH OF NEAREST STATION	CORRECTION –FEET
1013 and higher	Nil
1010	80
1005	220
1000	380
995	510
990	630
985	780
980	920
975	1080

7.31.2. EXAMPLE OF CORRECTIONS TO MINIMUM FLIGHT ALTITUDES AND LEVELS

Given: QNH = 975 hpa; Temperature = ISA - 10; Wind Speed = 30kts; published MEA=14,000ft

- Find: Minimum usable altitude and flight level

Step 1: QNH correction = +1,080 ft

Step 2: Temperature correction = 0.05 X 14,000 = + 700 ft

Step 3: Wind correction = +1,000 ft

- Solution: Minimum usable Altitude: 14,000 + 700 + 1000= 15,700 ft
- Solution: Minimum usable FL: 14,000+1080+700+1000=16,780 ft= FL 170 (rounded up).

7.31.3. ICAO COLD TEMPERATURE ERROR TABLE

- For more information refer to supplementary procedure on FCOM.

TABLE 17

TEMP ^{0}C	HEIGHT ABOVE AIRPORT (ft)												
	200	300	400	500	600	700	800	900	1000	1500	2000	3000	4000	5000
+10	10	10	10	10	20	20	20	20	20	30	40	60	80	90
- 0	20	20	30	30	440	40	550	50	60	90	120	1170	230	280
-10	20	30	40	50	60	70	80	90	100	150	200	290	390	490
-20	30	50	60	70	90	100	120	130	140	210	280	420	570	710
-30	40	60	80	100	120	140	150	170	190	280	380	570	760	950
-40	50	80	100	120	150	170	190	220	240	360	480	720	970	1210
-50	60	90	120	150	180	210	240	270	300	450	590	890	1190	1500

7.32. ALTERNATE AIRPORT REQUIREMENTS

7.32.1. An alternate airport, to which a flight may proceed when landing at the intended airport becomes inadvisable, will be selected for every flight except as provided below.

7.32.2. An alternative airport must meet operational requirements for the type of operation planned. Instrument approach and landing charts and other relevant information pertaining to the airport being selected must be available on board the aircraft. Special attention shall be paid to the runway length available for landing and takeoff, considering elevation, slope, temperature, wind meteorological minima, and other pertinent factors.

7.32.3. The selection of alternate airports shall be governed by the alternate minimum specified for each authorized airport as shown on Jeppesen instrument approach charts.

7.32.4. If, however, at the expected time of arrival, conditions at the alternate airport being considered are expected to be such that a circling approach will be necessary, the appropriate circling minimum, if higher than the respective alternate minimum, shall be applied. An alternate minimum based on the availability of either ILS or PAR, or both will only be applied if, at the expected time of arrival, conditions at the airport are expected to be such that these facilities will be in operation and wind direction and velocity are such that either a straight-in ILS landing or a PAR approach will be possible.

7.32.5. Alternate weather minima apply only for flight release and are not limiting for landing. Upon diversion to an alternate airport, landing minima are the same as if that airport were the intended destination of the flight.

7.33. TAKE OFF WEATHER PLANNING MINIMA

Refer to FOM 2.2 (7.5)

7.34. TAKE OFF ALTERNATE PLANNING MINIMA

7.34.1. If the meteorological conditions at the airport of takeoff are below the landing minima or if it would not be possible to return to the airport of departure for any reason, an alternate airport must be selected and specified in the Operational flight plan.

7.34.2. Aircraft having two engines: - An alternate airport located not more than one hour from the departure airport at normal cruising speed in still air with one engine inoperative. For maximum diversion distance in 60 minutes.

7.34.3. A take-off alternate shall have the following conditions:

a) The appropriate weather reports or forecasts or any combination thereof indicate that, during a period commencing one (1) hour before and ending one (1) hour after the estimated time of arrival at the aerodrome, the weather will be at or above the applicable AOM (Aerodrome operating minima).

b) When the only approaches available are non-precision and/or circling, the cloud ceiling must be taken into account.

c) Any limitation related to one engine inoperative operations must be taken into account.

d) The distance of a take-off alternate from the departure aerodrome is not to exceed the specified distance for the airplane type.

NOTE: The take-off alternate shall have CAT I minimum or applicable minima whichever is higher when a return to the departure point is not possible due to the operational capability of the aircraft, weather or any other unforeseen circumstances.

7.35. EN-ROUTE ALTERNATE AIRPORTS

7.35.1. No person may list an airport as an en-route alternate unless the appropriate weather reports or forecast or any combination thereof indicate that the weather conditions will be at or above the alternate weather minimums specified in the operations specifications for that airport are met when the flight arrives.

7.35.2. Flight crew members are required to use Jeppesen manual as guidance for proper identification of en-route alternates.

7.36. DESTINATION ALTERNATE PLANNING MINIMA

7.36.1. An adequate aerodrome may be usable for destination (except if the aerodrome is isolated) if the weather reports and/or forecasts indicate that, during a period commencing 1 hour before and ending 1 hour after the estimated time of arrival at the aerodrome, the weather conditions (RVR / visibility and for non-precision or circling approaches, ceiling at or above MDH) will be at or above the planning minima as follows:

TABLE 19

Type of approach	Planning minima
CAT II and CAT III	Cat I minima (RVR)
CAT I	Non-precision approach minima (ceiling/RVR)
Non-precision	Non-precision approach minima + (MDA + 200 ft. / RVR + 1000 m)
Circling	Circling minima

7.36.2. "Non-precision minima" mentioned in the table above, means the next highest minimum that is available in the prevailing wind and serviceability conditions; Localizer only approaches, if published, are considered to be "non-precision" in this context.

7.36.3. Tables publishing planning minima should indicate values that are likely to be appropriate on the majority of occasions (e.g. regardless of wind direction). Serviceability must, however, be fully taken into account.

NOTE: The planning minima for ETOPS enroute alternate aerodromes are defined in ETOPS.

7.36.4. A destination alternate shall be specified for flights prior to takeoff.

7.36.5. When the weather conditions for the destination and the first alternate airport are marginal at least one additional alternate must be designated.

7.36.6. If the current meteorological forecast for the airport of intended landing indicates that conditions, at estimated time of arrival, be at or above minima at the airport and runway concerned, at least one alternate is required.

7.36.7. If the destination forecast is below CAT I minima, verify that alternate weather forecasts are appropriate to the available approach means and at least at or better than CAT I minima.

7.36.8. Two usable destination alternates to be nominated if the weather reports or forecast for the destination indicate that from 1 hour before to 1hour after ETA it is below the applicable minima or no weather information for destination.

7.36.9. En-route alternate airport must have RFFS equivalent to that specified by ICAO as Category 4, or higher.

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SECTION 2.5 NORMAL OPERATIONS

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Section 2.5 Normal Operations

1.0. PURPOSE

The purpose of this normal operation procedure is to provide guidelines for compliance with normal operations.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. FLIGHT MANAGEMENT MODEL

Ethiopian airlines Flight Management Model depicts the standard aircraft handling sequence that is general in nature on all flight phases.

4.2. POLICY ON THE USE OF AUTOPILOT, AUTOTHROTTLE & FLIGHT DIRECTOR

4.2.1. Flight Crew should always use the highest available level of automation during all phases of flight. The PF must at all times be aware of the autopilot / flight director system mode engaged/change and the PF should cross-check mode control panel status. And if the use of automation results in a conflict between flight requirements and actual flight path, the PF must announce and discontinue the use of automation immediately. The discrepancy must be resolved before re-engagement automation.

4.2.2 Whenever flying the A/C manually it must be limited to flight director command unless the landing cannot be completed following the flight director guidance. In this case use PAPI, VASI and Runway centerline.

4.2.3 If a flight crew loses both Auto Pilot & flight director commands, the pilots can continue the approach by using Radar vectors or to the initial approach fix maintaining MSA. Thereafter follow the instrument approach procedure to the final approach fix and if visual follow PAPI or VASI and runway centerline to complete the landing. If visual reference is not available, the approach after the FAF may be continued using LOC, G/S or VOR guidance. If the approach is not stabilized a Go Around must be executed immediately. Raw Data approach can only be commenced for ILS, LOC & VOR approaches.

4.2.4 If a flight crew loses flight director guidance only, fly the A/C using the Autopilot at least to the point visual reference is obtained after the final phase (FAF) of the flight.

4.2.5. The autopilot must be engaged as early as practicable at or above the minimum engage altitudes specified in the FCOM provided that the A/C is well trimmed and flight director guidance is followed to enable accurate tracking & vertical path of the departure path.

4.2.6. Approaches with a disengaged automation should be carried out after due consideration is given to the level of fatigue, the likelihood of a go-around, the ATC environment and the present weather conditions. The PF must communicate his/her intent and use of automation.

4.2.7. The autopilot should be left engaged until approaching Minima during a non-precision approach or turning final on a visual approach.

4.2.8. Standard calls are to be used for mode selections, and all resultant changes are to be confirmed by announcing FMA changes as per FCOM.

5.0. DEFINITION

5.1. AVIATE: fly the aircraft in accordance with restrictions and limitations set forth in the OM;

- Protect the airplane
- Protect the controls
- Protect yourself…

5.2. NAVIGATE: guide the aircraft along the intended or appropriate route:

- Diversion
- Hold
- Approach

5.3. COMMUNICATE: verbalize intentions to other crew members and ATC, as applicable. Who do I need to speak with urgently?

5.4. ATZ - Aerodrome Traffic Zone

5.5. CFIT - Controlled Flight into Terrain

SECTION 2.5 NORMAL OPERATIONS

6.0. RESPONSIBILITY

It is the responsibility of each flight crewmember to comply with the operational activities contained in this section.

7.0. PROCEDURE

7.1. STERILE COCKPIT

7.1.1. The principle behind the Sterile Cockpit Policy is established to significantly reduce interruptions and distractions during the most critical phases of a flight. Therefore, no flight crew member may engage in, nor may any pilot-in-command permit any activity during this phase of flight which could distract any flight crew member from the performance of his duties, or which could interfere in any way with the proper conduct of those duties.

7.1.2. For the purpose of this requirement, an "activity" includes engaging in non-essential conversation within the cockpit and non-essential communication between the cabin and cockpit crew.

7.1.3. The **"Sterile Cockpit Policy**" comprises the following:

a) Cockpit door closed and locked.
b) Conversation limited to matters directly associated with aircraft operation.
c) Do not vacate an aircraft control seat below 10000 ft. above field elevation. If it is required to vacate the control seat for transfer of duties to another pilot flying crew member it must be done above 10,000 feet.
d) Visits to the flight deck are restricted to operationally essential Issues.
e) Cabin Crew on the flight deck or the cabin shall address active flight crew members only when necessary for the safe conduct of the flight.
f) Restriction of all activities except to flight operational matters.
g) One pilot shall have full access to the flight control and maintain constant vigilance during flight.
h) Headsets must be worn and boom mic shall be used during the Sterile Cockpit period. During cruise Captains' flight deck speaker may be used but the volume shall be kept at the minimum usable level to avoid interference with normal crew conversation with in the flight deck.
i) Time pressure, operational constraints, or shortened flight duration shall not be considered justification for deviation from sterile flight deck requirements. The requirement to maintain a sterile flight deck remains unchanged regardless of flight length, schedule constraints, or workload.

SECTION 2.5 NORMAL OPERATIONS

7.1.4. It is impossible for crew members outside the cockpit to know for certain when the sterile cockpit policy is being enforced. Therefore, the following requirements shall be adopted by Cabin Crew prior to calling the cockpit over the interphone:

a) Apart from the no-contact period, irrespective of the phase of flight, always inform flight crew immediately of any safety related situations.

b) From commencing push back (or engine start when no push back is required) through to top of climb, and once again from top of descent through to engine shutdown, no calls or visits except when required by the Cabin Secure/ready Procedure.

7.1.5. In order to enforce Sterile Cockpit Policy and avoid unnecessary interruption to the flight crew communication activities in the cockpit, flight crew shall initiate the first communication call to the cabin, after take-off. And before commencing descent flight crew should advise Cabin crew that descent is commenced.

7.1.6. Adherence to the Sterile Cockpit Policy should not affect:

a) The use of effective CRM practices by the flight crew.

b) Communication of emergency or safety related information by cabin crew.

7.2. CRITICAL PHASES OF FLIGHT ARE:

7.2.1. From commencing push back (or engine start when no push back is required), until passing 10,000 feet AGL on climb.

7.2.2. From 10,000 feet AGL on descent until engine shut down.

7.3. NO CONTACT PERIOD:

The time from Takeoff thrust application to gear retraction and from gear extension to Landing roll complete is the “NO CONTACT PERIOD”.

7.4. CABIN SECURED

7.4.1. Before (push back or Engine start) whichever is earlier the Team Leader Cabin Services shall inform the captain: “**Cabin Secured**” and during descent before passing 10,000’ AGL the Team Leader Cabin Services shall inform the Captain: “**Cabin Ready**”.

7.4.2. On ground when boarding is completed, passengers are seated, and all hand luggage are secured.

7.4.3. In flight when passengers are seated, all hand luggage are secured and galleys are secured.

7.4.4. **Aircraft Door Opening for Ventilation**

Cabin doors must not be opened for ventilation purposes unless proper ground safety measures are in place.

Procedure:

- If ventilation is required while on ground, a **stair unit must be securely positioned and connected** to the specific door before opening.
- The responsible crew (usually the Senior Cabin Crew Member or Ground Engineer) must ensure the area is clear and safe before door operation.

7.5. DOORS ARM AND DISARM

Before (pushback or Engine start) whichever is earlier the Team Leader Cabin Services must report to the PIC:

- "**Doors Armed"** right after all doors are closed and armed.
- **"Doors Disarmed**" right after all door are disarmed.

7.6. PASSENGER BRIEFING

7.6.1. Flight crew shall not commence a takeoff unless the passengers are briefed before takeoff. The PIC shall turn on required passenger information signs for each takeoff and landing. (ECARAS 8.9.2.16)

7.6.2. Extended overwater operations may not be commenced unless all passengers have been orally briefed on the location and operations of preservers, life rafts and other flotation means, including demonstration of the method of donning and inflating a life preserver. (ECARAS 8.9.2.18)

7.7. BEFORE TAKEOFF THE PASSENGERS SHALL BE BRIEFED

a) Smoking limitations and prohibitions (ECARAS 8.9.2.17).
b) The location and operation of emergency exits.
c) The use of seatbelts.
d) The location and use of any required emergency floatation means.
e) Fire extinguisher location and operation (ECARAS 8.9.2.17).
f) The placement of seatbacks in an upright position before takeoff and landing.
g) The use of oxygen and emergency oxygen equipment.
h) For extended over water operation, the location and operation of life preservers, life rafts, and other floatation means, including a demonstration of the method of donning and inflating a life preserver.

i) Availability of safety cards.

j) Location of emergency escape lighting.

7.8. BEFORE TAKEOFF ANNOUNCEMENT

As soon as the passengers' safety information is completed, and the final check of the cabin is done, the Team Leader Cabin Services shall inform the PIC "**cabin ready"** then the PIC instructs the first officer to alert the cabin "**cabin crew be seated for departure".**

7.9. AFTER TAKEOFF

After takeoff, immediately after the seatbelt sign is turned off, an announcement shall be made by the Team Leader Cabin Services that passengers should keep their seatbelts fastened, while seated, even when the seatbelt sign is off.

7.10 DUTY STATIONS, SEATBELTS AND SHOULDER-HARNESS

7.10.1. All essential flight crew shall not leave their assigned duty position in flight unless it is required for the performance of flight duties or to meet physiological needs.

7.10.2. All flight crew shall keep their seat belts fastened when at their assigned stations.

7.10.3. All flight crewmembers occupying a pilot's seat shall keep their safety harness (shoulder straps and seat belts) fastened during the take-off and landing phases of flight.

7.10.4. Other flight crew members shall keep their safety harness fastened during the take-off and landing phases of flight, unless the shoulder straps interfere with the performance of duties, in which case the shoulder straps may be unfastened but the seat belt shall remain fastened.

7.11 AUTHORISZED ACCESS TO FLIGHT DECK/JUMP SEAT

7.11.1. The flight crew compartment door shall be closed and locked. (ECARAS 7.10.1.2)

7.11.2. The flight crew compartment door shall be capable of being locked from within the compartment in order to prevent unauthorized access. (ECARAS 9.4.1.6)

7.11.3. In all Ethiopian flights there shall be approved means by which the cabin crew can discreetly notify the flight crew in the event of suspicious activity or security breaches in the cabin. (ECARAS 9.4.1.6)

7.11.4. All Ethiopian airplanes should be equipped with an approved flight crew compartment door (cockpit door) that is designed to resist penetration by small arms fire and grenade shrapnel and to resist forcible intrusions by unauthorized persons. This door shall be capable of being locked and unlocked from either pilot's station. (ECARAS 9.4.1.6)

SECTION 2.5 NORMAL OPERATIONS

7.11.5. The team leader cabin services must make sure that the flight deck door is closed from the moment passengers boarding is started until passengers have disembarked and it must be locked from the time all external doors are closed following embarkation until any such door is open for disembarkation, except when necessary to permit access or egress by authorized people.

7.11.6. Means should be provided for monitoring from either pilot's station the entire door area outside the flight crew compartment to identify persons requesting entry and to detect suspicious behavior or potential threat.

7.11.7. For security reasons, access to the flight deck should be limited to authorized personnel only.

7.11.8. The allocation of Flight Deck jump seats is at the absolute discretion of the captain and is only available to authorized people.

7.11.9. Authorized people are:

a) Crew members on official duty.
b) People flying with operational duties and have specific authorization issued by VP Flight Operations or Director Flying Operations.
c) A person who is authorized by the Regulating Authority shall be allowed to inspect the airplane and if necessary, remain in the flight deck during flight. Such authorized person shall have access to all the necessary flight crew, operations, maintenance and/or onboard documents.
d) Authorized ECAA flight inspector who is traveling on an official inspection flight.

7.11.10. Except when necessary to permit access to authorized people, the cockpit door shall remain closed and locked from the moment passenger boarding is completed and all aircraft exterior doors are closed until the time the aircraft doors are opened to disembark passengers.

7.11.11. All persons carried on the flight deck are made familiar with the relevant safety procedures.

NOTE: On flights to and from Israel flight deck door remains closed and locked from the time passengers are boarding to time disembarkation is completed.

7.12. FLIGHTS TO NORTH AMERICA AND EUROPE

7.12.1. No person shall be admitted to the flight deck of an aircraft other than:
A flight crew member.
a) A crew member performing their duties.

SECTION 2.5 NORMAL OPERATIONS

b) An inspector of the Ethiopian Civil Aviation Authority (ECAA); or

c) A person who has expertise related to the aircraft, its equipment or its crewmembers and who is required to be in the flight deck to provide a service to Ethiopian.

7.13. FLIGHT DECK JUMP SEAT BRIEFING

7.13.1. The occupation of a vacant flight crew seat requires certain instructions to the occupant regarding the following items for more information refer to FOM 1.5(7.1.2):

a) Handling of seat and seat harness.

b) Emergency exits.

c) Position, handling and time of use of the oxygen masks.

d) Position and use of the life vests (if required).

e) Behavior in emergency cases.

f) Not to touch any controls, switches, instruments or circuit breakers.

g) Sterile cockpit policy

7.14. FLIGHT DECK ENTRY PROTOCOL/REQUEST

7.14.1. The PIC shall ensure that the flight deck compartment door is locked at all times during operations (ECARAS 8.5.1.13). However, authorized crew members requiring access to the flight deck must use the following procedure:

a) Check that the galley and flight deck door area is clear of passengers and that the galley/cabin curtain is closed.

b) Use the intercom to establish contact with the flight crew. Self-identify by stating name and the established entry code. And confirm that entry to the flight deck is permitted. Confidentiality of the code must be respected by all.

c) Wait for the door to be unlocked.

d) Enter and close the door immediately.

e) When leaving the flight deck ensure the flight deck door area is clear of passengers (by using the view port or any other available means). Ensure the flight deck door is closed and locked after exiting.

7.15. FLIGHT DECK ACTIONS

7.15.1. On receiving an intercom request for flight deck access:

a) Confirm that the person requesting access is genuine by verifying the use of the correct code and the video surveillance display if installed.

b) Unlock the flight deck door.

c) Ensure that the door remains locked during the flight deck visit.

d) Once the visiting crew member has left the flight deck immediately confirm that the door is closed and locked.

7.16. FLIGHT CREW ABSENCE FROM THE FLIGHT DECK

7.16.1. Essential flight crew members shall remain in the flight deck except for the purpose of safety, physiological needs or when they are relieved for in flight rest for augmented crew.

7.16.2. Such absence is permitted in accordance with the following procedure:

a) The absence is restricted to the minimum time necessary.

b) Ensure when one pilot leaves the flight deck at cruising level, the remaining pilot is alert and has the situational awareness, will have the seatbelts and shoulder harnesses fully fastened, the headset on and has unobstructed access to flight controls.

c) Positive control transfers from PF to PM by verbal command of "You have control" and response of "I have control". The PF announces FMA and MCP/AFS panel when transferring control to the PM.

d) Prior to opening the flight deck door, ensure that the forward galley and flight deck door area is free of passengers by using the flight deck door viewer.

e) Confirm with cabin crew that the galley/cabin curtain is closed by intercom and summon a cabin crew to stay in the flight deck until the flight crew returns.

f) The flight deck door shall be locked whilst the crew member is absent.

g) The forward galley area must remain sterile, that is free of passengers, during the absence from the flight deck.

h) Flight deck crew members must not enter the flight deck if the flight deck entry and galley area are occupied by passengers.

NOTE: Flight crew member returning to the flight deck follows the entry request procedure as stated in this section under "**FLIGHT DECK ENTRY PROTOCOL"**.

7.16.3. Should the auto system fail, a cabin crew may occupy the cockpit observer seat during takeoff, climb, approach, landing and whenever only one pilot in the cockpit remaining.

7.16.4. The captain shall allow access to the flight deck to essential ground personnel while the aircraft is on ground unless limited by a prevailing safety issue.

7.16.5. Cabin crew should monitor the flight deck area to detect suspicious activity of any person or potential threat from the cabin. In such events, cabin crew shall alert the cockpit crew by pressing the "ALERT" button and inform them of the situation. For hijacking refer to FOM 3.2 (7.1.4'd').

7.17 CREW RESOURCE MANAGEMENT (CRM) BASICS

All ETHIOPIAN flight operations are based on the optimum use of Crew Resource Management. The principle of continuous mutual briefing and assistance shall be applied at all times. In normal cockpit work the Captain shall establish open communication between crew members in the cockpit and in the cabin as well as with ground personnel and Air Traffic Services.

The three areas of performance skills that must be continuously improved are:

$$\left.\begin{array}{l}\text{Technical}\\[1em]\text{Procedural}\\[1em]\text{Interpersonal}\end{array}\right\}\ \text{Competence for optimum performance}$$

7.17.1 TECHNICAL COMPTENCE

a) Manual Flying Skill

I. Ability to

- Control the airplane manually at all times
- Stabilize the airplane at all times
- Maintain horizontal and vertical profile
- Operate the airplane accurately and smoothly

II. Execution

- Apply appropriate pitch and power values
- Coordinate control inputs and trim
- Recognize trends by instrument scan and react as appropriate
- Adhere to applicable limitations and tolerances according to the FCOM and/or FCTM.

b) Knowledge of Systems

a. **System Design**

- Know the equipment and function
- Know the limitations
- Be familiar with the documentation

b. **Application**

- Know how to operate systems
- Know the behavior and interaction of systems.

c) Use of Automation

II. Handling

- Be able to employ all modes of automation
- Use optimum mode of operation
- Use automation to reduce workload

III. Monitoring

- Be aware of active mode of automation
- Be aware of mode changes
- Be flexible in changing the level of automation

7.17.2 PROCEDURAL COMPETENCE

a) Knowledge of procedures

I. Normal procedures

- Know normal procedures for all phases of flight
- Be thoroughly familiar with all relevant standard operating procedures

II. Abnormal Procedures

- Know how to handle abnormal procedures
- Know memory items
- Be familiar with relevant abnormal procedures

III. Adherence to Procedures (Discipline)

- apply required published procedures appropriately
- Perform procedures carefully and accurately
- Deviate from published procedures only if safety is in jeopardy

7.17.3. INTERPERSONAL COMPETENCE

a) Threat & Error Management

For safe operations, in addition to the technical task of flying, crews must accomplish four safety tasks:

I. Use proactive strategies to avoid committing errors.

II. Manage operational complexity, which translates into threat management. Manage threats of flights, not only environmental factors such as terrain, weather, and equipment malfunctions but also errors external to the Flight crew. (example, by ATC)

IV. Manage crew errors. Error is an inevitable result of human limitations such as fatigue and other physiological factors, limited memory and processing capacity, external stressors, poor group dynamics, and cultural influences.

V. Manage aircraft deviations, as undesired aircraft states (example, wrong configurations, speed, heading, etc.).

b) Communication

Generally, communication in aviation is the transfer of knowledge and also involves an aspect of social interaction. Crew member enhance safety by exchanging, encouraging, understanding information and assuring reception. Suggestion should never be discounted, discouraged or ignored. When ambiguities and/or uncertainties arise, they should be clearly addressed in order to resolve any misunderstanding.

I. Atmosphere

- Encourage open and honest communication
- Achieve a positive first impression
- Listen actively
- Consider suggestions

II. Information Transfer

- Share information

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- Assure reception
- Assure understanding

III. Information Management

- Clearly state plans and intentions
- Announce ambiguities
- Announce uncertainties
- Speak frankly about problems within the crew

c) Leadership and Teamwork

The Captain is the designated leader. Lead by the Captain, the crew can expect to achieve a safe and efficient outcome in a climate that is rational and free of intimidation. Social interaction conflicts have to be addressed and managed. Each crew member takes the initiative to be an active and constructive part of the team.

I. Command Ability

- Take the lead as a Captain
- Establish goals, control outcome and correct as necessary
- Evaluate and consider conditions suggested by others.

II. Team Ability

- Act as a constructive member of the team
- Take initiative
- Encourage others to cooperate
- Support other team members
- Seek ideas and views from others
- Present own point of view
- Provide appropriate feedback
- Propose alternative ideas if appropriate

III. Conflict Management

- Address and manage conflicts
- Achieve rational climate

- Avoid intimidation
- Adopt assertive behavior if appropriate and persist until attention of others is gained or corrective action is taken
- Accept appropriate criticism
- Discourage competition between crew members

d) Workload Management

Stress and errors are inherent factors of flight. Crew should aim to minimize their negative effect.

I. Task

- Prioritize operational tasks
- Distribute tasks appropriately
- Complete tasks appropriately
- Complete tasks in good time
- Use external and internal resources

II. Time

- Plan ahead
- Allocate time to task appropriately

III. Stress and error

- Aim to minimize negative effects of stress
- Aim to minimize effects of error

e) Situational Awareness and Decision Making

To achieve favorable outcome crew members should actively monitor execution and development of the situation.

I. Preparation

- Act with respect to time available
- Avoid distractions

- Anticipate factors affecting the flight
- Recognize factors affecting the flight

II. Processing

- Evaluate factors affecting the flight
- Choose appropriate course of action
- Monitor execution
- Monitor development of the situation

III. **Apply FOR DEC for complex decisions**

F-facts, **D**-decision

O-Options, **E-**execution,

R-Risks & benefits, C**-**check

IV. Interaction

- Involve others in the process
- Discuss discrepancies

7.18. TASK SHARING IN THE COCKPIT

7.18.1 The task sharing in the cockpit requires a clearly defined assignment of tasks to PF and PM with the aim to guarantee that the full attention of PF is concentrated on the primary task of piloting the airplane. The captain determines the assignment of PF and PM at the start of each flight.

a) PF Task Sharing:

- Control of airplane
- Observance of SOPs
- Compliance with flight safety releases
- Altitude and speed restrictions
- Airspace observation

- Correct use of checklists.

b) PM Task Sharing:

- Monitoring flight progress
- Assistance and supervision of PF
- Airspace observation
- Monitoring airplane systems
- Operating airplane systems in accordance with PF
- R/T communication and correct use of checklists.
- Keeping the necessary flight records.
- Setting, identifying and checking navigational aids according to the instructions of PF.

7.18.2 Whenever other activities or special events prevent the PF from focusing their full attention to piloting, they shall hand over to the PM with the call out, "**you have contro**l', who confirms the takeover with the reply **'I have control'**.

7.18.3 Task assignments to PF and PM shall be strictly observed.

7.18.4 Whenever a task is performed by the PF, the PM must be consulted or advised as applicable. Unnecessary interventions and interruptions shall be avoided at all times.

7.18.5 Whenever the captain, with due consideration of all relevant circumstances, decides:

a) That any phase of flight (i.e. take-off, landing) may be critical, they shall assign themselves as PF.

b) That in any portion of the flight it is a safer course of action to take over control of the airplane they shall do so, even if the Co-pilot originally had been assigned as PF.

7.19. COMPANY SOPS AND CALL OUTS

All ETHIOPIAN aircraft shall be operated in accordance with SOPs and FCOM/FCTM Call Outs.

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7.20. COCKPIT CHECK PROCEDURES (ECARAS 9.3.1.11)

7.20.1 Ethiopian shall issue to the flight crews and make available on each aircraft the checklist procedures approved by the Authority appropriate to the type and variant of aircraft.

7.20.2 Ethiopian shall ensure that approved procedures include each item necessary for flight crew members to check for safety before starting engines, taking off or landing and for engine and systems abnormalities and emergencies.

7.20.3. Ethiopian shall ensure that the checklist procedures are designed so that a flight crew member will not need to rely upon his memory for items to be checked.

7.20.4. Ethiopian shall make the approved procedures readily useable in the cockpit of each aircraft and the flight crew shall be required to follow them when operating the aircraft.

7.20.5. The use of checklists by the flight crew prior to, during and after all phases of flight operations and in abnormal and emergency situations is mandatory, to ensure compliance with:

a) Procedures contained in the OM (FOM, FCOM, SOP).
b) Provisions of the aircraft certificate of airworthiness.

7.20.6. No flight operation personnel shall deliberately violate flight operations policies and procedures.

7.21. CROSS CHECK OF CRITICAL ACTIONS BY THE FLIGHT CREW

7.21.1 All items that require cross check as per SOP’s shall be conducted as per FCOM.

7.21.2 In addition flight crew members should crosscheck and confirm critical actions during normal, abnormal and emergency situations, to include:

a) aircraft configuration changes (landing gear, wing flaps, speed brakes);
b) altimeter bug and airspeed bug settings, as applicable;
c) altimeter subscale settings;
d) altitude (window) selections;
e) transfer of control of the aircraft;
f) changes to the Auto Flight System (AFS)/Flight Management System (FMS) and radio navigation aids during the departure and or approach phases of flight;
g) Performance calculations or inputs, including AFS/FMS entries.
h) Weight/mass and balance calculations and associated AFS/FMS entries.

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7.22. LEVEL BUST

7.22.1. DESCRIPTION

a) A level bust is defined as any unauthorized vertical deviation of more than 300 feet from an ATC flight clearance. Within RVSM airspace this limit is reduced to 200 feet. Definitions applied by other organizations are similar but sometimes refer to a deviation of 300 feet or more. The level bust issue only relates to aircraft in controlled airspace or a designated ATZ outside controlled airspace and under either radar or procedural ATC control.

b) A Level Bust or Altitude Deviation occurs when an aircraft fails to fly at the level to which it has been cleared, regardless of whether actual loss of separation from other aircraft or the ground results.

c) A Level Bust can result in loss of separation between aircraft or between an aircraft and the terrain or a ground obstruction such as a mast (CFIT).

d) Level busts are becoming less dangerous because improvements in technology such as better TCAS and Mode S have improved the ability of controllers to safely manage any consequent loss of separation. Furthermore, the availability and proper use of ACAS provide a final safety net which significantly reduces the risk of a Mid-Air Collision, and Terrain Awareness and Warning System (TAWS) has also reduced the risk of a level bust resulting in a CFIT accident.

e) A potential loss of separation resulting from the ATC assigning an inappropriate altitude or flight level in a flight clearance does not constitute a level bust because no deviation from the flight clearance occurs; however, for completeness, examples of situations in which the action of the ATC could contribute to a level bust are listed in this article.

f) For complete information please refer the LEVEL BUST TOOL KIT.

7.22.2. TYPES OF LEVEL BUST

The following types exclude involuntary transient departure from acquired levels attributable to the effects of turbulence:

a) Flight crew accept and record a clearance correctly but don't follow the clearance (flight management error or technical fault)

b) Flight crew accept clearance correctly but sets it incorrectly without the error being picked up by the crew [flight management error]

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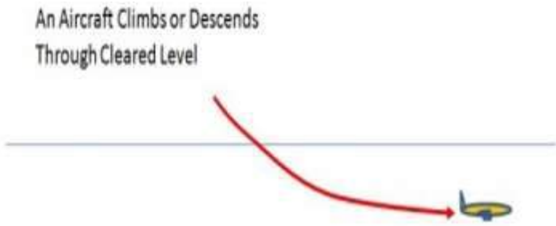
c) Flight crew reads back clearance incorrectly and this error is not picked up by ATC, so it is then recorded/set and followed [ATC error]

- Aircraft is unable to react fast enough to a late re-clearance and passes through new cleared level [ATC error].



- Aircraft follows clearance with the wrong altimeter subscale setting. [flight management error]

- aircraft departs a cleared flight level without clearance to do so [flight management error]

7.22.3 EFFECTS

a) Loss of Separation from other aircraft, which may result in collision.

b) Collision with an obstacle or the ground Controlled Flight into Terrain, especially as a result of having the wrong altimeter sub scale setting. This can happen when an aircraft descends in a low-pressure area with standard pressure (1013) set.

c) Injury, especially to cabin crew or passengers, occasioned by violent maneuvers to avoid collision with other aircraft or the ground.

7.22.4. DEFENSES

a) ALTITUDE AWARENESS AND ALTIMETRY

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In order to manage or mitigate potential risks related to the acceptance and maintenance of assigned altitudes. And to increase situational awareness and vigilance flight crew shall adhere to the following policy, guidance and procedures.

I. Automated or verbal altitude call outs must be acknowledged by all flight crew and there must be dual confirmation of the altitude to be maintained.

II. Flight crew must at all times be vigilant on altitude deviations in both auto flight and manual flights and correction must be made immediately. Significant deviations must be notified to ATC as soon as possible.

III. In order to avoid call sign confusion when receiving altitude clearance flight crew must at all times read back with proper call sign identification. When in doubt or disagreement confirmation must be sought from ATC.

IV. During first contact with ATC, while in flight, flight crew must report the cleared flight level unless specifically requested not to do so by ATC.

EXAMPLE: "Bole Tower Good morning ETH 831 climbing FL150.

V. Flight deck routines on multi-crew aircraft should include rigorous procedures for cross-checking/confirming cleared altitude, read back, and altitude set on FMS by the monitoring crew members, and includes callouts, such as "1000 feet to go" and "approaching level".

VI. Onboard aircraft equipment designed to warn of potential collision with other aircraft ACAS/TCAS or with the ground EGPWS.

VII. Ground-based equipment designed to warn of potential collision with other aircraft, such as Short term conflict alert (STCA), or provide more information on intentions of the aircraft. (e.g. Mode S)

VIII. Standardhttp://www.skybrary.aero/index.php/SOPs Operating Procedure, both on the flight deck and in the ATC unit, which details procedures to be followed to reduce the chance of level bust.

7.22.5. TYPICAL SENARIOS

a) Air- Ground Communications, for example:

I. The pilot miss-hears the level clearance, the pilot does not read back the clearance, and the ATC does not challenge the absence of a read back;

II. The pilot reads back an incorrect level, but the ATC does not hear the erroneous read-back and does not correct the pilot's read back or,

III. The pilot accepts a level clearance intended for another aircraft (call-sign confusion).

b) Pilot Induced Situation, for example:

I. One pilot is off frequency when clearance is given by ATC; although it is read back correctly, he/she miss-sets it on the FMS.

II. PM fails to spot that PF has set FL110 on the FMS not the FL100 that PM has (correctly) read back to ATC.

III. Flight crew used to a 6000ft Transition Altitude/Level set up a SID for a departure from an airport which has a 3000ft Transition Altitude and then fail, with a low barometric pressure, to change to standard pressure setting (1013) quickly enough.

IV. The flight crew read back what they think is correct believing that ATC will correct them if the readback is wrong, however ATC doesn't notice and it is not corrected. (ATC hear-back error but flight crew should have asked for the clearance again not guessed it)

V. A military fast jet without ACAS but with SSR Mode C/S exceeds its correctly acknowledged climb clearance and very rapidly causes multiple civil transport RA sequences at three different successive flight levels, because of the fast-jet's rate of climb, all consist of RAs which change too quickly to be capable of being actioned.

VI. Pilot understands and reads back the correct altitude or flight level but selects an incorrect altitude or flight level because of confusion of numbers with another element of the message (e.g. speed, heading or flight number), [proper cross-checking procedures on a multi-crew flight deck should mitigate against these]

c) Aircraft equipment designed to prevent a Level Bust (e.g. autopilot or altitude alert) does not operate as designed.

d) ATC-Induced Situations, for example:

I. ATC is interrupted or distracted and does not take action in time to rectify an impending loss of separation.

II. ATC assigns an incorrect altitude or reassigns a FL after the aircraft has been cleared to an altitude.

III. ATC's instructions are misunderstood because of inadequate English Proficiency, or use of standard phraseology, or speed of transmission;

IV. The altitude clearance is passed late, and re-clearance is not achievable without overshoot or undershoot.

V. The ATC issues an instruction for an altitude restriction when the aircraft is above the transition altitude (i.e., with altimeters set to standard pressure setting);

VI. The ATC issues a complex transmission containing more than two instructions (e.g., speed, altitude and heading).

e) Pilot-equipment interface.

The flight crew sets an incorrect altimeter pressure setting or mis-sets other aircraft technical equipment or fails to respond correctly to aural or visual warnings.

7.22.6. CONTRIBUTORY FACTORS

a) Factors which contribute to the risk of a level bust occurring:

I. Only one pilot/crew member on the ATC frequency;
II. ATC work load;
III. Interruption/distraction;
IV. Holding Patterns;
V. Airspace Procedure/Design;
VI. Pilot work-load;

b) Factors which increase the risk of collision following a level bust:

I. Volume of traffic
II. High rate of climb or descent

7.22.7. SOLUTIONS

a) effective Air-ground communications especially in terms of radio discipline and the particular importance of cross-checking and the read-back/hear-back process exercised by both pilots and ATC.
b) For flight crew, the overall principles of effective Crew Resource Management are particularly relevant as is the rigorous application of SOP related to altimeter subscale setting and to the receipt and pre-selection of altitude re-clearances. For ATCs, the principles of team resource management are important.
c) Mitigating the consequences of Level Busts is achieved by maximizing the effective use of both flight crew and Air Traffic Management (ATM) Safety Net.
d) For Flight crew, Situational Awareness (SA) with respect to other traffic in the vicinity can be enhanced by passive monitoring of other aircraft on the assigned frequency in use and by passively monitoring the TCAS display. If a corrective TCAS RA occurs then, it is essential that it is actioned as required, that ATC are advised of any deviation from clearance as soon as practicable and that any deviation is limited to the requirements of the RA.

7.23. RUNWAY/TAXIWAY INCURSION/EXCURSION AVOIDANCE

7.23.1 GENERAL

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a) The following recommendations and practices are ETHIOPIAN standard safety precaution that all flight crew are expected to take while accomplishing the task of moving an aircraft for the purpose of takeoff and bringing it to full stop at the gate to disembark passengers and/or cargo.

b) Flight crew should use a separate set of taxi charts to increase situational awareness of the area they are operating in and adhere to all clearances and cross-check all actions. While doing so, effects leading to errors like distraction, haste, fatigue and expectation bias should be taken into consideration.

c) Flight crew shall be vigilant in all areas of vulnerabilities and maintain situational awareness while operating in the airport environment, on the ground and in the air to ensure an awareness of the aircraft position relative to the airport surface and should also plan ahead in managing cockpit workload levels and prioritizing tasks.

d) Tasks that cause distraction (for example, engine starts while taxiing, FMS programming, Airborne Communication Addressing and Reporting System (ACARS) and company radio calls) should be delayed in active runways, crossing runways, in runways, or runway incursion hot spots.

e) Expect that early automatic terminal information service (ATIS) and arrival and/or departure information may be updated or changed. Properly pre-brief well in advance, expect and prepare for changes (load alternate arrival procedure, if possible).

f) When there is a high risk of an incursion the Flight Crew members has to be more vigilant and use all available resources such as heading indicators, airport diagrams, airport signs, markings, lighting, ATC and identify potential incursion areas or hot spots on airport diagrams and include in the pre-departure & arrival briefing to keep an A/C on its assigned flight and/or taxi route.

NOTE: Airport surfaces that could pose a higher risk of an incursion are depicted on 10-9x & 10-1Px charts as Hot Spot areas.

g) To avoid incursion and increase situational awareness the Flight Crew should;

I) Monitor clearances given to other aircraft.
II) Obtaining directions or progressive taxi instructions when taxi route in doubt.
III) Verify and cross check takeoff and landing runway.
IV) Verify takeoff and landing clearance.
V) Use aircraft lighting, heading indicators, Airport moving map if available and all items listed on ‘f’ above during taxi, runway crossing, takeoff and landing.
VI) Appropriate transponder uses at airports with ground surveillance radar.
VII) Appropriate use of TCAS when on the runway and holding in the takeoff position.

h) During reduced visibility operation do not exceed 10kts while taxiing and the First

Officer should make full use of taxi charts and call out ground speeds when indicating more than 10Kts to avoid the risk of possible incursion. Remember that in poor visibility, ground equipment, aircraft wingtips and tails may not be as readily seen as the taxiway lighting.

i) Use of landing lights at night in low visibility can be detrimental to the acquisition of visual references. Reflected lights from water droplets or snow may actually reduce visibility. Landing lights would therefore not normally be used in Category III weather conditions.

j) First Officers are not allowed to taxi an Aircraft under any circumstances.

7.23.2. TAXI OUT

During taxi out to improve situational awareness:

a) The First Officer shall monitor the progress of taxis.

b) Flight crew should be heads up and confirm all clearances to cross or hold short of any runway.

c) Clearance to follow another aircraft does not include a clearance to cross any runway.

d) Taxi speed should depend on existing conditions but should be a maximum of 30 knots on straight taxiways.

e) If departure runway is changed, flight crew should visually verify the correct FMS take off runway entry as part of the taxi and takeoff clearance process.

f) Flight crew should be aware that hold-short lines can be as far as 260 feet laterally from a runway. Because of Precision Obstacle Free Zone (POFZ) or when taxiway is not perpendicular to the runway, the hold-short line may be reached sooner than expected.

g) Do not accept line up and hold clearances unless the aircraft is ready for departure.

h) Before crossing any runway while taxiing to the assigned runway:

   I. Both pilots should scan the full length of the crossing runway for potential conflicts and report "clear left "and/or "clear right" as appropriate. The PM verifies the clearance and states "clear to cross". The PF acknowledges "clear to cross".

   II. Turn all exterior lights on (leave the landing lights off if they will adversely affect other pilots' vision) Upon clearing the runway, you may return to the previous lights configuration.

i) Upon entering and lining up the runway the PF should call out "Cleared on runway_ _ _, heading _ _ _ checked".

j) During line up and hold:

I. Align the aircraft off centerline during low visibility and night operations to contrast with runway lighting and markings.

II. Monitor traffic alert and collision avoidance system (TCAS) display for traffic approaching the runway.

III. Contact ATC if holding in line up position for more than 90 seconds or upon determination that potential conflict exists.

k) When clear for takeoff turn on landing lights and appropriate exterior lights.

7.23.3. LANDING ROLL

When the first officer (FO) makes the landing as the PF (recognize possible threat, which can include role reversal conflicts with normal cockpit duty orientation)

a) The flight crew should adhere to an established formal transfer of control procedure.

I. Ensure the PF and PM verbalize a positive transfer of control.

II. The captain announces, "**I have Control."** The FO replies, "**you have Control."**

III. There is no minimum speed for change of control on the runway. Directional control and braking effectiveness under existing meteorological conditions and aircraft configuration are primary concerns.

b) The FO should maintain runway centerline and should not attempt to steer the aircraft toward the turnoff taxiway or high-speed taxiway under normal conditions unless the captain so directs (based on a perceived need for consistent crosswind control in deceleration, or another significant concern).

c) The flight crew should refrain from acknowledging and accepting a clearance during high workload phases of the landing rollout.

d) The PM should not acknowledge ATC radio transmissions until they reach taxi speed and should not contact the company until all runways have been crossed and cleared.

e) Do not stop on the active runway after landing to regain orientation.

f) The captain should delay directing after landing flows and engine shutdowns when between two active runways unless stopped, with the parking brake set.

g) The PF should not exceed normal breaking to comply with and ATC requests for an early or expedited turn-off.

h) Flight crews should;

I. Use defensive fatigue management practices whenever circumstances can produce cumulative fatigue.

II. Assign utmost priority to verifying and adhering to a taxi clearance over nonessential communications and clean-up flows.

III. Use threat and error management to avoid fatigue-induced haste, which often results in critical omissions or inappropriate decisions.

7.23.4. TAXI IN

a) During taxiin to improve situational awareness the FO shall monitor the progress of taxis.

b) Flight crew should be heads up and confirm all clearances to cross or hold short of any runway.

c) Clearance to follow another aircraft does not include a clearance to cross any runway.

d) It is expected that crew will have briefed the taxi plan and/or taxi, and runway exit strategy in the top of descent briefing, emphasizing any runways that the aircraft is expected to cross after clearing the landing runway.

e) Flight crew should anticipate changes to previously received information either through Automatic Terminal Information Service (ATIS) or arrival /departure information. Thoroughly brief, well in advance with all the expected alternatives.

f) Taxi speed should depend on existing conditions but should be a maximum of 30 knots on straight taxiways.

g) After landing, do not change from tower frequency to ground control until directed to do so.

h) Before crossing the runway en route to an assigned gate:

I. Both pilots should scan the full length of the crossing runway for potential conflicts and report "**clear left "and/**or **"clear right"** as appropriate. The PM verifies the clearance and states "**clear to cross**". The PF acknowledges "**clear to cross**".

II. Turn all exterior lights on (leave the landing lights off if they will adversely affect other pilots' vision) Upon clearing the runway, you may return to the previous lights configuration.

III. Ensure the aircraft is completely clear of the runway (including the aircraft tail) immediately after landing.

7.24. LOGIPAD or COMPATIBLE OWN DEVICE (COD, AIRCRAFT EFB (CLASS II/III) AND INTRANET

7.24.1. GENERAL

SECTION 2.5 NORMAL OPERATIONS

a) In order to provide all personnel with the relevant information in the most convenient way and in a timely manner, Ethiopian preferably utilizes electronic data processing via e-mail, aircraft EFB (class II/III) and EFB (class I) (any compatible device) for a safe and efficient conduct of each flight.

b) All Flight Crew must ensure that they are aware of relevant information for the flight(s) by synchronizing (updating) their personal class I EFB before departure in ADD. Each Flight Crew has to carry their class I EFB on board the A/C.

c) To avoid gross error both pilots must perform calculations independently on their class I EFB or in their respective class II/III EFBs and compare the results.

d) For those aircraft equipped with class II/III EFB's incase of one EFB unserviceable or performance function unserviceable, the pilot shall use his class I EFB for performance calculation and for all other documents, and one Jespersen Airway Manual flight bag shall be used on the failed side of the EFB.

NOTE 1: Out of home base for A-350 A/C if one or two EFB fails Documentation shall provide a replacement for the failed EFB if available, otherwise the above on 'd' will be applicable.

NOTE 2: For en-route or from out station the pilot shall use his class I EFB. For the case of A-350 A/C, the class I EFB must be mounted on the designated place and shall be secured with the secure strap. For B-777/787 A/C terminal charts will be provided by the station manager to the operating crew if both Class III EFBs failed at outstation.

e) When departing with both EFB's (class II/III) unserviceable or the OPT/FLY SMART function in both EFB's unserviceable the class I EFB shall be used & the above 'd' will be applicable as well.

f) For those aircraft not equipped with class II/III EFB's, the flying crew shall use their class I EFB for performance calculation and other documents. In addition, one Jespersen Airway Manual flight bag shall be placed on both sides. If one of class I EFB is not operating, the aircraft can depart with one device. However, the calculation must be done by the PIC and the FO on the remaining class I EFB by resetting the data after the first entry. If both devices are not operating, dispatch office shall make performance calculations and send them to the flight. Although it is a remote possibility, the flight crew shall use all available aircraft documents/manuals to calculate the most conservative performance data in the event there is a complete loss of the performance function.

g) Guidance for the use of OPT/FLY SMART is available on the class I EFB. Additional operational instructions or changes to Operating Manuals will be given as FOB (Flight Operations Bulletin). How to use EFB is included on transition training.

h) All manuals (except that are placed in the A/C library in hard copy) are available in the EFBs.

i) For those aircraft not equipped with class II/III EFB's, two Jeppesen Airway manual flight bag is a standard requirement. It is the flight crews' responsibility to carry current company supplied or approved Class 1 EFB

j) The touch sensitive screen of the A/C EFB is meant to be used with the fingers only. Sharp objects such as pens can easily damage this surface. (Same consideration should be given to all touch sensitive surfaces such as those found on the ECL Cursor Control Device and other displays located throughout the aircraft).

NOTE: Crews must avoid fixation on the display or distraction from primary crew duties while using any EFB application.

7.24.2. AIRPORT MAP DISPLAY

a) The ND airport map display helps to enhance crew positional awareness when planning taxi routes and while taxiing. The system is not intended to replace normal taxi methods including the use of direct visual observation of the taxiways, runways, airport signs and markings and other airport traffic signals. Crew should always confirm the airport moving map database to be current. Prior to taxi, NOTAMS and airport charts (using EFB terminal charts or paper) should be consulted for the latest airport status to include closed taxiways, runways, construction, etc., since these temporary conditions are not shown on the airport map.

NOTE: Crews must avoid fixation on the display or distraction from primary crew duties while using the airport ND map display.

b) Use the ND airport map mode to provide enhanced positional awareness by:

I. Verifying taxi clearance and assisting in determining taxi plan. (both pilots)

II. Monitoring taxi progress and direction. (both pilots)

III. Alerting and updating the PIC with present position and upcoming turns and required stops.

c) The FO should be prepared to alert and update the PIC on present and next positions while taxiing at all times.

d) In flight, the ND airport plan mode may be used to aid in runway exit planning and anticipating the taxi route to the gate or parking spot.

SECTION 2.5 NORMAL OPERATIONS

NOTE: GPS position must be available to use the ND Airport Map.

e) MEL guide line shall be followed for action during a failed EFB or ND Map failure.

7.24.3. TERMINAL AND EN-ROUTE CHARTS

Electronic terminal charts may be used in place of paper charts. En-route charts are also available in the EFB. Crew should always confirm the terminal & En-route charts database to be current. Should the airplane be dispatched with one or both EFBs inoperative, the crew shall comply with the provisions of the MEL regarding the use of backup charts.

7.24.4. AIRCRAFT PERFORMANCE

a) An Onboard Performance Tool (OPT) or FLY SMART is incorporated into the EFB.

b) When all appropriate entries are made, the aircraft performance application provides runway specific performance information equivalent to AFM data or airline airport analysis. During approach preparation, the system can provide advisory landing distance information.

c) Both Captain and First officer are required to perform independent Take-off/Landing performance calculations and crosscheck the calculated data. Any discrepancy shall be resolved prior to Take-off/Landing.

NOTE: The flight crew has to check their NOTAMS prior to every departure & has to check serviceability of taxiways or runways (displaced threshold) and other facilities that affect the performance of the aircraft.

7.25. AIRPORT ANALYSIS/OPT/FLY SMART

7.25.1 Airport (takeoff) analysis charts shall only be issued by Mgr. Flight Operations Performance & Technical Support department as stated in FOPs Performance & Technical Support Working Manual 5.1. These charts shall provide take off performance calculations for those operations that are not provided by the onboard performance tool.

7.25.2 AIRPORT ANALYSIS/OPT/FLY SMART aerodrome specific performance data shall be based on periodic ICAO, or Jeppesen airport and obstacle database.

7.25.3 Mgr. Flight Operations Performance & Technical Support department shall frequently coordinate with Jeppesen OPSDATA for the update of AIRPORT ANALYSIS/OPT/FLY SMART to include short term changes in temporary runway restrictions and obstacles that could potentially affect the AIRPORT ANALYSIS/OPT/FLY SMART that are incorporated.

7.25.4 The AIRPORT ANALYSIS/OPT/FLYSMART shall include all departure, destination, and alternate airports taking into consideration limitation of:

a) Takeoff performance (accelerate-stop, close-in obstacles) that also ensures charting accuracy is accounted for, when necessary, in assessing takeoff performance in the event of a critical power unit failing at any point in the takeoff;

b) Maximum Brake Energy;

c) Field length;

d) Tire speed;

e) Climb performance (distant obstacles);

7.25.5. AIRPORT ANALYSIS/OPT/FLY SMART calculations shall consider all significant factors during all phases of flight including:

a) Aircraft weight (mass);

b) Operating procedures;

c) Pressure altitude;

d) Temperature;

e) Wind;

f) Runway gradient;

g) Runway contamination/braking action up on request by the flight crew otherwise flight crews shall apply necessary performance decrement from manufacturer QRH operations manual;

h) Obstacle data;

i) NOTAMS (including airport NOTAMS);

j) MEL/CDL information (If applicable);

k) Aircraft configuration/operation. Includes use of different flap setting, anti-ice usage and, when applicable, ice accretion.

l) Runway length used for aircraft alignment prior to takeoff, if applicable.

7.26. PERFORMANCE DETERMINATION

7.26.1. No person may operate an aircraft unless he determines or computes aircraft performance for all phases of flight considering environmental conditions and runway data information (e.g. runway length, airport elevation, slope, PCN, etc.).

7.26.2. Maximum structural weights (taxi, takeoff, landing, and Zero fuel) stipulated in respective aircraft AFM shall never be exceeded.

7.26.3. Take off performance is determined using the Airport analysis, the Onboard Performance Tool

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(OPT), FLY SMART or the AFM. Whenever practical the maximum assumed temperature method/flex thrust should be used for Takeoff.

7.26.4. Flight crew are required to assess Landing performance prior to arrival at destination or alternate airport to determine that sufficient landing distance with adequate safety margin exists on the run way of intended use in the conditions existing at the estimated time of arrival (ETA) in the aircraft configuration and with the means of deceleration that will be used for the landing.

7.26.5. Flight crew shall ensure that an approach is not continued below 1000 feet above airport elevation unless the PIC is satisfied that with the runway surface condition information available for the runway of intended use at the time of arrival with the A/C configuration and with the means of deceleration that will be used for the landing.

7.26.6. Below 1000 feet if the surface condition of the landing runway at the time of arrival is degraded from the anticipated or the calculated performance surface condition (the report can be from ATC or PIREP) the flight crew shall make a go-around, climb to a minimum safe altitude, calculate the performance again with the degraded surface condition, crosscheck the result, and if the performance shows that sufficient landing distance with adequate safety margin exists, the Flight can continue its approach for landing. If the report shows the same or improved condition the flight crew can complete their landing.

7.26.7. The PIC shall report the runway braking action special air-report (AIREP) when the runway braking action encountered is not as good as reported.

7.26.5 Flight crew are required to calculate go around performance with intended aircraft landing configuration.

7.26.6 Minimum cooling time limited weights as calculated from respective aircraft operations QRH and FPPM manuals or the On Board Performance Tool (OPT) or limitation for the required temperature shall never be exceeded.

NOTE: In the case of AIRBUS the temperature shall not exceed 300 degrees centigrade for takeoff and brake cooling time is included in the FCOM Hot brake non normal procedure.

7.26.7. All factors that significantly affect the performance of the aircraft such as, MEL/CDL limitations, condition of the runway i.e. presence of slush, standing water, snow and or ice, pressure correction (QNH) etc… runway cutback or reduced runway available, aircraft configuration, environmental conditions at the ETA (crosswind, tail wind, gust, rain, etc…), the use of (reverse thrust, manual or auto brake, speed brakes), non-normal or emergency events that may affect landing distance, any event or contingency that degrades must be considered when calculating the takeoff and landing performance. Stopping ability or increases landing distance or takeoff under the conditions present at the ETA.

NOTE: Runway braking action or the Canadian Runway Friction Index (CRFI) where applicable must be used when reported by ATC.

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7.26.8. For extended operations in very cold areas fuel freeze considerations shall be performed to prevent in-flight freezing of fuel. The considerations shall include fuel temperature monitoring for the specific fuels used in the operations.

7.26.9. Manufacturer manual and/or Flight planning system shall include information about en-route engine out performance data such as aircraft service ceiling and drift down altitude that assures terrain clearance.

7.27. PERFORMANCE PLANNING MANUAL (ECARAS 9.3.1.13)

For the use of the flight crew members and persons assigned on operational control functions during the performance of their duties, a performance planning manual acceptable to ECAA shall be provided. The performance planning manual shall be specific to the aircraft type and variant and shall contain adequate performance information to accurately calculate performance in all normal phases of flight operations.

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SECTION 2.6 OPERATIONAL IRREGULARITIES

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2.6. OPERATIONAL IRREGULARITIES

1.0 PURPOSE

The purpose of this operational irregularities procedure is to establish guidelines to deal with operations under irregularities that result in early departure, delay, consolidation of two or more flights, over flights, rerouting and cancellation of flights.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Director IOCC
- Flight Crewmembers
- Station Managers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. The following are defined as irregularities arising from the operation of a flight schedule. They may require the predisposition of the load or crew and/or the rescheduling of the aircraft on the route affected subsequent to the originally scheduled flight. Irregularities include, but are not limited to the following:

a) Early departure
b) Delay
c) Consolidation of two or more flights
d) Over flights
e) Rerouting a flight
f) Cancellation of a flight

4.1.1. The primary responsibility for all actions related to operational irregularities rests with the IOCC. In some special cases, such as at remote locations or in an emergency condition, the Captain may assume this authority and responsibility. In all cases, coordination with the various concerned company departments is absolutely essential and is to be accomplished as expeditiously as circumstances permit. Should there be any objections to the decisions of IOCC; the concerned department shall communicate such objections, together with an alternative to IOCC for consideration and appropriate action.

4.1.2 If authorization is required due to any irregularities the PIC can only be authorized through the proper channel from Flight Operations management (i.e. VP Flight Operations, Flight Operations Directors or respective chief pilots)

4.1.3. All pre-dispositions, decisions or actions made regarding operational irregularities should be based on the objective of ensuring that:

a) All passengers concerned will reach their final destination as expeditiously as possible.
b) All mail and cargo is handled expeditiously to ensure minimum delay to its destination.
c) Alternative actions are executed with minimum revenue loss and minimum additional cost to the company consistent with general passenger and cargo policies.

4.2. ADHERENCE TO SCHEDULE

Late arrival of services involves inconvenience to our customers, may temporarily disorganize ground handling, and may lead to loss of revenue. To reduce the possibility of late arrivals, early departures are encouraged subject to the conditions laid down in paragraph 4.3.2 below.

4.3. EARLY DEPARTURE OF FLIGHT

4.3.1 In general, company policy requires all flights to depart at published schedule departure time, however it is recognized that under certain special circumstances, early departure may be desirable.

4.3.2 An early departure of a flight is defined as any company flight that departs ahead of the published flight schedule departure time. Issues to be considered when considering an early departure include but are not limited to;

a) Avoid Meteorological conditions or a forecast which might endanger the aircraft or cause prolonged delay.
b) All passengers and other load have to be accommodated.
c) When all booked load is on hand, there is negligible prospect of further load, and destination stations will not be faced with a congestion of services.
d) Passengers will not be inconvenienced by lack of public transport or amenities during night or early morning hours, or by disruption of their arrangements.
e) Provide additional transit time at the next station to accommodate a critical loading or unloading requirement.
f) Minimize expected delays at stations with connecting flights.
g) Eliminate or minimize delays in arrival at the next station due to anticipated re- routing,

strong headwinds en-route, differences in-flight time caused by equipment substitution, or ATC delays after the start of taxi.

4.3.3. After coordination and agreement between the Captain and the local Station Manager, an early departure of up to 15 minutes prior to the published departure time is authorized without prior consent of Flight Control. Early departures in excess of 15 minutes should be preplanned and have the approval of Flight Control and all other interested company departments. However, in an extreme situation when the Captain deems it necessary to depart more than 15 minutes ahead of schedule he may authorize such a departure without prior approval from Flight Control, but upon his return to the home base he shall be required to justify his decision in writing.

4.3.4. In all cases, due consideration shall be given to the commercial aspects of the situation and potential loss of revenue shall be kept to a minimum.

4.3.5. An all-cargo flight may depart up to 30 minutes ahead of the published timetable if it has been determined that all reserved cargo and mail have been accommodated.

4.3.6. Commercial and charters should adhere to published departure times because of airport movement slots and handling arrangements.

4.3.7. For any deviation from the above authorization to depart early must be obtained in advance from Dir. Flying Operations or Dir. IOCC unless an emergency situation precludes the obtaining of such authorization.

5.0 DEFINITION

5.1 "Senior Official at the Airport" means the senior Ethiopian or Handling Agent's Official on duty at the time or Operational Planner.

6.0 RESPONSIBILITY

6.1. The primary authority and responsibility for all actions related to operational irregularities rests with the IOCC.

6.2. In the case of unforeseen delays, it is the responsibility of the appropriate department, Station managers or flight crew concerned to minimize the delay as much as possible.

7.0 PROCEDURE

7.1.1. DELAY OF FLIGHT

7.1.1 A Delay occurs when the schedule departure time is exceeded by more than 5 minutes. In general, the company's policy is that all company flights will depart at the published scheduled departure time.

SECTION 2.6 OPERATIONAL IRREGULARITIES

7.1.2 Company flights may be delayed intentionally to:

a) Avoid adverse meteorological conditions at the station of departure, en-route or at the station of destination.

b) Ensure connections for passengers and cargo from the fights operating behind schedule.

c) Take on board "aircraft on ground" (AOG) spare parts for company aircraft grounded en-route due to technical problems.

7.1.3 In the case of adverse meteorological conditions, the Captain is authorized to delay the departure of the flight without prior approval from Flight Control. However, whenever possible Flight Control should be informed of this decision so that they can coordinate this delay with the other company departments and give the Captain any assistance he may require in order to minimize the delay.

7.1.4 In all other cases, the local Station Manager is authorized to delay the flight up to 15 minutes beyond the scheduled departure time without prior approval from Flight Control. Such delays are only authorized in emergencies or certain extreme situations and will require written justification as soon as possible after the incident.

7.1.5 In case of late arrival or late positioning of the aircraft at a station, every effort will be made by all concerned to minimize the required ground or transit time, thereby reducing or eliminating the delay.

7.1.6 All delays (departures in excess of five minutes past scheduled departure time) shall be reported by both the Station manager and the PIC through delay report form or Email, for tabulation and possible further investigation.

7.2 REGAINING SCHEDULE

When services are operating behind schedule, every effort will be made to reduce ground transit time. The station manager at the airport will establish the new departure time with the PIC.

7.3. OVERFLIGHTS

An over flight is the intentional deletion of one or more scheduled intermediate stops due to operational, commercial or other reasons. All flights shall normally be operated in accordance with the published schedule, including scheduled stops. However, over flights may be arranged prior departure or it may be decided in flight.

7.3.1. OVERFLIGHTS ARRANGED PRIOR TO DEPARTURE

a) Generally, the authority for arranging an over fight prior to departure will reside with Flight Control and must always be properly coordinated with the other affected company departments. Maximum advance notice shall be given so that appropriate actions can be taken.

b) Over flight is authorized if due to: -

I. Operational conditions at next station are prohibitive or render a landing improbable (e.g. meteorological conditions, runway obstruction, etc.;

II. No revenue to and from next station (subject to approval of coordination);

III. Necessity to shorten delay on critical flights (subject to approval of coordination);

IV. Weather conditions;

V. If an early transit through the next station would be permissible (subject to approval of coordination);

VI. The effect on flight time limitations (subject to approval of coordination);

VII. Local connection for passengers carried beyond their destination and returning to the station over-flown (subject to approval of coordination);

7.3.2. OVERFLIGHTS ARRANGED DURING FLIGHT

a) If a decision to overfly a scheduled stop is made in flight, the Captain shall immediately pass this information to the station to be over flown, and to other stations as appropriate.

b) The final authority to arrange an over fight while in flight shall always reside with the PIC. Whenever possible, coordination with IOCC and the station(s) concerned shall be advised of the decision to overfly a particular station as soon as practicable after such a decision has been made.

c) Over flights arranged during flights are authorized if due to:

I. Operational conditions at next station (weather condition, runway obstruction etc.).

II. Technical defects detected in flight which cannot be repaired at next scheduled stop without excessive delay and which make it desirable to proceed to a station with better maintenance facilities.

d) If a scheduled stop is over flown for operational reasons or technical defects (i.e. the service is diverted owing to weather etc.) the Captain, through the senior official at the diversion airport shall decide with the senior official at the next scheduled stop, the best course of action to pursue, taking into consideration the following factors:

I) The practicability of returning to the scheduled stop when conditions permit;

II) Whether the load on board or to be picked up, justifies a return;

III) The alternative facilities available (i.e. surface or air transport) for positioning any joining load to the airport;

IV) Collecting disembarking load;

V) The effect on the round trip of the aircraft if the service waits a division airport to embark load or return to the scheduled stop.

SECTION 2.6 OPERATIONAL IRREGULARITIES

7.4. CONSOLIDATION OF FLIGHTS

7.4.1. A consolidation of flights as referred to herein is a combination of two or more flights either in whole or in certain segments of their routes. A consolidated flight shall be operated by only one aircraft according to the schedule of one of the combined flights.

7.4.2. Flights may be consolidated because of lack of equipment, lack of revenue, or for other operational requirements.

7.4.3. In general, consolidation is permissible only if the booked revenue load of the flights affected can be protected by the consolidated flight without rerouting. Otherwise, all affected flights shall be canceled, and a new flight with appropriate routing and schedule must be originated.

7.4.4. The authority for arranging the consolidation of two or more flights will always reside with IOCC and must always be properly coordinated with the affected company departments.

7.5. REROUTING OF FLIGHTS

7.5.1. Rerouting of a flight may include the addition of a stop or stops to regularly scheduled intermediate stops or a substitution of the final destination.

7.5.2. Flights may be rerouted for commercial or operational reasons.

7.5.3. The authority for the rerouting of a flight rests solely with IOCC but must be coordinated with the other affected company departments.

7.6. DIVERSION OF FLIGHTS EN-ROUTE

7.6.1. An en-route flight diversion occurs when the flight proceeds to a station other than its next planned destination, although no previous rerouting instructions have been given.

7.6.2. Insofar as possible, all company flights shall be operated in accordance with the published schedule, including scheduled stops. However, flights may be diverted due to:

- I) Operational conditions which prevent the completion of the planned operation.
- II) Technical problems requiring landing at a more suitable airport.
- III) A passenger or crew on board requiring immediate medical assistance.
- IV) Meteorological conditions either encountered or reported en-route, which require an alteration to the intended routing.

7.6.3. The Captain or the Team Leader Cabin Services shall inform the passengers over the public address system of the reasons for the diversion and give pertinent information regarding the diversion of the flight. If deplaning, passengers will be directed to take their personal belongings with them. No hand baggage should be left behind in the aircraft.

SECTION 2.6 OPERATIONAL IRREGULARITIES

7.6.4. If the airport has a company representative, he and the Captain shall be jointly responsible for the handling of the diversion in accordance with local procedures. If the airport has no company representative, the Captain shall be responsible for the coordination of all activities and shall assign duties to the team leader cabin services or other crew members as necessary.

NOTE: When a flight diverts to a non-maintenance station, the responsibility for the serviceability of the aircraft prior to departure rests with the flight crew in conjunction with Integrated Operations Control Center (IOCC).

7.6.5. A message providing notification of the ATA, anticipated time of delay, and disposition of passengers and load must be sent immediately upon arrival to each of the following:

- Destination station
- Departure station
- IOCC

7.6.6. Two cabin attendants shall escort the passengers to the lounge. The remaining cabin crew shall check the cabin to collect any hand baggage left behind and arrange the storage of the sealed-bar boxes. Arrangements are to be made for required catering services. The team leader cabin services shall be responsible to ensure:

a) In the case of night stops, assistance is given to the passengers to complete any custom/immigration requirements.
b) In case of lost baggage, the handling agent is informed, and appropriate action is taken.
c) The unaccompanied children are never left alone, and that a message is sent to the stations of the diversion.
d) The exact count of passengers is taken.
e) Passengers wanting to send a notification of the delay are given this assistance at company expense.
f) Assistance to any passenger wishing to transfer to another carrier. (A record must be kept of all such transfers).

7.7. CANCELLATION OF FLIGHT

7.7.1. Cancellation of flights is the deletion of a previously published schedule or non-scheduled flight or segment thereof before departure due to lack of equipment, extended delay, metrological conditions, crew shortages commercial reasons, or other causes.

7.7.2. Although it is the policy to operate in accordance with the published schedule, certain conditions may require the cancellation of a flight or segment thereof.

7.7.3. Such considerations may include but are not limited to;

a) Probable delay and other disruptions to subsequent operations.

b) The feasibility or necessity, from a traffic point of view, to terminate the contract of carriage.

c) Additional expense in operation, such as ferry flight or utilization or divert aircraft far in excess of a possible loss of revenue.

d) Weather conditions at airport of departure and/or destination that would cause serious delay of present and/or subsequent flights.

e) Lack of crew

7.7.4. The authority and responsibility for the cancellation of a flight generally rest on the IOCC. The exception is when the flight is originating at a remote location from which communication with IOCC is impossible. In this case, the authority is delegated to the Captain. In all cases coordination with the other affected departments of the airline is absolutely essential.

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SECTION 2.7 DIVERSION

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1.0 PURPOSE

The purpose of this diversion procedure is to establish guidelines to be followed during flight diversion.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Operational Planner
- Station Manager(s) at the diversion point
- Crew scheduling
- Personnel in Communication Department

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.04

4.1. DEPARTMENTS INVOLVED

4.1.1. The following departments and individuals are involved in the case of flight diversion:

a) Operational Planning (IOCC).
b) Station Managers or contract Representatives (Commercial Department) at the destination and diversion airports or at the nearest airport served by the company.
c) Crew Scheduling.
d) Communication Department.

SECTION 2.7 DIVERSION

4.2. COORDINATION REQUIREMENTS

4.2.1. Communications between the PIC and IOCC are usually the best means of coordinating the requirements of the flight and other departments regarding the disposition of the passengers, crew and airplane after the flight's arrival at the diversion airport. At remote stations where communications with IOCC may be difficult, the Captain must assume an active role in coordinating these requirements.

4.2.2. Early recognition of a potential diversion is important in dealing with such an occurrence, IOCC should notify the PIC of diversion airport preferences well in advance of a diversion.

4.2.3. These preferences effect the requirements of each department involved. When several flights are expected to divert from a particular area and it is possible to do so, a sufficient number of diversion airports are alerted to ensure adequate ground personnel and facilities to handle the diversions.

4.3 SELECTED AIRPORTS

4.2.4. Diversion Airports are selected on the basis of the airplane's remaining fuel range, airport facilities, weather conditions, passenger service capabilities, and crew and airplane scheduling requirements. If the diversion is the result of an airplane malfunction or a similar incident, safety factors may limit these considerations.

4.2.5. The airport selected as the diversion airport may not be designated as an alternate, but if not, the OFP must be amended to designate that airport as an alternate. Re-dispatch is not required for diversion to designated alternates.

NOTE: The en-route alternate airport selected for the diversion must be selected according to the criteria set in FOM 2.3 & Route Manual.

4.4 FLIGHT PLAN AND CLEARANCE

4.2.6. Before an airplane diverts, a flight plan must be filed to the diversion airport and an ATC clearance issued. The following flight plan information may be required and should be at hand when requesting this clearance.

a) Diversion airport
b) Route of flight
c) Altitude
d) Estimated time enroute
e) Endurance (hours and minutes)

4.2.7. The issuance of an ATC clearance to a diversion airport must not be interpreted as concurrence by the IOCC on the diversion.

SECTION 2.7 DIVERSION

4.5. IOCC

4.5.1. If diversion is required for scheduled Passenger Flight, Cargo/Charter/Non-Revenue Flight IOCC must be advised before making air turn backs unless there is a serious safety concern or emergency situation. This joint decision must be applied for the following situations.

a) Diversion to alternate
b) Technical or maintenance issue
c) Flight delays that affect subsequent flights
d) Ground handling or airport related disruptions
e) Any other situation that could impact onward operations.

4.5.2. After meeting dispatch requirements and as permitted by IOCC, the PIC should use the services of Flight Information Center (FIC) that contribute to the efficiency of the flight.

4.6. NOTIFICATION TO CABIN ATTENDANTS

4.6.1. The Team Leader Cabin Services should be advised of a potential diversion early enough to plan for cabin service and passenger accommodation. Team Leader Cabin Services should make passengers generally aware of the factors affecting the flight plan but may omit specific reference to a diversion until the flight actually diverts.

4.6.2. After a diversion has been initiated, the Team Leader Cabin Services should be advised of the diversion airport and ETA. This information should be given to station personnel after landing to facilitate commendation of passengers if the flight is canceled after landing.

5.0 DEFINITION

DIVERSION is a flight to any airport that is not the original destination designated in the dispatch release. The airport to which such a flight is diverted is the diversion airport.

6.0 RESPONSIBILITY

6.1. The Pilot-in-command is responsible to exercise his authority during flight irregularities.

6.2. The pilot-in-command is responsible to make appropriate communications to all concerned, as required with regard the flight condition.

7.0 PROCEDURE

7.1. MAXIMUM LANDING WEIGHT

7.1.1. Normally, the same landing gross weight (LGW) limits apply for an unscheduled landing that applies for landing at the filed destination airport. In an emergency the PIC may exceed the maximum LGW (including the structural limitation) if, in his judgment, doing so is safer than dumping fuel or otherwise meeting the LGW limitation.

7.1.2. Dumping fuel to meet LGW limitations is an option of the Captain's emergency authority. The Captain may exercise his emergency authority to exceed the maximum LGW if the landing is necessary as the result of an incident or a mechanical irregularity and he determines that such a landing would be safer than dumping fuel. The Captain may be expected to dump fuel or land above maximum LGW whenever:

a) Any malfunction or failure would render the airplane non airworthy.

b) Due to existing mechanical or other conditions, an expeditious landing would reduce a hazard.

c) The serious illness of a passenger or crewmember requires immediate medical attention.

NOTE: The maximum in-flight gross weight with landing flaps extended must be considered in the same context as the maximum structural LGW above.

7.1.3. For a landing at a gross weight above maximum LGW refer to the appropriate Airplane Flight Manual.

7.1.4. For reporting requirements when the maximum LGW is exceeded, see the Required Reports section on FOM 3.12.

7.1.5. Remember that a special inspection has to be carried out as prescribed in the maintenance manual before the next takeoff.

7.2. FUEL DUMPING/OVERWEIGHT LANDING

7.2.1. When deciding whether to dump fuel or not the Captain should consider the fact that fuel dumping as well as overweight landings are both non-normal procedures and should be treated as such, although neither of the two procedures involves any hazard if they are carried out in the correct manner. For fuel dumping policy refer to FOM 1.11 (4.1).

7.2.2. At the Captain's discretion it may be safer to land over-weight due to circumstances not connected with the fuel dumping procedure itself.

Examples: Flying in high-density traffic areas, ATC difficulties, risk of the aircraft flying through dumped fuel, or adverse weather at the landing airport.

NOTE: Overweight landings must be entered in the aircraft maintenance logbook and a proper inspection must be conducted before the next flight.

7.2.3. The PIC shall notify the appropriate ATC unit that fuel dumping is required and give the following information:

a) Position, time and level;

b) Reason for dumping fuel;

c) Plan of action

NOTE: All actions by the Captain relating to the dumping of fuel shall be governed by ATC instruction unless compliance is precluded by the nature of the emergency.

7.2.4. ATC shall also be informed at the termination of fuel dumping. Fuel dumping shall be conducted within the area selected by ATC. Whenever such an area cannot be assigned by ATC, the Captain shall select an area that in his opinion would not pose an undue hazard to persons and property.

a) Fuel dumping shall be done above 5000 feet above ground for better fuel vaporization.

b) Fuel dumping in a holding pattern shall be avoided.

c) Fuel dumping should not be performed near thunderstorm activity.

d) For detailed procedures refer to aircraft Operations and Flight Manuals.

7.3. AFTER LANDING

7.3.1. The PIC should confirm that provisions are adequate for passenger handling at the diversion airport.

7.3.2. If ground staff is insufficient to provide an acceptable level of customer service, the PIC may use his crew for customer service. Crew rest shall be based upon total duty time.

7.4. CREW AND AIRPLANE SCHEDULING

The PIC should forward his recommendations for scheduling the departure from the diversion airport.

7.5. DEPARTURE

7.5.1. For departure considerations from an airport where there is no company operations representative or handling agent, see Unassisted Revenue Departure below.

a) Scheduled Passenger Flight: OFP from IOCC is required to authorize the flight's departure.

b) All Other Flights: if communication with IOCC is possible a flight release is required to authorize the departure of a charter, non-revenue, or scheduled cargo flights. If communication with IOCC is not possible, the PIC may depart to complete the flight to its intended destination as soon as conditions permit.

7.6. UNASSISTED REVENUE DEPARTURE

7.6.1. The information provided below is to help flight crewmembers meet the pre-departure requirements of a revenue flight where there is no operations representative or handling agent to assist.

7.6.2. **The First Officer**:-

a) Is responsible for the completion of the required forms like load sheet, trim sheet and fuel sheet etc... These forms are in the Spare Operating Forms envelope in the airplane's library.

b) Before departure, a copy of each of the completed forms must be given to the personnel concerned.

7.6.3. **The PIC** is responsible to check all the required procedures are done accordingly before subsequent Flight.

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SECTION 2.8 HANDLING PASSENGERS

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1.0. PURPOSE

The purpose of this section is to establish the guideline as to how the passengers' safety, welfare, and comfort are met.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight and Cabin Crewmembers
- Manager Airport Services

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. It is Ethiopian policy not to inconvenience a large number of passengers to accommodate a few. However, in compassionate or urgent cases PICs may use their discretion to delay a flight departure. Unless there are operational reasons (e.g. deteriorating weather), PICs will delay a departure when officially requested to do so through the IOCC.

4.2. Flight and Cabin Crews must be on board whenever passengers are on board or embarking/disembarking at all stations. One member of Flight Crew will be sufficient to meet the above requirement.

5.0. DEFINITION

5.1. DISRUPTIVE PASSENGERS

5.1.1. Is one who endangers the safety of the aircraft, passengers or crew;

5.1.2. Is one whose conduct or mental or physical state is such as to cause discomfort or make him or her objectionable to other passengers or involve any hazard or risk to the passenger or to other persons or property.

5.1.3. Is one who fails to observe the instructions of Ethiopian staff.

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5.2. CATEGORIES OF PASSENGERS WITH DISABILITIES REQUIRING WHEELCHAIR

5.2.1. **WCHR:** Passengers who can ascend/descend steps and make their own way to/from their cabin seats but cannot walk long distances.

5.2.2. **WCHS:** Passengers who cannot ascend/descend steps but are able to make their own way slowly to/from their cabin seats.

5.2.3. **WCHC:** Passengers who are completely immobile and require wheelchairs to/from the aircraft and to/from their cabin seats.

6.0 RESPONSIBILITY

6.1. Passenger handling at an airfield is the responsibility of Ethiopian station staff and/or the contracted local handling agent.

6.2. PIC is responsible for passenger handling after acceptance of the release of the aircraft from an authorized member of the ground staff and the aircraft doors are closed.

6.3. Notwithstanding this, the PIC is to ensure that the combined number of adult and child passengers (excluding children who have not attained the age of two years) on board the aircraft does not exceed the number of certified passenger seats.

6.4. It is the responsibility of the Cabin Crew to bring to the PIC's attention cases where the amount of hand baggage in the cabin exceeds that which can be safely stowed in approved areas.

6.5. It is the responsibility of the ground staff to off-load hand baggage that cannot be safely stowed in the cabin. If space is available, it may be loaded in the aircraft holds at the PIC's discretion. The cabin is not reported to be secured until all hand baggage aboard the aircraft is safely stowed.

6.6. In the event that an aircraft experiences a prolonged delay after the passengers have boarded, the PIC having considered all relevant factors, shall decide whether they shall remain on board or disembark. The following points should be considered:

6.6.1. The expected duration of the delay.

6.6.2. Whether the presence of the passengers will hamper the task of maintenance personnel in the event of a technical delay.

6.6.3. The availability of suitable space in the terminal building to accommodate the passengers.

6.6.4. Any local restrictions on the temporary disembarkation of passengers.

6.6.5. Passenger inconvenience.

6.6.6. The PIC is ultimately responsible for the decision to exclude a passenger from carriage and the final decision must rest with him.

6.6.7. The Team Leader Cabin Services is responsible for unaccompanied minors until being handed over to the appropriate ground staff.

7.0 PROCEDURE

7.1 PASSENGER ANNOUNCEMENT

7.1.1. A public address (PA) system for passenger announcement, with the required specifications, shall be installed in all Ethiopian operations. (ECARAS 7.3.1.3)

7.1.2. Regarding flight progress and points of general interest shall be made at the discretion of the PIC, who shall also explain or direct the explanation of irregularities, including:

Delays, expected length of delay, and estimated arrival times.

a) Anticipated rough air. (Call attention to seat belt signs and requirements.)
b) Alternate airport landings and provisions for passengers.

7.1.3. No flight crew shall allow the movement of an aircraft on the surface, takeoff or land: (ECARAS 8.9.2.21)

a) When any food, beverage or tableware is located at any passenger seat and
b) Unless each food and beverage tray and seat back tray table is in stowed position.

7.1.4. No flight crew may allow the takeoff or landing of an aircraft unless each passenger seat back is in upright position. (ECARAS 8.9.2.20)

7.2. SEAT BELTS

7.2.1. Must be available for each passenger, and cabin crewmembers shall observe that passengers have fastened their seat belts whenever the "FASTEN SEAT BELT" signs are on. One child under the age of two and one adult may occupy one seat and seat belt in common. Seat belt signs shall be on during taxi, takeoff, descent, and landing and any time rough air is encountered or anticipated. Refer to procedure in turbulent weather under (FOM 2.18)

7.2.2. Cabin crew shall make announcement per CCOM regarding "Seat Belt Sign-On" and "Sign-Off".

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7.3. PORTABLE ELECTROINC DEVICES

The operation of portable radio transmitters or receivers is prohibited as some types have been known to cause interference with navigational equipment, for more information refer to FOM 1.11 (4.4). This does not restrict the use of;

- Portable voice recorders
- Hearing aids
- Heart pacemakers
- Electric shavers
- Small hand-held electronic calculators and those allowed by CCOM.

7.4. SMOKING IN THE CABIN

Smoking in Ethiopian aircraft is strictly prohibited. If a passenger is observed smoking in the cabin, cabin crew shall instruct accordingly and if necessary, write a report. Further, if the instructions from the cabin crew are not followed, the Team Leader Cabin Services shall give a formal warning and advise the PIC. The Team Leader Cabin Services must also write a report.

7.5. PASSENGER SMOKING IN THE TOILET AND TAMPERING WITH THE SMOKE DETECTOR

7.5.1. If a passenger is caught smoking in the toilet, Team Leader Cabin Services must advise the PIC and write a report.

7.5.2. Tampering with the smoke detector is a serious and hazardous matter and therefore if a passenger is caught tampering, it must be treated as a safety issue and the PIC must be notified immediately who in turn will notify IOCC to request for Police to meet the aircraft on arrival.

7.6. LIFE VEST DEMONSTRATION/BRIEFING

7.6.1. A life vest demonstration for passengers must made before takeoff if:

a) The takeoff or approach path is over water, and in the event of a mishap there is the possibility of a ditching.

b) The aircraft will be flying over water more than 50 nm away from shore.

c) The aircraft will be flying en-route over water beyond gliding distance from shore.

7.6.2. Detailed instructions regarding how and when a demonstration or briefing must be made are stipulated in the relevant instructions issued to Cabin Crewmembers.

7.6.3. Demonstrations/briefings may be replaced by a relevant video performance if such a system is available on the respective aircraft type.

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7.7. OXYGEN MASK DEMONSTRATION

The passenger oxygen mask demonstration is compulsory for all flights. It will be carried out in accordance with the relevant instructions issued to Cabin Crew either by an actual demonstration or by the use of an approved video.

7.8. CARRIAGE OF THERAPEUTIC OXYGEN BOTTLES

7.8.1. The type of therapeutic oxygen bottles available are continuous flow bottles which use a mask or demand type flow cannula/nasal tube.

7.8.2. Only six bottles of oxygen are permitted to be carried on all Ethiopian flights. Necessary entry shall be made in the Tech Log specifying number of oxygen bottles used and their locations for attention of crew and for removal of bottles from the stowage location on return to ADD.

7.8.3. Carriage of extra portable oxygen bottles, same as those permanently installed on the airplanes, as loose items in the cabin is prohibited.

7.8.4. The therapeutic oxygen bottle using a demand type outlet with a cannula/nasal tube is used exclusively on long haul flights only when six therapeutic oxygen bottles (with constant flow outlet using a mask) will not meet the passenger’s requirements.

7.8.5. The number and type of therapeutic oxygen bottles will depend on the oxygen flow requirements of the passenger, but the number of therapeutic oxygen bottles onboard must not exceed six on any single sector and all bottles will be of identical type. Intermixing of bottles with continuous flow mask or with demand flow cannula/nasal tubes is not allowed.

7.8.6. Per FAA acceptance criteria, passengers are allowed to carry portable oxygen concentrator (POC) onboard Ethiopian flights. Cabin crew shall check **“The manufacturer of this POC has determined this device conforms to all applicable FAA acceptance criteria for POC carriage and use on board aircraft”** label on the POC in red letter.

7.8.7. Passenger shall bring an adequate number of fully charged batteries on board based on the battery manufacture’s estimate of the hours of battery life while the device is in use and the information provided in the physician’s statement, to power the device for not less than 150% of the expected maximum flight duration.

7.8.8. Passenger must present at the airport a physician’s statement (MEDIF) from their physician that is issued not more than 10 days prior to departure of the flight. Ethiopian medical doctor may not need to approve the MEDIF and it shall be returned to the passenger.

NOTE: passenger with POC shall not occupy an emergency exit seat.

7.9. PASSENGER UPGRADING

7.9.1. Once in the aircraft, a passenger may only be upgraded to a higher class compartment upon payment of the difference in fare cost. Payment is by credit card only. There must also be space available.

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7.9.2. The PIC may authorize upgrades for operational reasons or whenever he feels it to be in the best interest of the Company. In all such cases details of the upgrading must be submitted on a Captain's Report annotated "Copy to VP".

7.10. CABIN RATE OF DESCENT

For passenger comfort, avoid rates of cabin descent greater than 500 ft. per minute.

7.11. CABIN SERVICE

7.11.1. Unless otherwise informed by the PIC, the Team Leader Cabin Services is permitted to initiate cabin service after:

a) The seat belt signs have been switched off following takeoff, or
b) "Cabin crew cleared for duties" intercom call from the Flight Crew has been received.
c) If the Seat Belt signs are ON, hot liquids shall not be served to passengers.

7.12. CARRIAGE OF DIPLOMATS AND DIPLOMATIC MAIL SECURITY SCREENING

7.12.1. Diplomatic bags/pouches are inviolable and shall not be opened or detained.

7.12.2. They are exempted from pre-boarding security screening by any means, including X-ray and metal detection equipment.

7.12.3. Diplomatic couriers convey diplomatic bags/pouches under conditions outlined in Article 27 of the Vienna Convention, 1961. Diplomatic couriers are identified by:

a) A document indicating their status, issued by the state which they serve; and
b) A document (waybill) indicating the number of packages which comprise the consignment or diplomatic bag/pouch for which they are responsible. Each package has a label bearing a serial number, which identifies it with the waybill.

7.12.4. The person's, or personal baggage's, of diplomatic and/or military couriers are not exempted from screening procedures.

7.13. CHECK-IN PROCEDURES FOR DIPLOMATIC BAGS

7.13.1. The maximum allowable weight per package per seat is 75kgs.

7.13.2. Bags are to be placed on the seat cushion and must be fully secured with the seat belt, plus extension belts. The bags must not be stowed above the seat back rest level.

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7.14. TRANSIT PASSENGERS

Where local authorities or other circumstances do not allow transit passengers to disembark, the cabin crew will be required to carry out a transit security search, which involves conducting a passenger carry-on bag match with passengers on board.

7.15. TRANSIT OR TURN-AROUND STOPS WITH NO CREW CHANGE

During such stops where passengers disembark, they must do so with all their personal effects. All cabin crew will remain onboard and at least one of the Flight Crew has to be in the Flight Deck. Once passengers, caterers, cleaners, etc. have disembarked, the cabin crew will conduct a transit security search. Following completion of these procedures, passengers will be allowed to board the aircraft.

7.16. TRANSIT OR TURN-AROUND STOPS WITH A CREW CHANGE

7.16.1. During such stops where passengers disembark, they must do so with all their personal effects. The inbound cabin crew must not leave the aircraft until the new crew has physically arrived and a crew-to-crew hand-over has been conducted. For the Flight Crew at least one of them shall be in the Flight Deck until one of the new Flight Crew assumes control of the A/C.

7.16.2. Once passengers, caterers, cleaners etc. have disembarked, the next leg operating cabin crew must conduct a transit security search.

7.16.3. Upon re- boarding, station staff will reconcile the names of transit passengers with the Transit Passenger Manifest. Should any discrepancies occur, the discrepancy must be reported to the PIC. He shall then order baggage identification for security reasons.

7.17. UNACCOMPANIED MINORS AND INFANT

7.17.1. Children between the age of 5 to end of 11 years can be accepted for travel without accompanying adults and they are called unaccompanied minors.

7.17.2. The number of unaccompanied minors without escort shall be limited to four (4) on international flights and one (1) on domestic flights. The team leader cabin services is responsible for every UM in the cabin during normal and abnormal situation.

7.17.3. Unaccompanied minors will be handed over to the Team Leader Cabin Services by a member of ground staff on boarding.

7.17.4. It is the Team Leader’s Cabin Services responsibility to ensure that no unaccompanied minors are inadvertently disembarked at transit stations.

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7.17.5. In case of flight disruption at a transit station, unaccompanied minors remain the responsibility of the Team Leader Cabin Services until being handed over to the appropriate ground staff.

7.17.6 An infant is a baby under 2 years old and cannot travel unaccompanied. She/he has to be seated with her/his guardian. However, if a passenger is traveling with two infants, only one will be allowed to share the seat of the guardian and the other will have to have another adjacent seat belt with an adult.

7.18. CONSUMPTION OF ALCOHOLIC BEVERAGES

No passenger may drink any alcoholic beverage aboard the aircraft unless it has been served to him by a Cabin Crewmember. No alcoholic beverage shall be served to a passenger who appears to be intoxicated.

7.19. REFUSAL OF CARRIAGE/REMOVAL OF PASSENGER

7.19.1. If at any time prior to boarding an aircraft, ground staff reasonably determines that carriage or onward carriage should be refused to a disruptive passenger:

a) That passenger may be denied a boarding card.
b) If a boarding card has been issued, ground staff will take such steps as are appropriate and reasonable in the circumstances to prevent the passenger boarding the aircraft.
c) Any checked baggage of the passenger should be retrieved and made available to the passenger.

7.19.2. Flight Operations may not allow the transportation of special situation passengers except: (ECARAS 9.3.1.8)

a) As provided in this manual and
b) With the knowledge and concurrence of the PIC

7.20. PASSENGER-EXCLUSION FROM AIRCRAFT

7.20.1. The sole recourse of any person refused carriage or removed en-route for any reason specified shall be the recovery of the refund value of the unused portion of his ticket.

7.20.2. Ethiopian will refuse carriage or onward carriage when, in the exercise of reasonable discretion, it is considered that a passenger:

a) Manner or conduct is improper or inappropriate.
b) Does not observe instructions of authorized Air Line Officials.

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c) Are apparently under the influence of alcohol or narcotics.

d) Are apparently suffering from an infectious or contagious disease(s). crew members are also responsible to evaluate a traveler with a suspected communicable disease, based on the presence of certain other signs or symptoms.

e) Are apparently of unsound mind.

f) Are women in the late stages of pregnancy unless in possession of a doctor's certificate stating fitness for travel.

g) Are persons not in possession of the requisite documents.

7.20.3. During flight if a passenger is suspected with communicable disease the PIC has to report promptly to ATC the following information:

a) aircraft identification,

b) departure aerodrome,

c) destination aerodrome,

d) estimated time of arrival,

e) number of persons on board,

f) number of suspected case (s) on board and

g) Nature of the public health risk, if known.

7.20.4 Ethiopian may be under serious liability if a passenger is excluded for reasons, which ultimately prove to have been unjustifiable or unreasonable. Before taking such action particularly on long journeys, a doctor shall, if possible be consulted. In the absence of medical opinion, in no circumstances shall it be indicated that a passenger is considered to come within categories 7.20.2 (c), (d) (e) or (f) above.

7.20.5 Responsibility for exclusion of passengers' rests with the manager airport services. He/she that will confer with the aircraft Captain, as necessary.

7.20.6 In the event of a passenger's behavior coming within any of the above categories during flight, the Captain shall take such action, as he deems necessary, to ensure said passenger neither endangers the aircraft, its occupants, nor causes annoyance or inconvenience. If the case is such that disembarkation at the next point of landing, which may not be the passenger's destination, is considered, the Captain should obtain written statements, giving names and addresses, from independent witnesses, preferably passengers, confirming the behavior and attitude of the passenger. These statements shall be handed to the duty officer at the next point of landing.

7.20.7 If it is alleged that the passenger is under the influence of alcohol, details of any drink supplied during flight shall be given to the Duty Officer.

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7.21. LIABILITY

Ethiopian may suffer serious liability if a passenger is excluded for a reason that is later unreasonable. However Ethiopian officials must exclude any passenger who is obviously likely to endanger himself, or the aircraft or its contents, or to cause substantial annoyance or inconvenience to other passengers.

7.22. PASSENGER MISCONDUCT

7.22.1. GENERAL

a) Cabin attendants should be alert to the personal behavior of any passenger who could threaten the welfare of any other passenger or crew member. The captain is to be advised at the first indication of the potential problem.

b) In the early stages of a disruptive passenger incident onboard the aircraft (whether at the stand, during taxiing or in flight), the Team Leader Cabin Services and the Senior Cabin Crew Member should be involved, and the PIC should be kept fully informed of what is happening.

c) The PIC will delegate the Team Leader Cabin Services and the Senior Cabin Crew Member his authority to deal with a disruptive passenger incident. They may request the assistance of suitably able-bodied passengers, as required. **Under no circumstances should either of the pilots become directly involved in any contact or confrontation with a disruptive passenger while any form of conflict is being resolved or handled.**

d) Cabin Crew should be alert to the personal behavior of any passenger who could threaten the welfare of any other passenger or crewmember. The PIC is to be advised before a significant problem develops.

e) Where circumstances allow, before taking final action under these procedures, the concerned staff should display the appropriate form of notice to the disruptive passenger. A verbal warning must be given to the disruptive passenger.

f) The following acts or conditions violate the law. Any person violating any of these or any other law or regulation may be removed from the flight and prosecuted to the extent of the law. The Captain is to be notified immediately when such an act or condition is discovered or suspected. A report should be submitted accordingly upon completion of the flight.

   I. Apparent intoxication.
   II. Being obviously under the influence of drugs (except under proper medical care).
   III. Threatening another passenger or a crewmember with physical violence.
   IV. Indecent exposure or proposals.
   V. Theft.

VI. Carrying an unauthorized deadly or dangerous weapon, either concealed or unconcealed.

VII. Interfering with the safety of a crewmember or the aircraft.

VIII. Conveying false information concerning the flight and its safety (including bomb threat).

IX. Committing or attempting to commit an act of aerial piracy.

g) The Captain must submit a written report on any act of passenger misconduct.

h) The PIC is responsible for ensuring that this policy is applied and that appropriate written reports are submitted by members of the crew. For this reason, the Team Leader Cabin services should ensure that any incidents of this nature are fully documented by themselves and by any crewmembers who are involved in an incident or are witness to it.

i) As there are many different types of incidents, the PIC shall use his discretion in deciding when an incident should be referred to the Authorities.

7.22.2. ABUSIVE OR SUGGESTIVE LANGUAGE

a) Although this type of behavior is disturbing and offensive, it need not necessarily be a hazard to ground staff, crew or to the safety or security of the aircraft. As long as it does not create the threat of violence or physical attack, it is not considered an assault.

b) The objective in dealing with incidents of this nature is to calm the passenger down and avoid any confrontation.

c) Because such a passenger might make himself objectionable or cause discomfort to other passengers, the incident might result in the refusal or carriage or onward carriage of the passenger, the off-loading of the passenger and/or legal action being taken against the passenger.

7.22.3. REFUSAL TO COMPLY WITH INSTRUCTIONS FROM CABIN CREW

a) The refusal of a passenger to comply with instructions from Cabin Crew constitutes disruptive behavior and might, according to the circumstances, endanger the safety and security of the aircraft, crew, the passenger concerned and other passengers.

b) Examples of such incidents would be a passenger refusing to fasten his seat belt for take-off or landing, and a passenger refusing to fasten his seat belt during the time that the “seat belt” sign is illuminated in the aircraft.

c) Because the refusal of the passenger might endanger the safety or security of the aircraft, the passenger concerned, and other passengers and crew, it might result in the off-loading of the passenger and/or legal action against the passenger being taken.

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7.22.4. PASSENGER RESTRAINING DEVICES

a) Passenger restraining devices are placed on all Ethiopian aircraft & will be placed in the cabin.

b) Whenever the restraining devices are used, the complete details must be recorded on the Captain's Report.

c) The authority to use these restraining devices in-flight rests solely with the PIC of the aircraft. Their use is restricted solely to occasions where a passenger or passengers, by their conduct, are considered likely to endanger the aircraft, crew or other passengers.

d) The Flight Crew must be neither physically involved in the use or application of these devices, nor present in the passenger cabin while the restraint process is in progress.

e) When suitably restrained, the passenger should be isolated if space permits. He/she should be kept under observation at all times. The crew may consider using a staff passenger for this purpose if one is on board, or any other responsible person. The authority for this must first be obtained from the PIC.

7.22.5. UNRULY PASSENGER CONTROL IN-FLIGHT

a) Should any passenger become unruly during flight, the Captain shall always be the one responsible for restoring order in the cabin. He may delegate the Team Leader Cabin services to achieve control by talking with the unruly individual, but if that is not effective, the captain shall use every available means to ensure safety of the flight. This may include sending warning to the passenger, putting the seat belt sign on and announcing turbulence ahead over the PA, if this is not still effective he may have to use more positive measures, such as restraining the passenger with restraining device, belts, ropes, tie raps, blankets. The team Leader Cabin services may request the assistance of able-bodied passengers if necessary to restrain the unruly passenger.

If this should become ineffective and it is obvious that it will endanger the safety of the flight diverting to the nearest alternate airport can be considered. But under no circumstances shall the flight crew leave the flight deck until the safety of the aircraft and the passenger is assured. The captain should notify law enforcement officials so that they can meet the flight and take custody of the unruly passenger(s) upon arrival.

b) Refer to the unruly passenger handling guidance found in the flight folder and security manual.

7.22.6. INDECENT EXPOSURE

a) These incidents are rare but considered serious.

b) The passenger concerned must be instructed to cease the offence and must be informed, if the incident occurs in flight, that the incident will be reported to the Police at the next place of landing.

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c) Because such a passenger might take himself objectionable or cause discomfort to other passengers the incident might result in the off-loading of the passenger and/or legal action being taken against the passenger.

7.22.7. DRUNKNESS AND DRUGS

a) Drunkenness is often an underlying cause of disruptive behavior. If a passenger exhibits signs of drunkenness during the flight, crewmembers are empowered to refuse to serve him any further alcohol and to remove from his tray any unconsumed alcohol.

b) Drugs and the combination of drugs and alcohol might also be causes of disruptive behavior. Should crew reasonably consider that a passenger is under the influence of drugs, they are empowered to refuse to serve any alcohol to that passenger and to remove from his tray any unconsumed alcohol.

c) Because such a passenger might make himself objectionable or cause discomfort to other passengers and/or endanger the safety or security of the aircraft, the passenger concerned, and other passengers and crew, the incident might result in the off-loading of the passenger and/or legal action being taken against the passenger.

7.22.8. COURT PROCEEDINGS

a) To facilitate legal action being taken at ADD or overseas stations it might be necessary for crew and staff members involved in the incident to give evidence and attend court.

b) Giving evidence and attending court might involve time off from work. It might be necessary to travel to a court outside ADD. Time off for this purpose will be treated as duty time. Travel to court for this purpose will be treated as duty travel.

c) It is appreciated that giving evidence and attending court can be stressful and disruptive, but, in the interest of preventing future disruptive incidents, this further involvement of crew and staff members concerned makes a considerable contribution to airline safety and benefits all airlines and their employees in the long run.

7.23. OFF-LOADING

7.23.1. If the PIC of the aircraft reasonably determines that carriage or onward carriage should be refused to a disruptive passenger, he is authorized:

a) If the incident of disruptive behavior takes place whilst the aircraft is on the ground, to return the aircraft to the stand and off-load the passenger concerned, or

b) If the incident of disruptive behavior takes place in flight, to off-load the passenger at the next scheduled place of landing or

c) In extreme situations, to divert to the nearest airport and off-load the passenger at that airport.

SECTION 2.8 HANDLING PASSENGERS

7.23.2. If the PIC of the aircraft wishes to off-load a disruptive passenger for any of the reasons mentioned in the paragraph above, he should request the Police to meet the aircraft and assist in the off-loading process. Any checked baggage of the passenger should be retrieved from the aircraft and made available to the passenger.

7.23.3. A Captains Report must be completed whenever any passenger is excluded from a flight. It must state the complete details of the incident, the name and address of the excluded person(s) and names and addresses of at least two independent witnesses who are willing to give evidence if required to do so.

7.24. OFF-LOADING MISSING PASSENGER BAGGAGE

7.24.1. Whenever a passenger fails to show up at the boarding gate after he/she had previously checked in with baggage, then Offloading of baggage belonging to the missing passenger will commence 10 minutes before ETD.

7.24.2. When a passenger checks in pooled baggage as part of a family or group, and then, for whatever reason, does not board the aircraft or leaves the aircraft, the following measures must be taken:

a) All bags checked-in by the group must be off-loaded from the aircraft and lined-up for identification.
b) The remaining passengers of the family or group who are travelling (on board) must physically identify their own bags by touching them. The identified baggage should be re-loaded on the aircraft. Any bags to be off-loaded must be separated and will be returned to the MHB (Mis-handled Baggage) section.
c) At ADD the re-screening of the identified, pooled baggage is not a requirement as a procedure.

7.24.3. There is no requirement for the flight crew to sight the off-loaded baggage.

7.24.4. If the passenger is located before this process is completed, then he/she may be accepted for the flight.

7.24.5. If the passenger is located after the baggage offloading is completed, then he/she will not be accepted for the flight. At all airports other than ADD, these procedures may differ in accordance with local regulations.

7.24.6. The Dispatcher/Load Control Agent will adjust weights on the Load-sheet via Load sheet re-issue.

7.24.7. The Load Control Agent will obtain a signature from the Captain on the Missing Passenger report.

SECTION 2.8 HANDLING PASSENGERS

7.25. CARRIAGE OF DEPORTEES/ INADMISSIBLE PASSENGERS

7.25.1. GENERAL

a) An inadmissible passenger (INAD) is a person who is refused entry by the Immigration Authorities. This category of passenger will generally not pose any threat to flights and therefore are not limited to any number, and do not need special authorization to travel. as long as these INAD passengers are willing to **voluntarily** return home and may not pose potential risk to flight safety.

b) A deportee (DEPO) is a person who has entered a country and who some time later is formally ordered by the authorities to be removed from that country.

c) A prisoner passenger is a person who is charged with a criminal offence and is wanted by the governmental authorities of another country or is being sent home to the home country for trial/conviction.

d) Prisoners are escorted at all times and remain under close physical supervision and custody of escorts. Prisoners will usually be in some form of physical restraint.

e) Individuals or mass deportation of persons from countries like Saudi Arabia and others can be accepted unaccompanied (without escort) when the deportees are over stayers or have gone through the prison system or are directly being deported, etc. In all cases the deportees should be willing to voluntarily return and do not pose any harm in flight or any risk to the flight safety.

f) Up to three accompanied (involuntary) deportees may travel on any ET flight with prior notification and approval by the Regional Director Sales & Services. However, the Pilot in Command may refuse carriage to the deportee/s depending on the behavior and aggressiveness of the passengers in question. At least two escorts for one involuntary deportee shall be a requirement.

7.25.2. HANDLING ON BOARD

a) In all cases the PIC must be advised that the INAD, DEPO, or Prisoner passengers are on board, with or without an escort. He has the authority to refuse to carry DEPO or Prisoner passengers provided he has strong reasons with a valid justification to do so. Persons who physically resist boarding the aircraft will be excluded from carriage. The PIC will file an ASR when off-loading/refusing DEPO/Prisoner passengers.

b) Arrangements shall be made that DEPO or Prisoner passengers will be boarded ahead of passengers and seated as discretely as possible at the rear of the aircraft. The escort may carry plastic restraining devices and will use them if required. Otherwise, such persons will be treated with the same courtesy and tact as all other passengers on board.

c) To avoid INAD, DEPO, or Prisoner passengers destroying their travel documents and claiming asylum on arrival, which could result in Ethiopian being fines by the Immigration Control Authorities at the destination, the Departure Station must ensure that tickets and passports are handed over to the Team Leader Cabin Services of the flight for his/her retention.

7.25.3. NOTIFICATION

a) Obtain the deportation or removal order form the authorities notifying the airline the reason for deportation. The deportation order along with travel documents shall be handed over to the Cabin team leader if the deportee/INAD is not accompanied or to the escorting police if the deportee/INAD is accompanied. Due to mandatory requirement of security check, the deportation order and travel documents shall upon request be shown or handed over to police/immigration authorities at transit and/or destination points of the deportee/INAD passengers.

b) For safety reasons, documents shall not be given in front of the INAD/Deportee and the passenger shall not be told that his/her documents are with the Cabin crew. Notify the Captain and the Cabin Team Leader that an inadmissible passenger/involuntary deportee is onboard and his/her seat number.

c) Station of Departure handover the travel document to the Cabin Team Leader upon her / his signature and retain the original signed form.

d) This form should be prepared in five copies by the Station of Departure.

e) The Cabin Team Leader handover the travel document to the Ground Staff/ Cabin Team Leader at transfer station upon her / his signature and retain the second signed form.

f) The Ground Staff / the Cabin Team Leader at transfer station handover the travel document to the Cabin Team Leader of the flight on which the deportee passenger travels to the final destination upon her/his signature.

g) The Cabin Team Leader handover the travel document to the Ground Staff at the final destination upon her / his signature (if there are no transfer stations, item 'f', and 'g' above will not be applicable).

7.25.4. DISEMBARKATION

a) It is the PIC's responsibility to ensure that a deportee is not permitted to disembark at any point within the jurisdiction of the deporting country, unless ordered to do so by that country's authorities. When an aircraft lands at another station within the jurisdiction of the deporting country or returns to the point of departure, the PIC will inform the station staff and the authorities of the INAD, DEPO, or Prisoner passenger's presence. Station staff must ensure that the authorities' instructions for custody of these passengers, either on board the aircraft or elsewhere pending re-embarkation, are carried out.

b) The PIC can permit "DEPO" and "INAD" passengers wishing to disembark at any intermediate point outside the deporting country's jurisdiction to do so, provided the authorities at the station concerned permit them to disembark.

SECTION 2.8 HANDLING PASSENGERS

7.26. PASSENGERS WITH REDUCED MOBILITY (PRM)

7.26.1. GENERAL

a) Passengers on a wheel chair, on a stretcher, who are blind, with impaired hearing, who are pregnant are called PRM.

b) passengers with reduced mobility categories are subdivided into the following groups:

I. Ambulatory Passengers able to reach an emergency exit during an evacuation without assistance.

II. Passengers requiring assistance in order to reach an Non Ambulatory emergency exit during an evacuation. A non-ambulatory passenger is one who is not able to board and deplane from an aircraft unassisted or who is not able to move about the aircraft unassisted.

c) A non-ambulatory passenger may travel alone except in the case of a non-ambulatory passenger cannot feed himself or manage his own bodily functions in the toilet, an accompanying able bodied person/attendant must be provided.

d) The PIC must be advised about the carriage of all non-ambulatory disabled passengers or other passengers who could require special assistance in the event of an evacuation.

7.26.2. HANDLING AND BOARDING OF PRM

a) Passengers able to reach an emergency exit without assistance shall be seated in aisle seats. Such seats should be near the floor type emergency exits/doors but not in the seat blocks immediately adjacent to exits/doors.

b) Two non-ambulatory passengers should not be seated directly across the aisle from each other.

c) Cabin Crew are considered to be able to care for comfort, safety and assistance in an emergency.

7.26.3. MULTIPLE NON-AMBULATORY PRM INCLUDING GROUPS

a) For each individual trip, the number of non-ambulatory disabled passengers to be carried should be communicated to the crew prior to boarding. The maximum number permitted as well as the provision of extra able bodied persons/attendants, including seating arrangements, must be determined after evaluating all safety considerations.

b) Any required accompanying able bodied persons/attendants in addition to the standard Cabin Crew will be responsible for the passengers’ comfort, as well as their safety and assistance during emergency evacuation.

c) Where appropriate, groups will be subdivided into smaller groups (depending on the aircraft type) and shall be seated in areas specifically designated for the purpose. Further information is available in the Services Manual.

7.26.4. EVACUATION PROCEDURES

a) Cabin Crew are responsible for the evacuation of all passengers able to reach the emergency exits without assistance.

b) Passengers requiring assistance will be evacuated by their escorts, Cabin/Flight Crew, and able-bodied passengers, depending on the situation.

7.26.5. CARRIAGE OF HANDICAPPED PASSENGERS

Assistance must be provided in the transportation and seating of handicapped passengers. They are categorized as either ambulatory, who do not require assistance to board and walk about on the aircraft or non-ambulatory, who requires assistance to board and walk about on the aircraft. It is recommended that handicapped passengers be seated in such a manner that in the event of an emergency evacuation, they can leave the aircraft, either unassisted or assisted, by the safest and most expedient route while not slowing the evacuation.

7.26.6. ORDERING OF WHEELCHAIRS FOR PASSENGERS

a) The normal procedure for ordering wheelchairs is through Ethiopian Reservations, who will ensure that details are relayed to stations and shown on the Passenger Information List (PIL).

b) Each Ethiopian wide-body aircraft is equipped with a wheelchair that can be used to move passengers within the aircraft. When boarding passengers with disabilities requiring wheelchair, it will be necessary for station staff to liaise with the Cabin Crew in order for the wheelchair to be removed from its stowage position and positioned to the appropriate door prior to the arrival of the passenger.

c) It is emphasized that the wheelchair is intended for use within the passenger cabins and it must not be removed from the aircraft.

d) Cabin Crews are not authorized to carry passengers to/from toilets etc. This function is the responsibility of the accompanying able-bodied escort.

e) Passengers with disabilities requiring Wheelchair will be accepted for carriage, without escort, if they are able to feed themselves and take care of their bodily functions.

f) There are no restrictions on the number of WCHR and WCHS passengers that may be carried on a flight. However, restrictions apply for WCHC passengers.

e) The number of WCHC accepted on an aircraft is controlled by the number of seats available for WCHC in that cabin or class of operation. The number of escorts (able-bodied passengers) available to look after individuals and group WCHC passengers.

f) However, during flight situations may arise making it necessary for additional wheelchairs to be available on arrival. In this case the process for ordering wheelchairs is as follows:

I. Last Minute WCHR Request at Outstations

- SCCM (Senior Cabin Crew Member) to inform the Captain (giving the passengers' name). The Captain should then forward this additional request ahead.
- Flight Crews are not to include wheelchair requests that are already listed on the PIL. The Team Leader Cabin Services will have this information.

II. Last Minute WCHR Request at ADD Only

- If a passenger becomes ill during flight or has an obvious disability that prevents him/her from walking, the SCCM will complete the "Additional Wheelchair Request" form as soon as possible to the Customer handling unit.
- Wheelchair requests already on the PIL must NOT be included. This is most important, as it would cause the order to be unnecessarily increased, thereby diverting resources from other flights.

NOTE: When advising stations for wheelchair requirements at landing, use of the correct code will assist Ground Staff in providing appropriate equipment. Code categories are:

7.26.7. CARRIAGE OF PASSENGERS WITH DISABILITIES (THOSE REQUIRING WHEELCHAIR)

Class of Services	All Aircraft Type for International & Domestic Flights					
	Business Class			Economy Class		
	WCHR	WCHS	WCHC	WCHR	WCHS	WCHC
No. of Individual WCH case Passenger per Flight	No restriction	No restriction	2	No restriction	No restriction	5
No. of WCH case passenger in group per flight	No restriction	No restriction	5	No restriction	No restriction	8

a) The above table does not apply on flight to/from EU and US.

b) The check-in agent is responsible for advising to transit and destination stations the presence of WCHC passengers on board the flight and location of wheelchairs and the need to arrange special assistance.

7.26.8. CARRIAGE OF PASSENGERS ON A STRETCHER

SECTION 2.8 HANDLING PASSENGERS

a) Only one passenger requiring a stretcher is allowed per flight.

b) A passenger requiring a stretcher shall be accompanied by one able-bodied adult attendant who will take care of the passenger throughout the journey. If the condition of the patient demands and/or the authorized physician so requires, the passenger shall be accompanied by nurse or physician.

c) The stretcher shall be installed and fitted by maintenance personnel on the right or left rear end of the applicable class of service.

7.26.9. CARRIAGE OF PREGNANT PASSENGERS

a) Expectant mothers are normally not regarded as incapacitated. However certain restrictions apply, these are given below:

I. MEDICAL CERTIFICATE OR LETTER:

1. No medical certificate or letter is required up to 28 weeks of an uncomplicated single pregnancy (normal pregnancy).
2. Common complications in pregnancy that require a medical certificate or letter can include gestational diabetes, pre-eclampsia and eclampsia, placenta praevia, intrauterine growth retardation and premature rupture of membranes.
3. Medical clearance is needed after 28 weeks of pregnancy by her attending doctor (no need from ET medical services).
4. Travel is permitted until 34 weeks of pregnancy for flights more than 2 hours and 36 weeks for flights 2 hours and less provided that she fills pregnancy certificate and has medical clearance from Doctor.
5. The certificate or letter must be returned to the passenger after verification, as it might be required at down-line stations. If required, a photocopy may be retained.

II. ACCEPTANCE:

In addition to the certificate requirement, there are certain limitations as the pregnancy nears its end. These are:

1. Uncomplicated single pregnancies - accept up to end of the 36th week.
2. Travel on urgent medical, complicated or compassionate reasons, after these limiting periods, must be approved by ET Medical Services.

7.27. BIRTH

7.27.1. If a child is born during flight, the aid of a physician or nurse should be requested from among the passengers. In the absence of such aid, female Cabin Crew will assist.

SECTION 2.8 HANDLING PASSENGERS

7.27.2. The PIC shall be notified by the team leader cabin services in case of imminent birth.

7.27.3. The PIC shall notify the next airfield in advance, requesting an ambulance to meet the aircraft. On arrival, in conjunction with the team leader cabin services, he shall complete a report in duplicate containing the following items:

a) Date and time of birth in hours and minutes.

b) Place of birth (given in late/long).

c) Sex.

d) Full name of the parents (including maiden name of mother).

e) Nationality of the parents or former nationality for displaced persons, as well as the place of birth.

f) Home address of the parents.

g) Witnesses of birth (full names and home addresses). Signature of the PIC and 2 other crew members.

7.27.4. The original of this report is to be handed over to the local police authorities, and the copy included with the ship's papers in the trip folder, attached to the Report.

7.28. MENTALLY DISTURBED (INSANE) PASSENGERS

7.28.1. Mentally disturbed passengers have to be escorted.

7.28.2. The accompanying escort shall carry all necessary restraining devices to be used in case the passenger needs to be restrained.

NOTE: If a passenger becomes mentally disturbed in-flight and he/she is traveling alone cabin crew members must give undivided attention to the passenger's movement especially when he/she wants to use the lavatory.

7.29. ILLNESS/INJURY

7.29.1 GENERAL HANDLING OF SICK PASSENGERS

a) The preservation of life and maintenance of the health of passengers whilst in Ethiopian care is paramount.

b) Passengers who have sickness before check-in to any of Ethiopian flights must always obtain safe-to-fly certificate from Ethiopian Company Doctor before they are accepted and checked-in to Ethiopian flights.

c) If the passenger is originating from outstations and they bear safe-to-fly certificates from another Doctor, the certificate has to be sent to Addis Ababa and endorsed by Ethiopian Company Doctor before the passenger is allowed to board.

d) No station manager or Ground Handling company is allowed to Check-in and board sick passengers without a medical certificate from Ethiopian Company Doctor.

e) Permanent physical disabilities shall not be considered as sickness, except that such passengers may be required to be accompanied by assistants, only when required.

f) Sick passengers with safe-to-fly medical certificates from Ethiopian Company Doctor may not be accepted to board the flight or may be offloaded from the flight if the physical condition of the sick passenger is substantially different or has deteriorated from what is described in the medical certificate in the judgment of the Pilot in Command (PIC) and the Team Leader Cabin Services.

7.29.2. PASSENGERS WHO BECOME SICK AFTER CHECK-IN BUT BEFORE BOARDING

a) If a passenger becomes sick after he/she checked-in on an Ethiopian flight but before boarding the flight, the passenger has to be checked by a medical doctor at the Airport before he/she is allowed to board the aircraft. The Ethiopian station manager or his delegate shall;

I. allow the passenger to board the flight only if he gets confirmation or opinion from the medical doctor that the passenger is safe to fly, or

II. off load the passenger if the medical doctor has doubts about the conditions of the passenger, or his opinion does not support the passenger to fly.

7.29.3 PASSENGERS WHO BECOME SICK AFTER BOARDING THE AIRCRAFT BUT BEFORE THE DOOR IS CLOSED

a) The Ethiopian station manager or his/her delegate shall make the sick passenger to be seen and checked by a medical doctor either while the passenger is on-board or at the Airport medical facility,

I. If the doctor determines he/she is safe to fly, the passenger shall be allowed to continue the flight.

II. If the doctor determines he/she is not safe to fly, or is doubtful about the passenger conditions, or further examination of the sick passenger would delay the flight considerably, the passenger shall be offloaded and the flight released.

b) A PIC has the right to refuse or accept an invalid or incapacitated passenger, if he believes the passenger:

I. Is likely to be at risk by the journey

II. Constitutes a flight safety hazard

III. Might cause inconvenience or discomfort to others.

SECTION 2.8 HANDLING PASSENGERS

c) When making onload /offload decisions for passengers with medical needs, various support options exist to enable the Captain to make an informed choice. If any doubt exists, pilots should seek help or clarification from medical services or a doctor as follows:

I. Airport Medical Clinic

II. An Ethiopian doctor, contactable through the station manager.

7.29.4. PASSENGERS WHO BECOME SICK AFTER PUSHBACK AND DURING TAXIING

a) The Team Leader Cabin Services shall make a careful assessment of the general condition of the sick passenger by talking to the passenger himself/herself, and makes best judgment on the passengers ability to complete the flight.

b) If the Team Leader Cabin Services believes it would be unsafe for the life of the passenger to continue with the flight, she/he shall inform the PIC and the flight may return to the gate. Then, the Ethiopian station manager or his/her delegate shall offload the sick passenger and dispatch the flight.

c) If the Team Leader Cabin Services believes the condition of the sick passenger is not serious, no further action is required, and the flight shall continue.

7.29.5. PASSENGER SICKNESS IN FLIGHT

The Team Leader Cabin Services shall;

a) Provide all the necessary first aids to the sick passenger, Cabin Crew may open the Emergency Medical Kit without first seeking permission from the PIC. Team Leader Cabin Services must make an entry in the Cabin/IFE Log and inform the Captain when the seal on the Emergency Medical Kit has been broken, so that the Captain may make an entry in the Report.

b) Cabin Crew members shall administer first aid in cases of injury and/or illness encountered by passengers. Such cases shall be handled in a calm assuring manner and will not be discussed with other passengers.

c) Under no circumstances shall an injection be given to passengers by a Cabin crewmember. The only medication they may be given is that provided by Ethiopian, either in the first aid kits or supplied along with the passenger amenities and stowed in the galleys.

d) All Ethiopian aircraft are equipped with the "Forerunner Heart Stream Defibrillator", for use by Cabin Crew as a supplementary treatment to cardio-pulmonary resuscitation (CPR). The defibrillator is kept in the same stowage as the emergency medical kit on each aircraft.

e) Solicit medical assistance from passengers on board who may have a medical profession (he/she must verify the medical qualification of any one volunteering to assist).

SECTION 2.8 HANDLING PASSENGERS

f) Inform the condition of the passenger to the PIC and update him/her regularly.

7.29.6. IF THE CONDITION OF THE PASSENGER DETERIORATES AND BECOMES CRITICAL

a) The Team Leader Cabin Services shall inform same to the PIC.

b) The PIC shall contact the IOCC Duty Manager through radios (VHF, HF, Satcom, ACARS) as found convenient. If IOCC could not be contacted, the PIC shall attempt to contact FOPs senior management for guidance.

c) IOCC Duty Manager shall contact the Ethiopian Company Doctor right away and establish contact with the Team Leader Cabin Services for advice on additional medical support for the sick passenger. If the condition of the sick passenger becomes stable or better, the flight shall continue to its destination.

d) If the sick passenger condition does not improve and becomes more critical, the IOCC Duty Manager shall consult FOPs senior management for advice.

e) After assessing the situation, FOPs senior management (VP or Directors) shall determine if the flight has to continue or make diversion to the nearest airport, and advise same directly to the PIC or through IOCC Duty Manager.

7.29.7. IF THE PIC IS AUTHORIZED TO MAKE FLIGHT DIVERSION

a) He/she shall select the nearest suitable airport where the patient can be assisted better, and the Flight can be supported well.

I. PIC should not normally give any assurance that Ethiopian will accept any responsibility for the costs of providing medical care. The only exception is where passengers have suffered injury as a result of their flight with Ethiopian (e.g. injury caused by clear air turbulence, pressurization failure, etc.), where Ethiopian would accept full responsibility for the costs.

II. At locations where there is no Ethiopian representative the PIC should consider the following options.

- Ask a relative or friend of the passenger to stay with the sick passenger.
- Find a volunteer of the same nationality or with appropriate language skills to remain with the sick passenger.

b) When requesting assistance from the airfield of intended landing, the following details shall be relayed if possible:

I. Name of passenger.

II. Illness (if known) or injury.

III. Request for a doctor and/or ambulance.

c) IOCC Duty Manager shall immediately make all the necessary arrangements at the Diversion airport through the Ethiopian country manager responsible for the region.

d) The sick passenger shall be offloaded at the diversion station, and the aircraft resumes the flight to minimize the inconvenience on the rest of the passengers.

7.30. PRESUMED DEATH ON BORAD: POLICY AND PRCEDURES

7.30.1 Ethiopian Cabin crews are expected to carry out Cardio Pulmonary Resuscitation (CPR) on a casualty in the event of a sudden cardiac arrest on board. The aim of this policy is to provide flight and cabin crew with clear guidelines on when it is appropriate to stop the resuscitation effort and how to handle a 'presumed death on board' situation.

7.30.2. CPR can be stopped in any of the following cases:

a) A doctor is on board and willing to oversee resuscitation procedures carried out by the crew. The doctor is qualified to pronounce death and will do so when appropriate. **The doctor should not 'certify' death as this will be done by the relevant medical authority at the destination (regardless of whether there is a diversion or not).**

b) If there is no doctor on board then the cabin crew may stop the resuscitation effort in the following cases:

1. The captain advises of any adverse conditions or limitations which would threaten flight safety.
2. Signs of life are obvious e.g. coughing, breathing, movement.
3. CPR has been given for 45 minutes with a 'no shock advised' command on the defibrillator and no signs of life.
4. The aircraft lands and care is transferred to the emergency medical services on the ground.

NOTE: Cabin crew should carry out CPR for 45 minutes, including the operation of the defibrillator. If after that time there are no signs of life, the resuscitation effort can be stopped after discussion with the captain and family members/travelling companions.

7.30.3. Crew should look for the following signs that are suggestive of death:

a) Complete absence of movement and/or responses.

b) Absence of breathing and signs of circulation.

c) Body looks pale and gradually becomes bluish and cool to the touch.

d) Pupils may become dilated and not respond to light.

7.30.4. If the cabin crew member is satisfied that all resuscitation efforts have failed, and the passenger has not responded they should implement the procedures described on 7.31 below.

7.31. APPARENT DEATH IN FLIGHT

7.31.1. If a person appears to be dead the Captain is to be notified immediately. The aid of a physician or nurse should be requested. A person may be declared dead only by a physician or nurse. If a physician or nurse on board the airplane positively declares death, the flight may continue to destination.

7.31.2. A passenger declared dead by a physician or nurse shall be secured in a seat with seat belt and covered with a blanket. A deceased passenger may be moved only if death occurred when the passenger was not seated. It is advisable that as few passenger as possible be made aware of the incident. If possible, passengers should be relocated away from the area.

7.31.3. Pertinent information, including the time of the incident, is to be noted in the Flight Service Log.

7.31.4. The Team Leader Cabin Services shall perform all the required actions per the Cabin Crew Operations Manual.

7.31.5. The PIC shall notify the IOCC Duty Manager through radio or ACARS.

7.31.6. The IOCC Duty Manager shall inform the destination station manager to make advance preparations for removal of the body from the aircraft upon arrival.

7.31.7. If death has occurred, the Captain shall notify the destination airport and shall continue the flight to its destination airport unless he/she believes there are sufficient reasons (Safety, Passenger issues etc.) to divert to the nearest airport. In this latter case, the PIC has to get authorization from FOPs senior management or IOCC Duty Manager, using the same procedure as defined above. Normally, the body will remain at the diversion airport because of the need to complete formalities both at the diversion airport and at the intended destination.

7.31.8. If illness, injury, or death occurs to a passenger or crewmember, the circumstances must be described in detail and the actions taken should be included on the Report form upon return to home base. The Team Leader Cabin Services attendance shall submit a report on the nature or extent of a passenger injury or illness to the Captain. This report shall be submitted in conjunction with the Flight Crew Report form.

7.32. CARE AND STOWAGE OF THE BODY

7.32.1. This is covered in detail in the cabin crew manual. Relevant points for flight crew are;

a) Note the time when all active treatment has stopped.

b) If available, a cabin crew member should collect the mortality kit from the Emergency Medical Kit.

c) The zipper of the body bag must not be completely closed; the head must be exposed. The head may be covered with a light blanket.

d) The body should preferably be moved to an area of privacy e.g. Galley. It may be more convenient to stow the body near an exit during cruise. Where the body obstructs an exit or other available equipment, it may be placed in a seat.

e) Do not place the body in the lavatory.

f) Do not place the body in the Crew Rest Compartment.

7.33. FORMALITIES

7.33.1. Inform IOCC, who can start making arrangements with the Airport Services Manager.

7.33.2. In all cases the Ethiopian Airport Services Manager or his/her deputy at the airfield of landing should be informed via the Ground Handling Agent that there is a 'presumed death on board' case, who in turn, will notify the authorities.

7.33.3. Crew should liaise with the Airport Services Manager to complete all required documentation.

7.33.4. The cabin crew must record when all active resuscitation ceased.

7.33.5. Captains must document the incident on ASR.

7.33.6. A Doctor or Airfield Medical Officer should be available on arrival to certify death. The body should remain on board until all passengers have de-planned. Where passengers are required to remain on board, use discretion. Station staff will advise.

7.33.7. The Airport Services Manager is responsible for checking that the 'Death Certificate' requirements are complied with.

7.34. MEDICAL ASSISTANCE ON ARRIVAL

7.34.1. When a medical problem develops in flight and the captain deems it advisable to have medical assistance available on arrival, a message should be addressed to the Station Manager at the arrival station. It should fully describe the person's condition.

7.34.2. Medical expenses other than those for injury aboard the airplane are the passengers' responsibilities. Accordingly, requests for medical aid on arrival should indicate that it is a passenger request whenever this is the case.

SECTION 2.8 HANDLING PASSENGERS

7.35. ASSISTANCE ON ARRIVAL AT ADD

Flight Crew members are required to provide the following information to the local ATC and IOCC;

- Name of patient
- Age
- Sex
- General condition (e.g. conscious, unconscious, disorientated, violent, etc.)
- Chief complaint (e.g. difficulty in breathing, chest pain, bleeding, vomiting, abdominal pain, fever, chills, etc.)

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SECTION 2.9 TRANSPORTATION OF SPECIAL CARGO REGULATIONS

SECTION 2.9 TRANSPORTATION OF SPECIAL CARGO REGULATIONS

SECTION 2.9 TRANSPORTATION OF SPECIAL CARGO REGULATIONS

1.0 PURPOSE

1.1 HAZARDOUS MATERIALS

The purpose of this procedure is to establish guidelines on handling of hazardous materials and/or dangerous goods.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Director Flight Training & Standards

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

Dangerous goods may be carried provided that the indicated net quantities are not exceeded and the applicable packing instructions according to the IATA Dangerous Goods Regulations are strictly adhered to. As part of aircraft library, the latest IATA Dangerous Goods Regulation Manual and Emergency Response Guidance for aircraft incidents involving dangerous goods shall have to be referred.

4.1. No person shall carry dangerous goods in an aircraft registered or operated in Ethiopia except: with the written permission of the authority and subject to any condition the authority may impose in granting such permission and in accordance with the technical instructions for the safe transport of dangerous goods by air issued by the council of international civil aviation organization and with any variations to those instructions that the authority may from time to time mandate and provide notification to ICAO. (ECARAS 8.5.1.27)

4.2. Dangerous goods must not be loaded in the aircraft or carried in passenger or as crew or as checked-in or carry-on baggage, except those allowed in the IATA dangerous goods manual sub section 2.3.

4.3. For Explosives of Division 1.4S, shipped as cargo and for Sporting weapons and ammunition permitted in or as checked baggage.

4.4. Details on provisions for Dangerous Goods carried by passengers or crew refer to IATA dangerous goods manual sub section 2.3.A.

4.5. Weapons and Ammunition that is not in the possession of law enforcement officers or other authorized person must be stowed in an area that is inaccessible to any person & shall not be carried inside cabin and/or flight deck. Such items shall be accepted for transportation only when placed inside cargo hold.

4.6. Law enforcement officers and other authorized persons are allowed to carry weapons in the passenger cabin. If weapons are going to be carried in the passenger cabin the following procedures will be followed;

1. NISS will review and allow carriage of weapons by individuals in the cabin compartment.
2. The PIC shall be notified prior to the departure of a flight.
3. The number of authorized armed persons on board the aircraft will be notified to the PIC.
4. The individuals name & seat location on the aircraft should also be notified to the PIC.
5. For more information, refer to Ethiopian Airlines Security Program Manual.

4.7. NOTIFICATION TO PILOT-IN-COMMAND (NOTOC)

4.7.1. The pilot-in-Command should be informed as early as practicable before departure of the aircraft, with written information (NOTOC) concerning dangerous goods that are to be carried except those items listed on IATA Dangerous Goods Table 9.5.A. For multi sector flights involving a crew change each station issuing a NOTOC shall prepare a separate NOTOC for each station en-route. The NOTOC must specify at least the following:

a) The air waybill number;
b) The proper shipping name (supplemented with the technical name (s) if appropriate and UN number;
c) The class or division, and subsidiary risk (s) corresponding either to the subsidiary risk label (s) applied of in the case of class 2 and in the case of class 1, the compatibility group;
d) The sample shown on the dangerous goods transport document;
e) The number and exact loading location of packages;
f) The net quantity or gross mass if applicable, of each package, except that this does not apply to radioactive material or other dangerous goods where the net quantity or gross mass if not required on the dangerous goods transport document;
g) For radioactive materials the number of packages, over pack or freight containers, their category, and their transport index if applicable and their exact loading location;
h) Whether the package must be carried on cargo aircraft only;
i) The aerodrome which the package (s) is to be unloaded; and

j) Where applicable, an indication that the dangerous goods are being carried under state authorization.

k) Emergency Drill Code.

4.7.2. The information to the pilot in command must also include information that there is no evidence that any damaged or leaking packages have been loaded on the aircraft.

4.7.3. The written information to the pilot-in-command must be readily available to the pilot in command during flight.

4.7.4. The information to the pilot in command should be presented on a dedicated form and should not be by means of air waybills, dangerous goods transport documents, invoice, etc.

4.8. CLASSIFICATION OF DANGEROUS GOODS

4.8.1. CLASSES AND DIVISIONS

a. CLASS 1 - EXPLOSIVES

- Class 1 is divided into six divisions:

DIVISION 1.1	SUBSTANCES HAVING MASS EXPLOSION HAZARD
DIVISION 1.2	PROJECTION HZARD BUT NOT A MASS EXPLOSION HAZARD
DIVISION 1.3	SUBSTANCES HAVING FIRE HAZARD AND EITHER A MINOR BLAST OR MINOR PROJECYION HAZARD OR BOTH, BUT NOT A MASS EXPLOSION HAZARD
DIVISION 1.4	SUBSTANCES HAVING NO SIGNIFICANT HAZARD (ONLY A SMALL HAZARD)
DIVISION 1.5	VERY INSENSITIVE SUBSTANCES, HAVING A MASS EXPLOSION HAZARD
DIVISION 1.6	EXTREMELY INSENSITIVE ARTICLES WHICH DO NOT HAVE A MASS EXPLOSION HAZARD

b. CLASS 2 - GASES, COMPRESSED, LIQUEFIED, DISOLVED UNDER PRESSURE OR DEEPLY REFRIGERATED

c. CLASS 3 - FLAMMABLE LIQUIDS

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d. CLASS 4 - FLAMMABLE SOLIDS; SUBSTANCES LIABLE TO SPONTANEOUS COMBUSTION; SUBSTANCES WHICH, ON CONTACT WITH WATER, EMIT FLAMMABLE GASES

- Class 4 is divided into three divisions:

DIVISION 4.1 FLAMMABLE SOLIDS

DIVISION 4.2 SUBSTANCES LIABLE TO SPONTANEOUS COMBUSTION

DIVISION 4.3 SUBSTANCES WHICH, ON CONTACT WITH WATER, EMIT FLAMMABLE GASES (DANGEROUS WHEN WET)

e. CLASS 5 - OXIDIZING SUBSTANCES AND ORGANIC PEROXIDES

- Class 5 is divided into two divisions:

DIVISION 5.1 OXIDIZING SUBSTANCES

DIVISION 5.2 ORGANIC PEROXIDE

f. CLASS 6 - POISION (TOXIC) AND INFECTIOUS SUBSTANCES

- Class 6 is divided into two divisions:

DIVISION 6.1 POISONOUS SUBSTANCES

DIVISION 6.2 INFECTIOUS SUBSTANCES

g. CLASS 7 – RADIOACTIVE MATERIALS

h. CLASS 8 - CORROSIVES

i. CLASS 9 – MISCELLANEOUS DANGEROUS GOODS

4.9. TRANSPORTATION OF SPECIAL CARGO

Whenever there is a plan that Special Cargo is going to be carried on board Ethiopian flight, the flight crew should be informed in advance so that they will take the necessary precaution during loading and during flight.

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4.10. FORBIDDEN GOODS

4.10.1 Explosives that initiate or decompose when subjected to a temperature of 75^{0}c (167^{0}f) for 48 hours.

4.10.2 Explosives containing both chlorates and ammonium salts.

4.10.3 Explosives containing mixtures of chlorates with phosphorus.

4.10.4 Solid explosives, which are classified as extremely sensitive to mechanical shock.

4.10.5 Liquid explosives, which are classified as moderately sensitive to mechanical shock.

4.10.6 Any article or substance, which is liable to produce a dangerous evolution of heat or gas under the conditions normally encountered in air transport.

5.0 DEFINITION

NOTOC-means notice to the captain

Dangerous Goods: Articles or substances which are capable of posing a risk to health, safety, property or the environment and which are shown in the list of dangerous goods in the ICAO Technical Instructions (see definition below) or which are classified according to those Instructions.

Dangerous Goods Accident: An occurrence associated with and related to the transport of dangerous goods which results in fatal or serious injury to a person or major property damage.

Dangerous Goods Incident: An occurrence, other than a dangerous goods accident, associated with and related to the transport of dangerous goods, not necessarily occurring on board an aircraft, which results in injury to a person, property damage, fire, breakage, spillage, leakage of fluid or radiation or other evidence that the integrity of the packaging has not been maintained. Any occurrence relating to the transport of dangerous goods which seriously jeopardizes an aircraft, or its occupants is deemed to constitute a dangerous goods incident.

Dangerous Goods Transport Document: A document specified by the ICAO Technical Instructions for the Safe Transportation of Dangerous Goods by Air. It is completed by the person who offers dangerous goods for air transport and contains information about those dangerous goods. The document bears a signed declaration indicating that the dangerous goods are fully and accurately described by their proper shipping names and UN numbers (if assigned) and that they are correctly classified, packed, marked, labeled and in a proper condition for transport.

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6.0 RESPONSIBILITY

6.1. ETHIOPIAN AIRLINES RESPONSIBILITIES

Ethiopian Airlines must not accept a package or over pack containing dangerous goods unless it has been inspected, found to be properly marked and labeled and determined that there are no holes, leakage or other indications that its integrity has been compromised.

6.2 All Ethiopian staff with responsibilities in relation to the carriage of Dangerous Goods are trained in accordance with the requirements. Flight crew members are not responsible for duties and responsibilities related to the preflight inspection of accessible dangerous goods.

6.3 Flight Crew duties & responsibility regarding Dangerous Goods;

6.3.1 Signature of NOTOC to indicate receipt of information.

6.3.2 If an inflight emergency occurs, as soon as the situation permits advice to the appropriate Air Traffic Services Unit the details of the dangerous Goods on board. The PIC should give the information as listed on the NOTOC and as a minimum state:

- Proper shipping names of goods transported and UN number.
- The Class and Division.
- For Class 1 articles the compatibility group and subsidiary risks.
- The gross weight and location on board the aircraft.

And finally follow the Emergency Response Guidance for Aircraft Incidents Involving Dangerous Goods which contains general guidance on the procedures to be used for each class and Division of Dangerous Goods involving spillage, leakage or fire.

7.0. PROCEDURE

7.1**.** The PIC is to be notified by Marketing and IOCC whenever a dangerous article is carried onboard the airplane. This notification shall normally be accomplished by a hazardous materials manifest that requires a signature by the PIC prior to departure.

7.2. MAGNETIZED MATERIALS

Magnetized materials can be carried only in the aft hold.

7.3. BATTERY DRIVEN WHEELCHAIRS

7.3.1. Before loading wheelchairs with a dry battery or with a battery filled with a non- spillable gel it must be ensured that:

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a) The battery is securely attached to the wheelchair

b) The battery is disconnected

c) The battery poles are insulated in order to prevent short- circuits.

7.3.2 So called "high tech" wheelchairs often have integrated dry batteries, which cannot be disconnected at the battery terminals. They have a removable control unit, which if it is removed, disconnects the battery

7.4. DRY ICE

7.4.1. A maximum of 200kgs of dry ice per package is allowed.

7.4.2. Transit and destination stations must ventilate the compartments before entering live animals and dry ice must not be loaded together in the same compartment.

7.4.3. Dry ice may be carried for cooling perishable goods or as cargo. The following are limitation of **UN1845 DRY ICE** per aircraft for shipments.

Aircraft	Dry ice capability (per aircraft)	Compartment restrictions
Passenger Aircraft		
B777-300	1000KG	Total weight in FWD + AFT + BLK
B777-200	1000KG	
A350-900	1000KG	
B787-9	800KG	
B787-8	800KG	
B767-200	400KG	
B737-8 MAX	250KG	
B737-800	250KG	
B737-700	250KG	
Q400	20KG	Total in BLK
Freighter Aircraft		
B737-800F	250KG	Total in MAIN DECK +FWD +AFT
B777-200	5700KG	Total in MAIN DECK + FWD + AFT + BLK

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NOTE:

1. The maximum total weight of Dry ice loadable on ET777F is 5700KG (1,000Kgs in the Lower deck and 4,700kgs in Main deck).
2. The maximum total weight of Dry ice loadable on ET passenger flight is 1,000kgs (no compartment wise weight restriction applies) and this limit can never be exceeded.
3. Ethiopian B787-8 passenger aircraft under registration **ET-ASG, ET-ASH** and **ET-ASI** are not equipped with FCAC system (Ventilation and Temperature controlling system) and maximum allowed limit is 400KGs.
4. Ethiopian A350-900 passenger aircraft under registration **ET-ATQ, ET-ATR, ET-ATY, ET-AUA, ET-AUB, ET-AUC** and **ET-AVB** are not equipped with ventilation and temperature controlling system and maximum allowable limit is 400KGs.
5. When dry ice is used as a refrigerant for other than dangerous goods loaded in a unit load device the quantity limits per package shown in columns J and L section for DRY ICE do not apply. In such case, the unit load device must be identical to the operator and must allow the venting of the carbon dioxide gas to prevent a dangerous buildup of pressure. (special provision A151)

7.5. RADIOACTIVE MATERIAL

7.5.1 There are three categories of radioactive material. For radioactive material of category I (RRW) there are no special restrictions. However, for the carriage of radioactive material of category II and III (RRY) the following restrictions have to be observed:

a) The permitted total of transport indices must not be exceeded

b) The material must be loaded in the correct loading area

c) The loading height (distance from compartment floor to top of packages) is limited in lower compartments of passenger aircraft.

7.5.2. The actual amount to transport indices for each package must be entered on the hazard labels of the respective package by the shipper (except RRW). If several packages with radioactive material are carried on the same flight, the total amount of transport indices of all packages shall be taken as a basis.

7.5.3. Packages with radioactive material of category II and III (RRY) without entry transport indices on the label or with broken seals (if any) must not be carried.

7.5.4 Radioactive materials must be restrained so that movement is not possible under any circumstances during flight.

7.5.5. The transport of fissile materials is not allowed on Ethiopian aircraft.

7.5.6. The respective load limitations for radioactive materials onboard an aircraft are shown on the following tables.

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TABLE 1

AIRCRAFT		MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE			
	LOADING HEIGHT	FORWARD HOLD		AFT HOLD	
B737/700-800		Net section 11	Net section 13	Net section 41	Net section 43
	HT 1	1.0	1.0	1.0	1.0
	HT 2	3.0	3.0	3.0	3.0

NOTE: For minimum distance between packages in different net sections: please refer to the current Table of separation of Radioactive Material-Passenger and cargo Aircraft in IATA Dangerous Goods Regulations Manual.

TABLE 2

AIRCRAFT		MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE			
	LOADING HEIGHT	FORWARD HOLD		AFT HOLD	
757 (PAX)		Net section 1	Net section 2	Net section 3	Net section 4
	HT 1	1.0	1.0	1.0	1.0
	HT 2	3.0	3.0	3.0	3.0

NOTE: For minimum distance between packages in different net separations: please refer to the current Table of Separation of Radioactive Material-Passenger and Cargo Aircraft in IATA Dangerous Goods Regulations Manual.

TABLE 3

AIR CRAFT		MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE			
	LOADING HEIGHT	FORWARD HOLD		AFT HOLD	
757F		Net section 1	Net section 2	Net section 3	Net section 4
	HT 1	1.0	18.0	300.0	300.0
	HT 2	3.0	18.0	300.0	300.0

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NOTE: For minimum distance between packages, please refer to the current Table of Separation of Radioactive Material-Cargo Aircraft only in IATA Dangerous Goods regulations manual.

TABLE 4

AIRCRAFT	LOADING HEIGHT	MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE			
		FORWARD HOLD		AFT HOLD	
B767F		Positions 11P 11L, 11R together	Position 23P 24L, 24R together	Positions 31L, 31R together	Positions 43L, 43R together
	HT 1	1.0	1.0	1.0	1.0
	HT 2	3.0	3.0	3.0	3.0
	HT 3	8.0	8.0	8.0	7.0

NOTE: For minimum distance between packages in different net sections: please refer to the current table of separation of Radioactive Material-Passenger and Cargo Aircraft in IATA Dangerous Goods Regulations Manual.

TABLE 5

AIRCRAFT	LOADING HEIGHT	MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE				
		FORWARD COMPARTMENT				
B787-800		Positions 11P	Position 12P	Position 13P	Position 21P	Position 22P
	HT 1	1	0	1	0	1
	HT 2	4	0	0	4	0
	HT 3	8	0	0	0	8

TABLE 6

AIRCRAFT	LOADING HEIGHT	MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE
		AFT COMPARTMENT

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		Position 31P	Position 32P	Position 41P	Position 42P
B787-800	HT 1	1	0	1	0
	HT 2	4	0	0	4
	HT 3	8	0	0	0

NOTE: For minimum distance between packages in different net sections: please refer to the current Table of separation of Radioactive Material-Passenger and cargo Aircraft in IATA Dangerous Goods Regulations Manual.

TABLE 7

AIRCRAFT	**LOADING HEIGHT**	**MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE**					
		FORWARD COMPARTMENT					
B777-200 LR		Position 11P	Position 12P	Position 13P	Position 21P	Position 22P	Position 23P
	HT 1	1	0	1	0	1	0
	HT 2	4	0	0	4	0	0
	HT 3	8	0	0	0	8	0

TABLE 8

AIR CRAFT	**LOADING HEIGHT**	**MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE**						
		AFT COMPARTMENT						
B777-200 LR		Positions 31L & R	Positions 32L & R	Positions 33L & R	Positions 41L & R	Positions 42L & R	Positions 43L & R	Positions 44L & R
	HT 1	1	0	1	0	1	0	1
	HT 2	4	0	0	4	0	0	4
	HT 3	8	0	0	0	8	0	0

NOTE: For minimum distance between packages in different net sections: please refer to the current Table of separation of Radioactive Material Passenger and Cargo Aircraft in IATA Dangerous Goods Regulations Manual.

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TABLE 9

AIRCRAFT	LOADING HEIGHT	MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE FORWARD COMPARTMENT							
B777-300 ER		Position 11P	Position 12P	Position 13P	Position 21P	Position 22P	Position 23P	Position 24P	Position 25P
	HT 1	1	0	1	0	1	0	1	0
	HT 2	4	0	0	4	0	0	4	0
	HT 3	8	0	0	0	8	0	0	0

TABLE 10

AIRCRAFT	LOADING HEIGHT	MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE AFT COMPARTMENT					
B777-300 ER		Position 31P	Position 32P	Position 33P	Position 34P	Position 41P	Position 42P
	HT 1	1	0	1	0	1	0
	HT 2	4	0	0	4	0	0
	HT 3	8	0	0	0	8	0

NOTE: For minimum distance between packages in different net sections: please refer to the current Table of separation of Radioactive Material-Passenger and Cargo Aircraft in IATA Dangerous Goods Regulations Manual.

TABLE 11

AIR CRAFT	LOADING HEIGHT	MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE MAIN DECK													
B777-200 LRF		A L/R	B L/R	C L/R	D L/R	E L/R	F L/R	G L/R	H L/R	J L/R	K L/R	L L/R	M L/R	P L/R	R
	HT-1	0	0	0	0	15*	15*	15*	15*	15*	15*	15*	15*	15*	15*
	HT-2	0	0	0	0	15*	15*	15*	15*	15*	15*	15*	15*	15*	15*
	HT-3	0	0	0	0	15*	15*	15*	15*	15*	15*	15*	15*	15*	15*

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TABLE 12

AIRCRAFT		MAXIMUM T.I AND ALLOWED LOADING POSITIONS/ARE									
	LOADING HEIGHT	LOWER DECK									
B777-200 LRF		11P	12P	13P	21P	22P	23P	31P	32P	41P	42P
	HT-1	0	0	0	0	0	30*	30*	30*	30*	30*
	HT-2	0	0	0	0	0	30*	30*	30*	30*	30*
	HT-3	0	0	0	0	0	30*	30*	30*	30*	30*

NOTE: For minimum distance between packages, please refer to the current Table or Separation of Radioactive Material-Cargo Aircraft only in IATA Dangerous Goods Regulations Manual.

* Total TI at each position may be increased to the point, where the sum does not exceed the 150.

Total TI in cargo aircraft must not exceed 300.

TRANSPORTATION OF RADIOACTIVE MATERIAL ON Q400 AIRCRAFT IS NOT ALLOWED.

TABLE 13

AIRCRAFT		MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE				
	LOADING HEIGHT	FORWARD COMPARTMENT				
B787-900		TBD	TBD	TBD	TBD	TBD
	HT 1	TBD	TBD	TBD	TBD	TBD
	HT 2	TBD	TBD	TBD	TBD	TBD
	HT 3	TBD	TBD	TBD	TBD	TBD

TABLE 14

AIRCRAFT		MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE
	LOADING HEIGHT	AFT COMPARTMENT

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B787-900		TBD	TBD	TBD	TBD
	HT 1	TBD	TBD	TBD	TBD
	HT 2	TBD	TBD	TBD	TBD
	HT 3	TBD	TBD	TBD	TBD

TABLE 15

AIRCRAFT		**MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE**													
	LOADING HEIGHT	**MAIN DECK**													
A350-900		TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	HT-1	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	HT-2	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	HT-3	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

TABLE 16

AIRCRAFT		**MAXIMUM T.I AND ALLOWED LOADING POSITIONS /ARE**													
	LOADING HEIGHT	**MAIN DECK**													
A-350-900		TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
		TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
		TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
		TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

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7.6. MARKING LABELING

It is important that marking and labeling be done in accordance with the specifications of IATA Dangerous Goods Regulations and must conform, in shape, color, format, symbol and text, to the specimen design given in this section. The shipper is responsible for all necessary marking and labeling of each over pack containing dangerous goods, in compliance with IATA Dangerous Goods Regulations.

7.7. DANGEROUS GOODS MANAGEMENT

7.7.1 Ethiopian shall not transport dangerous goods unless approved by ECAA. Ethiopian shall comply with the provisions contained in the ICAO Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO Doc. 9284) on all occasions when dangerous goods are carried, irrespective of whether the flight is wholly or partly within outside the territory of Ethiopia. When dangerous goods are to be transported outside the territory of Ethiopia, Ethiopian shall review and comply with the appropriate variations noted by contracting states contained in the ICAO Technical Instructions. (ECARAS 9.5.1.3)

7.7.2. Articles and substances are excluded from dangerous goods provided they are: (ECARAS9.5.1.3)

a) Required to be aboard the aircraft for operating reasons.
b) Carried as catering or cabin service supplies.
c) Carried for use in flight as veterinary aid.
d) Carried for use in flight for medical aid for a patient, provided.

I. Gas cylinders have been manufactured particularly for that purpose.
II. Drugs, medicines and other medical matter are under the control of trained personnel during the time when they are in use in the flight.
III. Equipment containing wet cell batteries is kept and secured in upright position to prevent spillage of the electrolyte.
IV. Proper provision is made to stow and secure all the equipment during takeoff and landing and at all other times when deemed necessary by the PIC.
V. They are carried by passengers or crewmembers.

7.7.3. Ethiopian shall take all reasonable measures to ensure that articles and substances that are specifically identified by name or generic description in the ICAO Technical Instructions as being forbidden for transport under any circumstances are not carried on any aircraft. (ECARAS 9.5.1.4)

7.8. DANGEROUS GOODS LABELING AND MARKING (ECARAS 9.5.1.7)

The operator shall ensure that packages, over packs, and freight containers are labeled and marked as specified in the ICAO Technical Instructions.

7.9. DANGEROUS GOODS TRANSPORT DOCUMENT (ECARAS 9.5.1.8)

Ethiopian shall ensure dangerous goods are accompanied by dangerous goods transport document unless otherwise specified by the ICAO Technical Instructions.

7.10. HANDLING OF DANGEROUS GOODS INCIDENTS

7.10.1 Ethiopian shall not carry any dangerous goods in an aircraft registered in Ethiopia or operated in Ethiopia except with the written permission from the Authority and in accordance with the ICAO Technical Instructions. (ECARAS 8.5.1.27)

7.10.2 Any dangerous goods accidents and incidents shall be reported to the appropriate authority of the state of the operator and the state in which the accident or incident occurred within 72 hours of the event. (ECARAS 9.5.1.15)

7.10.3 In the event of accident or serious incident of an aircraft carrying dangerous goods as cargo, the operator must provide information, without delay, to emergency services responding to the accident or serious incident about the dangerous goods on board, as shown on the NOTOC.

7.10.4 The appropriate authority of the state of the operator and the state in which the accident or incident occurred shall also be notified as soon as possible.

7.10.5 In the event of incident of an aircraft carrying dangerous goods as cargo, the operator must provide information, if requested to do so, without delay, to emergency services responding to the accident or serious incident about the dangerous goods on board, as shown on the NOTOC.

7.10.6 Follow the appropriate aircraft emergency procedures for fire or smoke removal.

- No smoking sign … ON
- Consider landing as soon as possible
- Consider turning off non-essential electrical power
- Determine source of smoke/fire/fumes
- For dangerous goods incidents in the passenger cabin, see cabin crew checklist and co-ordinate cockpit/cabin crew actions.
- Determine emergency response drill code (Emergency Response Guidance)

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- Use guidance from aircraft emergency response drills chart to help deal with incident.
- If time available, notify ATC of at least UN number of any dangerous goods being carried

7.11. AFTER LANDING

7.11.1 Disembark passengers and crew before opening any cargo compartment doors

Inform ground personnel/emergency services of nature of item and where stowed. Make appropriate entry in maintenance log.

7.12. LIVE ANIMALS

7.12.1 The Captain must be advised of the species, location and quantity of all live cargo on board the aircraft. The flight crew must be notified via a "special load notification to captain" (NOTOC) or similar form (in printed format or by electronic means) of any live animal load and of the required actions (hold temperature, light, ventilation, etc…).

7.12.2. Animals instinctively fear the strange environment encountered during transportation. Therefore, in transporting live animals, there are a number of basic principles with which Ethiopian and the shipper should comply, as these affect the welfare and comfort of the animal. This in turn, has a bearing on the animal's behavior during air transportation and is, therefore, of the utmost importance to all concerned.

7.12.3. Ethiopian will accept animal carriage only in a suitable clean container, which must be leak-proof and escape-proof. The container shall be constructed in such a manner that which will permit handling staff to give the necessary attention to the animal without risk of the animals harming them. For general transport purposes, Ethiopian will carry animals only in closed containers; however, carriage in open stalls must be especially arranged. Adequate litter must be provided to absorb excrement. The use of straw as absorbent material should be avoided because of the restrictions upon the importation of straw imposed by a large number of countries. At least one IATA LIVE animals label or tag, must be properly completed, and attached to each live animal container, unless otherwise stated in the individual container requirements. In addition, orientation labels (THIS WAY UP) shall be placed, where possible, on all four sides of the container to indicate the upright position of the container.

7.12.4. Ethiopian shall accept only animals that appear to be in good health and condition and fit to travel to the final destination.

7.12.5. Ethiopian requires the shipper to declare when animals are pregnant or have given birth in the last 48 hours. Mammals that are declared to be pregnant will not be accepted unless accompanied by a veterinary certificate certifying that the animal is fit to travel and there is no risk of birth occurring during the journey. Animals should be loaded as close as possible to the aircraft, scheduled departure. They should not be left standing unnecessarily in the open or standing in the aircraft when the departure is delayed. Animals should not be stowed in the vicinity of other cargo that can do them harm: such as shipments packed in dry ice, cryogenic liquids, poisons class “B”, irritants so that there is sufficient air for normal breathing. Other cargo should not block the ventilation apparatus of containers. Precautions should be taken to prevent other cargo from falling onto the animal containers. Unnecessary tilting and jolting of containers should be avoided.

7.12.6. In addition, live animals should not be loaded in close proximity of consignments of Argon, liquid Nitrogen, liquid in non-pressurized containers and poisonous articles.

7.12.7. The PIC should be advised of the species, location and quantity of all live cargo on board the aircraft.

7.12.8. Different governments throughout the world have different regulations regarding advanced import permit, veterinary health certificate, veterinary examination, quarantine, transshipment requirements or prohibition restrictions, which may also include the food provided for the animals, therefore before live animal carriage is performed the above requirements should be addressed.

7.13. PETS ON PASSENGER FLIGHTS

7.13.1. Pets can be carried in the passenger cabin or cargo hold depending on their weight and size.

7.13.2. A maximum of 2 pets in economy and 1 in C-9 are accepted in the passenger cabin in a container size not more than 55X40X50cms and weight not to exceed 8 KGS including the transport container.

7.13.3. Dogs trained to assist the blind or deaf, if accompanying a passenger who is dependent on the dog, may be carried in the passenger cabin if the dog is properly harnessed and muzzled and accompanied by a valid health and vaccination certificate and entry permit and other documents required by countries of departure, transit and entry.

7.13.4. The animal may leave the container during flight, provided it is carried on the owner’s lap and returned into the container during meal, liquor services, taxi, takeoff and landing.

7.13.5. The container must be placed in front of the passenger seat; the adjacent aisle must be unobstructed.

7.13.6. Sales office has confirmed the reservation for the carriage in the cabin. However, if a passenger brings a pet without prior reservation such pet may be accepted if it is in a container.

7.13.7. Station personnel must notify the PIC. The PIC may refuse carriage of a live animal if any of the above is not complied with, or if the safety of the flight is affected. In such an event he is required to submit a written report to Director Flying Operations.

7.13.8. For flights to and from USA, refer to Nondiscrimination on the basis of Disability Manual onboard the aircraft.

7.14. VALUABLES GOODS

7.14.1. Details for carriage of valuable goods on Ethiopian flight are contained in Load Control Section 09 of Sales and Services Manual. However, A portion of that policy is included in this section for the information of the flight crew.

7.14.2. Valuable goods shall be loaded last, just prior to closing of cabin door during preparation for takeoff, in the appropriate section of the aircraft.

7.14.3. At transit stations valuable goods should be removed by station agent for safe custody and reloaded again just prior to closing of cabin door during preparation for take-off, in the appropriate section of the aircraft.

7.14.4. At destination station valuable goods shall be kept in safe custody of concerned station agents until delivery.

7.14.5. The Pilot-in-Command should be informed whenever valuable goods are to be carried on board their flight.

7.15. AIRCRAFT LOADING AND HANDLING MANUAL (ECARAS 9.3.1.15)

Aircraft loading and handling manual, acceptable to ECAA, shall be provided for the use of the flight crew members, ground handling personnel assigned on operational control functions during their duties. This manual shall be specific to the aircraft type and variant and shall contain the procedures and limitations for serving and loading of the aircraft.

SECTION 2.9 TRANSPORTATION OF SPECIAL CARGO REGULATIONS

7.16. SUBDUING LARGE ANIMALS

7.13.2. The following procedure should be used on a cargo flight when a large animal becomes uncontrollable and could endanger the flight.

a) All persons on board shall be instructed to don oxygen masks and use 100 percent oxygen.

b) Turn the weather radar off. (Arcing in the radar set may occur at high cabin altitudes).

c) Raise the cabin altitude to 25,000 ft.

7.13.3. The animal should be unconscious in approximately two minutes. Re-pressurize the airplane after that animal has been restrained.

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SECTION 2.10 RULES OF THE AIR

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SECTION 2.10 RULES OF THE AIR

1.0. PURPOSE

The purpose of these rules of the air procedure is to establish guidelines to comply with the rules of the air contained in ICAO Annex 2.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight crewmembers

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. TERRITORIAL APPLICATION

The Rules of the Air shall apply to airplanes bearing the nationality and registration marks of ICAO Contracting States, wherever they may be, to the extent that they do not conflict with the rules published by the State having jurisdiction over the territory over flown.

4.2. COMPLIANCE

4.2.1. The operation of an airplane, either in flight or in the maneuvering area of an airport shall be in compliance with the General Rules and, in addition when in flight, with either:

a) Visual Flight Rules, or
b) Instrument Flight Rules

4.2.2. All Ethiopian commercial flights shall be conducted under an IFR flight plan in accordance with an IFR clearance and if an instrument approach is used or required it shall be done in accordance with the approach procedures approved or accepted by the state in which the airport of intended landing is located. In this case dispatchers are responsible to file IFR flight plan with the appropriate ATS unit and Flying crew are responsible to obtain IFR clearance in order to confirm the flight is conducted with all of the ATS Services within the controlled airspace.

4.2.3. In addition, Ethiopian conducts certain portion of commercial flights under VFR.

SECTION 2.10 RULES OF THE AIR

NOTE: A pilot may elect to fly in accordance with instrument flight rules in visual meteorological conditions, or he may be required to do so by the appropriate authority.

4.3. PREFLIGHT ACTION

4.3.1. Before beginning a flight, the Captain shall familiarize himself/herself with all available information appropriate to the intended operation. Preflight action for flights away from the vicinity of an airport and for all IFR flights shall include a careful study of available current weather report and forecasts, taking into consideration fuel requirements and alternative courses of action if a flight cannot be completed as planned.

4.3.2. The captain of any airplane registered in the States prior to commencement of flight shall ensure, prior to flight, that:

a) The flight can be safely completed in accordance with the applicable rules and regulations.

b) Each member of the flight crew will comply with the prescribed check system to be used.

4.4. FLYING OVER CITIES, POPULATED AREAS, OR OVER AN OPEN AIR ASSEMBLY OF PERSONS

4.4.1. No airplane shall be permitted to fly at heights below those prescribed by the Civil Aviation Authority in conjunction with other competent authorities in the State except in cases of emergency or by a special permission from such authority.

4.4.2. No airplane shall be flown over cities, populated areas, or over an open air assembly of persons unless it is at such a height as that in the event of an emergency a safe landing can be made without undue hazard to persons or property on the surface this does not apply for normal takeoff or landing. Permission from the Civil Aviation Authority based on approval of General Manager of CAA and the applicable State.

4.4.3. Except when necessary for takeoff or landing, or except by permission from the appropriate authority, a VFR flight shall not be flown:

a) Over the congested area of cities, over settlement, or over open air assembly of persons at a height less than 1000 ft (300 m) above the highest obstacle within a radius of 2000 ft (600 m) from the airplane.

b) Else-where than as specified above, at a height less than 500 ft (150m) above the ground or water.

4.5. DROPPING OR SPRAYING

Nothing shall be dropped or sprayed from an airplane in flight, except in cases of emergency or by permission from the Civil Aviation Authority and approved by other competent authorities in the state.

4.6. TOWING

No airplane or other object shall be towed by an airplane except by permission from the Civil Aviation Authority and approval of another competent authority in the state.

4.7. ACROBATIC OR FORMATION FLIGHTS

No airplane shall be flown acrobatically in formations or in parades except by a prior permission from the Civil Aviation Authority and approval of other competent authorities in the State of operations.

4.8. PROHIBITED AREAS AND RESTRICTED AREAS

4.8.1. Airplanes shall not be flown in prohibited or restricted areas, the particulars of which have been duly published, except in accordance with the conditions of the restrictions or by permission of the state over whose territory the areas are established. The Civil Aviation Authority may prohibit or restrict the flight or airplanes without regard to nationality:

a) Over designated areas in a territory of the State due to military reasons or for reasons of public interest.
b) Over the territory of the State or any part thereof, immediately and temporarily, in exceptional circumstances or for reasons of public interest.

4.8.2. The Civil Aviation Authority may designate danger areas that shall be publicly announced.

4.8.3. If the Captain becomes aware that the airplane under his command is flying over a prohibited area, he shall immediately notify the appropriate air control service unit and shall comply strictly with the instructions issued to him thereby. If he is not able to make such notification, he shall land promptly in the nearest airport in the State outside of the prohibited area and submit a detailed report of the incident.

4.8.4. Whenever an airplane is warned by a competent authority for having flown over a prohibited area, it shall promptly comply with the instructions issued by said authority in extreme circumstances, offending aircraft may be forced to land.

SECTION 2.10 RULES OF THE AIR

4.9. COLLISION AVOIDANCE

4.9.1. PROXIMITY

An airplane shall not be operated in such proximity of other airplanes as to create a collision hazard.

4.9.2. RIGHT-OF-WAY

The airplane having the right-of-way shall maintain its heading and speed, but nothing in these rules shall relieve the Captain from the responsibility of taking action necessary to avert collision. An airplane that is obliged by the following rules to give way to another shall avoid passing over or under the other or crossing ahead of it unless well clear.

4.9.3. APPROACHING HEAD-ON

When two airplanes are approaching head-on or approximately so, and there is danger of collision, each shall alter its course to the right.

4.9.4. CONVERGING

When two aircraft are converging at approximately the same level, the aircraft that has the other on its right shall give way, except in the following cases:

a) Power-driven, heavier-than air airplanes shall give way to airships, gliders, and balloons.
b) Airships shall give way to gliders and balloons.
c) Gliders shall give way to balloons.
d) Power-driven aircraft shall give way to aircraft that are seen to be towing other aircraft or objects.

4.9.5. OVERTAKING

An airplane that is being overtaken has the right-of-way, and the overtaking airplane, whether climbing, descending or in horizontal fight, shall keep out of the way of the other airplane by altering its course to the right, and no subsequent change in the relative positions of the two airplanes shall absolve the overtaking airplane from this obligation until it has completed passing and is clear.

4.9.6 LANDING

An airplane in flight or operating on the ground shall give way to other aircraft landing or in the final stages of an approach to land. When two or more airplanes are approaching airport for the purpose of landing, airplanes at the higher level shall give way to those at the lower level, but the latter shall not take advantage of this rule to cut in front of or over take another which is in the final stages of an approach to land.

4.9.7. EMERGENCY LANDING

An airplane that is aware that another is compelled to land shall give way to that airplane. It is important that vigilance for the purpose of detecting potential collisions be not relaxed on board an airplane in flight and while operating on the maneuvering area of an airport.

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

6.1. The Pilot-in-Command is the primarily responsible person to comply with the rules and regulations stated in this section of the Manual.

6.2. The Captain of an airplane shall be directly responsible of its operation in accordance with the applicable Rules of the Air, except that he may depart from these rules in circumstances that render such departure absolutely necessary in the interest of safety. He shall notify the competent authorities of any such departure.

7.0 PROCEDURE

Refer to Jeppesen Airway Manual for the procedural differences between ICAO and local airspace rules where applicable.

7.1. OPERATION OF AIRPLANE IN THE VICINITY OF AN AIPORT

7.1.1. The Captain of an airplane operating within the area of an airport or in other areas adjacent there to, shall be responsible for the compliance with those rules relating to the use of the airport and air traffic.

7.1.2. An airplane operated on or in the vicinity of an airport shall:

a) Observe other airport traffic for the purpose of avoiding collision.
b) Conform with or avoid the pattern of traffic formed by other airplanes in operation.
c) Make all turns to the left when approaching for a landing and after taking off unless otherwise instructed.
d) Land and take off into the wind unless safety, the runway configuration or air traffic considerations determine that a different direction is preferable.

SECTION 2.10 RULES OF THE AIR

NOTE: Additional rules may apply in airport traffic zones.

7.2. SIMULATED INSTRUMENT FLIGHT

7.2.1. An airplane shall not be flown under simulated instrument flight conditions unless:

- Fully functioning dual controls are installed in the airplane, and a competent pilot occupies a control seat to act as safety pilot for the person who is flying under simulated instrument conditions. The safety pilot shall have adequate vision forward and to each side of the airplane or a competent observer in communication with the safety pilot shall occupy a position in the airplane from which his field or vision adequately supplements that of the safety pilot.

7.3. AIR TRAFFIC CONTROL SERVICES AND CLEARANCES

7.3.1 Air traffic services applicable to aircraft operating under IFR within controlled airspace shall include:

a) Maintenance of minimum separation standards;

b) Traffic advisory information;

c) Terrain or obstruction alerting;

d) Strategic route planning;

e) Automatic flight plan closure at airports with functioning control towers.

7.3.2 An air traffic control clearance shall be obtained prior to operating any Ethiopian flight. A request for such clearance shall accompany the submission of a flight plan to ATC.

7.3.3 The use of standard radio phraseology when communicating with ATC is mandatory at all times. This includes, as a minimum;

a) To use call sign whenever addressing ATC and also during read back.

b) Whenever ATC gives clearance it must be accepted and read-back for confirmation.

7.3.4. Whenever a clearance is given at least two flight crew members Shall monitor and confirm clearances to ensure a mutual (flight crew) understanding of clearances accepted, not to miss or misunderstood the given clearances that could pose a safety risk to the flight;

a) Whenever the operation is conducted-in traffic congested areas, mountainous terrain and high work load environments.

b) The clearance includes heading, altitude/flight level, route/waypoint changes, frequency changes during critical phases of flight and instructions for any operation on or near a runway.

c) Instruction for holding short of a runway is given.

7.3.5 If an ATC clearance is not satisfactory to the Captain, he may request and, if practicable, will be issued an amended clearance.

7.3.6 Flight crew members shall clarify clearances with ATC whenever any flight crew member is in doubt regarding the clearance or instruction received.

7.3.7 Whenever an airplane has requested a clearance involving priority, a report explaining the necessity for such priority shall be submitted, if requested, by the appropriate ATC unit.

7.3.8 A flight plan may cover only that portion of the flight which is subject to air traffic control.

7.3.9 A clearance may cover only part of a current flight plan, as indicated in a clearance limit or by reference to specific maneuvers such as taxiing or landing.

7.4. ADHERENCE TO ATC CLEARANCE

7.4.1. PIC shall obtain an ATC clearance before operating a controlled flight, or a portion of a flight as a controlled flight. (ECARAS 8.8.2.1) When an ATC clearance has been obtained, no PIC may deviate from the clearance, except in an emergency. (ECARAS 8.8.2.2)

7.4.2. All pilots of a VFR flight shall obtain and comply with ATC clearances and maintain a listening watch before and during operations: (ECARAS 8.8.3.5)

a) Within Classes B, C, and D airspace
b) As part of aerodrome traffic at controlled aerodromes and
c) Under Special VFR

7.4.3. Except as provided below, an airplane shall adhere to the current flight plan (or the applicable portion thereof) submitted for a controlled flight unless a request for a change has been made and a corresponding clearance obtained from the appropriate ATC unit.

7.4.5. The Captain shall be responsible for compliance with all ATC clearances and instructions, and he shall not be permitted to depart from their requirements except in cases of emergency necessitating immediate action, in which event, as soon as circumstances permit after taking emergency measures, he shall notify the appropriate ATC unit of the action taken and, when necessary, obtain an amended clearance.

7.4.5. In the event that a flight has deviated from an ATC clearance, the Captain must take the following action:

a) Deviation from Track: If the airplane is off track, action shall be taken forthwith to adjust the heading of the airplane to regain track as soon as practicable.

b) Changes in Estimated Times: ATC shall be notified as soon as possible of any revised estimate of time of arrival at the next reporting point if such revised ETA varies by more than three minutes from the estimate previously provided. The time variance requiring such notification may differ for particular ATC or regional authorities.

7.5. POSITION REPORT

7.5.1. Unless exempted by ATC, a controlled flight shall report as soon as possible to the appropriate ATC unit the time and level of passing each designated compulsory reporting point together with any other required information. Position reports shall similarly be made in relation to additional points when requested by ATC. In the absence of designated reporting points, position reports shall be made at intervals prescribed by ATC.

7.5.2. An IFR flight operating outside controlled airspace and required by ATC to submit a flight plan and comply with the applicable rules, shall report its position as specified above.

NOTE: Airplanes electing to use the air traffic advisory service while operating IFR within specified advisory airspace are, expected to comply with the provisions of section on Air Traffic control clearances of this chapter except that the flight plan and changes thereto are not subjected to clearances, and two-way communications shall be maintained with the unit providing the air traffic advisory service.

7.6. TERMINATION OF CONTROL

When a controlled flight has landed, or is no longer subject to air traffic control service, the appropriate air traffic control unit shall be notified as soon as possible.

7.7. COMMUNICATIONS

7.7.1. An airplane operated as a controlled flight shall maintain continuous listening watch on the appropriate radio frequency of, and establish necessary two-way communication with, the appropriate ATC unit,

7.7.2. When so prescribed by ATC, an IFR flight operating within specified areas or along specified routes outside controlled airspace shall maintain a listening watch on the appropriate radio frequency and establish two-way communication, as necessary, with the ATC unit providing flight information service.

7.7.3. The use of SELCAL or a similar automatic signaling device satisfies the requirement to maintain a listening watch.

7.7.4. Flight crew communication with ATC shall be in the standard phraseologies as prescribed in the Jeppesen manual.

SECTION 2.10 RULES OF THE AIR

7.8. VISUAL FLIGHT RULES

7.8.1. VISUAL METEOROLOGICAL CONDITIONS

No person may operate an aircraft under VFR when the flight visibility is less than, or at a distance from the clouds that is less than that prescribed, or the corresponding altitude and class of airspace in the following table:

Airspace and VMC Minimums*			
Airspace	A***B C D E	F	G
		Above 900m (3,000ft) AMSL or above 300m (1,000 ft) above terrain, whichever is higher	At and below 900m (3,000 ft) AMSL or 300m (1,000 ft) above terrain, whichever is higher
Distance from cloud	1,500 m horizontally 300m (1,000 ft) vertically		Clear of cloud and in sight of the traffic
Flight visibility	8 km at and above 3,050 m (10,000 ft) AMSL		5km**

*When the height of the transition altitude is lower than 3,050 in (10,000 ft) AMSL, FL 100 should be used in lieu of 10,000 ft.

** When so prescribed by the appropriate ATC authority:

- lower flight visibilities to 1,500 m may be permitted for flights operating: at speeds that, in the prevailing visibility, will give adequate opportunity to observe other traffic or any obstacles in time to avoid collision; or in circumstances in which the probability of encounters with other traffic would normally be low, e.g. in areas of low volume traffic and for aerial work at low levels.

- Helicopters may be permitted to operate in less than 1,500 m flight visibility, if maneuvered at a speed that will give adequate opportunity to observe other traffic or any obstacles in time to avoid collision.

*** The VMC minima in Class A airspace are included for guidance to pilots and do not imply acceptance of VFR flights in Class A airspace.

7.8.2 No person may land or takeoff an aircraft under VFR from an aerodrome located within a control zone, or enter the aerodrome traffic zone or traffic pattern airspace unless the:

a) Reported ceiling is at least 450 m (1,500 ft); and

b) Reported ground visibility is at least 5 km; or, except when a clearance is obtained from ATC.

7.8.3. No person may land or takeoff an aircraft or enter the traffic pattern under VFR from an aerodrome located outside a control zone, unless VMC conditions are at or above those indicated on the above table.

7.8.4. SPECIAL VFR OPERATIONS

(a) No person may conduct a Special VFR flight operation to enter the traffic pattern, land or takeoff an aircraft under Special VFR from an aerodrome located in Class B, Class C, Class D or Class E airspace unless authorized by an ATC clearance, the aircraft remains clear of clouds and the flight visibility is at least 1.5 km (1 statute mile).

(b) No person may conduct a Special VFR flight operation in an aircraft between sunset and sunrise unless the PIC is current and qualified for IFR operations and the aircraft is qualified to be operated for IFR flight.

7.8.5. VFR CRUISING ALTITUDES

(a) Each person operating an aircraft in level cruising flight under VFR at altitudes above 900 m (3,000 ft) from the ground or water, shall maintain a flight level appropriate to the track as specified in the table of cruising level.

(b) Paragraph (a) does not apply when otherwise authorized by ATC, when operating in a holding pattern, or during maneuvering in turns.

7.8.6 A T C CLEARANCES FOR VFR FLIGHTS

(a) Each pilot of a VFR flight shall obtain and comply with ATC clearances and maintain a listening watch before and during operations within Classes B, C and D airspace, as part of aerodrome traffic at controlled aerodromes and Under Special VFR.

7.8.7 VFR FLIGHTS REQUIRING ATC AUTHORISATION

(a) Unless authorized by the appropriate ATC authority, no pilot may operate in VFR above FL 200 or at transonic and supersonic speeds.

(b) ATC authorization for VFR flights may not be granted in areas where a VSM of only 300m (1,000 ft) is applied above FL 290.

(c) No person shall operate night VFR flight between sunset and sunrise.

7.9. INSTRUMENT FLIGHT RULES

7.9.1. Ethiopian shall not (ECARAS 7.2.1.4)

a) Operate an airplane under IFR over route that cannot be navigated by reference to visual landmarks, unless the airplane is equipped with navigation equipment in accordance with the requirements of air traffic service in the area(s) of operation.

b) Conduct single pilot IFR operations unless the airplane is equipped with an autopilot with at least altitude hold and heading mode.
c) Operate an airplane under IFR unless it is equipped with an audio selector panel accessible to each flight crewmembers.
d) Conduct single pilot IFR or night operations in commercial air transport operations unless the airplane is equipped with a headset with boom microphone or equivalent and a transmit button on the control wheel.

7.10. CHANGE FROM VFR FLIGHT TO IFR FLIGHT

7.10.1. The pilot of an airplane wishing to change from visual flight rules to instrument flight rules shall:

a) If a flight plan was submitted, communicate necessary changes of the flight plan to ATS,
b) Submit a flight plan to ATS and obtain ATC clearance using HF RTF at least 20 min or VHF RTF 10 min prior to the point of entry into the control zone or control area. Thereafter, they should comply with ATC clearance for the entire flight in the control zone.

7.10.2. When it becomes evident that, due to weather deterioration, flight in Visual Meteorological Conditions (VMC) in accordance with the current flight plan will not be practicable, the pilot of an airplane operated as a controlled VFR flight shall:

a) Request an amended clearance enabling the airplane to continue in VMC to destination or to an alternative airport, or to leave the controlled airspace (instrument/visual) concerned.
b) If no clearance can be obtained, continue to operate in VMC and notify ATC of any action being taken either to leave the controlled airspace (instrument/visual) or to land at the nearest suitable airport.
c) If operating within a control zone, request authorization to operate as a special VFR flight.
d) Request clearance to operate in accordance with instrument flight rules.

7.11. CHANGE FROM AN IFR FLIGHT TO A VFR FLIGHT

7.11.1. The captain of an airplane electing to change the status of its flight from compliance with instrument flight rules to compliance with visual flight rules shall, notify ATC that the IFR flight is canceled and give the changes to be made to its current flight plan.

7.11.2. When an airplane operating under instrument flight rules is flown in or encounters visual meteorological conditions, the IFR flight plan shall not be canceled unless it is anticipated and intended that the flight will continue for a reasonable period of time in uninterrupted visual meteorological conditions.

7.11.3. Ethiopian shall not operate under VFR when the flight visibility is less than, or at a distance from the cloud that is less than that prescribed, or the corresponding altitude and class of airspace. (ECARAS 8.8.3.1)

7.11.4 The reported cloud ceiling must be above the Minimum Sector Altitude, and the reported visibility shall not be less than 5000 m.

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SECTION 2.11 NATIONAL REGULATIONS–OPERATIONS SPECIFICATIONS

SECTION 2.11 NATIONAL REGULATIONS–OPERATIONS SPECIFICATIONS

1.0 PURPOSE

This section establishes guidelines on how operational specifications, under national regulations, are conducted.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. AIR OPERATOR CERTIFICATE (AOC)

4.1.1. All flight crew shall have adequate knowledge of the operations approved under the Air Operator Certificate (AOC), this includes but not limited to:

a) The approaches authorized by the State;
b) Ceiling and visibility requirements for takeoff, approach, and landing;
c) Allowance for inoperative ground components;
d) Wind limitations and required runway lighting.

4.2. BASIC EN-ROUTE AUTHORIZATIONS AND LIMITATIONS

4.2.1. A route guide, approved by ECAA, shall be provided for the use by flight crewmembers and persons assigned on operational control functions during the performance of their duties as a guide and aeronautical charts. These route guide and aeronautical charts shall be current and appropriate for the proposed types and areas of operations to be conducted. (ECARAS 9.3.1.20)

4.2.2. Except as specified in this section, the company is authorized to operate over any of the following routes:

SECTION 2.11 NATIONAL REGULATIONS–OPERATIONS SPECIFICATIONS

a) Any route or route segment published by the Civil Aviation Authorities or foreign government provided that such foreign routes conform to ICAO standards (PANS-OPS). Flights planned and initially released over these routes may be operated within controlled airspace predicated upon air traffic control radar vectoring service.

b) Additionally, flights may be planned and operated within controlled airspace over direct or other routes predicated upon VOR/VORTAC facilities provided:

I. Such routes lie within the advertised service volume of the facilities used.

II. Operations are conducted at least 2000 ft above the terrain at or above the MEA if one is established.

c) Operations shall be conducted over the routes defined in approved aeronautical publications. Off airways direct routes have a width of four nautical miles on either side of the course between the points defining such routes.

4.2.3. The basic route authorizations do not apply when:

a) It is necessary to avoid potentially hazardous meteorological conditions.

b) The flight is proceeding to an alternate airport.

c) The flight is otherwise cleared by air traffic control.

4.3. OPERATIONS AT AIRPORTS WITHOUT AIRPORT TRAFFIC CONTROL TOWER SERVICE

4.3.1. Operations will not be conducted without airport traffic control tower service unless flights are furnished local traffic advisory service through at least one of the following services:

a) An air/ground radio communications facility located in a position from which the operator is capable of observing local traffic and issuing traffic advisories. The air/ground radio communications facility is not required under the following circumstances:

I. If an airport is equipped with an air traffic control tower, the air/ground radio communications facility is not required during periods that the control tower is not in operation provided that data consisting of active runway identification, wind direction and velocity, and reported traffic activity for that airport is obtained from the flight service station or UNICOM serving that airport.

II. Flight operations may be conducted during periods of temporary (not over 48 hours) equipment failure of the airline air/ground radio communication facility provided that data regarding active runways, wind direction and speed, and reported traffic for that airport is obtained from a flight service station serving that airport.

b) Subject to Civil Aviation Authority approval, a flight service station may be used in lieu of the airline air/ground radio communications facility specified above to provide local traffic advisory information to airline flights. Except as authorized above, flight service stations authorized to provide this service must be physically located on the airport and have a view of the airport environment at least equal to that of an airline air/ground radio communications facility.

4.3.2. Flights departing from airports without airport traffic control tower service shall follow VFR procedures for class G airspace and request ATC clearance after airborne using HF RTF at least 20 min or VHF RTF 10 min prior to the point of entry into the control zone or control area. Thereafter, they should comply with ATC clearance for the entire flight in the control zone.

4.3.2. Flights departing from airports with a control tower shall get a clearance (direct routing and level) before departure and upon arrival at these airports, if the prevailing weather condition is above the required VFR minima, they may request visual approach to the airfield and comply with VFR procedures in uncontrolled airspace (class G).

4.3.3. A/C shall make radio broadcast to aware other traffics in vicinity and shall receive ATC clearance before departure when out of controlled airspace and before 10 minutes prior to entry to controlled airspace.

4.3.4. Weather and airfield information for these airports can be obtained from company representative at these stations. VFR weather conditions must be respected before operating to uncontrolled airspace.

4.4. AIRPORT AUTHORIZATIONS AND LIMITATIONS

Except as authorized by the Civil Aviation Authority, the company will not use any airport other than those specified in Chapter 2.2 unless the instrument approach procedures and the IFR minima for that particular airport are specified herein. The instrument approach procedures and IFR minima prescribed or approved by the government of the country in which the airport is located are applicable, provided that such procedures and minima meet the criteria specified in ICAO (PANS-OPS) documents.

4.5. Treat all GPWS/EGPWS alerts as genuine alerts. During night or IMC conditions apply the procedure immediately. During daylight VMC conditions with terrain and obstacles clearly in sight the alert may be considered cautionary.

4.6. To determine differences in rules and procedures for any airspace of intended use, like differences between prevailing or local airspace rules and ICAO airspace rules, please refer to Jeppessen for different local airspace rules and etc.

SECTION 2.11 NATIONAL REGULATIONS–OPERATIONS SPECIFICATIONS

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

6.1. It is the primary responsibility of a pilot-in-command not to depart an aircraft with defect(s) that will affect the airworthiness of the aircraft.

6.2. The pilot-in-command is fully responsible for the compliance of policies and procedures stated herein.

7.0 PROCEDURE

7.1. AIRPLANE INSTRUMENTS AND EQUIPMENT

7.1.1. Airplanes shall be equipped with suitable instruments and with navigation equipment appropriate to the route to be flown.

7.1.2. No Ethiopian aircraft may be operated unless it is equipped with the required instruments and equipment appropriate to the type of flight operation conducted and the route being flown. (ECARAS 7.2.1.1)

7.1.3. All Ethiopian aircraft shall be equipped with flight instruments which will enable the flight crew to: (ECARAS 7.2.1.1)

a) Control the flight path of the aircraft
b) Carry out any required procedural maneuvers
c) Observe the operation limitations of the aircraft in the expected operating condition.

7.1.4. If any instrument or item of equipment required by this section for a particular operation becomes inoperative en-route, the PIC shall comply with the procedures in the airplane Operations Manual for such an occurrence.

7.2. MINIMUM FLIGHT INSTRUMENTS (ECARAS 7.2.1.2)

7.2.1 No Ethiopian aircraft may be operated unless it is equipped with the following flight instruments:

a) An airspeed indicating system calibrated in knots, miles per hour or kilometers per hour
b) Sensitive pressure altimeter calibrated in feet
c) An accurate timepiece indicating the time in hours, minutes, and seconds

d) A magnetic compass

e) Any other equipment as prescribed by the Authority

7.2.2 Authority is required from ECAA to takeoff an aircraft with an inoperative instrument(s) or equipment. (ECARAS 8.2.1.5)

7.2.3. The company may not operate an airplane unless that airplane:

a) Is a registered civil airplane of Ethiopia and carries an appropriate certificate of airworthiness issued by the Civil Aviation Authority.

b) Is in an airworthy condition and meets the applicable airworthiness requirements of Ethiopia, including those relating to identification and equipment.

c) Is operated using as approved weight and balance control system

d) Is operated in accordance with the operating limitations of the approved airplane Flight Manual.

7.2.4. No flight crew shall commence a flight unless the required equipment

a) Meets the minimum performance standard, all operational and airworthiness requirements and the relevant provisions of ICAO Annex 10.

b) Is installed such that the failure of any single unit required for either communication or navigation or both purposes will not result in the inability to communicate and/or navigate safely on the route being flow.

c) Is in operable condition for the kind of operation being conducted, except as provided in the MEL. (ECARAS 7.1.1.4)

7.3. ENGINE INSTRUMENTS (ECARAS 7.6.1.1)

7.3.1. Unless the Authority allows or requires different instrumentation for turbine engine powered airplanes to provide equivalent safety, no person may operate without the following engine instruments:

a) A means for indicating fuel quantity in each fuel tank

b) An oil pressure indicator for each engine

c) An oil temperature indicator for each engine

d) A manifold pressure indicator for each altitude engine

e) A tachometer for each engine

7.4. INSTRUMENTS FOR OPERATIONS IN DAY IFR (ECARAS 7.2.1.4)

7.4.1. In addition, all aircraft, when operated in IFR or when the aircraft cannot be maintained in

SECTION 2.11 NATIONAL REGULATIONS–OPERATIONS SPECIFICATIONS

a descent altitude without reference to one or more flight instruments, shall be equipped with: (ECARAS 7.2.1.4)

a) An airspeed indicating system
b) A sensitive pressure altimeter calibrated in feet
c) A turn and slip indicator
d) Attitude indicator (artificial horizon) which:

I. Operates independently of any other attitude indicating systems,
II. Is powered continuously during normal operations and
III. After a total failure of the normal electrical generating system, is automatically powered for a minimum of 30 minutes from a source independent of the normal electrical generating system. (ECARAS 7.2.1.5)

e) A heading indicator (directional gyroscope)
f) A means of indicating whether the supply of power to the gyroscopic instruments is adequate.
g) A means of indicating in the flight crew compartment the outside air temperature.
h) A rate of climb and descent indicator.

NOTE: Instruments mentioned in c), d) and e) may be met by combination of instruments or by integrated flight instrument systems.

7.5. INSTRUMENTS FOR OPERATIONS IN DAY VFR (ECARAS 7.2.1.3)

7.5.1. In all Ethiopian operations, each pilot's station shall have separate flight instruments as follows:

a) An airspeed indicator calibrated in knots, miles per hour or kilometers per hour
b) A sensitive pressure altimeter calibrated in feet
c) A vertical speed indicator
d) A turn and slip indicator, or a turn coordinator incorporating a slip indicator
e) An attitude indicator
f) Any other equipment as required by the Authority

7.6. AIRCRAFT LIGHTS AND INSTRUMENT ILLUMINATION (ECARAS 7.5.1.2)

7.6.1. All Ethiopian aircrafts shall be equipped with:

a) Two landing lights
b) Anti-collision light system
c) Illumination for all flight instruments and equipment that are essential for the safe operation of the aircraft
d) Lights in all passenger compartments
e) A flashlight for each crew member station
f) Navigation/position lights

7.7. REQUIRED EQUIPMENT

7.7.1. When flying over water away from land surface suitable for making an emergency landing at a distance of more than 185 km (100 NM) for single-engine airplanes, and more than 370km (200NM) for multi-engine airplanes capable or continued flight with one engine inoperative:

a) Lifesaving rafts in sufficient numbers to carry all persons on board, stowed so as to facilitate their ready use in emergency, provided such life-saving equipment including means of sustaining life as is appropriate to the flight to be undertaken.
b) equipment for making the pyrotechnical distress signals
c) All Ethiopian operated airplanes must have life jackets for each occupant.

7.8. MNPS

7.8.1. For flights in Minimum Navigation Performance Specifications airspace (MNPS), all aircraft shall be equipped with navigation equipment that:

a) Continuously provides information to the flight crew of adherence to or departure from track to the required degree of accuracy at any point along that track. No Ethiopian aircraft may be operated unless it is equipped with navigation equipment that will enable it to proceed in accordance with its operational flight plan and the requirements of air traffic services. (ECARAS8.2.1.4), (ECARAS7.4.1.1)
b) The navigation equipment required shall be visible and usable by either pilot seated at his duty station.
c) For more information, refer to MNPSA manual.

7.9. EQUIPMENT FOR RVSM OPERATIONS

7.9.1. Two Independent Altitude Measurement Systems; that Indicates to the flight crew the Flight Level being flown;

7.9.2. One Secondary Surveillance Radar (SSR) Altitude Reporting Transponder; that automatically reports pressure altitude.

7.9.3. An Altitude Alert System; which alerts the flight crew when a deviation occurs from the selected Flight Level. The threshold for the alert shall not exceed ±90 m (300 feet);

7.9.4. An Automatic Altitude Control System; that Automatically maintains a selected Flight Level;

NOTE: In the event of turbulence or equipment failure after entry into RVSM airspace refer to Jeppesen manual for contingency procedures of different airspaces.

7.10. ALTITUDE DISPLAY LIMITS FOR RVSM & NON RVSM OPERATIONS

7.10.1. For RVSM airspace, standby altimeters do not meet altimeter accuracy requirements. The maximum allowable in-flight difference between Captain and First Officer altitude displays is 200 feet. The maximum allowable on-the-ground altitude display differences are found in the respective aircraft limitation part of the FCOM.

NOTE: Please refer to Jeppessen for specific local area and contingency procedures in case of instrument degradation through a flight.

7.10.2. For NON-RVSM airspace, altitude readout must be within 60 feet below 5000 feet and vary linearly to 100 feet at 10,000 feet & there should be only 75 feet difference between Captain & F/O with the field elevation. The maximum allowable in-flight difference between captain & F/O altitude display is 100 feet at FL100 & vary linearly to 200 feet at FL280. Cross check with standby altimeter in case of mismatch.

7.11. ACAS II (TCAS II VERSION 7.1)

7.11.1. All turbine engine aircraft with a maximum certificated T/O mass in excess of 5,700 kilograms, or authorized to carry more than 19 passengers, shall be equipped with ACAS II/TCAS II version 7.0. The installed equipment shall be approved by ECAA. (ECARAS7.7.1.7)

NOTE: All Ethiopian aircraft are equipped with version 7.1 except Q-400.

7.11.2. All aircraft shall be equipped with a pressure reporting transponder.

7.11.3. Flight crews shall follow TCAS RA commands using the established procedures even if it is in conflict with ATC instructions. If doing so would jeopardize the safe operation of the aircraft, the crew may perform otherwise based on the Captain’s good judgment.

Refer to Flight crew operations manual/QRH.

7.11.4. ALTITUDE REPORTING TRANSPONDER (ECARAS 7.4.1.5)

No person may operate an aircraft in commercial air transport operations unless it is equipped with a pressure-altitude reporting transponder that operates in accordance with the requirements of Ethiopia air traffic services and ICAO Annex 10.

SECTION 2.11 NATIONAL REGULATIONS–OPERATIONS SPECIFICATIONS

7.11.5. ALTITUDE ALERTING SYSTEM (ECARAS 7.7.1.4)

All Ethiopian airplanes shall be equipped with an altitude alerting system.

7.12. AIRBORNE WEATHER RADAR SYSTEM

Each pressurized aircraft when carrying passengers, shall be equipped with an airborne weather radar system when operating in areas where thunderstorms or other potentially hazardous weather conditions regarded as detectable with airborne weather radar, may be expected to exist along the route either at night or under instrument meteorological conditions.

7.13. WARNING SYSTEM FOR ANY DANGEROUS LOSS OF PRESSURIZATION

Any pressurized aircraft intended to be operated at flight altitudes above 25000 feet for which the individual certificate of airworthiness is first issued on or after 1 July 1962, shall be equipped with a device that provides positive warning to the pilot of any dangerous loss of pressurization.

7.14. GPWS

7.14.1. All aircraft with a maximum certificated T/O mass in excess of 5,700 kilograms, or authorized to carry more than 9 passengers, shall be equipped with a GPWS system.

7.14.2. Each GPWS shall automatically provide timely and distinctive warning to the flight crew of the following circumstances: (ECARAS 7.7.1.5)

a) Excessive descent rate.
b) Excessive terrain closure rate.
c) Excessive altitude loss after takeoff or go-around.
d) Unsafe terrain clearance while not in landing configuration (gear not locked down, flaps not in a landing position).
e) Excessive descent below the instrument glide path.

7.15. EGPWS

7.15.1. Effective 1 January 2007, all aircraft the Operator shall ensure aircraft in its fleet with a maximum certificated takeoff mass in excess of 5700 Kg, or authorized to carry more than nine passengers, shall be equipped with a GPWS that has a forward-looking terrain avoidance function.

7.15.2. Whenever there is a possibility of adverse weather or terrain and obstacles near the intended flight path, one pilot should monitor the weather radar display and the other pilot should monitor the terrain display. The purpose of these actions is to increase terrain and obstacle awareness and to help prevent Controlled Flight Into Terrain (CFIT).

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The weather and terrain displays should be used during night or IMC operations, during departures and approaches conducted below the MSA, when operating near terrain or obstacles, during an emergency descent, when landing at the nearest suitable airport, and at all times in non-radar environments. The selected range for weather and terrain displays must be reasonable to identify potential threats during takeoff and landing, with an optimal range of 40 NM or less. For RNP AR operations, circling approaches, and visual approaches, the TERR function must be selected on both NDs unless weather monitoring is required on one side.

7.16. LOOK AHEAD WINDSHEAR WARNING SYSTEM

All Aircraft with a maximum certificated takeoff mass in excess of 5700kg, or authorized to carry more than nine passengers, should be equipped with a forward-looking wind shear warning system.

7.17. FLIGHT DATA RECORDER

7.17.1. No person shall operate an aircraft unless it is equipped with flight data recorder capable of recording the aural environment of the flight deck during the flight time. This recorder shall be constructed, located and installed so as to provider maximum practical protection for the recordings in order that the information may be preserved, recovered & transcribed. (ECARAS 7.8.1.2)

7.17.2. Engraving metal foil and photographic film are obsolete recording media and are no longer acceptable for use in Flight Data Recorders. All aircraft shall not be equipped with this type of Flight Data Recorder.

7.17.3. All aircraft with a maximum certificated take-off mass in excess of 5,700 kilograms shall be equipped with a Flight Data Recorder (FDR) Type I or II as follows: (ECARAS 7.8.1.2)

7.17.4. The FDR shall be capable of recording the last 25 hours of aircraft operation at a minimum. The FDR shall be capable of recording time, altitude, airspeed, normal acceleration and heading as a minimum. The FDR shall not be of analog type utilizing frequency modulation (FM). Ethiopian preserves Flight recorder records and associated flight recorders to the extent possible in the event that the airplane involved in an accident or incident. The FDR shall be a type that is in accordance with ECARAS 7.8.2.1(A), ECARAS 7.8.2.2, ECARAS 7.8.2.3, ECARAS 7.8.2.4 and ICAO Annex 6.3.1

7.18. COCKPIT VOICE RECORDER

7.18.1. All turbine aircraft with a maximum certificated takeoff mass of over 5,700 kilograms shall be equipped with a Cockpit Voice Recorder that records the aural environment on the flight deck during flight and is capable of retaining recorded information for the last 30 minutes of its operation as a minimum.

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7.18.2. A CVR shall not be turned off during flight time. The CVR shall be de-activated upon completion of flight time following an accident and shall not be re-activated before the disposition as determined in accordance with ICAO Annex 13.

7.19. DATA LINK COMMUNICATION

7.19.1. The following requirements shall be in place if Ethiopian utilizes data link for aircraft communications.

a) Ensure all aircraft for which the individual certificate of airworthiness is first issued after 1 January 2005 that utilize data link communications and are also required to carry a CVR, are equipped with an FDR that records all data link communications to and from the aircraft; the minimum recording duration shall be equal to the duration of the CVR and correlated to the recorded cockpit audio,

Or

b) Have a process to ensure the time and content of all data link messages between the airline and aircraft utilized for international flights, and for which the individual certificate of airworthiness is first issued after 1 January 2005, are recorded and retained.

* * * * *

SECTION 2.12 FUEL & OIL POLICY

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1.0PURPOSE

The purpose if this fuel and oil policy procedure is to establish guidelines that enable adherence to fuel and oil handling policy.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Manager Flight Operations Performance & Technical Support
- Flight Control

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. FUEL & OIL CONSUMPTION RECORDS

4.1.1. Fuel records will be passed to the Manager Flight Operations Performance & Technical Support for analysis and entry to the approved Fuel Monitoring System.

4.1.2. Oil carriage and consumption will be recorded in the Aircraft Technical Log and retained in Maintenance, Repair & Overhaul Division.

4.2. OIL

4.2.1. While the engine oil contents must obviously be sufficient to cover the same elements as those for the fuel, it will be sufficient for the pilot in command to ensure that before flight the oil contents have been serviced in accordance with the manufacturer's recommendations, and between flights that no excess oil consumption has taken place.

4.2.2. While computing oil required for a flight any other condition that might cause an increase in oil consumption shall also be considered.

SECTION 2.12 FUEL & OIL POLICY

4.3 FUEL

4.3.1. The PIC is delegated the authority to ensure a flight is not commenced unless the usable fuel required is on board the aircraft and is sufficient to complete the planned flight safely.

4.3.2 The following factors are considered as a guidance when computing the fuel required for a flight:

a) Current Meteorological reports, or a combination of current reports and forecasts;

b) The anticipated aircraft weight;

c) The individual aircraft fuel consumption;

d) Current aircraft performance monitoring program if available or data provided by the manufacturer

e) Expected routings and flight levels.

f) NOTAMS

g) Applicable ATS procedures, restrictions & Possible variations in ATC routings & anticipated traffic delays;

h) Procedures prescribed for en-route loss of pressurization or failure of one engine;

i) MEL/CDL adjustments;

j) Any anticipated operational constraints (weather, de-icing, slots, etc...)

k) Any other conditions that may delay landing of the aircraft or increase fuel consumption.

4.3.3. Determine the fuel uplift accuracy by comparing the on board quantity measurement against supplier truck reading. The truck reading in volume (E.g. Liters) must be converted into mass using the actual Specific Gravity provided by the fuel supplier. (The maintenance staff, administering fueling, is responsible to ensure accurate truck reading and specific gravity measurement is provided to the crew).

a) The normal acceptable difference should not be above 4% in the up lifted fuel mass.

NOTE: Whenever a discrepancy of more than 4% exists, it should be entered in the A/C maintenance Logbook. And same will be noted on the Maintenance provided fuel slip.

4.3.4. If fuel is consumed during a flight for purposes other than originally intended during pre-flight planning, the PIC shall ensure that such flight is not continued without a re-analysis and, if applicable, adjustment of the planned operation to ensure sufficient fuel remains to complete the flight safely.

SECTION 2.12 FUEL & OIL POLICY

4.4 MINIMUM FUEL SUPPLY FOR VFR FLIGHTS (ECARAS 8.6.2.14)

4.4.1. No person may commence a flight in an airplane under VFR unless, considering the wind and forecast weather conditions, there is enough fuel to fly to the first point of intended landing and assuming normal cruising speed;

a) For flights during the day or night, for at least 45 minutes thereafter; and
b) For international flights, for at least an additional 15% of the total flight time calculated for cruise flight.

4.5 MINIMUM FUEL SUPPLY FOR IFR FLIGHTS (ECARAS 8.6.2.15)

4.5.1. No person may commence a flight under IFR unless there is enough fuel supply, considering weather reports and forecasts to;

a) Fly to the first point of intended landing execute an instrument approach;
b) Execute a missed approach and fly from that aerodrome to the most critical (in terms of fuel consumption) alternate aerodrome, if required; and
c) Fly thereafter at normal cruising speed

I. In turbofan or turbojet airplane, for 30 minutes at a holding speed at 450m (1500feet) above the aerodrome, plus a reserve for contingencies specified by the operator and approved by the Authority.

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

6.1. IN FLIGHT FUEL MANAGEMENT

6.1.1. The Captain shall manage fuel in flight to ensure that minimum fuel upon landing will be the Final Reserve Fuel (30 minutes of fuel).

6.1.2. The final decision for the amount of fuel and oil to be carried is the responsibility of the PIC to ensure the safety of the flight.

7.0 PROCEDURE

7.1. PRE-FLIGHT FUEL PLANNING

7.1.1. Fuel calculation for all Ethiopian international flights shall be done based on ECON cruise, long range cruise or minimum time (variable cost index).

SECTION 2.12 FUEL & OIL POLICY

7.1.2. The expected operating conditions, to which the text of this policy refers, encompass the parameters and assumptions on which the calculations of the operational flight plan are based.

7.1.3. A flight shall not be commenced unless, taking into consideration the expected operating conditions. The aircraft shall carry sufficient fuel to ensure that it can safely complete the flight considering:

a) Taxi fuel;
b) Trip fuel
c) Contingencies;
d) Destination Alternate fuel;
e) Final reserve fuel (Holding fuel);
f) Extra fuel

NOTE: (Refer to FOM 2.13 for isolated aerodrome fuel planning).

7.2. TAXI FUEL

7.2.1. Taxi fuel is planned to cover ground maneuvers from engine start to the beginning of takeoff roll and APU consumption, where applicable.

7.2.2. The planned amount of taxi fuel shall realistically take into account local condition. Where no local experience is available standard amount shall be used as specified below.

A350 -900	A350-1000	B777	B787	B-767	B-757	B737-8, MAX8, 737F	B737NG	Q400
600 kgs	700 Kgs	700 Kgs	300 Kgs	300 kgs.	200 kgs.	200 kgs.	150 kgs.	50 kgs.

7.3. TRIP FUEL

7.3.1. Fuel required flying from the aerodrome of departure to the aerodrome to which the flight is planned based on anticipated operating conditions.

7.3.2. This amount shall include:

a) Take off (taking in to account the departure procedure).
b) Climb fuel- fuel required for the selected climb procedure to initial cruising level.
c) Cruise Fuel- Fuel required flying from top of climb to Top of descent.

NOTE: Step climb and step descent will be considered.

d) Descent Fuel- fuel required to descent from last cruising level to initial approach altitude then to a point where the approach procedure is initiated.

e) Approach Procedure and Landing Fuel- an amount of fuel required to execute an instrument approach and then land at the destination aerodrome.

7.4. CONTINGENCY FUEL

7.4.1. It is an amount of fuel, which must be carried to cover unexpected deviations from Planned Operating Conditions.

7.4.2. It corresponds to **3%ERA** (en-route alternate) contingency fuel planning which is similar to in-flight re-planning in that it requires the mandatory selection in the OFP of an ERA located along the second part of the trip and before the destination aerodrome. The designation of ERA is predicated on the qualitative and quantitative assumption that, even if the **3% ERA** contingency fuel is used before reaching the planned destination, there would be sufficient fuel on board to land at the ERA aerodrome with final reserve fuel on board.

7.4.3. Criteria & process for performance-based **3% En-route Alternate (ERA)** Contingency Fuel Planning.

a) Contingency fuel based on 3% ERA shall not be lower than the amount specified on the table below.

TABLE 1: MINIMUM CONTINGENCY FUEL

AEROPLANE TYPE	MINIMUM CONTINGENCY FUEL IN KGS.
1. A350-900	700
2. A350-1000	900
3. B-777	900
4. B-787	600
5. B-767	600
5. B-757	400
6. B-737/800	400
7. B-737/700	400

b) Flight Watch shall monitor the actual fuel consumption rate of the specific airplane utilizing 3% ERA contingency fuel using Flight Watch monitoring program.

c) Flight watch shall provide the PIC with the operational information concerning meteorological conditions, ground facilities, NOTAMS & services at the destination & alternate aerodromes. Refer to FOM 2.1 (7.1)

d) Before reaching the destination aerodrome, the expected fuel remaining on arrival at the destination is less than the required alternate fuel plus the final reserve fuel, the Captain must determine either to divert to **3% ERA** aerodrome or Proceed to the destination, in accordance with FOM 2.12 (7.12.4).

e) An aerodrome shall be selected for the purpose of **3% ERA** contingency fuel when the appropriate meteorological reports or forecasts, or any combination thereof, indicate that, during a period commencing one hour before and ending one hour after the estimated time of arrival at the **3% ERA** aerodrome, the meteorological conditions will be at or above the planning minima in accordance with the TABLE below.

TABLE 2

TYPE OF APPROACH	**PLANNING MINIMA**
CAT II and CAT III	Cat I minima (RVR)
CAT I	Non-precision approach minima (ceiling/RVR)
Non-precision	Non-precision approach minima + (MDA + 200 ft. / RVR + 1000 m)
Circling	Circling minima

NOTE 1: The period of use of the **3% ERA** aerodrome shall also be specified on the OFP.

NOTE 2: There is no fuel calculation linked to the location of the ERA. The location of the ERA in the defined circle allows by definition a safe landing at the ERA if diversion happens from cruise level during the second half of the trip.

f) Ensure the **3% ERA** aerodrome is located within a circle having a radius equal to 20% of the total flight plan distance, the center of which lies on the planned route at a distance from the destination aerodrome of 25% of the total flight plan distance, or at-least 20% of the total flight plan distance plus 50nm, whichever is greater, all distances are to be calculated in still air conditions.

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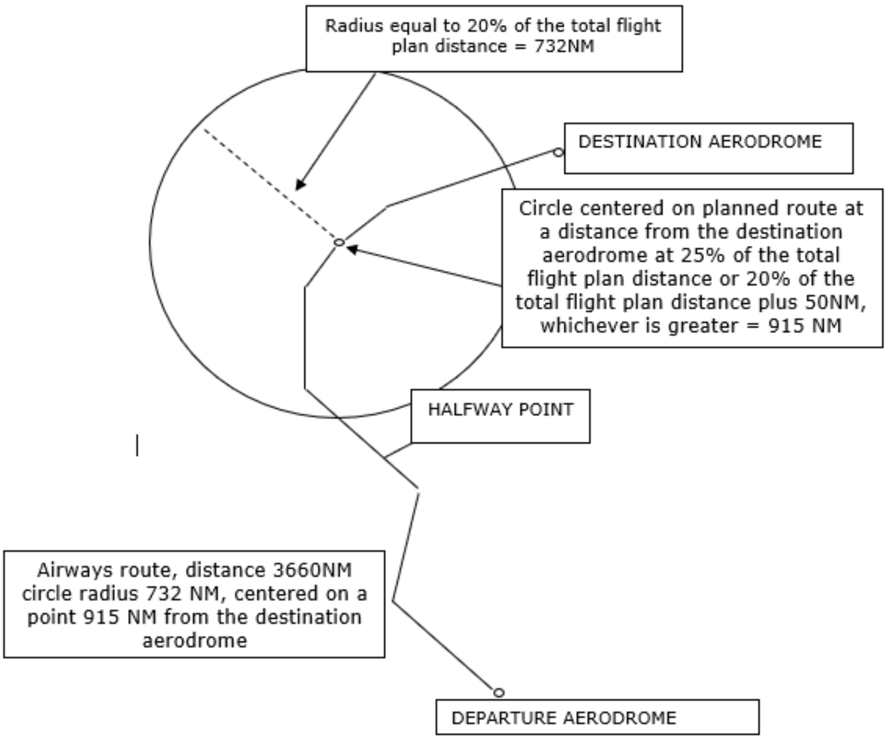

7.4.4. If the above criteria listed on 7.4.3 isn't satisfied for **3%ERA** contingency fuel planning, **5%** of the trip fuel calculated at the last hour fuel flow or of the fuel required from the point of inflight re-planning based on the consumption rate used to plan the trip fuel.

7.4.5. An amount to fly for five minutes at holding speed at 1500 feet above the destination aerodrome in standard conditions is the minimum contingency fuel in any case.

7.5. DESTINATION ALTERNATE FUEL

7.5.1 It is fuel required to fly from the planned aerodrome of destination to the alternate aerodrome specified in the operational flight plan based on the following operating conditions.

a) Missed approach instead of T/O.

b) LRC/ECON Cruise speed schedule.

c) Cruise FL depending only on ground distance maximum FL 370.

d) Descent from top of descent to the point where the approach is initiated.

e) Executing an approach & landing at the destination alternate airport.

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f) If two destination alternates are required, alternate fuel shall be sufficient to proceed to the alternate which requires the greater amount of alternate fuel.

7.6. FINAL RESERVE FUEL (HOLDING FUEL)

7.6.1 Fuel required to fly for 30 minutes at holding speed at 1500 ft above the alternate aerodrome under standard temperature conditions.

7.6.2 The calculation assumes clean configuration in the holding pattern and aircraft weight as specified in the AOM.

7.6.3 For the purpose of increasing fuel state awareness and determining an approximate final reserve value, whereas actual values will be computed in the OFP, the below amount on TABLE 2 provides a reference estimate for each fleet type and variant in operation calculated based on the maximum landing weight.

TABLE 3: APPROXIMATE FINAL RESERVE FUEL VALUE

AEROPLANE TYPE	FINAL RESERVE FUEL IN KGS.
A-350/900	2500
A-350/1000	3000
B-777/300 ER	3354
B-777/200 LR	3013
B-777/200 F	3406
B-787/800	2222
B-787/900	2350
B-767/300	2214
B-757/200	1715
B-737/800 & B-737/800F W (WINGLET)	1225
B-737/800 SFP (SHORT FIELD PACKAGE)	1210
B-737-8MAX	1100
B-737/700	1082
Q-400	421

7.7. EXTRA FUEL

7.7.1. Extra fuel covers anticipated deviations from the planned operating condition.

7.7.2. Extra fuel is any amount of fuel in excess of the planned take-off fuel, which is carried to cover anticipated deviations from expected operating conditions, for economic reasons fuel

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cost saving (tankering fuel)/supply difficulties.

NOTE: Fuel Tankering is not recommended if the takeoff or landing runway is contaminated or expected to be so.

7.7.3. Additional extra fuel shall be considered for possible approach delays. Subject to revision, initially 15 minutes of extra fuel (calculated at 10000 ft with the last hour fuel burn) will be carried for all anticipated low visibility operations. This fuel for dispatch planning purposes may be an average amount that can be determined from fleet data. Dispatch will calculate this figure and include it into OFP.

7.7.4. The amount of extra fuel is subject to the discretion of the pilot-in-command and shall be calculated using the last hour fuel flow.

7.7.5 The final decision on the fuel to be carried rests with the Captain. It is mandatory to write the amount of extra fuel uplifted on the OFP, in addition the reason for the extra uplift should be explained through appropriate Captain's report to DIR. FLYING OPS.

7.7.6 Extra fuel in excess of 1000 kgs should be included in the operational flight plan as extra fuel.

7.7.7. Flight dispatch will indicate in the remark section of the dispatch release where extra fuel shall be uplifted.

7.8. DIVERSION FUEL

7.8.1. It is the sum of the Alternate Fuel plus Final Reserve Fuel.

7.8.2. During the preflight phase of the flight, if the diversion fuel is less than the value given in the **TABLE 4** below, additional fuel shall be planned by the flight dispatch to achieve the minimum diversion fuel value. Once in-flight, this minimum diversion fuel value is not applicable, which includes re-planning of a flight.

TABLE 4: MINIMUM DIVERSION FUEL

AEROPLANE TYPE	MINIMUM DIVERSION FUEL IN KGS.
A-350	4500
B-777	5000
B-787	4000
B-767	4500
B-757	4000
B-737	2500
Q-400	N/A

SECTION 2.12 FUEL & OIL POLICY

7.9. TURBO PROP AIRCRAFT

7.9.1. A turbo propeller powered aircraft shall not be released for flight or take-off unless, considering wind and weather conditions expected, it has enough fuel: -

a) To fly to and land at the airport to which it is dispatched.
b) Thereafter, to fly to and land at the most distant alternate airport specified in the dispatch release and;
c) Thereafter, to fly for 45 minutes at normal cruising fuel consumption.

7.9.2 No person may release a turbo propeller aircraft to an airport for which an alternate is not specified in this Chapter of this manual, unless it has enough fuel. Considering wind and other weather conditions expected to fly to the airport and there after fly for at least three hours at normal cruising fuel consumption.

7.10. MINIMUM TAKE-OFF FUEL

7.10.1. The sum of trip, contingency, alternate and final reserve fuel shall never be less than the minimum fuel for any takeoff if published in the AOM (Aircraft Operations Manual).

7.10.2. For dispatching an aircraft for an ETOPS flight, the dispatcher must determine, for the considered route, both a standard and an ETOPS fuel planning. The highest of both fuel requirements shall be considered as being the minimum required block fuel for the flight.

7.11. IN FLIGHT MONITORING OF FUEL

7.11.1. In general, the following fuel quantities shall always be available on board.

a) Trip fuel for the remaining portion of the flight
b) Alternate fuel
c) Final reserve fuel (holding fuel)

7.11.2. If a flight was planned using re-dispatch procedures the following fuel quantities shall be available at the re-dispatch point:

a) Trip fuel to the re-dispatch destination
b) Contingency fuel to the re-dispatch destination
c) Alternate fuel
d) Final reserve fuel (holding fuel)

7.11.3. Whilst in flight the flight crew shall continuously monitor the fuel situation for the remaining part of the flight taking into consideration changes in the operational status of the aerodrome of intended landing and alternate aerodromes, as well as deviations from the flight plan resulting in changes of aircraft range.

7.11.4. If, as a result of an inflight fuel check, the expected fuel remaining on arrival at the destination is less than the required alternate fuel plus final reserve fuel (hold), the Captain must take into account the traffic and the operational conditions prevailing at the destination aerodrome, along the diversion route to an alternate aerodrome and at the destination alternate aerodrome, when deciding to proceed to the destination aerodrome or to divert. Refer recommendation "Fuel Management Approaching Destination Aerodrome" below.

a) When anticipating a diversion factors influencing the fuel required to the alternate aerodrome shall be reconsidered, i.e. Wind, Expected flight level & Runway in use and associated procedures.

b) It is permissible to continue the flight towards destination when an en-route fuel check predicts less than the alternate and the final reserve fuel remaining at destination aerodrome if the following conditions are met:

I. At least final reserve fuel will remain at touchdown.

II. When one runway is available at destination and destination alternate is within 30 minutes the meteorological conditions prevailing are such that, for the period from ETA until 1hr after, meets a ceiling of at least 2000 feet or circling height + 500 feet, whichever is greater and a visibility of at least 5Km. When two separate runways are available at destination and the meteorological conditions prevailing are such that for the period from ETA until 1hr after meets the following table minima.

TABLE 5

TYPE OF APPROACH	PLANNING MINIMA
ILS	Non-precision approach minima (ceiling/RVR or VIS)
Non-precision	Non-precision approach minima + (MDA + 200 ft. / RVR + 1000 m or VIS + 1000m)
Circling	Circling minima

7.11.4. Whenever the flight crew foresees a shortage of fuel in flight that might endanger the safe conduct of the remaining portion of the flight, flight plan changes should be initiated as soon as practical to take the necessary precautions for a refueling stop or diversion.

7.11.5. It is the Captain's responsibility to plan and operate the flight ensuring that the fuel remaining at touchdown is not less than;

a) For Commitment to Destination = Final Reserve Fuel.

b) For Destination = Alternate fuel Plus Final Reserve Fuel.

c) For Alternate = Final Reserve Fuel

7.12. MINIMUM FUEL ADVISORY

7.12.1. During Low Fuel operation as soon as there is any doubt as to the safe conduct of a flight, immediately request assistance from ATC.

7.12.2. When, having committed to land at a specific airport, the PIC calculates that any change to the existing clearance to that airport may result in landing with less than planned final reserve fuel; by declaring "**MINIMUM FUEL".**

7.12.3. When the calculated usable fuel predicted to be available upon landing at the nearest airport where a safe landing can be made is less than the planned final reserve fuel the captain shall declare emergency; by declaring **MAYDAY, MAYDAY, MAYDAY, FUEL**. The remaining fuel onboard must be reported in minutes.

7.12.4. Whenever a landing is made with less than final reserve fuel the Captain shall make a written report.

7.13. CHANGE OF ALTERNATE WHEN APPROACHING DESTINATION AERODROME

7.13.1. When approaching destination aerodrome any airport authorized as alternate airport may be considered for a possible diversion as long as the latest information available indicates that meteorological conditions will, during a period of +/-1 hour of ETA, be at or above straight in or circling minimum as applicable.

7.13.2. The minimum fuel required for such diversion shall consist of alternate fuel plus final reserve fuel.

7.14. FUEL MANAGEMENT APPROACHING DESTINATION AERODROME

7.14.1. If, due to unexpected delays encountered while approaching destination aerodrome, the remaining fuel reaches the amount specified as the minimum fuel for alternate plus final reserve, the PIC needs to request the expected delay and the flight may be continued to the destination aerodrome if this course of action provides the same or a higher degree of safety than the initiation of a diversion to the alternate aerodrome, considering especially:

a) The meteorological conditions at destination and alternate at decision time and for the expected time of landing;

b) The number and state of suitable runways at destination and alternate and approach aids available;

c) The expected approach time issued or anticipated.

SECTION 2.12 FUEL & OIL POLICY

7.14.2. In any event, the expected time of arrival at destination aerodrome shall be such that the remaining fuel after landing will not be less than final reserve fuel

- If/when a decision to consume fuel intended for flight to the alternate, such a decision shall be understood as a "commitment to stay".

7.15. RE-DISPATCH PROCEDURES

7.15.1 A flight may be planned via re-dispatch point (any point along the route) to any aerodrome contained in the list of airports.

7.15.2. The amount of usable fuel, on board for departure, shall be the greater of (a) or (b) below.

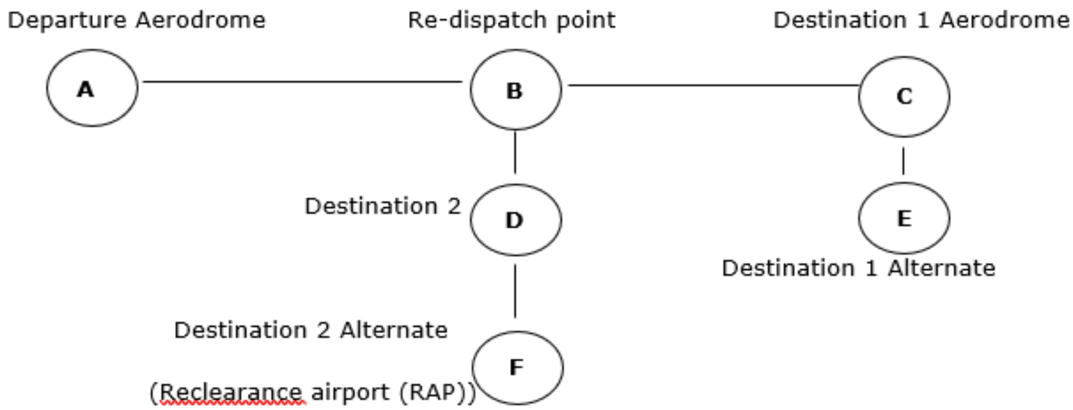
a) The sum of:

- Taxi fuel
- Trip fuel to the destination 1 (planned destination) aerodrome via the re-dispatch point
- Contingency fuel equal to not less than 5% of the estimated fuel consumption from the re-dispatch point to the destination 1 (planned destination) aerodrome
- Alternate fuel
- Final reserve fuel
- Extra fuel

b) The sum of:

- Taxi fuel
- Trip fuel to destination 2 (RAP) aerodrome via the re-dispatch point
- Contingency fuel from departure aerodrome to the destination 2 (RAP) aerodrome
- Alternate fuel
- Final reserve fuel
- Extra fuel

SECTION 2.12 FUEL & OIL POLICY



7.15.3 In order to proceed to the planned destination aerodrome, the PIC must ensure that the usable fuel remaining at the re-dispatch point is at least the total of:

a) Trip fuel from the re-dispatch point to the destination aerodrome plus

b) Contingency fuel equal to 5% of the trip fuel from the re-dispatch point to the planned destination plus

c) Planned destination aerodrome alternate fuel and

d) Final reserve fuel

7.15.4 Airports closed due to night curfew or other reasons will not be planned as re- dispatch destination.

NOTE: For more information on re-dispatch operational flight plan summary refer to Flight Operations Route Manual Section 1.

7.16. FUEL FREEZING POINT DETERMINATION

7.16.1. The variation of the freezing point of a fuel mixture is not linear. Therefore, the only reliable way to obtain an accurate freeze point of a mixture of fuels is to make an actual freeze point measurement. When this is not possible, consider the freezing point of the mixture to be the same as the highest freezing point when the fuel type in lowest quantity reaches 10% of the mixture. Determination of the fuel freezing point of fuel mixtures may be particularly a concern when operating transatlantic or transpacific routes and when very low OATs are expected as the aircraft will have to continuously cope with the mixture of JET A generally delivered in USA and JET A1 elsewhere.

SECTION 2.12 FUEL & OIL POLICY

Determination of fuel freezing point:

a) More than 90% of the final fuel is uplifted at departure airport:

I. Use actual measured value of the uplifted fuel, if available, otherwise, use -40°C for Jet A; -47°C for Jet A 1.

b) Less than 90% of the final fuel is uplifted at departure airport:

I. Use -40°C.

7.16.2. Mixing all Mixing all the residual JET A with all the refuel JET A1 to achieve maximum dilution is not considered practical. To practically achieve the best dilution, all the JET A should be placed in the inner wing tanks as these have the largest volume (by transfer of outer tanks JET A fuel into the inner tanks either during the previous flight or on ground before refueling).

7.16.3. Placing all the JET A into the inner wing tanks potentially enables a maximum compartmental structure of a wing tank and the fact that the residual JET A fuel will start at the inboard end of the tank, the concentration of JET A will be greater near the tank's inboard end.

7.16.4. The poor dilution of the JET A in the inner wing tank and its concentration near the inboard end of the tank has a potentially positive consequence. This is because the fuel near the inboard end of the inner wing tank tends to be consumed first by the engines. Thus, the concentration of the remaining JET A fuel on board, later in flight, when low fuel temperatures might be encountered in the case of low OATs, will be less than at takeoff. This gives a higher confidence margin that low concentrations of JET A in JET A1 will have a freeze point similar to JET A1 and can thus be treated as JET A1 with respect to a cold fuel alert.

7.16.5. For in-flight fuel management of fuel freezing, refer to FCOM.

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SECTION 2.13 ISOLATED AERODROME

SECTION 2.13 ISOLATED AERODROME

SECTION 2.13 ISOLATED AERODROME

1.0 PURPOSE

The purpose of this section is to provide clear guidelines for flight crew and dispatchers for Isolated Aerodrome operation.

1.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Dispatchers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

5.0 DEFINITION

5.1 Isolated Aerodrome: The destination airport can be considered as an isolated airport, if the fuel required to the nearest adequate destination alternate airport is more than two hours at normal cruise consumption above the destination airport including final reserve fuel. (for turbo prop airplane isolated aerodrome fuel is the amount of fuel required to fly for 45 minutes plus 15% of the flight time planned to be spent at cruising level, including final reserve fuel, or two hours, whichever is less).

5.2. Point of Safe Return (PSR): is the point of last possible diversion to an en-route alternate. This point is to be determined on each flight to an isolated aerodrome operation. While this point can be calculated and specified in the OFP, such a calculation does not typically take in to account any extra fuel, or the real time changes in fuel consumption that will occur after departure. The actual PSR will therefore often be reached later in the flight than the point originally calculated in the OFP. So, the flight crew can calculate the actual position of the PNR by using the FMS in accordance with 7.1.4 below.

SECTION 2.13 ISOLATED AERODROME

6.0 RESPONSIBILITY

6.1. The Dispatcher is responsible to prepare the correct document according to the guidelines given below.

6.2. The final decision for the amount of fuel to be carried is the responsibility of the PIC to ensure the safety of the flight.

7.0 PROCEDURE

7.1. PRE-FLIGHT FUEL PLANNING

7.1.1. The aircraft shall carry sufficient fuel to ensure that it can safely complete the flight.

7.1.2. The amount of usable fuel onboard for departure shall be the greater of (a) or (b):

a) The sum of:

- Taxi fuel;
- Trip fuel from the departure aerodrome to the destination aerodrome;
- 5% Contingency fuel from departure aerodrome to the destination aerodrome;
- Extra fuel if required; but not less than fuel to fly for two hours at normal cruise consumption above the destination aerodrome. This shall not be less than the final reserve fuel.

b) The sum of:

- Taxi fuel;
- Trip fuel from the departure aerodrome to the PNR along the planned route;
- Trip fuel from PNR to the last available en-route alternate aerodrome;
- 5% Contingency fuel from departure aerodrome to the last available en-route alternate aerodrome;
- Extra fuel if required; but not less than fuel to fly for 30 minutes at holding speed at 1500 feet above the last available en-route alternate aerodrome elevation in standard conditions. This shall not be less than the final reserve fuel.

7.1.4. Weather Requirements: Forecast weather conditions 1 hour before to 2 hours after planned arrival time at destination shall be at or above the planning minima shown below.

SECTION 2.13 ISOLATED AERODROME

TABLE 1

Approach Facility	Ceiling	Visibility
Precision Approach (CAT 1, 2 & 3)	Authorized DH/DA plus an increment of 200ft	Authorized visibility plus an increment of 800 meters
Non-Precision Approach of Circling	Authorized MDH/MDA plus an increment of 400ft	Authorized visibility plus an increment of 1500 meters

7.1.4. Fuel Requirements on OFP: The Alternate fuel on the OFP shows blank instead on the XHLD DEST an hour and thirty minutes fuel will be shown and the final reserve fuel (30 minutes) remains the same as the normal OFP fuel planning.

7.1.5 In flight, when flying to an isolated Aerodrome in order to proceed to the destination aerodrome, the PIC must ensure that the usable fuel remaining at the PNR is at least the total of:

a) Trip fuel from the PNR to the destination aerodrome;

b) Contingency fuel from the PNR to the destination aerodrome;

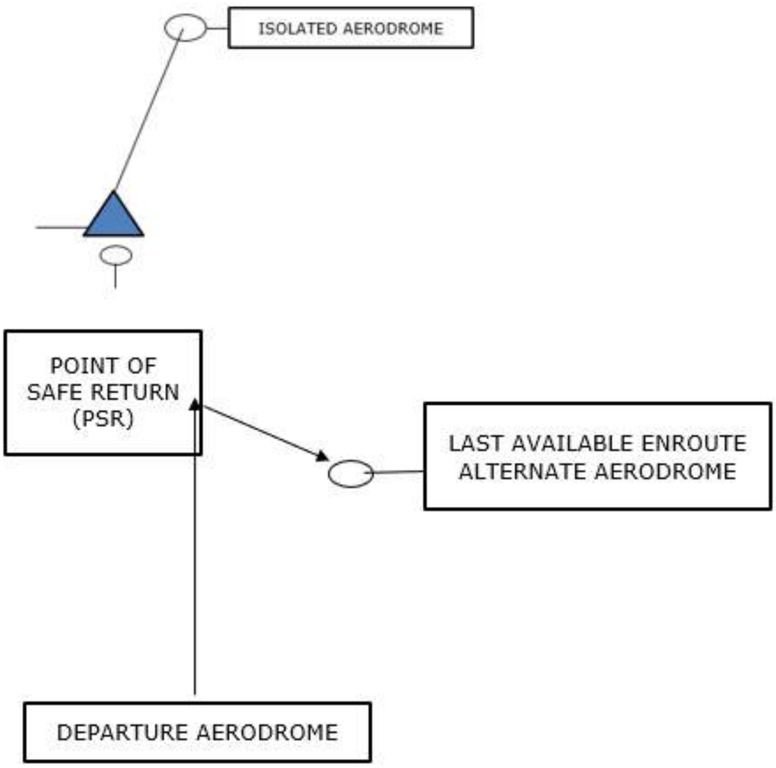
c) Fuel required to fly for 2 hours at normal cruise consumption above the destination aerodrome. This shall not be less than final reserve fuel and

d) The expected weather conditions at the destination aerodrome comply with those specified for planning minima as shown above on Table-1 of this section or the applicable minima of the isolated aerodrome and the weather trend shows a continuous improvement.

e) Traffic and other operational conditions indicate that a safe landing can be made at the estimated time of use.

NOTE: The fuel required to divert to the last available en-route alternate aerodrome at the actual PNR shall not be less than the trip fuel and 30 minutes of fuel upon landing.

SECTION 2.13 ISOLATED AERODROME



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SECTION 2.14 FUELING AND DEFUELING OF AIRCRAFT

SECTION 2.14 FUELING AND DEFUELING OF AIRCRAFT

SECTION 2.14 FUELING AND DEFUELING OF AIRCRAFT

PURPOSE

The purpose of this section is to establish guidelines for compliance with fueling and defueling of an aircraft.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Cabin Crewmembers
- Maintenance (fueling) Technician

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. At all Ethiopian domestic and international airports whose local authorities have not recommended it otherwise, passengers may remain on or allowed to move onto or from an aircraft and light maintenance work may be carried out (works not liable to produce sparks and not executed in the safety area) during pressure fueling operation provided that the following standard and special precautions are taken.

a) During fueling/defueling with an engine running, no passenger shall remain onboard, embark or disembark.

b) No maintenance work shall be carried out on the a/c.

4.2 STANDARD PRECATUIONS FOR FUELING/DEFUELING OPERATIONS

4.2.1. All fueling/defueling operations must be conducted by a qualified Flight or Ground Technician or by one flight crewmember. He/she shall coordinate all aspects with the fuel supplier company and when required with the aircraft crew.

4.2.2. The PIC and fueling technician must confirm the presence of the fire truck unless the state aerodrome requirement states otherwise.

4.2.3. Fueling technicians must know the most expeditious means to call fire brigade.

SECTION 2.14 FUELING AND DEFUELING OF AIRCRAFT

4.2.4. Approved flameproof electrical equipment in the vicinity of the aircraft may be left on if essential, other equipment should be switched off.

4.2.5. Personnel movement within the safety area shall be reduced to a minimum; however, under no circumstances shall:

a) Passengers remain or walk in the safety area.

b) Electrical connection be stopped or established.

a) Photographic flash bulbs and electronic flash equipment be used within 3 meters of the filling and venting points on the aircraft or fueling equipment.

b) Anybody smokes in the area.

4.2.6. In the absence of anti-static additive, the fueling rate must be half of the normal rate.

4.2.7. No fueling operations shall take place during storms with high risk of static discharges.

4.2.8. Do not operate HF radios during refueling/defueling operations.

4.2.9. Do not operate the weather radar within 50 feet of fueling/defueling operations or fuel spills.

4.2.10. Avoid switching on or off electrical or electronic circuits or carry on maintenance work liable to cause ignition within the safety area.

4.2.11. All equipment with operating engines within the safety area must be of an approved type.

4.2.12. Nailed boots or shoes shall not be worn during the fueling operation because of the possibility of sparks.

4.2.13. Fuel spillage must be prevented. The danger associated with fuel spillage is the formation of vapor; usually it is not easy to have vapors of jet A1 at ambient temperatures up to 37.8 ^{0}C. Fuel spillage must be cleaned up before engine start. Small spills of more than 3 meters long in any direction or 4.5meter square area and of a continuing nature must be covered with foam. Ground slope or wind may carry spilled fuel to possible ignition sources. Spilled fuel shall be removed by use of absorbent or neutralizing material:

NOTE: a) Use of sand should be avoided where jet aircraft is involved.

b) Equipment used to remove spilled fuel must not produce any spark.

4.2.14. If during fueling/defueling, the presence of fuel vapor is detected in the cabin, the flight deck and/or Ground Technician shall be informed immediately. The fueling or defueling process and all other activities using electrical equipment (e.g. cleaning) shall stop immediately.

4.2.15. As soon as fuel spillage and/or fumes and vapors are detected in the cabin the following actions must be taken immediately:

a) Alert fuel supervisor.

b) Stop all activity taking place in the danger areas.

c) Call airport fire brigade;

d) Don't start or shut down engines and equipment.

e) Fuel trucks must not be moved until danger is eliminated.

f) Personnel and vehicles not involved must be kept out of the danger area at a distance of 20 meters.

g) Location of spill, direction of flow, wind etc., will determine the escape route.

h) The aircraft must be towed away from the area.

4.2.16. The maintenance supervisor at location of spillage shall take complete charge of ramp area and emergency action.

4.2.17. If the truck delivery pressure exceeds the pressure listed, maintenance shall be advised immediately and given the following information:

a) Number of nozzles in use.

b) Maximum pressure observed and where pressure was measured.

c) Duration of overpressure.

d) Quality of fuel in tanks.

e) Overflow occurrence or not

4.2.18. Unless otherwise advised by maintenance, the tank(s) exposed to the excessive fueling pressure shall be opened and a complete structural inspection made. All fueling system components malfunctioning shall be corrected prior to releasing the aircraft for service.

5.0 DEFINITION

N/A

6.0. RESPONSIBILITY

6.1. The fueling technician is responsible for following the procedures and precautions mentioned in this section.

6.2. The PIC who is in-charge of the flight is responsible for assuring the compliance with the procedure stated herein.

7.0. PROCEDURE

7.1. FUELING/DEFUELING WHILE PASSENGERS ON BOARD, EMBARKING OR DISEMBARKING

7.1.1. If during fueling/defueling when passengers remain on board, are boarding or disembarking, it has to be ensured that the fueling/defueling process is supervised by a qualified person who should maintain two-way communication by the aircraft inter-communication system or maintain visual contact with the cockpit crew to monitor passenger safety and fueling operations. One of the flight crew should be in the cockpit monitoring the communication throughout fueling or defueling.

NOTE: During fueling/defueling when passengers remain on board, are boarding or disembarking, it has to be ensured the fueling/defueling process is supervised by qualified person who should two-way communication intercom system or shall stay with in the vicinity of the cockpit crew where he/she can respond to all pilot calls.

7.1.2. The no smoking sign is turned on and the seat belt sign must be switched off.

7.1.3. The flight crew must inform the cabin crew of the beginning and ending of fueling/defueling. The flight crew can indicate that fueling has been completed by turning on the fasten seat belt sign.

7.1.4. Lavatories shall be locked during the entire process of fueling/defueling.

7.1.5. A fire truck must be present (applies only if it is a country or aerodrome requirement).

7.1.6. Notify to the fire service the location of passenger on a stretcher as applicable.

7.1.7. In principle for all aircraft types, two exit doors shall be opened (one main exit door shall be forward of the wing) and passenger steps or jet ways shall be positioned at these doors. All other exit doors on the opposite side of the fueling/defueling of the aircraft exterior must remain free of obstacles within an area of at least 12 meters from the aircraft.

7.1.8. On the side of fueling/defueling activity, one exit door must be free of obstacles on the ramp within an area of at least 12 meters from the aircraft.

7.1.9. The fueling/defueling process must be interrupted immediately if it is observed that any of the safety regulations are not adhered to.

7.1.10. The following special safety regulations have to be strictly adhered to:

SECTION 2.14 FUELING AND DEFUELING OF AIRCRAFT

a) If no passenger step/jet way is available with passengers on board, the aircraft may be fueled or defueled from one side only with the following restrictions:

I. No loading or unloading of cargo.

II. No catering and cleaning.

III. Emergency exits and aisles shall not be blocked.

IV. No vehicles/servicing equipment shall be positioned within the range of all emergency exit doors to ensure an immediate operation of the emergency exit chutes.

V. All doors should be armed for evacuation

b) If only one passenger step/jet way is available with passengers on board, the aircraft may be fueled or defueled from one side only with the following restrictions:

I. Passenger step/jet way should be attended by a qualified cabin crew for passengers boarding or disembarking to be carried out in a controlled manner, gathering in the entrance area shall be avoided.

II. One exit door on the side of Passenger step/jet way shall be closed and attended by a qualified cabin crew throughout the refueling/defueling process and the area outside this door shall be unobstructed, the exit door should not be armed, unless required in an emergency for evacuation.

III. In the cabin, the required emergency exits as well as the aisles must never be blocked by unattended baggage, catering or cleaning equipment.

IV. One aft door on the non-refueling side must be attended by a qualified cabin crew throughout the refueling/defueling process the exit door should not be armed, unless required in an emergency for evacuation.

NOTE: For Q-400 A/C one qualified cabin crew attend both aft doors.

c) If two passenger steps/jet ways are available with passengers on board, the aircraft may be fueled or defueled from one side only with the following restrictions:

I. Passenger step/jet way should be attended by a qualified cabin crew for passengers boarding or disembarking to be carried out in a controlled manner and gathering in the entrance area shall be avoided.

II. In the cabin the required emergency exits as well as the aisles must never be blocked by unattended baggage, catering or cleaning equipment.

III. One aft door on the non-refueling side must be attended by a qualified cabin crew throughout the refueling/defueling process the exit door should not be armed, unless required in an emergency for evacuation.

d) If two side fueling/defueling is required with passengers on board, the above procedures listed on b and c apply. However, both aft doors must be attended by qualified cabin crews throughout the refueling/defueling process and the area outside these doors shall be unobstructed and the exit door should not be armed, unless required in an emergency for evacuation.

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SECTION 2.15 SIGNALS

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SECTION 2.15 SIGNALS

PURPOSE

The purpose of this section is to establish guidelines on how visual warnings are disseminated and acknowledged.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight crewmembers
- Marshallers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. VISUAL WARNINGS-UNAUTHORIZED FLIGHT NEAR CERTAIN AREAS

4.1.1. By day and night, a series of projectiles discharged at intervals of 10 seconds, each showing bursts or red and green lights or stars, will indicate to an unauthorized airplane that is flying in or is about to enter a restricted, prohibited, or danger area and is required to take remedial action.

4.1.2. The universal aviation signals shall have only the meaning indicated in the implementing standard. (ECARAS 8.8.2.11)

NOTE: These signals can be emitted either from the ground or from another airplane.

SECTION 2.15 SIGNALS

TABLE 1

LIGHT AND PYROTECHNIC SIGNALS FOR AIRPORT TRAFFIC		
LIGHT TOWARDS AIRPLANE CONERNED	MEANING TO AIRPLANE IN FLIGHT	MEANING TO AIRPLANE ON GROUND
Steady Green	Cleared to land	Cleared for take off
Steady Red	Give way to other Aircraft and continue Circling	Stop
Series of Green flashes	Return for landing	Cleared to Taxi
Series of red flashes	Airport unsafe: Do not Land	Taxi clear of landing Area in use
Series of white flashes	Land at this airport and to apron	Return to starting point on the airport
Red pyrotechnic Light	Notwithstanding any Previous instructions Do not land for the time being	
+ Clearance to land and to taxi will be given in due course		

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

It is the pilot-in-command's primary responsibility to comply with the policy and procedure stated in this section.

7.0 PROCEDURE

7.1. ACKNOWLEDGEMENT – AIRPLANE IN FLIGHT

7.1.1. During the hours of daylight, the pilot of an airplane in flight shall acknowledge by rocking the airplane's wings.

NOTE: This signal should not be expected on the base and final legs of the approach.

7.1.2. During the hours of darkness, the pilot shall acknowledge by flashing the airplane's landing lights on and off twice or, if not so equipped, by switching its navigation lights on and off twice.

SECTION 2.15 SIGNALS

7.2. ACKNOWLEDGEMENT – AIRPLANE ON THE GROUND

During the hours of daylight, the pilot shall acknowledge by moving the airplane's ailerons or rudder. During the hours of darkness, the pilot shall acknowledge by twice flashing on and off the airplane's landing lights or, if not so equipped, by twice switching on and off its navigation lights.

7.3. VISUAL GROUND SIGNALS FOR AIRPORT TRAFFIC

Signals for the control of airport traffic and marshaling signals are obtained from the Air Traffic Control section of the Jeppesen Airway Manual.

7.4. SIGNALS FROM MARSHALLER TO AIRPLANE

7.4.1. These signals are designed for use by the marshaller, with his hands or illuminated wands (at night) as necessary to facilitate observation by the pilot and facing the airplane in a position forward of the left wing tip within view of the pilot.

7.4.2. The meaning of the relevant signals remains the same if illuminated wands or torchlight are held. Airplane engines are numbered from left to right number one engine being the left outer engine.

7.5. SIGNALS FROM PILOT TO MARSHALLER

These signals are designed for use by a pilot in his cockpit with hands plainly visible to the marshaller and illuminated as necessary to facilitate observation by the marshaller.

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SECTION 2.16 ETOPS POLICY

SECTION 2.16 ETOPS POLICY

SECTION 2.16 ETOPS POLICY

SECTION 2.16 ETOPS POLICY

1.0 PURPOSE

The purpose of this section is to provide guidelines for the performance of flights under ETOPS requirements.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. GENERAL

4.1.1. This document represents ETHIOPIAN AIRLINES ENTERPRISE guidance for all ETOPS flights. They are in compliance with the regulations set by the ECAA. The primary reference used herein is the FAA AC No: 120-42B.

4.1.2. Operations beyond the threshold distance determined in accordance with (ECARAS8.6.2.10) shall not be conducted unless approved to do so by the Authority (ECARAS 8.6.2.11).

NOTE: Currently ETOPS (Extended Twin Operations) is replaced by EDTO (Extended Diversion Time Operations) which applies to Twins, Tries and Quads.

5.0 DEFINITIONS

5.1. ETOPS Operations is applied to all flights conducted under ECAA Technical Directive Chapter 6.1, FAR part 121, JAR Ops 1 or equivalent, in a twin-engine aircraft over a route that contains a point further than 60 minutes flying time from an adequate airport at the selected one-engine-out diversion speed schedule in still air and ISA conditions. It is based on single-engine flying time to an adequate airport (up to 180 minutes) depending on the aircraft type. It requires specific operational procedures and appropriate authority approval.

5.2. ADEQUATE AIRPORT

5.2.1. An Airport is considered adequate when it satisfies the aircraft performance requirements applicable at the expected landing weight. A list of the adequate airports that will be considered for a specific area of operation must be presented to the local operational authorities as part of the operational approval, this list then becomes part of the Operational specifications. It is worth noting that it is not necessary to meet the runway pavement requirements normally to be considered for the regular use of an airport. In accordance with the provisions of the ICAO Convention - Annex 14 and ICAO Airport Manual (Document 9157 - AN/91), the aircraft ACN (Aircraft Classification Number) is allowed to exceed the runway PCN (Pavement Classification Number), when an airport is used in case of emergency. The amount of possible exceedance can be obtained from the above referenced ICAO document or from each individual national or local airport authority. Ethiopian will use no more than 100% exceedance.

5.2.2. The following should also be met at the expected time of use:

a) Availability of the airport.
b) Over flight and landing authorizations.
c) Capability of ground operational assistance (ATC, Meteorological and Air Information Services Offices, Lighting).
d) Availability of navaids such as ILS, VOR, NDB (at least one compatible navaid must be available for an instrument approach).
e) Airport category for rescue and firefighting (ICAO Doc 9137 - AN/898 Part 1).

5.2.3. The following criteria may also be considered:

a) Capability of technical assistance.
b) Capability of handling and catering (fuel, food, etc.,)
c) Ability to receive and accommodate the passengers.
d) Other particular requirements of Ethiopian Airlines.

5.3. SUITABLE AIRPORT

5.3.1. A suitable airport for dispatch purposes is an airport confirmed to be adequate which satisfies the ETOPS dispatch weather requirements in terms of CEILING and VISIBILITY minima (refer to weather reports and forecasts) within a validity period. This period begins one hour before the earliest Estimated Time of Arrival (ETA) at the airport and ends one hour after the latest ETA. In addition, cross-wind forecasts should be acceptable for the same validity period.

5.3.2. Field conditions should also ensure that a safe landing can be accomplished with one engine and/or airframe system inoperative (refer to possible NOTAMs SNOWTAMs, approach procedure modification).

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5.4. DIVERSION / EN-ROUTE ALTERNATE AIRPORT

5.4.1. A "diversion" airport, also called "en-route alternate" airport, is an adequate / suitable airport to which a diversion can be accomplished.

5.4.2. An aerodrome shall not be selected as an ETOPS en-route alternate aerodrome unless the appropriate weather reports or forecasts, or any combination indicate that during a period commencing ONE hour before and ONE hour after the expected time of arrival at the aerodrome, the weather conditions will be at or above the planning minima in accordance with the operator’s ETOPS approval. (ECARAS 8.6.2.12)

5.5. MAXIMUM DIVERSION TIME

5.5.1. The maximum diversion time from an en-route alternate airport is granted by the ECAA. At present, the maximum diversion time for **ETHIOPIAN A350, B777, B787 and B767 is 180 minutes, but for B737 is 120 minutes.**

5.5.2. It is only used for determining the area of operation, and therefore is not an operational time limitation for conducting a diversion. Prevailing weather conditions or other factors can influence actual diversion time.

NOTE: Except for ETOPS beyond 180 Minutes, actual diversion time may exceed the authorized diversion time as long as the flight is conducted within the authorized ETOPS Area of Operation.

5.6. MAXIMUM DIVERSION DISTANCE

5.6.1. The maximum diversion distance is the distance covered in still air and ISA conditions within the maximum diversion time at the selected one-engine-out diversion speed schedule and at the associated cruise altitude including the descent from the initial cruise altitude to the diversion cruise altitude (also called drift-down). It is used for dimensioning the area of operations.

5.6.2. For ETHIOPIAN ETOPS operations for maximum diversion distance refer area of operation on **TABLE 1** in this section.

5.7. ETOPS AREA OF OPERATIONS

The ETOPS area of operations is the area in which it is authorized to conduct a flight under ETOPS regulations and is defined by the maximum diversion distance from an adequate airport or set of adequate airports. It is represented by circles centered on the adequate airports, the radius of which is the defined maximum diversion distance.

5.8. ETOPS ENTRY POINT (EEP)

The ETOPS Entry Point is the point located on the aircraft's outbound route at ONE hour flying time, at the selected one-engine-out diversion speed schedule (in still air and ISA conditions), from the last adequate airport prior to entering the ETOPS segment. It marks the beginning of the ETOPS segment.

5.9. ETOPS SEGMENT

The ETOPS segment starts at the EEP and finishes when the route is back and remains within the 60-minute area from an adequate airport. An ETOPS route can obtain several successive ETOPS segments separated from each other.

5.10. EQUAL TIME POINT (ETP)

An Equal time Point is a point on the aircraft route which is located at the same flying time from two suitable diversion airports. The ETP position can be determined using operational flight plan that features such capability mathematically or graphically on navigation or plotting chart.

5.11. CRITICAL POINT (CP)

The Critical Point is the point on the route which is critical with regard to the ETOPS fuel requirements if a diversion has to be initiated from that point. The CP is usually, but not always (depending on the configuration of the area of operations), the last ETP within the ETOPS segment. Note that, the last ETP is not necessarily the ETP between the last two alternate airports).

5.12. ONE-ENGINE OUT DIVERSION SPEED

5.12.1. The one-engine-out diversion speed is a Mach/IAS speed combination selected by the operator and approved by the operational authority. The Mach is selected at the beginning of the diversion descent down to the transition point where the Indicated Airspeed (IAS) takes over.

5.12.2. The one-engine-out diversion speed for the intended area of operations shall be a speed within the certified operating limits of the aircraft which minimum maneuvering speed and VMo/MMo (maximum certified operating speed) considering that the remaining engine thrust is at Maximum Continuous Thrust (MCT) or less.

5.12.3. This diversion speed is used in:

a) Establishing the area of operations.
b) Establishing the critical fuel scenario for the single-engine diversion.
c) Establishing the net level-off altitude to safely clear any en-route obstacle by the appropriate margin as specified in applicable operational rules.

5.12.4. This speed is used in case of diversion following an engine failure. However, the pilot in

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command has the authority to deviate from this planned speed after assessment of the actual emergency situation.

5.12.5. For Ethiopian, ETOPS speed is indicated in **TABLE 1** in this section.

5.13. ETOPS EXIT POINT (EXP)

The ETOPS Exit Point is the point located on the aircraft's route, where the aircraft has been flying in an ETOPS segment and enters an area of one hour flying time at the selected one-engine-out diversion speed schedule (in still air and ISA condition) to an adequate airport. It marks the end of that particular ETOPS segment.

6.0. RESPONSIBILITY

6.1. The pilot-in-command is primarily responsible for compliance with ETOPS procedures/requirements.

6.2. Each flight crew is responsible to be aware of ETOPS requirements.

6.3. It is the responsibility of the PIC to ensure that the required en route alternates for ETOPS are selected and specified in ATC flight plans in accordance with the ETOPS diversion time approved by the Authority. (ECARAS 8.6.2.12)

7.0. PROCEDURE

7.1. ETOPS ALTERNATE AIRPORT: RESCUE AND FIGHTING SERVICE

7.1.1. Except as provided in paragraph 7.1.2 of this section, the following rescue and firefighting service (RFFS) must be available at each airport listed as an ETOPS alternate airport in a dispatch or fight release.

a) For ETOPS UP TO 180 minutes, each designated ETOPS alternate airport must have RFFS equivalent to that specified by ICAO as Category 4, or higher.

b) For ETOPS BEYOND 180 minutes, each designated ETOPS alternate airport must have RFFS equivalent to that specified by ICAO category 4, or higher. In addition, the aircraft must remain within the ETOPS authorized diversion time from an adequate airport that has RFFS equivalent to that specified by ICAO category 7, or higher.

7.1.2. If the equipment and personnel required in paragraph 7.1.1 of this section are not immediately available at an airport, the certificate holder may still list the airport on the dispatch or fight release if the airport's RFFS can be augmented to meet paragraph 7.1.1 of this section from local firefighting assets. A 30-minute response time for augmentation is adequate if the local assets can be notified while the diverting airplane is enroute. The augmenting equipment and personnel must be available on arrival of the diverting airplane and must remain as long as the diverting airplane needs RFFS.

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7.2. AREA OF OPERATIONS

7.2.1. ETOPS operations are allowed within a well-defined area. The size of this area depends on:

a) The maximum diversion time.
b) The selected one-engine-out diversion speed schedule.
c) The number and location of the selected adequate diversion airports.

7.2.2. The area of operations is determined in still air and ISA conditions considering the relevant aircraft performance with one engine inoperative and the remaining engine being at MCT or less. Therefore, if the area of operation is determined once, it does not need to be reassessed for each flight (considering the en-route weather forecast or the aircraft performance depending on the take-off weight) unless one or more adequate diversion airports happen to be unsuitable.

7.2.3. The aircraft performance level considered for the calculation is associated with a unique aircraft weight which is called the aircraft reference weight.

7.2.4. Ethiopian fleet ETOPS considerations

TABLE 1

Aircraft type	Max Diversion distance NM			Reference Wt (Tones)	Company Speed
	60 Mins	120Mins	180Mins		
B737	413	806	N/A	65	.79/290
B738 & B738F	408	800	N/A	73	.79/290
B737-8MAX	415	814	N/A	77	.79/310
B757 (PW2040)	432	853	N/A	105	.80/310
B763 (PW4062)	429	852	1273	175	.79/310
B777LR/F	421	830	1239	320	.84/300
B777-3ER	419	825	1231	320	.84/300
B787-800	440	857	1275	204	.85/310
B787-900	435	846	1258	234	.85/310
A350-900	427	834	1241	250	.83/330
A350-1000	432	848	1265	290	.83/330

7.3. AIRCRAFT REFERENCE WEIGHT

7.3.1. The concept of reference weight has evolved with time. Previously, according to CAA regulations (CAP 513), the aircraft reference weight was the aircraft weight after two hours of flight considering a take-off at the maximum take-off weight. At present, JAA and FAA have agreed not to define reference weight, but to leave the operator free to determine its own reference weight considering the ETOPS route structure. This weight should be as realistic as possible and submitted for approval to the authority.

7.3.2. It is suggested that the aircraft reference weight should be defined as the highest of the estimated gross weight values at the critical points of the various routes being considered within the given area of operation. The computation will be done considering a take-off at the maximum take-off weight (structural or runway limitation) and standard speed schedule in still air and ISA conditions.

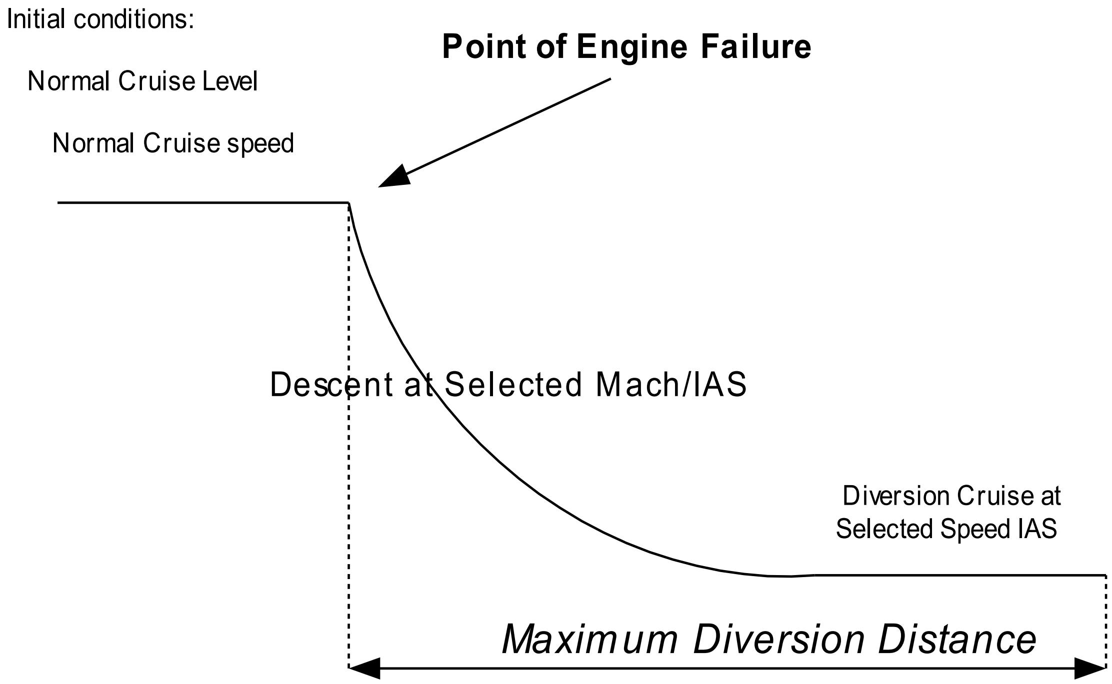
7.3.3. Whenever applicable, the above computation should be conducted considering that a given route may be supported by different set of declared en-route alternates (thus resulting in different CP locations). The reference weight used will be as indicated in the table above under Maximum Diversion Speed.

7.4. DIVERSION SPEED SCHEDULE AND MAXIMUM DIVERSION DISTANCE

7.4.1. Using the aircraft reference weight and the selected one-engine-inoperative diversion speed schedule, it is possible to determine the optimum diversion cruise flight level, and the best True Air Speed (TAS).

7.4.2. The resulting TAS at the diversion flight level, combined with the maximum diversion time allowed, provides the maximum diversion distance. However, an agreed interpretation of the regulations is to take benefit of the descent (during which the TAS is higher than during the diversion cruise) to increase the maximum diversion distance as represented in the following figure.

7.4.3. Diversion Profile



7.5. ETOPS FUEL REQUIREMENTS

7.5.1. Unlike the area of operation, which is determined in still air and ISA conditions, the fuel planning must consider the expected meteorological conditions along the considered routes (forecast wind component and temperature).

7.5.2. For dispatching an aircraft for an ETOPS flight, the dispatcher must determine, for the considered route, both a standard and an ETOPS fuel planning. The highest of both fuel requirements shall be considered as being the minimum required block fuel for the flight.

7.6. PERFORMANCE FACTOR

7.6.1. For determining a dependable fuel planning, the operator should always consider the latest updated aircraft performance factor. For ETOPS operation the determination becomes mandatory or a blanket 5% degradation factor is to be applied.

7.6.2. The performance factor reflects the airframe/engines deterioration with time and is used to determine the actual fuel consumption. It is determined by the processing of in-flight manual (or automatic) recordings of engines and aircraft parameters. For a brand new aircraft whose performance is equivalent to the baseline, the performance factor is equal to 1 (one). The performance factor should be defined for each individual aircraft within the operator's fleet.

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7.7. ETOPS FUEL PLANNING

7.7.1. For ETOPS operations, a specific ETOPS fuel planning (also called Critical Fuel Reserves in the regulations) should be established.

7.7.2. The ETOPS fuel planning is split into two parts:

a) The first part corresponds to a standard fuel scenario from the departure airport to the Critical Point (CP) and

The second part corresponds to the critical fuel scenario from the CP to the diversion airport.

7.8. CRITICAL FUEL SCENARIO

7.8.1. This scenario based on the failure case occurring at the CP and requiring a diversion. The point of occurrence is so-called Critical because in terms of fuel planning a diversion at this point is the least favorable.

7.8.2. The diversion profile is defined as follows:

a) descent at a pre-determined speed to the required diversion flight level,
b) diversion cruise at a pre-determined speed,
c) normal descent down to 1,500ft above the diversion airport,
d) 15 minutes holding at this altitude,
e) first approach (IFR) and go-around,
f) Second approach (VFR) and landing.

7.8.3. The ETOPS critical fuel scenario is based on the separate study of three failure cases, occurring at the critical point, with their respective diversion profiles.

a) Aircraft depressurization

I. Emergency descent at Vmo/Mmo speed-brakes extended down to FL100 or MORA whichever is higher,
II. Diversion cruise performed at Long Range Cruise (LRC) speed.

b) Engine failure and aircraft depressurization:

I. Emergency descent at Vmo/Mmo (speed brakes extended) down to FL100 or MORA whichever is higher,

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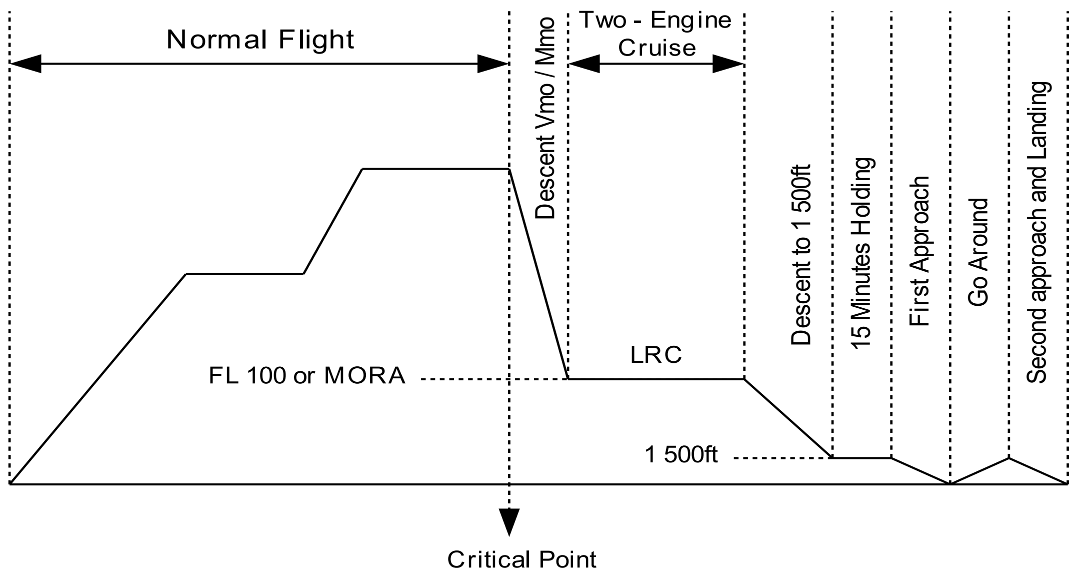
II. Diversion cruise at the speed schedule adopted for the determination of the area of operation.

c) Engine Failure

I. Descent and Diversion using fixed speed strategy down to the maximum altitude capability.

7.8.4. DEPRESSURIZATION FLIGHT PROFILE



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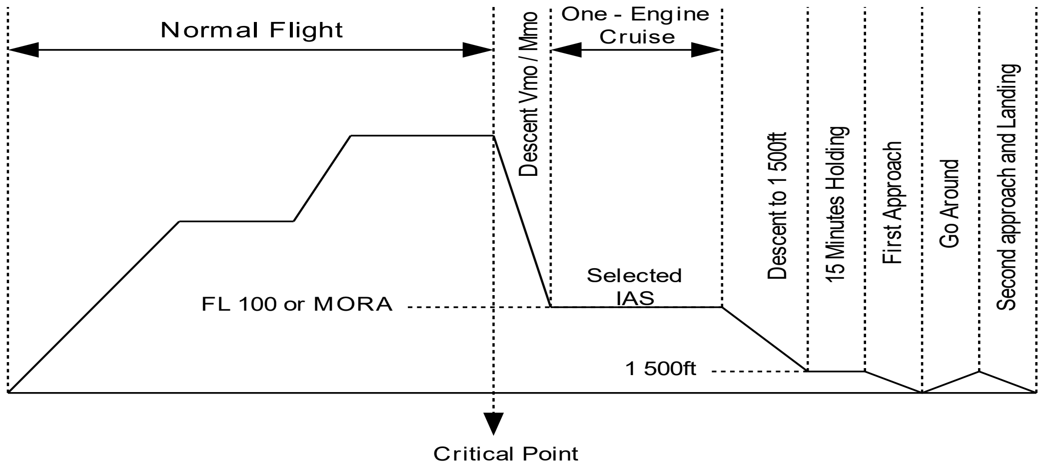
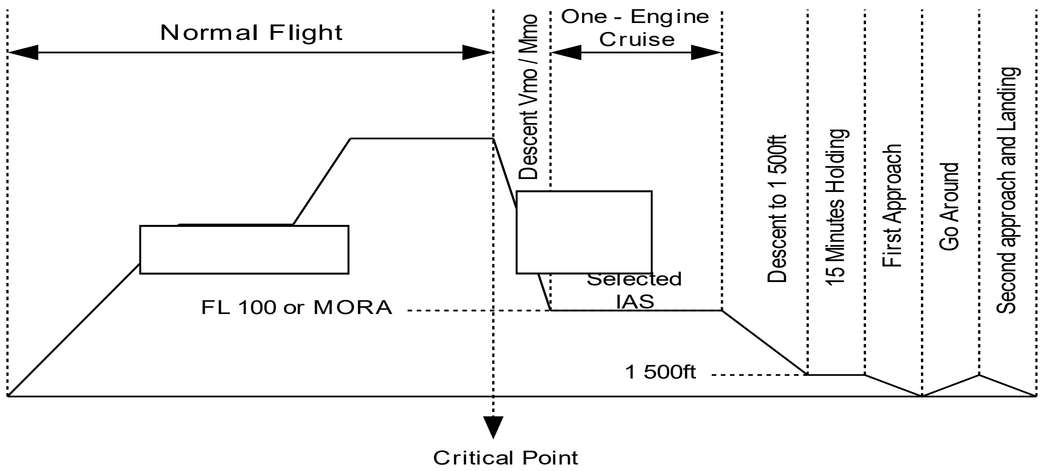
7.8.5 ONE ENGINE OUT DEPRESSURIZATION FLIGHT PROFILE



7.8.6. ONE ENGINE OUT FLIGHT PROFILE



7.8.7. However, flight above FL100 may be desired or required. This is allowed if the aircraft is equipped with supplemental oxygen for the maximum diversion time for the flight crew and a required percentage of passengers in accordance with applicable regulatory requirements and could be mandatory in case of obstacles. In this case the diversion cruise may be allowed at a level up to FL140.

7.8.8. For each scenario, the required block fuel must be computed in accordance with the operator's ETOPS fuel policy and with the regulatory ETOPS critical fuel reserves described in the following section.

7.8.9. Depending on the strategy and on the one engine-out speed selected for the single- engine diversion scenario, one of these two scenarios will result in the highest fuel requirement.

7.8.10. The scenario resulting in the highest fuel requirement is referred to as the ETOPS critical fuel scenario, the associated block fuel requirement is referred to as the ETOPS critical fuel.

7.9. FUEL RESERVES

7.9.1. ETOPS regulations require the addition of specific fuel reserves to the ETOPS diversion fuel.

7.9.2. For the computation of the ETOPS critical fuel reserves and of the completed ETOPS critical fuel planning, the diversion fuel shall include the following provisions:

a) Fuel burn- from the CP to the diversion airport (understood to be 1500 ft overhead the airport), and
b) 15 minutes holding at 1500ft at minimum maneuvering speed
c) 5% fuel mileage penalty or a demonstrated performance factor, effect of any MEL item.
d) Effect of engine / nacelle and wing anti-ice adjusted for exposure time to forecast icing.
e) Effect of ice accumulation on the unheated surfaces of the aircraft
f) APU fuel consumption, if required as a power source (MEL).
g) Increase fuel required by 0.8% per 10°c above ISA.
h) Increase diversion fuel by 5% to account for wind errors.

7.9.3. **TABLE 3** summarizes fuel burn penalties for ETOPS diversion through known icing conditions at company selected ETOPS diversion speed. Burn penalties are a percentage increase over the baseline (no ice drag and no anti-ice system) fuel burn.

When icing conditions are forecast, use the greater of:

a) **TAI only** (Thermal Anti-Ice) Fuel for engine and wing anti-ice for the **entire time** during which icing is forecast
b) **(TAI+ Ice drag)** The effect of airframe icing during **10%** of the time during which icing is forecast, including ice accumulation on unprotected surfaces, and the fuel used by engine and wing anti-ice during this period.

7.9.4. Example on icing fuel adjustment for B787-800 A/C is shown below.

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If icing forecasted for 25% of the diversion time the calculation will be as follows:

TAI FUEL FACTOR = 1%*0.25*100% =0.25%

TAI+ICE DRAG FUEL FACTOR= 27%*0.25*10% =0.675%

THEREFORE, **TAI+ICE DRAG (0.675%) is the higher fuel penalty**

TABLE 2

Aircraft Type	TAI only	TAI + ICE DRAG
B737-700	12%	34%
B737-800 & B737-800F	9%	28%
B737-8MAX	7%	45%
B757	7%	11%
B763	8%	13%
B777 LR/F	3%	7%
B777-3ER	3%	8%
B787-800	1%	27%
B787-900	1%	26%
A350	3%	7%

7.9.5. The following considers non-mandatory factors:

a) Effect of the demonstrated performance factor for all standard and ETOPS fuel requirement computations,

b) Carriage of contingency fuel from the departure to the CP, as dictated by the specific aspects of the route or the operator's fuel policy, when computing the ETOPS critical fuel planning. The fuel factors to be considered for standard and ETOPS fuel plans (before and after the CP) are summarized in the two tables below:

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TABLE 3

Fuel Factors	Standard Fuel Planning	ETOPS Fuel Planning	
		From Departure to CP	From CP to Diversion
Performance Factor	X	X	X
Contingency Fuel	X	X	X
Effect of MEL	X	X	X
Effect of Wing Nacelle and Wing Anti Ice			X
Effect of Ice accumulation			X
Provision for weather avoidance			If diversion more than 138 minutes

7.9.6. The complete ETOPS critical fuel planning for the ETOPS critical fuel scenario (i.e. from the departure to the CP and then from the CP to the diversion airport) must be compared to the standard fuel planning (i.e. from the departure to the destination and destination alternate) computed in accordance with the company fuel policy and applicable operational requirements. The highest of both fuel requirements shall be considered as the required block fuel for the flight. Therefore, the pilot is then assured of safely completing the flight whatever the flight scenario is (normal flight or diversion).

7.10. ETOPS DISPATCH WEATHER MINIMA

7.10.1. The most up-to-date information provided to the flight crew indicates that conditions at identified enroute alternate airports will be at or above the below minima listed on Table 4 below for the operation at the estimated time of use.

TABLE 4

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ETOPS Alternate Minimum

Approach Facility Configuration[1]	Alternate Airport IFR Weather Minimum Ceiling[2]	Alternate Airport IFR Weather Minimum Visibility[3]
For airports with at least one operational navigational facility providing a straight-in non-precision approach procedure or Category I precision approach, or, when applicable, a circling maneuver from an instrument approach procedure.	Add 400 ft. to the MDA (H) or DA (H), as applicable.	Add 1600m to the landing minimum.
For airports with at least two operational navigational facilities, each providing a straight-in approach procedure to different suitable runways.	Add 200 ft. to the higher DA (H) or MDA (H) of the two approaches used.	Add 800m to the higher authorized landing minimum of the two approaches used.
One usable authorized category II ILS Instrument Approach procedure (IAP)	300 feet	1200m or RVR 4000 feet(1200m)
One usable authorized category III ILS Instrument Approach procedure (IAP).	200 feet	800m or RVR 1800 feet (550m)

[1]When determining the usability of an IAP, wind plus gust must be forecasted to be within operating limits, including reduced visibility limits, and should be within the manufacturer's maximum demonstrated crosswind value.

[2]Conditional forecast elements need not be considered, except that **PROB40 or TEMPO** condition below the lowest applicable operating minima must be taken into account.

NOTE: **DETERIORATION** will be taken in to account if below applicable operating minima, but **IMPROVEMENT** is to be disregarded.

[3]When dispatching under the provisions of the MEL, those MEL limitations affecting instrument approach minima must be considered in determining ETOPS alternate minima.

7.10.2. Circling minima are normally not taken into account for ceiling minima. However, if the weather forecast requires the consideration of a circling approach, add 400ft to the published circling minima.

7.10.3. Period of validity:

a) For each alternate airport, the dispatch weather minima must be ensured during

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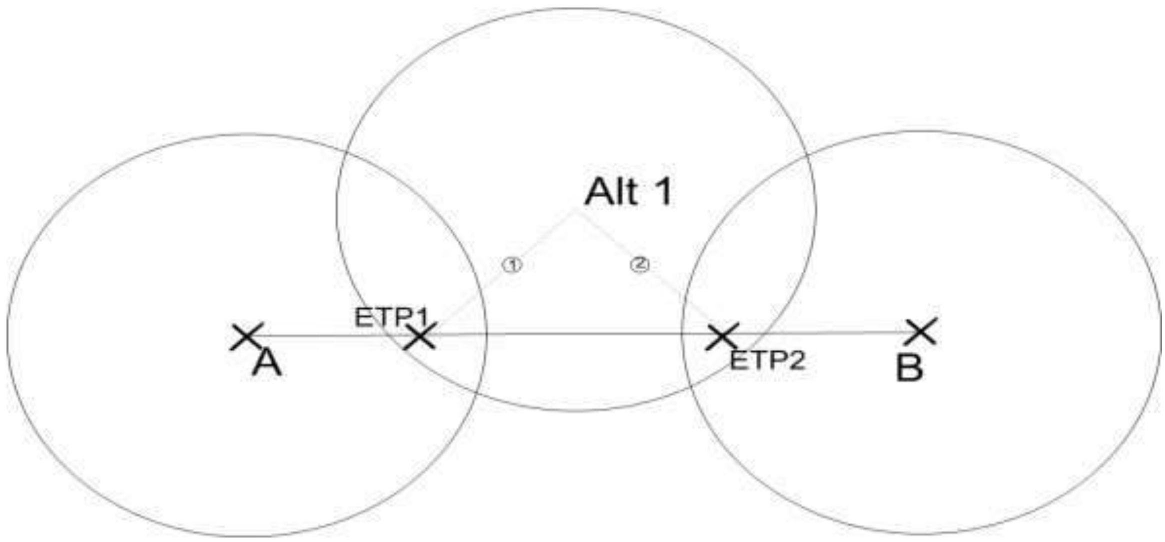
a certain time period. This period of validity begins one hour before the earliest ETA at this airport and ends one hour after the latest ETA.

b) The earliest ETA to Alt1: Diversion from ETP1 to Alt1 with no failure (e.g. medical emergency) All Engines Operating (AEO) speed (e.g. VMO)

c) The latest ETA to Alt1: Diversion from ETP2 to Alt1 following loss of pressurization-(FL100 or MORA whichever is higher) One Engine Inoperative (OEI) or All Engine Operative (AEO) speed (e.g. LRC).

7.10.4. The validity period can be illustrated by the following figure:



a) Estimated flight time from: Departure Airport to ETP1 = T1

Departure Airport to ETP2 = T2

ETP1 to Alternate airport = TA1

ETP2 to Alternate airport = TA2

Departure time = DT

b) Period of validity start: (DT+T1+TA1) -1 Hour

c) Period of validity end: (DT+T2+TA2) +1 Hour

7.11. PRE-FLIGHT - COCKPIT PREPARATIONS

7.11.1. Flight crew should note that there are additional MEL restrictions for ETOPS Operations.

7.11.2. All FMS preparations are to be cross-checked by both pilots as for non ETOPS operations, all track and distance between waypoints are to be verified. ETOPS waypoints such as ETPs shall not be added to the legs page by entering a LAT/Long waypoint from a operational flight plan or using an along track waypoint, it may be entered in to the FIX page. Diversion airports may also be added to the Fix page as the Flight progresses.

7.11.3. On the A-350,777/787, B767/757 and B737 an ETA may be entered on the ETA / ALT line on the fix page for reference.

7.12. IN-FLIGHT ETOPS PROCEDURES

7.12.1. BEFORE ETOPS ENTRY POINT:

a) The crew must make every effort to obtain weather forecasts or reports & check the adequacy of the ETOPS en-route alternates. The ETOPS dispatch minima do not apply when airborne.

b) If weather forecasts are lower than the normal crew minima & if any conditions are identified that would preclude a safe approach & landing at unidentified ETOPS alternate airports during the estimated time of use, then re-routing is required, or turn back if no route at the authorized distance from an en-route alternate airport can be used.

7.12.2. AFTER ETOPS ENTRY POINT:

a) The crew should continue to update the weather forecasts and reports for en-route alternates including ETOPS alternates. There is no requirement to modify the normal course of the flight if en-route alternate weather degrades below normal minima.

b) As for a normal flight, the crew must make every effort to keep themselves informed on the weather at the destination and the destination alternate.

7.12.3. FUEL MONITORING

a) The procedures normally used as per airline policy is also applicable for ETOPS. This is true even for flights where ETOPS fuel planning is the limiting factor. There are no requirements in the ETOPS rules to reach the CP with the Fuel On board (FOB) being at least equal to the fuel required by the critical fuel scenario.

b) This means that CP should not be considered as a re-clearance point. Therefore, if during the flight it appears that the estimated FOB at the CP will be lower than the fuel required by the critical fuel scenario, there is no requirement to make a diversion, provided the estimated fuel at destination is above the minimum required to divert to the destination alternate.

c) Normal rules apply; however, it is recommended that if the CP is regularly over flown with a FOB lower than the fuel required by the critical fuel scenario, the appropriate corrective actions should be taken in the way the required fuel is determined at dispatch (i.e. increase performance factor, route reserves, etc.)

7.13. APU HIGH ALTITUDE START

7.13.1. It is a requirement, essential to the continuation of ETOPS Operations approval, that we constantly provide the reliability of the APU. This standard procedure carried out as part of maintenance program.

7.13.2. When a flight is assigned by maintenance, the crew will be asked to perform an inflight start check. The start attempt should be initiated before top of descent (normally at least 15 minutes prior to top of descent) or at such time that will ensure a 2-hour cold soak at altitude before the start attempt. Once a successful start is achieved, it must be shut down at the commencement of descent until after landing and the maintenance Log shall be annotated "APU High altitude start accomplished". Three consecutive start attempts are permitted.

NOTE: For B737 aircraft the APU must be started and running prior to reaching a point farther than 60 minutes from an adequate alternate airport and kept running for the duration of the ETOPS flight.

7.14. DIVERSION DECISION-MAKING

7.14.1. Re-routing or diversion decisions consider the following:

a) Loss of MNPS or RVSM capability, before entering the area (as applicable), weather minima at diversion airport(s) going below the company / crew en-route minima, before reaching the EEP, or diversion airport(s) becoming unsuitable for any reason.
b) Failure cases requiring a diversion to the nearest airport (cases leading to a LAND ASAP).
c) Excessive fuel consumption, (check inflight fuel monitoring on FOM 2.12).

7.14.2. Whatever the one engine-inoperative speed schedule assumed in the determination of the area of operation, the crew is free to adopt the strategy it considers the most appropriate after assessment of the overall situation. This means that in conducting the diversion the application of the pre-planned speed strategy is not mandatory.

7.14.3. The final decision belongs to the crew who should choose the safest course of action, to include diversion airport, as well as appropriate diversion speed and altitude.

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7.14.4. In case of a cargo fire, diversion to the nearest suitable airport is mandatory, whatever is the performance, in term of protection time, of the fire extinguishing system.

7.15. IN-FLIGHT DIVERSION STRATEGIES

7.15.1. This section provides the single engine performance data to be used for the conduct and monitoring of the flight following an engine failure.

7.15.2. The diversion strategy (descent and cruise speed schedule) will be selected as a function of the prevailing operational factors (e.g. obstacles clearance requirements and/or ETOPS operations).

7.15.3. Depending on the prevailing operational constraints, the most appropriate diversion strategy shall be selected, out of the following options.

TABLE 5

	STANDARD STRATEGY	OBSTACLE STRATEGY	FIXED SPEED STRATEGIES
DESCENT TO CEILING	company SPD MCT	Drift down SPD MCT	Company SPD MCT
CRUISE	LRC Ceiling	If Obstacle Not Cleared: Maintain Min. Maneuvering speed	Check for Maximum Altitude Capability
	LRC Speed	If Obstacle Cleared: Revert to standard Strategy	Company Speed
DESCENT TO DESTINATION	Thrust Idle / Normal Speed		

7.15.4. The use of these strategies is at the discretion of the pilot. As a guideline, the considerations are as follows:

a) Standard strategy gives you the best fuel burn-off but the maximum diversion time required.

b) Fixed Speed Strategy will give you the minimum possible diversion times at the expense of fuel.

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c) Obstacle strategy should be used until clear of all obstacles.

7.15.5. The determination of the area of operation is based on a diversion at a selected single-engine Mach/IAS speed schedule (in still air and ISA conditions). In practice this speed can vary between Minimum Maneuvering Speed and Vmo / Mmo. The aircraft has been designed, flight tested and certified to safely fly within this range of speeds even with one engine inoperative. The choice of the ETOPS diversion is made by Ethiopian as a function of its route structure and associated constraints. Therefore, a diversion at high speed will maximize the maximum diversion distance and hence the area of operation, whereas a diversion at low speed will reduce the maximum diversion distance during the allowed maximum diversion time while permitting a higher diversion altitude. But, at the same time, this will minimize the fuel consumption. For non-ETOPS operations, in case of an engine failure, either the standard strategy or the obstacle clearance strategy is considered for diversion. The standard strategy corresponds to a descent at a speed shown on **TABLE 1** in this section down to an altitude close to the LRC ceiling, and a diversion cruise at LRC speed. These speeds are called Company Speeds.

7.15.6. The obstacle clearance strategy corresponds to a drift-down at Minimum Maneuvering speed (EO FMC speed) until the obstacles are cleared. Once the obstacles are cleared, the standard strategy is applied. For ETOPS operations, in case of an engine failure, if a pilot chose to fly either the standard strategy or the obstacle clearance diversion strategy, the associated diversion speed, respectively LRC speed and minimum maneuvering speed, which are substantially low speeds, would restrict the maximum diversion distance. Consequently, it may result in a restricted area of operation, reducing operational capabilities. Therefore, for ETOPS operations, higher one-engine-inoperative diversion speeds will extend the area of operations.

7.15.7. The typical ETOPS diversion strategy is now called "Fixed Speed Strategy" in order to differentiate it from the standard obstacles strategies. The word "fixed" is used to emphasize the fact that a selected speed schedule is followed during both the diversion descent and cruise phases, (except in case of cabin pressurization loss). Whereas standard and obstacle strategies consider during descent Company Speeds and minimum maneuvering respectively and during diversion cruise LRC speed which is a function of the aircraft weight and flight altitude (if terrain is not a factor).

7.15.8. In addition, it must be ensured that the net flight path and net ceiling for the selected ETOPS diversion speed clear any en-route obstacle by the appropriate margin as specified in the applicable operational regulations.

7.15.9. The following procedure is for guidance only. Even though all possible contingencies cannot be covered, this will provide guidance in such cases as: inability to maintain assigned flight level due to weather (severe turbulence), aircraft performance problems, or pressurization failures. This is also applicable primarily when a rapid descent, turn back, or diversion is required. The pilot's judgment will determine the specified sequence of actions to be taken having regard to the prevailing circumstance.

7.15.10. If unable to continue a flight as planned for NAT airspace:

a) A reversed clearance should be obtained, prior to initiating any action.
b) Broadcast your call sign, your position (including the ATS route designator or the track code as appropriate), intentions, at frequent intervals on 121.5 MHZ (with 123.45 as a back-up frequency).
c) Until a revised clearance is obtained, the specified NAT contingency procedures should be complied with.
d) Special procedures; this procedure is intended to take the A/C safety off the assigned track in the OTS in an expeditious manner until the revised clearance is obtained. Therefore, while accomplishing this procedure, it is mandatory to obtain an ATC clearance in the process of leaving the NAT airspace.

7.15.11. Initial Action for NAT airspace:

a) Leave the assigned route or track by turning 45 (degree) to the right or left whenever possible to establish an offset track of 15NM. The direction of the turn should be determined by the position of the A/C relative to any organized route or track system. (e.g. Whether the aircraft is outside, at the edge or within the system).
b) Other factors which affect the direction of turn are: direction to alternate airport, terrain clearance, and levels allocated on adjacent route or tracks.

7.15.12. Subsequent Action for NAT airspace:

a) Aircraft able to maintain its assigned flight level shall, once established on the offset track as specified above:

   I. Climb or descend 500 ft. if below flight level 410.
   II. Climb 1000 ft. or descend 500 ft. if at flight level 410.

b) Aircraft unable to maintain assigned flight level should;

   I. Minimize the descent while acquiring the 15 NM offset track and expedite descent to a level not used by the NAT traffic (BELOW 285).
   II. Before commencing any diversion across the flow of adjacent traffic aircraft should leave the NAT airspace.

NOTE: refer to MNPSA manual for more information

7.16. ETOPS FLIGHT MESSAGE

7.16.1. The Dispatcher constructs the flight message, which provides the crew with all available

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and relevant current reports and/or information relating to airport conditions, serviceability of navigational facilities, both electronic and visual, and weather reports and forecasts for the complete route. Warnings of forecasts hazardous weather conditions such as CAT, thunderstorms, and low altitude wind shear are also given. The Dispatcher is responsible for ensuring that latest NOTAMs are reviewed for serviceability of navigational aids both en-route and at all airports listed in the plan.

7.16.2. The ETOPS Flight Message is a seven-section document prepared by the Ethiopian Airlines Dispatch Section and transmitted (normally by check to the point of departure of an ETOPS flight).

a) Section One - NOTAMs

- All NOTAMs relevant to the route are listed.

b) Section Two - Weather

- This section lists latest available actual and forecast weather conditions for all en-route alternates, in addition to those for destination and destination alternate airports.

c) Section Three – Operational flight plan (OFP)

- In addition to the standard OFP route information, an ETOPS flight plan indicates ETOPS Entry/Exit Points (EEP/EXP), En-route alternates, Critical Point (CP ETOPS) and any extra fuel required for the ETOPS section (ETOPS ADJUSTED). Individual breakdown of en-route alternate requirements are shown in addition to earliest and latest times of availability of these alternatives. A section indicating fuel requirements from the CP is also included.

d) Section Four - Warnings

- This section indicates any Navigational Warnings company originated warnings, or other forecast or known hazard that could affect the flight.

e) Section Five – Company Notices

- Any company administrative or operational notice of a non-hazardous nature is included in this section.

f) Section Six – Additional Information

- This section conveys any other relevant messages to the crew that may not have been previously addressed.

g) Section Seven – ETOPS Release

- ETOPS release by the Flight Dispatch Officer is required by law, and thus this section, which presents a summary of all parameters of the flight including the names of the crew and the Dispatcher. The release statement is an essential and legal requirement of the flight message. The Dispatcher by putting his name to these documents, takes a shared responsibility with the Pilot in Command for the release of the flight. However, ultimate responsibility remains with the aircraft's Pilot in Command who can override any decisions made by the Dispatcher.

7.17. APPROVED ETOPS ROUTES

7.17.1. Ethiopian airlines approved ETOPS routing and description are all contained in the Flight Operations Route manual (FORM).

7.17.2. All applicable routes and ETOPS alternates are carefully depicted in the Flight Operations Route manual (FORM) En-route alternates are based on the preferred routing.

7.17.3. The choice of en-route alternates listed in the FORM is not limiting. When necessary, the Pilot in Command may nominate other adequate airports as en-route alternates in flight.

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SECTION 2.17 COLD WEATHER OPERATIONS

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SECTION 2.17 COLD WEATHER OPERATIONS

1.0. PURPOSE

The purpose of this cold weather operations procedure is to provide guidelines on how operations in cold weather is conducted

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. Take-off and Landing performance data for fluid contaminated runways are given in FCOM.

4.2. Aircraft performance on wet or contaminated runways as no accurate correlation can be made between the aircraft friction coefficient on a given runway and the reported friction coefficient or braking action, the performance data given in the FCOM have been established for given depths of water or contaminant (slush, snow).

NOTE: Therefore, the only way to determine the applicable take-off and landing performance is to obtain the depth and type of contaminant.

4.3. Ethiopian may not operate an aircraft in IFR or VFR at night unless that aircraft is equipped with two independent static pressure systems/ports vented to outside atmospheric pressure so that they will be least affected by airflow variation or moisture or other foreign matter. (ECARAS 7.10.1.10)

4.4. If surface mean wind speeds of **60** kts or above is reported, takeoff or landing is not authorized and the airfield must be considered closed.

5.0 DEFINITION

- **VMC**— Visual Meteorological Conditions.
- **IMC**— Instrument Meteorological Conditions.

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5.1. FRICTION COEFFICIENT: The friction coefficient is defined as the ratio of the maximum available tire friction force and the vertical load acting on the tire. This coefficient is named "Mu" or "μ".

5.2. AIRPLANE REFERENCE FIELD LENGTH is the minimum field length required for take-off at maximum certificated take-off mass, sea level, standard atmospheric conditions, still air and zero runway slope, as shown in the appropriate airplane flight manual prescribed by the certificating authority or equivalent data from the airplane manufacturer. Field length means balanced field length for airplanes, if applicable, or take-off distance in other cases.

6.0 RESPONSIBILITY

It is the responsibility of each flight crew to know the procedures mentioned hereunder in order to conduct operations in cold weather

7.0 PROCEDURE

7.1. ADVERSE RUNWAY CONDITIONS

7.1.1 For takeoff and landing performance purpose runway surfaces are categorized as Dry, Slippery or contaminated. A runway is considered to be:

- **a) DRY** when neither slippery nor contaminated.
- **b) SLIPPERY** when due to the presence of a shallow depth of water-based material on the runway, the stopping capability is reduced below that available on the runway when it is dry, or the runway is covered by a solid water-based material which inherently has a lower co-efficient of friction than a dry runway surface. (e.g. ice)
- **c) CONTAMINATED** when due to the presence of a significant depth of water-based material on the runway the stopping capability is reduced below that available on the runway when it is dry. Precipitation drag occurs.
- d) **Frost** is a deposit of small white ice crystals formed on the ground or other surfaces when the temperature falls below freezing.

7.1.2. A runway is considered to be contaminated, slippery or icy when more than 25 percent of the runway surface area (whether in isolated areas or not) with in the required length and width being used, is covered by water-based material.

- i. Water-based materials which can result in slippery or contaminated runways are defined in the following table.
- ii. Takeoff is not allowed on icy runways.
- iii. Maximum takeoff thrust must be used on contaminated runways.
- iv. The available clearance or treated runway width shall not be less than 30m.

7.1.4. takeoff or landing is not allowed on runways covered with more than:

- ½ inch (13mm) of standing water, slush, wet snow.

- 4 inch (102mm) of dry snow.

TABLE 1

Water-Based Material on Runways	
Term	**Definition**
Damp	A runway is damp when the surface is not dry, but when the water on it does not give a shiny appearance.
Wet	A runway is wet when the surface has a shiny appearance due to a thin layer.
Standing Water	Water of a substantial depth.
Slush	Water saturated with snow which spatters when stepping firmly on it.
Wet Snow	Snow, which if compacted by hand, will stick together and tend to form a snowball.
Dry/Loose Snow	Snow which can be blown of loose, or if compacted by hand will fall apart again upon release.
Compacted Snow	Snow which has been compressed.
Icy	Ice

7.2. GLOBAL REPORTING FORMAT (GRF)

The following table is an abbreviation version of the Matrix for runway condition assessment in terms of Takeoff and Landing performance assessment. The runway condition descriptions and codes are aligned with control/braking action reports.

TABLE 2

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Runway Condition Assessment Matrix		
Runway surface Description	**Runway Condition Code**	**Control/ pilot report of runway Braking Action**
Dry	6	---
Frost Wet (dampness or water up to & including 1/8 "(3mm) depth) **Up to & including 3mm depth** Slush Dry Snow wet Snow	5	Good
-15^{0}C and colder OAT: Compacted Snow	4	Good to Medium
Slippery when wet (wet runway) Dry or Wet Snow (any depth) over compacted Snow **More than (3mm) depth** Dry Snow Wet Snow **Warmer than 15^{0}C OAT**: Compacted Snow	3	Medium
More than 1/8" (3mm) Standing Water Slush	2	Medium to poor
Ice	1	Poor
Wet Ice Water on top of compacted Snow Dry Snow or Wet Snow over Ice	0	Nil

7.2.1. The ICAO Global Reporting Format (GRF) is a globally-harmonized methodology for runway surface condition assessment and reporting that it is intended to be the only such reporting format for international aviation, with the objective of reducing runway excursions, thus improving the safety of airport operations.

7.2.2. The ICAO Global Reporting Format for runway surface conditions (GRF) mitigates the risk of runway excursions by enabling a harmonized assessment and reporting of runway surface conditions and an improved flight crew assessment of take-off and landing performance.

7.2.3. RCR is a comprehensive standardized report relating to runway surface condition(s) and its effect on the airplane landing and takeoff performance.

7.2.4. Reporting runway surface conditions has two parts; Airplane performance section and Situational awareness section.

a) Airplane performance section has the following;

Item A- Airplane Location Indicator (example, for Paris “LFPG”)

Item B- Date and Time Assessment (example, April 28 1600UTC will be “28041600”)

Item C- Lower Runway Designator Number (example, “09R”)

Item D- Runway Condition Code (Each Runway Third) (example, “5/4/3” wet Runway, compacted snow, slippery when wet)

Item E- Percent Coverage (Each Runway Third) (example, 25/50/NR)

TABLE 3

Assessed percent	Reported Percent
Less than 10%; or Runway condition is “DRY” (Item G) & RWYCC=6 (Item D); or If the percent is not reported	NR
10.25%	25
26-50%	50
51-75%	75
76-100%	100

Item F- Depth of Loose Contaminant (Each Runway Third) (example, “04/05/04” 4mm in depth, 5 mm in depth and 4mm in depth);

- For STANDING WATER, 04 (4mm) is the minimum depth value at and above which the depth is reported. (from 3mm and below, the runway third is considered WET).

- For SLUSH, WET SNOW & DRY SNOW, 03(3mm) is the minimum depth value at and above which the depth is reported.
- Above 4mm for STANDING WATER & 3mm for SLUSH, WETSNOW & DRY SNOW an assessed value is reported and significant change relates to observed change from this assessed value.

Item G- Condition Description for Each Third (example, "wet runway, compacted snow, slippery when wet"

Item H- Width of Runway to which the RWYCCs Apply (example, "40")

b) Situational Awareness Section

Item I- Reduced Runway Length (example; "RWY 09R REDUCED TO 2800)

Item J- Drifting Snow on the Runway (example; "DRIFTING SNOW" if it exists)

Item K- Loose Sand on the Runway (example; "RWY 09R LOOSE SAND" if it exists)

Item L- Chemical Treatment on the Runway (example; "RWY 09R CHEMICALLY TREATED" if it exists)

Item M- Snow Banks on the Runway, if present, distance from runway centerline (m) followed by: L", "R" or "LR" as applicable. (example; "RWY 09R SNOWBANK R20 FM CL. If it exists)

Item N- Snow Banks on the Taxiway (example; "TWY K SNOWBANK". If it exists)

Item O- Snow Banks adjacent to the Runway (example; "RWY 09L ADJACENT SNOWBANKS. If it exists)

Item P- Taxiway Conditions (example; "TWY B POOR".)

Item R- Apron Conditions (example; "APRON NORTH POOR")

Item S- Measured Friction Coefficient

Item T- Plain Language Remarks

7.2.5 DISSEMINATION OF INFORMATION

Through the AIS (Aeronautical Information Services) and ATS (Air Traffic Services); when the runway is wholly or partially contaminated by standing water, snow, slush, ice or frost, or is wet associated with the clearing or treatment of snow, slush, ice or frost.

Through the ATS only; when the runway is wet, not associated with the presence of snow, slush, ice or frost.

NOTE: AIS is disseminated through SNOWTAM and ATS is disseminated through Voice or ATIS.

7.3. SNOWTAM FORMAT AND RUNWAY CONDITION DECODING (New Provisions)

7.3.1. SNOWTAM is a special series NOTAM given in a standard format providing a surface condition report notifying the presence or cessation of hazardous conditions due to snow, ice, slush, frost, standing water, or water associated with snow, slush, ice, or frost on the movement area.

7.3.2. As of 5 November 2020, the maximum validity of SNOWTAM is 8 hours.

7.3.3. A SNOWTAM cancels the previous SNOWTAM. When a new SNOWTAM is issued for a specific aerodrome that has another valid SNOWTAM, the new one automatically replaces the older SNOWTAM (there is no need to refer the older SNOWTAM in the new SNOWTAM)

7.3.4. The letters used to indicate items (A to T; third column of the SNOWTAM template) are only used for reference purpose and should not be included in the messages.

7.3.5. Mandatory information's in SNOWTAM are Item A, B, C, D, G (For "G" mandatory only when RWYCC is reported 1 to 5; not needed if RWYCC is 0 or 6)

EXAMPLE: LFPG 04281600 09R 5/4/4 SLUSH/COMPACTED SNOW/COMPACTED SNOW

7.3.6. Normally SNOWTAM will have the data designator, for SNOWTAM it will be "SW" then geographical designator for states example LF=FRANCE, EG=United Kingdom, etc., (Refer DOC 7910 for more), then SNOWTAM serial number in a four-digit group (the SNOWTAM serial number resets at the beginning of each calendar year example, with SNOWTAM "0001" on January 01 at 0000UTC). Then after Items (A to T) continues.

EXAMPLE: SWLF0212 LFPG 04281600….

7.3.7. In the case of an error to a SNOWTAM message previously disseminated with the same serial number=COR.

7.3.8. When a SNOWTAM is reported on more than one runway of the same aerodrome for which the SNOWTAM is issued, B to H (Airplane Performance Section) should be reported.

7.4. SNOWTAM FORMAT AND RUNWAY CONDITION DECODING (Old Provisions)

A) EDDF B) 11070620 C) 10 D)2200 E)40L F)4/5/4 G)20/10/10 H)30/35/30 MUM J)30/5L k) TES L L) TOTAL M) 0900 N) NO P) YES 12 S)11070920 T) FIRST 300M RWY 10 COVERED BY 50 MM SNOW, RWY CONTAMINATION 100%

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EDDF: Aerodrome Location indicator

11190830: Month, day, hour & minutes in UTC.

10: The two digits correspond to the runway designator (e.g. 09, 27 etc.). In the case of parallel runways, the "Left" runway is indicated by the designator only (e.g. 09L as 09) while the "Right" runway has 50 added so that 09R becomes 59 and 27R becomes 77. Whenever all runways are affected the figure group 88 will be used.

NOTE: 99 may sometimes appear as the first two digits. This does not purport to be a runway indicator but means that the information is a repetition of the last message because no new message has been received in time for transmission.

2200: Cleared runway length, if less than published length (m).

40L: Cleared runway width, if less than published width (if offset left or right of centerline add "L: or "R").

4/5/4: Deposits over total runway length (observed on each third of the runway starting from threshold having the lower runway designation number).

NIL (0)-Clear and Dry	**4**- Dry Snow	**7**- Ice
1- Damp	**5**- Wet Snow	**8**- Compacted Snow
2- Wet	**6**- Slush	**9**- Frozen Ruts or Ridges
3- Rime or Frost		

20/10/20: Mean depth (mm) for each third of total runway length.

The depth of deposit is indicated by two digits in accordance with the following scale:

00 < 1 mm; **01**= 1mm; **02**=2mm; etc.; **10**=10mm etc.; **15**=15mm; etc.; **20**=20mm; etc. up to; **90**=90mm

NOTE: Code 91 is not used thereafter the depth is indicated as follows:

92=10cm; **93**=15cm; **94**=20cm; **95**=25cm; **96**=30cm; **97**=35cm; **98**=40cm or more

99=runway or runways non-operational due to snow, slush, ice, large drifts or runway clearance, but depth nor reported.

//= depth of deposit operationally not significant or not measurable.

30/35/30: Friction measurement on each third of runway and friction measurement device.

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TABLE 4

Measured or calculated coefficient	Estimated surface friction	Braking action
0.40 and above	5- Good	95
0.39-0.36	4- Medium/Good	94
0.35-0.30	3- Medium	93
0.29-0.26	2- Medium/poor	92
0.25 and below	1- Poor	91
9	9- Unreliable	99
Braking action not reported, runway not operational, aerodrome closed etc.		//

NOTE 1: Where braking action is assessed at number of points along a runway, the mean value will be transmitted or. if operationally significant, the lowest value.

NOTE 2: If measuring equipment does not allow measurement of friction with satisfactory reliability, which may be the case when a runway is contaminated by wet snow, slush, or loose snow, aquaplaning code **99** will be used.

J) 30/5L: Critical snowbanks. If present, insert height (cm) / distance from the edge of runway in meters followed by left (**L**) or right (**R**) side or both sides (**LR**) as viewed from the threshold.

K) YES L: If the runway lights are obscured inset **yes** followed by **L** or **R** or both **LR** as viewed from the threshold.

L) TOTAL: When further clearance will be undertaken, insert length (m)/ width (m) to be cleared or if full dimensions' insert TOTAL.

M) 0900: Anticipated time of completion (UTC).

N) NO: Taxi way conditions; "**NO**" if no taxi way serving the associated runway are available.

P) YES12: If snowbanks more than 60 cm, insert "**YES**" followed by the lateral distance parting the snowbanks in meters.

R) -: apron conditions; enter "**NO**" if apron unusable.

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S) 11070920: Next planned observation/measurement is for (month/day/hour/minute in UTC).

T) First 300M RWY 10covered by 50 mm: plain Language remarks (including contaminated coverage and other significant information like sanding or de-icing).

NOTE: In METAR format the extent of runway contamination is expressed as a single digit in accordance with the following scale;

1 - less than 10% of runway contaminated (covered)

2 - 11% to 25% of runway contaminated (covered)

5 – 26% to 50% of runway contaminated (covered)

9 – 51% to 100% of runway contaminated (covered)

/ - not reported (e.g. due to runway clearance in progress).

7.5. SNOWTAM DECODER METAR FORMAT

a) The first two digits indicate the runway designator.
b) The third digit indicates the runway deposits.
c) The fourth digit indicates the extent of runway contamination.
d) The fifth and sixth digits indicate the depth of deposit.
e) The seventh and eighth digits indicate the friction co-efficient or Braking Action.

NOTE: the occasion may arise when a new report or a valid report is not available in time for dissemination with the appropriate METAR message. In this case, the previous runway state report will be repeated, as indicated by code **99** in place if runway designator.

EXAMPLE:

99421594- Dry snow covering 11% to 25% of the runway: depth 15mm; braking action medium to good.

- **14//99//-** Runway 14 non –operational due to runway clearance in progress.
- **14//////-** Runway 14 contaminated but reports are not available or are not updated due to aerodrome closure or curfew, etc.
- **88//////-** All runways are contaminated by reports are not available or are not updated due to aerodrome closure of curfew, etc.
- **14CLRD//-**runway 14 contaminations has ceased to exit.
- (No further reports will be sent unless recontamination occurs).

7.6. RUNWAY FRICTION CHARACTERISTICS

7.6.1. Ethiopian shall have a performance data control system, containing current performance data for each aircraft, route and airport it uses, approved by ECAA for obtaining, maintaining and distributing to appropriate personnel. The system approved by the Authority shall provide current obstacle data for departure and arrival performance calculations. (ECARAS 9.3.1.14)

7.6.2. The stopping performance of aircraft is to a great degree dependent on the available friction between the aircraft tires and the runway surface, their landing and take-off speeds. In some conditions the runway length required for landing or take-off could be critical in relation to the runway length available.

7.6.3. Adequate runway friction characteristics/braking action is mainly needed for three distinct purposes:

a) deceleration of the aircraft after landing or a rejected take-off;
b) directional control during the ground roll on take-off or landing, in particular in the presence of cross-wind, asymmetric engine power or technical malfunctions;
c) Wheel spin-up at touchdown.

7.6.4. To compensate for the reduced stopping and directional control capability for adverse runway conditions (such as wet or slippery conditions) performance corrections are applied in the form of:

a) Runway length increment;
b) Reduction in allowable take-off or landing weight;
c) Reduction of allowable cross-wind component.

7.6.5. Various systems are used to measure the runway friction coefficient/conditions:

- Skiddometer High pressure tire (SKH)
- Skiddometer Low pressure tire (SKL)
- Surface Friction Tester (SFT)
- Mu-meter (MUM)
- Diagonal braked vehicle (DBV)
- Tapley meter (TAP)
- James Brake Decelerometer (JBD)
- Brake meter- Dynometer (BRD)
- Grip Tester (GRT)

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7.6.6. The results of the friction measuring equipment do not generally correlate with each other for all surface conditions and no correlation has been established between these results and the stopping performance of an aircraft.

7.6.7. The only perfect way of measuring the friction coefficient "Mu" for a specific aircraft is by using that specific aircraft braking system on the surface concerned.

7.6.8. When friction measurement is not available but can be only estimated, the pilot is informed only of the estimated braking action reported as "good", "medium", "poor", "unreliable (nil)" or a combination of these terms.

7.6.9. Pilots should treat reported braking action measurements with caution and interpret them conservatively.

7.6.10. Friction measurements or braking action estimation may be reported:

a) In plain language by the tower.
b) By the routine weather broadcast.
c) By SNOWTAM.

7.6.11. ATC issues the latest braking action report for the runway in use to each arriving and departing aircraft. Pilots shall also be prepared to provide a descriptive runway condition report to ATC after landing if the braking action is degraded.

7.6.12. Meteorological observations in connection with knowledge of previous runway conditions will, in many cases, permit a fair estimate to be made of braking action.

7.6.13. On snow- or ice-covered runways not treated with, e.g. sand, the coefficient of friction varies from as low as 0.05 to 0.30. It is very difficult to state exactly how and why the runway conditions vary. The braking action is very much dependent upon the temperature especially near the freezing point. However, when it is freezing, the braking action could be fairly good, it will so remain if the temperature decreases but if the temperature rises to the freezing point or above, the braking action will decrease rapidly. Sometimes very low friction coefficient values occur when humid air is drifting in over an icy runway even though the temperature may be well below the freezing point.

7.6.14. Some of the various conditions which are expected to influence the braking action are given below:

7.7. FRICTION COEFFICIENT BETWEEN 0.10 AND 0.30 (POOR-MEDIUM/POOR)

- Slush or rain on snow- or ice-covered runway;
- Runway covered with wet snow or standing water;
- Change from frost to temperature above freezing point;

- Change mild to frost (not always);
- The type of ice which is formed after long periods of cold;
- A thin layer of ice formed:
- By frozen ground having been exposed to humidity or rain at 0°C or above;
- When due to radiation, e.g. when the sky clears, the runway surface temperature drops below freezing point and below the dew point (this ice formation can take place very suddenly and occur while the reported air temperature may still be quite a few degrees above the freezing point.)

7.8. FRICTION COEFFICIENT BETWEEN 0.25 AND 0.35 (MEDIUM/POOR-MEDIUM)

Snow-covered runways at temperatures below freezing point, exposed to sun;

- Slush-covered runway.

7.9. FRICTION COEFFICIENT BETWEEN 0.35 AND 0.45 (MEDIUM/GOOD-GOOD)

7.9.1. Snow-covered runways which have not been exposed to temperatures higher than about -2°C to -4°C.

7.9.2. Damp or wet runway without risk of hydroplaning (less than 3 mm water depth).

7.10. GUIDELINES FOR OPERATIONS ON SLIPPERY SURFACES

7.10.1. GENERAL CONSIDERATION

a) The use of thrust reversers is mandatory on contaminated runways.
b) The two most important variables confronting the pilot when runway coefficient of friction is low and/or conditions for hydroplaning exist are length of runway and crosswind magnitude.
c) The total friction force of the tires is available for two functions - braking and cornering. If there is a crosswind, some friction force (cornering) is necessary to keep the aircraft on the center line. Tire cornering capability is reduced during braking or when wheels are not fully spun up. Locked wheels eliminate cornering. Therefore, in crosswind conditions, a longer distance will be required to stop the aircraft.

7.11. RUNWAYS COVERED WITH SNOW, SLUSH OR STANDING WATER

7.11.1. Pilots should be aware of the problem imposed by snow slush and standing water on runways. i.e. retardation force during acceleration and deterioration of braking action.

7.11.2. In addition to the detrimental effect of snow, slush and standing water on acceleration and braking action, the possibility of severe damage to the aircraft from such impingement shall be taken into account.

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7.11.3. Depth limit and correction are published in the respective AOM.

7.11.4. The pilot in command shall be responsible to acquire information on the condition of the runway by every reasonable means at his disposal. If there is any uncertainty, the final decision shall rest with the pilot in command.

7.11.5. Due consideration shall be given to all other relevant factors such as anticipated wind and characteristics of runway concerned i.e. length, width, slope and type of surface.

7.11.6. Refer to each Aircraft Operations Manual or Flight Crew Training Manual and FOM for cross wind guidance.

7.12. TAXIING

7.12.1. Aircraft may be taxied at the captain’s discretion on ramps and taxiways not cleared of snow and slush. More power than normal may be required to commence and continue taxi so care should be taken to avoid jet blast damage to buildings, equipment and other aircraft.

7.12.2. Be aware of the possibility of ridges or ruts of frozen snow that might cause difficulties. The boundaries/edges of maneuvering areas and taxiway should be clearly discernible. If in doubt, request "Follow me" guidance.

7.12.3. When executing sharp turns while taxiing or parking at the ramp, remember that braking and steering capabilities are greatly reduced with icy airport conditions; reduce taxi speed accordingly.

7.12.4. Slat/flap selection should be delayed until immediately before line up to minimize contamination.

7.13. TAKE-OFF

7.13.1. Severe retardation may occur in slush or wet snow.

7.13.2. In most cases, lack of acceleration will be evident early on the takeoff run. Maximum permissible power must be used from the start.

7.13.3. Large quantities of snow or slush, usually containing sand or other anti-skid substances may be thrown into the engines, static ports and onto the airframe. Pod and engine clearance must be watched when the runway is cleared and snow is banked at the sides of runways or taxiway.

7.14. LANDING

7.14.1. Pilots should be aware that where rain, hail, sleet or snow showers are encountered on the approach or have been reported as having recently crossed the airfield, there is a high probability of the runway being contaminated.

7.14.2. The runway state should be checked with ATC before commencing or continuing the approach. Very often a short delay is sufficient to allow the runway to drain or the contaminant to melt.

7.14.3. Use of reverse thrust on landing on dry snow in very low temperatures will blow the dry snow forward especially at low speed. The increase in temperature may melt this snow and form clear ice on re-freezing on static ports.

7.14.4. The required landing field length for dry runways is defined as 1.67 times the demonstrated dry landing distance. For wet runways, this landing distance requirement is increased by 15%.

7.14.5. The required landing field length for contaminated runways is defined as 1.15 times the demonstrated contaminated landing distance.

7.14.6. The shortest stopping distances on wet runways occur when the brakes are fully applied as soon as possible after main wheel spin up with maximum and immediate use of reverse thrust. Landing on contaminated runways without antiskid should be avoided. It is strongly recommended to use the Auto brake (if available) provided that the contaminant is evenly distributed.

7.14.7. The factors and considerations involved in landing on a slippery surface are quite complex and depending on the circumstances, the pilot may have to make critical decisions almost instinctively. The following list of items summarizes the key points to be borne in mind. Several may have to be acted upon simultaneously.

7.14.8. Limit crosswind components when runway conditions are marginal and runway length short.

a) Establish and maintain a stabilized approach.

b) Consider the many variables involved before landing on a slippery runway.

c) Landing weather forecast

I. Aircraft weight and approach speed
II. Landing distance required
III. Hydroplaning (aquaplaning) speed
IV. Condition of tires
V. Brake characteristics (anti-skid, Auto brake mode)
VI. Wind effects on the directional control of the aircraft on the runway
VII. Runway length and slope

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VIII. Glide path angle

d) Do not exceed VAPP/ calculated VREF + wind increment at the threshold. an extended flare is more likely to occur if excess approach speed is present.

e) Be prepared to go around.

f) Flare the aircraft firmly at the 1000 ft. aiming point. Avoid buildup of drift in the flare and runway consuming float. A firm landing, by facilitating a prompt wheel spin up, also ensures efficient antiskid braking.

g) Select reverse thrust as soon as possible.

h) Get the nose of the aircraft down quickly. Do not attempt to hold the nose off for the aerodynamic braking. Aim to have the nose wheel on the ground by the time reverse thrust reaches the target level.

i) If the Auto brake is not available, and if remaining runway length permits, allow the aircraft to decelerate to less than dynamic hydroplaning speed before applying wheel brakes. If however maximum braking is required apply and hold full brake pedal deflection. Continue to apply rudder inputs while braking. The brakes are the primary means for stopping the aircraft but if necessary the full reverse thrust may be maintained until the aircraft is fully stopped.

j) Excessive braking in crosswinds will lead to the aircraft drifting away from the centerline.

k) Keep the aircraft aligned with the runway centerline. Use rudder and aileron inputs.

l) As rudder effectiveness decreases, reduce aileron deflection proportionately.

CAUTION: Do not allow large deviations from the runway heading to develop as recovery can become very difficult. Use of the nose wheel steering is not recommended. Under slippery conditions, the nose wheels must be closely aligned with the aircraft track or they will scrub.

m) If directional or lateral control difficulties are experienced, disconnect the auto brake, if necessary, reduce reverse thrust levels symmetrically, and regain directional control with rudder, aileron and differential braking. Once under control, reapply manual braking and increase symmetrical reverse levels as required while easing the aircraft back towards the runway centerline.

7.14.9. After landing in heavy slush do not retract the slats and flaps. Allow ground personnel to clear ice and slush from slats and flaps before full retraction. Taxi with caution to parking area as flaps extended provides a much-reduced ground clearance.

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7.15. ETHIOPIAN MAXIMUM RECOMMENDED CROSSWIND & TAIL WIND VALUES

7.15.1 CAPTAINS CROSS WIND LIMITATIONS

Maximum cross wind values indicated in the FCOM or AFM or OEM is considered as a limitation for captains.

7.15.2 INEXPERIENCED CAPTAINS CROSS WIND LIMITATIONS

TABLE 5

Dry	---	25
Good	0.40 and Above	25
medium/good	0.36 to 0.39	20
medium	0.30 to 0.35	15
medium/poor	0.26 to 0.29	10
poor	0.25Bnd below	Not Allowed
unreliable	-	Not Allowed

NOTE: This may be increased to values indicated in individual FCOM or AFM as limitation per the following conditions:

a) If the flight is diverted to alternate and any further delay cannot be made
b) When up on arrival at destination there is critical fuel condition
c) If the pilot determines at the time, given the situation, it is the safest course of action.

7.15.3. FIRST OFFICERS CROSS WIND LIMITATIONS

TABLE 6

Dry	---	20 kts
Good	0.40 and above	15 kts
Medium/good	0.36 to 0.39	5-10 kts
Medium	0.30 to 0.35	Less than 5 kts
Medium/poor	0.26 to 0.29	Not Allowed
poor	0.25 and below	Not Allowed
unreliable	-	Not Allowed

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NOTE: a first officer may carry out takeoff and landing if the runway is not contaminated and the above cross wind limitation is respected.

7.15.4. Maximum tailwind values indicated in the FCOM or AFM is considered as a limitation for T/off and Landing. Refer to FOM 1.10(4.1.6 d).

7.16. MINIMUM RUNWAY WIDTH

7.16.1. The code is composed of two elements which are related to the airplane performance characteristics and dimensions. Element 1 is a number based on the airplane reference field length and element 2 is a letter based on the airplane wing span and outer main gear wheel span.

7.16.2. The code number for element 1 shall be determined from **TABLE-7** column selecting the code number corresponding to the higher value of the airplane reference field lengths of the airplanes for which the runway is intended.

7.16.3. The code letter for element 2 shall be determined from **TABLE-7** column 3, by selecting the code letter which corresponds to the greatest wing span, or the greatest outer main gear wheel span, whichever gives the more demanding code letter of the airplanes for which the facility is intended.

TABLE 7: AERODROME REFERENCE CODE

Code element 1		Code element 2		
Code number (1)	Airplane reference Field length (2)	Code letter (3)	Wing span (4)	Outer main gear wheel span[a] (5)
1	Less than 800m	A	Up to but not including 15m	Up to but not including 4.5m
2	800m up to but not including 1200m	B	15m up to but not including 24m	4.5m up to but not including 6m
3	1200m up to but not including 1800m	C	24m up to but not including 36m	6m up to but not including 9m
4	1800m and over	D	36m up to but not including 52m	9m up to but not including 14m
		E	52m up to but not including 65m	9m up to but not including 14m
		F	65m up to but not including 80m	14m up to but not including 16m

a) Distance between the outside edges of the main gear wheels.

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7.16.4. The width of a runway shall be not less than the appropriate dimension specified in the following tabulation.

TABLE 8: CODE LETTER

Code number	A	B	C	D	E	F
1*	18m	18m	23m	-	-	-
2*	23m	23	30m	-	-	-
3	30m	80	30m	45m	-	-
4	-	-	45m	45m	45m	60m

* The width of a precision approach runway shall be not less than 30m where the code number is 1 or 2.

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SECTION 2.18 A/C GROUND DE-ICING/ANTI-ICING

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SECTION 2.18 A/C GROUND DE-ICING/ANTI-ICING

1.0 PURPOSE

The purpose of this A/C deicing/anti-icing procedure is to establish guidelines to the technical airworthiness of the airplane as a result of ice formation.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- The Person who releases the aircraft as airworthy
- Ground crew/technician(s) who perform the de-icing/ant-icing

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1 Airplane preparation for service begins and ends with a thorough inspection of the airplane exterior. The airplane and specially its surfaces providing lift, controllability and stability must be aerodynamically clean. Otherwise, safe operation is not possible.

5.0. DEFINITIONS

Active frost is a condition when frost is forming. It occurs when airplane surface temperature is at or below 0°C (32°F) and at or below dew point.

Anti-icing is a precautionary procedure, which provides protection against the formation of frost or ice and the accumulation of snow on treated surfaces of the aircraft, for a limited period of time (holdover time).

Anti-icing code describes the quality of the treatment the aircraft has received and provides information for determining the holdover time.

Anti-icing fluid: a) mixture of water and Type I fluid, b) premix Type I fluid, c) Type II fluid, Type III fluid or Type IV fluid, d) mixture of water and Type II fluid, Type III fluid or Type IV fluid.

Check is an examination of an item against a relevant standard by a trained and qualified

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person.

Clear ice is a coating of ice, generally clear and smooth, but with some air pockets. It is formed on exposed objects at temperatures below, or slightly above, freezing temperature, with the freezing of super-cooled drizzle, droplets or raindrops.

Cold soak Effect: Even in ambient temperature between -2°C and at least +15°C, ice or frost can form in the presence of visible moisture or high humidity if the aircraft structure remains at 0°C or below. Anytime precipitation falls on a cold-soaked aircraft, while on the ground, clear icing may occur. This is most likely to occur on aircraft with integral fuel tanks,

after a long flight at high altitude. Clear ice is very difficult to visually detect and may break loose during or after takeoff. The following can have an effect on cold soaked wings: Temperature of fuel in fuel cells, type and location of fuel cells, length of time at high altitude flights, quantity of fuel in fuel cells, temperature of the loaded fuel and the time since re-fueling.

Compacted snow: Snow which has been compressed into a solid mass that resists further compression and will hold together or break up into chunks if picked up. Specific gravity: 0.5 and over.

Contamination: in this chapter refers to all forms frozen or semi-frozen moisture such as frost, snow, ice, or slush.

Contamination check is check of airplane surfaces for contamination to establish the need for de-icing

Contaminated runway: A runway is considered to be contaminated when more than 25% of the runway surface area (whether in isolated areas or not) within the required length and width being used is covered by the following:

- Surface water more than 3 mm (0.125 in) deep, or slush, or loose snow, equivalent to more than 3 mm (0.125 in) of water; or

- Snow which has been compressed into a solid mass which resists further compression and will hold together or break into lumps if picked up (compacted snow); or Ice, including wet ice

Continuous: Intensity changes gradually, if at all.

Damp runway: A runway is considered damp when the surface is not dry, but when the moisture on it does not give it a shiny appearance.

De-icing is a procedure by which frost, ice, slush or snow is removed from the aircraft in order to provide clean surfaces. This may be accomplished by mechanical methods, pneumatic methods, or the use of heated fluids.

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De/Anti-icing is a combination of the two procedures, de-icing and anti-icing, performed in one or two steps. A de-/anti-icing fluid, applied prior to the onset of freezing conditions, protects against the buildup of frozen deposits for a certain period of time, depending on the fluid used and the intensity of precipitation. With continuing precipitation, holdover time will eventually run out and deposits will start to build up on exposed surfaces. However, the fluid film present will minimize the likelihood of these frozen deposits bonding to the structure, making subsequent de-icing much easier.

Dew point is the temperature at which water vapor starts to condense.

Dry runway: A dry runway is one which is neither wet nor contaminated and includes those paved runways which have been specially prepared with grooves or porous pavement and maintained to retain "effectively dry" braking action, even when moisture is present.

Fluids (de-icing and anti-icing):

- **De-icing fluids are:**
  a) Heated water
  b) Newtonian fluid (ISO or SAE or AEA Type I in accordance with ISO 11075 specification)
  c) Mixtures of water and Type I fluid
  d) Non-Newtonian fluid (ISO or SAE or AEA Type II or IV in accordance with ISO 11078 specification)
  e) Mixtures of water and Type II or IV fluid.

De-icing fluid is normally applied heated to ensure maximum efficiency.

- Anti-icing fluids are:
  a) Newtonian fluid (ISO or SAE or AEA Type I in accordance with ISO 11075 specification)
  b) Mixtures of water and Type I fluid
  c) Non-Newtonian fluid (ISO or SAE or AEA Type II or IV in accordance with ISO 11078 specification)
  d) Mixtures of water and Type II or IV fluid.
- Anti-icing fluid is normally applied unheated on clean aeroplane surfaces.

Dry snow: Snow which can be blown if loose or, if compacted by hand, will fall apart upon release; specific gravity: up to but not including 0.35. Dry snow is normally experienced when temperature is below freezing and can be brushed off easily from the airplane.

Freezing conditions are conditions in which the outside air temperature is below +3°C (37.4F) and visible moisture in any form (such as fog with visibility below 1.5 km, rain, snow, sleet or ice crystals) or standing water, slush, ice or snow is present on the runway.

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Freezing drizzle (METAR code: FZDZ) is a fairly uniform precipitation composed exclusively of fine drops - diameter less than 0.5 mm (0.02 inch) - very close together which freeze upon impact with the ground or other objects.

Freezing fog (METAR code: FZFG) is a suspension of numerous tiny super cooled water droplets which freeze upon impact with ground or other exposed objects, generally reducing the horizontal visibility at the earth's surface to less than 1 km (5/8 mile).

Freezing rain (METAR code: FZRA) is a precipitation of liquid water particles which freezes upon impact with the ground or other exposed objects, either in the form of drops of more than 0.5 mm (0.02 inch) diameter or smaller drops which, in contrast to drizzle, are widely separated.

Friction coefficient: Relationship between the friction force acting on the wheel and the normal force on the wheel. The normal force depends on the weight of the airplane and the lift of the wings.

Frost is a deposit of ice crystals that form from ice-saturated air at temperatures below 0°C (32°F) by direct sublimation on the ground or other exposed objects.

Glaze ice or rain ice is a smooth coating of clear ice formed when the temperature is below freezing, and freezing rain contacts a solid surface. It can only be removed by deicing fluid; hard or sharp tools should not be used to scrape or chip the ice off as this can result in damage to the airplane.

Grooved runway: see dry runway.

Hail (METAR code: GR) is a precipitation of small balls or pieces of ice, with a diameter ranging from 5 to 50 mm (0.2 to 2.0 inches), falling either separately or agglomerated.

Hear Force is a force applied laterally on an anti-icing fluid. When applied to a Type II or IV fluid, the shear force will reduce the viscosity of the fluid; when the shear force is no longer applied, the anti-icing fluid should recover its viscosity. For instance, shear forces are applied whenever the fluid is pumped, forced through an orifice or when subjected to airflow. If excessive shear force is applied, the thickener system could be permanently degraded, and the anti-icing fluid viscosity may not recover and may be at an unacceptable level.

Hoar frost (a rough white deposit of crystalline appearance formed at temperatures below freezing point) usually occurs on exposed surfaces on a cold and cloudless night. It frequently melts after sunrise; if it does not, an approved de-icing fluid should be applied in sufficient quantities to remove the deposit. Generally, hoar frost cannot be cleared by brushing alone.

Holdover time is the estimated time anti-icing fluid will prevent the formation of frost or ice and the accumulation of snow on the protected surfaces of an airplane, under (average) weather conditions mentioned in the guidelines for holdover time. The ISO/SAE

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specification states that the start of the holdover time is from the beginning of the anti-icing treatment.

Ice Pellets (METAR code PE) is a precipitation of transparent (sleet or grains of ice) or translucent (small hail) pellets of ice, which are spherical or irregular, and which have a diameter of 5 mm (0.2 inch) or less. The pellets of ice usually bounce when hitting hard ground.

Icing conditions may be expected when the OAT (on the ground and for takeoff) or when TAT (in flight) is at or below 10°C, and there is visible moisture in the air (such as clouds, fog with low visibility of one mile or less, rain, snow, sleet, ice crystals) or standing water, slush, ice or snow is present on the taxiways or runways. (AFM definition)

Intermittent: Intensity changes gradually, if at all, but precipitation stops and starts at least once within the hour preceding the observation.

Icy runway: A runway is considered icy when its friction coefficient is 0.05 or below.

Light freezing rain is a precipitation of liquid water particles which freezes upon impact with exposed objects, in the form of drops of more than 0.5 mm (0.02 inch) which, in contrast to drizzle, are widely separated. The measured intensity of liquid water particles are up to 2.5mm/hour (0.10 inch/hour) or 25 grams /dm2 /hour with a maximum of 2.5 mm (0.10 inch) in 6 minutes.

Lowest Operational Use Temperature (LOUT): The lowest operational use temperature (LOUT) is the higher (warmer) of the lowest temperature at which the fluid meets the aerodynamic acceptance test (according to AS5900) for a given type (high speed or low speed) of airplane or the freezing point of the fluid plus the freezing point buffer of 10 °C (18 °F) for Type I fluid and 7 °C (13 °F) for Type II, III or IV fluids. For applicable values refer to the fluid manufacturer's documentation.

Moderate and heavy freezing rain: Precipitation of liquid water particles which freezes upon impact with the ground or other exposed objects are widely separated.

Negative Buffer is the freezing point above the Outside Air Temperature (OAT).

Non-Newtonian fluids have characteristics that are dependent upon an applied force. In this instance it is the viscosity of Type II and IV fluids which reduces with increasing shear force. The viscosity of Newtonian fluids depends on temperature only.

One step de-/anti-icing is carried out with an anti-icing fluid, typically heated. The fluid used to de-ice the airplane remains on airplane surfaces to provide limited anti-ice capability.

Precipitation: Liquid or frozen water that falls from clouds as rain, drizzle, snow, hail, or sleet.

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Precipitation intensity is an indication of the amount of precipitation falling at the time of observation. It is expressed as light, moderate or heavy. Each intensity is defined with respect to the type of precipitation occurring, based either on rate of fall for rain and ice pellets or visibility for snow and drizzle. The rate of fall criteria is based on time and does not accurately describe the intensity at the time of observation.

Radiational cooling a process by which temperature decreases, due to an excess of emitted radiation over absorbed radiation. On a typical calm clear night airplane surfaces emit long wave radiation; however, there is no solar radiation (shortwave) coming in at night and this long wave emission will represent a constant net energy loss. Under these conditions the airplane surface temperatures may be up to 4°C or more below that of the surrounding air.

Rain or high humidity (on cold soaked wing) water, visible moisture or humidity forming ice or frost on the wing surface, when the temperature of the airplane wing surface is at or below 0°C (32°F).

Rain (METAR code: RA) is a precipitation of liquid water particles either in the form of drops of more than 0.5 mm (0.02 inch) diameter or of smaller widely scattered drops.

Rain and snow: Precipitation in the form of a mixture of rain and snow.

Rime Ice (a rough white covering of ice deposited from fog at temperature below freezing). As the fog usually consists of super-cooled water drops, which only solidify on contact with a solid object, rime may form only on the windward side or edges and not on the surfaces. It can generally be removed by brushing, but when surfaces, as well as edges, are covered it will be necessary to use an approved de-icing fluid.

Saturation is the maximum amount of water vapor allowable in the air. It is about 0.5 g/m3 at - 30°C and 5 g/m3 at 0°C for moderate altitudes.

SIGMET is information issued by a meteorological watch office concerning the occurrence, or expected occurrence, of specified en-route weather phenomena which may affect the safety of airplane operations.

Sleet is a precipitation in the form of a mixture of rain and snow. For operation in light sleet it treats as light freezing rain.

Slush is water saturated with snow, which spatters when stepping firmly on it. It is encountered at temperature around 5°C.

Snow (METAR code SN): Precipitation of ice crystals, most of which are branched, star shaped or mixed with unbranched crystals. At temperatures higher than about -5°C (23°F), the crystals are generally agglomerated into snowflakes.

Snow grains (METAR code: SG) is a precipitation of very small white and opaque grains of ice. These grains are fairly flat or elongated. Their diameter is less than 1 mm (0.04

inch). When the grains hit hard ground, they do not bounce or shatter.

Snow pellets (METAR code: GS) is a precipitation of white and opaque grains of ice. These grains are spherical or sometimes conical. Their diameter is about 2 to 5 mm (0.1 to 0.2 inch). Grains are brittle, easily crushed; they bounce and break on hard ground.

Note: For hold over purposes treat snow grains as snow

Super cooled water droplets are a condition where water remains liquid at negative Celsius temperature. super cooled drops and droplets are unstable and freeze upon impact.

Thin hoar frost is a uniform white deposit of fine crystalline texture, which is thin enough to distinguish surface features underneath, such as paint lines, markings, or lettering.

Two step de-icing/anti-icing consists of two distinct steps. The first step (de-icing) is followed by the second step (anti-icing) as a separate fluid application. After de-icing a separate overspray of anti-icing fluid is applied to protect the relevant surfaces, thus providing maximum possible anti-ice capability.

Visible moisture: Fog, rain, snow, sleet, high humidity (condensation on surfaces), ice crystals or when taxiways and/or runways are contaminated by water, slush or snow.

Visual meteorological conditions: Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, equal to or better than specified minima.

Wet snow: Snow which, if compacted by hand, will stick together and tend to or form a snowball. Specific gravity: 0.35 up to but not including 0.5. Wet snow is normally experienced when temperature is above freezing and is more difficult to remove from the airplane structure than dry snow being sufficiently wet to adhere.

Wet runway: A runway is considered wet when the runway surface is covered with water, or equivalent, less than or equal to 3 mm or when there is sufficient moisture on the runway surface to cause it to appear reflective, but without significant areas of standing water.

6.0 RESPONSIBILITY

6.1 MAINTENANCE RESPONSIBILITY

6.1.1 The information report (deicing/anti-icing) given to the cockpit is part of the technical airworthiness of the airplane. The person releasing the airplane is responsible for the

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performance and verification of the results of the of the de-icing/anti-icing treatment. The final authority of accepting the performed treatment lies, however, with the Captain.

6.1.2 Each crew planning to operate an aircraft in conditions where frost, ice, or snow may reasonably be expected to adhere to the aircraft shall: (ECARAS 9.3.1.22)

a) Use only aircraft adequately equipped with for such conditions.

b) Ensure flight crew is adequately trained for such conditions.

c) Have an approved ground de-icing and anti-icing program

6.2. OPERATIONAL RESPONSIBILITY

The transfer of operational responsibility, in icing conditions, takes place at the moment the airplane starts moving by its own power.

6.3. MAINTENANCE / GROUND CREW DECISION

The ground crew member, charged with the responsibility of deicing/anti-icing duties of the airplane should be clearly identified. He should check the airplane for the need to de-ice, based on his own judgment, initiate de-/anti-icing, if required, and he is responsible for the correct and complete de-icing and/or anti-icing of the airplane.

6.4. PILOTS DECISION

6.4.1. As the final decision rests with the PIC, his request will supersede the ground crew member's judgment not to de-/anti-icing. As the Captain is responsible for the de-/anti-icing condition of the airplane during ground maneuvering prior to takeoff, he can request another de-/anti-icing application with a different mixture ratio to have the airplane protected for a longer period against accumulation of precipitation. He can simply request a repeat application.

6.4.2. Therefore, the Captain should take into account forecasted or expected weather conditions, taxi conditions, taxi times, holdover time and other relevant factors. The Captain must, when in doubt about the aerodynamic cleanliness of the airplane, perform (or have performed) an inspection or simply request a further de-/anti-icing. Even when responsibilities are clearly defined and understood, sufficient communication between flight and ground crews is necessary. Any observation considered valuable should be mentioned to the other party to have redundancy in the process of decision making.

7.0 PROCEDURE

7.1. IMPORTANCE OF COMMUNICATION

7.1.1. To ensure optimum results during de-/anti-icing operations, a good level of communication between ground and flight crews is essential.

7.1.2. Any observations or points significant to the flight or ground crew should be exchanged between them. These observations may be concerning the weather or airplane-related circumstances or other factors critical to the dispatch of the flight. Several incidents have shown that increased awareness of one part of the flight/ground crew team could have avoided a critical situation.

7.1.3. As a minimum requirement the information communicated to the Captain must include, the details of when the airplane was de-iced and the quality of treatment (type of fluid). This is summarized by the anti-icing code.

REMEMBER: Uncertainty should not be resolved by transferring responsibility. The only satisfactory answer is clear communication.

7.2. CONDITIONS WHICH CAUSE AIRPLANE ICING

7.2.1. WEATHER-RELATED CONDITIONS

Weather conditions dictate airplane de-/anti-icing on the ground. Icing conditions on the ground can be expected when air temperatures fall below freezing and when moisture or ice occurs in the form of either precipitation or condensation. Precipitation may be rain, sleet or snow. Frost can occur due to the condensation of fog or mist. To these weather conditions must be added further phenomena that can also result in airplane ice accretion on the ground.

7.2.2. AIRPLANE-RELATED CONDITIONS

The concept of icing is commonly associated only with exposure to inclement weather. However, even if the OAT is above freezing point, ice or frost can form if the airplane structure is below 0° C (32° F) and moisture or relatively high humidity is present. With rain or drizzle falling on sub-zero structure, a clear ice layer can form on the wing upper surfaces when the airplane is on the ground. In most cases this is accompanied by frost on the underwing surface.

7.3. CHECKS TO DETERMINE THE NEED TO DE-ICE / ANTI-ICE

7.3.1. THE CLEAN AIRPLANE CONCEPT

a) Why de-ice/anti-ice on ground? The airplane performance is certified based upon an uncontaminated or clean structure. Ice, snow or frost accumulations will disturb the airflow, affecting lift and drag and also increasing weight. The result on performance can be dramatic. Final responsibility for the maintenance of clean airplane concept rests with the Captain of the flight.

b) All critical surfaces of an airplane must be clean of any surface contamination such as ice, snow or frost. T/off will not commence unless the critical surfaces are clear of any deposit that might adversely affect the performance and/or controllability of the aircraft.

c) Airplane preparation for service begins and ends with a thorough inspection of the airplane exterior. The airplane and especially its surfaces providing lift, controllability and stability must be aerodynamically clean. Otherwise, safe operation is not possible.

7.3.2. EXTERNAL INSPECTION

a) A member of the flight crew or where applicable a qualified ground agent must conduct an inspection of the airplane to visually sight all critical parts of the airplane before T/off, this inspection must be performed from vantage points offering a clear view of these parts.

b) Must inspect areas include:

I. Wing surfaces including leading edges
II. Horizontal Stabilizers upper and lower surface and elevators
III. Vertical Stabilizer and rudder
IV. Fuselage
V. Air data probes
VI. Static vents
VII. Angle-of-attack sensors
VIII. Control surface cavities
IX. Engines
X. Propellers
XI. Generally, intakes and outlets
XII. Landing gear and wheel bays

NOTE: If cabin crew members see any icing build up on any of the airplane's surfaces, they shall inform the flight crew.

7.4. CLEAR ICE PHENOMENON

7.4.1. Under certain conditions, a clear ice layer or frost can form on the wing upper surfaces when the airplane is on the ground. In most cases this is accompanied by frost on the under-wing surface. Severe conditions occur with precipitation when sub-zero fuel is in contact with the wing upper surface skin panels. The clear ice accumulations are very difficult to detect from ahead of the wing or behind during walk-around, especially in poor lighting and when the wing is wet. The leading edge may not feel particularly cold.

7.4.2. The clear ice may not be detected from the cabin either because wing surface details show through.

7.4.3. The following factors contribute to the formation of intensity and the final thickness of the clear ice layer:

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a) Low temperature of fuel that was added to the airplane during the previous ground stop and/or the long airborne time of the previous flight resulting in a situation that the remaining fuel in the wing tanks is below 0° C.

b) Abnormally large amount of remaining cold fuel in wing tanks causing the fuel level to be in contact with the wing upper surface panels as well as the lower surface, especially in the wing tank area.

c) Temperature of fuel added to the airplane during the current ground stop, adding (relatively) warm fuel can melt dry, falling snow with the possibility of re-freezing.

7.4.4. Drizzle/rain and ambient temperatures around 0°C on the ground is very critical. Heavy freezing has been reported during drizzle/rain even at temperatures of 8 to 14° C (46 to 57° F). The use of thermal leading edge anti-icing may melt falling dry snow that refreezes later.

7.4.5. The area's most vulnerable to freezing are:

a) The wing root area between the front and rear spars,

b) Any part of the wing that will contain unused fuel after flight,

c) The areas where different structures of the wing are concentrated (a lot of cold metal), such as areas above the spars and the main landing gear doubles plate.

7.5. GENERAL CHECKS

7.5.1. A recommended procedure to check the wing upper surface is to place high enough steps as close as possible to the leading edge and near the fuselage and climb the steps so that you can touch a wide sector of the tank area by hand. If clear ice is detected, the wing upper surface should be de-iced and then re-checked to ensure that all ice deposits have been removed.

NOTE: It must always be remembered that below a snow / slush / anti-icing fluid layer there can be clear ice.

7.5.2. During checks on ground, electrical or mechanical ice detectors should only be used as a back-up advisory. They are not a primary system and are not intended to replace physical checks.

7.5.3. Ice can build up on airplane surfaces when descending through dense clouds or precipitation during an approach.

7.5.4. When ground temperatures at the destination are low, it is possible that when flaps are retracted accumulations of ice may remain undetected between stationary and moveable surfaces. It is therefore important that these areas are checked prior to departure and any frozen deposits removed.

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7.5.5. Under freezing fog conditions, it is necessary for the rear side of the fan blades to be checked for ice build-up prior to start-up. Any deposits discovered should be removed by directing air from a low flow hot air source, such as a cabin heater, onto the affected areas.

7.5.6. When slush is present on runways, inspect the airplane when it arrives at the ramp for slush/ice accumulations. If the airplane arrives at the gate with flaps in a position other than fully retracted, those flaps which are extended must be inspected and, if necessary, deiced before retraction.

7.5.7. The flight crew operating manual for individual airplane types may allow take-off with a certain amount of frost on certain parts of the airplane (refer to the individual FCOM).

7.5.8. It is important to note that the rate of ice formation is considerably increased by the presence of an initial depth of ice. Therefore, if icing conditions are expected to occur along the taxi and take-off path, it is necessary to ensure that all ice and frost is removed before flight. This consideration must extend the awareness of flight crew to include the condition of the taxiway, runway and adjacent areas since surface contamination and blown snow are potential causes for ice accretion equal to natural precipitation.

7.6. THE PROCEDURES TO DE-ICE AND ANTI-ICE AN AIRPLANE

When airplane surfaces are contaminated by frozen moisture, they must be de-iced prior to dispatch. When freezing precipitation exists and there is a risk of precipitation adhering to the surface at the time of dispatch, airplane surfaces must be anti-iced. If both anti-icing and de-icing are required, the procedure may be performed in one or two steps. The selection of a one or two step process depends upon weather conditions, available equipment, available fluids and the holdover time required to be achieved. When a large holdover time is expected or needed, a two-step procedure using undiluted fluid should always be considered for the second step.

7.7. DE-ICING

Ice, snow, slush and frost shall be removed from airplane surfaces prior to dispatch or anti-icing. Ice, snow, slush or frost may be removed from airplane surfaces by heated fluids or mechanical methods or any other approved methods such as infrared de-icing which is being developed. For maximum effect, fluids shall be applied close to the airplane surfaces to minimize heat loss. Different methods to efficiently remove frost, snow, and ice are described in detail in the ISO method specification.

7.8. GENERAL DE-ICING FLUID APPLICATION STRATEGY

7.8.1. The following guidelines describe effective ways to remove snow and ice. However, certain airplane may require unique procedures to accommodate specific design features. The relevant airplane maintenance or servicing manuals should be consulted.

- **Wings/horizontal stabilizers/elevators**: Spray from the tip towards the root, from the highest point of the surface camber to the lowest.

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- **Vertical surfaces**: Start at the top and work downward.

- **Fuselage**: Spray along the top centerline and then outboard; avoid spraying directly onto windows.

- **Nose/Radome Area and Flight Deck Windows**: Type I fluid/water mixture or manual methods of removal are recommended.

- **Landing gear and wheel bays**: Keep application of de-icing fluid in this area to a minimum. It may be possible to mechanically remove accumulations such as blown snow. However, where deposits have bonded to surfaces they can be removed using hot air or by careful spraying with hot de-icing fluids. It is not recommended to use a high-pressure spray.

  De-icing fluid shall not be sprayed directly onto brakes and wheels.

- **Engines**: Deposits of snow should be mechanically removed (for example using a broom or brush) from engine intakes prior to departure. Any frozen deposits that may have bonded to either the lower surface of the intake, the fan blades or propellers shall be removed by hot air or other means recommended by the engine manufacturer.

7.9. ANTI-ICING

7.9.1. Applying anti-icing protection means that ice, snow or frost will, for a period of time, be prevented from adhering to, or accumulating on, airplane surfaces. This is done by the application of anti-icing fluids.

7.9.2. Anti-icing fluid should be applied to the airplane surfaces when freezing rain, snow or other freezing precipitation is falling and adhering at the time of airplane dispatch.

7.9.3. For effective anti-icing protection, an even film of undiluted fluid is required over the airplane surfaces which are clean or which have been de-iced. For maximum anti-icing protection undiluted, unheated Type II or IV fluid should be used. The high fluid pressures and flow rates normally associated with de-icing are not required for this operation and, where possible, pump speeds should be reduced accordingly. The nozzle of the spray gun should be adjusted to give a medium spray. The anti-icing fluid application process should be continuous and as short as possible.

7.9.4. Anti-icing should be carried out as near to the departure time as is operationally possible in order to maintain holdover time. In order to control the uniformity, all horizontal airplane surfaces must be visually checked during application of the fluid. The amount required will be a visual indication of fluid just beginning to drip off the leading and trailing edges.

7.9.5. Most effective results are obtained by commencing on the highest part of the wing section and covering from there towards the leading and trailing edges. On vertical surfaces, start at the top and work down. Surfaces to be protected during anti-icing are:

- Wing upper surface
- Horizontal stabilizer upper surface
- Vertical stabilizer and rudder
- Fuselage depending upon amount and type of precipitation

7.9.6. Ant-icing treatment shall always cove the entire wing, the entire vertical stabilizer/rudder and horizontal stabilizer/elevator on both sides of the airplane.

NOTE: Type I fluids have limited effectiveness when used for anti-icing purposes. Little benefit is gained from the minimal holdover time generated.

7.10. GENERAL AEROPLANE REQUIREMENTS AFTER DE-ICING/ANTI-ICING

7.10.1. Wings tail and control surfaces shall be free of ice, snow, slush and frost except that a coating of frost may be present on wing lower surfaces in areas cold soaked by fuel between forward and aft spars.

7.10.2. Pitot heads and static ports shall be clear of ice, frost, snow and fluid.

7.10.3. Engine inlets, exhaust nozzles, cooling intakes, control system probes and ports shall be clear of ice and snow. Engine fan blades or propellers shall be clear of ice, frost and snow.

7.10.4. Air conditioning inlets and exits shall be clear of ice, frost and snow.

7.10.5. Landing gears and landing gear doors shall be unobstructed and clear of ice, frost and snow.

7.10.6. Fuel tank vents shall be clear of ice, frost and snow.

7.10.7. Fuselage shall be clear of ice, slush and snow. Frost may be present in accordance with the manufacturer’s documentation.

7.10.8. Nose/radom area and flight deck windows shall be clear of snow, slush or ice prior to departure.

7.11. LIMITS AND PRECAUTIONS

7.11.1. AIRPLANE RELATED LIMITS

a) The use of Type II or IV fluids in 100% concentration or 75/25 mixture is limited to airplane with a rotation speed (VR) higher than 85kt. This is to assure the sufficient flow-off of the fluid during take-off.

b) One-step deicing/anti-icing is performed with a heated anti-icing fluid. The fluid used to de-ice the airplane remains on the airplane surfaces to provide limited anti-ice capability.

c) The correct fluid concentration shall be chosen with regard to desired holdover time and is dictated by OAT and weather conditions (Refer to the tables at the end of this section).

d) Two-step de-icing/anti-icing (when the first step is performed with de-icing fluid) requires selection of the correct fluid with regard to ambient temperature (refer to the tables at the end of this section). After de-icing, a separate overspray of ant-icing fluid shall be applied to protect the relevant surfaces thus providing maximum possible ant-ice capability.

e) The second step shall be performed before first step fluid freezes. When fluid with a negative buffer is used for the first step and/or when treating composite surfaces, freezing may happen quickly.

f) When re-freezing occurs following the initial treatment, both first and second steps must be repeated.

7.11.2. TEMPERATURE LIMITS

a) When performing two-step de-icing / anti-icing, the freezing point of the heated fluid used for the first step must not be more than 3°C above ambient temperature.

b) The freezing point of the Type I fluid mixture used for either one-step de-icing / anti-icing or as the second step in a two-step operation shall be at least 10°C below the ambient temperature/OAT.

c) Type II, III and IV fluids used as de-icing / anti-icing agents have a lower temperature application limit of -25°C. The application limit may be lower, provided that a 7°C buffer is maintained between the freezing point of the undiluted fluid and the outside air temperature. Freezing points are provided in the fluid manufacturer's documentation.

7.11.3. APPLICATION LIMITS

a) Under no circumstances shall an airplane that has been anti-iced receive a further coating of anti-icing fluid directly on top of the contaminated film. In continuing precipitation, the original anti-icing coating will be diluted at the end of the holdover time and re-freezing could begin. Also, a double anti-ice coating should not be applied because the flow-off characteristics during take-off may be compromised.

b) Should it be necessary for an airplane to be re-protected prior to the next flight, the external surfaces must first be de-iced with a hot fluid mix before a further application of anti-icing fluid is made. Refer to the tables at the end of this section for the application.

c) Ensure that any fluid remaining from previous treatment is flushed off.

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7.12. FLUID APPLICATION

7.12.1. The fluids used should be limited to those complying respectively with standards AMS 1424B/ISO 11075 and AMS 1428C/ISO 11078 for Type I, Type II and Type IV. With specific regard to the application of Type IV fluids, and indeed Type II fluids, special care needs to be taken. Repeated application in dry conditions, as a preventive measure, may leave a residue that when exposed to precipitation can re-hydrate. This takes the form of a high freeze point gel in aerodynamically quiet areas of the airplane. This gel could lead to the restricted movement of control surfaces. To date this has only been reported on airplane types with unpowered flight controls and has not been reported on airplane with powered flight controls. Therefore, the airplane should be frequently cleaned of any residue and/or de-iced using a heated Type I fluid or hot water prior to the application of Type II or Type IV fluids (two step processes).

7.12.2. For de/anti-icing activities the following standards should be followed:

a) ISO 11076 airplane de-icing/anti-icing methods with fluids.
b) SAE ARP 4737E airplane de-icing/anti-icing methods with fluids.
c) AEA recommendations for the de-icing/anti-icing of airplane on ground, in order to fully benefit from the longer hold over times of Type IV fluids, they must be used undiluted. Diluted Type IV is only tested to the same specification as a Type II fluid.

7.12.3. **The airplane must always be treated symmetrically** - that is, left-hand and right-hand side shall receive the same and complete treatment when anti-icing. De-icing only may be local but still symmetrical. Aerodynamic problems could result if this requirement is not met. During anti-icing and de-icing, the moveable surfaces shall be in a position as specified by the airplane manufacturer.

7.12.4. **Engines-** are usually not running or are at idle during treatment. Air conditioning should be selected OFF. The APU may be run for electrical supply but the bleed air valve should be closed. All reasonable precautions must be taken to minimize fluid entry into engines, other intakes / outlets and control surface cavities. Do not spray de-icing / anti-icing fluids directly onto exhausts or thrust reversers.

7.12.5. De-icing / anti-icing fluid **should not be** directed into the orifices of pitot heads, static vents or directly onto angle-of-attack sensors.

7.12.6. Do not direct fluids onto flight deck or cabin windows because this can cause cracking of acrylics or penetration of the window sealing.

7.12.7. All doors and windows must be closed to prevent:

- Galley floor areas being contaminated with slippery de-icing/anti-icing fluids.
- Upholstery becoming soiled.

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7.12.8. Any forward area from which fluid may blow back onto windscreens during taxi or subsequent take-off should be free of fluid residues prior to departure. If Type II or IV fluids are used, all traces of the fluid on flight deck windows should be removed prior to departure, particular attention being paid to windows fitted with wipers.

7.12.9. De-icing/anti-icing fluid can be removed by rinsing with clear water and wiping with a soft cloth. Do not use the windscreen wipers for this purpose. This will cause smearing and loss of transparency. Landing gear and wheel bays must be kept free from build-up of slush, ice or accumulations of blown snow.

7.12.10. Do not spray de-icing fluid directly onto **hot wheels** or **brakes**. When removing ice, snow or slush from airplane surfaces, care must be taken to prevent it entering and accumulating in auxiliary intakes or control surface hinge areas, i.e. remove snow from wings and stabilizer surfaces forward towards the leading edge and remove from ailerons and elevators back towards the trailing edge.

7.12.11. Do not open any door until all ice has been removed from the surrounding area

7.12.12. A functional flight control check using an external observer may be required after deicing / anti-icing. This is particularly important in the case of an airplane that has been subjected to an extreme ice or snow covering.

7.12.13. During anti-icing and de-icing, the moveable surfaces shall be in a position as specified by the aircraft manufacturer.

7.13. FINAL CHECK BEFORE AIRPLANE DISPATCH

7.13.1. No airplane shall be dispatched for departure after a de-icing / anti-icing operation unless the airplane has received a final check by a responsible authorized person.

a) The inspection must visually cover all critical parts of the airplane and be performed from points offering sufficient visibility of these parts (e.g. from the de-ice itself or another elevated piece of equipment). It may be necessary to gain direct access to physically check (e.g. by touch) to ensure that there is no clear ice on suspect areas.

b) This final check shall cover wings, horizontal stabilizer, vertical stabilizer and fuselage. This check shall also include any other parts of the airplane on which a de-icing/anti-icing treatment was performed according to the requirements during the contamination check.

7.14 PRE TAKEOFF CHECK

7.14.1. When freezing precipitation exists, it may be appropriate to check aerodynamic surfaces just prior to the airplane taking the active runway or initiating the take-off roll in order to confirm that they are free of all forms of frost, ice, slush and snow.

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a) After the proper inspections, de-icing and anti-icing procedures and verifications, the airplane is ready to taxi. The Captain shall continually monitor the environmental situation after the de-icing/anti-icing treatment has been carried out. There can, however, be delays before takeoff and the fluid may be contaminated during precipitation. In this case a pre-takeoff check should be performed. The Captain shall assess, proper to takeoff, whether the applied holdover time is still appropriate. This check is normally performed from inside the aircraft as a visual check. If the visual check is insufficient a **pre-takeoff contamination** check should be performed.

7.15 PRE-TAKEOFF CONTAMINATION CHECK

This is a check of the critical surfaces for contamination. This check shall be performed when the condition of the critical surfaces of the airplane cannot be effectively assessed by a pre-takeoff check or when the applied holdover time has been exceeded. This check is normally performed from outside the airplane. The alternate means of compliance to a pre-takeoff contamination check is a complete de-icing / anti-icing re-treatment of the airplane.

7.16. FLIGHT CREW NOTIFICATION AND RECORDING

7.16.1. No airplane shall be dispatched for departure after a de-icing / anti-icing operation unless the flight crew has been notified of the type of de-icing / anti-icing operation performed. The ground crew must make sure that the flight crew has been informed. The flight crew should make sure that they have the information.

7.16.2. This information includes the results of the final inspection by qualified personnel, indicating that the airplane critical parts are free of ice, frost and snow. It also includes the necessary anti-icing codes to allow the flight crew to estimate the holdover time to be expected under the prevailing weather conditions.

7.17. ANTI-ICING CODES

7.17.1. The Anti-Icing Code must be recorded and communicated to the Captain by referring to the last step of the procedure and in the sequence as listed below:

a) Type of fluid, i.e. Type I, Type II, Type III or Type IV

b) Concentration of fluid within the fluid/ water mixture expressed as percentage by volume for Type II, Type III and Type IV fluids (no requirement for Type I fluid)

CAUTION: When informing about fluid concentration, fluid proportion always comes first.

c) Local Time (hours/ minutes) at the beginning of the final de-icing/ anti-icing step.

d) The holdover time starts:

I. With a one-step procedure at the first application of the anti-icing fluid,

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II. With two-step procedure at the beginning of the second step (anti-icing).

EXAMPLE: A de-icing/ anti-icing procedure whose last step is the use of a mixture of 75% of a Type II fluid and 25% water beginning at 13:35 Local Time is notified and recorded as follows:

Anti-Icing Code TYPE II/75, 13:35

7.18. MANDATORY STATEMENTS

Make sure that the Captain receives the information that all de-icing/ anti-icing operations are now complete, airplane is clean and that all personnel and equipment are clear before reconfiguration or movement of airplane.

7.19. RECORDING

For all airplanes the Anti-Icing Code shall be entered into the Flight Operations De/Anti-icing Log sheet.

NOTE: all times shall be entered in UTC.

7.20. HOLDOVER TIME GUIDELINES

7.20.1. Holdover time is obtained by anti-icing fluids remaining on the airplane surfaces. With a one-step deicing/anti-icing operation the holdover time begins at the start of the application of the treatment and with a two-step operation at the start of the final (anti-icing) step. Holdover time will have effectively run out when frozen deposits start to form/accumulate on treated airplane surfaces.

7.20.2. Due to their properties, Type I fluids form a thin liquid wetting film, which provides limited holdover time, especially in conditions of freezing precipitation. With this type of fluid, no additional holdover time would be provided by increasing the concentration of the fluid in the fluid/water mix. Type II, III, and IV fluids contain a pseudo plastic thickening agent, which enables the fluid to form a thicker liquid wetting film on external airplane surfaces.

7.20.3. This film provides a longer holdover time especially in conditions of freezing precipitation. With this type of fluid additional holdover time will be provided by increasing the concentration of the fluid in the fluid/water mix, with maximum holdover time available from undiluted fluid.

7.20.4. The tables under this section give an indication as to the time frame of protection that could reasonably be expected under conditions of precipitation. However, due to the many variables that can influence holdover time, these times should not be considered as minimums or maximums as the actual time of protection may be extended or reduced, depending upon the particular conditions existing at the time.

7.20.5. The lower limit of the published time span is used to indicate the estimated time of protection during moderate precipitation and the upper limit indicates the estimated time of protection during light precipitation.

CAUTION: Heavy precipitation rates or high moisture content, high wind velocity or jet blast may reduce holdover time below the lowest time stated in the range. Holdover time may also be reduced when airplane skin temperature is lower than OAT. Therefore, the indicated times should be used only in conjunction with a pre-takeoff check.

CAUTION: Surface coatings currently available that may be identified as ice phobic or hydro phobic, enhance the appearance of airplane external surfaces and/or lead to fuel savings.

Since these coatings may affect the fluid wetting capability and the resulting fluid thickness of de-icing/anti-icing fluid, they have the potential to affect holdover time and aerodynamics. For more information, consult the aircraft manufacturers.

NOTE 1: Certain fluids may be qualified according to fluid specifications but may not have been tested during winter to develop the holdover time guidelines specified in this document. Holdover time guidelines in this document are not applicable to these fluids.

NOTE 2: For use of holdover time guidelines consult Fluid Manufacturer Technical Literature for minimum viscosity limits of fluids as applied to airplane surfaces.

NOTE 3: A degraded Type II, Type III, or Type IV fluid may be used, provided the holdover time guidelines for Type I fluids are used.

7.20.6. A Type II, Type III, or Type IV fluid is considered to be degraded if the viscosity is below the minimum limit as provided by the fluid manufacturer.

NOTE: Holdover time guidelines can also be obtained for individual fluid products and these “brand name” holdover times will be found to differ from the Tables published here.

If an airline decides to use these bran tables, it shall refer to the FAA or TC documentation, particularly for the application of the “light” and “very light” snow columns.

7.20.7. The generic holdover timetable indicates times that can be used for all certified fluids. Type I Fluid /Water Mixture is selected so that the Freezing Point of the mixture is at least 10 °C (18 °F) below actual OAT.

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7.20.8. The table is read by first verifying the outside air temperature, then the form of precipitation, the time cell to use is where these two parameters cross. For example, the times to use are between 0:02 minutes and 0:05 minutes for a temperature that shows between –3 °C and –6 °C, and weather condition of a snowfall precipitation. It is up to the captain to decide on which time is usable. If the de-icing crew is asked to give this information to the flight crew, it is essential to give the time span (e.g. 2-5 min.). The table indicates times that are valid for both composite and metallic airplane surfaces.

7.20.9. De-icing/anti-icing fluids used during ground de-icing are not intended for and do not provide – protection during flight.

7.20.10. The generic holdover timetable indicates times that can be used for all certified fluids. Type-II and –IV tables have a similar layout, except for the times indicated. The logic of interpreting the table and the layout for type III is much the same as Type II and IV. The responsibility for the application of these data remains with the user. Anti-icing fluid used during ground de-icing/anti-icing are not intended for and do not provide protection during flight. Thickened fluid can be used as a de-icing fluid when sufficiently diluted but a 100% mixture is usually not used for de-icing purposes.

7.20.11. The table is read by first verifying the outside air temperature, then the form of precipitation and then the concentration (%) of fluid used. The time cell to use is where these parameters cross. It is important not to confuse the concentration of fluid used because a false reading can lead to a dramatic error in holdover times. For example, the Table shows times to use between 0:20 minutes and 0:40 minutes for a temperature between –3 °C and –14 °C, a concentration of 100/0 fluid. Mixture and a weather condition of snowfall precipitation. It is up to the captain to decide on which time is usable. If the de-icing crew is asked to give this information to the flight crew, it is essential to give the time span (e.g. 20-40 min.).The difference in fluid mixture depends on the procedures used by each de-icing provider. Some providers only offer one concentration (e.g. 100%) when others mix the glycol with water (e.g. 75% glycol and 25% water). The table indicates times that are valid for both composite and metallic airplane surfaces.

7.20.12. The generic holdover timetable for frost conditions indicates times that can be used for all qualified fluids in frost conditions, therefore there is no other weather selection available. The idea to read the time is the same as for the other tables but the difference is that here are all fluid types in the same table. Note that there is no fluid concentration for Type-I, only for the thickened fluids the lower limit of the published time span is used to indicate the estimated time of protection during moderate precipitation and the upper limit indicates the estimated time of protection during light precipitation. Anti-icing fluid used during ground de-icing/anti-icing are not intended for and do not provide protection during flight.

NOTE: The table indicates times that are valid for both composite and metallic airplane surfaces.

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7.21. PILOT'S TECHNIQUES

The purpose of this section is to deal with the issue of ground de-icing/anti-icing from the pilot's point of view. The topic is covered in the order it appears on cockpit checklists and is followed through, step by step, from flight preparation to take-off. The focus is on the main points of decision-making, flight procedures and pilot techniques.

7.21.1. RECEIVING AIRPLANE

When arriving at the airplane, advice from local ground maintenance staff may be considered because they may be more familiar with local weather conditions. If there is nobody available or if there is any doubt about their knowledge concerning de-icing/anti-icing protocol, pilots have to determine the need for de-icing/anti-icing by themselves. If the prevailing weather conditions call for protection during taxi, pilots should try to determine «off block time» to be in a position to get sufficient anti-icing protection regarding holdover time.

NOTE: This message should be passed to the de-icing/anti-icing units, the ground maintenance, the boarding staff, dispatch office and all other units involved.

7.21.2. COCKPIT PREPARATION

a) Before treatment, avoid pressurizing or testing flight control systems.

b) Try to make sure that all flight support services are completed prior to treatment to avoid any delay between treatment and start of taxiing.

c) During treatment observe that:

I. Engines are shut down or at idle

II. APU may be used for electrical supply, bleed air OFF

III. Air conditioning should be OFF

IV. All external lights of treated areas must be OFF.

d) Consider whether communication and information with the ground staff is/has been adequate. A specific item included in the normal cockpit preparation procedures is recommended. The minimum requirement is to receive the anti-icing code in order to calculate the available protection time from the holdover timetable.

e) Do not consider the information given in the holdover timetables as precise. There are several parameters influencing holdover time. The time frames given in the holdover timetables consider very different weather situations world-wide. The view of the weather is rather subjective; experience has shown that a certain snowfall can be judged as light, medium or heavy by different people. If in doubt, a pre-take-off contamination check should be considered.

f) As soon as the treatment of the airplane is completed, proceed with engine starts. Regarding responsibility and decision see section.

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7.21.3. TAXIING

a) During taxi the flight crew should observe the intensity of precipitation and keep an eye on the airplane surfaces visible from the cockpit. Ice warning systems of engines and wings or other additional ice warning systems must be considered. Sufficient distance from the preceding airplane must be maintained as blowing snow or jet blasts can degrade the anti-icing protection of the airplane.

b) The extension of slats and flaps should be delayed, especially when operating on slushy areas. However, in this case slat/flap extension should be verified prior to take off.

7.21.4. TAKE-OFF

Recommendations given in FCOM of individual airplane types regarding performance corrections (effect of engine bleeds) or other procedures applied when operating in icing conditions should be considered. The Captain shall not commence take off unless the critical surfaces are clear of any deposits which might adversely affect the performance and/or controllability of the airplane.

7.21.5. GENERAL REMARKS

a) In special situations flight crews must be encouraged not to allow operational or commercial pressures to influence decisions. The minimum requirements have been presented here, as well as the various precautions.

CAUTION: If there is any doubt as to whether any critical surface is contaminated do not take off.

b) As in any other business, the key factors to keep procedures efficient and safe are awareness, understanding and communication. If there is any doubt or question at all, ground and flight crews must communicate with each other.

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7.22. HOLDOVER TIME GUIDELINE TABLES (APPLICABLE 2025-2026)

ACTIVE FROST HOLDOVER TIME (HOT) GUIDELINES WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.

The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation.

CAUTIONS

- The responsibility for the application of these data remains with the user.
- Fluids used during ground de/anti-icing do not provide in-flight icing protection.

SECTION 2.18 A/C GROUND DE-ICING/ANTI-ICING

TABLE 1: ACTIVE FROST HOLDOVER TIMES FOR SAE TYPE I, TYPE II, TYPE III, AND TYPE IV FLUIDS[1]

Outside Air Temperature[2,3,4]	Type I Aluminum	Type I Composite
-1°C and above (30°F and above)	0:45	0:35
below -1 to -3°C (below 30 to 27°F)		
below -3 to -10°C (below 27 to 14°F)		
below -10 to -14°C (below 14 to 7°F)		
below -14 to -21°C (below 7 to -6°F)		
below -21 to -25°C (below -6 to -13°F)		
below -25°C (below -13°F)		

Outside Air Temperature[3,4]	Concentration Fluid/Water By % Volume	Type II	Type III[5]	Type IV
-1°C and above (30°F and above)	100/0	8:00	2:00	12:00
	75/25	5:00	1:00	5:00
	50/50	2:00	0:30	3:00
below -1 to -3°C (below 30 to 27°F)	100/0	8:00	2:00	12:00
	75/25	5:00	1:00	5:00
	50/50	1:30	0:30	3:00
below -3 to -10°C (below 27 to 14°F)	100/0	8:00	2:00	10:00
	75/25	4:00	1:00	5:00
below -10 to -14°C (below 14 to 7°F)	100/0	6:00	2:00	6:00
	75/25	1:00	1:00	1:00
below -14 to -21°C (below 7 to -6°F)	100/0	3:00	2:00	6:00
below -21 to -25°C (below -6 to -13°F)	100/0	2:00	2:00	4:00
below -25°C (below -13°F)	100/0	No Holdover Time Guidelines Exist		

NOTES

1. To use the HOTs in this table, ensure that the fluid and dilution being used is listed in the List of Qualified Fluids Tested for Anti-Icing Performance and Aerodynamic Acceptance table (Table 55 - Table 58). Any restrictions on the use of the fluid have to be identified and applied.
2. Type I Fluid / Water Mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature.
3. Ensure that the lowest operational use temperature (LOUT) is respected.
4. Changes in outside air temperature (OAT) over the course of longer frost events can be significant; the appropriate holdover time to use is the one provided for the coldest OAT that has occurred in the time between the de/anti-icing fluid application and takeoff.
5. To use the Type III fluid frost holdover times, the fluid brand being used must be known. AllClear AeroClear MAX must be applied unheated.

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 308.

HOLDOVER TIME (HOT) GUIDELINES FOR SAE TYPE I FLUIDS
WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.

The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation.

CAUTIONS

- The responsibility for the application of these data remains with the user.
- The time of protection will be shortened in heavy weather conditions, heavy precipitation rates, or high moisture content.
- High wind velocity, jet blast or blowing snow may reduce holdover time below the lowest time stated in the range.
- The holdover time may be reduced when aircraft skin temperature is lower than outside air temperature.
- Fluids used during ground de/anti-icing do not provide in-flight icing protection.

SECTION 2.18 A/C GROUND DE-ICING/ANTI-ICING

TABLE 2: HOLDOVER TIMES FOR SAE TYPE I FLUID ON CRITICAL AIRCRAFT SURFACES COMPOSED PREDOMINANTLY OF ALUMINUM[1]

Outside Air Temperature[2,3]	Freezing Fog, Freezing Mist[4], or Ice Crystals[5]	Snow mixed with Freezing Fog[6]	Very Light Snow, Snow Grains or Snow Pellets[7,8]	Light Snow, Snow Grains or Snow Pellets[7,8]	Moderate Snow, Snow Grains or Snow Pellets[7,8]	Freezing Drizzle[9]	Light Freezing Rain	Moderate Snow mixed with Rain[10,11]	Rain on Cold-Soaked Wing[11]	Other[12]
-3°C and above (27°F and above)	0:11 - 0:17	0:06 - 0:11	0:18 - 0:22	0:11 - 0:18	0:06 - 0:11	0:09 - 0:13	0:04 - 0:06	0:02 - 0:02	0:02 - 0:05	
below -3 to -6°C (below 27 to 21°F)	0:08 - 0:13	0:05 - 0:08	0:14 - 0:17	0:08 - 0:14	0:05 - 0:08	0:05 - 0:09	0:04 - 0:06			
below -6 to -10°C (below 21 to 14°F)	0:06 - 0:10	0:04 - 0:06	0:11 - 0:13	0:06 - 0:11	0:04 - 0:06	0:04 - 0:07	0:02 - 0:05	CAUTION: No holdover time guidelines exist		
below -10°C (below 14°F)	0:05 - 0:09	0:02 - 0:04	0:07 - 0:08	0:04 - 0:07	0:02 - 0:04					

NOTES

1 These holdover times apply to aircraft with critical surfaces constructed predominantly or entirely of aluminum materials that have demonstrated satisfactory use of these holdover times. Takeoff after the longest applicable holdover time has been exceeded is not permitted for Type I fluids.
2 Type I fluid / water mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature.
3 Ensure that the lowest operational use temperature (LOUT) is respected.
4 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
5 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
6 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
7 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
8 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
9 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
10 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
11 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
12 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3300.

TABLE 3: HOLDOVER TIMES FOR SAE TYPE I FLUID ON CRITICAL AIRCRAFT SURFACES COMPOSED PREDOMINANTLY OF COMPOSITES[1]

Outside Air Temperature[2,3]	Freezing Fog, Freezing Mist[4], or Ice Crystals[5]	Snow mixed with Freezing Fog[6]	Very Light Snow, Snow Grains or Snow Pellets[7,8]	Light Snow, Snow Grains or Snow Pellets[7,8]	Moderate Snow, Snow Grains or Snow Pellets[7,8]	Freezing Drizzle[9]	Light Freezing Rain	Moderate Snow mixed with Rain[10,11]	Rain on Cold-Soaked Wing[11]	Other[12]
-3°C and above (27°F and above)	0:09 - 0:16	0:03 - 0:06	0:12 - 0:15	0:06 - 0:12	0:03 - 0:06	0:08 - 0:13	0:04 - 0:06	0:01 - 0:01	0:01 - 0:05	CAUTION: No holdover time guidelines exist
below -3 to -6°C (below 27 to 21°F)	0:06 - 0:08	0:02 - 0:05	0:11 - 0:13	0:05 - 0:11	0:02 - 0:05	0:05 - 0:09	0:04 - 0:06	CAUTION: No holdover time guidelines exist		
below -6 to -10°C (below 21 to 14°F)	0:04 - 0:08	0:02 - 0:05	0:09 - 0:12	0:05 - 0:09	0:02 - 0:05	0:04 - 0:07	0:02 - 0:05			
below -10°C (below 14°F)	0:04 - 0:07	0:02 - 0:04	0:07 - 0:08	0:04 - 0:07	0:02 - 0:04					

NOTES

1 These holdover times apply to newer aircraft with critical surfaces constructed predominantly or entirely of composite materials. Takeoff after the longest applicable holdover time has been exceeded is not permitted for Type I fluids.
2 Type I fluid / water mixture must be selected so that the freezing point of the mixture is at least 10°C (18°F) below outside air temperature.
3 Ensure that the lowest operational use temperature (LOUT) is respected.
4 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
5 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
6 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
7 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
8 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
9 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
10 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
11 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
12 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3300.

HOLDOVER TIME (HOT) GUIDELINES FOR SAE TYPE II FLUIDS
WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.

The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation.

CAUTIONS

- The responsibility for the application of these data remains with the user.
- The only acceptable decision-making criterion, for takeoff without a pre-takeoff contamination inspection, is the shorter time within the applicable table cell.
- The time of protection will be shortened in heavy weather conditions, heavy precipitation rates, or high moisture content.
- High wind velocity, jet blast or blowing snow may reduce holdover time below the lowest time stated in the range.
- The holdover time may be reduced when aircraft skin temperature is lower than outside air temperature.
- Fluids used during ground de/anti-icing do not provide in-flight icing protection.
- Whenever frost or ice occurs on the lower surface of the wing in the area of the fuel tank, indicating a cold-soaked wing, the 50/50 dilution of Type II or IV is not to be used for the anti-icing step because fluid freezing may occur.

TABLE 4: GENERIC HOLDOVER TIMES FOR SAE TYPE II FLUIDS[1]

Outside Air Temperature[2]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[3], or Ice Crystals[4]	Snow mixed with Freezing Fog[5]	Snow, Snow Grains or Snow Pellets[6,7]	Freezing Drizzle[8]	Light Freezing Rain	Moderate Snow mixed with Rain[9,10]	Rain on Cold-Soaked Wing[10]	Other[11]
-3°C and above (27°F and above)	100/0	0:55 - 1:50	0:20 - 0:40	0:30 - 0:55	0:35 - 1:05	0:25 - 0:35	0:05 - 0:05	0:07 - 0:45	
	75/25	0:40 - 1:10	0:15 - 0:25	0:15 - 0:30	0:25 - 0:40	0:15 - 0:25	0:03 - 0:03	0:04 - 0:25	
	50/50	0:15 - 0:30	0:05 - 0:10	0:07 - 0:15	0:09 - 0:15	0:06 - 0:09			
below -3 to -8°C (below 27 to 18°F)	100/0	0:30 - 0:45	0:15 - 0:30	0:20 - 0:40	0:20 - 0:45	0:15 - 0:20			
	75/25	0:25 - 0:55	0:09 - 0:15	0:10 - 0:25	0:15 - 0:30	0:09 - 0:20			
below -8 to -14°C (below 18 to 7°F)	100/0	0:30 - 0:45	0:10 - 0:25	0:15 - 0:30	0:20 - 0:45[12]	0:15 - 0:20[12]	CAUTION: No holdover time guidelines exist		
	75/25	0:25 - 0:55	0:07 - 0:15	0:09 - 0:20	[12]	[12]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:20	0:01 - 0:05	0:02 - 0:07					
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:20	0:00 - 0:02	0:01 - 0:03	0:15 - 0:30	0:09 - 0:20			
below -25°C (below -13°F)	100/0	0:15 - 0:20	0:00 - 0:00	0:00 - 0:01					

NOTES

1 To use the HOTs in this table, ensure that the fluid and dilution being used is listed in the Type II Fluids Tested for Anti-Icing Performance and Aerodynamic Acceptance table (Table 56). Any restrictions on the use of the fluid have to be identified and applied.
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
10 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
12 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 5: TYPE II HOLDOVER TIMES FOR ABAX ECOWING AD-2

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:20 - 3:00	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:40 - 1:15	0:40 - 1:40	0:30 - 0:45	0:07 - 0:07	0:09 - 1:25	
	75/25	1:15 - 1:25	0:20 - 0:40	1:45 - 2:00	0:55 - 1:45	0:25 - 0:55	0:35 - 1:05	0:20 - 0:30	0:03 - 0:03	0:04 - 0:50	
	50/50	0:15 - 0:30	0:05 - 0:10	0:35 - 0:40	0:15 - 0:35	0:07 - 0:15	0:09 - 0:15	0:06 - 0:09			
below -3 to -8°C (below 27 to 18°F)	100/0	0:45 - 2:30	0:25 - 0:45	2:00 - 2:00	1:00 - 2:00	0:30 - 1:00	0:25 - 1:10	0:20 - 0:30			
	75/25	0:35 - 1:55	0:20 - 0:35	1:40 - 2:00	0:50 - 1:40	0:25 - 0:50	0:15 - 0:55	0:20 - 0:35			
below -8 to -14°C (below 18 to 7°F)	100/0	0:45 - 2:30	0:20 - 0:40	1:45 - 2:00	0:55 - 1:45	0:30 - 0:55	0:25 - 1:10[11]	0:20 - 0:30[11]			
	75/25	0:35 - 1:55	0:20 - 0:35	1:35 - 2:00	0:50 - 1:35	0:25 - 0:50	[11]	[11]	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:40	0:01 - 0:05	0:20 - 0:30	0:07 - 0:20	0:02 - 0:07	0:15 - 0:55	0:20 - 0:35			
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:40	0:00 - 0:02	0:09 - 0:15	0:03 - 0:09	0:01 - 0:03					
below -25 to -27°C (below -13 to -17°F)	100/0	0:15 - 0:40	0:00 - 0:00	0:05 - 0:07	0:01 - 0:05	0:00 - 0:01					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 6: TYPE II HOLDOVER TIMES FOR AVIATION XI'AN HIGH-TECH CLEANWING II

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	0:55 - 1:50	0:20 - 0:40	1:35 - 1:55	0:55 - 1:35	0:30 - 0:55	0:35 - 1:05	0:25 - 0:35	0:07 - 0:07	0:10 - 0:55	
	75/25	0:50 - 1:20	0:20 - 0:35	1:20 - 1:40	0:45 - 1:20	0:25 - 0:45	0:35 - 1:00	0:20 - 0:30	0:05 - 0:05	0:07 - 0:50	
	50/50	0:35 - 1:00	0:10 - 0:20	0:50 - 1:05	0:25 - 0:50	0:15 - 0:25	0:20 - 0:40	0:10 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:45 - 1:50	0:15 - 0:30	1:20 - 1:35	0:40 - 1:20	0:25 - 0:40	0:30 - 0:55	0:20 - 0:25			
	75/25	0:40 - 1:45	0:20 - 0:35	1:20 - 1:35	0:45 - 1:20	0:25 - 0:45	0:35 - 0:40	0:20 - 0:25			
below -8 to -14°C (below 18 to 7°F)	100/0	0:45 - 1:50	0:15 - 0:25	1:05 - 1:20	0:35 - 1:05	0:20 - 0:35	0:30 - 0:55[11]	0:20 - 0:25[11]	CAUTION: No holdover time guidelines exist		
	75/25	0:40 - 1:45	0:20 - 0:35	1:20 - 1:35	0:45 - 1:20	0:25 - 0:45	0:35 - 0:40[11]	0:20 - 0:25[11]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:20 - 0:50	0:09 - 0:20	0:45 - 1:00	0:25 - 0:45	0:15 - 0:25					
below -18 to -25°C (below 0 to -13°F)	100/0	0:20 - 0:50	0:05 - 0:10	0:30 - 0:35	0:15 - 0:30	0:07 - 0:15					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:30 - 4:00	0:45 - 1:10	2:00 - 2:00	1:35 - 2:00	1:00 - 1:35	1:20 - 2:00	0:45 - 1:25	0:08 - 0:08	0:10 - 1:30	
	75/25	1:50 - 2:45	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:10 - 1:30	0:30 - 0:55	0:04 - 0:04	0:06 - 0:50	
	50/50	0:55 - 1:45	0:09 - 0:20	0:45 - 0:55	0:25 - 0:45	0:10 - 0:25	0:20 - 0:30	0:10 - 0:15			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 1:45	0:35 - 1:00	2:00 - 2:00	1:15 - 2:00	0:45 - 1:15	0:35 - 1:30	0:25 - 0:45			
	75/25	0:25 - 1:05	0:20 - 0:40	1:45 - 2:00	0:55 - 1:45	0:30 - 0:55	0:25 - 1:10	0:20 - 0:35			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 1:45	0:30 - 0:50	1:50 - 2:00	1:05 - 1:50	0:40 - 1:05	0:35 - 1:30[11]	0:25 - 0:45[11]			
	75/25	0:25 - 1:05	0:15 - 0:30	1:20 - 1:40	0:40 - 1:20	0:20 - 0:40	0:25 - 1:10[11]	0:20 - 0:35[11]	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:50	0:06 - 0:20	1:10 - 1:40	0:25 - 1:10	0:08 - 0:25					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:50	0:02 - 0:07	0:30 - 0:40	0:10 - 0:30	0:03 - 0:10					
below -25 to -29°C (below -13 to -20°F)	100/0	0:30 - 0:50	0:01 - 0:05	0:20 - 0:30	0:07 - 0:20	0:02 - 0:07					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 8: TYPE II HOLDOVER TIMES FOR CRYOTECH POLAR GUARD® II

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:50 - 4:00	0:50 - 1:25	2:00 - 2:00	1:55 - 2:00	1:05 - 1:55	1:35 - 2:00	1:15 - 1:30	0:10 - 0:10	0:15 - 2:00	
	75/25	2:30 - 4:00	0:30 - 1:05	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	1:40 - 2:00	0:40 - 1:10	0:07 - 0:07	0:09 - 1:40	
	50/50	0:50 - 1:25	0:07 - 0:20	1:10 - 1:35	0:25 - 1:10	0:10 - 0:25	0:20 - 0:45	0:09 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 2:30	0:35 - 1:05	2:00 - 2:00	1:25 - 2:00	0:50 - 1:25	0:35 - 1:35	0:35 - 0:45			
	75/25	0:40 - 1:30	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	0:25 - 1:05	0:35 - 0:45			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 2:30	0:30 - 0:50	2:00 - 2:00	1:10 - 2:00	0:40 - 1:10	0:35 - 1:35[11]	0:35 - 0:45[11]			
	75/25	0:40 - 1:30	0:20 - 0:45	2:00 - 2:00	0:55 - 2:00	0:25 - 0:55	[11]	[11]			CAUTION: No holdover time guidelines exist
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:50	0:08 - 0:25	1:35 - 2:00	0:35 - 1:35	0:10 - 0:35	0:25 - 1:05	0:35 - 0:45			
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:50	0:03 - 0:10	0:40 - 0:55	0:15 - 0:40	0:04 - 0:15					
below -25 to -30.5°C (below -13 to -23°F)	100/0	0:25 - 0:50	0:02 - 0:05	0:25 - 0:30	0:07 - 0:25	0:02 - 0:07					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 9: TYPE II HOLDOVER TIMES FOR ESSPO CHEMICALS D.O.O NORDWING PG2

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:40 - 3:10	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:05 - 1:50	0:50 - 1:05	0:15 - 0:15	0:20 - 1:45	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:05 - 2:20	0:25 - 0:45	2:00 - 2:00	1:00 - 2:00	0:30 - 1:00	0:50 - 1:30	0:25 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:05 - 2:20	0:20 - 0:40	1:45 - 2:00	0:50 - 1:45	0:25 - 0:50	0:50 - 1:30[11]	0:25 - 0:50[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:20 - 0:35	0:10 - 0:25	1:15 - 1:40	0:30 - 1:15	0:15 - 0:30					
below -18 to -25°C (below 0 to -13°F)	100/0	0:20 - 0:35	0:06 - 0:15	0:50 - 1:05	0:20 - 0:50	0:08 - 0:20					
below -25 to -30°C (below -13 to -22°F)	100/0	0:20 - 0:30	0:03 - 0:07	0:25 - 0:30	0:09 - 0:25	0:04 - 0:09					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 10: TYPE II HOLDOVER TIMES FOR KILFROST ABC-K PLUS

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:15 - 3:45	0:45 - 1:15	1:00 - 1:40	1:50 - 2:00	1:00 - 1:25	0:15 - 0:15	0:20 - 2:00	
	75/25	1:40 - 2:30	0:25 - 0:50	0:35 - 1:10	1:25 - 2:00	0:50 - 1:10	0:10 - 0:10	0:15 - 2:00	
	50/50	0:35 - 1:05	0:05 - 0:10	0:07 - 0:15	0:20 - 0:30	0:10 - 0:15			
below -3 to -8°C (below 27 to 18°F)	100/0	0:30 - 1:05	0:40 - 1:10	0:55 - 1:30	0:25 - 1:00	0:15 - 0:35			
	75/25	0:25 - 1:25	0:25 - 0:50	0:35 - 1:05	0:20 - 0:55	0:09 - 0:30			
below -8 to -14°C (below 18 to 7°F)	100/0	0:30 - 1:05	0:40 - 1:05	0:50 - 1:25	0:25 - 1:00[11]	0:15 - 0:35[11]			
	75/25	0:25 - 1:25	0:25 - 0:50	0:35 - 1:05	[11]	[11]		CAUTION: No holdover time guidelines exist	
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:55	0:01 - 0:05	0:02 - 0:07					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:55	0:00 - 0:02	0:01 - 0:03	0:20 - 0:55	0:09 - 0:30			
below -25 to -29°C (below -13 to -20°F)	100/0	0:30 - 0:55	0:00 - 0:00	0:00 - 0:01					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 11: TYPE II HOLDOVER TIMES FOR KILFROST ICE CLEAR II

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:25 - 2:25	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:00 - 1:35	0:40 - 1:05	0:10 - 0:10	0:15 - 2:00	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:05 - 2:35	0:30 - 0:50	2:00 - 2:00	1:10 - 2:00	0:40 - 1:10	0:30 - 1:15	0:35 - 0:55			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:05 - 2:35	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:35 - 1:05	0:30 - 1:15[11]	0:35 - 0:55[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			CAUTION: No holdover time guidelines exist
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 0:45	0:10 - 0:20	0:55 - 1:05	0:30 - 0:55	0:15 - 0:30					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 0:45	0:06 - 0:10	0:30 - 0:35	0:15 - 0:30	0:08 - 0:15					
below -25 to -28°C (below -13 to -18°F)	100/0	0:35 - 0:45	0:05 - 0:09	0:25 - 0:30	0:10 - 0:25	0:06 - 0:10					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 12: TYPE II HOLDOVER TIMES FOR MKS DEVO CHEMICALS COREICEPHOB TYPE II

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:55 - 2:45	0:30 - 1:00	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	1:10 - 2:00	0:45 - 1:00	0:09 - 0:09	0:15 - 1:35	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	1:05 - 1:45	0:15 - 0:35	1:35 - 1:55	0:45 - 1:35	0:25 - 0:45	0:50 - 1:15	0:25 - 0:30			
below -3 to -8°C (below 27 to 18°F)	100/0	0:45 - 1:25	0:25 - 0:45	1:50 - 2:00	1:00 - 1:50	0:30 - 1:00	0:30 - 1:10	0:25 - 0:35			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:45 - 1:25	0:20 - 0:35	1:30 - 1:50	0:50 - 1:30	0:25 - 0:50	0:30 - 1:10[11]	0:25 - 0:35[11]	CAUTION: No holdover time guidelines exist		
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:25	0:08 - 0:15	0:35 - 0:40	0:20 - 0:35	0:10 - 0:20					
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:25	0:03 - 0:05	0:15 - 0:15	0:07 - 0:15	0:04 - 0:07					
below -25 to -27°C (below -13 to -17°F)	100/0	0:15 - 0:25	0:02 - 0:04	0:10 - 0:10	0:05 - 0:10	0:03 - 0:05					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 13: TYPE II HOLDOVER TIMES FOR NEWAVE AEROCHEMICAL FCY-2

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:15 - 2:25	0:25 - 0:40	0:30 - 0:55	0:35 - 1:05	0:25 - 0:35	0:06 - 0:06	0:08 - 0:45	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:45 - 1:30	0:15 - 0:30	0:20 - 0:40	0:20 - 0:45	0:15 - 0:20			
	75/25	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:45 - 1:30	0:10 - 0:25	0:15 - 0:30	0:20 - 0:45[11]	0:15 - 0:20[11]			
	75/25	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:35	0:01 - 0:05	0:02 - 0:07					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:35	0:00 - 0:02	0:01 - 0:03					
below -25 to -28°C (below -13 to -18°F)	100/0	0:25 - 0:35	0:00 - 0:00	0:00 - 0:01					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303

TABLE 14: TYPE II HOLDOVER TIMES FOR ROMCHIM ADD-PROTECT NG TYPE II

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:10 - 2:25	0:25 - 0:55	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	0:50 - 1:20	0:35 - 0:50	0:05 - 0:05	0:07 - 1:10	CAUTION: No holdover time guidelines exist
	75/25	1:00 - 1:50	0:20 - 0:40	1:55 - 2:00	0:55 - 1:55	0:25 - 0:55	0:40 - 1:15	0:25 - 0:40	0:05 - 0:05	0:07 - 0:55	
	50/50	0:25 - 0:55	0:10 - 0:20	0:55 - 1:05	0:30 - 0:55	0:15 - 0:30	0:20 - 0:35	0:10 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 1:35	0:20 - 0:40	1:50 - 2:00	0:50 - 1:50	0:25 - 0:50	0:35 - 1:10	0:25 - 0:35			
	75/25	0:55 - 1:25	0:15 - 0:30	1:25 - 1:45	0:40 - 1:25	0:20 - 0:40	0:25 - 1:05	0:20 - 0:30			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 1:35	0:15 - 0:30	1:25 - 1:50	0:40 - 1:25	0:20 - 0:40	0:35 - 1:10[11]	0:25 - 0:35[11]			
	75/25	0:55 - 1:25	0:10 - 0:25	1:05 - 1:25	0:30 - 1:05	0:15 - 0:30	0:25 - 1:05[11]	0:20 - 0:30[11]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:20	0:01 - 0:05	0:20 - 0:30	0:07 - 0:20	0:02 - 0:07					
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:20	0:00 - 0:02	0:09 - 0:15	0:03 - 0:09	0:01 - 0:03					
below -25 to -28°C (below -13 to -18°F)	100/0	0:15 - 0:20	0:00 - 0:00	0:05 - 0:07	0:01 - 0:05	0:00 - 0:01					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

TABLE 15: TYPE II HOLDOVER TIMES FOR ROMCHIM ADD-PROTECT TYPE II

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:40 - 3:30	0:20 - 0:45	1:55 - 2:00	1:00 - 1:55	0:30 - 1:00	0:40 - 1:35	0:25 - 0:45	0:06 - 0:06	0:09 - 0:50	
	75/25	0:40 - 1:10	0:15 - 0:25	1:00 - 1:10	0:30 - 1:00	0:15 - 0:30	0:25 - 0:40	0:15 - 0:25	0:04 - 0:04	0:05 - 0:25	
	50/50	0:20 - 0:35	0:07 - 0:15	0:30 - 0:35	0:15 - 0:30	0:09 - 0:15	0:10 - 0:30	0:08 - 0:10			
below -3 to -8°C (below 27 to 18°F)	100/0	0:30 - 0:45	0:15 - 0:30	1:20 - 1:40	0:40 - 1:20	0:20 - 0:40	0:25 - 0:50	0:20 - 0:30			
	75/25	0:30 - 0:55	0:09 - 0:15	0:40 - 0:50	0:25 - 0:40	0:10 - 0:25	0:20 - 0:30	0:15 - 0:20			
below -8 to -14°C (below 18 to 7°F)	100/0	0:30 - 0:45	0:15 - 0:25	1:05 - 1:20	0:35 - 1:05	0:15 - 0:35	0:25 - 0:50[11]	0:20 - 0:30[11]			
	75/25	0:30 - 0:55	0:07 - 0:15	0:35 - 0:40	0:20 - 0:35	0:09 - 0:20	0:20 - 0:30[11]	0:15 - 0:20[11]	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:25	0:01 - 0:05	0:20 - 0:30	0:07 - 0:20	0:02 - 0:07					
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:25	0:00 - 0:02	0:09 - 0:15	0:03 - 0:09	0:01 - 0:03					
below -25 to -28°C (below -13 to -18°F)	100/0	0:15 - 0:25	0:00 - 0:00	0:05 - 0:07	0:01 - 0:05	0:00 - 0:01					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 3303.

HOLDOVER TIME (HOT) GUIDELINES FOR SAE TYPE III FLUIDS WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.
The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation.

CAUTIONS

- The responsibility for the application of these data remains with the user.
- The only acceptable decision-making criterion, for takeoff without a pre-takeoff contamination inspection, is the shorter time within the applicable table cell.
- The time of protection will be shortened in heavy weather conditions, heavy precipitation rates, or high moisture content.
- High wind velocity, jet blast or blowing snow may reduce holdover time below the lowest time stated in the range.
- The holdover time may be reduced when aircraft skin temperature is lower than outside air temperature.
- Fluids used during ground de/anti-icing do not provide in-flight icing protection.

TABLE 16: TYPE III HOLDOVER TIMES FOR ALLCLEAR AEROCLEAR MAX APPLIED UNHEATED ON LOW SPEED AIRCRAFT[1]

Outside Air Temperature[2]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[3], or Ice Crystals[4]	Snow mixed with Freezing Fog[5]	Very Light Snow, Snow Grains or Snow Pellets[6,7]	Light Snow, Snow Grains or Snow Pellets[6,7]	Moderate Snow, Snow Grains or Snow Pellets[6,7]	Freezing Drizzle[8]	Light Freezing Rain	Moderate Snow mixed with Rain[9,10]	Rain on Cold-Soaked Wing[10]	Other[11]
-3°C and above (27°F and above)	100/0	0:45 - 1:55	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40	0:25 - 0:50	0:14 - 0:25	0:04 - 0:04	0:05 - 0:40	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -10°C (below 27 to 14°F)	100/0	0:50 - 1:40	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40	0:25 - 0:45	0:15 - 0:25	CAUTION: No holdover time guidelines exist		
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -10 to -16°C (below 14 to 3°F)	100/0	0:40 - 1:45	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40					

NOTES

1 These holdover times are for aircraft conforming to the SAE AS5900 low speed aerodynamic test criterion. Fluid must be applied unheated to use these holdover times. No holdover times exist for this fluid applied heated. If uncertain whether the aircraft conforms to the low, middle, or high speed aerodynamic test criterion, no holdover time guidelines exist below -16°C (3°F).
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type III fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
10 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 326.

TABLE 17: TYPE III HOLDOVER TIMES FOR ALLCLEAR AEROCLEAR MAX APPLIED UNHEATED ON MIDDLE SPEED AIRCRAFT[1]

Outside Air Temperature[2]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[3], or Ice Crystals[4]	Snow mixed with Freezing Fog[5]	Very Light Snow, Snow Grains or Snow Pellets[6,7]	Light Snow, Snow Grains or Snow Pellets[6,7]	Moderate Snow, Snow Grains or Snow Pellets[6,7]	Freezing Drizzle[8]	Light Freezing Rain	Moderate Snow mixed with Rain[9,10]	Rain on Cold-Soaked Wing[10]	Other[11]
-3°C and above (27°F and above)	100/0	0:45 - 1:55	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40	0:25 - 0:50	0:14 - 0:25	0:04 - 0:04	0:05 - 0:40	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -10°C (below 27 to 14°F)	100/0	0:50 - 1:40	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40	0:25 - 0:45	0:15 - 0:25			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -10 to -20.5°C (below 14 to -5°F)	100/0	0:40 - 1:45	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40					

NOTES

1 These holdover times are for aircraft conforming to the SAE AS5900 middle speed aerodynamic test criterion. Fluid must be applied unheated to use these holdover times. No holdover times exist for this fluid applied heated. If uncertain whether the aircraft conforms to the low, middle, or high speed aerodynamic test criterion, no holdover time guidelines exist below -16°C (3°F).
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type III fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
10 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 326.

TABLE 18: TYPE III HOLDOVER TIMES FOR ALLCLEAR AEROCLEAR MAX APPLIED UNHEATED ON HIGH SPEED AIRCRAFT[1]

Outside Air Temperature[2]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[3], or Ice Crystals[4]	Snow mixed with Freezing Fog[5]	Very Light Snow, Snow Grains or Snow Pellets[6,7]	Light Snow, Snow Grains or Snow Pellets[6,7]	Moderate Snow, Snow Grains or Snow Pellets[6,7]	Freezing Drizzle[8]	Light Freezing Rain	Moderate Snow mixed with Rain[9,10]	Rain on Cold-Soaked Wing[10]	Other[11]
-3°C and above (27°F and above)	100/0	0:45 - 1:55	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40	0:25 - 0:50	0:14 - 0:25	0:04 - 0:04	0:05 - 0:40	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -10°C (below 27 to 14°F)	100/0	0:50 - 1:40	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40	0:25 - 0:45	0:15 - 0:25			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A		CAUTION: No holdover time guidelines exist	
below -10 to -25°C (below 14 to -13°F)	100/0	0:40 - 1:45	0:13 - 0:30	1:20 - 1:45	0:40 - 1:20	0:18 - 0:40					
below -25 to -35°C (below -13 to -31°F)	100/0	0:25 - 1:00	0:07 - 0:16	0:45 - 1:00	0:20 - 0:45	0:10 - 0:20					

NOTES

1 These holdover times are for aircraft conforming to the SAE AS5900 high speed aerodynamic test criterion. Fluid must be applied unheated to use these holdover times. No holdover times exist for this fluid applied heated. If uncertain whether the aircraft conforms to the low, middle, or high speed aerodynamic test criterion, no holdover time guidelines exist below -16°C (3°F).
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type III fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
10 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 51 provides allowance times for ice pellets and small hail for SAE Type III fluids, applied unheated).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 326.

HOLDOVER TIME (HOT) GUIDELINES FOR SAE TYPE IV FLUIDS WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.

The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation.

CAUTIONS

- The responsibility for the application of these data remains with the user.
- The only acceptable decision-making criterion, for takeoff without a pre-takeoff contamination inspection, is the shorter time within the applicable table cell.
- The time of protection will be shortened in heavy weather conditions, heavy precipitation rates, or high moisture content.
- High wind velocity, jet blast or blowing snow may reduce holdover time below the lowest time stated in the range.
- The holdover time may be reduced when aircraft skin temperature is lower than outside air temperature.
- Fluids used during ground de/anti-icing do not provide in-flight icing protection.
- Whenever frost or ice occurs on the lower surface of the wing in the area of the fuel tank, indicating a cold-soaked wing, the 50/50 dilution of Type II or IV is not to be used for the anti-icing step because fluid freezing may occur

TABLE 19: GENERIC HOLDOVER TIMES FOR SAE TYPE IV FLUIDS[1]

Outside Air Temperature[2]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[3], or Ice Crystals[4]	Snow mixed with Freezing Fog[5]	Very Light Snow, Snow Grains or Snow Pellets[6,7]	Light Snow, Snow Grains or Snow Pellets[6,7]	Moderate Snow, Snow Grains or Snow Pellets[6,7]	Freezing Drizzle[8]	Light Freezing Rain	Moderate Snow mixed with Rain[9,10]	Rain on Cold-Soaked Wing[10]	Other[11]
-3°C and above (27°F and above)	100/0	1:15 - 2:15	0:20 - 0:40	1:50 - 2:00	0:55 - 1:50	0:25 - 0:55	0:35 - 1:10	0:15 - 0:30	0:06 - 0:06	0:08 - 1:05	CAUTION: No holdover time guidelines exist
	75/25	1:25 - 2:40	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:40 - 1:15	1:00 - 1:20	0:30 - 0:50	0:07 - 0:07	0:09 - 1:20	
	50/50	0:30 - 0:55	0:07 - 0:20	1:00 - 1:10	0:25 - 1:00	0:10 - 0:25	0:15 - 0:40	0:09 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:15 - 0:35	0:20 - 0:35	1:35 - 2:00	0:45 - 1:35	0:25 - 0:45	0:25 - 1:00	0:20 - 0:25			
	75/25	0:40 - 1:20	0:25 - 0:50	1:50 - 2:00	1:05 - 1:50	0:30 - 1:05	0:20 - 1:05	0:15 - 0:25			
below -8 to -14°C (below 18 to 7°F)	100/0	0:15 - 0:35	0:15 - 0:30	1:25 - 1:50	0:45 - 1:25	0:20 - 0:45	0:25 - 1:00[12]	0:20 - 0:25[12]			
	75/25	0:40 - 1:20	0:20 - 0:45	1:45 - 2:00	0:55 - 1:45	0:25 - 0:55	0:20 - 1:05[12]	0:15 - 0:25[12]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:30	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:30	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25°C (below -13°F)	100/0	0:15 - 0:30	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 To use the HOTs in this table, ensure that the fluid and dilution being used is listed in the Type IV Fluids Tested for Anti-Icing Performance and Aerodynamic Acceptance table (Table 58). Any restrictions on the use of the fluid have to be identified and applied.
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
10 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids and Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail.
12 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330

TABLE 20: TYPE IV HOLDOVER TIMES FOR ABAX ECOWING AD-49

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:20 - 4:00	0:45 - 1:25	2:00 - 2:00	1:55 - 2:00	1:00 - 1:55	1:25 - 2:00	1:00 - 1:25	0:08 - 0:08	0:10 - 1:55	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:20 - 1:35	0:35 - 1:05	2:00 - 2:00	1:30 - 2:00	0:45 - 1:30	0:25 - 1:25	0:20 - 0:25			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:20 - 1:35	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:40 - 1:15	0:25 - 1:25[11]	0:20 - 0:25[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:40	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:40	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25 to -26°C (below -13 to -15°F)	100/0	0:25 - 0:40	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 21: TYPE IV HOLDOVER TIMES FOR AFLRUS LLC GREEN FLO TYPE 4

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:15 - 2:30	0:20 - 0:40	1:50 - 2:00	0:55 - 1:50	0:25 - 0:55	0:35 - 1:10	0:15 - 0:30	0:06 - 0:06	0:08 - 1:10	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:50 - 2:35	0:20 - 0:35	1:35 - 2:00	0:45 - 1:35	0:25 - 0:45	0:35 - 1:00	0:20 - 0:30			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:50 - 2:35	0:15 - 0:30	1:25 - 1:50	0:45 - 1:25	0:20 - 0:45	0:35 - 1:00[11]	0:20 - 0:30[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 1:10	0:10 - 0:25	1:20 - 1:45	0:35 - 1:20	0:15 - 0:35					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 1:10	0:10 - 0:25	1:10 - 1:30	0:30 - 1:10	0:15 - 0:30					
below -25 to -30°C (below -13 to -22°F)	100/0	0:25 - 1:00	0:04 - 0:10	0:35 - 0:40	0:15 - 0:35	0:06 - 0:15					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 22: TYPE IV HOLDOVER TIMES FOR ALAB INTERNATIONAL PROFLIGHT EG4

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:05 - 4:00	0:45 - 1:25	2:00 - 2:00	1:50 - 2:00	1:00 - 1:50	1:25 - 2:00	0:40 - 1:00	0:08 - 0:08	0:10 - 2:00	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	2:30 - 3:55	0:45 - 1:25	2:00 - 2:00	1:50 - 2:00	1:00 - 1:50	1:05 - 2:00	0:50 - 1:35			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	2:30 - 3:55	0:45 - 1:25	2:00 - 2:00	1:50 - 2:00	1:00 - 1:50	1:05 - 2:00[11]	0:50 - 1:35[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 1:35	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 1:35	0:15 - 0:35	1:30 - 1:55	0:45 - 1:30	0:20 - 0:45					
below -25 to -26°C (below -13 to -15°F)	100/0	0:35 - 1:35	0:15 - 0:30	1:25 - 1:45	0:40 - 1:25	0:20 - 0:40					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 23: TYPE IV HOLDOVER TIMES FOR ALAB INTERNATIONAL PROFLIGHT PG4

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:25 - 2:15	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:05 - 1:25	0:40 - 0:50	0:10 - 0:10	0:15 - 1:20	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:05 - 2:20	0:25 - 0:50	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	0:45 - 1:10	0:35 - 0:45			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:05 - 2:20	0:20 - 0:45	2:00 - 2:00	1:00 - 2:00	0:30 - 1:00	0:45 - 1:10[11]	0:35 - 0:45[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 0:50	0:10 - 0:25	1:15 - 1:40	0:35 - 1:15	0:15 - 0:35					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 0:50	0:07 - 0:15	0:50 - 1:00	0:20 - 0:50	0:09 - 0:20					
below -25 to -29°C (below -13 to -20°F)	100/0	0:35 - 0:50	0:05 - 0:15	0:40 - 0:50	0:15 - 0:40	0:07 - 0:15					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 24: TYPE IV HOLDOVER TIMES FOR ALLCLEAR CLEAR IV FLIGHT

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:35 - 2:50	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:05 - 1:45	0:50 - 1:10	0:15 - 0:15	0:20 - 1:15	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:50 - 1:55	0:25 - 0:45	1:55 - 2:00	1:00 - 1:55	0:30 - 1:00	0:50 - 1:25	0:40 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:50 - 1:55	0:20 - 0:40	1:40 - 2:00	0:50 - 1:40	0:25 - 0:50	0:50 - 1:25[11]	0:40 - 0:50[11]	CAUTION: No holdover time guidelines exist		
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:45	0:10 - 0:20	1:00 - 1:10	0:30 - 1:00	0:15 - 0:30					
below -18 to -23°C (below 0 to -9°F)	100/0	0:25 - 0:45	0:08 - 0:15	0:40 - 0:50	0:20 - 0:40	0:10 - 0:20					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 25: TYPE IV HOLDOVER TIMES FOR ALLCLEAR CLEARWING EG

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:50 - 3:15	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:10 - 1:35	0:30 - 1:00	0:08 - 0:08	0:10 - 1:30	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:35 - 3:45	0:25 - 0:55	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	1:05 - 1:30	0:30 - 1:00			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:35 - 3:45	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	1:05 - 1:30[11]	0:30 - 1:00[11]			CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:55 - 2:00	0:15 - 0:35	1:35 - 2:00	0:45 - 1:35	0:20 - 0:45					
below -18 to -25°C (below 0 to -13°F)	100/0	0:55 - 2:00	0:09 - 0:20	0:55 - 1:10	0:25 - 0:55	0:15 - 0:25					
below -25 to -29°C (below -13 to -20°F)	100/0	0:55 - 2:00	0:07 - 0:15	0:45 - 0:55	0:20 - 0:45	0:10 - 0:20					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 26: TYPE IV HOLDOVER TIMES FOR ASGLOBAL 4FLITE EG

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:35 - 3:15	0:25 - 0:45	2:00 - 2:00	1:00 - 2:00	0:30 - 1:00	0:40 - 1:10	0:20 - 0:35	0:06 - 0:06	0:08 - 1:05	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:25 - 2:45	0:20 - 0:40	1:50 - 2:00	0:55 - 1:50	0:25 - 0:55	0:40 - 1:10	0:20 - 0:35			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:25 - 2:45	0:20 - 0:35	1:35 - 2:00	0:50 - 1:35	0:25 - 0:50	0:40 - 1:10[11]	0:20 - 0:35[11]	CAUTION: No holdover time guidelines exist		
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:50 - 1:25	0:15 - 0:35	1:35 - 2:00	0:45 - 1:35	0:20 - 0:45					
below -18 to -25°C (below 0 to -13°F)	100/0	0:50 - 1:25	0:15 - 0:30	1:20 - 1:40	0:35 - 1:20	0:20 - 0:35					
below -25 to -30°C (below -13 to -22°F)	100/0	0:30 - 1:05	0:09 - 0:20	0:55 - 1:05	0:25 - 0:55	0:10 - 0:25					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 27: TYPE IV HOLDOVER TIMES FOR ASGLOBAL 4FLITE PG

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:50 - 3:15	0:40 - 1:10	2:00 - 2:00	1:35 - 2:00	0:50 - 1:35	1:10 - 1:35	0:45 - 1:05	0:15 - 0:15	0:15 - 1:20	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:05 - 1:55	0:30 - 0:50	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	0:55 - 1:10	0:35 - 0:55			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:05 - 1:55	0:20 - 0:40	1:40 - 2:00	0:55 - 1:40	0:30 - 0:55	0:55 - 1:10[11]	0:35 - 0:55[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:45	0:15 - 0:25	1:05 - 1:20	0:35 - 1:05	0:15 - 0:35					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:45	0:07 - 0:15	0:35 - 0:45	0:20 - 0:35	0:09 - 0:20					
below -25 to -26°C (below -13 to -15°F)	100/0	0:30 - 0:45	0:06 - 0:15	0:35 - 0:45	0:20 - 0:35	0:08 - 0:20					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 28: TYPE IV HOLDOVER TIMES FOR AVIAFLUID AVIAFLIGHT EG

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:30 - 3:05	0:30 - 0:50	1:55 - 2:00	1:10 - 1:55	0:40 - 1:10	1:05 - 2:00	0:30 - 0:50	0:09 - 0:09	0:10 - 2:00	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:20 - 3:00	0:25 - 0:45	1:45 - 2:00	1:00 - 1:45	0:35 - 1:00	0:55 - 1:30	0:35 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:20 - 3:00	0:25 - 0:40	1:35 - 1:55	0:55 - 1:35	0:30 - 0:55	0:55 - 1:30[11]	0:35 - 0:50[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 1:45	0:20 - 0:40	1:40 - 2:00	0:50 - 1:40	0:25 - 0:50					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 1:45	0:15 - 0:30	1:20 - 1:35	0:40 - 1:20	0:20 - 0:40					
below -25 to -31°C (below -13 to -24°F)	100/0	0:35 - 1:05	0:07 - 0:15	0:35 - 0:45	0:20 - 0:35	0:09 - 0:20					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 29: TYPE IV HOLDOVER TIMES FOR AVIAFLUID AVIAFLIGHT PG

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:15 - 4:00	0:40 - 1:15	2:00 - 2:00	1:40 - 2:00	0:55 - 1:40	2:00 - 2:00	1:10 - 1:55	0:15 - 0:15	0:20 - 2:00	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:05 - 2:10	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:35 - 1:05	0:35 - 1:55	0:45 - 1:05			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:05 - 2:10	0:20 - 0:35	1:30 - 1:50	0:50 - 1:30	0:25 - 0:50	0:35 - 1:55[11]	0:45 - 1:05[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:20 - 0:35	0:10 - 0:20	0:50 - 1:00	0:25 - 0:50	0:15 - 0:25					
below -18 to -25°C (below 0 to -13°F)	100/0	0:20 - 0:35	0:05 - 0:09	0:25 - 0:30	0:15 - 0:25	0:06 - 0:15					
below -25 to -25.5°C (below -13 to -14°F)	100/0	0:20 - 0:35	0:05 - 0:09	0:25 - 0:30	0:10 - 0:25	0:06 - 0:10					

NOTES

1. Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2. Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3. Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5. To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6. Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7. Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8. These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9. No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10. Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11. No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 30: TYPE IV HOLDOVER TIMES FOR CHEMCO CHEMR EG IV

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:05 - 3:35	0:25 - 1:00	2:00 - 2:00	1:15 - 2:00	0:35 - 1:15	0:45 - 1:40	0:25 - 0:40	0:07 - 0:07	0:09 - 1:45	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:25 - 3:40	0:25 - 1:00	2:00 - 2:00	1:15 - 2:00	0:35 - 1:15	1:00 - 1:35	0:35 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:25 - 3:40	0:25 - 1:00	2:00 - 2:00	1:15 - 2:00	0:35 - 1:15	1:00 - 1:35[11]	0:35 - 0:50[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:40 - 1:25	0:15 - 0:30	1:25 - 1:45	0:40 - 1:25	0:20 - 0:40					
below -18 to -25°C (below 0 to -13°F)	100/0	0:40 - 1:25	0:15 - 0:30	1:25 - 1:45	0:40 - 1:25	0:20 - 0:40					
below -25 to -27°C (below -13 to -17°F)	100/0	0:40 - 1:25	0:15 - 0:30	1:25 - 1:45	0:40 - 1:25	0:20 - 0:40					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 31: TYPE IV HOLDOVER TIMES FOR CHEMCO CHEMR NORDIK IV

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:15 - 4:00	0:40 - 1:20	2:00 - 2:00	1:45 - 2:00	0:55 - 1:45	1:20 - 2:00	0:55 - 1:20	0:15 - 0:15	0:25 - 2:00	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:50 - 4:00	0:40 - 1:20	2:00 - 2:00	1:45 - 2:00	0:55 - 1:45	1:15 - 2:00	0:45 - 1:20			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:50 - 4:00	0:40 - 1:20	2:00 - 2:00	1:45 - 2:00	0:55 - 1:45	1:15 - 2:00[11]	0:45 - 1:20[11]			CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:40 - 1:30	0:35 - 1:10	2:00 - 2:00	1:35 - 2:00	0:50 - 1:35					
below -18 to -25°C (below 0 to -13°F)	100/0	0:40 - 1:30	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:35 - 1:05					
below -25 to -29°C (below -13 to -20°F)	100/0	0:40 - 1:30	0:20 - 0:40	1:50 - 2:00	0:55 - 1:50	0:30 - 0:55					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 32: TYPE IV HOLDOVER TIMES FOR CHONGQING JOBA CHEMICAL FW-IV

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:15 - 4:00	0:35 - 1:15	2:00 - 2:00	1:40 - 2:00	0:50 - 1:40	1:30 - 2:00	0:45 - 1:05	0:10 - 0:10	0:15 - 2:00	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	2:30 - 4:00	0:30 - 1:00	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	0:50 - 2:00	0:35 - 1:10			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	2:30 - 4:00	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:35 - 1:15	0:50 - 2:00[11]	0:35 - 1:10[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			CAUTION: No holdover time guidelines exist
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 1:40	0:20 - 0:40	2:00 - 2:00	0:55 - 2:00	0:25 - 0:55					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 1:40	0:15 - 0:30	1:20 - 1:45	0:35 - 1:20	0:15 - 0:35					
below -25 to -29°C (below -13 to -20°F)	100/0	0:35 - 1:40	0:10 - 0:25	1:10 - 1:30	0:30 - 1:10	0:15 - 0:30					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 33: TYPE IV HOLDOVER TIMES FOR CLARIANT SAFEWING MP IV LAUNCH

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	4:00 - 4:00	0:50 - 1:20	2:00 - 2:00	1:45 - 2:00	1:05 - 1:45	1:30 - 2:00	1:00 - 1:40	0:10 - 0:10	0:15 - 1:40	
	75/25	3:40 - 4:00	0:45 - 1:20	2:00 - 2:00	1:45 - 2:00	1:00 - 1:45	1:40 - 2:00	0:45 - 1:15	0:08 - 0:08	0:10 - 1:45	
	50/50	1:25 - 2:45	0:20 - 0:35	1:25 - 1:40	0:45 - 1:25	0:25 - 0:45	0:30 - 0:50	0:20 - 0:25			
below -3 to -8°C (below 27 to 18°F)	100/0	1:00 - 1:55	0:40 - 1:05	2:00 - 2:00	1:30 - 2:00	0:55 - 1:30	0:35 - 1:40	0:25 - 0:45			
	75/25	0:40 - 1:20	0:40 - 1:10	2:00 - 2:00	1:30 - 2:00	0:50 - 1:30	0:25 - 1:10	0:25 - 0:45			
below -8 to -14°C (below 18 to 7°F)	100/0	1:00 - 1:55	0:35 - 1:00	2:00 - 2:00	1:20 - 2:00	0:50 - 1:20	0:35 - 1:40[11]	0:25 - 0:45[11]	CAUTION: No holdover time guidelines exist		
	75/25	0:40 - 1:20	0:35 - 1:00	2:00 - 2:00	1:25 - 2:00	0:45 - 1:25	0:25 - 1:10[11]	0:25 - 0:45[11]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:50	0:05 - 0:15	1:15 - 1:45	0:20 - 1:15	0:06 - 0:20					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:50	0:02 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -25 to -28.5°C (below -13 to -19°F)	100/0	0:30 - 0:50	0:01 - 0:04	0:20 - 0:30	0:06 - 0:20	0:01 - 0:06					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 34: TYPE IV HOLDOVER TIMES FOR CLARIANT SAFEWING MP IV LAUNCH PLUS

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:55 - 4:00	0:40 - 1:35	2:00 - 2:00	2:00 - 2:00	0:55 - 2:00	2:00 - 2:00	1:00 - 2:00	0:15 - 0:15	0:20 - 2:00	
	75/25	3:55 - 4:00	0:35 - 1:25	2:00 - 2:00	1:55 - 2:00	0:50 - 1:55	2:00 - 2:00	1:20 - 1:25	0:15 - 0:15	0:20 - 1:50	
	50/50	1:15 - 1:50	0:15 - 0:35	1:35 - 2:00	0:45 - 1:35	0:20 - 0:45	0:25 - 1:00	0:15 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 2:15	0:35 - 1:15	2:00 - 2:00	1:40 - 2:00	0:45 - 1:40	0:25 - 1:35	0:25 - 0:40			
	75/25	0:40 - 2:00	0:30 - 1:05	2:00 - 2:00	1:30 - 2:00	0:35 - 1:30	0:20 - 1:05	0:20 - 0:30			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 2:15	0:30 - 1:05	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	0:25 - 1:35[11]	0:25 - 0:40[11]			
	75/25	0:40 - 2:00	0:25 - 0:55	2:00 - 2:00	1:15 - 2:00	0:30 - 1:15	0:20 - 1:05[11]	0:20 - 0:30[11]	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:50	0:05 - 0:20	1:15 - 1:50	0:25 - 1:15	0:07 - 0:25					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:50	0:02 - 0:07	0:30 - 0:45	0:09 - 0:30	0:03 - 0:09					
below -25 to -29°C (below -13 to -20°F)	100/0	0:25 - 0:50	0:01 - 0:04	0:20 - 0:30	0:06 - 0:20	0:02 - 0:06					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 35: TYPE IV HOLDOVER TIMES FOR CRYOTECH POLAR GUARD® ADVANCE

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:50 - 4:00	0:50 - 1:25	2:00 - 2:00	1:55 - 2:00	1:05 - 1:55	1:35 - 2:00	1:15 - 1:30	0:10 - 0:10	0:15 - 2:00	
	75/25	2:30 - 4:00	0:30 - 1:05	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	1:40 - 2:00	0:40 - 1:10	0:07 - 0:07	0:09 - 1:40	
	50/50	0:50 - 1:25	0:07 - 0:20	1:10 - 1:35	0:25 - 1:10	0:10 - 0:25	0:20 - 0:45	0:09 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 2:30	0:35 - 1:05	2:00 - 2:00	1:25 - 2:00	0:50 - 1:25	0:35 - 1:35	0:35 - 0:45			
	75/25	0:40 - 1:30	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	0:25 - 1:05	0:35 - 0:45			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 2:30	0:30 - 0:50	2:00 - 2:00	1:10 - 2:00	0:40 - 1:10	0:35 - 1:35[11]	0:35 - 0:45[11]	CAUTION: No holdover time guidelines exist		
	75/25	0:40 - 1:30	0:20 - 0:45	2:00 - 2:00	0:55 - 2:00	0:25 - 0:55	0:25 - 1:05[11]	0:35 - 0:45[11]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:50	0:08 - 0:25	1:35 - 2:00	0:35 - 1:35	0:10 - 0:35					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:50	0:03 - 0:10	0:40 - 0:55	0:15 - 0:40	0:04 - 0:15					
below -25 to -30.5°C (below -13 to -23°F)	100/0	0:25 - 0:50	0:02 - 0:05	0:25 - 0:30	0:07 - 0:25	0:02 - 0:07					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 36: TYPE IV HOLDOVER TIMES FOR CRYOTECH POLAR GUARD® XTEND

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:30 - 4:00	0:50 - 1:30	2:00 - 2:00	2:00 - 2:00	1:05 - 2:00	2:00 - 2:00	1:00 - 1:50	0:15 - 0:15	0:20 - 1:45	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:00 - 1:50	0:40 - 1:10	2:00 - 2:00	1:35 - 2:00	0:50 - 1:35	0:35 - 1:40	0:50 - 0:55			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:00 - 1:50	0:35 - 1:00	2:00 - 2:00	1:20 - 2:00	0:45 - 1:20	0:35 - 1:40[11]	0:50 - 0:55[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:40	0:15 - 0:30	1:20 - 1:40	0:40 - 1:20	0:20 - 0:40					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:40	0:05 - 0:10	0:30 - 0:40	0:15 - 0:30	0:06 - 0:15					
below -25 to -29°C (below -13 to -20°F)	100/0	0:25 - 0:40	0:03 - 0:06	0:20 - 0:25	0:09 - 0:20	0:04 - 0:09					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 37: TYPE IV HOLDOVER TIMES FOR DOW INC. UCAR ENDURANCE™ EG106 ADF/AAF

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:05 - 3:10	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:10 - 2:00	0:50 - 1:15	0:15 - 0:15	0:20 - 2:00	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:50 - 3:20	0:25 - 0:50	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	0:55 - 1:50	0:45 - 1:10			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:50 - 3:20	0:25 - 0:45	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	0:55 - 1:50[11]	0:45 - 1:10[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 1:05	0:15 - 0:35	1:45 - 2:00	0:50 - 1:45	0:25 - 0:50					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 1:05	0:15 - 0:30	1:30 - 1:55	0:40 - 1:30	0:20 - 0:40					
below -25 to -29°C (below -13 to -20°F)	100/0	0:30 - 1:05	0:15 - 0:30	1:20 - 1:45	0:40 - 1:20	0:20 - 0:40					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 38: TYPE IV HOLDOVER TIMES FOR DOW INC. UCAR™ FLIGHTGUARD™ AD-49

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:20 - 4:00	0:45 - 1:25	2:00 - 2:00	1:55 - 2:00	1:00 - 1:55	1:25 - 2:00	1:00 - 1:25	0:08 - 0:08	0:10 - 1:55	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:20 - 1:35	0:35 - 1:05	2:00 - 2:00	1:30 - 2:00	0:45 - 1:30	0:25 - 1:25	0:20 - 0:25			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:20 - 1:35	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:40 - 1:15	0:25 - 1:25[11]	0:20 - 0:25[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:40	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:40	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25 to -26°C (below -13 to -15°F)	100/0	0:25 - 0:40	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 39: TYPE IV HOLDOVER TIMES FOR ESSPO CHEMICALS D.O.O NORDWING PG4

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:35 - 3:10	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:35 - 1:15	1:05 - 1:45	0:50 - 1:00	0:15 - 0:15	0:20 - 1:45	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 2:25	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	0:55 - 1:30	0:35 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 2:25	0:20 - 0:45	1:55 - 2:00	0:55 - 1:55	0:30 - 0:55	0:55 - 1:30[11]	0:35 - 0:50[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:45	0:10 - 0:25	1:20 - 1:45	0:35 - 1:20	0:15 - 0:35					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:45	0:06 - 0:15	0:50 - 1:10	0:20 - 0:50	0:09 - 0:20					
below -25 to -30°C (below -13 to -22°F)	100/0	0:20 - 0:35	0:04 - 0:09	0:30 - 0:40	0:15 - 0:30	0:05 - 0:15					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 40: TYPE IV HOLDOVER TIMES FOR INLAND TECHNOLOGIES ECO-SHIELD®

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:15 - 2:40	0:35 - 1:00	2:00 - 2:00	1:20 - 2:00	0:45 - 1:20	0:40 - 1:30	0:35 - 0:40	0:09 - 0:09	0:15 - 1:35	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:10 - 2:35	0:30 - 0:55	2:00 - 2:00	1:10 - 2:00	0:40 - 1:10	0:50 - 1:25	0:30 - 0:40			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:10 - 2:35	0:25 - 0:50	1:55 - 2:00	1:05 - 1:55	0:35 - 1:05	0:50 - 1:25[11]	0:30 - 0:40[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 1:00	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 1:00	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25 to -25.5°C (below -13 to -14°F)	100/0	0:30 - 1:00	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 41: TYPE IV HOLDOVER TIMES FOR JSC RCP NORDIX DEFROST ECO 4

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:30 - 2:40	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:35 - 1:15	1:05 - 1:30	0:40 - 1:05	0:10 - 0:10	0:15 - 1:10	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 2:35	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:35 - 1:05	0:50 - 1:20	0:35 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 2:35	0:25 - 0:45	2:00 - 2:00	1:00 - 2:00	0:30 - 1:00	0:50 - 1:20[11]	0:35 - 0:50[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:50	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:50	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25 to -25.5°C (below -13 to -14°F)	100/0	0:30 - 0:50	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 42: TYPE IV HOLDOVER TIMES FOR JSC RCP NORDIX DEFROST NORTH 4

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:10 - 4:00	0:30 - 1:00	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	1:05 - 2:00	0:30 - 0:50	0:06 - 0:06	0:09 - 1:55	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	2:40 - 4:00	0:30 - 1:00	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	1:05 - 2:00	0:40 - 1:00			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	2:40 - 4:00	0:30 - 1:00	2:00 - 2:00	1:25 - 2:00	0:40 - 1:25	1:05 - 2:00[11]	0:40 - 1:00[11]		CAUTION: No holdover time guidelines exist	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:45 - 1:55	0:07 - 0:20	0:50 - 1:05	0:25 - 0:50	0:10 - 0:25					
below -18 to -25°C (below 0 to -13°F)	100/0	0:45 - 1:55	0:03 - 0:10	0:40 - 0:55	0:15 - 0:40	0:05 - 0:15					
below -25 to -26°C (below -13 to -15°F)	100/0	0:45 - 1:55	0:01 - 0:06	0:25 - 0:35	0:08 - 0:25	0:02 - 0:08					

NOTES

1. Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2. Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3. Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5. To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6. Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7. Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8. These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9. No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10. Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11. No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 43: TYPE IV HOLDOVER TIMES FOR KILFROST ABC-S PLUS

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:10 - 4:00	0:55 - 1:35	2:00 - 2:00	2:00 - 2:00	1:15 - 2:00	1:50 - 2:00	1:05 - 2:00	0:15 - 0:15	0:25 - 2:00	
	75/25	1:25 - 2:40	0:30 - 0:55	2:00 - 2:00	1:15 - 2:00	0:45 - 1:15	1:00 - 1:20	0:30 - 0:50	0:08 - 0:08	0:10 - 1:20	
	50/50	0:30 - 0:55	0:15 - 0:25	1:00 - 1:10	0:30 - 1:00	0:15 - 0:30	0:15 - 0:40	0:15 - 0:20			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 3:30	0:50 - 1:25	2:00 - 2:00	1:50 - 2:00	1:05 - 1:50	0:25 - 1:35	0:20 - 0:30			
	75/25	0:45 - 1:50	0:30 - 0:50	1:50 - 2:00	1:05 - 1:50	0:40 - 1:05	0:20 - 1:10	0:15 - 0:25			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 3:30	0:45 - 1:15	2:00 - 2:00	1:45 - 2:00	1:00 - 1:45	0:25 - 1:35[11]	0:20 - 0:30[11]	CAUTION: No holdover time guidelines exist		
	75/25	0:45 - 1:50	0:25 - 0:45	1:45 - 2:00	1:00 - 1:45	0:35 - 1:00	0:20 - 1:10[11]	0:15 - 0:25[11]			
below -14 to -18°C (below 7 to 0°F)	100/0	0:40 - 1:00	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:40 - 1:00	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25 to -28°C (below -13 to -18°F)	100/0	0:40 - 1:00	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 44: TYPE IV HOLDOVER TIMES FOR MKS DEVO CHEMICALS COREICEPHOB TYPE-IV PG

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:20 - 3:50	0:35 - 1:15	2:00 - 2:00	1:40 - 2:00	0:45 - 1:40	1:25 - 2:00	0:50 - 1:20	0:09 - 0:09	0:10 - 1:40	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:15 - 0:35	0:25 - 0:55	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	0:40 - 1:30	0:20 - 0:35			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:15 - 0:35	0:20 - 0:45	2:00 - 2:00	0:55 - 2:00	0:25 - 0:55	0:40 - 1:30[11]	0:20 - 0:35[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A		CAUTION: No holdover time guidelines exist	
below -14 to -18°C (below 7 to 0°F)	100/0	0:15 - 0:30	0:01 - 0:06	0:30 - 0:45	0:09 - 0:30	0:02 - 0:09					
below -18 to -25°C (below 0 to -13°F)	100/0	0:15 - 0:30	0:00 - 0:02	0:10 - 0:20	0:03 - 0:10	0:01 - 0:03					
below -25 to -29°C (below -13 to -20°F)	100/0	0:15 - 0:30	0:00 - 0:01	0:07 - 0:10	0:02 - 0:07	0:00 - 0:02					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 45: TYPE IV HOLDOVER TIMES FOR NEWAVE AEROCHEMICAL FCY 9311

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	1:55 - 4:00	0:25 - 0:55	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	1:10 - 2:00	0:40 - 1:05	0:09 - 0:09	0:15 - 1:25	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:35 - 2:05	0:20 - 0:40	1:50 - 2:00	0:55 - 1:50	0:30 - 0:55	0:35 - 1:20	0:20 - 0:35			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:35 - 2:05	0:20 - 0:35	1:35 - 2:00	0:50 - 1:35	0:25 - 0:50	0:35 - 1:20[11]	0:20 - 0:35[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 0:55	0:10 - 0:20	1:00 - 1:15	0:30 - 1:00	0:15 - 0:30					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 0:55	0:05 - 0:10	0:35 - 0:40	0:15 - 0:35	0:07 - 0:15					
below -25 to -29.5°C (below -13 to -21°F)	100/0	0:30 - 0:55	0:05 - 0:10	0:30 - 0:40	0:15 - 0:30	0:06 - 0:15					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 53 provides allowance times for Type IV PG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 46: TYPE IV HOLDOVER TIMES FOR NEWAVE AEROCHEMICAL FCY-EGIV

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:35 - 4:00	0:25 - 0:55	2:00 - 2:00	1:10 - 2:00	0:35 - 1:10	1:20 - 2:00	0:40 - 1:05	0:10 - 0:10	0:15 - 2:00	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:25 - 3:25	0:20 - 0:45	2:00 - 2:00	1:00 - 2:00	0:25 - 1:00	0:50 - 2:00	0:45 - 1:05			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:25 - 3:25	0:20 - 0:40	1:55 - 2:00	0:50 - 1:55	0:25 - 0:50	0:50 - 2:00[11]	0:45 - 1:05[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:35 - 1:55	0:15 - 0:30	1:35 - 2:00	0:40 - 1:35	0:15 - 0:40					
below -18 to -25°C (below 0 to -13°F)	100/0	0:35 - 1:55	0:09 - 0:20	1:10 - 1:35	0:30 - 1:10	0:15 - 0:30					
below -25 to -29°C (below -13 to -20°F)	100/0	0:35 - 1:55	0:08 - 0:20	1:00 - 1:20	0:25 - 1:00	0:10 - 0:25					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 52 provides allowance times for Type IV EG fluids in ice pellets and small hail).
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 47: TYPE IV HOLDOVER TIMES FOR NEWAVE AEROCHEMICAL FCY-EGIV PLUS

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	3:05 - 4:00	0:30 - 1:00	2:00 - 2:00	1:20 - 2:00	0:40 - 1:20	1:15 - 2:00	0:40 - 0:55	0:10 - 0:10	0:15 - 2:00	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:35 - 3:15	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	1:15 - 2:00	0:30 - 0:45			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:35 - 3:15	0:20 - 0:40	2:00 - 2:00	0:55 - 2:00	0:25 - 0:55	1:15 - 2:00[11]	0:30 - 0:45[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 1:10	0:15 - 0:30	1:30 - 1:55	0:40 - 1:30	0:20 - 0:40					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 1:10	0:09 - 0:20	1:05 - 1:25	0:30 - 1:05	0:15 - 0:30					
below -25 to -29°C (below -13 to -20°F)	100/0	0:30 - 1:10	0:08 - 0:20	1:00 - 1:15	0:25 - 1:00	0:10 - 0:25					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than "moderate". No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of "moderate" precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a "heavy" precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 48: TYPE IV HOLDOVER TIMES FOR SHAANXI CLEANWAY CLEANSURFACE IV

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:30 - 4:00	0:30 - 1:15	2:00 - 2:00	1:40 - 2:00	0:40 - 1:40	1:20 - 2:00	0:40 - 1:10	0:09 - 0:09	0:15 - 1:50	
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	0:55 - 2:05	0:20 - 0:45	2:00 - 2:00	1:00 - 2:00	0:25 - 1:00	0:30 - 1:30	0:25 - 0:35			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	0:55 - 2:05	0:15 - 0:35	1:45 - 2:00	0:45 - 1:45	0:20 - 0:45	0:30 - 1:30[11]	0:25 - 0:35[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	CAUTION: No holdover time guidelines exist		
below -14 to -18°C (below 7 to 0°F)	100/0	0:25 - 0:35	0:10 - 0:20	1:05 - 1:20	0:30 - 1:05	0:15 - 0:30					
below -18 to -25°C (below 0 to -13°F)	100/0	0:25 - 0:35	0:05 - 0:10	0:35 - 0:45	0:15 - 0:35	0:07 - 0:15					
below -25 to -30°C (below -13 to -22°F)	100/0	0:20 - 0:30	0:04 - 0:09	0:30 - 0:35	0:15 - 0:30	0:06 - 0:15					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

TABLE 49: TYPE IV HOLDOVER TIMES FOR XINJIANG ZHONGTIAN LIYANG AVIATION CLEARICE-IV

Outside Air Temperature[1]	Fluid Concentration Fluid/Water By % Volume	Freezing Fog, Freezing Mist[2], or Ice Crystals[3]	Snow mixed with Freezing Fog[4]	Very Light Snow, Snow Grains or Snow Pellets[5,6]	Light Snow, Snow Grains or Snow Pellets[5,6]	Moderate Snow, Snow Grains or Snow Pellets[5,6]	Freezing Drizzle[7]	Light Freezing Rain	Moderate Snow mixed with Rain[8,9]	Rain on Cold-Soaked Wing[9]	Other[10]
-3°C and above (27°F and above)	100/0	2:25 - 4:00	0:25 - 0:50	2:00 - 2:00	1:05 - 2:00	0:30 - 1:05	1:05 - 1:55	0:35 - 0:55	0:09 - 0:09	0:15 - 1:30	CAUTION: No holdover time guidelines exist
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	50/50	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -3 to -8°C (below 27 to 18°F)	100/0	1:20 - 2:10	0:20 - 0:40	1:50 - 2:00	0:55 - 1:50	0:25 - 0:55	0:55 - 1:35	0:35 - 0:50			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -8 to -14°C (below 18 to 7°F)	100/0	1:20 - 2:10	0:15 - 0:35	1:35 - 2:00	0:45 - 1:35	0:20 - 0:45	0:55 - 1:35[11]	0:35 - 0:50[11]			
	75/25	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
below -14 to -18°C (below 7 to 0°F)	100/0	0:30 - 1:00	0:15 - 0:30	1:15 - 1:35	0:40 - 1:15	0:20 - 0:40					
below -18 to -25°C (below 0 to -13°F)	100/0	0:30 - 1:00	0:15 - 0:25	1:10 - 1:25	0:35 - 1:10	0:15 - 0:35					
below -25 to -29°C (below -13 to -20°F)	100/0	0:30 - 1:00	0:10 - 0:25	1:05 - 1:20	0:30 - 1:05	0:15 - 0:30					

NOTES

1 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
2 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0°C (32°F) and below.
3 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
4 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm that the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0°C (32°F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
11 No holdover time guidelines exist for this condition below -10°C (14°F).

CAUTIONS

- The cautions that apply to the holdover times in the table above can be found on page 330.

ALLOWANCE TIMES TABLES FOR WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.

The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation.

CAUTIONS

- The responsibility for the application of these data remains with the user.
- Fluids used during ground de/anti-icing do not provide in-flight icing protection.
- Allowance time cannot be extended by an inspection of the aircraft critical surfaces.

TABLE 50: LIST OF FLUIDS VALIDATED FOR USE WITH ALLOWANCE TIMES[1]

Manufacturer	Fluid Name	Applicable Allowance Times (ATs)
Type III Fluids		
AllClear Systems LLC	AeroClear MAX	ATs for Type III Fluids
Type IV EG Fluids		
AFLRUS LLC	Green Flo Type 4	Fluid has not been validated[2]
ALAB International	PROFLIGHT EG4	Fluid has not been validated[2]
AllClear Systems LLC	ClearWing EG	ATs for Type IV EG Fluids
ASGlobal	4Flite EG	Fluid has not been validated[2]
AVIAFLUID International Ltd	AVIAFlight EG	ATs for Type IV EG Fluids
CHEMCO Inc.	ChemR EG IV	ATs for Type IV EG Fluids
CHEMCO Inc.	ChemR Nordik IV	ATs for Type IV EG Fluids
Chongqing Joba Chemical Co., Ltd	FW-IV	ATs for Type IV EG Fluids
Dow Inc.	UCAR ENDURANCE™ EG106 ADF/AAF	ATs for Type IV EG Fluids
JSC RCP Nordix	Defrost NORTH 4	ATs for Type IV EG Fluids
Newave Aerochemical Co. Ltd.	FCY-EGIV	ATs for Type IV EG Fluids
Newave Aerochemical Co. Ltd.	FCY-EGIV PLUS	Fluid has not been validated[2]
Xinjiang Zhongtian Liyang Aviation	Clearice-IV	Fluid has not been validated[2]
Type IV PG Fluids		
ABAX Industries	ECOWING AD-49	ATs for Type IV PG Fluids
ALAB International	PROFLIGHT PG4	Fluid has not been validated[2]
AllClear Systems LLC	Clear IV Flight	Fluid has not been validated[2]
ASGlobal	4Flite PG	ATs for Type IV PG Fluids
AVIAFLUID International Ltd	AVIAFlight PG	ATs for Type IV PG Fluids
Clariant Produkte (Deutschland) GmbH	Safewing MP IV LAUNCH	ATs for Type IV PG Fluids
Clariant Produkte (Deutschland) GmbH	Safewing MP IV LAUNCH PLUS	ATs for Type IV PG Fluids
Cryotech Deicing Technology	Polar Guard® Advance	ATs for Type IV PG Fluids
Cryotech Deicing Technology	Polar Guard® Xtend	ATs for Type IV PG Fluids
Dow Inc.	UCAR™ FLIGHTGUARD™ AD-49	ATs for Type IV PG Fluids
ESSPO CHEMICALS D.O.O	Nordwing PG4	Fluid has not been validated[2]
Inland Technologies Inc.	ECO-SHIELD®	ATs for Type IV PG Fluids
JSC RCP Nordix	Defrost ECO 4	ATs for Type IV PG Fluids
Kilfrost Limited	ABC-S Plus	ATs for Type IV PG Fluids
MKS DevO Chemicals	COREICEPHOB TYPE-IV PG	Fluid has not been validated[2]
Newave Aerochemical Co. Ltd.	FCY 9311	ATs for Type IV PG Fluids
Shaanxi Cleanway Aviation Chemical Co., Ltd	Cleansurface IV	Fluid has not been validated[2]

NOTES

1. Allowance times are for use with undiluted (100/0) Type III, Type IV EG, and Type IV PG fluids only. No allowance times exist for Type II fluids.
2. No allowance times exist for this fluid at the time of publication as the allowance times have not yet been validated.

TABLE 51: ALLOWANCE TIMES FOR SAE TYPE III FLUIDS[1,2]

<table>
<tr><th rowspan="2">Precipitation Types or Combinations and Applicable METAR Codes[4]</th><th colspan="4">Outside Air Temperature</th></tr>
<tr><th>Above 0°C
(Above 32°F)</th><th>0 to -5°C
(32 to 23°F)</th><th>Below -5 to -10°C
(Below 23 to 14°F)</th><th>Below -10°C[3]
(Below 14°F)</th></tr>
<tr><td>Light Ice Pellets (or Small Hail[5,6])
-PL</td><td>10 minutes</td><td>10 minutes</td><td>10 minutes</td><td rowspan="7">Caution: No allowance times currently exist</td></tr>
<tr><td>Light Ice Pellets (or Small Hail[5,6]) Mixed with Light Snow
-PLSN, -SNPL</td><td>10 minutes</td><td>10 minutes</td><td>10 minutes</td></tr>
<tr><td>Light Ice Pellets (or Small Hail[5,6]) Mixed with Light, Moderate or Heavy Freezing Drizzle
-PLFZDZ, -FZDZPL, FZDZPL, +FZDZPL</td><td></td><td>7 minutes</td><td>5 minutes</td></tr>
<tr><td>Light Ice Pellets (or Small Hail[5,6]) Mixed with Light, Moderate or Heavy Drizzle
-PLDZ, -DZPL, DZPL, +DZPL</td><td>7 minutes</td><td></td><td></td></tr>
<tr><td>Light Ice Pellets (or Small Hail[5,6]) Mixed with Light Freezing Rain
-PLFZRA, -FZRAPL</td><td></td><td>7 minutes</td><td>5 minutes</td></tr>
<tr><td>Light Ice Pellets (or Small Hail[5,6]) Mixed with Light Rain
-PLRA, -RAPL</td><td>7 minutes</td><td></td><td></td></tr>
<tr><td>Moderate Ice Pellets (or Small Hail[5,6])
PL</td><td>5 minutes</td><td>5 minutes</td><td>5 minutes</td></tr>
</table>

NOTES

1 These allowance times are for use with undiluted (100/0) fluids applied unheated on aircraft with rotation speeds of 100 knots or greater. To use the allowance times in this table, ensure the fluid being used is listed in the List of Fluids Validated for Use with Allowance Times (Table 50).

2 Takeoff is allowed up to 90 minutes after start of fluid application if the precipitation stops at or before the allowance time expires and does not restart. Takeoff is not permitted if the OAT decreases during the 90 minutes in conditions of light ice pellets mixed with either: light, moderate or heavy freezing drizzle; light, moderate or heavy drizzle; light freezing rain; or light rain.

3 Ensure that the lowest operational use temperature (LOUT) is respected.

4 Applicable METAR codes include only Ice Pellet conditions (i.e., PL). While allowance times can be used when Small Hail is reported instead of Ice Pellets, the corresponding METAR codes for Small Hail are not listed in the table.

5 The reporting of Small Hail by METAR varies by country, which can result in different HOTs or allowance times when the same METAR code is used in different regions. For details, refer to the section regarding Hail, Small Hail, Ice Pellets, Snow Grains and Snow Pellets in TP 14052E - *Guidelines for Aircraft Ground Icing Operations*.

6 If the METAR does not report an intensity for small hail, use the “moderate ice pellets or small hail” allowance times. If the METAR reports an intensity with small hail, the condition with the equivalent intensity can be used, e.g. if light small hail is reported, the “light ice pellets” allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the “light ice pellets mixed with light snow” allowance times.

CAUTIONS

- The cautions that apply to the allowance times in the table above can be found on page 362.

TABLE 52: ALLOWANCE TIMES FOR SAE TYPE IV ETHYLENE GLYCOL (EG) FLUIDS[1,2]

Precipitation Types or Combinations and Applicable METAR Codes[5]	Outside Air Temperature				
	Above 0°C[3] (Above 32°F)	**0 to -5°C[3]** (32 to 23°F)	**Below -5 to -10°C[3]** (Below 23 to 14°F)	**Below -10 to -16°C[3]** (Below 14 to 3°F)	**Below -16 to -22°C[3,4]** (Below 3 to -8°F)
Light Ice Pellets (or Small Hail[6,7]) -PL	70 minutes	70 minutes	70 minutes	50 minutes	30 minutes
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light Snow -PLSN, -SNPL	50 minutes	50 minutes	30 minutes	25 minutes	
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light, Moderate or Heavy Freezing Drizzle -PLFZDZ, -FZDZPL, FZDZPL, +FZDZPL		40 minutes	30 minutes		
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light, Moderate or Heavy Drizzle -PLDZ, -DZPL, DZPL, +DZPL	40 minutes				
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light Freezing Rain -PLFZRA, -FZRAPL		40 minutes	30 minutes		
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light Rain -PLRA, -RAPL	40 minutes			**Caution: No allowance times currently exist**	
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light Freezing Rain and Light Snow -PLFZRASN, -PLSNFZRA, -FZRAPLSN, -FZRASNPL, -SNPLFZRA, -SNFZRAPL		20 minutes			
Light Ice Pellets (or Small Hail[6,7]) Mixed with Light Rain and Light Snow -PLRASN, -PLSNRA, -RAPLSN, -RASNPL, -SNPLRA, -SNRAPL	20 minutes				
Moderate Ice Pellets (or Small Hail[6,7]) PL	35 minutes	35 minutes	35 minutes	15 minutes	10 minutes
Moderate Ice Pellets (or Small Hail[6,7]) Mixed with Moderate Snow PLSN, SNPL	25 minutes	25 minutes	15 minutes	10 minutes	
Moderate Ice Pellets (or Small Hail[6,7]) Mixed with Light, Moderate or Heavy Freezing Drizzle PLFZDZ		20 minutes	10 minutes		
Moderate Ice Pellets (or Small Hail[6,7]) Mixed with Light, Moderate or Heavy Drizzle PLDZ	20 minutes		**Caution: No allowance times currently exist**		
Moderate Ice Pellets (or Small Hail[6,7]) Mixed with Moderate Rain PLRA, RAPL	15 minutes				

NOTES

- The notes that apply to the allowance times in the table above can be found on page 366.

CAUTIONS

- The cautions that apply to the allowance times in the table above can be found on page 362.

TABLE 52 (CONT'D): ALLOWANCE TIMES FOR SAE TYPE IV ETHYLENE GLYCOL (EG) FLUIDS

NOTES

1 These allowance times are for use with undiluted (100/0) EG-based fluids. To use the allowance times in this table, ensure the fluid being used is listed in the List of Fluids Validated for Use with Allowance Times (Table 50).
2 Takeoff is allowed up to 90 minutes after start of fluid application if the precipitation stops at or before the allowance time expires and does not restart. Takeoff is not permitted if the OAT decreases during the 90 minutes in conditions of light ice pellets mixed with either: light, moderate or heavy freezing drizzle; light, moderate or heavy drizzle; light freezing rain; light rain; light rain and light snow; or light freezing rain and light snow, as well as in conditions of moderate ice pellets mixed with either: light, moderate or heavy freezing drizzle; light, moderate or heavy drizzle; or moderate rain.
3 No allowance times exist for EG-based fluids when used on aircraft with rotation speeds less than 100 knots.
4 Ensure that the lowest operational use temperature (LOUT) is respected.
5 Applicable METAR codes include only Ice Pellet conditions (i.e., PL). While allowance times can be used when Small Hail is reported instead of Ice Pellets, the corresponding METAR codes for Small Hail are not listed in the table.
6 The reporting of Small Hail by METAR varies by country, which can result in different HOTs or allowance times when the same METAR code is used in different regions. For details, refer to the section regarding Hail, Small Hail, Ice Pellets, Snow Grains and Snow Pellets in TP 14052E - *Guidelines for Aircraft Ground Icing Operations*.
7 If the METAR does not report an intensity for small hail, use the "moderate ice pellets or small hail" allowance times. If the METAR reports an intensity with small hail, the condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

TABLE 53: ALLOWANCE TIMES FOR SAE TYPE IV PROPYLENE GLYCOL (PG) FLUIDS[1,2]

Precipitation Types or Combinations and Applicable METAR Codes[6]	Outside Air Temperature				
	Above 0°C[3] (Above 32°F)	**0 to -5°C[3]** (32 to 23°F)	**Below -5 to -10°C[3]** (Below 23 to 14°F)	**Below -10 to -16°C[4]** (Below 14 to 3°F)	**Below -16 to -22°C[4,5]** (Below 3 to -8°F)
Light Ice Pellets (or Small Hail[7,8]) -PL	50 minutes	50 minutes	30 minutes	30 minutes	20 minutes
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light Snow -PLSN, -SNPL	40 minutes	40 minutes	15 minutes	15 minutes	
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light, Moderate or Heavy Freezing Drizzle -PLFZDZ, -FZDZPL, FZDZPL, +FZDZPL		25 minutes	10 minutes		
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light, Moderate or Heavy Drizzle -PLDZ, -DZPL, DZPL, +DZPL	25 minutes				
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light Freezing Rain -PLFZRA, -FZRAPL		25 minutes	10 minutes		
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light Rain -PLRA, -RAPL	25 minutes			**Caution: No allowance times currently exist**	
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light Freezing Rain and Light Snow -PLFZRASN, -PLSNFZRA, -FZRAPLSN, -FZRASNPL, -SNPLFZRA, -SNFZRAPL		20 minutes			
Light Ice Pellets (or Small Hail[7,8]) Mixed with Light Rain and Light Snow -PLRASN, -PLSNRA, -RAPLSN, -RASNPL, -SNPLRA, -SNRAPL	20 minutes				
Moderate Ice Pellets (or Small Hail[7,8]) PL	15 minutes	15 minutes	10 minutes	10 minutes	
Moderate Ice Pellets (or Small Hail[7,8]) Mixed with Moderate Snow PLSN, SNPL	15 minutes	15 minutes	5 minutes	5 minutes	
Moderate Ice Pellets (or Small Hail[7,8]) Mixed with Light, Moderate or Heavy Freezing Drizzle PLFZDZ		10 minutes	7 minutes		
Moderate Ice Pellets (or Small Hail[7,8]) Mixed with Light, Moderate or Heavy Drizzle PLDZ	10 minutes		**Caution: No allowance times currently exist**		
Moderate Ice Pellets (or Small Hail[7,8]) Mixed with Moderate Rain PLRA, RAPL	10 minutes				

NOTES

- The notes that apply to the allowance times in the table above can be found on page 668.

CAUTIONS

- The cautions that apply to the allowance times in the table above can be found on page 362.

TABLE 53 (CONT'D): ALLOWANCE TIMES FOR SAE TYPE IV PROPYLENE GLYCOL (PG) FLUIDS

NOTES

1 These allowance times are for use with undiluted (100/0) PG-based fluids applied on aircraft with rotation speeds of 100 knots or greater. To use the allowance times in this table, ensure the fluid being used is listed in the List of Fluids Validated for Use with Allowance Times (Table 50).

2 Takeoff is allowed up to 90 minutes after start of fluid application if the precipitation stops at or before the allowance time expires and does not restart. Takeoff is not permitted if the OAT decreases during the 90 minutes in conditions of light ice pellets mixed with either: light, moderate or heavy freezing drizzle; light, moderate or heavy drizzle; light freezing rain; light rain; light rain and light snow; or light freezing rain and light snow, as well as in conditions of moderate ice pellets mixed with either: light, moderate or heavy freezing drizzle; light, moderate or heavy drizzle; or moderate rain.

3 No allowance times exist for PG-based fluids when used on aircraft with rotation speeds less than 100 knots.

4 No allowance times exist for PG-based fluids when used on aircraft with rotation speeds less than 115 knots.

5 Ensure that the lowest operational use temperature (LOUT) is respected.

6 Applicable METAR codes include only Ice Pellet conditions (i.e., PL). While allowance times can be used when Small Hail is reported instead of Ice Pellets, the corresponding METAR codes for Small Hail are not listed in the table.

7 The reporting of Small Hail by METAR varies by country, which can result in different HOTs or allowance times when the same METAR code is used in different regions. For details, refer to the section regarding Hail, Small Hail, Ice Pellets, Snow Grains and Snow Pellets in TP 14052E - *Guidelines for Aircraft Ground Icing Operations*.

8 If the METAR does not report an intensity for small hail, use the "moderate ice pellets or small hail" allowance times. If the METAR reports an intensity with small hail, the condition with the equivalent intensity can be used, e.g. if light small hail is reported, the "light ice pellets" allowance times can be used. This also applies in mixed conditions, e.g. if light small hail mixed with light snow is reported, use the "light ice pellets mixed with light snow" allowance times.

SUPPLEMENTAL GUIDANCE FOR WINTER 2025-2026

The HOT Guidelines are provided for information and guidance purposes. The HOT Guidelines on their own do not change, create, amend or permit deviations from regulatory requirements.

The HOT Guidelines may use mandatory terms such as “must”, “shall” and “is/are required” so as to convey the intent of meeting regulatory requirements and SAE Standards, where applicable. The term “should” is to be understood, unless an alternative method of achieving safety is implemented that would meet or exceed the intent of the recommendation

SECTION 2.18 A/C GROUND DE-ICING/ANTI-ICING

TABLE 54: SNOWFALL INTENSITIES AS A FUNCTION OF PREVAILING VISIBILITY

Visibility		Day		Night	
Statute Miles	**Meters**	**Below -1°C** Below 30°F	**-1°C and above** 30°F and above	**Below -1°C** Below 30°F	**-1°C and above** 30°F and above
≤1/4 (≤3/8)	**≤400** (≤600)	Heavy	Heavy	Heavy	Heavy
1/2 (>3/8 to ≤5/8)	**800** (>600 to ≤1000)	Moderate	Heavy	Heavy	Heavy
3/4 (>5/8 to ≤7/8)	**1200** (>1000 to ≤1400)	Moderate	Moderate	Heavy	Heavy
1 (>7/8 to ≤1 1/8)	**1600** (>1400 to ≤1800)	Light	Light	Moderate	Heavy
1 ¼ (>1 1/8 to ≤1 3/8)	**2000** (>1800 to ≤2200)	Light	Light	Moderate	Moderate
1 ½ (>1 3/8 to ≤1 5/8)	**2400** (>2200 to ≤2600)	Light	Light	Moderate	Moderate
1 ¾ (>1 5/8 to ≤1 7/8)	**2800** (>2600 to ≤3000)	Light	Light	Light	Light
2 (>1 7/8 to ≤2 ¼)	**3200** (>3000 to ≤3600)	Very Light	Very Light	Light	Light
2 ½ (>2 ¼ to ≤2 ¾)	**4000** (>3600 to ≤4400)	Very Light	Very Light	Light	Light
3 (>2 ¾ to ≤3 ¼)	**4800** (>4400 to ≤5200)	Very Light	Very Light	Very Light	Light
≥3 ½ (>3 ¼)	**≥5600** (>5200)	Very Light	Very Light	Very Light	Very Light

NOTES

- The METAR/SPECI reported visibility or flight crew observed visibility will be used with this visibility table to establish snowfall intensity for Type I, II, III and IV holdover time guidelines, during snow, snow grain, or snow pellet precipitation conditions. This visibility table will also be used when snow, snow grains, or snow pellets are accompanied by blowing or drifting snow, or when snow is mixed with ice crystals, freezing fog, rain or drizzle in the METAR/SPECI.
- The use of Runway Visual Range (RVR) is not permitted for determining visibility used with the holdover tables.
- Some METARs contain tower visibility as well as surface visibility. Whenever surface visibility is available from an official source, such as a METAR, in either the main body of the METAR or in the Remarks ("RMK") section, the preferred action is to use the surface visibility value.
- If the visibility is being reduced by snow along with form(s) of obscuration such as fog, haze, smoke, etc., use of the table above may overestimate the actual snowfall intensity. However, use of the snowfall intensity being reported in accordance with the Environment and Climate Change Canada Manual of Surface Weather Observations (MANOBS), may underestimate the actual snowfall intensity as it does not directly correlate to the snowfall intensities used when determining holdover times. Use of the visibility table in all snow conditions with or without obscuration is recommended

.

Note:

Holdover and allowance times have been adjusted to 76 percent of standard times.
This adjustment is for use when flaps/slats are deployed prior to de/anti-icing.
Standard holdover and allowance times can be used if flaps and slats are deployed as close to departure as safety allows.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

1.0 PURPOSE

The purpose of this section is to establish guidelines on how to perform operations under adverse weather conditions.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Air Traffic Controller

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1 GENERAL

4.1.1. Flights through areas with known or forecast thunderstorms, severe turbulence wind shear, or volcanic ash should be avoided whenever possible due to various hazards involved, e.g. hail, lightning strikes, gusts, up and downdrafts with subsequent altitude or attitude changes and high “g” loads, etc.

4.1.2. Do not take off during heavy thunderstorm activity at the departure airfield. Delay the approach or divert to an alternate airfield rather than penetrate a severe thunderstorm in the approach area.

4.1.3. Strong winds may reach a magnitude where ground handling and operation, including taxi, may become unsafe or even impossible.

4.1.4. If **surface** mean wind speeds of **60** kts or above is reported, takeoff or landing is not authorized, and the airfield must be considered closed.

4.1.5. Mutual information on development and position of thunderstorms between pilots and ATC, as well as a careful weather watch is of great importance for the early and adequate avoidance of severe weather areas.

4.1.6. Use all available weather and wind shear detection systems, both in the aircraft and ground based. Use all ATC and Pilot reports.

a) Pilots are encouraged to improve their own standards of judgment based on visual cues. They should also be aware of the extreme limitations of weather radar in producing a reliable picture regarding the existence, location and intensity of wind shear. Radar and wind shear detection systems fitted to aircraft can only paint weather and moisture, whereas wind shear may occur some considerable distance from any such visible weather.

b) At some airfields low level wind shear alert systems attempt to provide warning in the event of horizontal changes in wind direction and velocity exceeding certain values, and rapid surface pressure changes, which help to detect cold front passages and thunderstorm gust fronts.

c) In the case of wind shear along the approach path, pilot reports still represent the main source of information.

4.1.7. With strong shears, aircraft can experience large fluctuations of airspeed and lift in very short periods.

4.1.8. Pilots should be cautious whenever wind shear is expected.

4.1.9. Turbulence and wind shear present a potential hazard during takeoff and climb out, and approach and landing.

4.1.10. Immediate corrective action, in accordance with FCOM procedures, to avoid high sink rates close to the ground is of vital importance.

5.0. DEFINITION

- **Light Turbulence** - Slight discomfort and light oscillations.
- **Moderate Turbulence** - Moderate changes in aircraft attitude or altitude accompanied by small variations in air speed. Walking is difficult, loose objects move around.
- **Severe Turbulence** - Abrupt changes in aircraft attitude or altitude. Aircraft may be out of control for short periods accompanied by large variations in airspeed. Occupants are forced violently against seat belts. Loose objects are tossed around. Aircraft handling affected.
- **Strong Turbulence** - intermittent jolts.
- **Turbulence -** is meteorologically defined as a disturbed flow of air with embedded irregular whirls, eddies or waves. An aircraft in turbulent flow is subjected to irregular and random motion while more or less maintaining its intended flight path.
- **Wind Shear -** is meteorologically defined as the local variation of wind velocity in a given (but changeable) direction. Wind shear in aviation equates to the rate of change of wind velocity along the path of an aircraft.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

NOTE: Wind shear, with or without turbulence, alters the lift force acting on an aircraft, resulting in significant sinking or rising motion. It is thus categorized as:

- Increased performance shear caused by increasing headwind or decreasing tailwind component or vertical updrafts.
- Decreased performance shear caused by decreasing headwind or increasing tailwind component or vertical downdrafts.

6.0. RESPONSIBILITY

6.1**.** The captain is Responsible to comply with the procedures stated hereunder.

7.0 PROCEDURE

7.1. WAKE TURBULENCE

7.1.1. The danger of wake turbulence is greatest during the critical stages of flight on take-off or landing, and all captains are reminded of the need to allow adequate interval between their own and preceding heavier airplane for any such turbulence to dissipate.

7.1.2. Avoid flight below and behind a large A/C path, however if a large A/C is observed above and on the same track (meeting or overtaking), adjust your position laterally preferably upwind in Coordination with ATC.

NOTE: For recognition purpose, A380-800 airplanes will include the expression 'Super' immediately after the call sign in the initial radiotelephony contact between such airplanes and ATS units.

7.2. AIRPLANE CATEGORY

TABLE 1

Category	ICAO and flight plan
Heavy (H)	>136000kgs
Medium (M)	>7000kgs and < 136000kgs
Light (L)	< 7000kgs

NOTE: The wake turbulence category of an airplane should be indicated on the flight plan (item 9) as J (A380-800), H, M or L.

7.3. WAKE VORTEX SEPARATION

7.3.1. Wake turbulence separation minima are the spacing between airplanes, determined either by time or distance, to be applied so that airplanes do not fly through the wake of a preceding airplane within the area of maximum vortices. Where the separation minima required for IFR flights are greater than the recommended separation for wake turbulence, the IFR separation minima will be applied.

7.3.2. Although air traffic controllers will normally be responsible for the application of the appropriate separation for departing or arriving airplane and the need to observe particular intervals when following airplane of a higher wake turbulence category, captains shall ensure that the appropriate separations according to the scenarios described in the paragraphs that follow are applied.

7.3.3. Acceptance of a visual approach clearance is an acknowledgement that the pilot will ensure safe takeoff and landing intervals and accepts the responsibility of providing his own wake turbulence separation

7.4. SUCCESSIVE AIRPLANES ON INTERMEDIATE APPROACH

7.4.1. The following wake turbulence separation minima are applied in the intermediate approach segment:

a) nm between a heavy and a medium airplane following or crossing behind at the same level or less than 1,000ft below;

b) 6 nm between a heavy and light airplane following or crossing behind at the same level or less than 1,000ft below;

c) As per final approach minima (as shown on **TABLE 2**) for airplanes following or crossing behind an A380-800 at the same level of less than 1000ft below.

NOTE: The intermediate approach phase is specific to each Individual Instrument Approach Procedure.

7.5. SUCCESSIVE AIRPLANES ON FINAL APPROACH

7.5.1. The wake turbulence separation minima in the table below are applied to airplanes on final approach when:

a) An airplane is operating directly behind another airplane at the same altitude or less than 1000ft below: or

b) An airplane is crossing behind another airplane, at the same altitude or less than 1000ft below; or

c) Both airplanes are using the same runway or parallel runways separated by less than 760 meters.

NOTE: When parallel runways separated by less than 760 meters are in use, such runways are considered to be a single runway for wake turbulence reasons and normal wake turbulence separation minima apply to landing and departing airplanes respectively.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

TABLE 2

LEADING AIRPLANE	FOLLOWING AIRPLANE	MINIMUM DISTANCE (NM)
A380-800 A380-800 A380-800 A380-800	A380-800 Heavy Medium Light	4 6 7 8
Heavy Heavy Heavy Heavy	A380-800 Heavy Medium Light	4 4 5 7
Medium	A380-800 Heavy Medium Light	* * 3 6
Light Light Light Light	A380-800 Heavy Medium Light	* * * *

NOTE: * indicates that separation for wake turbulence reasons is not necessary.

7.6. AIRPLANES DEPARTING IN THE SAME DIRECTION

7.6.1. Wake turbulence separation minima on departure are applied by measuring airborne times between successive airplanes.

7.6.2. The wake turbulence separation minima in the **TABLE 3** below are applied when airplanes are using:

a) The same runway; or

b) Parallel runways separated by less than 760 m; or

c) Crossing runways if the projected flight path of the second airplane will cross the projected flight path of the first airplane at the same altitude or less than 1,000ft below; or

d) Parallel runways separated by 760m or more, if the projected flight path of the second airplane will cross the projected flight path of the first airplane at the same altitude or less than 1000ft below.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

TABLE 3

<table>
<tr><th>LEADING AIRPLANE</th><th>FOLLOWING AIRPLANE</th><th colspan="2">MINIMUM WAKE TURBULENCE SEPARATION AT THE TIME AIRPLANES ARE AIRBORNE</th></tr>
<tr><td rowspan="2">A380-800</td><td>Heavy (inc A380-800)</td><td>2 minutes</td><td rowspan="4">Departing from the same position

or

From a parallel runway separated by less than 760 meters</td></tr>
<tr><td>Medium

light</td><td>2 minutes</td></tr>
<tr><td>Heavy</td><td>Medium

Light</td><td>2 minutes</td></tr>
<tr><td>Medium</td><td>Light</td><td>2 minutes</td></tr>
</table>

NOTE 1: If departing from an intermediate point on the same runway or from an intermediate point of a parallel runway separated by less than 760 meters add one minute for the minimum wake turbulence separation on the above table.

NOTE 2: For wake turbulence separation purposes, airplane carrying out a touch-and-go are considered as making a departure from an intermediate point on the runway.

7.7. AIRPLANES DEPARTING IN OPPOSITE DIRECTION

7.7.1. A wake turbulence separation of 2 minutes will be applied between a Medium or Light airplane and a Heavy airplane and between a Medium airplane and a Llight airplane whenever the heavier airplane is making a low or missed approach and the lighter airplane is:

a) Utilizing an opposite-direction runway for take-off; or

b) Landing on the same runway in the opposite direction; or on a parallel opposite direction runway separated by less than 760 meters.

NOTE: If A380-800 is making a low missed approach the wake turbulence separations of 3 minutes will be applied.

7.8. DISPLACED LANDING THRESHOLD

7.8.1. If the projected flight paths are expected to cross, a wake turbulence separation of 2minutes will be applied when operating on a runway with a displaced threshold when:

a) A departing Medium or Light airplane follows a Heavy arrival of a departing Light airplane follows a Medium arrival; or

b) An arriving Medium or Light airplane follows a heavy airplane departure, or an arrival Light airplane follows a departing medium airplane; or

c) A departing Heavy airplane follows an A380-800 arrival; or

d) An arrival Heavy airplane follows an A380-800 departure.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

NOTE: If it is A380-800 in place of heavy aircraft above (a and b) a wake turbulence separation of 3 minutes will be applied

7.9. WIND SHEAR

7.9.1. Pilots must be alert to the possibility of wind shear during departure when studying weather information indicating:

a) Thunderstorm cells are in the vicinity of the airfield at a distance of 15 nm or less.
b) Frontal speeds exceeding 30 kts are evident.
c) The presence of high base convective clouds with high surface temperatures and large dew point spread.
d) Strong temperature inversions.

7.9.2. If presence of wind shear is confirmed delay takeoff or don't continue an approach **(AVOIDANCE)**

7.9.3. If wind shear is suspected, the following action should be considered: **(PRECAUTIONS)**

a) Selection of the most favorable runway considering length, obstacles and climb out direction.
b) Use of maximum thrust for takeoff etc…

7.9.4. If wind shear is encountered in flight, follow wind shear escape maneuver. **(RECOVERY)**

7.9.5. More information for the above **AVOIDANCE**, **PRECAUTIONS**, and **RECOVERY** of wind shear refer to FCOM, QRH & SOP.

7.9.6. Thunderstorms shall be avoided:

a) Visually by staying well clear of cumulonimbus clouds.
b) By using the airborne weather radar to find the most suitable corridor.
c) Strong echoes shall be avoided by 10 nm or more. This is most important at 20,000ft and above and for circumnavigation of echoes that have prominent scallops or other protrusions.
d) By requesting radar vectors from ATC.

7.9.7. Whenever possible avoid:

a) Flight in cirrus clouds if thunderstorm activity is reported along the route, as they may be hiding anvil tops and reduces the effectiveness of the airborne weather radar.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

b) Flight at or near the freezing level where the heaviest icing and hail must be expected.

c) Altitudes between 10,000 ft and 25,000 ft, as they will provide the roughest ride even outside active storm centers.

d) Flying below the overhang of CB clouds. This is the area where heavy hail can be expected.

7.9.8. During approach, good monitoring of thrust, attitude, airspeed and vertical speed as well as prompt action are the best insurance against the effects of wind shear or microburst. Always be alert for differences of actual wind readouts compared to reported surface winds.

7.9.9. If wind shear is detected, or sudden speed loss or a downburst is encountered, or if there is any doubt about the ability to re-establish a correct approach path, conduct the appropriate FCOM wind shear procedures if required, and execute a missed approach.

7.10. TURBULENCE

7.10.1. If flying in a thunderstorm or severe turbulence area is anticipated or unavoidable, the following preparations should be made:

7.10.2. Monitor airborne weather radar closely.

a) Select SEAT BELTS signs ON.

b) The Captain must provide the cabin crew with a recommendation regarding the suspension of cabin service and an estimate of the severity and duration of the turbulence. Food and drink service shall be minimized or even stopped, depending on the expected degree of turbulence.

c) If moderate or greater turbulence is anticipated or experienced, the Captain shall inform the cabin crew to cease all cabin service, have service carts stowed and cabin crew take their seats.

d) If such turbulence is imminent, the Captain shall advise cabin crew directly via the following PA announcement: "**CABIN CREW TAKE YOUR SEATS**".

e) It is recommended to use the words "bumpy" or "rough air" in place of "turbulence" when making passenger announcements.

f) Secure all loose items in the cockpit and fasten the shoulder harness.

g) Switch on cockpit lighting to high intensity to avoid being dazzled by lightning in the thunderstorm.

h) Fly at the recommended turbulence penetration speed and conduct the Turbulence Penetration procedures required by the FCOM.

7.10.3. Flight crewmembers shall maintain communication and coordination with cabin crewmembers as to the safe operation of the aircraft prior to takeoff roll, during turbulence, prior to landing and other times such as refueling/de-fueling etc…

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

7.10.4. The flight deck crew has to notify the team leader cabin services if turbulence is anticipated and at any time when seat belt sign is on to provide a recommendation regarding the suspension of cabin services, intensity and duration of turbulence.

7.10.5. If no communication from the flight deck, the Team Leader Cabin Services shall establish communicate with the cockpit crew and request for instruction on courses of action to follow.

7.10.6. When the seat belt sign is on the team leader cabin services has to coordinate with the flight deck crew regarding appropriate announcement advising the passengers of the situation.

7.10.7. If a reasonable amount of time has elapsed with no turbulence and the fasten seat belt sign remains on, the Team Leader Cabin Services should initiate contact with the flight deck crew via the interphone in order to determine resuming duties.

7.10.8. At maximum cruise altitude, the margin between low speed and high speed buffet is small and any increase of "g" loads, whether caused by maneuvering or by turbulence, may lead to serious difficulties. This shall be considered when trying to climb over a turbulent region. Avoid altitudes approaching maximum cruise altitude.

7.10.9. Should control be partially lost due to severe turbulence, resulting in a steep dive, the following recommendations may be helpful for a successful recovery:

a) Use speed brakes to prevent a rapid speed build-up.
b) Do not retract speed brakes until recovery is achieved.

7.11. ASSESSMENT AFTER FLIGHT THROUGH TURBULENCE OR WIND SEAR

7.11.1. After passing through a thunderstorm or turbulent area, various aircraft systems should be checked functionally as far as possible:

a) Flight and engine instruments.
b) Pitot and static heating.
c) Radio and navigation equipment.
d) Compass readings.

7.11.2. In order to ensure that a technical inspection for damage is carried out after the aircraft has been exposed to abnormal stresses, e.g. severe turbulence, lightning strikes, etc., an entry shall be made in the aircraft Technical Log. Any exceedence of engine or airframe limitations must be noted in the log.

7.12. REPORTING TO ATC

Whenever any significant turbulence or wind shear is encountered, particularly below 500 feet AGL including climb out and enroute phases, ATC shall be informed immediately, giving position, altitude, wind velocity and direction above and below the shear level, if available, or observed airspeed changes, etc.

7.13. VOLCANIC ASH

7.13.1 **Report of Volcanic Activity** Remain clear of reported volcanic activity. Encounters with volcanic ash can be extremely hazardous. Meteorological Services provide segments for volcanic ash.

CAUTION: Remain upwind of volcano if possible to avoid flight through volcanic ash. Volcanic ash may be undetectable at night of in IMC.

7.13.2 **Ash detection in Flight** Detecting volcanic ash may be difficult; however the following indications may appear in the vicinity of volcanic ash:

a) Poor radio reception due to static electricity,
b) Smoke or dust appearing in the flight deck, occasionally in concentrations sufficient to hinder breathing,
c) Acrid odor similar to electrical smoke,
d) Multiple engine malfunctions, such as surges, increasing EGT, torching, flameout, etc…
e) At night, St. Elmo's fire/ static discharges may be observed around the windshield accompanied by a bright orange glow in the engine inlets.

7.13.3 When encountering visible ash cloud the following generic procedures are recommended; however, flight crew should reference the specific Aircraft Operating Manual/Emergency Checklist for detailed guidance:

a) Turn on continuing ignition.
b) declare an emergency.(**MAYDAY**,**MAYDAY**,**MAYDAY**)
c) Do not climb in order to over-fly the ash cloud.
d) Reduce power to idle in order to provide additional engine stall margin and lower turbine temperature.
e) Try to escape the ash cloud by turning 180^0 and descending (if terrain clearance permits).
f) Monitor altitude versus airspeed.
g) Turn on all accessory air bleeds including all air conditioning packs, nacelles, and wing anti-ice. This will provide an additional engine stall margin by reducing engine pressure.

7.13.4 Flight through volcanic ash cloud should be avoided because of the extreme hazard to the

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

engines and the airframe. The ash cloud may extend for several hundred miles and eruptions may send ash up to 40,000ft, neither ash clouds nor volcanic ash can be detected by aircraft weather radar. Use all available sources of information to remain clear, or to exit areas of volcanic ash; consider reversing course if required. Refer to the applicable pilot's handbook for specific procedures.

7.13.5. En Route Operations. In order to maximize the likelihood of avoiding an ash cloud or any other airborne hazardous material while en route, pilots must take advantage of all available transmitted information, via either voice transmission or message transmission from area control centers.

7.13.6 Pilots, especially those operating in remote areas, should familiarize themselves with the locations of active volcanoes along their route. Since, thin ash clouds are often difficult to observe during daylight hours and impossible to see at night, the previously mentioned signs may be the first indication that the aircraft is entering a cloud and the pilot should begin escape maneuvers.

7.13.7. If an aircraft does encounter volcanic ash, the pilot must understand the options available and make tactical decisions in a timely manner. During an eruption, controller workload will increase considerably due to the need to alert other aircraft of the airborne hazardous material and the need to reroute en-route aircraft around the contaminated airspace.

7.13.8 Over-flight or Under-flight of closed No Fly Zone (NFZ) areas due to ash concentrations may be permitted (for over-flight; the NFZ areas must be considered as mountain range, with decision point, drift-down and depressurization escape routes as required).

7.13.9 MEl Considerations for NFZ over-flight:

a) **Depressurization:** Both packs and bleeds must be serviceable prior to departure. Additionally, APU and APU bleed must be serviceable.
b) **Engine:** Ignition Starting & Continuous Relight. Additionally, Pneumatic started & Valve System.
c) **CAT III:** Aircraft must be CAT II/III & Auto-land capable (in case of abrasive damage to flight deck windows).

7.14. VOLCANIC GAS/ VOLCANIC GASEOUS CLOUDS

NOTE: Recent input from representatives of the FAA, the Volcanic Ash Advisory Center (VAAC) and Volcano Observatories have helped to clarify procedural issues recently encountered during volcanic eruptions.

7.14.1. Some volcanic events produce not only ash, but sulfurous gas which generally is invisible, but can be detected by certain satellite technology. When mixed with significant amount of moisture the gas may be cloudlike and can be visible to the naked eye. It may have a brownish-reddish tint and both gas and gas clouds may maintain the sulfur like smell.

SECTION 2.19 ADVERSE WEATHER OPERATION POLICY

7.14.2 Satellites also have the ability to track ash clouds that have reached high levels in the atmosphere. When detected, the appropriate VAAC will issue a Volcanic Advisory which is disseminated to ATC or the ATS provider. (In the United States, the National Weather Service will issue SIGMETS based on the information provided by the VAAC). When satellite technology is no longer able to validate the presence of ash, SIGMETS are withdrawn or cancelled. NO SIGMETS are issued for volcanic gas of gas clouds, with the exception being Australia.

7.14.3 There is currently no guidance for avoidance of a gaseous cloud associated with volcanic activity other than remaining well clear of volcanic activity. When planning and operating flights, Ethiopian will make every effort to avoid the gaseous cloud associated with volcanic activity. If however, a flight encounters a gaseous cloud the flight crew should report the incident to ATC and the company. Encounters with actual volcanic ash must be reported to ATC & the company.

7.14.4. Ethiopian shall not conduct operation on routes that covers large active volcanic areas and in the terminal areas of airports in the vicinity of active volcanoes.

7.14.5. Refer to aircraft FCOM/QRH and FCTM for specific aircraft guidance.

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SECTION 2.20 ALL WEATHER OPERATIONS

SECTION 2.20 ALL WEATHER OPERATIONS

SECTION 2.20 ALL WEATHER OPERATIONS

SECTION 2.20 ALL WEATHER OPERATIONS

1.0 PURPOSE

The purpose of this procedure is to establish guidelines for all weather operations.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Crewmembers

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1. The limits of the aircraft are indicated in the Aircraft Flight Manual of each aircraft type. The limit of usable landing aids is the value of the DH/DA for the approach landing category. These limits or operating minima must not be less than those imposed by the concerned state and according to the type of flight.

4.2. When the visibility for landing is less than 800 meters, approach and landing are not authorized unless RVR report is available for runway of intended use.

4.3. The operational standards set in this document are in accordance with ICAO DOC 9365-AN/910 "Manual of All Weather Operations".

4.4. No Ethiopian flight shall be conducted in accordance with VFR unless available current meteorological reports, or a combination of current and forecasts, indicate that the meteorological conditions along the route, or that part of the route to be flown under VFR will, at the appropriate time, allow VFR operations. (ECARAS 8.6.2.4)

5.0 DEFINITIONS

5.1. **All Weather Operations (AWO)** consists in operating an aircraft in low visibility conditions. The term AWO includes Low Visibility Take-Off (LVTO), landing Category II (CAT II), landing Category III (CAT III) and Low Visibility Taxi (LV TAXI). Weather limitations (visibility) applied for AWO are called minima.

5.1.1. **Alert Height (AH):** A height above the highest elevation in the touchdown zone, above which a Category III approach would be discontinued and missed approach initiated, if a

SECTION 2.20 ALL WEATHER OPERATIONS

failure occurred in one of the required redundant operational systems in the airplane or in the relevant ground equipment.

NOTE: Below Alert Height, the approach, flare, touchdown and roll out may be safely accomplished following any failure in the airplane or associated category III systems. However, if **NO AUTOLAND, AUTOPILOT, AUTOPILOT DISC, SPEEDBRAKE EXTENDED, AUTHOTHROTTLE DISC OR LOC/GS DEVIATION ALERT** displayed (in the case of **AIRBUS if NO AUTOLAND** light illuminated) and if there is no sufficient visual cue to continue the approach go around must be initiated.

NOTE: Alert height for A350, B757/767, and B777/787/CATIIIB operation is 200 feet.

5.1.2. **Alert Height** is based on characteristics of an aircraft and its particular fail operational airborne category III system.

NOTE: Alert Height has no significance for CAT II approaches.

5.1.3. **Decision Height (DH):** A specified height at which a missed approach must be initiated if the required visual reference to continue the approach to land has not been established.

NOTE: A Decision Height (DH) of 20ft shall be used whenever DH is required for CAT IIIB approaches.

5.1.4 **Fail-Operational Flight Control System**: A flight control system is fail-operational if, in the event of a failure below alert height, the approach, flare and landing can be completed automatically. In the event of a failure, the automatic landing system will operate as a fail-passive system.

5.1.5. **Fail-Passive Flight Control System:** A flight control system is fail-passive if, in the event of a single failure, there is no significant out-of-trim condition or deviation of flight path or attitude, but the landing is not completed automatically. For a Fail-Passive flight control system, the pilot assumes control of the airplane after a failure.

5.1.6. **Low Visibility Procedures (LVP):** Procedures applied at an aerodrome for the purpose of ensuring safe operations during Category II and III approaches and Low Visibility Takeoffs. Low Visibility Procedures are put into force at airfields authorized for CAT II/III operations when the RVR falls below 550m and/or the cloud base falls below 200ft.

5.1.7. **Low Visibility Takeoff (LVTO):** A take-off on a runway where the RVR is less than 400m.

5.1.8. The physical value of Decision Height is defined as the distance between the wheels of the aircraft, with the aircraft on the glide slope, and the highest elevation of the runway in the touchdown zone. This height is read on the radio altimeter.

5.1.9. **Radio Altimeter:** aircraft equipment which makes use of the reflection of radio waves from the ground to determine the height of the aircraft above the surface.

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5.1.10. **Runway Visual Range (RVR):** is the range over which the pilot of an airplane on the centerline of the runway can see the runway surface markings or the lights delineating the runway or identifying its centerline.

6.0 RESPONSIBILITY

6.1. The pilot-in command is primarily responsible to adhere with the policy and procedure stated in the section of the manual.

7.0. PROCEDURE

7.1. RVR MEASUREMENTS AND MINIMUMS

7.1.1. CAT II and III operations require rapidly updated and reliable report of the visibility conditions which a pilot may expect to encounter in the touchdown zone and along the runway. RVR measurements replace the use of Reported Visibility Values (RVV) which is not appropriate for conditions encountered during the final approach and landing in low visibility, because the visibility observation are often several miles away from the touchdown zone of the runway.

7.1.2. For CAT II and III operations, the RVR measurements as provided by a system of calibrated transmissometers and account for the effects of ambient background light and the intensity of runway lights. Transmissometers systems are strategically located to provide RVR measurements associated with three basic portions of a runway. In addition, on some ling runways additional one transmissometer might be available at the end of the runway

a) The touchdown zone (TDZ),
b) The mid-runway portion (MID),
c) The rollout portion or stop end,
d) The Far End RVR, if available.

7.1.3. OTHER THAN CATEGORY II/III RVR REQUIREMENTS

a) RVR 550m (1800ft) or higher-

Touchdown if available controlling. If touchdown is inoperative, midpoint is required and controlling.

b) RVR 400m (1400ft) or higher- (Special Authorization CAT I)

Touchdown is required and controlling. As installed midpoint and rollout are advisory.

7.1.4. CATEGORY II RVR Requirements;

RVR 500m (1600ft) or less –

Touchdown required and controlling. Midpoints or rollout is also required and advisory.

If MID point is reported, it shall not be less than 125m (400ft).

If ROLLOUT is reported, it shall not be less than 75m (300ft).

7.1.5. CATEGORY III RVR Requirements;

a) For CAT. IIIA approaches the minimum required RVR;

TDZ: - 200m (700FT), MID: - 125m (400ft), ROLLOUT: - 75m (300ft)

b) For CAT. IIIB approach the minimum required RVR;

TDZ, MID, & ROLLOUT 75m (300ft)

c) for all CAT III operations the following shall be considered;

- All available RVR reports are required and controlling, where four RVR reporting systems are installed (i.e. TDZ, MID, ROLLOUT and FAR END SENSORS) the FAR END RVR report is advisory information to the pilots or may be substituted for the ROLLOUT RVR report if that is not available.
- During fail operational landing system with fail passive or fail operational rollout system, any one of RVR reporting system may be temporarily inoperative. The other two are required and controlling.
- During fail passive landing system with fail passive or fail operational rollout system, either MID or ROLLOUT reporting system may be temporarily inoperative. The minimum TDZ RVR required will be 200m (700ft), MID RVR report, if available, shall not be lower than 125m (400ft) and if ROLLOUT RVR reported it shall not be lower than 75m (300ft)
- During fail operational or fail passive landing system, if rollout system fails, TDZ and MID RVR reports shall not be lower than 200m (700ft).
- During fail operational landing system with fail passive rollout system, TDZ and MID RVR reports if available shall not be lower than 125m (400ft).

7.2. DECISION HEIGHT (DH) CONCEPT

7.2.1. Decision height is the wheel height above the runway elevation by which go-around must be initiated unless adequate visual reference has been established and the aircraft position and approach path have been assessed as satisfactory to continue the approach and landing in safety. In this definition, runway elevation means the elevation of the highest point in the touchdown zone. DH recognition must be by means of height measured by radio-altimeter.

7.2.2. Decision height is a specified point in space at which a pilot must make an operational decision. The pilot must decide if the visual references are adequate to safely continue the approach have been established.

7.2.3. If the visual references have not been established as per FOM 2.2 (7.18), a go-around must be executed.

7.2.4. If the visual references have been established, the approach can be continued.

7.2.5. However, the pilot may always decide to execute a go-around if sudden degradations in the visual references or sudden flight path deviation occur.

7.2.6. The DH is measured by means of radio-altimeter. When necessary, the published DH takes into account the terrain profile before runway threshold.

7.2.7. It should be stressed that the DH is the lowest limit of the decision zone in limiting conditions.

7.2.8. The PF shall:

a) be assessing the visual references;
b) come to this zone prepared for a go around but with no pre-established judgment;
c) Make a decision according to the quality of the approach and the way the visual references develop as DH is approached.

7.2.9. The DH shall not be lower than:

a) The minimum DH specified in the AFM,
b) The minimum height to which the precision approach aid can be used without the required visual reference
c) The OCH for the category of the aircraft
d) The DH to which the flight crew is authorized to operate state published minimum

7.3. CAT II & III OPERATIONS

7.3.1. No Ethiopian aircraft shall be operated in CAT II & III unless: (ECARAS 8.8.1.9)

a) The flight crew of the aircraft holds the appropriate authorizations and ratings.
b) Each flight crew member has adequate knowledge of and familiarity with the

aircraft and the procedures to be used.

c) The instrument panel in front of the pilot has the appropriate instrumentation for the type of flight control guidance system that is being used.

7.3.2. Jeppesen/state published minima are applicable to all approved Ethiopian fleet.

7.3.3. For all CAT II/III operations Flight crew shall set RA value as a minimum.

7.3.4. Unless otherwise authorized by ECAA, Ethiopian may not operate a civil aircraft in a CAT II or III operations unless each ground component required for that operation and the related airborne equipment is installed and operating. (ECARAS 8.8.1.9)

NOTE: For flights into North America, please refer to the US Operations Specification for more information & limitations.

7.4. CAT II OPERATIONS

7.4.1. A CAT II approach is a precision instrument approach and landing with decision height lower than 200ft (60m) but not lower than 100ft (30m) and a runway visual range not less than 300m (1000ft).

7.4.2. The main objective of CAT II operations is to provide a level of safety equivalent to other operations, but in more adverse weather conditions and lower visibility.

7.4.3. CAT II weather minima have been established to provide sufficient visual references at DH to permit a manual landing or a missed approach to be executed.

7.5. VISUAL REFERENCES FOR CAT II OPERATIONS

7.5.1. A pilot shall not continue a precision approach CAT II below the DH unless the required visual references are attained and can be maintained.

7.5.2. Refer to FOM 2.2 (7.18) for visual reference.

7.6. INSTRUMENTS AND EQUIPMENT FOR CATEGORY II OPERATIONS (ECARAS7.2.1.6)

7.6.1. The instruments and equipment listed hereunder shall be installed and approved for CAT II operations.

a) Two localizer and glide slope receiving system.

b) Two gyroscopic pitch and bank indicating system.

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c) Two airspeed indicators.

d) Two vertical speed indicators.

e) Warning systems for immediate detection by the pilot of system faults in items a, b, c & d above.

f) Dual controls.

g) An externally vented static pressure system with an alternate static pressure source.

h) A windshield wiper.

i) A heat source for each air speed system pitot tube.

j) One self-monitoring radio altimeter with dual display.

k) Two sensitive altimeters adjustable for barometric pressure having markings at 20 feet intervals. A marker beacon receiver that provides distinctive oral and visual indications of the outer and middle marker.

l) A communications system that does not affect the operation of at least one of the ILS systems.

m) A flight control guidance system that consists of either an automatic approach coupler a flight director system.

n) A flight director system must display computed information as steering command in relation to an ILS localizer and, on the same instrument, either computed information as pitch command in relation to an ILS glide slope or basic ILS glide slope information. An automatic approach coupler must provide at least automatic steering in relation to an ILS localizer.

7.7. CAT III OPERATIONS

7.7.1. A CAT III operation is a precision approach lower than CAT II minima. CAT III is divided in three sub-categories: CAT III A, CAT III B, CAT III C, associated with three minima levels (CAT III A is associated with highest minima and CAT III C with lowest minima).

7.7.2. A category III A approach is a precision instrument approach and landing with a decision height lower than 100ft (30m) and a runway visual range not less than 700ft (200m).

7.7.3. A category III B approach is a precision instrument approach and landing with no decision height or a decision height lower than 50ft (15m) and a runway visual range less than 700ft (200m), but not less than 250ft (75m).

7.7.4. A category III C approach is a precision instrument approach and landing with no decision height and no runway visual range limitation.

7.7.5. The main objective of CAT III operations is to provide a level of safety equivalent to other operations but in the most adverse weather conditions and associated visibility.

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7.7.6. In contrast to other operations, CAT III weather minima do not provide sufficient visual references to allow a manual landing to be performed. The minima only permit the pilot to decide if the aircraft will land in the touchdown zone and to ensure safety during roll-out. Therefore, an automatic landing system is mandatory to perform CATIII operations. Its reliability must be sufficient to control the aircraft to touchdown and roll-out to a safe taxi speed in CATIII operations.

7.7.7. The automatic landing system is only equipment capable of providing automatic control of the aircraft during the approach and landing and is not influenced by adverse weather conditions. This system is mandatory for all CAT III Operations.

7.7.8. It is common to practice automatic landing in good visibility, however when conducting practice cat III procedures ground controllers must be informed, ILS performance must be sufficient, and ILS signals must be protected.

7.8. VISUAL REFERENCES FOR CAT III OPERATIONS

7.8.1. For CAT IIIA and IIIB operations a pilot must not continue an approach below DH unless the required visual references are attained and can be maintained.

7.8.2. Refer to FOM 2.2 (7.18) for visual reference.

7.9. INSTRUMENTS AND EQUIPMENT FOR CATEGORY III OPERATIONS (ECARAS7.2.1.7)

7.9.1. **Airborne systems for CAT IIIA Minima** are not less than RVR 200m (700ft): The following equipment in addition to the instrument and navigation equipment required by IFR flight and CAT II operations is the minimum aircraft equipment required for CAT IIIA:

a) A redundant flight control or guidance system demonstrated in accordance with international acceptable criteria which include:

   I. A Fail Operation or Fail Passive automatic landing system as least to touch down.
   II. A Fail Operational or Fail Passive manual flight guidance system providing suitable head-up or head-down command guidance.
   III. A hybrid system, using automatic landing capability as the primary means of landing at least to touch down; or
   IV. Other systems that can provide an equivalent level of performance and safety.

b) An automatic throttle or automatic thrust control system that meets approved criteria as specified in the AFM.

c) At least two independent navigation receivers/sensors providing lateral and vertical position or displacement information.

d) At least two approved radio altimeter systems that meet the performance requirements criteria as specified in the AFM.

e) Failure detection, annunciation, and warning capability as determined in the AFM.

f) Missed approach guidance provided by one or more of the following:

I. Attitude displays that include suitable pitch attitude markings or pre-established computed pitch command display;

II. An approved flight path angle displays; or

III. An automatic or flight guidance go-around capability

g) Suitable forward and side flight deck visibility for each pilot as specified in the AFM.

h) Suitable windshield rain removal, ice protection, or defog capability as specified in the AFM.

7.9.2. **Airborne system for CAT IIIB minima** less than RVR 200m (700ft) but not less than RVR 75m (250ft): The following equipment in addition to the instrument and navigation equipment required by IFR flight and CAT II and CAT IIIA operations is the minimum aircraft equipment required for CAT IIIB plus the following extra equipment requirements:

a) A redundant flight control or guidance system demonstrated in accordance with international acceptable criteria including:

I. A Fail Operational landing system with a Fail Operational or Fail Passive automatic rollout system; or

II. A Fail Passive landing system, limited to touch down zone RVR not less than 200m (700ft), with Fail Passive rollout provided automatically or by a flight guidance system providing suitable head-up or head-down guidance; or

III. A Fail Operational Hybrid automatic landing and rollout system with comparable manual flight guidance system, using automatic landing capability as the primary means of landing; or

IV. Other systems that can provide an equivalent level of performance and safety.

b) Refer to “b” to “h” stated on 7.9.1. above.

7.9.3. **Airborne system for CAT IIIC minima** less than RVR 75m (300ft): The following equipment in addition to the instrument and navigation equipment required by IFR flight and CAT II, cat IIIA and CAT IIIB operations is the minimum aircraft equipment required for CAT IIIC:

a) A Fail Operational Automatic Flight Control System, or manual flight guidance system designed to meet fail operational system criteria, or a hybrid system in which both the fail-passive automatic system and the monitored manual flight guidance components provide approach and flare guidance to touch down, and in combination provide full fail operational capability; and

b) A fail operational automatic, manual, or hybrid rollout control system

7.10. RVR for CAT III APPROACH VS. FLIGHT CONTROL SYSTEMS AND DH

TABLE 1

Approach Category	Decision Height	RVR as a function of Automatic Landing System Status			
		Fail Passive	Fail Operational		
			Without rollout system	With rollout system	
				Fail Passive	Fail Operational
IIIA	Less than 100ft	200 m*	200 m	200 m	200 m
IIIB	Less than 50 ft	Not authorized	Not Authorized	125 m	75 m
IIIB	No DH	Not Authorized	Not Authorized	Not authorized	75 m

* For operations to actual RVR values less than 300 m, a go-around is assumed in the event of an autopilot failure at or below DH.

7.11. PRE-FLIGHT PREPARATION FOR CAT II OR III

7.11.1. In addition to normal flight preparation, the following planning and preparation must be performed when CAT II or CAT III approaches are envisaged.

7.11.2. Review NOTAMS to make sure that the destination airport meets CAT II or III requirements.

a) Runway and approach lighting,

b) Radio navaid availability.

c) RVR equipment availability, etc.

7.11.3. For fuel planning refer FOM 2.12 (7.7.3.)

7.11.4. Check that the weather forecast at destination is within airline and crew operating minima. If the forecast is below CAT I minima, verify that alternate weather forecasts are appropriate to the available approach means and at least at or better than CAT I minima.

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7.11.5. Crew qualification and currency must be reviewed (both CAPT and F/O must be qualified and current)

7.11.6. When the aircraft logbook is available, confirm that no write-ups were entered during previous flights that would affect equipment required for CAT II/III operations. All unsatisfactory auto land operations shall be recorded in the aircraft maintenance log, in addition to the AUTO APPROACH AND AUTOLAND PERFORMANCE REPORT form.

7.11.7. Aircraft capable of CAT II/III shall be placarded as CAT II/III compliant. Check that required equipment for CAT II/III approach are operative. The required equipment list is given in the respective aircraft Minimum Equipment List and in the AWO CAT II/III Approach Briefing Checklist in the cockpit.

7.12. APPROACH PREPARATION FOR CAT II OR III

7.12.1. On receiving the weather, the Captain shall brief the Low Visibility approach per the normal and low visibility approach briefing flow, in addition this flight crew has to review the following items:

a) anti-icing,
b) runway surface conditions,
c) Use of auto brake (Use minimum to medium auto brake if the runway is dry and moderate to high auto brake 3 or 4 if the runway is wet).
d) Taxi chart and exits.
e) Review the go-around procedure,
f) fuel status, and
g) Any other items deemed necessary for that particular approach.
h) Use Jeppesen CAT II/III approach charts.
i) Set applicable DH on radio altimeters. Set pressure altimeter to CAT I decision altitude.
j) Check aircraft status to ensure that the aircraft capability has not been degraded
k) For Auto land performance, refer to the Aircraft Operations Manual of each type.

7.12.2. Prior to descent, advise ATC that the aircraft intends to make a CAT II/III approach at destination, who will check the status of the ILS and lighting and protect the sensitive areas from incursion by aircraft or vehicles. Such an approach may not be undertaken until the clearance has been received. Before the outer marker, RVR values from TDZ, MID (and ROLLOUT when available), must be transmitted. The approach chart will define the required minimum values.

7.12.3. When an aerodrome is operating to CAT II/III, minima special ATC procedures are introduced which permit CAT II/III equipped aircraft to gain priority over the other holding aircraft.

7.12.4. For CAT IIIB approaches with AH, auto throttles must be available and for other CAT II/IIIA approaches they should be used if serviceable.

7.12.5. Check and adjust pilot seats to ensure optimum visual acuity. In the correct position, pilots should be looking along the plane surface of the top of the glare shield panel.

7.12.6. Use of landing lights: At night in low visibility conditions, landing lights can be detrimental to the acquisition of visual references. Reflected light from water droplets or snow may actually reduce visibility. Landing lights would therefore not normally be used in CAT II or CAT III weather conditions due to resulting glare.

7.12.7. Start the APU on passing / leaving FL 100 / 10,000 ft. whenever carrying out an actual, potential or practice low visibility approach. The APU will provide automatic electrical source redundancy. (Not required on A350, B787 and B777).

7.13. COMMENCING THE APPROACH FOR CAT II OR III OPERATIONS:

7.13.1. On first contact with the Approach Controller, request CAT II or III approach. Protection applied in CAT II/III conditions may be relaxed if the visibility improves to CAT I or better; the initial call prompts ATC to pass the current operational status.

7.13.2. When cleared to leave the holding pattern, Captain must take over as PF.

7.13.3. Should RVR fall below minimum after the aircraft has left the holding area, the Captain may at his discretion continue the approach to the final approach fix or Approach Ban point but must discontinue the approach by that point if the RVR is still below minimum (for **APPROACH BAN** refer FOM 2.2 (7.19)).

7.14. FINAL APPROACH FOR CAT II OR III OPERATIONS:

7.14.1. On first contact with Tower, advise that the aircraft is making a CAT II/III approach. Normal spacing is 8-10 nautical miles.

7.14.2. A/C must be configured for landing. Select gear down, set intermediate flap when glide slope alive and landing flaps just before glide slope capture, and adjust to final approach speed. Do not use low noise approach or high-speed approach profiles, except at airfields that specify otherwise.

7.14.3. During the final approach, the First Officer must concentrate his attention entirely on the Forward Instrument panel from localizer capture until the end of the rollout or missed approach and call out '**no flare**' or '**no roll out'** if applicable.

7.14.4. From the Outer Marker/FAF down, the captain should have his right hand on the thrust levers to permit quick actuation of the Go Around switch if required.

7.14.5. The Captain should guard the rudders all the time and be prepared to use them if required

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when automatic control is lost at the termination of G/A mode or following a disconnect.

7.14.6. On CAT II, IIIA or IIIB approaches with a DH, the Captain should start seeking, visual cues from the "approaching minimums" callout.

7.14.7. At DH, the Captain must respond promptly to the "Minimum" call by announcing his intentions (unless the intention to land was made before reaching DH due to early acquisition of sufficient visual references). The Captain's response should be either "LANDING/CONTINUE" or "GO AROUND".

7.14.8. On a CAT III approach the call at AH is "ALERT".

7.15. MISSED APPROACH FOR CAT II OR III OPERATIONS:

7.15.1. If the controlling RVR is reported to decrease below the applicable minima prior to reaching the DH, and if the aircraft is at or passed the approach ban point when this report is received, the approach may be continued to DH. At the DH, if the pilot has not established the required visual reference a missed approach will be initiated.

7.15.2. In CAT III operations with DH, the condition required at DH is that there should be visual references which confirm that the aircraft is over the touchdown zone. Go-around is mandatory if the visual references do not confirm this. For CAT III approaches with no DH visual reference is not required.

7.15.3. Loc deviation greater than 1/3 dot or Glideslope deviation greater than ½ dot in the 1000 feet AGL and excess ILS deviation and the pointer flashing execute a missed approach. PM should call out the deviation. e.g. "localizer" and the response & action should be go around.

7.15.4. Sustained airspeed deviation greater than + or – 5kts in the Decision Region (300 to 100ft AGL) a missed approach should be initiated.

7.15.5. If the decision to continue has been made and the visual references subsequently become insufficient (for the appropriate category), or the flight path deviates unacceptably, go around must be initiated (a go-around initiated below the DH, whether auto or manual, may result in ground contact).

7.15.6. If the visual references are lost after touchdown, a go-around should not be attempted. The roll-out should be continued with the autopilot(s) in ROLL-OUT mode down to taxi speed.

7.15.7. If after leaving 1000 ft. AGL, on a CAT IIIB approach, the RVR falls below that specified for landing then the approach may be continued to land.

7.15.8. GA mode is available throughout the approach. If touchdown occurs after the automatic GA is initiated, the GA will continue. If necessary to Go-Around, the Captain should call

"Go Around" and execute the go around according to the established procedure.

7.16. AFTER LANDING FOR CAT II OR III OPERATIONS:

7.16.1. After touchdown the spoilers deploy, automatic braking commences, and the nose is lowered. The captain is to apply reverse thrust. Allow the autopilots and auto brakes to remain engaged until a safe stop is ensured.

7.16.2. Pilot intervention should only take place if it is clearly evident that pilot action is required.

7.16.3. When ready to turn off the runway, the Captain should disconnect the autopilot and taxi clear of the runway. Be aware of the boundary markings of the sensitive area.

7.17. FAILURES AND ASSOCIATED ACTIONS

7.17.1. In general, there are three possible responses to the failure of any system, instrument or element during the approach.

- **a) CONTINUE** the approach to the planned minima.
- **b) REVERT** to higher minima and proceed to a new DH (above 1000 ft)
- **c) GO AROUND** and reassess the capability.

7.17.2. The nature of the failure and the point of its occurrence will determine which response is appropriate.

7.17.3 As a general rule, if a failure occurs above 1000ft AGL the approach may be continued reverting to a higher minimum, provided the appropriate conditions are met.

7.17.4. Below 1000ft AGL as a principle, the occurrence of any failure implies a go-around. Another approach may then be undertaken to the appropriate minima for the given condition. It has been considered that below 1000ft, not enough time is available for the crew to perform the necessary switching, to check system configuration and limitations and brief for minima, however the PIC has the authority to continue the approach if he believes that he has enough time to change the minima and the briefing is already covered for such happenings.

7.18. ABNORMAL PROCEDURES

7.18.1. INCAPACITATION

a) In the event of the captain's incapacitation on a CAT II/III approach after passing IAF, the First Officer is authorized to continue the approach provided the auto approach and auto land system performance is satisfactory.

b) Incapacitation action should be delayed until after completion of the landing run, unless the incapacitated pilot is obstructing the flight controls.

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7.18.2. AUTOPILOT STATUS CHANGES

Refer to FCOM and FCTM.

7.18.3. AUTOPILOT DISCONNECT

In the event of a total autopilot disconnect before Alert Height or Decision Height, an immediate manual Go-Around should be initiated. Below Alert Height or Decision Height, the landing may be continued if the required visual conditions exist.

7.18.4. ENGINE FAILURE

a) For B757/767 aircraft CAT II/III approaches mustn't be planned with one engine inoperative but may be continued for CAT II/III approach and auto land if engine failure occurs after the automatic engagement of the autopilot below 1500 feet RA and landing flap has been selected.

b) For A350, B787/777 aircraft CAT II/III approach and auto land with one engine inoperative may be planned if PIC believes that it is the safest course of action.

NOTE: The above procedure doesn't apply for ETOPS flights.

c) If the above conditions listed on (a & b) are not met or if unable to accomplish CAT II/III approach and Autoland execute missed approach and proceed to alternate airport, if the reported visibility/RVR at destination is CAT I or better, a single engine approach procedure down to CAT I minima may be flown.

7.19. LOW VISIBILITY ATC PROCEDURES

7.19.1. At Airfields authorized for CAT II/III operations when RVR falls below 550 meters or when the cloud base falls below 200ft, or when the weather, is expected to deteriorate rapidly, "Low Visibility Procedures" are put into operation. This means that the runway environment and the ILS sensitive areas are protected to avoid localizer and/or glide slope disturbance by intruding vehicles or aircraft. Approach separation standards are increased to 8-10 nm to allow time for the preceding aircraft to clear the sensitive area before the next landing aircraft passes the final approach fix.

7.19.2. This inevitably leads to delay and extended holding, which should be allowed for in the fuel planning stages. Sensitive area boundaries, known as reiffer lines, extended up to 137 meters either side of runway centerline.

7.19.3 Flight crew shall have RVR reporting for approach and landing operations to be conducted with any RVR minima less than 800meters. A conversion of meteorological visibility to RVR is not used to establish any required approach and landing RVR minimum less than 800metres.

7.19.4 CAT II/III holding point marker boards, situated clear of the sensitive area, are clearly marked and may be supplemented by flashing yellow lights. After landing, aircraft should clear the runway without delay and pilots should look out for the flashing yellow lights to determine whether they have departed the sensitive area. When taxing for take-off, care must be exercised not to go beyond the CAT II/III holding marker board unless cleared to enter the runway for take-off. Special CAT II/III instructions for ground movements of aircraft are often found on the Jeppesen chart (10-1 series).

7.20. LOW VISIBILITY TAXI PROCEDURE

7.20.1. The centerline taxiway lighting with its reduced spacing (generally 30m, but reduced to 15m to 7m on bends), facilitates taxiing in, low visibility. Do not exceed 10kts during taxiing and the First Officer should make full use of taxi charts and call out ground speeds when indicating more than 10Kts to avoid the risk of possible incursion. Remember that in poor visibility, ground equipment, aircraft wingtips and tails may not be as readily seen as the taxiway lighting.

a) Be fully aware of other aircraft taxiing in your vicinity.

b) Observe CAT II/III holding point markings/lights.

c) If at any time you are unsure of your ground position or miss a turn, advise ATC immediately.

7.20.2 Taxiing below RVR of 75 m or less by visual guidance only is not allowed.

7.21. LOW VISIBILITY TAKE OFF (LVTO) PROCEDURE

7.21.1. The captain shall carry out all low visibility takeoffs.

7.21.2. Confirm that the aircraft is lined up on the runway centerline lights and not the edge lights.

7.21.3. Check that the number of centerline lights is consistent with the reported RVR.

7.21.4. Make sure ILS is tuned & identified, and the localizer is centered before takeoff.

7.21.5. Use centerline lights or markings for directional control during the takeoff roll and stopping after a rejected takeoff. As speed increases, the streamlining effect of these improves, and directional control becomes easier.

7.21.6. Takeoff minima stated in Jeppesen chart 10-9A/20-9A category C & D is applied for all low visibility takeoffs.

7.21.7. Takeoff shall be rejected if visual reference is lost below 80kts, 100kts for Airbus, use center line light bumps and Localizer deviation indicator to keep the A/C center.

7.21.8. Use of assumed temperature or de-rate is authorized when low visibility procedures are in effect.

7.21.9. For LVTO the following instruments are required apart from the MEL; Windshield Wipers for both the Captain and First officer, Window heat system, Antiskid system, Thrust Reverser, ILS Receiver as a minimum.

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7.21.10. Weather conditions RVR 500m (1600) or greater, Touchdown RVR data is required and controlling. However, it can be substituted by MIDPOINT RVR if the touchdown RVR is inoperative. All the other RVR data is advisory.

7.21.11. When weather conditions are below RVR 500m (1600ft) down to 75m (300ft), where only two RVR sensors are available, both are required and controlling. Where three RVR sensors are installed on the takeoff runway, the TDZ, MIDPOINT & ROLLOUT RVR values are controlling when operational. However, the failure of any one RVR will not affect operations, provided the remaining two RVR sensors report values at or above the appropriate minima.

7.21.12. Furthermore, the pilot assessment can replace RVR/Visibility value for the first part of the takeoff roll which is the TDZ RVR, provided that the TDZ is inoperative and the rest of RVR reports meet the minimum required 150m RVR value, or the TDZ RVR report can be a minimum of 125m RVR and lights spacing of centerline of the runway shall be 15m or less and HIRL shall be 60M or less.

7.22. APPROVED AIRFIELDS FOR LVP OPERATIONS

All Runways approved for All Weather Operations will have a Low Visibility Approach Chart in the Jeppesen Airways Manual.

7.23. PRACTICE, EXPERIENCE AND RECENCY REQUIREMENT

7.23.1. EXPERIENCE FOR REDUCED VISIBILITY APPROACH / AUTOLAND

a) After successful completion of low visibility simulator training the PIC or the trainee captain has to fly a minimum of 20 sectors on the type including line flying under supervision (Route Check), to practice CAT II/III approaches in the actual aircraft.

b) The following guidelines apply to the practice of Cat II/III procedures in line service.

   I. When carrying out a practice low visibility approach and auto land using a non-guarded facility (Low Visibility Procedures not in force) or using a CAT I facility, raw data and visual cues must be closely monitored to ensure that an unsafe situation does not develop.

   II. The practice of CAT II/III approach must be confined to "good weather conditions, (better than ceiling 600 ft/visibility 3200 m). In conditions lower than this, CAT I procedures must be observed. An auto land may be executed following a CAT I approach observing the safeguards in (1) above.

c) At the captain's discretion, when conditions are favorable, an automatic approach/ landing with auto the throttle disconnected below 1500ft RA can be practiced, with or without an autopilot disconnect below 100ft RA.

d) A minimum of 5 CAT II/III approaches must be practiced before executing actual CAT II/III approach.

7.23.2. PILOT IN COMMAND EXPERIENCE

SECTION 2.20 ALL WEATHER OPERATIONS

a) Until a PIC has 15 sectors performing PIC duties in the aircraft type (which include 5 approaches to landing using CAT I or CAT II procedures), he/she may not plan for or initiate an instrument approach when the DH or MDA is less than 100m (300 feet) or the visibility is less than 1.5km. (ECARAS 8.10.1.29)

NOTE: The PIC can fly CAT I or II operations after he has flown 15 sectors as a PIC on type. However, he can fly CAT I up to his 15 sectors flight with the above limitations.

b) Until a PIC has 20 sectors performing PIC duties in the aircraft type (which include 5 approaches to landing using CAT III procedures), he/she may not plan for or initiate an instrument approach when the DH or MDA is less than 30m (100 feet) or the visibility is less than 350m RVR (1200 feet). (ECARAS 8.10.1.29)

NOTE: The PIC can fly CAT III Operation after he has flown 20 Sectors as a PIC on type.

7.23.3. FIRST OFFICER EXPERIENCE

40 sectors as a First Officer in CAT II/III approved aircraft after completion of final line check on type.

7.23.4. RECENCY REQUIREMENT

a) Once authorized to operate to Low Visibility Minimum, it is the individual pilot's responsibility to maintain his proficiency by carrying out a minimum of 5 CAT II/III approaches, actual or practice, between simulator checks.

b) Pilots are also responsible for recording the 5 practice/actual CAT II/III approaches on their personal low visibility approach record card (LARC). At least one such approach shall be carried out every 60 days.

c) Low Visibility Operations will be part of each simulator recurrent / proficiency check.

d) At least one LVTO to the lowest applicable minima shall be flown during the conduct of proficiency check.

7.24. CONTINUOUS MONITORING

7.24.1. In order to detect any undesirable trends before they become hazardous, operations must be continuously monitored by using flight crew reports.

7.24.2. The following information must be retained for a period of 12 months:

a) The total number of approaches, by airplane type, where the airborne equipment was utilized to make satisfactory, actual or practice approaches to the applicable CAT II/III minima; and

b) Reports of unsatisfactory approaches and/or auto lands, by aerodrome and airplane registration, in the following categories:

I. Airborne equipment faults;
II. Ground facility difficulties;
III. Missed approaches because of ATC instructions; or
IV. Other reasons (specify)

7.24.3. The following procedure is established to monitor the performance of the automatic landing system of each aircraft.

a) Flight crew must submit AUTO APPROACH AND AUTOLAND PERFORMANCE REPORT for all practice/actual CAT II/III approaches. Every practice or actual approach using Low Visibility (CAT II/III) procedures is to be recorded by annotation on the Airplane Flight Records as A/L (P) or A/L (A) in the box adjacent to the right of NT for the sector concerned.
b) All unsatisfactory auto lands shall be recorded in the maintenance log;
c) Upon receiving such reports, the concerned Chief Pilot must communicate with Flight Operations Quality Assurance to identify the probable cause by using data from FDM;
d) The respective Chief Pilot will declare the aircraft limitation until satisfactory performance is assured.
e) Following appropriate action, the performance of auto land must be rechecked for satisfactory performance and AUTO APPROACH AND AUTOLAND PERFORMANCE REPORT shall be submitted by the flight crew.

7.25. CAT II/III APPROACH BRIEFING CHECKLIST FOR B737

7.25.1 Before commencing a CAT II/III approach the crew will check and brief the following:

a) Both pilots are CAT II/III qualified and current
b) PIC shall be the PF
c) Aircraft maintenance status (Approach Category Status Placard)
d) Landing weight must be at or below maximum certified landing weight
e) B737-800 verify that the autoland status advisory messages are not shown
f) Destination and alternate weather
g) Use of CAT II/III approach chart
h) Runway status, CAT II/III, surface condition, use of auto brake, taxi chart and exits
i) Fuel status and action after go around
j) Radio altimeter – CAT II/III minima

SECTION 2.20 ALL WEATHER OPERATIONS

k) Use of anti-ice and exterior lights

l) Request ATC for CAT II/III approach*

m) Flap 40 landing is recommended, set to Vref + 5kts only

n) Start APU at 10,000ft AGL (FL100)

o) Adjust seats

* Not required for practice approach, if a decision is made to cancel the autoland for personal preferences, it should be done before 400ft RA prior to auto-trimming.

7.26. CATII/III AIRCRAFT REQUIREMENTS FOR B737

TABLE 2

REQUIRED EQUIPMENT	CAT II	CAT IIIA
Autopilot	2	2
ILS Receiver	2	2
Radio Altimeter	2	2
Decision Height Indicator	2	2
IRS	2	2
Autothrottle	1	1
Independence source of electrical power	2	2
Hydraulic system	2	2
Windshield wipers	With	With
Control column stab trim switches	With	With
Display source	2	2
TR unit	2	2
Bus Tie Breaker	Closed	Closed

7.27. AUTOLAND LIMITATION FOR B737

TABLE 3

WIND DIRECTION	WIND COMPONENT
Head wind	25kts
Cross wind	20kts

SECTION 2.20 ALL WEATHER OPERATIONS

Tail wind	10kts
Max G/S angle	3.25°
Min G/S angle	2.5°
Flaps	30 or 40 degrees
No reported wind shear condition	

7.27.1. B737-700 Disengage the autopilot after touchdown. Control the airplane manually.

7.23.4. B737-800 Disengage the autopilot before turning off the runway.

7.23.5. AFDS rollout mode performance cannot be assured when used on contaminated runways. Therefore, sufficient visual cue is required during rollout. If an Autoland is accomplished on contaminated runway, the pilot must be prepared to disengage the autopilot and take over manually should rollout directional control become inadequate.

7.23.6. For approach and Go Around alerts refer to SOP.

7.23.7. For Category III approach, the maximum crosswind limitations at touchdown is 15 kts or AFM's limitations whichever is more restrictive.

7.28. CAT II/III APPROACH BRIEFING CHECKLIST FOR B757/767

7.28.1. Before commencing a CAT II/III approach the crew will check and brief the following.

a) Both pilots are CAT II/III qualified and current
b) PIC shall be the PF
c) Aircraft maintenance status (Approach Category Status Placard)
d) Landing weight must be at or below maximum certified landing weight
e) Destination and alternate weather
f) Use of CAT II/III approach chart.
g) Runway status, CAT II/III, surface condition, use of auto brake, taxi chart and exits
h) Fuel status and action after go around
i) Alternate airport on ROUTE 2
j) Radio altimeter – CAT II/III minima (For CATIIIB with DH set 20'RA)
k) Altimeter orange bug – CAT I minima
l) Use of anti ice and exterior lights
m) Request ATC for CAT II/III approach*
n) Start APU at 10,000ft AGL (FL100)
o) Check ASA

SECTION 2.20 ALL WEATHER OPERATIONS

p) Adjust seats

* Not required for practice approach

7.29. CAT II/III AIRCRAFT EQUIPMENT REQUIREMENTS FOR B757/767

TABLE 4

REQUIRED EQUIPMENT	CAT II ASA LAND 2	CAT IIIA ASA LAND 2	CATIIIB ASA LAND 3
Autopilot	2	2	3
EADI displays from separate symbol generators	2	2	2
ILS Deviation	2	2	2
ILS Receiver	2	2	3
Radio Altimeter	2	2	3
Autoland Status Annunciators	1*	2	2
Decision Height Indication	2	2	2
Inertial Reference Unit (in NAV mode), associated with engaged autopilots	2	2	3
Autothrottle System	None	None	With
Independent Source of Electrical Power	2	2#	3**
Hydraulic System	3	3	3
Rollout guidance	None	None	With
Windshield Wipers	With	With	With
Control Column stab trim switches	Captain's	Captain's	Captain's

NOTE 1: * First officer ASA.

NOTE 2: ** APU must be operational for CAT IIIB

NOTE 3: # At least two generators should be available

NOTE 4: All EICAS warning messages except autopilot disc, approach can be continued.

NOTE 5: If rollout isn't engaged do not autoland.

NOTE 6: If flare isn't engaged do not autoland.

SECTION 2.20 ALL WEATHER OPERATIONS

7.30. AUTO LAND LIMITATIONS FOR B757/767

TABLE 5

Wind Direction	component
Head wind	25kts
Tail wind	10kts
Cross wind	25kts *
Max G/S angle	3.25 degrees
Min G/S angle	2.5 degrees
Flaps	25 or 30 degrees (with one or both engine operating)
No reported wind shear condition	

*** AFDS rollout mode performance cannot be assured when used on contaminated runways. Therefore, sufficient visual cue is required during rollout. If an autoland is accomplished on a contaminated runway, the pilot must be prepared to disengage the autopilot and take over manually should rollout directional control become inadequate.**

***** For Category III approach, the maximum crosswind limitations at touchdown is 15 kts or AFM's limitations whichever is more restrictive.

7.31. CAT II/III APPROACH BRIEFING CHECKLIST FOR B777/787

7.31.1. Before commencing a CAT II/III approach the crew will check and brief the following.

a) Both pilots are CAT II/III qualified and current
b) PIC shall be the PF
c) Aircraft maintenance status (Approach Category Status Placard)
d) Landing weight must be at or below maximum certified landing weight
e) Destination and alternate weather
f) Use of CAT II/III approach chart
g) Runway status, CAT II/III, surface condition, use of auto brake, taxi chart and exits
h) Fuel status and action after go around
i) Alternate airport on ROUTE 2
j) Radio altimeter – CAT II/III minima (For CAT IIIB use DH 20'RA as applicable)

SECTION 2.20 ALL WEATHER OPERATIONS

k) Use of anti-ice and exterior lights
l) Request ATC for CAT II/III approach*
m) Adjust seats

* Not required for practice approach

NOTE 1: * refers to: At least F/O side ASA must be serviceable.

NOTE 2: All EICAS warning messages except autopilot disc, approach can be continued.

NOTE 3: If rollout isn't engaged do not autoland.

NOTE 4: If flare isn't engaged do not autoland.

NOTE 5: If speed brake extended above alert height (200 feet) retract the speed brake & continue the approach.

7.32. AUTOLAND LIMITATIONS FOR B787

TABLE 6

Wind Direction	component
Head wind	25kts
Tail wind	15kts
Cross wind	25kts *
Max G/S angle	3.25 degrees
Min G/S angle	2.5 degrees
Flaps	20, 25 or 30 degrees (with one or both engines operating)
No reported wind shear condition	

7.33. AUTOLAND LIMITATIONS FOR B777

TABLE 7

Wind Direction	component
Head wind	25kts
Tail wind	15kts
Cross wind	25kts *

Max G/S angle	3.25 degrees
Min G/S angle	2.5 degrees
Flaps	20 or 30 degrees (with one or both engines operating)
No reported wind shear condition	

*** AFDS rollout mode performance cannot be assured when used on contaminated runways. Therefore, sufficient visual cue is required during rollout. If an autoland is accomplished on a contaminated runway, the pilot must be prepared to disengage the autopilot and take over manually should rollout directional control become inadequate.**

***** For Category III approach, the maximum crosswind limitations at touchdown is 15 kts or AFM's limitations whichever is more restrictive.

7.34. CAT II/III APPROACH BRIEFING CHECKLIST FOR A350

3.34.1. Before commencing a CAT II/III approach the crew will check and brief the following.

a) Both pilots are CAT II/III qualified and current
b) PIC shall be the PF
c) Aircraft maintenance status (Auto- approach capability)
d) Destination and alternate weather
e) Use of CAT II/III approach chart
f) Runway status, CAT II/III, surface condition, use of auto brake or BTV, taxi chart and exits
g) Fuel status and action after go around
h) Alternate airport on SEC
i) Radio altimeter – CAT II/III minima (For CAT IIIB use DH 20'RA as applicable)
j) Use of anti-ice and exterior lights
k) Request ATC for CAT II/III approach*****
l) Adjust seat

***** Not required for practice approach

7.35. CAT II/III EQUIPMENT FOR A350

Prior to initiating an ILS CAT II/III approach with auto land, the crew should check that the equipment or functions listed in the following table are available:

SECTION 2.20 ALL WEATHER OPERATIONS

TABLE 8

Required Equipment not Monitored by FG	**LAND2**	**LAND3 SINGLE**	**LAND3 DUAL**
AP pb, or captain sidestick pushbutton, or F/O sidestick pushbutton	2	2	2
Autoland light	1	1	1
AP OFF alert	Available	Available	Available
DH Indication	1 for PM	1 for PM	1 for PM
Attitude indication on PFD	2	2	2
Attitude indication on standby instrument	1	1	1
Windshield Wiper, of rain repellent	1 for PF	1 for PF	1 for PF
Windshield heating	1 for PF	1 for PF	1 for PF

7.35.1. In addition, the FG monitors other required equipment, and computes the approach capability. The loss of a required equipment, monitored by the FG, is announced by an assigned alert (e.g. NAV RA SYS A+B+C FAULT), or by the AUTO FLT APPROACH CAPABILITY DOWNGRADED alert.

The following table presents equipment, monitoring by the FG, required to perform an ILS CAT II or CAT III Approach with Autoland:

SECTION 2.20 ALL WEATHER OPERATIONS

TABLE 9

Required Equipment Monitored by FG	LAND2	LAND3 SINGLE	LAND3 DUAL
AP	1 AP engaged	1 AP engaged	2 Aps engaged
A/THR	Not required	Active	Active
PRIM	1	1	1
SEC	SEC 2 or SEC 3	SEC 2 or SEC 3	SEC 2 or SEC 3
North reference	MAG	MAG	MAG
Slat and Flap Systems	(SLAT SYS 1 and FLAP SYS 1) or (SLAT SYS 2 and FLAP SYS 2) *(4)*	(SLAT SYS 1 and FLAP SYS 1) or (SLAT SYS 2 and FLAP SYS 2) *(4)*	(SLAT SYS 1 and FLAP SYS 1) or (SLAT SYS 2 and FLAP SYS 2) *(4)*
Elevator	1	1	1
Rudder actuation, rudder position and rudder trim	Available	Available	Available
Normal or Alternate law	Available	Available	Available
Hydraulic circuit	1	1	1
AFS CP, or CAPT AFS CP backup, F/O AFS CP backup	1	1	1
PFD display	2	2	2
FWS, including the audio function	1	1	2
FCDC	1	1	2
Braking Systems (BCS)	Not required	Not required	1
Symmetrical braking	Not required	Not required	Available
Antiskid	Not required	Not required	Available and ON
Nosewheel Steering	Not required	Not required	Available
Ground spoilers	Available	Available	Available
Radio Altimeter	1	2	2
MMR tuned with ILS frequency	2	2	2
ADR	2	2	2

SECTION 2.20 ALL WEATHER OPERATIONS

TABLE 10

Required Equipment Monitored by FG	LAND2	LAND3 SINGLE	LAND3 DUAL
IRS in NAV mode	2	2	3
Angle of Attack	2	2	3
Sideslip angle	2	2	2
Engine	1	1	1
AC busbar	*(3)*	*(3)*	4
Supply of AC busbars	*(3)*	*(3)*	2 independent AC busbars
Supply of DC busbars	*(3)*	*(3)*	2 independent DC emergency busbars

(1) In LAND3 DUAL, the nosewheel steering must be to receive AP guidance orders.

(2) In addition, the tuning of both ILS must be consistent.

If the turning of both ILS in MMR is not consistent, the **NAV LS TUNING DISAGREE** alert triggers.

(3) In accordance with the failure, and the associated equipment loss, capability downgrade may occur.

(4) The approach and landing capability downgrades to APPE1 if all of the following are applicable:

- A failure to the loss of the Slat system (i.e. F/CTL SLAT SYS 1+2 FAULT) or to the loss of the Flap system (i.e. F/CTL Flap SYS 1+2 FAULT)
- The configuration is not set CONF 3 or CONF FULL.

NOTE: If rollout or flare isn't engaged do not Autoland.

SECTION 2.20 ALL WEATHER OPERATIONS

7.36. AUTOLAND LIMITATIONS FOR A350

TABLE 11

Wind Direction	component
Head wind	40kts
Tail wind	10kts
Cross wind	30kts
Max G/S angle	3.5 degrees
Min G/S angle	2.5 degrees
Flaps	CONF 3 or CONF FULL (with one or both engines operating)
No reported wind shear condition	

NOTE: Wind limitation is based on the surface wind reported by the tower. If the ND indicates a current wind speed that exceeds the above-noted Autoland limitations, but the tower reports a surface wind speed within limitations, then the flight crew can perform an automatic landing. If the tower reports a surface wind speed beyond the limits, the flight crew can only perform a CAT I or CAT II automatic approach without Autoland.

- For Category III approach, the maximum crosswind limitations at touchdown are 15 kts or AFM's limitations whichever is more restrictive.

7.37 FAILED OR DOWNGRADED EQUIPMENT-EFFECT ON LANDING MINIMUMS

TABLE 12

Failed to Downgraded Equipment	Effect on Landing Minimums				
	CAT IIIB	CAT IIIA	CAT II	CAT I	Non-Precision
ILS stand-by transmitter	Not allowed		No effect		
Outer marker	No effect if replaced by published equivalent position				Not applicable
Middle Marker	No effect				No effect unless used as MAP
TDZ RVR assessment system	May be temporarily replaced with Midpoint RVR if approved by the			No effect	

SECTION 2.20 ALL WEATHER OPERATIONS

<table>
<tr><td></td><td colspan="3">State of the Aerodrome. RVR may be reported by human observation.</td><td></td></tr>
<tr><td>Midpoint or Rollout RVR</td><td colspan="4">No effect</td></tr>
<tr><td>Anemometer for runway in use</td><td colspan="4">No effect if other ground source available</td></tr>
<tr><td>Ceilometer</td><td colspan="4">No effect</td></tr>
<tr><td>Approach Lights</td><td colspan="2">Not allowed for operations with DH> 50ft</td><td>Not allowed</td><td>Minimums as for NALS</td></tr>
<tr><td>Approach lights except the last 210m</td><td colspan="2">Not effect</td><td>Not allowed</td><td>Minimums as for BALS</td></tr>
<tr><td>Approach lights except the last 420m</td><td colspan="2">Not effect</td><td>Not effect</td><td>Minimums as for IALS</td></tr>
<tr><td>Standby power for approach lights</td><td colspan="4">No effect</td></tr>
<tr><td>Whole runway light system</td><td colspan="3">Not allowed</td><td>Day: Minimums as for NALS

Night: Not allowed</td></tr>
<tr><td>Edge light, threshold lights and runway end lights</td><td colspan="4">Day only

Night: Not allowed</td></tr>
<tr><td>Centerline lights</td><td colspan="2">Day: RVR 300m

Night: Not allowed</td><td>Day:

RVR 300m

Night:

RVR 550m</td><td>No effect</td></tr>
<tr><td>CL lights spacing increased to 30m</td><td>RVR 150m</td><td colspan="3">No effect</td></tr>
<tr><td>TDZ lights</td><td>Day: RVR 200m

Night: RVR 300m</td><td colspan="2">Day: RVR 300m

Night: RVR 550m</td><td>No effect</td></tr>
<tr><td>Standby power for runway lights</td><td colspan="3">Not allowed</td><td>No effect</td></tr>
<tr><td>Taxiway light system</td><td colspan="4">No effect – except delays due to reduced movement rate</td></tr>
</table>

SECTION 2.20 ALL WEATHER OPERATIONS

7.38. APPROACH LIGHT SYSTEMS

TABLE 13

Class of Facility	Length, Configuration and Intensity of Approach Lights
FALS (Full Approach Light System)	**ICAO:** CAT I precision approach lighting system (HIALS 720m or more), distance coded centerline, barrette centerline.
IALS (Intermediate Approach Light System)	**ICAO:** simple approach lighting system (HIALS 420m-719m), simple source, barrette.
BALS (Basic Approach Light System)	Any other approach lighting system (HIALS 210m-419m, MIALS or ALS of 210m or more).
NALS (No Approach Light System)	Any other approach lighting system (HIALS, MIALS or ALS less than 210m) or no approach lights.

* * * *

SECTION 2.21 ACARS

SECTION 2.21 ACARS

SECTION 2.21 ACARS

1.0 PURPOSE

The purpose of this section is to provide standard operating procedures as well as area of responsibility, guidance, information and the steps to be followed for uplink and downlink of LOAD SHEET, WIND DATA, POSITION REPORT, FLIGHT PLAN ROUTE and ATC COMMUNICATIONS to and from the aircraft via the ACARS for the flight crew to increase utilization of ACARAS to maximum level in cost effective manner.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- IOCC
- Line Maintenance
- Planning and Engineering

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.02

4.1 CLC shall use ACARS to uplink the final load sheet to the cockpit whenever the ACARS communication with the aircraft is available in working conditions.

4.2. Flight Watch shall monitor aircraft in operation using ACARS position reports primary.

4.3. It is ET's standard procedure to uplink last minute flight plan changes to the aircraft FMC via ACARS.

4.4. Upon request from flying cockpit crew the dispatch office shall uplink data such as route, wind, weather and other operational data.

SECTION 2.21 ACARS

5.0. DEFINITION

ACARS: Aircraft communication addressing, and reporting system is a system that is installed on aircraft for up linking and downlinking various data. The system interacts with the ground system via VHF and satellite communication data transmission systems.

AM: ACARS MANAGER is communication system that is used to process messages and transmit them from DM to aircraft and from the aircraft to DM and other flight operation ground systems through the ACARS data link service provider network.

DM: DISPACH MANAGER is a flight planning system used to generate operational flight plan consisting of the route data (FPN) that the pilot is going to use to navigate the aircraft for a given flight. In addition, DM down loads the predicted wind data for the flight route.

DOWNLINK: Transmit data from aircraft to the ground system.

FDE: Flight deck effect.

FRM: Fault Reporting Manual.

UPLINK: Transmit data from ground system to the aircraft.

6.0 RESPONSIBILITY

6.1 The Flight Crews are responsible to do the following:

6.1.1. Use ACARS data link communication at all time of the flight.

6.1.2. Report maintenance faults, observed technical remarks, Emergency Situation.

6.1.3. Consult ground staff (e.g. MCC, FMT) for any appropriate action during maintenance fault in flight.

6.1.4. Provide fault codes in all the downlink free text messages if applicable by referring the Fault Reporting Manual (FRM) located in the flight deck.

6.1.5. Pilots report irregularity (turn back, re-route, diversion) or Emergency Situations such as Hijack/7500 report, Mayday repot for decision making through free text to ADDAMET SITA address.

7.0 PROCEDURE

7.1 FLIGHT CREW SHALL:

7.1.1 Set VHF #3 in data mode immediately after the aircraft is powered on and leave it in data mode at all times.

7.1.2. Shall initialize the ACARS system immediately with the flight in the ACARS /FMC CDU by entering the flight number, origin and destination.

7.1.3. Report any ACARS device including communication devices malfunction by writing on

SECTION 2.21 ACARS

maintenance logbook and Captain's report.

7.1.4. Request the dispatchers via VHF/HF radio or ACARS free text to uplink the route data (FPN)

7.1.5. Monitor and check for uplink data Weather, NOTAM, Flight Plan, Load Sheet, Performance,

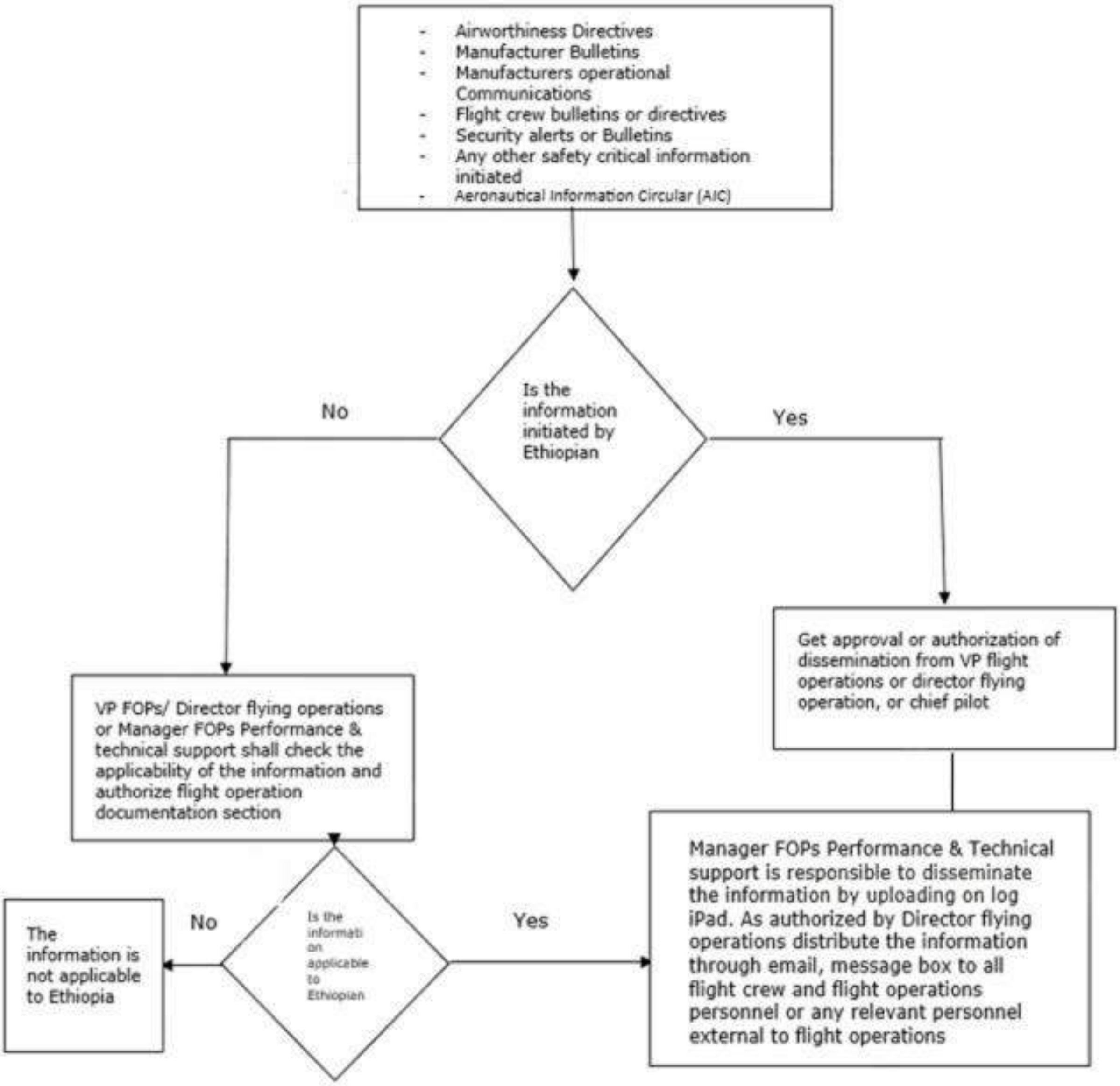
7.1.6. Request all necessary data using ACARS like Route, Weather, NOTAM.

a) Upon receiving the Route Data/Weather/NOTAM,
b) Shall load the received route data into the FFMX by Pressing the ACCEPT button.
c) Shall fix the route data in the FMC with the pertinent SID and STAR waypoints.

7.1.7. request load sheet 30 minutes prior to ETD on the format applicable to each aircraft type.

* * * * *

APPENDIX 2.1


- Airworthiness Directives
- Manufacturer Bulletins
- Manufacturers operational Communications
- Flight crew bulletins or directives
- Security alerts or Bulletins
- Any other safety critical information initiated
- Aeronautical Information Circular (AIC)
Is the information initiated by Ethiopian
No
Yes
Get approval or authorization of dissemination from VP flight operations or director flying operation, or chief pilot
VP FOPs/ Director flying operations or Manager FOPs Performance & technical support shall check the applicability of the information and authorize flight operation documentation section
Manager FOPs Performance & Technical support is responsible to disseminate the information by uploading on log iPad. As authorized by Director flying operations distribute the information through email, message box to all flight crew and flight operations personnel or any relevant personnel external to flight operations
Is the informati on applicable to Ethiopian
No
Yes
The information is not applicable to Ethiopia

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CHAPTER 3 TABLE OF CONTENT

CHAPTER 3 TABLE OF CONTENT

SECTION 3.1 CREW DUTIES AND RESPONSIBILITIES – EMERGENCY

1.0. PUPOSE

The purpose of this crew duties and responsibilities during emergency procedure is to establish guidelines to flight crewmembers on how to execute their duties and responsibilities during emergency flight.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Crewmembers
- Traffic Controller

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.03

4.1. EMERGENCY FLIGHT MANAGEMENT

4.1.1. In all flight operations, the Captain designated by the Company is in command, regardless of what seat he is occupying or who is handling the flight controls. The sole exception to this rule is incapacitation of the Captain, in which case the sequence of command described under "Incapacitation" section takes precedence.

4.1.2. It is Ethiopian Airlines policy to make landing at the nearest airport when an engine fails or during engine fire.

4.1.3. It is Ethiopian airlines policy that no Ethiopian airlines airplane may takeoff at such a mass where there is no suitable landing aerodrome within 60 minutes at any point along the intended route unless that airplane can continue to a suitable landing aerodrome for safe landing.

4.1.4. Engine failure escape procedures are published in the Logipad (OPT) and shall be strictly adhered to in the event of an engine failure, either before or after V1. These procedures shall be executed in accordance with the individual aircraft Standard Operating Procedures (SOP), along with guidance from the Aircraft Flight Crew Training Manual (FCTM) or its equivalent documentation.

4.1.5. If an engine fails or is shut down on a two-engine airplane, landing shall be made at the nearest suitable airport considering safety only. Any airport of which the pilot has or can obtain the necessary information and which he considers safe under prevailing circumstances may be used.

4.1.6. If two or more suitable airports are at the same flying time from the position of the aircraft, the station having maintenance facility should have priority.

5.0. DEFINITION

N/A

6.0 RESPONSIBILITY

6.1. PILOT IN COMMAND

6.1.1. The decision to reject the takeoff is the sole responsibility of the captain.

6.1.2. The Pilot-in-command has a full responsibility to take appropriate measures during emergencies.

6.1.3. Each flight crewmember has the responsibility to inform/alert the pilot-in-command whenever unusual condition occurs.

6.1.4. Flight Crewmembers share responsibilities mentioned in this section of the Manual.

6.1.5. It is the Captain’s responsibility not to allow a flight to continue to any airport to which it has been released if in his opinion the flight cannot be completed safely, unless, in his opinion, there is no safer procedure. In that event, continuation toward that airport is an emergency situation.

6.1.6. For emergency authority of the captain refer FOM 1.5.

7.0 PROCEDURE

The Aircraft Operations Manual incorporates aircraft emergency procedures. This section contains general information applicable to all company aircraft covered in the Aircraft Operations Manual.

7.1. REJECTED TAKE-OFF

7.1.1. The decision to reject the takeoff is the sole responsibility of the captain.

7.1.2. A rejected takeoff is a maneuver performed during the takeoff roll to expeditiously stop the aircraft on the runway.

7.1.3. As airspeed approaches V1, the effort required to stop the aircraft on the runway for RTO approaches maximum. After V1, it may not be possible to stop the aircraft on the runway. The decision to reject a takeoff must be made prior to V_1 so that the maneuver can be initiated no later than V_1 and must be accompanied by the timely execution of the rejected takeoff maneuver procedures.

7.1.4. Prior to V_1, a takeoff shall be rejected in the event of engine failure, engine fire (fire warning), predictive windshear warning or if the aircraft is unsafe or unable to fly.

7.1.5. First Officers, when making the takeoff, shall not abandon control of the aircraft until the Captain announces, "REJECT" and makes a positive input on the controls.

NOTE: For A350, the Captain announces "STOP"

7.2. COMMUNICATIONS DURING EMERGENCY

The Captain must ensure that the flight crew, cabin attendants, supernumeraries, passengers, Air Traffic Control and company dispatch are promptly notified of essential information. There should be no reluctance to declare an emergency. Request for Assistance can be rescinded at any time and the emergency canceled later if appropriate.

7.3. MONITORING OF GUARD FREQUENCY (121.5) AND INFLIGHT BROADCAST PROCEDURES (IFBP) FREQUENCY

7.3.1. Constant monitoring of the guard frequency 121.5 MHz throughout the entire flight is mandatory. Under normal conditions VHF set no. 2 shall be tuned to 121.5 MHz.

7.3.2. Monitoring of 121.5 MHz is particularly important as follows;

a) On long-range over-water flights or on fights that require the carriage of an emergency locater transmitter (ELT), except during those periods when aircraft are carrying out communications on the VHF channels, or when airborne requirement limitations or flight deck duties do not permit simultaneous guarding of two channels;

b) If required by the applicable authorities, in areas or over routes where the possibility of military intercept or other hazardous situations exit.

c) If required by the applicable authorities, the appropriate common frequency used for inflight broadcast communication, e.g. Africa, designated airspace without ATC coverage. (Refer to Jeppesen for different airspace requirements).

d) In airspace without ATC coverage inflight broadcast procedures (IFBP) frequency may be used for inflight communication.

7.4. PASSENGERS BRIEFING DURING EMERGENCY

The Captain shall explain the situation in a calm, professional manner so as to instill confidence that crewmembers know exactly what they are doing. This is most important in preventing panic. The briefing should include instructions as appropriate depending on the circumstances.

7.5. RECOGNIZING AN EMERGENCY

7.5.1. Generally, the earlier an emergency is recognized and handled appropriately, the fewer hazards exist. Nearly all emergencies are dealt with more easily when they are recognized early.

7.5.2. Each crewmember shall be alert to unusual conditions and report them to the Captain. Such reports are essential in dealing effectively with a potential or actual emergency. Cabin attendants are to be encouraged in this regard. They are to report promptly any unusual sound, smell, and appearance of vibration of the aircraft. Should such a report concern an item that proves to be of little or no consequence, a flight crewmember should, nonetheless, give a full and considerate explanation to the cabin attendant making the report.

7.6. CREW COORDINATION DURING EMERGENCY

7.6.1. Standard verbiage, terminology, signals and verbal commands must be used for communication between flight crew and cabin crew during normal, abnormal and emergency situations. As in normal operation, effective action during an emergency depends on the crewmembers functioning as a team. The unexpected and critical nature of emergencies requires the disciplines which are the basis of effective crew action during an emergency.

7.6.2. When an emergency or abnormal happens from the cabin, one of the cabin crew will inform the flight deck immediately. Then cabin crew will start their action as per their CCOM in a deliberate and coordinated way from the flight deck. In an emergency it is important to establish communications without delay between the flight deck and cabin crewmembers. All crewmembers should be kept aware of the problem and the plan that follows as time and good judgment permit.

7.6.3. When an emergency occurs, the following items must be considered, in sequence:

a) FLIGHT MANAGEMENT

One pilot must fly the aircraft. Usually, this is the pilot flying at the time, but the Captain may elect either to fly the aircraft himself or to have the other pilot fly it. Assuming control of the aircraft does not relieve the Captain of the responsibility for directing crew action.

b) IDENTIFYING THE EMERGENCY

The crewmember who first recognizes the emergency should announce it in a firm, clear voice, for example, "Engine Failure," or "Wheel Well Fire." If the fire bell is ringing, it should be silenced promptly without command. The Captain should confirm the condition and then direct the required crew action.

NOTE: during critical phases of flight and when flying manually the pilot monitoring shall silence the fire bell.

c) EMERGENCY EVACUATION ASSIGNMENTS

The duties specified in the evacuation assignments are to be accomplished when ordered by the Captain. Each crewmember must be able to accomplish the duties specified in the evacuation assignments for his or her stations and also be familiar with the assignments of other crewmembers.

d) ABNORMAL SITUATIONS

Conditions like abnormal airframe vibration, abnormal landing configuration, etc… flying crew shall complete their procedures or prepare the A/C for approach and then communicate with the cabin crew as per the standard.

e) CHECKLIST OR MEMORY ITEMS

When a non-normal situation is evident, at the discretion of the Captain/Pilot-flying, both crewmembers systematically and without delay accomplish all recall items in their area of responsibility. When executing abnormal/emergency procedures specified in the type specific FCOM, flight crew should verbally confirm (dual response) before the execution of any critical aircraft system controls. Such procedures should at a minimum address:

I. engine thrust levers;
II. fuel master or control switches;
III. engine fire handles or switches
IV. engine or APU fire extinguisher switches or cargo fire arm switch;
V. IDG/CSD disconnect switch;
VI. Auto throttle arm switch.

f) USING THE CHECKLIST

The Captain/Pilot-flying calls for non-normal checklist when:

I. The flight path is under control,

II. The aircraft is not in a critical stage of flight,

III. All recall items are complete.

NOTE: Pilot monitoring, in two-man crew should first verify that each recall item has been accomplished. The checklist is normally read aloud during such verification, but the Pilot-flying is not required to respond except for items that are not in agreement with the checklist. Challenge and response items should be read aloud.

g) REVIEWING AND PLANNING

After completion of the procedures or as per the requirement of the specific checklist, the PIC shall brief the team leader cabin services the abnormal and/or emergency situations happened on the flight as per the below standard which is called TEST;

T: - Advise the type of emergency or abnormal situations (what exactly happened on the flight).

E: - Advise if evacuation is needed after landing or to wait for the PIC’s command.

S: - Advise if there are any special procedures to be followed (example if the A/C is not going to vacate the runway, A/C proceeding to remote stands because of the emergency, gear extension may be well ahead of the norm, engine vibration etc …).

T: - Advise time remaining for landing.

Then the PIC will make a PA with a reassuring voice that the emergency is handled as per the procedure and that they are continuing, diverting or returning back to the departure in ---- minutes and further information is given to the passengers after landing.

Then if time permits the company will be advised by the flying crew.

Before Approach all the information and procedures shall be reviewed. A plan for subsequent action should be prepared and thoroughly understood by each crewmember concerned.

* * * * *

SECTION 3.2 HIJACKING AND BOMB THREAT

1.0 PURPOSE

The purpose of this aerial piracy procedure is to establish guidelines to be followed when an aircraft is under armed threat during flight.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Cabin Attendants
- IOCC

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.03

4.1. UNLAWFUL INTERFERENCE

4.1.1. An aircrew that is being subjected to unlawful interference shall endeavor to notify the appropriate ATC unit, and if able pass any significant associated circumstances, any deviations from the current flight plan necessitated thereby in order to enable the ATC unit to give priority to the airplane.

NOTE: In any case, ATC units will endeavor to recognize any indication of unlawful interference and will respond promptly to requests by affected aircrews. Information pertinent to the safe conduct of the flight will continue to be transmitted, and necessary action will be taken to expedite the conduct of all phases of the flight.

4.1.2. When an aircraft is under armed threat during a flight, it is extremely important that all crewmembers adopt a manner and attitude that will avoid alarming or frightening the hijackers or the passengers. All crewmembers must remain calm regardless of circumstances and must convey an air of composure to others. The ability to remain cool, think clearly, and operate methodically requires the knowledge of what to do under given circumstances, and for this reason, the procedural guidelines detailed in this section are established.

5.0 DEFINITION

5.1. Hijackers fall into one of the following seven types:

5.1.1 **Criminals** who might use aircraft hijacking to extort money, to gain the release of prisoners, to escape justice and to avoid extradition. In almost all circumstances they are well armed and very determined.

5.1.2 **Mentally Imbalanced** who are usually seeking publicity. They are extremely dangerous and have to be handled with great care.

5.1.3 **Refugees** who are usually escaping from repressive regimes. They invariably value their own lives and respect the lives of others.

5.1.4 **Terrorists** are well organized and trained, armed and disciplined. They usually have a political aim. They likely to be fanatical and will pursue their aim aggressively. Terrorists ate likely to care little for their own personal safety. Many terrorist inspired hijacks have involved acts of extreme violence against passengers and crew. These hijackers may be actively assisted by the Authorities of State where they have ordered the aircraft to fly.

5.1.5 Deportees

5.1.6 Inadmissible Passengers

5.1.7 Adventurers

6.0 RESPONSIBILITY

6.1. Crewmembers are responsible to be on alert for any incident/accident or circumstances which might potentially develop in to a hijacking incident.

6.2. Cabin crewmembers are responsible to make frequent checks of the cabin during flight.

6.3. Cabin crewmembers are responsible to closely observe passengers for any kind of suspicious activity or security breaches in the passenger cabin which might affect the safety of the flight.

6.4. Pilot-in-command is responsible to comply with the required actions listed here under.

7.0. PROCEDURE

7.1. HIJACKING/ UNLAWFUL SEIZURE OF AIRCRAFT

7.1.1. GENERAL

a) The circumstances surrounding a hijacking/unlawful seizure of an aircraft are highly variable; hence it is not possible to provide specific information to flight crew. However, the safety of aircraft and its occupants must be of paramount consideration and any occurrence must be dealt in accordance with the captain’s judgment of the prevailing circumstances.

b) Unlawful seizure of interference with an aircraft is a crime and will be dealt with the police or security forces in the same manner as any crime of violence.

c) The captain should anticipate the police or security forces who have the necessary power of arrest and entry on premises and property without warrant, will begin to exercise their powers and their authority to control the future course of events as soon as the incident is reported. From this point, responsibility of the captain diminishes, and he adheres to the lawful instructions of the police or security forces to the extent that he considers it to be consistent with the safety of the passengers and crew.

d) Some hijackers may desire to die under spectacular circumstances. They may seem to be confused. They may fail or refuse to name a destination or persist in ordering the flight to a destination that it is impossible to reach. They may create highly unstable situations, changing orders as the flight progresses.

e) The crew should attempt to determine the hijacker's intended destination, a hijacker with no firm destination or a clearly impossible destination in mind may be considering suicide. This person creates a high-risk situation. A hijacker with a firm, reasonable destination in mind probably creates a situation of less immediate risk.

7.1.2. QUICK REFERENCE FOR FLIGHT CREW

a) Set the transponder to the appropriate code and/or relay the code through the radio in accordance with OM.

b) transmit all information gathered to ATC;

c) turn the seatbelt on to curtail passenger movement;

d) Instruct passengers through the public address system to remain seated and cooperate;

e) Keep flight deck door closed;

f) Use all reasonable measures to ensure hijackers don't access the flight deck;

g) Consider landing at the next available airport;

h) On the ground, maintain lockdown of the flight deck door. Do not open the door until corroboration from the authorities is received that it is safe to do so;

i) Consider pulling fire switches and disconnecting generators, thereby making the aircraft non-flyable. This substantially reduces the likelihood of the aircraft being forced back into the air, flown by on-board crew, and it precludes flight of the aircraft by hijackers with pilot training. This decision shall be determined by the circumstances and risk faced from the perpetrator/s, prioritizing the safety of the passengers, the crew and the general public.

j) Upon balancing the various aspects of the threat situation of the specific incident, it may, in the judgment of the PIC, be in the overall interest of passengers, crew, and others, for the flight crew may consider escaping from the aircraft. This decision shall also be determined by the PIC and the security personnel on the ground upon evaluation of the situation.

k) After landing, the captain should attempt to stall for time and try to negotiate the disembarkation of the passengers and cabin crew members

7.1.3. QUICK REFERENCE GUIDE FOR CABIN CREW:

a) Keep the flight deck door locked and secured at all times.

b) Remain calm and poised, comply with the initial demands of the hijackers, avoiding conflict and arguments'

c) Attempt to identify the type of hijackers and the type of weapons they hold, and they claim to hold; attention should be paid to accomplices who might not have disclosed themselves yet;

d) If hijacker insists on going into the cockpit advice the captain from the aft intercom Use the **alert call** and standard phraseology "**PLEASE OPEN THE COCKPIT DOOR IMMEDIATELY."**

e) Never open the flight deck door;

f) Keep passenger manifest and other flight related paperwork away from the hijackers;

g) Efforts should be made to keep a sense of normalcy in the cabin;

h) Establish effective communication with the hijacker to defuse emotion.

i) Refrain from taking any type of forceful action.

j) Restrict the amount of crew dealing with each hijacker to one or two;

k) Divert hijacker's attention by stating that the blame goes to local authorities if demands are not met;

l) Apply effective division of responsibilities;

m) If crew member encounter passenger requiring medical assistance locate medically qualified passengers and seek approval from the hijacker(s) to attend the sick passenger;

n) Keep passengers calm and quit;

o) Be aware of passengers who might take individual initiative against the hijackers;

p) Seek approval from hijackers to provide service to passengers;

q) Prepare for a prolonged incident by allocating cutlery and cups to passengers, negotiating for passenger's use of the toilets and rationing meals and beverages;

r) Maintain the hygiene and sanitary conditions of the cabin;

s) Deny alcohol to passengers and hijackers, unless the latter forcefully demand it;

t) On the ground negotiate with hijackers to request toilet and water service and fresh supplies of water and food;

u) Negotiate the release of the elderly, the sick and minors.

7.1.4. GUIDELINES IN DEALING WITH HIJACKERS

a) If information is received that suspected or declared hijacker is on-board take-off, the aircraft should be returned to the terminal. The crew will not attempt to evaluate or search suspicious persons. This will be done by trained security personnel.

b) Any crew member who gets the opportunity shall inform the Captain on the interphone system of a hijacker(s) presence in the cabin by using **alert call** and using the standard phraseology "**PLEASE OPEN THE COCKPIT DOOR IMMEDIATELY**" using the aft intercom.

c) If possible, the hijacker(s) should be discouraged from dealing with the captain and efforts shall be made to keep them from entering the flight deck. Always remember that there may be other hijackers, who remain seated among the passengers so as not to disclose themselves.

d) If the hijacker(s) is in the flight deck, crew should endeavor to communicate the situation to ATC. Generally, hijackers are aware of the need for communication although they may be suspicious and demand that communications are monitored. He should be informed that no resistance will be offered, although he should be instructed not to touch any aircraft controls, system or instruments.

e) If the hijacker(s) demands endanger the flight, the consequences of such demand should be explained in a manner which does not aggravate the situation.

f) Full account should be taken of the probability of the hijacker(s) being in a highly emotional state on mind. Pilots are advised to refrain from unnecessary conversation or actions which may irritate the hijacker(s).

g) If the hijacker doesn't speak or understand English, the captain should endeavor to land the aircraft using the pretext:

I. We do not have enough fuel to get to …

II. The airport is not large enough at … for this aircraft

III. The aircraft has a mechanical problem and we cannot proceed to …

IV. The weather is not good enough for a safe landing

V. What you are requesting is not possible. We must land first …

VI. We do not have the required information for that airport necessary to conclude a safe operation.

h) Crew members should not disagree with the hijacker(s); rather every endeavor should be made to relieve his anxiety in order to maintain an effective dialogue.

i) It is important to try and establish that the hijacker(s) does in fact have a weapon. Some hijackings have been attempted without weapon.

j) Crew members should not attempt to use force unless it is certain that such action will be successful.

k) After landing, the Captain should attempt to stall for time and try to negotiate the disembarkation of the passengers and flight attendants.

7.1.5. POST HIJACK

After any incident of this nature, crew is likely to face official enquiries and severe media and press no statement should be made without permission of the CEO/VP Flight Operations.

7.2. TRANSPONDER CODES

7.2.1. Pilots of hijacked aircrafts, to indicate the associated message should use the following signals. Air traffic controllers are familiar with this signal and their meanings. Refer to **TABLE 1** below.

7.2.2. Controllers shall acknowledge receipt of code 7500 or code 7700 by transmitting the following "(Call Sign), this is (name of facility) on 7500) (or 7700). What is your altitude (or heading)?" This will indicate to the pilot that the controller is aware of his signal and that it will be relayed to appropriate authorities. It is possible that the controller will not notice the code immediately and will not acknowledge receipt.

7.2.3. Pilots who decide to change from code 7500 to code 7700 should remain on code 7500 until three minutes have elapsed or until an acknowledgement of code 7500 has been received from the controller whichever is sooner.

7.2.4. Pilots who retract flaps after having squawked code 7700 should return to and remain on code 7500, if possible.

7.2.5. An aircraft squawking code 7700 and not in radio contact with the ground will be considered by ATC to have an in-flight emergency (in addition to hijacking), and the emergency procedure in appropriate ATC handbooks shall be followed. In these cases, notification of other concerned authorities shall include information that the aircraft was observed to have displayed hijack code as well as the emergency code.

7.2.6. Above all, cooperation and compliance with the hijacker's demands are essential in safeguarding the lives of many passengers. Again, the entire crew must remain calm, alert IOCC and ATC, comply, and use common sense.

TABLE 1

PILOT SIGANL	PILOT MESSAGE
CODE 7500. See Note 1	Being hijacked
CODE 7500 followed by CODE 7700. See Note 2	Situations appear desperate. Want armed intervention.
Leaves flaps down after landing.	Situation still desperate. Want armed intervention and aircraft immobilized.

Retracting flaps after having squawked Code 7700. Return to Code 7500. See Note 3	Leave alone – do not intervene.

7.3. THE INTEGRATED OPERATIONS CONTROL CENTER (IOCC)

7.3.1. The IOCC is the main coordination center for all requests, advice, and information in all flight emergency situations.

7.3.2. Duty managers will alert SITA and ATC so as to provide interference free communication when possible.

7.3.3. Duty managers will also alert appropriate company personnel and the police, requesting surveillance as other action.

7.4. SPECIAL EMPHASIS FOR CREW MEMBERS

Experience gained from the past hijackings indicates that there are certain things an air crewmember can do to reduce the threat to the safety of a flight under control of hijackers. Critical situations have developed which have caused serious concern for the safety of both passengers and crew. To each of those situations, certain common actions were taken by the aircrew and personnel on the ground, because communications were either sketchy or nonexistent, each party involved acted almost independently without knowledge of what the other party was doing or wanted done.

7.5. COMMUNICATION PROCEDURES

7.5.1. Probably the single most critical aspect of dealing with hijacking is communication. Crew procedures should cover every foreseeable situation. Each crewmember should be fully familiar with the standard hijack code signals so as to assure proper transmittal of messages in actual hijack situations.

7.5.2. Should an in-flight hijacking or diversion occur, to the extent permitted by circumstances, flight crewmembers should transmit as much information as possible to the ground. Ground stations have been advised not to reply to each transmission unless requested.

7.5.3. If minimum verbal communications are feasible, standard hijack signals should be transmitted on the air-ground frequency in use at the time.

7.6. FLIGHT CREW

Historically during past hijackings and diversions, flight crew especially Captains, have been called upon to make decisions with little or no information upon which to base the decision. Almost without exceptions, Captains have been successful in returning such situations to low key. The following is a special item for the crews if the hijackers happen to be in the cockpit. Pull voice recorder circuit breaker upon landing and leave it pulled until the aircraft arrives at its final destination.

7.7. BOMB THREAT

7.7.1. Upon receiving notification of the suspected placement of a bomb on board the aircraft, the company shall notify the Captain of the situation and advise him to take the emergency action he considers necessary under the circumstance. Upon receiving notification of a suspected bomb placement on his aircraft, the Captain shall follow the sequential steps in this section. This will afford the greatest structural strength and resistance to internal explosive force and reduce the risk of passenger and crew exposure.

7.7.2. When a bomb threat has been made with reference to a specific passenger's luggage, and the Captain is advised that the passenger and his luggage will be removed, the flight may continue without further inspection.

7.7.3. Any passenger who has given false information amounting to a sabotage threat may not be allowed to continue on the flight but should be turned over to local authorities. If local authorities refuse to accept custody, the passenger in question will be refused transportation onboard the aircraft.

7.8. SEARCH PROCEDURES-GROUND

In order to ensure a successful search of ground facilities in the event of a bomb threat, it is very important that these facilities be maintained in a neat and orderly manner. During day-to-day usage of a facility for example, the Flight Control offices, all items used in the area should have a designated place. It is recommended that even personal objects such as lunch boxes and brief cases be stored in specific locations. Orderly storage practice will enable supervisors or trained investigator to quickly detect anything foreign, such as a homemade bomb.

7.9. SEARCH TEAMS

An important phase of search procedures is the selection of search teams. For this purpose, it is better to rely on persons with continual access to the area under investigation, such as supervisors and other responsible employees, than officials such as police and bomb disposal technician who are unfamiliar with the normal arrangement.

7.10. COMPANY PROCEDURES

7.10.1. Upon receipt of warnings or threats directed at a controlled access area, company personnel shall:

a) Notify the area supervisor.

b) Notify area security personnel who shall prohibit further access to the area.

7.10.2. If no specific time element has been given, or if warning time permits, the responsible supervisor shall conduct a search of the area. When a decision to evacuate has been made, (personnel should be directed to remove all personal property at the same time evaluation does not give time to collect). The area should then be thoroughly searched by the appropriate search team.

7.11. UNATTENDED AIRCRAFT

7.11.1. Whenever aircraft are parked and unattended, the following precautions must be taken to prevent unauthorized access:

a) The aircraft shall be secured by closing all exit and access doors. Both fore and aft air stairs (if installed) shall be retracted when leaving an aircraft unattended overnight.

b) Loading bridges, passenger ramps, and service carts shall be removed or demobilized.

c) Open access areas such as wheel wells, engine access panels, engine inlets, and cargo and equipment compartments must be inspected after an aircraft has been left unattended. Qualified maintenance personnel prior to moving the aircraft should perform such inspection.

7.12. ATTENDED AIRCRAFT

7.12.1. Aircraft, while parked at the terminal or in gate areas, must also be protected from access by unauthorized persons even when attended.

7.12.2. The following precautions must be taken unless modified by airport or safety regulations:

a) Doors leading directly to passenger loading areas and passenger gates shall be equipped with locking devices and kept locked while not in active use.

b) The access doors to the ramp shall be locked while not in active use unless local ordinance or fire regulations prohibit such action.

7.12.3. All company employees granted access to company aircraft shall prominently display their identification badges.

7.12.4. Company employees shall be instructed to challenge all unidentified individuals and restrict their access to public areas only.

7.12.5. Except for flight crewmember, boarding and deplaning passengers, and identified aircraft service personnel, the ramp shall be kept free of sightseers, visitors, and other unauthorized persons. Polite but firm policing of the ramp is mandatory.

7.12.6. Company personnel shall supervise cleaning crew on board the aircraft.

7.12.7. The station, to prevent sabotage, shall control containers and commissary supplies of all kinds.

7.12.8. Vehicular movement on the ramp is to be prohibited except for company or contract vehicles necessary for servicing the aircraft.

7.12.9. During the times when no service is scheduled for a station, the concourse or waiting room is to be secured, if possible.

7.13. SPECIFIC THREAT

7.13.1. When the company receives a bomb threat directed against a particular aircraft or flight, the following actions shall be taken to determine whether any explosive, incendiary devices or weapons are aboard the aircraft involved.

a) Conduct a security inspection (aircraft search) on the ground before the next flight of the aircraft or, if the aircraft is then in flight, immediately after its next landing.

b) If the aircraft is being operated on the ground, advise the Captain to immediately submit the aircraft for a security inspection.

c) If the aircraft is in flight, advise the Captain to take the emergency action he considers necessary under the circumstances.

d) Upon receipt of information concerning a suspected act of aerial piracy, the company shall immediately notify appropriate aeronautical authorities and police agencies.

7.14. AIRCRAFT SEARCH PROCEDURE ON GROUND

7.14.1. The aircraft search procedure should be exercised whether the aircraft is parked or in operation on the ground. The primary goal is avoidance of casualties and property damage.

7.14.2. Notify the Captain so he can immediately submit the aircraft for security inspection.

7.14.3. Place suspected aircraft in a designated isolated area at least 100 meters from other aircraft, personnel, and facilities.

7.14.4. Tow or taxi the aircraft to a designated isolated area and remove passengers and personnel at least 100 meters from the aircraft, baggage and air freight location.

7.14.5. De-plane passengers and all of their personal effects.

7.14.6. Designate ground crews to immediately unload the aircraft and move baggage and cargo (air freight) 150 meters away from the aircraft but within the hot cargo area. Passengers are to leave the hot cargo area and proceed directly to the terminal.

7.14.7. The search shall be composed of technical personnel who are familiar with the suspected aircraft. For the protection of personnel, the search area should include blast and fragmentation protection between individual baggage search positions.

Separating these positions with barricades minimizes the number of personnel exposed to the potential hazards of an explosion during search operations. In no case should all the baggage and passenger from a threatened aircraft be brought together in one location.

7.14.8. One team on the inside and another team on the outside shall search the aircraft. Each team shall consist of a member of the disposal squad and a company representative familiar with the aircraft. Both teams shall follow an appropriate checklist to avoid duplication of their efforts.

7.14.9. With all aircraft equipment shutoff and the aircraft doors closed, the search crew should stand still inside the aircraft for a few seconds with their eyes closed to attempt to detect any mechanical timer noise within the aircraft. If no mechanical timer can be detected, the aircraft must be systematically searched for anything that does not belong to the aircraft. Search should begin at floor level in areas accessible the public and continue through less accessible sections of the aircraft.

7.14.10. Both searches will follow a checklist to the aircraft involved in order to complete a rapid and thorough search without duplication of efforts; these checklists are included in the following pages.

7.14.11. If a suspected item is located, it should not be touched. The Emergency Control Officer should be notified immediately so a qualified bomb disposal support unit may be requested.

7.14.12. Search personnel should take no further action except to evacuate the area around the aircraft to a minimum radius of 100 meters and request fire fighting support to stand by in the event of explosion and fire prior to the time the bomb disposal team declares the device safe.

7.14.13. After the devise is declared safe or after disposal the search must continue until the checklist has been completed to ensure that no other devises exist and aircraft cleared of any potential secondary device.

7.14.14. In the event of an explosion, provisions for first aid to injured personnel should be arranged first followed by established damage control procedures. After the effects of the explosion are under control, applicable search procedures should continue to ensure that no other explosive devices exist.

7.14.15. As each compartment of the aircraft is searched, it should be conspicuously marked with tape or some other suitable means to avoid duplication of effort.

7.14.16. When search has been completed, the aircraft must be secured against subsequent contamination until the next scheduled boarding time.

7.14.17. All cargo and baggage is subject to inspection. Points or origin and destination are to be verified and each item of luggage is to be matched with its owner and removed from the area. All unidentified luggage remaining, and any air cargo shall be searched by the bomb disposal squad or refused for shipment.

7.14.18. When the checklist is completed the aircraft shall be kept free of all personnel until it is released for flight. Only the VP Flight Operations or Dir. Flying Operations have authority to release such an aircraft for flight.

7.15. CHECKLIST FOR AIRCRAFT SEARCH

7.15.1. INTERIOR

Moving from front to rear, the search team can examine the:

a) Flight Station: (Cockpit)

I. Around & under all seats
II. Seat & wall packs and storage bins
III. Rudder pedal areas
IV. Escape rope compartments
V. Jump seat area and coat Hanger compartments
VI. Jeppesen bags, A/C Library.

b) Forward and aft lavatories:

I. Waste containers
II. Cabinet storage areas & dispensers
III. Toilet cans
IV. Removable wall panels

c) Galleys:

I. Behind and under galley inserts
II. Drawers and containers
III. Waste containers
IV. Over-galley doors & compartments
V. Coffee machine

d) Entrance areas & service doors:

I. Emergency chute covers
II. Storage compartments on bulkhead
III. Flight attendants' jump seats

e) Cabin area:

I. Seat backs and under side cushions

II. Floor area under seats

III. Between seats and windows

IV. Ceiling access compartments

V. Floor access compartments

VI. Ceiling vents

f) Overhead hat-racks

I. Pillows, blankets etc.

II. Oxygen bottles and masks

III. Life raft stowage

7.15.2. EXTERIOR

Moving clockwise the search shall examine the:

a. Nose gear & wheel well
b. Nose area access panels and compartments
c. Radar compartment
d. External power receptacle
e. Air door area
f. Forward baggage compartments
g. Pneumatic connections
h. Leading edge & fuel service panels
i. Engine (s)
   I. Cowl doors
   II. Nose cowl interior
   III. Air inlets and other access doors
   IV. Tail pipe & thrust reverses area
j. Main gear & wheel wells
k. Center gear as applicable
l. AFT Baggage

7.16. RECOMMENDED LEAST RISK BOMB LOCATION (LRBL)

7.16.1. To avoid the activation of an altitude –sensitive bomb, the cabin altitude should not exceed the value at which the bomb is discovered. To reduce the effects of the explosion, the aircraft should fly as long as possible with approximately 1 PSI differential pressure, to help the blast to go outwards. 1 PSI differential pressure corresponds to a 2500 feet difference3 between the aircraft and the cabin altitude. The flight crew maintains the differential pressure with the manual pressure control.

7.16.2. On all Ethiopian aircraft LRBL is RIGHT AFT SERVICE DOOR.

7.16.3. RECOMMENDED CABIN LRBL PROCEDURES

STEP	ACTION	PURPOSE
1	If possible DESCEND to reduce pressure differential between the cabin and ambient air pressure to zero as soon as possible, LAND and TAXI to a REMOTE SITE and DEPLANE the aircraft immediately. DISCONTINUE use and RESTOW all portable electronic devices. PERFORM ARRIVAL ACTIONS 3, 4, 5, 17 & 18. **NOTE:** On average, it takes ~30 min. to construct the LRBL stack at the appropriate door on the aircraft using steps 2 through19 below.	To get passengers and crew away from the hazard. If landing and deplaning within 30 min. not possible, the LRBL should be constructed. Reducing pressure differential reduces risk to aircraft and occupants.
2	If unable to land within 30 min, DESCEND to attain no more than 1PSI pressure differential while MAINTAINING the existing cabin altitude until the suspect device is secured at the LRBL. 1PSI corresponds to a 2500 feet difference between the a/c and the cabin altitude **NOTE:** Reducing the pressure differential without altering the cabin altitude prior to movement of the suspect device to the LRBL will reduce risk to the airframe.	To avoid strongly amplifying the effects of an explosion. Maintaining the existing cabin altitude will reduce the risk of pressure sensitive device activation.
3	CONTACT ATC and company operations office as appropriate and request ASSISTANCE from EXPLOSIVES EXPERTS.	To get expert advice directly from the duty Aviation Explosive Security Specialists.

4	After considering the aircraft's capabilities and the distance to the nearest suitable airport, CONFIGURE the aircraft for landing, and RESTRICT MANEUVERING to a minimum. This may not be possible in all instances due to the distance to the nearest suitable airport.	In the event of detonation, the systems for lowering landing gear and other landing aids could be damaged.
5	MOVE passengers at least four rows away from the location of the suspect device. In aircraft with passenger cabins or crew rest areas over the location of a suspect device in another passenger cabin, VACATE those areas as well. ENSURE all passengers are seated with their seatbelts fastened, and seat backs and tray tables in their full, upright positions. DISCONTINUE use of all portable electronic devices. **NOTE:** On full flights it may be necessary to double up passengers. **WARNING:** A suspect device in flight should, when directed by the pilot in command, be removed to the proper LRBL site using only proper LRBL procedures contained in the sequential steps (6 through 15) below. **CAUTION:** If movement of the device is not directed or is not possible, placement of materials around the object may be beneficial. Consult with the duty aviation explosives expert in these instances.	Distance from an explosion is one of the best protective measures for safety. Placing seat backs and tray tables in their full, upright positions will provide additional protection.

SECTION 3.2 HIJACKING AND BOMB THREAT

6	VERIFY the designated LRBL for your aircraft. DIRECT crew to PERFORM LRBL procedures as described in the sequential steps below. **WARNING:** 1. Alternate locations must not be used without consulting with an aviation explosives expert. Never take a suspected device to the flight deck. a) Do not cut or disconnect any wires and do not open or attempt to gain entry to internal components of a closed or concealed suspect device. Any attempt may result in an explosion. Booby-trapped closed devices have been used on aircraft in the past. **NOTE:** T-4's if on board have been trained in LRBL procedures and may (or may not) be able to assist in LRBL procedures depending upon the scenario.	To obtain the correct sequence of steps to reduce risk in the event of a suspect device.
7	SLIDE a stiff thin card, such as the emergency information card, underneath the suspect device. If there is no resistance under the suspected device, LEAVE THE CARD IN PLACE. **CAUTION:** In the unlikely event that the card cannot be slipped under the suspect device, it may indicate that an anti-lift switch is present and that the suspect device cannot be moved. **NOTE:** No anti-lift activated device has ever been reported onboard an aircraft in flight. If the suspect device cannot be moved to the LRBL, consult with the duty aviation explosives expert on the ground to develop appropriate alternative emergency procedures.	To check for an anti-lift switch, prevent detonation if an anti-lift switch is used in the suspect device, and to assist in maintaining positive control in the position found during movement to the LRBL.
8	If emergency equipment (e.g. PBE, fire extinguishers, etc.) is located in close proximity to the LRBL and they can be easily removed, REMOVE and STOW them in an alternate location away from the area around the LRBL. If possible, DISABLE all nonessential power (e.g. galley equipment) to the areas in the vicinity of the LRBL.	Removing oxygen bottles and securing galley power will reduce missile and secondary fire ignition hazards. PBE, fire extinguishers and other emergency equipment will remain accessible in event needed.

SECTION 3.2 HIJACKING AND BOMB THREAT

9	PREPARE the LRBL BEFORE moving the suspect device. CONSTRUCT a platform of baggage out to the aisle, from the floor up to the mid-height of the door at the LRBL. Wetted material (e.g. blankets or clothes) should make up the last 10in (25 cm) of this platform. PLACE a single, thin sheet of plastic (e.g. trash bag) on top of the wetted materials. **CAUTION:** Do not omit the plastic sheet, as the suspect device could get and possibly short circuit electronic components causing inadvertent device actuation.	Baggage will dissipate blast forces that could otherwise damage the floor structure or critical systems beneath it, and also reduce or prevent fragments and fire. Wetted materials will also greatly reduce the chance of cabin fires. The plastic sheet prevents short circuiting electronic components.
10	POSITION a 'suspect device location indicator line' from the location on the platform where you will place the suspect device, EXTENDING outward into the aisle. **WARNING:** Do not physically tie or otherwise attach the suspect device location indicator line to the suspect device. **NOTE:** A suspect device location indicator line is a 6 to 8 feet (1.8 to2.4m) line (e.g. neckties, headset cord, or belts connected together) preferably of contrasting color that helps the responding bomb squad find the precise location of the suspect device within the LRBL stack constructed.	To pinpoint the location of the suspect device within the LRBL stack, enabling aircrew to avoid unnecessary exposure, and enabling the responding bomb squad to find the suspect device with minimal time/disturbance.
11	If the suspect device can be relocated, MOVE it in the position found to the prepared LRBL base with the card in place beneath it. STABILIZE it on top of the plastic sheet above the 10 in (25 cm) of wetted materials and CENTER it against the inside surface of the door. **NOTE:** Though very sensitive vibratory-activated devices are not expected onboard aircraft in flight, less sensitive types requiring a 90° change in attitude have been encountered. Keeping the suspect device in the position found reduces the chance of an accidental detonation.	To transport the suspect device to the LRBL without tilting it from the attitude found.

12	POSITION the suspect device in the attitude found, as close to the center of the door as measured from side to side and top to bottom as possible, at the LRBL site and PLACE an additional single thin sheet of plastic over the suspect device. **WARNING:** Ensure that the suspect device is near the center of the LRBL door on a properly constructed LRBL stack. Placing a suspect device on the cabin floor will significantly increase risk. **CAUTION 1:** Ensure that the suspect device, when placed on the stack against the door, is above the slide pack but not against the door handle, and if possible, avoid placement in the view port. **CAUTION 2:** Do not omit the plastic sheet, as the suspect device could get wet and possibly short circuit electronic components causing inadvertent device activation.	Placement of the suspect device at the mid height and the fore and aft center of the door, above the slide pack enables successful directing of the blast, smoke, fire and fragmentation outside of the aircraft, rather than into the cabin area. The plastic sheet prevents short circuiting electronic components.
13	SATURATE soft blast-attenuating materials (e.g. blankets or clothes with water or any other non-flammable liquid and CAREFULLY PACK at least 10 in (25 cm) of wetted material around and on top of the suspect device.	Wetted materials will lessen the thermal effects, absorb energy, and assist in directing the blast, smoke and fragmentation outward, and reduce the chance of cabin fires.
14	ARRANGE luggage and blast-attenuation materials in a manner that prevents excessive weight from being applied directly on the suspect device and FILL the entire remaining area above the suspect device with soft blast-attenuating materials (e.g. seat cushions) up to the cabin ceiling and out to the aisle.	Baggage will dissipate blast forces that could otherwise damage the overhead or critical systems nearby, and also reduce or prevent fragments and fire.
15	SECURE the LRBL stack in place using seatbelts, headset cords, ties or other appropriate materials.	To ensure the LRBL stack stays in place during the remainder of the flight.
16	Cabin crew NOTIFY flight deck crewmembers that the suspect device is SECURE at the LRBL.	LRBL procedure is complete, continue flight to landing.

17	Upon notification from the cabin crew that the suspect device is at the LRBL, CONTINUE DESCENT and PREPARE FOR APPROACH AND LANDING. TAXI to a REMOTE SITE, and DEPLANE passengers as soon as possible using exits that maximize the distance from the suspect device. **WARNING:** Do not taxi to an airport passenger terminal with a suspect device onboard. **NOTE:** If applicable, consider disarming the door directly across from the LRBL door to enable the response teams to open this door from the outside if desired during their procedures.	To reduce passenger and crew exposure to the suspect device.
18	ADVISE the tower to inform crash, fire and rescue and bomb squad personnel on the location of the LRBL and the suspect device and recommend aircraft access points.	To assist first responders in avoiding hazard areas, by apprising them of suspect device location, areas to avoid, and recommended access points.

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SECTION 3.2 HIJACKING AND BOMB THREAT

SECTION 3.3 SIGNALS FOR USE IN THE EVENT OF INTERCEPTION

1.0. PURPOSE

The purpose of this section is to establish guidelines for compliance with signals during an aircraft interception.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- VP Flight Operations
- Flight crewmembers

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.04

5.0. DEFINITION

N/A

6.0. RESPONSIBLITY

6.1. The Captain is responsible to deactivate the CVR to preserve cockpit voice data following an accident or serious incident.

6.2. Vice President Flight operations is responsible to authorize reactivation of the cockpit voice recorder after deactivation by the pilot-in-command.

6.3. The PIC is responsible to perform additional duties and responsibilities listed hereunder.

7.0. PROCEDURE

7.1. SIGNALS INITIATED BY INTERCEPTING AIRCRAFT AND RESPONSES BY INTERCEPTED AIRCRAFT.

SERIES	INTERCEPTING	MEANING	INTERCEPTED AIRCRAFT	MEANING

SECTION 3.3 SIGNALS FOR USE IN THE EVENT OF INTERCEPTION

	AIRCRAFT SIGNALS		RESPONDS	
1	DAY or NIGHT – Rocking aircraft and flashing navigational lights at irregular intervals (and landing lights in the case of a helicopter) from a position slightly above and ahead of, and normally to the left of, the intercepted aircraft (or to the right if the intercepted aircraft is a helicopter) and, after acknowledgement, a slow level turn, normally to the left, (or to the right in the case of a helicopter) onto the desired heading. **NOTE:** Meteorological conditions or terrain may require the intercepting aircraft to reverse the positions and direction of turn given above in series 1. If the intercepted aircraft is not able to keep pace with the intercepting aircraft, the latter is expected to fly a series of racetrack patterns and to rock the aircraft each time it passes the intercepted aircraft.	You have been intercepted. Follow me.	DAY or NIGHT Rocking aircraft, flashing navigational lights at irregular intervals and following. NOTE: Additional action to be taken by intercepted aircraft is prescribed in this section.	Understood, will comply.
2	DAY or NIGHT – An abrupt breakaway maneuver from the intercepted aircraft consisting of a climbing turn of 90 degrees or more without crossing	You may proceed.	DAY or NIGHT – Rocking the aircraft.	Understood, will comply.

	the line of flight of the intercepted aircraft.			
3	DAY or NIGHT – Lowering landing gear (if fitted), showing steady landing lights and over-flying runway in use or, if the intercepted aircraft is a helicopter, over flying the helicopter landing intercepting helicopter makes a landing approach, coming to hover near to the landing area.	Land at this aerodrome.	DAY or NIGHT – Lowering landing gear, (if fitted), showing steady landing lights and following the intercepting aircraft and, if, after over flying the runway in use or helicopter landing area landing is considered safe, proceeding to land.	Understood, will comply.
4	DAY or NIGHT – Raising landing gear (if fitted) and flashing landing lights while passing over runway in use or helicopter landing area at a height exceeding 300m (1,000’) but not exceeding 600m (2,000’) (in the case of a helicopter, at a height exceeding 50m (170’) but not exceeding 100m (330’) above the aerodrome level and continuing to circle runway in use or helicopter landing area. If unable to flash landing lights, flash any other light available.	Aerodrome you have designated is inadequate.	DAY or NIGHT – If it is desired that the intercepted aircraft follow the intercepting aircraft to an alternate aerodrome, the intercepting aircraft raises its landing gear (if fitted) and uses the Series 1 signals prescribed for intercepting aircraft. If it is decided to release the intercepted aircraft, the intercepting aircraft uses the Series 2 signals prescribed for intercepting aircraft.	Understood, follow me. Understood, you may proceed.

5	DAY or NIGHT – Regular switching on and off of all available lights but in such a manner as to be distinct from flashing lights.	Cannot comply.	DAY or NIGHT – Use Series 2 signals prescribed for intercepting aircraft	Understood.
6	DAY or NIGHT – irregular flashing of all available lights	In distress.	DAY or NIGHT – USE series 2 signals prescribed for intercepting aircraft	Understood.

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SECTION 3.3 SIGNALS FOR USE IN THE EVENT OF INTERCEPTION

SECTION 3.4 OXYGEN REQUIREMENTS AND USE DURING EMERGENCY

1.0 PURPOSE

The purpose of this section is to establish guidelines for requirements and use of oxygen during emergency.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Cabin Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.05

4.1 Flights in a pressurized aircraft shall not be commenced unless a sufficient amount of stored breathing oxygen is carried to supply all crew members and passengers in the event of loss of pressurization for any period that the atmospheric pressure in any compartment occupied by people would be less than 10,000 feet (700 HPA).

4.2 All aircraft operated above 25,000 feet shall be equipped with a quick donning oxygen mask for each flight crew member.

4.3 All oxygen masks shall be correctly stowed within the stowage provided, so that, they can be available for immediate use.

4.4 Before the takeoff a flight each flight crew member shall personally, during the initial preflight, check his oxygen equipment to ensure that the oxygen mask is functioning properly, and that the oxygen supply and pressure are adequate for use as described in the AFM.

4.5 When an airplane certificated to operate at flight altitudes up to and including flight level 250, and unable to descend safely within 4 minutes to 14,000 feet, the A/C shall be provided with automatically deployable oxygen equipment immediately available to each occupant. Oxygen must be available for 30 minutes period for at least 10 percent of the passenger cabin occupants.

5.0 DEFINITION

N/A

5.1. USING OXYGEN

5.1.1. **Supplemental oxygen**: - oxygen that may be necessary because the amount of oxygen available in the cabin is insufficient to support vital human functions.

5.1.2. **Protective oxygen:** - oxygen that may be necessary to support vital human functions because the air in the cabin is contaminated or unusable. Oxygen on board the aircraft may be used for either purpose.

6.0 RESPONSIBILITY

6.1. The Pilot-in-command is responsible for compliance with the procedures stated in this section.

6.2. The Team Leader Cabin Services is responsible for accomplishing the duties mentioned herein.

6.3. The PIC shall ensure that breathing oxygen and masks are available to crewmembers in sufficient quantities for all flights. In no case shall the minimum supply of oxygen on board the aircraft be less than that prescribed by ECAA. (ECARAS 8.5.1.27)

7.0 PROCEDURE

It is of primary importance to determine that a person in need of oxygen is breathing. If he is not, mouth-to-mouth resuscitation should be used to restore breathing. Oxygen is of no benefit to a person who is not breathing. Instructions in the method of mouth-to-mouth resuscitation are found in the first aid kit.

7.1. OXYGEN DEFICIENCY

7.1.1. The most noteworthy effects of oxygen deficiency are:

a) Slow and confused mental processes, perhaps without the victim's awareness.

b) First vision and later hearing loss.

c) An urge to breathe rapidly and deeply. (This urge should be resisted because of other physiological changes that may result.)

d) Increasing distress that can, if not alleviated, result in unconsciousness or death.

7.1.2. The time of useful consciousness shown below is the period during which an individual is capable of coordinated action for self-preservation.

TABLE 1

TIME OF USEFUL CONSCIOUSNESS		
ALTITUDE	**AFTER A RAPID DECOMPRESSION**	**AFTER REMOVING AN OXYGEN MASK**
25,000 feet	$2^{1/2}$ minutes	5 minutes
30,000 feet	30 seconds to 1 minute	2 minutes
40,000 feet and above	18 seconds	18 seconds

7.1.3. The time of useful consciousness is generally never less than 18 seconds because this is the shortest time required for oxygen starved blood from the lungs to reach the brain and depress brain functions, recovery after administering oxygen requires nearly the same length of time.

7.2. OXYGEN PRECAUTIONS

7.2.1. When oxygen is used from any source for whatever reason, the following precautions are to be observed:

a) Oxygen must not be used near a spark, a flame, or lighted tobacco.

b) While anyone on the flight deck is using oxygen, smoking is not permitted on the flight deck.

c) Mineral base oils (used in some cosmetics and lip balms) must not come in contact with or near oxygen equipment.

NOTE: Smoking must not be permitted while passenger oxygen is being used.

7.2.2. For precautions while using portable oxygen bottles, see Portable Oxygen Bottles in this section.

7.2.3. When portable oxygen is no longer being used, the oxygen supply valve (s) should be closed to preserve the remaining oxygen supply and reduce the hazard of fire.

7.3. PASSENGER OXYGEN SYSTEM

7.3.1. The passenger oxygen system provides oxygen for use by occupants of the main cabin if the need arises. The quantity of oxygen (pressure) required in the system depends on the number of cabin occupants, the length of the route, and the minimum en route altitude. It must be sufficient for the following usages:

7.3.2. When the oxygen masks drop from overhead units in the cabin, each cabin occupant has to pull a mask toward him/her, cover hi/hers nose and mouth with it, and breathe normally.

7.3.3. Oxygen does not flow through a mask until it has been pulled from the position to which it was released.

7.3.4. When the procedures have been accomplished to restore a safe cabin altitude, the Captain should advise the Supervisor Cabin Attendants of the plan for conducting the remainder of the flight.

7.4. CREW OXYGEN SYSTEM

Refer to FCOM, QRH and SOP.

7.5. PORTABLE OXYGEN IN THE CABIN

7.5.1. In addition to oxygen available in the aircraft's oxygen system, portable oxygen bottles are provided for use by passengers and cabin attendants. The usable capacity of each bottle is 300 liters at a pressure of 1800 PSI. There are two constant flow

outlets, one at four liters per minute (labeled HI) and the other at two liters per minute (labeled LO). Two disposable oxygen masks are provided with each portable oxygen bottle. The oxygen valve is opened and closed with a yellow knob at the top of the cylinder. An over-the-shoulder carrying strap is provided with each bottle.

7.5.2. If more than one bottle is to be used, the passenger must be instructed to watch the pressure gage, and another bottle should be requested prior to 100 PSI.

7.5.3. The Cabin Attendant must determine that the mask has been properly attached to either the 2 or 4-liter outlet, as required by the passenger. The following precautions are to be observed in using portable oxygen bottles:

a) Portable oxygen bottles may be used for an unscheduled emergency or for planned or unplanned medical purposes.

b) When not in use, portable bottles will be properly stowed.

c) For planned medical applications, additional bottles stored on board for such purposes will be used first.

d) Connecting or disconnecting of oxygen dispensing equipment and portable oxygen bottles while any passenger is aboard is prohibited. This does not prohibit turning oxygen flow ON or OFF or changing the hose between the 2 and 4-liter outlets.

e) Before the end of the flight the Team Leader cabin services shall record the location and number of oxygen bottle (s) used in-flight in the cabin maintenance logbook.

7.5.4. Any passenger using portable oxygen bottle must be in an assigned seat.

7.5.5. The passenger and the oxygen equipment must not restrict access to or use of any regular or emergency exit or the aisle.

7.5.6. No person shall smoke whenever a passenger who is in possession of portable oxygen bottle regardless of whether or not the oxygen is being used.

7.6. USING PORTABLE OXYGEN

7.6.1. The mask is to be connected properly to the desired bottle outlet (HI OR LO). The HI outlet should be used initially until the actual flow requirement can be determined,

then the oxygen valve is opened slowly using the yellow knob. The user should confirm oxygen flow, place the mask over the nose and mouth with the restraining band around the head, pull the ends of the band until the mask is held comfortably secure, press the metal strip over the nose and that the mask fits the face, and breathe normally.

7.6.2. Facing the gauge, the right side is the HI outlet for adults and the left side is the LOW outlet for children.

7.6.3. For the location of portable oxygen bottles, see the emergency equipment location diagram in the Emergency chapter of the Aircraft Operations Manual.

7.7. UNPRESSURIZED AIRCRAFT

All unpressurized aircraft operated at flight altitudes where the cabin altitude will be greater than 10,000 feet (less than 700 HPA) shall be equipped with oxygen storage and dispensing apparatus in accordance with the requirements of the TABLE-2 below as appropriate for the route to be flown & also ensures the aircraft can continue at a pressure altitude that will allow continued safe flight and landing.

7.8. PRESSURIZED AIRCRAFT

All aircraft operated at flight altitudes greater than 10,000 feet (less than 700 HPA), but pressurized to maintain a cabin altitude of less than 10,000 feet (greater than 700 HPA), shall be capable of descending to an altitude after a loss o0f pressurization that will allow continued safe flight and landing and are equipped with oxygen storage and dispensing apparatus in accordance with the requirements of the TABLE-2 below as appropriate for the route to be flown & also ensures the aircraft can continue at a pressure altitude that will allow continued safe flight and landing.

TABLE 2: MINIMUM REQUIREMENTS FOR SUPPLEMENTAL OXYGEN

SUPPLY FOR	DURATION AND CABIN PRESSURE ALTITUDE
1. All occupants of flight deck seats on	Entire flight time when the cabin pressure

flight deck duty	altitude exceeds 13000 ft. and entire flight time when the cabin pressure altitude exceeds 10000 ft. but does not exceed 13000 ft. after the first 30 minutes at those altitudes.
2. All safety minimum cabin crew members	Entire flight time when the cabin pressure altitude exceeds 13000 ft. and entire flight time when the cabin pressure altitude exceeds 10000 ft. but does not exceed 13000 ft. after the first 30 minutes at those altitudes.
3. 100% of passengers	Entire flight time when the cabin pressure altitude exceeds 15000 ft.
4. 30% of passengers	Entire flight time when the cabin pressure altitude exceeds 14000 ft. but does not exceed 15000 ft.
5. 10 % of passengers	Entire flight time when the cabin pressure altitude exceeds 10000 ft. but does not exceed 14000 ft. after the first 30 minutes at these altitudes.

7.9. LOSS OF CREW OXYGEN

7.9.1. If the supply of oxygen in the airplane's crew oxygen system should become depleted while en-route, the following temporary measures are to be observed;

a) Immediately descend to FL 250.

b) Obtain portable oxygen bottles with disposable masks (from the cabin) one for each flight deck crewmember and secure them in a position where they are readily available.

c) Ensure that the MSA has to be 10,000 feet or less in case of depressurization

d) Diversion to the nearest suitable airport is considered if there is any operational issue which warrants diversion or the procedures listed on "a, b & c" couldn't be applied.

* * * * * *

SECTION 3.4 OXYGEN REQUIREMENTS AND USE DURING EMERGENCY

SECTION 3.5 DECOMPRESSION

1.0 PURPOSE

The purpose of this Decompression procedure is to establish guidelines on how to manage rapid decompression.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crew members
- Cabin Crew members

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.03

It is the policy of Ethiopian Airlines to make an emergency descent whenever a rapid decompression occurs to keep passengers safe and avoid uncontrollability of the aircraft.

5.0 DEFINITION

Decompression occurs whenever cabin altitude exceeds the pre-set altitude in an uncontrolled way.

5.1. RAPID DECOMPRESSION - OBJECTIVE SIGNS

5.1.1. There is always a remote possibility of a rapid loss of cabin pressure in any pressurized aircraft. The signs of rapid decompression are:

a) A rush of air
b) Loud bang
c) Rapid drop in temperature
d) Cabin filled with dust, debris, loose objects
e) Moisture will condense in the form of fine mist
f) Noise level will increase considerably

5.2. RAPID DECOMPRESSION - SUBJECTIVE SIGNS

5.2.1. The effects of rapid decompression could be serious to crew and passengers in a few seconds. The physiological effects on a person are due to a lack of oxygen and the expansion of gases trapped in the body cavities following the fall in pressure. They are usually accompanied by the following signs, which might be of short duration but are still dangerous:

a) There is a sudden expansion of the chest as air is blown out through the nose and mouth causing difficulties in breathing
b) Cold sensation
c) Sinuses and ears may feel full momentarily
d) Speaking will be difficult
e) Abdominal distension sufficient to cause discomfort or pain
f) Possible dental pain experienced
g) Cheek & lip fluttering
h) Possible dizziness experienced (hypoxia)

6.0 RESPONSIBILITY

6.1. The Captain-in-command has the responsibility to perform the emergency procedures for decompression per QRH.

6.2. Cabin attendants are responsible to abide by the flight captain’s command.

7.0. PROCEDURE

7.1. AT DECOMPRESSION

7.1.1. During slow decompression timely remedial action such as descent to a lower level can be taken, with little or no chance of causing damage to the cabin or its occupants.

7.1.2. However, due to various reasons, a rapid decompression might occur and will require an emergency descent by the flight crew and immediate action by the cabin crew, in this case Flight crew accomplish the emergency procedures for decompression /rapid descent as stipulated in the QRH appropriate to the aircraft type.

7.1.3. If a loss of cabin pressure occurs at night cabin crew shall, turn on all cabin lights to awaken any sleeping passengers.

7.1.4. Note: For Airbus "Emergency Descent" Captain announces, **“ATTENTION THIS IS THE CAPTAIN, PULL DOWN AND USE YOUR OXYGEN MASK”.**

7.1.5. Emergency must be declared. **(MAYDAY)**

7.1.6. Cabin crew put on the nearest available oxygen mask, sit down, and fasten seat belt or hold on a seat.

7.1.7. Loss of cabin pressure becomes a matter of immediate concern when the oxygen masks drop from overhead units in the cabin. Each crewmember must begin using oxygen immediately.

7.1.8. Having briefed the passengers before takeoff regarding the use of oxygen, cabin attendants should not attempt to minister to passengers until advised by the Captain that it is safe to do so. Such an attempt could result in the loss of consciousness.

7.1.9. To avoid interrupting essential interphone communications between cabin crewmembers and flight deck, Cabin Attendants shall not use the interphone until called from the flight deck.

7.2. IMMEDIATELY FOLLOWING DECOMPRESSION

7.2.1. When reaching at safe altitude Flight deck crew shall advise the cabin crews & passengers that emergency descent is over & to breath normally through PA.

7.2.2. The captain contacts the Team Leader Cabin Services through the interphone or calls her to the flight deck to get a preliminary briefing on the situation in the cabin.

7.2.3. Cabin crew should transfer from drop mask to portable oxygen bottles, if required in consultation with the PIC.

7.2.4. After being advised that the cabin altitude is safe and briefed by the PIC cabin attendants shall to visit each passenger and provide first aid oxygen to those suffering from hypoxia. Passengers should be advised to keep their activity to a minimum to reduce their requirement for oxygen. The lavatories and lounge areas should be checked for persons who need first aid oxygen.

7.2.5. Ensure the 'No Smoking' sign is respected.

7.2.6. Keep door areas clear.

7.2.7. Report any injury or damage to Captain.

7.2.8. Lack of oxygen that occurs when cabin altitude exceeds 10,000 feet causes loss of judgement, less self-criticism and over confidence. During pressurized flight, cabin altitude is usually less than 8,000 feet and the average individual feel no ill effects. However, expectant mothers, invalids and the elderly suffer at lower altitude and cabin crew should keep a careful watch on these passengers. Should the cabin altitude exceed 10,000 feet passenger should remain seated and the serving of food stopped. Liaison between the cabin and flight deck must be maintained to confirm satisfactory operation of oxygen equipment. The NO SMOKING sign shall be ON whenever the oxygen is in use. It is imperative that flying staffs are conversant with the oxygen equipment on the aircraft type and method of operation.

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SECTION 3.5 DECOMPRESSION

SECTION 3.6 INCAPACITATION – EMERGENCY

1.0 PURPOSE

This section establishes guidelines how to handle flight during incapacitation situation.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Crewmembers
- Cabin crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.03

5.0 DEFINITION

Incapacitation of a crewmember is defined as any condition that affects the health of a crewmember during the performance of duties assigned to him which renders him incapable of performing the assigned duties.

The definition includes either total or partial incapacitation which does not allow the fulfillment of duties in the "normal" way.

5.1. TYPES OF INCAPACITATION

5.1.1. **Obvious incapacitation:** means total functional failure and loss of capabilities. Generally, this can be easily detectable and can be a prolonged condition. Among the possible causes are heart disorders, severe brain disorders, severe internal bleeding etc.

5.1.2. **Subtle incapacitation:** This may be considered a more significant safety hazard, because it is difficult to detect, and the effects can range from partial loss of function to complete unconsciousness. Possible causes may be minor brain seizures, low

blood sugar (hypoglycemia) other medical disorders or preoccupation with personal problems. Since the crewmember concerned may not be aware of or capable of rationally evaluating his situation, this type of incapacitation is very dangerous.

SECTION 3.6 INCAPACITATION – EMERGENCY

6.0 RESPONSIBILITY

6.1. The Team Leader Cabin Services is responsible for following instructions from the Flight Crew.

7.0. PROCEDURE

7.1. FLIGHT CREW INCAPACITATION

7.1.1. STEP ONE

a. Take over control of the aircraft by announcing "I HAVE CONTROL"

b. Engage autopilot manage the flight.

c. Reach out and lock the inertial seat harness lock.

d. Declare emergency or urgency as applicable, advise:

   I. For captain incapacitation declare "**MAYDAY**" and captain incapacitated.

   II. For first officer incapacitation declare "**MAYDAY**" and first officer incapacitated

e. Over the PA advise specific cabin crew to come to the flight deck immediately. (e.g. "**Cabin crew B in the flight deck**" spoken twice)

f. If possible have the incapacitated crewmember removed from his seat or move his seat fully back to prevent obstruction of flight controls. The help of other crewmembers may be required.

g. If required, set COM and NAV to your side.

7.1.2. STEP TWO

a. Have the cabin crew or passengers take care of the incapacitated crewmember (ask if doctor or other medical person is onboard)

b. Arrange a landing as soon as practical after considering all pertinent factors.

c. Arrange medical assistance after landing – giving as many details about the condition of the affected person.

7.1.3. STEP THREE

a. Prepare for landing (cockpit and cabin), but do not press for a hasty approach.

b. Perform approach checklist earlier than normal (request assistance from other crewmembers or "capable" persons.

NOTE: Discretely check if a company qualified pilot is available on board to replace the unconscious one and continue the flight.

c. Request radar vectoring and make an extended approach, where possible, to reduce work load.

d. Do not change seats – fly the aircraft from your normal position.

e. Organize work after landing

7.2. CABIN CREW INCAPACITATION

Refer to CCOM for cabin crew incapacitation.

7.3. SUCCESSION OF COMMAND

7.2.1. If the Captain is incapacitated for any reason, command evolves to the next senior crewmember in the following sequence:

a) Assigned Captain
b) Relief Captain (if applicable)
c) First officer
d) Relief first officer (if applicable)
e) Team Leader Cabin Services
f) Senior Cabin Crew II
g) Senior Cabin Crew I
h) Cabin crew
i) Junior cabin crew

7.4. GENERAL

In flight pilot incapacitation is a significant safety hazard and has caused many accidents. Incapacitations have occurred more frequently than other emergencies in all age groups and during all phases of flight. There are many forms of incapacitation ranging from obvious sudden death to a lingering and difficult to detect partial loss of functions.

7.5. RECOGNITION OF INCAPACITATION

7.5.1. Flight crew must suspect possible incapacitation at any time if the crew does not respond to two verbal communications.

SECTION 3.6 INCAPACITATION – EMERGENCY

7.5.2. Early recognition of incapacity is of utmost importance. A silent collapse will hardly be detected during normal activities (for instance during the cruise phase of a flight), as communications may sometimes be reduced to a minimum. This requires that all crewmembers monitor each other very closely. Some symptoms of an incapacitation are:

a. Incoherent speech;

b. Strange behavior;

c. Irregular breathing;

d. Pale fixed facial expression;

e. Jerky motions that is either delayed or too rapid.

7.5.3. If any of these are present, incapacitation must be suspected, and action taken to check the state of the crewmember.

* * * * *

SECTION 3.6 INCAPACITATION – EMERGENCY

SECTION 3.7 FIRE OR SMOKE

1.0 PURPOSE

This section provides guidelines on how to extinguish cabin fire or smoke.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers
- Cabin crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.03

4.1. Flight crew reset of a tripped fuel pump circuit breaker or fuel pump control circuit breaker is prohibited. In flight, flight crew reset of any other tripped circuit breakers is not recommended. However, a tripped circuit breaker may be reset once, after a short cooling period (approximately 2 minutes), if in the judgment of the captain, the situation resulting from the circuit breaker trip has a significant adverse effect on safety. On the ground, flight crew reset of any other tripped circuit breaker should only be done after maintenance has determined that it is safe to reset the circuit breaker.

4.2. Flight crew cycling (pulling and resetting) of circuit breakers to clear a non-normal situation is not recommended, unless directed by a non-normal checklist.

4.3. When a non-normal checklist directs the flight crew to attempt only one reset of a switch per flight, a second reset of the switch should not be done until maintenance has cleared the malfunction.

4.4. If there is a popped out circuit breaker cabin crew members must inform the cabin supervisor and the PIC immediately whether it is on ground or inflight and follow the PIC's instructions on the operating procedure of popped out circuit breakers.

4.5. Cabin crew shall report any incident of fire and/or smoke to the captain. When notifying the flight crew be precise and maintain continuous communication. Specify:

- Location (area of smoke origin)

- Density (amount of fire or smoke)
- Color
- Odor
- Action being taken
- Whether fire is extinguished or not
- And cabin situation in general
- The Team Leader Cabin Services shall inform the flight crew as to the number of fire extinguishers used.
- All Ethiopian airplanes are equipped with a built-in fire detection & suppression system.

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

The Pilot-in-command is responsible to follow procedures stated in the affected Aircraft Operations Manual and act accordingly.

7.0 PROCEDURE

7.1. CLASSIFICATION OF FIRE

7.1.1. There are several types of fire that can occur in an airplane. Cabin crew should be able to identify each type and determine the most effective extinguishing agent to use. Types of fire which might be encountered on board and around an airplane are classified as:

a) Class "A" Fire- Flammable Solid

Any object that might catch ignition and be set on fire requiring the cooling effect of water (e.g. material wood, paper, cushions, etc.) is classified under class "A" fire. It is safe to use any type of extinguishers against such fire wherever water is not available.

b) Class "B" Fire- Liquid Fire

Liquid fire involves flammable substances that are usually lighter than water (e.g. oil, fuel, paint, kerosene). Water and water glycol fire extinguisher should not be used to fight such fire, as water will only help it spread and expand. Concentration should be on the exclusion of oxygen.

c) Class "C" Fire- Electrical Fire

Fire involving electrical equipment is usually the result of a short circuit. It is essential to cut the electrical source of ignition and exclude the oxygen. Beware of using water against such fire to prevent electric shocks. If there is no other alternative, water glycol extinguishers could be used in short shots.

d) Class "D" Fire- Metal Fire

Metal fire involves certain combustible metals (e.g. magnesium, titanium, potassium, sodium). These metals burn at high temperatures and give off sufficient oxygen to support combustion. They may react violently with water or other chemicals and must be handled with care.

7.2. CABIN FIRE OR SMOKE

Any fire inside the fuselage is to be considered as emergency and treated as such until it is certain that it has been extinguished. The Captain is to be notified of any such fire, regardless of its origin. The most common of these fires are electrical and grease fires in the galley and fires resulting from careless use of cigarettes and matches. In the event of fire or smoke in the cabin, appropriate procedure stated in the Aircraft Operation Manual should be followed.

7.3. CABIN AIR CONDITIONING SMOKE OR FUMES

7.3.1. The Captain is to be notified immediately if smoke or other vapor is detected coming from any of the cabin air supply vents. It is possible that the flight crew might otherwise remain unaware of the condition for an appreciable time.

7.3.2. No Ethiopian aircraft may be operated unless each lavatory is equipped with a smoke detector system that provides:

a) A warning light in the cockpit or
b) A warning light or audio warning in the passenger cabin which would be readily detected by a cabin crew member, taking into consideration the positioning of cabin crew members throughout the passenger compartment during various phases of flight.

7.4. FLIGHT DECK AIR CONTAMINATION

7.4.1. Depending on the type and degree of air contamination, it may be desirable to have all persons on the flight deck don oxygen masks and use 100% oxygen. However, if it is apparent that the air on the flight deck is or could become contaminated, at least one pilot at the controls shall use 100% oxygen. He shall remain on oxygen until it has been established that there is no longer a threat from contaminated air to any person on the flight deck.

7.4.2. Goggles need to be used to maintain adequate vision or for personal safety or comfort in the presence of irritating smoke or fumes. Goggles are to be donned after the oxygen mask to ensure a tight goggle fit.

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SECTION 3.8 ENGINE FAILURE

1.0 PURPOSE

The purpose of this section is to provide guidelines on how to manage engine failure.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight Crewmembers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.03

4.1. Flight crew reset of a tripped fuel pump circuit breaker or fuel pump control circuit barker is prohibited. In flight, flight crew reset of any other tripped circuit breakers is not recommended. However, a tripped circuit breaker may be reset once, after a short cooling period (approximately 2 minutes), if in the judgment of the captain, the situation resulting from the circuit breaker trip has a significant adverse effect on safety. On the ground, flight crew reset of any other tripped circuit breaker should only be done after maintenance has determined that it is safe to reset the circuit breaker.

4.2. Flight crew cycling (pulling and resetting) of circuit breakers to clear a non-normal situation is not recommended, unless directed by a non-normal checklist.

4.3. When a non-normal checklist directs the flight crew to attempt only one reset of a switch per flight, a second reset of the switch should not be done until maintenance has cleared the malfunction.

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- Location (area of smoke origin)
- Density (amount of fire or smoke)
- Color
- Odor
- Action being taken
- Whether fire is extinguished or not
- And cabin situation in general
- The Team Leader Cabin Services shall inform the flight crew as to the number of fire extinguishers used.
- All Ethiopian airplanes are equipped with a built-in fire detection & suppression system.

5.0 DEFINITION

N/A

6.0 RESPONSIBILITY

The Pilot-in-command is responsible to follow procedures stated in the affected Aircraft Operations Manual and act accordingly.

7.0 PROCEDURE

7.1. CLASSIFICATION OF FIRE

7.1.1. There are several types of fire that can occur in an airplane. Cabin crew should be able to identify each type and determine the most effective extinguishing agent to use. Types of fire which might be encountered on board and around an airplane are classified as:

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7.3.4. No Ethiopian aircraft may be operated unless each lavatory is equipped with a smoke detector system that provides:

c) A warning light in the cockpit or

d) A warning light or audio warning in the passenger cabin which would be readily detected by a cabin crew member, taking into consideration the positioning of cabin crew members throughout the passenger compartment during various phases of flight.

7.4. FLIGHT DECK AIR CONTAMINATION

7.4.3. Depending on the type and degree of air contamination, it may be desirable to have all persons on the flight deck don oxygen masks and use 100% oxygen. However, if it is apparent that the air on the flight deck is or could become contaminated, at least one pilot at the controls shall use 100% oxygen. He shall remain on oxygen until it has been established that there is no longer a threat from contaminated air to any person on the flight deck.

7.4.4. Goggles need to be used to maintain adequate vision or for personal safety or comfort in the presence of irritating smoke or fumes. Goggles are to be donned after the oxygen mask to ensure a tight goggle fit.

7.5 ENGINE FAILURE AFTER V1

1. Recognition

- Monitor engine instruments and aircraft behavior.
- Identify engine failure through:
  - Abnormal engine indications
  - Asymmetrical thrust
  - Aircraft yaw or roll

2. Continue Takeoff

- Do **not** attempt to reject takeoff after V1.
- Maintain directional control using rudder and aileron.
- Rotate at VR and lift off.

3. Initial Climb

- Pitch to appropriate attitude for engine-out climb.
- Maintain V2 (takeoff safety speed).
- Positive rate of climb → Gear up.

4. Secure the Engine

- Follow the engine failure memory items (QRH/ECAM/EICAS).
- Confirm engine failure with both pilots before shutdown.
- Complete engine shutdown and secure procedures.

SECTION 3.8 ENGINE FAILURE

5. Assess the Situation

- Check aircraft performance and controllability.
- Review weather, terrain, and fuel status.
- Communicate with ATC and declare an emergency (MAYDAY).

6. Decide on Return or Diversion

- Return to departure airport if safe and practical.

7. Divert to the nearest suitable airport if conditions require.

- Consider:
  - Runway length and surface
  - Weather and visibility
  - Airport rescue and firefighting category (RFF)
  - Terrain and obstacles

8. Prepare for Approach and Landing

- Review approach charts and landing performance.
- Brief the approach and missed approach procedure.
- Configure aircraft early and stabilize approach by 1000 ft AGL.

9. Post-Landing

- Evacuate if necessary.

Notify company and complete required reports.

* * * * *

SECTION 3.8 ENGINE FAILURE

SECTION 3.8 ENGINE FAILURE

SECTION 3.9 EMERGENCY RADIO COMMUNICATIONS

1.0 PURPOSE

This section establishes guidelines for radio communications during an emergency.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight crewmembers
- Cabin attendants
- ATC
- Flight Control
- Ramp personnel

4.0 . POLICY

Refer to Management Policy Manual chapter 9 section 9.03

4.1. For local radio communication failure procedures in different countries, refer to the EMERGENCY section of the Jepessen Airway Manual.

5.0 DEFINITION

MAYDAY: Signal for distress

PAN PAN: Signal for urgency

6.0 RESPONSIBILITY

6.1. The Pilot-in-command is responsible to use the appropriate signal and make announcement/communication accordingly.

6.2. It is the pilot-in-command’s responsibility to get the appropriate assistance from any available source when the position of the aircraft is doubtful.

7.0. PROCEDURE

7.1. EMERGENCY FREQUENCIES

7.1.1. If attempts to contact ATC & Flight Control on assigned frequencies and networks are unsuccessful, the following frequencies may be used to alert stations of the emergency.

TABLE 1

FREQUENCY	EFFECTIVE RANGE	GUARDED BY
VHF/AM 121.5 MHz	Line of sight	All military towers, VHF DF stations radar facilities, most communication stations, ocean carriers and military aircrafts.
HF/AM 2182 MHz	**Day**: Less than 300 miles **Night**: Less than 1000 miles	Most commercial coast stations and some ships.

7.1.2. The emergency frequencies are not to be used for routine messages; they should be kept clear to serve their intended function.

7.2. EMERGENCY MESSAGE ADDRESSES

7.2.1. By ICAO agreement, a ground station receiving an aircraft distress message is to render assistance using every possible means. This includes notifying all persons and agencies that could assist and also notifying those who would know of the emergency or who could participate in subsequent search and rescue operations.

7.2.2. These include:

a) Other aircraft in the area
b) Surface vessels
c) Air trafic services
d) Air carrier offices
e) Search and rescue facilities
f) Other radio stations.

7.3. EMERGENCY MESSAGE SIGNALS

There are two categories of aircraft emergency messages:

7.3.1. International Distress Signal "**MAYDAY**"

a) A distress message concerns an aircraft in grave and imminent danger and in need of immediate assistance. It has the highest priority. The distress message content should include as much of the following as time permits or is appropriate:

I. "**MAYDAY, MAYDAY, MAYDAY**."
II. Radio call sign (repeated three times)
III. Nature of the emergency
IV. Intentions
V. Position or estimated position (stating which).

b) A distress call may not be immediately acknowledged transmission should be made initially on the primary frequency. If the distress call is not acknowledged within 15 seconds, it should be repeated on the secondary frequency. If there is still no response, the message should be repeated alternatively on the primary and secondary frequency until it is acknowledged.

c) After establishing communications with any station, all transmissions should be directed to that station until the primary guard is assumed by another station.

d) When an emergency occurs, the pilot who cannot otherwise establish communications without delay may alert a ground radar facility to the emergency by setting the transponder to Mode A, Code 7700 (distress). Thereafter, radio communications should be established with ATC as soon as possible.

NOTE: In certain areas, use of emergency code 7700 may result in removal of the SSR return from ATC radar screens in case where the ground equipment is not provided with automatic means for its immediate recognition

7.3.2. International Urgency Signal "**PAN PAN**"

An urgent message concerns the safety of an aircraft, ship, or other vehicle or of some person on board or in sight. It is second in priority to a distress message. The initial call should be on the primary frequency to the primary station. It should include signal, "PAN, PAN, PAN," followed by the standard call to the station. If necessary, this call may be directed to "All Stations." After establishing communications with a station, the message with details should follow.

7.4. POSITION DOUBTFUL

7.4.1. If there is doubt as to the position of the airplane but not to a degree that affects safety, the phrase "position doubtful" should be added as a remark following the normal position report. This alerts ATC to the possible requirement to increase traffic separation. Acknowledging a doubtful position is encouraged to maintain an adequate separation and minimize exposure to violation.

7.4.2. When the position of the airplane is doubtful or unknown, the Captain should consider requesting assistance from any source, including ADF stations. ADF approach procedure is not to be conducted in IMC unless the Captain has declared an emergency.

7.4.3. When radio communications fail on an assigned frequency, the pilot shall attempt contact on another frequency appropriate to the route or on 121.5 MHZ.

7.5. NAVIGATION ERRORS

7.5.1. If a navigation error develops as described below, all pertinent information is to be recorded on the flight plan or the chart used as the official record of the flight.

7.5.2. A track deviation between 20 and 30 miles is considered a significant error. The heading must be altered promptly to rejoin the assigned track within 100 miles along track.

7.5.3. A track deviation of more than 30 miles is considered a gross error. An amended ATC clearance is required to rejoin the assigned track or another track.

7.6. AIR GROUND COMMUNICATIONS FAILURE

7.6.1. When radio communications fail on an assigned frequency, the pilot shall attempt contact on another frequency appropriate to the route or on 121.5 MHZ.

7.6.2. When communications with the assigned station are not possible. Messages should be relayed via any other station (including another airplane) with which communications can be established.

7.6.3. If the attempts specified above fail the aircraft station shall transmit its message twice on the designated frequencies preceded by the phrase "**TRANSMITTING BLIND**"

7.6.4. HF communications, SELCAL satisfies the requirement to maintain a listening watch.

7.7. TWO-WAY COMMUNICATIONS FAILURE

7.7.1 An aircraft operated as a controlled flight shall maintain continuous air-ground voice communication watch on the appropriate communication channel of, and establish two-way communication as necessary with, the appropriate air traffic control unit, except as may be prescribed by the appropriate ATS authority in respect of aircraft forming part of aerodrome traffic at a controlled aerodrome.

7.7.2. If a communication failure precludes compliance with 7.1.1, the aircraft shall comply with the communication failure procedures in 7.7.5 below, and with such of the following procedures as are appropriate. The aircraft shall attempt to establish communications with the appropriate air traffic control unit using all other available means. In addition, the aircraft, when forming part of the aerodrome traffic at a controlled aerodrome, shall keep a watch for such instructions as may be issued by visual signals.

7.7.3. If in visual meteorological conditions, the aircraft shall continue to fly in visual meteorological conditions land at the nearest suitable aerodrome and report its arrival by the most expeditious means to the appropriate air traffic control unit.

7.7.4. If in instrument meteorological conditions or when the pilot of an IFR flight considers it inadvisable to complete the flight in accordance with 7.1.3 the aircraft shall:

a. unless otherwise prescribed on the basis of regional air navigation agreement, in airspace where radar is not used in the provision of air traffic control, maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 20 minutes following the aircraft's failure to report its position over a compulsory reporting point and thereafter adjust level and speed in accordance with the filed flight plan;

b. In airspace where radar is used in the provision of air traffic control, maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 7 minutes following, the time the last assigned level or minimum flight altitude is reached or the time the transponder is set to Code 7600 or the aircraft's failure to report its position over a compulsory reporting point, whichever is later, and thereafter adjust level and speed in accordance with the filed flight plan;

c. when being radar vectored or having been directed by ATC to proceed offset using RNAV without a specified limit, rejoin the current flight plan route no later than the next significant point, taking into consideration the applicable minimum flight altitude;

d. proceed according to the current flight plan route to the appropriate designated navigation aid or fix serving the destination aerodrome and, when required to ensure compliance with "e" below, hold over this aid or fix until commencement of descent;

e. commence descent from the navigation aid or fix specified in d. at, or as close as possible to, the expected approach time last received and acknowledged; or, if no expected approach time has been received and acknowledged, at, or as close as possible to, the estimated time of arrival resulting from the current flight plan;

f. complete a normal instrument approach procedure as specified for the designated navigation aid or fix; and

g. land, if possible, within thirty minutes after the estimated time of arrival specified in "e" or the last acknowledged expected approach time, whichever is later.

7.7.5. Air-Ground Communications Failure

a) When an aircraft station fails to establish contact with the aeronautical station on the designated frequency, it shall attempt to establish contact on another frequency appropriate to the route. If this attempt fails, the aircraft station shall attempt to establish communication with other aircraft or other aeronautical stations on frequencies appropriate to the route. In addition, an aircraft operating within a network shall monitor the appropriate VHF frequency for calls from nearby aircraft.

b) If the attempts specified above fail, the aircraft station shall transmit its message twice on the designated frequency, preceded by the phrase "TRANSMITTING BLIND" and, if necessary, include the addressee(s) for which the message is intended.

7.8. RECEIVER FAILURE

7.8.1. If unable to establish communications because of receiver failure, the Captain should transmit required messages on the normal frequency using the phrase.

7.8.2. "**Transmitting blind due to receiver failure**". The message should be repeated and followed by the estimated time of the next transmission.

7.8.3. At a controlled airport, the aircraft shall keep a watch for such instructions as may be issued by visual signals. These light signals from the tower regulate airport operations:

TABLE 2

	MEANING	
LIGHT SIGNALS	**ON THE GROUND**	**IN-FLIGHT**
Steady – Green	Cleared for Take-off	Cleared to land
Flashing Green	Cleared to Taxi	Return for landing
Steady Red	Stop	Give way to other airplane
Series of Red Flashes	Taxi clear of the runway in use	The airport is UNSAFE. Do not land*.
Series of White Flashes	Return to the starting point on the aerodrome.	Land at this aerodrome and proceed to apron*.

Series of Green Flashes	Cleared to taxi	Return for landing*.
Red pyrotechnic		Not with standing any previous instruction do not land for the time being.
Alternate Red and Green	Airport Exercise extreme caution	Exercise extreme caution

* Clearance to land and taxi will be given in due course.

7.9. SUSPENDING TRANSMISSION

If it is necessary to suspend transmission, the pilot should notify the station concerned beforehand. He should include the time at which he expects to resume communications. The pilot shall advise the station when communications have returned to normal.

7.10. GROUND COMMUNICATIONS

Ground communications and other approved communications services are provided for the Flight Operations and Commercial Department business through the Aeronautical Fixed Telecommunications Network (AFTN) SITA. At remote stations where communication with Flight Control is not possible, the Captain must contact company through whatever means is available.

7.11. COMMUNICATIONS AND NAVIGATION FAILURE

7.11.1. The following procedures apply if combined communications and navigation failures make it impossible or inadvisable to comply with the loss of communications procedures. The triangular patterns indicated below are intended to alert ATC and military radar facilities that assistance is required.

7.11.2. With receiver only operating, the pilot flying shall monitor 121.5 MHZ and fly a right-hand, triangular pattern with one-minute legs and 120-degree turns at 1.5^0/second. At least two patterns should be flown before resuming course. A radar controller observing such a pattern shall respond on 121.5 MHZ.

7.11.3. With transmitter and receiver inoperative, the pilot flying shall fly a left-hand triangular pattern as described above. A radar controller observing such a pattern shall dispatch an escort if possible. If at night or in IMC, the landing lights should be turned ON to aid the escort aircraft. When contact is established the pilot should follow the escort. (See Interception Signals in FOM 3.3.)

* * * *

SECTION 3.9 EMERGENCY RADIO COMMUNICATIONS

SECTION 3.10 EMERGENCY PHASES

SECTION 3.10 EMERGENCY PHASES

1.0 PURPOSE

The purpose of this section is to establish guidelines for an emergency phase of operation.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- CEO
- VP Flight Operations
- Flight crewmembers
- Cabin crewmembers
- Air traffic controllers

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.03

When an aircraft is known or considered to be in a state of emergency ATC units will, declare one of the following emergency phases as appropriate:

4.1. UNCERTAINTY PHASE

4.1.1. No communication has been received from an aircraft within a period of thirty minutes after the time a communication should have been received or from the time an unsuccessful attempt to establish communication with such aircraft as first made, whichever is the earlier; or

4.1.2. When an aircraft fails to arrive within thirty minutes of the estimated time of arrival last notified to or estimated by ATC units whichever is the later; Except that doubt exists as to the safety of the aircraft and its occupants.

4.2. ALERT PHASE

4.2.1. Following the uncertainty phase, subsequent attempts to establish communication with the aircraft or inquiries to other relevant sources have failed to reveal any news of the aircraft; or

4.2.2. When an aircraft has been cleared to land and failed to land within five minutes of the estimated time of landing and communication has not been reestablished with the aircraft; or

4.2.3. When Information has been received which indicates that the operating efficiency of the aircraft has been impaired, but not to the extent that a forced landing is likely; Except that evidence exists that would allay apprehension as to the safety of the aircraft and its occupants.

4.3. DISTRESS PHASE

4.3.1. Following the alert phase and further unsuccessful attempts to establish communication with the aircraft and wider Spread unsuccessful inquiries point to the probability that the aircraft is in distress; or

4.3.2. When the fuel on board is considered to be exhausted, or to be insufficient to enable the aircraft to reach safely, or

4.3.3. When information is received which indicates that the operating efficiency of the aircraft has been impaired to the extent that a forced landing is likely; or

4.3.4. When information is received, or it is reasonably certain that the aircraft is about to make or has made a forced landing. Except when there is reasonable certainty that the aircraft and its occupants are not threatened by grave and imminent danger and do not require immediate assistance.

5.0. DEFINITION

- **SAR-** Search and Rescue

6.0 RESPONSIBILITY

6.1. The Operational Planner is responsible to notify VP Flight Operations and Director Flying Operations receipt of emergency message or absence of communication within the last 30 minutes.

6.2. The ATC Authority is responsible to notify the Rescue Coordination Center and set in motion the search terms as defined by ICAO Annex 12 Search and Rescue.

6.3. VP Flight Operations is responsible to liaison between the Rescue Coordination Center (RCC) and Flight Control to ensure maximum coordination of the company and RCC teams.

7.0 PROCEDURE

7.1. NOTIFICATION (AIRBORNE)

Whenever a serious failure, defect, mechanical hazard, or other irregularity which may subsequently result in an emergency being declared occurs during flight, the Captain shall, if possible, notify the appropriate ATC unit and the nearest company facility by the fastest means available. The Operational Planner or Station Manager, upon receiving information that one of the preceding emergency phases has been declared involving a company aircraft, shall initiate the appropriate alarm message in accordance with the following notification plans:

7.3. NOTIFICATION (GROUP)

7.3.1. When the UNCERTAINTY phase has been declared, the following notification procedure shall apply:

a) The Operational Planner will confirm that ATC has received the emergency message from the aircraft, or no communication within 30 minutes of the last due communication, he shall notify the VP Flight Operations, Director Flying Operations and the Manager Flight Control.

b) If Distress is declared, the ATC Authority shall notify the Rescue Coordination Center and set in motion the search terms as defined by ICAO Annex 12 Search and Rescue.

c) VP Flight Operations shall provide liaison between the Rescue Coordination Center (RCC) and Flight Control to ensure maximum coordination of the company and RCC teams.

7.4. EN-ROUTE AND STATION EMERGENCIES

7.4.1. The appropriate ATC/SAR unit in close coordination with the designated company flight watches office will handle en-route emergencies. During the various emergency phases ATC units will make every effort to establish communication with the aircraft and/or obtain information of its progress from other sources.

7.4.2. The responsible Operational Planner or Station Manager shall supply the appropriate ATC/SAR units with all available information essential for the effectiveness of search and rescue operations throughout subsequent emergency phases.

7.4.3. This shall include:

a) Pertinent information included in the flight plan;

b) Endurance at time of last report;

c) Number of crew and passengers aboard;

d) Emergency equipment carried on board the aircraft;

e) Color and distinctive marks of aircraft when deemed necessary;

f) Any other essential information pertinent to the safety of the aircraft.

7.4.4. A chronological log shall be maintained by the responsible Operational Planner or Station Manager, indicating details of the emergency situation during the various phases, actions taken, and results. An alarm message shall be initiated in accordance with the applicable notification plan.

7.4.5. When an emergency or accident occurs involving an aircraft on the ground or on landing or takeoff, the Operational Planner shall initiate the alert as required by the state in which the accident occurs. At stations where contract personnel perform the duties of the Operational Planner, the Station Manager shall assume this responsibility.

7.4.6. Company employees should not make unauthorized statements to the news media or to any individual not directly associated with the search and rescue unit.

7.4. TESTIMONY OR STATEMENT FOLLOWING AN ACCIDENT

No crewmember is allowed to give any information, documents or materials to any person or agency without prior authorization from the CEO/VP Flight Operations.

7.5. SECURITY OF THE AIRCRAFT FOLLOWING AN ACCIDENT

7.5.1. When all that is possible has been done concerning the welfare of the passengers and crew, the Captain should take whatever precautions he can to safeguard the aircraft and its contents.

7.5.2. After an accident the captain is responsible to take precautions to safeguard the aircraft and its contents (historical records) until the takeover of an investigation team.

7.5.3. Following an accident, it is important to enlist the cooperation of airport officials and the police to ensure that vital evidence is not lost by unnecessary interference with wreckage before the arrival of the official investigating team.

7.6. SIGHTING AIRCRAFT WRECKAGE

7.6.1. The sighting of aircraft wreckage should be reported as soon as practical to the nearest air-to-ground communications facility. The report should include:

a) Location of the wreckage.

b) Type of aircraft if known evidence of survivors.

7.6.2. If circumstances permit, the flight should remain at the scene as a guide to search and rescue efforts.

7.7. OBSERVING AN ACCIDENT

7.7.1. When a Captain observes that either another aircraft or a surface craft is in distress, he shall, unless he is unable to or in the circumstances of the case, considers it unreasonable or unnecessary:

a) Keep in sight of the craft in distress until such time as his presence is no longer necessary or until he is no longer able to remain in the vicinity of the distressed craft.

b) Report to the rescue coordination center or ATC unit as much of the following information as possible:

I. Type, identification, and condition of aircraft in distress;

II. Position expressed in geographical coordinates or in distance and true bearing from a distinctive landmark or radio fix;

III. Time of observation expressed in hours and minutes UTC;

IV. Number of persons observed;

V. Whether persons have been seen to abandon craft in distress;

VI. Number of persons observed to be afloat;

VII. Apparent physical condition of survivors.

c) Act as instructed by the rescue coordination center or the ATC unit.

7.7.2. If the Captain of the first aircraft to reach the accident is unable to establish communication with the RCC or ATC unit, he shall take charge of activities of all other aircraft that arrive until such time by mutual arrangement he hands over control to the aircraft best able to provide communication in the prevailing circumstances.

7.8. SIGNALS WITH SURFACE CRAFT

7.8.1. When it is necessary for an aircraft to direct a surface craft to the place where an aircraft is in distress, the aircraft shall do so by transmitting precise instruction by means at its disposal. If such instructions cannot be transmitted (or when necessary for any other reason) instructions shall be given by using the following procedures performed in sequence:

a) Circle the surface craft at least once;

b) Cross the projected course of the surface craft close ahead at low altitude rocking the wings or opening and closing the throttle or propeller pitch;

c) Heading in the direction in which the surface craft is to be directed.

NOTE: According to international SAR commendations, a large percentage of ships are now equipped with VHF/AM 121.5 and according to the rules, when they are circled by an aircraft, they switch on VHF 121.5 and 2182 HF/AM and attempt voice contact.

7.8.2. When the Captain considers that assistance of the surface craft is no longer required, he shall so advise the surface craft by crossing its wake close astern at low altitude and; rock the wings or opening and closing the thrust lever or changing the propeller pitch.

7.8.3. The Captain should expect maritime signals in response as follows:

a) For acknowledging receipt of signals:

I. The hoisting of the "code pennant" (vertical red and white stripes) close up (meaning understood);

II. The flashing of a succession of "T's" by signal lamp in Morse code.

b) For inability to comply:

I. The hoisting of the international flag "N" (a blue and white checkered square);

II. The flashing of a succession of "N's" in Morse code.

7.9. SIGNAL CODE FOR USE BY SURVIVORS

7.9.1. When the symbols described in the following table are used, they have the meaning indicated. Insofar as is possible, the following instructions shall be followed:

a) Form symbols by any available means, such as strips of fabric, aircraft parts, pieces of wood, stones, etc. tramp trenches in sand or snow. Build fences of any materials that will make shadows to form the symbols. Provide as much color contrast as possible between material used and the background;

b) Make symbols not less than 2.5 meters (8 feet) long and as conspicuous as possible;

c) Lay symbols exactly as depicted to avoid confusion with other symbols;

d) Make every effort to attract attention by other means, such as radio, flares, smoke, or reflected light.

7.10. GROUND/AIR VISUAL SIGNAL CODE FOR USE BY SURVIVORS

TABLE 1

NO.	MESSAGE	CODE SYMBOL

1	Require assistance	V
2	Require medical assistance	X
3	No or Negative	N
4	Yes or Affirmative	Y
5	Proceeding in this direction	↑

7.11. SIGNAL CODE BY GROUND SEARCH PARTIES

TABLE 2

NO.	MESSAGE	CODE SYMBOL
1	Operation completed	LLL
2	We have found all personnel	LL
3	We have found only some personnel	++
4	We are not able to continue. Returning to base	XX
5	Have divided into two groups. Each proceeding in direction indicated	← / →
6	Information received that aircraft is in this direction	→ →
7	Nothing found. Will continue to search	NN

7.9.2. When a ground signal has been displayed and is understood, the aircraft shall acknowledge the signal by:

a) During the hours of daylight, by rocking the wings;

b) During the hours of darkness, turning the landing lights or navigation lights on and off twice.

7.9.3. Absence of any of the above signals indicates that the ground signal has not been received/ understood.

* * * * *

SECTION 3.10 EMERGENCY PHASES

SECTION 3.11 EMERGENCY LANDING OR DITCHING

1.0 PURPOSE

The purpose of this Emergency landing or Ditching procedure is to establish guidelines to be followed during emergency landing or ditching situations.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight crewmembers
- IOCC
- Cabin Attendants

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.03

4.1. LANDING AT UNAUTHORIZED AIRPORT

4.1.1. Only in an emergency may a flight land at an airport not authorized for use in normal operation. In this context an emergency airport is an airport that is not authorized, has no refueling provisions is not alternate airport or that is not an airport designed as a destination or alternate airport in a flight release.

4.1.2. In the event that the captain decides landing at an unauthorized airport is the safest course of action, the Captain must notify IOCC.

5.0 DEFINITION

UNAUTHORIZED AIRPORT: is an airport that is not authorized, has no refueling provisions and is not alternate airport or that is not an airport designed as a destination or alternate airport in a flight release.

6.0 RESPONSIBILITY

6.1. The Pilot-in-command is responsible to follow and comply with the procedures stated in this section.

6.2. Cabin attendants are responsible to perform activities listed in this section and CCOM.

7.0. PROCEDURE

7.1. FLIGHT DECK PREPARATION FOR CRASH LANDING/ DITCHING

7.1.1. When the decision has been made to ditch, complete the ditching preparation checklist as outlined in each aircraft's Operations Manual and in the cabin attendant's manual.

7.1.2. Survival after a forced landing at sea depends to a great extent on how rapidly rescue is affected. Therefore, it is important that ground stations are advised as soon as possible of any occurrence that could result in a ditching.

7.1.3. Once the decision has been made to ditch, the Captain should consider taking advantage of the ditching assistance provided by ocean station vessels if any are operating nearby. They are normally able to illuminate a sea-lane for ditching at night and to provide radar vectors to a ditching near the vessel when weather is a factor. The Captain should also consider requesting another aircraft to remain in the area as an escort. If an ocean station vessel is not in the area, an aerial escort can be helpful in directing rescuers to the ditching aircraft. The Captain should determine the full assistance potential from any station available.

7.1.4. SEND DISTRESS SIGNALS

Transmit MAYDAY, current position, course, speed, altitude, situation, intention, time and position of intended touchdown, type of aircraft, and request Airport Surveillance Radar (ASR) intercept using prevailing air to ground frequency. If available Set transponder code 7700 and, if practical, determine the course to the nearest ship or landfall.

7.1.5. ADVISE CREW AND PASSENGERS

Alert crew and passengers are to prepare for ditching. Assign life raft positions and order all loose equipment in the aircraft secured. Put on life vest, shoulder harness and seat belts. Do not inflate life vest until after exiting the aircraft.

7.1.6. FUEL BURN-OFF

Consider burning off fuel prior to ditching if emergency permits. This will provide greater buoyancy and a lower VREF. However, do not reduce fuel to a critical amount, as ditching with power improves ability to properly control touchdown.

7.1.7. PASSENGER CABIN PREPARATION

Confer with cabin personnel either by interphone or by personal report to cockpit to assure that passenger cabin preparations for ditching are complete.

7.2. CABIN PREPARATION FOR CRASH LANDING/ DITCHING

7.2.1. The amount of preparation that can be made for an emergency landing or ditching depends on the time available before landing/ditching. Preparations must be as thorough as time permits. The following should be accomplished:

a) For a ditching, complete the preceding announcements, as outlined.
b) Review the general evacuation plan, including the operation of the seat belts, doors, slides, emergency exits, life vests, lifelines, and life rafts.
c) Assign primary and alternate exits to passengers.
d) Assign competent passengers to aid infants, children, and the handicapped.
e) Place hand luggage, loose items that cannot be stowed securely, and potentially dangerous personal articles in lavatories and lock the doors from the outside.
f) Turn off the galley power.
g) Secure the galley and class divider curtains open.

7.2.2. Before a ditching, all available time should be used to refine the passenger's knowledge of evacuation equipment and procedures.

7.2.3. At least two competent passengers should be assigned to each exit with instructions to assist in positioning life rafts and in evacuating passengers from the aircraft. Life rafts should not be removed from their stowage areas nor should any exit be opened until the aircraft has come to a complete stop.

7.3. ANTICIPATED CRASH LANDING/DITCHING

7.4.1. Press **ALERT CALL** (for A350**; EMER pb**) or PIC calls over PA "Team Leader to the flight deck" and brief Team Leader Cabin Services the following items:

a) Type of emergency.
b) If evacuation is required.
c) Time available and if there is any special instructions to be followed.

7.4.2. Flight crew will announce to the passengers through PA:

a) Type of emergency.
b) Time available.
c) To fasten their seat belts and prepare for an emergency water landing (crash landing).

d) To follow cabin crew instruction.

7.4.3. Depending on time availability cabin attendants brief and demonstrate information to the passengers according to CCOM.

7.3.4. 500 FEET OR 1 MINUTE BEFORE IMPACT

Captain or a designated crewmember will announce: **"CABIN CREW, BE SEATED".**

NOTE: When this announcement is made at night, the Cabin lights are to be turned off.

7.3.5. 200 FEET BEFORE IMPACT

Captain or a designated crewmember will announce: **"BRACE FOR IMPACT".**

7.4. UNANTICIPATED CRASH LANDING/DITCHING

Since most of unanticipated accidents happen during takeoff and landing, the fight crew has no time to make detailed passenger announcement, therefore the flight crew will announce **"brace, brace, brace"** to make passengers and crew members aware of the emergency situation.

7.5. DITCHING FINAL

7.4.4. Report final position, select flaps as outlined in the Aircraft Operations Manual appropriate for the existing emergency or non-normal condition.

7.4.5. Maintain airspeed at VREF. Use appropriate bug setting for flaps selected. Maintain 200 to 300 fpm rate of descent. Plan to touchdown on windward side and parallel to waves or swells, if possible. To accomplish flare and touchdown, rotate smoothly to touchdown attitude.

7.4.6. Maintain airspeed and rate of descent with thrust. After contact with the water, reduce thrust to idle.

7.6. DETERMINING THE DITCHING HEADING

7.5.1. Weather and sea conditions should be determined from weather sources if available. Normally, there are primary swells, each moving in a different direction. During daylight, the primary swells can be observed best at an altitude between 10,000 and 12,000 feet. Secondary swells become visible at lower altitudes.

7.5.2. The sea is nearly always rougher than it seems to be at altitude.

a) If the surface wind is more than 35 knots, the ditching should be made into the wind, regardless of the direction of the swells. However, a ditching into the upslope of a swell should be avoided.

b) **If the surface wind is less than 35 knots**, the best ditching heading is determined by whichever of the two headings below results in the greater headwind component, thus minimizing forward speed on initial contact with the water:

I. A heading parallel to the primary swells and down the secondary swells.

II. A heading parallel to the secondary swells and down the primary swells.

c) **If the wind is calm**, the ditching heading should be in accordance with (b), above. If there are one or more Secondary swells and a wind factor, the best ditching heading can be difficult to choose.

7.7. LANDING TECHNIQUE FOR DITCHING

7.7.1. If possible, the ditching should be accomplished while engine power is available to permit maneuvering to a favorable touchdown area. During a ditching at night, the landing lights should be used to illuminate the surface of the area.

7.7.2. The final approach and landing should be made with the gear up and full flaps, using VREF as target speed. (The landing gear warning should be silenced. (Pull the circuit breaker.)

7.7.3. Touchdown should be made with the slowest forward speed and the lowest descent rate consistent with adequate control. The pitch attitude should be about 10 degrees nose up. This provides optimum planning action and load distribution over the bottom of the fuselage. The wings should be level with the surface of the sea, not with the horizon. (See Aircraft Operations Manual for the type procedure.)

7.7.4. The aircraft should not be allowed to fall through to touchdown from a stall. This would result in a severe impact, which could collapse the bottom of the fuselage followed by an abrupt deceleration caused by the nose burying itself in the sea.

7.7.5. If time permits prior to an emergency landing that may require an aircraft evacuation, Flight Attendants shall brief the passengers on the exits to be used. Consideration should be given to the type of emergency in progress, and passengers must be instructed not be use certain exits if they are liable to be hazardous.

7.8. EVACUATION EXECUTION

7.8.1. The standard aircraft evacuation announcement is made by the Captain or designated crewmember:

7.8.2. **"EVACUATE, EVACUATE, EVACUATE LEAVE THE AIRCRAFT IMMEDIATELY THROUGH (LEFT/RIGHT/BOTH SIDE OF AIRCRAFT)."**

7.8.3. Additional instructions should be included as condition dictates.

7.8.4. When the order, "CABIN CREW AT YOUR STATION" is given, all cabin attendants should be seated and fasten their seat belts securely.

7.8.5. The bracing position is to be assumed on the command, "BRACE FOR IMPACT." Initial contact can be expected to occur in approximately 10 seconds.

7.8.6. Whether on land or water, an emergency landing can result in sudden and violent forces acting on the aircraft before it comes to a complete stop. Seat belts must not be unfastened until the aircraft has come to a complete stop.

7.8.7. After the aircraft has come to rest, proceed to assigned duty positions for evacuation and evacuate as soon as possible.

7.8.8. During an evacuation, the Captain shall assist the evacuation process at the forward cabin and the F/O will assist passengers outside the aircraft guiding them towards assembly point. For specific cabin crew assignments in an emergency refer to the CCOM.

7.8.9. The need for most emergency evacuations comes about with little or no warning. As a result, such an evacuation is conducted without the benefit of a specific briefing. This means that its success depends on each crewmember's knowledge of evacuation assignments and related considerations.

7.8.10. Whether evacuation is planned or unplanned, the Captain is to see that the command to evacuate and any other specific instructions are issued to cabin occupants as soon as possible. If the need for an emergency evacuation cannot be determined immediately, the following announcement should be made to prevent any unwanted evacuation, either by a cabin attendant or a passenger: **"CABIN CREW AT YOUR STATION."** The Captain must have the assurance that each cabin attendant, when anticipating a requirement to evacuate the aircraft, will wait for the Captain's direction.

7.8.11. Any abnormal occurrence on the ground that does not require an evacuation but that could be of concern to cabin occupants, such as a sudden stop or an unusual sound, it should be explained promptly to prevent any undesirable reaction on the part of the cabin attendants or passengers. After such an occurrence and before continuing, the Captain should check the door warning lights for assurance that an evacuation has not begun.

7.8.12. If no crewmember on the flight deck has provided the necessary direction, the team leader cabin services should attempt to contact a flight crewmember in the cockpit, either by interphone or in person to obtain instructions. An evacuation must not be initiated while the aircraft is moving.

7.8.13. When the command to evacuate is given, the evacuation must be conducted aggressively. The hazard to passengers and crew may increase as time passes. Variation from standard procedures should be avoided unless a specific condition clearly dictates a nonstandard course of action.

7.8.14. The suitability of an exit should be evaluated before it is used. Some considerations are height above the ground, its proximity to a fire, or its relationship to waves or the water line. If an exit is not safe to use, passengers should be directed to another exit. As each passenger reaches an exit, they must be urged to move rapidly through it and onto the slide or, in the event of ditching, into the raft or the water.

7.8.15. While the primary objective of an aircraft evacuation is to deplane all occupants promptly, consideration should also be given to removing the evacuation items such as fire extinguishers, first aid kits, flashlights, megaphones, and the emergency radio. The consideration becomes increasingly important if the evacuation is accomplished away from an airport, in a remote area, or in extreme temperature.

7.8.16. For land emergency landing, two competent persons should be assigned to precede other passengers down each slide and instructed to hold the slide in position and assist other passengers. This precaution is important in strong and gusty winds.

7.8.17. If the evacuation is conducted on land, passengers should be instructed to clear the bottom of the slide, move without delay as far away as possible, and remain assembled in-group. Passengers deplaning ahead of the wing should be directed behind the tail. This is intended to reduce the exposure of persons to fuel or fuel explosion.

7.8.18. After ditching when the aircraft has come to a stop, the life rafts should be removed from their stowage areas, carried to usable exits, and latched. Immediately following inflation, the rafts should be:

a) Attached to the wing life line or to some section of the aircraft structure using the inflation handle and Boarded by at least two full-sized persons.

b) If possible, passengers should be transferred directly from the aircraft into the life rafts. The importance of doing so depends on the sea state, the weather and the extent of personal injuries.

7.8.19. If unable to board the rafts directly from inside the aircraft, passengers should be instructed to inflate their life vests as they leave the aircraft. Persons on the wing should hold on to the wing lifeline. Those in the wing should hold on to a line to avoid drifting or being washed away. The first competent person to board the raft should ensure inflations, assist in the boarding of other persons and hold the raft away from damaged aircraft structure.

7.8.20. The center deck support should be inflated as soon as possible. Persons in the raft should assist other persons on board, preferably at a boarding station. It may be quite difficult for a person to board the raft from the water without help.

7.8.21. Persons in the raft should sit with their backs against the rail and their feet towards the center. Sharp objects, including shoes, that could damage the raft should be removed or otherwise disposed of. Persons should move on hands and knees, and unnecessary movement should be restricted.

7.8.22. When all persons are aboard the raft, the lanyard may be cut. The sea anchor should be deployed promptly. All loose equipment should be secured when it is in use so that it is not lost or washed overboard.

7.8.23. Rafts should be tied together only in a very calm sea. This would provide a larger target for search and rescue units and permit optimum distribution of rations and equipment. If there is any appreciable wave motion, rafts should be separated to avoid the hazards of collision and upset.

7.9. POST EVACUATION

7.9.1. Assuring that all passengers are out of the aircraft Disconnect slide/rafts from exit doors. Be careful not to rip or puncture slides. Avoid drifting or blowing into or under parts of the aircraft. Remain clear of fuel contaminated water.

7.9.2. As soon as the evacuation has been completed the Captain should direct activities as necessary to ensure the general welfare of the passengers and crew. Injured passengers should be given all available aid and comfort. If there is a physician or a trained nurse among the passengers, he or she should be asked to render professional assistance.

7.9.3. On land, when it becomes apparent that the condition of the aircraft is not hazardous, it may be re-boarded to remove items that could improve the passengers' comfort, such as first aid kits, life rafts, beverages, food, shoes, coats, and blankets.

7.9.4. All crewmembers should mix with the passengers and demonstrate an interest in them, reassuring them that arrangements for their welfare are underway. Crewmembers should not group together or disassociate themselves from the passengers.

7.9.5. When rescue assistance arrives, the Captain should oversee the orderly transfer of responsibility for the care of passengers from his own and his crew's jurisdiction to an adequate number of competent personnel. Until this is accomplished, the primary responsibility of the Captain and his crew is the welfare of the passengers. This must have priority over all other duties and responsibilities.

7.10. SURVIVAL

7.10.1. Ethiopian may not operate an aircraft across land areas which have been designated by ECAA as areas in which search and rescue would be difficult unless equipped with enough survival kits for the number of occupants appropriate for the route to be flown. (ECARAS 7.9.1.5)

7.10.2. Survivors should remain in the vicinity of the aircraft unless there is definite reason to believe that the search and rescue efforts will not locate it.

7.10.3. First aid instructions are in a survival booklet, which, together with a first aid kit, is in the necessary kit attached to each life raft.

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SECTION 3.11 EMERGENCY LANDING OR DITCHING

SECTION 3.12 EMERGENCY EQUIPMENT

1.0 PURPOSE

The purpose of this section is to provide guidelines on how to use the under listed emergency equipment during emergency.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Flight crewmembers
- Cabin crewmembers
- Medical Director

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.03

It is Ethiopian Airlines policy to have the following emergency equipment on aircraft for compliance with during emergency.

4.1. EMERGENCY LOCATOR TRANSMITER (ELT)

4.1.1. On all international flights, Ethiopian shall ensure aircraft utilized for such flights, are equipped with ELTs as follows:

I. For aircraft with more than 19 passenger seats, a minimum of two ELTs that operate on 121.5 MHz and 406 MHz simultaneously, one of which shall be automatic;

II. For aircraft with 19 passenger seats or less, a minimum of one automatic ELT that operates on 121.5 MHz and 406 MHz simultaneously.

4.1.2. Each Ethiopian aircraft used in international flights shall be equipped with a minimum of two emergency locator transmitters (ELTs), one of which shall be automatic; all ELT transmitters shall operate on 121.5 MHz and 406 MHz simultaneously,

4.1.3. The Operator should ensure all aircraft are equipped with a minimum of one automatic ELT that operates on 121.5 and 406 MHz simultaneously.

4.2. FLASH LIGHTS (TORCHES)

Each aircraft that is intended to be operated at night shall be equipped with a flashlight (torch) at each flight crew member station.

4.3. LAVATORY FIRE EXTINGUISHER

No Ethiopian airplane may be operated unless each lavatory is equipped with a built-in fire extinguisher that will automatically discharge into each disposal receptacle upon occurrence of a fire. (ECARAS 7.9.1.8)

4.4. PORTABLE FIRE EXTINGUISHERS

4.4.1. Portable fire extinguishers, of a type that will minimize the hazard of toxic gas concentration, shall be provided on board each aircraft.

4.4.2. At least one shall be located in:

a) The pilots compartment and
b) Each passenger compartment that is separate from the pilot's compartment and that is not readily accessible to the flight crew.
c) Each galley not located on the main passenger deck.

4.5. DRY CHEMICAL FIRE EXTINGUISHER

4.5.1. To operate, the user shall:

a. Hold the bottle erect;

b. Aim at base of flame, and

c. Lift handle;

d. Press the trigger.

4.5.2. The dry chemical extinguisher is used for fires on flammable liquids, greases, etc., where a blanketing effect is essential, and for fires in live electrical equipment, when the use of an electrically nonconductive extinguishing agent is of first importance.

4.5.3. Discharge in the cockpit is not permitted. Discharges directly on persons should be avoided due to the possibility of suffocation.

4.6. WATER FIRE EXTINGUISHER

4.6.1. To operate, the user shall:

a) Hold the bottle erect;

b) Turn handle clockwise as far as possible, thus charging the bottle with CO_2;

c) Direct at base of flame, and

d) Press trigger to discharge;

e) To charge in flight: unscrew handle, replace CO_2 cylinder, and reinstall handle. The extinguisher is now ready for use.

4.6.2. The water extinguisher is used for fires in ordinary combustible materials (fabric, paper, etc.) where quantities of water are highly effective. DO NOT use on electrical fires. Avoid using on grease and oil fires.

4.6.3. Antifreeze compound has been added to the water, making it unfit for drinking.

4.7. PROTECTIVE BREATHING EQUIPMENT (PBE)

4.7.1. All aircraft with a maximum certificated takeoff Mass exceeding 5,700 kilograms or having a maximum seating configuration of more than 19 seats shall be equipped with Crew Protective Breathing Equipment (PBE) as follows:

a) Equipment to protect the eyes, nose and mouth of each flight crew while on flight duty and to provide oxygen for a period of not less than 15 minutes. The supply for PBE may be provided by the supplemental oxygen present on board.

b) Sufficient portable PBE to protect the eyes, nose and mouth of all required cabin crew and to provide breathing gas for a period of not less than 15 minutes.

4.7.2. PBE intended for flight crew use shall be conveniently located on the flight deck and be easily accessible for immediate use by each required flight crew at their assigned duty station.

4.7.3. PBE intended for cabin crew use shall be installed adjacent to each required cabin crew duty station.

4.7.4. An additional PBE shall be provided at or adjacent to the hand held fire extinguisher except that, where the fire extinguisher is located inside a cargo compartment, the PBE shall be stowed outside but adjacent to the entrance to that compartment.

4.8. LIFE VESTS (JACKETS)

4.8.1. Each aircraft must carry one life jacket or any equivalent flotation device for each person on board and should be easily available from their seat when:

a) Flying over water and at a distance of more than 50NM (93km) away from the shore;

b) Flying en route over water beyond gliding distance from the shore;

c) Taking off or landing at aerodromes where the takeoff path or approach is so disposed over water that is in the event of a mishap, ditching would be likely.

4.8.2. For each infant, the flotation devise should be an infant specific equipped with a survivor locator light.

4.9. USING THE LIFE VEST

4.9.1. The life vest is to be inflated only after leaving the aircraft. If a life vest is inflated inside the aircraft, it could hamper the wearer's use of an emergency exit. If the cabin is partially submerged, it may be impossible to exit the aircraft wearing an inflated vest.

4.9.2. The life vest, with both lobes inflated, should keep a person upright with his head out of the water in normal sea. With a fully inflated life vest, boarding the life raft is difficult and ability to swim is reduced. The best swimming strokes while wearing a life vest are the sidestroke, breaststroke, and elementary backstroke. A good swimmer can make better progress in an un-inflated life vest.

4.10. LIFE VEST DONNING FOR CHILDREN

4.10.1. Open the life vest completely by extending the back flap fully. Place the vest over the child's head with the arms and legs through the strap loops.

4.10.2. Pull the two yellow PULL-tabs until the straps fit securely between the legs. Tie the straps around the waist. For a small child, one tube will provide ample flotation and the other can be held in reserve.

4.10.3. For a description of the light on the life vest, see Life Vest and Life Raft Lights in this section.

4.10.4. For the location of life vests see the Emergency Equipment Location diagram in the Emergency Chapter of the Aircraft Operations Manual.

4.11. LIFE RAFTS (ECARAS 7.9.1.18)

4.11.1. All Ethiopian aircraft shall have lifesaving rafts with a sufficient capacity to carry all persons on board in the event of the loss of one raft of the largest capacity.

4.11.2. All lifesaving rafts shall be stowed so as to facilitate their ready use in an emergency.

4.11.3. Life rafts shall be equipped with the following life sustaining equipment:

a) An electric survivor locator light
b) A survivor kit
c) A pyrotechnic signaling device and
d) An ELT

4.11.4. The rafts consist of twin buoyancy rails with the deck supported between the rails and bonded together to form an integral unit. The rails, when inflated, form an elliptical shape and are offset to facilitate boarding. An inflated step is provided at each boarding station. The outer sides of the rails are marked with arrows to indicate the direction to the nearest boarding station. The tip and bottom are identical so that the raft will be right side up regardless of which side turns up during inflation. The deck is made of a rubberized fabric material, and care should be taken to avoid ripping with high heel shoes or other sharp objects.

4.11.5. Stencils on the raft and instructions identify all inflation valves and attachments to the raft. The following are stenciled on both sides of the deck:

- International Morse Code
- Canopy erection instructions
- Location of the sea anchor
- Instructions for repairing the raft
- Instruction for using the air pump and inflation valves

4.11.6. Upon actuation, two air bottles fully inflate the raft rails within 10 seconds. The center deck support is a torus, which is inflated with a pump provided for that purpose.

4.11.7. The canopy provides protection from sun and weather. A centermost and support poles around the edge at the life raft rail support it. The canopy can also be used to collect to the offset rail design of the raft the canopy is oval. Alignment arrows are marked on the canopy and raft to assist in erection the canopy properly.

4.11.8. The carrying case encloses the entire raft. Two carrying handles are attached to the case. The inflation cord handle is under the red flap. The accessory kit is an integral part of the carrying case. It is attached to a rail and, after raft inflated, its location is indicated on the rail. The minimum contents of the survival kit are shown below.

ITEMS	QTY.	ITEMS	QTY.
Instruction	2	Compass	1

Books			
Survival book	1	Sponge	1
Food ration kits	6	Flares, MK-13	4
Repair clamp	1	First aid kit	1
Pliers	1	Water container (Plastic)	1
Sea dye marker	3		
Jackknife	1	Flashlight (2-cell)	1
Signal whistle	1	Desalting kits	6
Signal Mirror	1	Flash light	
		Batteries	2

4.11.9. Two people using the long straps around the circumference of the case or the two end handles should carry the life raft in its carrying case from its stowage area. One person can drag the raft using one of the end handles. To avoid inadvertently opening the snaps and dumping the raft out of the carrying case (which could inflate it prematurely), the raft should not be carried with the snap side down. The un-inflated raft should be placed outside the aircraft holding the end of the carrying case near the inflation lanyard handle. The raft is inflated by pulling the handle and lanyard completely from the container.

4.11.10. Two 75-foot heaving lines are located in pockets in the tip and bottom rails. One mooring line, a 20-foot length of nylon cord, is provided. These lines (or escape ropes) must not be used to secure the raft to the aircraft, as this will cause the raft to sink if the aircraft sinks. The inflation lanyard is designed to break if the aircraft to prevent it from drifting away.

4.11.11. There may be some minor differences between one raft and another. The company will provide training in the use of this equipment, prior to assigning crewmembers to flight duties. See Aircraft Operations Manual for specific storage procedures.

4.12. VEST AND RAFT LIGHTS

4.12.1. Lights on life vests and life rafts are the same type and batteries and light units are interchangeable. The light is powered by a water-activated battery, which is submerged when the life vest or life raft is in use. Fluids other than water can be used; however, the brightness of the light may be affected. Illumination is produced about two seconds after the battery is submerged in water. The useful operational life is approximately 10 hours. To discontinue operation and conserve battery life, the battery must be removed from the water, and the water in the battery shaken or blown out.

4.12.2. These lights should be operated only during periods of darkness. There are four lights on each life raft, each with a battery. Two lights are located on each rail, upper and lower, with the batteries located on the rail. This permits the operation of two lights, regardless of which side of the raft is up. Submerged batteries may be removed from their pockets to conserve battery life. Batteries on the top rail (for the submerged lights) may be removed and used as spares.

4.12.3. The emergency transmitter is a fully automatic device which, when activated, transmits an Omni directional signal as an aid to location during search and rescue.

4.12.4. When the transmitter is in the stowed position a whip antenna is folded from one end, back along the case of the transmitter. The antenna is held in the stowed position with water-soluble tape. After approximately 30 seconds of immersion, the tape dissolves and releases the antenna. A 60-foot lanyard attached to the transmitter and wrapped around the case, may be used to attach the transmitter to the life raft.

4.12.5. The transmitter operates on two frequencies simultaneously, 121.5 MHz and 406 MHz The transmitted signal, which has line-off-sight characteristics, is a swept tone, identifying it as an emergency signal.

4.12.6. The battery is dry and therefore inert until immersed in water or other aqueous fluid. The transmitter is activated automatically after immersion. Activation by salt water is almost immediate. Fresh water or other fluid, or immersion in a container of fluid, may delay activation by as much as 15 minutes. Operating life of the transmitter is in excess of 48 hours.

4.13. CRASH AXE

4.13.1. All Ethiopian airplane shall be equipped with a crash axe appropriate for effective use in that type of airplane. It must be stored in a place not visible to passengers on board the airplane. (ECARAS 7.9.1.10)

4.13.2. At least one crash axe should be placed on in the flight deck.

4.14. MEGAPHONES

4.14.1. Portable battery-powered megaphones are provided for use during emergencies to aid voice communications with passengers or rescuers.

4.14.2. One megaphone shall be available for aircraft with seating capacity of 60 – 100 passengers and two megaphones for aircraft with seating capacity of more than 100 passengers.

4.14.3. For the location of megaphones see the Emergency Equipment Location diagram in the Emergency Chapter of the Aircraft Operations Manual.

4.15. FIRST AID KITS

4.15.1. According to regulatory requirements, each passenger aircraft must be equipped with first-aid-kit. First aid kits must be located near exits so that they are easily accessible to cabin crew, and, in view of the possible use of medical supplies outside the airplane in an emergency situation.

a) One kit for aircraft with less than 100 passengers seats

b) Two kits for aircraft with 100 to 199 passengers seats

c) Three kits for aircrafts with 200 to 299 passengers seats

d) Four kits for aircraft with 300 or more passengers seats

4.15.2 In addition to the above requirements all cargo aircraft must be equipped with one first aid kit that is readily accessible to the flight crew and supernumeraries.

4.16. MEDICAL KIT/ DOCTORS KIT

4.16.1. According to regulatory requirements, each passenger aircraft with more than 100 passenger seats must be equipped with one medical kit.

a) Only a qualified medical professional can utilize the content.

b) The captain has to be informed in case of usage.

4.17 AUTOMATIC EXTERNAL DEFIBRILLATOR (AED)

Automatic External Defibrillator is provided on flights with duration of equal to or more than 5 hours.

4.18. SURVIVAL KIT

Survival kit is a kit that contains equipment for survival and search/rescue aids for use by aircraft crew& passengers.

5.0 DEFINITION

N/A

6.0. RESPONSIBILITY

6.1. The Pilot-in-command has the responsibility to check the availability and serviceability of the aforementioned emergency equipment, as applicable under his responsibility

6.2. The assigned cabin crew is responsible for checking the availability of the emergency equipment in their assigned location and their serviceability, as applicable, under her responsibility

7.0 PROCEDURE

N/A

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SECTION 3.12 EMERGENCY EQUIPMENT

SECTION 3.13. INCIDENT/ ACCIDENT & MANDATORY REPORTABLE OCCURRENCE

1.0. PURPOSE

The purpose of this procedure is to establish guidelines for Accident \Incident or occurrence reporting

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERONS AFFECTED

- Employees of the Airline
- Employees of ECAA

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.03

The Accident/Incident or Occurrence Reporting System is established to assist ETHIOPIAN and Ethiopian Civil Aviation Authority in early identification of potential hazards and system deficiencies, and to assist in the assessment of associated hazards or system deficiencies.

5.0. DEFINITION

5.1. AIRCRAFT ACCIDENT Means an occurrence associated with the operation of an aircraft which takes place between the time any person boards the aircraft with the intention of flight and all such persons have disembarked, and in which any person suffers death or serious injury, or in which the aircraft receives substantial damage.

5.2. SERIOUS INJURY-Means an injury which is sustained by a person in an accident and which:

- requires a stay in the hospital for more than 48 hours commencing within seven days from the date on which the injury was received; or
- results in a fracture of any bone (except simple fractures of fingers, toes or nose); or
- Involves lacerations which cause nerve, muscle or tendon damage or severe hemorrhage; or
- involves injury to any internal organ; or

- involves second or third degree burns or any burns affecting more than five percent of the body surface; or
- Involves exposure to infectious substances on injurious radiation.

5.3. **SUBSTANTIAL DAMAGE** Means damage or failure which adversely affects the structural strength, performance, or flight characteristics of the aircraft, and which would normally require major repair or replacement of the affected component.

5.4. AIRCRAFT INCIDENT is any aviation occurrence involving, but not limited, an aircraft where:

- Engine failure, fire, flameout, shutdown or significant malfunction,
- Smoke or fire occurs,
- Difficulties in controlling the aircraft in flight are encountered due to any aircraft system malfunction, weather phenomena, wake turbulence, operations outside the approved flight envelope or uncontrolled vibrations,
- Rejected take-off,
- Go-around (not related to weather), rejected landing,
- The aircraft fails to remain within the landing and take-off area, lands with one or more landing gear retracted or drags a wing tip, tail or engine pod,
- Landing gear malfunction,
- A hard landing - a landing deemed to require a "hard landing check",
- Overweight landing,
- Any crew member is unable to perform their flight duties as a result of incapacitation,
- Decompression, explosive or otherwise, occurs that necessitates an emergency descent,
- A diversion,
- A fuel shortage occurs that necessitates a diversion or requires approach and landing priority at the destination of the aircraft,
- The aircraft is refueled with the incorrect type of fuel or contaminated fuel,
- A collision or risk of collision with any other aircraft or with any vehicle, terrain or obstacle occurs, including a collision or risk of collision that may be related to Air Traffic Control procedures or equipment failures,
- Taxiway incursion or excursion,
- The aircraft receives a Traffic Alert and Collision Avoidance System (TCAS) Resolution Advisory,
- Air misses,

- A flight crew member declares an emergency or indicates any degree of emergency that requires priority handling by an Air Traffic Control Unit or by Crash, Fire, or Rescue Services,
- Toxic gases or corrosive materials leak from any area aboard the aircraft,
- Turbulence encounter - an encounter resulting in injury to occupants or deemed to require a "turbulence check" of the aircraft,
- Lightning strike,
- Cracked or shattered cockpit window(s),
- Significant loss of braking action,
- Operation of any primary warning or caution system associated with the aircraft systems or equipment unless: the crew conclusively established that the condition was false at the time it occurred; or, the indication was confirmed as false immediately after landing,
- Failure of emergency system or equipment, including any exit door and lighting, to perform satisfactorily,
- Unauthorized incursion or operating irregularity involving vehicles, pedestrians or animals,
- Failure of a navigational aid, approach aid, communications system, airport lighting, power failure or any other system breakdown which has an adverse effect upon flight safety or major impact on operations,
- Criminal action - hijacking, bomb threat, riot, sabotage, or breach of aviation/airport security,
- Unavailability of a runway due to any obstructions or foreign object that results in a major impact on airport operations,
- Bird strikes,
- Missing aircraft reports, Search and Rescue activation, Emergency Locator Transmitter (ELT) activation,
- Labor action affecting operational capability,
- Item dropped from an aircraft,
- Regulatory infractions which have immediate safety implications involve commercial Air Operators or may generate media attention,
- Environmental emergencies such as a fuel spill, hazardous chemical or radioactive spill on airport property,
- Death or serious injury to employees or members of the public while onboard an aircraft,
- Any occurrence which may generate a high degree of public interest or concern, or could be a direct interest to specific Foreign Civil Aviation Authorities,

6.0. RESPONSIBILITY

6.1. IMMEDIATE REPORTING

6.1.1. ETHIOPIAN shall forward the operations Accident/Incident or Occurrence Report for those occurrences classified as Critical and High Profile Events immediately after the event by available information transmitting media to ECAA no later than 24 hours from the time of the occurrence. Dir. Corporate QA, SMS, EMS, Industrial safety and ERP shall report to ECAA the results of their investigation of the occurrence using appropriate Operations Occurrence Report form.

6.1.2. The ECAA reserves the right to conduct a parallel or independent investigation concerning all occurrences which may influence or have influenced aviation safety.

6.2. INFORMATION

6.2.1. The reporting of accidents/incidents under ETHIOPIAN system is a mandatory requirement.

6.2.2. ECAA encourages voluntary confidential reporting to the same criteria across the whole spectrum of the Civil Aviation Operations. Without prejudice to the proper discharge of its responsibilities, the ECAA will not disclose the name of the person submitting a voluntary confidential report or of a person to whom it relates unless, in either case, the person authorizes disclosure.

7.0. PROCEDURE

The reporting of all accidents/incidents will assist the ECAA in responding to events that may have safety implications to all Air Operators, or Civil Aviation Transportation System.

7.1. REPORTING OF INCIDENTS

7.1.1. Make a written report, when appropriate, to the Dir. Flying Operations for any irregularity for which the company could be held liable.

7.1.2. The Captain is responsible for the reporting of safety related occurrences and incidents/ accidents such as near miss (Air Traffic Incident Report), bird strike, aircraft accident or damage report, report on faulty radio communication, nonvisual and visual navigational aids, using the forms provided in the flight folder to ECAA, Dir. Flying Operations and Director corporate Safety and QA as soon as possible.

7.1.3. The time and place of each dumping estimated quantity of fuel dumped shall be recorded in the technical log and Captain’s Report

7.1.4. Captain must submit written report on any act of passenger misconduct.

7.1.5. Whenever possible, the Captain must advise the Operational Planner and/or ECAA immediately, in the event of any accident or incident resulting in damage to the airframe, property or passengers.

7.1.6. It is essential that incident reports are completed as soon as possible after landing and dispatched by the quickest means possible to the VP Flight Operations.

7.1.7. The Captain must submit incident reports on the form, which is provided to cover all system failures, in which the safety of the aircraft was or could have been endangered. Examples of such failures are:

- Hydraulic System Failure.
- Pneumatic system Failure.
- Landing Gear Systems failure.
- Flight Control System defect.
- Pressurization Failure.
- Emergency landing
- Deviation from ATC procedures
- Any external part of the aircraft becoming detached.
- Lightning Strike
- Bird Strike
- Near miss
- Fire
- Engine Shutdown (intentional or unintentional)
- Severe Turbulence/ Wind shear
- Hard Landing
- Overweight Landing
- Leaving the paved surface during taxi, take-off or landing.
- Collision with vehicle, equipment or obstructions.
- When injury or damage is caused to persons, property or other aircraft either on the ground or in flight.
- Dangerous Goods related occurrences.

7.1.8. PIC shall submit, without delay, an air traffic incident report whenever an aircraft in flight has been endangered by:

a) A near collision with another aircraft or object

b) Faulty air traffic procedures or lack of compliance with applicable procedures by ATC or by the flight crew

c) A faulty ATC facility.

7.1.9. GPWS alerts and Warnings, whether spurious or genuine must be reported.

7.2. REPORTING OF ACCIDENTS

7.2.1. Any aviation occurrence where, at any time during the period commencing when the first person boards an aircraft for the purpose of flight and ending when the last person disembarks from the aircraft after the flight:

a) Any person sustains serious injury or fatal injury that is not self- inflicted or caused by natural causes as a result of that person:

I. Being in the aircraft,

II. Coming into direct contact with any part of the aircraft, including any part that may have detached from the aircraft, or

III. Being directly exposed to jet blast/propeller wash of the aircraft;

b) Aircraft sustains damage or structural failure adversely affecting the structural strength, performance or flight characteristics of the aircraft normally requiring major repair or replacement of any affected component part.

7.2.2. The PIC shall notify the nearest appropriate authority, by the quickest available means, of any accident involving his/her aircraft that results in serious injury or death of any person, or substantial damage to the aircraft or property/equipment.

7.3. AIRPLANE INCIDENT OR ACCIDENT

Incidents (operational or non-operational) or airplane accidents must be reported immediately, by the quickest means possible to the V.P. Flight Operations.

7.4. CRITICAL AND HIGH PROFILE EVENTS

a) Aircraft evacuation due to potential hazard

b) Dangerous Goods Spill affecting safety

c) Aircraft Hijacking

d) Hostage Taking

e) Terrorist activities

f) Bomb Threats/Bomb Found

7.5. MECHANICAL RELIABILITY REPORTING

7.5.1. The company shall report to the Civil Aviation Authority the occurrence or detection of each failure, malfunction, or defect involving:

a) Fires during flight and weather the related fire warning system functioned properly.
b) Fires during flight not protected by a fire warning system.
c) False fire warning during flight.
d) Engines exhaust system that caused damage during flight to the engine adjacent structure, equipment, or components.
e) An aircraft component that cause an accumulation or circulation of toxic or noxious smokes or vapor in the crew or passenger compartment.
f) Engine shutdown in flight because of flame out, external damage to the engine, foreign objects ingestion, or icing.
g) Shutdown of more than one engine.
h) A fuel or fuel dumping system that affected fuel flow or caused hazardous leakage.
i) A landing gear extension or retraction in flight, or the opening or closing of landing gear doors in flight.
j) Loss of brake actuating force in an airplane in motion on the ground.
k) Cracks, permanent deformation of structure, or corrosion of airplane structure in excess of manufacturer's specified limits.
l) The failure of any airplane system or component necessitating emergency action during flight.

7.6. ADDITIONAL REPORTABLE ITEMS

7.6.1. The following incidents shall have an entry made in the airplane technical log:

a) Static discharges
b) Lightning/hail strikes
c) Bird strikes or bird ingestion
d) Use of emergency equipment
e) Possible snow, slush, or water damage to the airplane
f) Severe turbulence, wind shear, wake turbulence or severe weather encounters

g) Overweight or hard landings

h) Spills of hazardous corrosives

i) Spills of radioactive materials

7.7. SUPPLEMENTARY INFORMATION

Following the initial notification, additional information should be forward as it becomes available.

7.8. MANDATORY WRITTEN REPORTS

7.8.1. The importance of submitting a precise and detailed report cannot be over emphasized; as such reports provide a means of protecting passengers, crewmembers, and the company from any legal issues that might arise.

7.8.2. Reports are not limited to those listed; if a situation arises the Captain considers worthy of a written report, he should submit one.

7.8.3. Notwithstanding such determination, the following must be reported:

7.9. SABOTAGE THREAT

Include the source of the threat, whether by persons on board or by radio, and the action taken. If the flight is continued, give the reason for doing so.

7.10. LANDING AT AN UNAUTHORIZED AIRPORT

This is not the same as landing at an alternate airport; therefore, the sequence of events leading to the decision to land at an unauthorized airport must be given.

7.11. CABIN FIRE

Report the source and intensity of the fire and the extent of damage incurred. Did Cabin Attendants follow proper procedures? Was the flight crew alerted? Was their help necessary in extinguishing the fire?

7.12. INCAPACITATION

Report the type of incapacitation (e.g., loss of mental or physical function, illness, injury, or death) and phase of flight during which it occurred. Outline the course of action taken and state whether or not the flight was affected.

7.13. FLIGHT CONTROL SYSTEM MALFUNCTION/FAILURE

Report all flight control system malfunctions or failures and the effect on flight safety.

7.14. DIVERSION

Weather conditions are to be reported at the destination and alternate airports at time of diversion. Report any unusual problems encountered, time on the ground and whether the flight was able to continue. If the flight was canceled, were ground handling facilities adequate?

7.15. COMMUNICATIONS FAILURE

Report time of failure and procedures followed. If radio failure leads to other incidents, such as a near miss or navigational errors, include such details in the report.

7.16. NAVIGATIONAL ERRORS

Deviations approved by ATC due to weather or other reasons need not be reported. Deviations from flight plan due to radio failure or human error must be reported. Any deviation that would be cause for an ATC violation must be reported.

7.17. DECLARATION OF EMERGENCY

7.17.1. If either the Captain or the Flight Controller exercises his emergency authority, that individual must keep ATC and Flight Control fully informed of the progress of the flight. A written report of any deviation from established procedures or regulations is required.

7.17.2. A report of an incident or airplane accident will satisfy this requirement. The Captain of a flight given priority by ATC in an emergency must submit a written report of that emergency, if requested by ATC.

7.18. EXCEEDING MAXIMUM LANDING WEIGHT

Landing at a weight in excess of maximum landing weight requires a written report as well as an entry in the Technical Log. This report should describe the landing (rough or smooth) and indicate whether side loads were imposed on the landing gear during touchdown.

7.19. IN FLIGHT ENGINE SHUTDOWN

If an engine is shut down due to fire or malfunction, including precautionary shutdown, the Captain, in addition to providing a written report, shall advise ATC of the inflight stoppage and keep the receiving station or ATC advised of the flight's progress.

7.20. LANDING WITH AN ENGINE INOPERATIVE

7.20.1. Landing with an engine inoperative calls for a written report. If the flight lands anywhere other than the nearest suitable airport, the Captain must report in writing reasons for his selection, indicating why such action was as safe as, or safer than, landing at the nearest suitable airport.

7.20.2. The Captain shall report each stoppage of engine rotation in flight to the appropriate ATC facility as soon as practicable and shall keep the facility fully informed of the progress of the flight.

7.20.3. Additionally, the Captain, when appropriate, shall inform the appropriate company tracking facility via SITA, AFTN, or direct communication of the engine shutdown and his intended actions.

7.20.4. If the Captain lands at an airport other than the nearest suitable airport in point of time, he shall, upon completing the trip, submit a written report to the VP Flight Operations stating his reasons for determining that landing elsewhere other than at the nearest suitable airport was a safe course of action.

7.20.5. Within 10 days after the Captain returns to his home base, the VP Flight Operations shall submit a complete report to the Civil Aviation Authority.

7.21. REJECTED TAKEOFF

All rejected takeoffs shall be reported. If the rejected takeoff is the result of, or results in, an airplane accident, see "Airplane Incident or Accident" in this section for reporting requirements.

7.22. COCKPIT VOICE RECORDER

The flight crew shall not intentionally switch off the cockpit voice recorder (CVR). The Captain is authorized to deactivate the CVR to preserve cockpit voice data following an accident or serious incident. An entry should be made in the airplane maintenance log requesting removal of the voice recorder and shipment to V.P Flight Operations. The CVR shall remain deactivated until removal or authorized to reactivate by V.P Flight operations.

7.23. FLIGHT DATA RECORDER

In the event an airplane involved in an accident or incident, the preservation of all related flight recorder records and, if necessary, the association flight recorders, and their retention in safe custody pending their disposition as determined in accordance with Annex 13.

7.24. RESPONSIBILITY FOR NOTIFICATION

7.24.1. The Station Manger must report any aircraft incident or accident, which occurs at an online station, to Flight Operations immediately. The Captain of the flight involved shall render full assistance to the local Station Manager in preparing the notification.

7.24.2. This includes reminding him, if necessary, that such notification is required. The Captain is authorized to send a supplementary message if he thinks that it would add to the clarity or completeness of the report, and he is obligated to submit his own written report.

7.24.3. The Captain must:

a) Report all hazardous flight conditions to the appropriate ATC without delay.

   Examples include but not limited to severe weather condition, volcanic ash observed encountered, security breaches undesirable navigational aid performance, wild life, lasers, aircraft performance (e.g. loss of navigational capability etc…), inadequacy of navigational facilities or undesirable navigational aid performance, braking action or other irregularity in navigational or ground facilities, ELT broadcast, unmanned free balloons, etc…

b) Notify the nearest Authority, by the quickest available means, of any accident or serious incident resulting in injury, death, or substantial aircraft damage.

c) Notify the appropriate local Authority without delay in the event of any emergency situation that necessitated action in violation of local regulations and/or procedures;

d) Submit, if required by the State of occurrence, a report to the appropriate local Authority and also to the Authority of the State of the Operator.

NOTE: Ethiopian encourages any employee to report hazards or contributing factors to safety related accidents, incidents and events. The information reported as well as confidentiality of the reporter are of the utmost importance. Once the report has been received management will decide on any appropriate risk assessment action whether potential or real hazard can cause harm in any way to people, assets or the environment.

7.25. DANGEROUS GOODS ACCIDENT/ INCIDENT REPORTING PROCEDURES

In the event of a Dangerous Goods accident or Dangerous Goods incident, a report must be sent to the ECAA within 72 hours, unless exceptional circumstances prevent this. If necessary, a subsequent report shall be made as soon as possible giving all the details that were not known at the time the first report was sent. If a report has been made verbally, written confirmation shall be sent as soon as possible. Any type of accident or incident must be reported irrespective of whether the Dangerous Goods were in cargo, mail, stores, passengers' baggage of crew baggage.

A report must be made on any occasion when undeclared or miss-declared dangerous good are discovered in cargo, mail or stores. Such a report must be made to the ECAA and the state in which this occurred.

A report must be made on any occasion when Dangerous Goods that are not permitted are discovered in passengers' baggage or on their person after the passenger has checked in for the flight concerned. Such a report must be made it the appropriate authority of the state in which this occurred.

The first and any subsequent report should be submitted on the Ethiopian ASR form and should be as precise as possible and contain such of the following data that is relevant:

a) Date of the incident of accident or the flight date;
b) Location, the flight number and flight date;
c) Description of the goods and the reference number of the air waybill, pouch, baggage tag, ticket, etc.;
d) Proper shipping name (including the technical name, if appropriate) and UN/ID number, when known;
e) Class or division and any subsidiary risk;
f) Type of packaging, and the packaging specification marking on it;
g) Quantity of dangerous goods;
h) Name and address of the shipper, passenger, etc.
i) Any other relevant details;
j) Suspected cause of the incident or accident;
k) Action taken;
l) Any other reporting action taken; and
m) Name, title, address and telephone number of the person making the report.

Copies of relevant documents and any photographs taken should be attached to the report.

7.26. CONTENT AND ADDRESSEE

7.26.1. Written reports required in this section shall be submitted to the V.P. Flight Operations within 24 hours following the reported event. The Flight Crew Report form shall be used for this purpose. Written reports should include only as much of the following information as applies and shall not be delayed or withheld because some information may be unavailable:

a) Flight Description:

I. Flight designation
II. Date
III. Names as shown on the Flight Crew Report form plus others if applicable.
IV. Number of passengers
V. Stations of departure and arrival
VI. Dangerous articles or magnetized materials carried on board.

b) Airplane Description:

I. Airplane registration number
II. Gross weight and center of gravity
III. Flight profile: stage of flight, true airspeed, and altitude.
IV. Configuration: landing gear, flaps and speed brakes.
V. Airplane inoperative component (s) or malfunction.

c) Weather:

I. En-route weather: temperature, turbulence, lighting, and icing.
II. Airport weather: ceiling, visibility, temperature, wind direction and velocity, and altimeters setting.

d) Airport and NAVAIDS:

I. Runway used
II. Runway condition: clear, wet, or slippery
III. NAVAIDS used
IV. NAVAIDS available

7.27. DESCRIPTION OF THE OCCURRENCE

7.27.1. The occurrence is to be described accurately and objectively, including any information that could have a bearing:

a) Time (UTC) and location
b) Emergency equipment used (ground and airplane) and the extent to which it served its intended purpose.

c) Passenger announcements, briefings and timing and use of the SEAT BELT and NO SMOKING signs.

d) Emergency evacuation conducted or not (if conducted, the exits used and the estimated evacuation time).

e) Personal injury (each person's injury)

7.27.2. This description should include whether prescribed procedures were applied. If they were applied, were they effective? And reason if they were not applied.

7.28. CREW RESTRICTION

No crewmember involved in an operational incident or airplane accident may continue a flight or originate any other flight without receiving specific authority to do so from the V.P. Flight Operations.

7.29. OPERATIONS INCIDENT/ACCIDENT REPORT FORM

Part A. Distribution

____

____

____

Part B. Report Data

1. Reported By: ______________________________

2. Date of accident/incident: ____/____/____

3. Time of accident/incident: ___________ am/pm

4. Facility/Location of Incident: ______________

5. Reporting Number: ____________ (YY-MM-DD-##)

o Initial

o Follow-up

o Correction

(Continue to Part C)

Part C. Narrative Summary

What happened? Tell us how the accident/incident/hazard occurred. Examples: "Damaged fuel tank."; "Weapon found in carry-on luggage", "Vehicle ran into left wing of aircraft"

Names and addresses of witnesses to the accident/incident/hazard.

Name Address Phone

_________________ _______________ ___________________

(Continue to Part D)

(COMPLETE PARTS D, E, F and M FOR ALL INCIDENTS, THEN AS APPLICABLE)

Part D. Categorization of Incident (Note: Incident may fit in more than one category)

- Hazardous Material Incident
- Request for Assistance
- Weapons Incident
- Safeguards and Security Incident
- Transportation Incident
- Natural Disaster
- Industrial Accident
- Other

Part E. External Notifications/Inquiries

1. State/Local Officials Notified: _____________________ Other Federal Agencies

2. Notified: ______________________

3. Press Release/Press Advisory: _____________

4. Other: ______________________________

Part F. Significant Actions Taken

Alert Issued:

♦ Alert I □ ♦ Alert II □ ♦ Alert III □

Environmental Contractor Deployed

TSA alerted

FAA alerted

Investigation ordered

Follow-up date required by ____/____/____

Other: ____

Part G. Damage/Consequences

1. Casualties Involved

   a. Fatalities: __________

   b. Injuries: __________

2. Property Damage Estimate

   a. Airport Property: __________ Other: __________

   b. For Transportation Incident

3. Type of Vehicle Involved:

♦ Truck □ ♦ Aircraft □ ♦ Mower □ ♦ Tug □ ♦ Other: ______________

Part H. Hazardous Material Incident

1. Response Level

   ♦ Internal ☐ ♦ External ☐

2. Weather Conditions:

   a. Precipitation: _________

   b. Other: _________

3. Type/Amount of Material Released: ___________________________________

4. National Response Center Notified

   ♦ Yes ☐ ♦ No ☐

Part J. Request for Assistance

1. Emergency Assistance:

   a. Name of Requesting Organization/Individual: _________________

   b. TAA Department Responding: _______________________________

   ♦ Onsite ☐ ♦ Offsite ☐

   c. Additional resources Alerted: ________________________________

2. Other Assistance Request:

   a. Requesting Organization/Individual: _________________________

   b. Nature of request: ____

3. Recommended Actions: ____

Part K. Safeguards and Security Incident

1. Nature of Incident:

   a. Malevolent threat or act involving weapons:

b. Threat or act involving improvised explosive device or dispersion device:

c. Threat or act of malevolence involving TAA facility or activity:

i. ♦ Bomb threat ☐ ♦ terrorism ☐ ♦ sabotage ☐

ii. Protest: ♦ ☐ Civil ☐ ♦ Disturbance ☐

iii. Hijacking or threatened hijacking of an aircraft

d. Local Law Enforcement Officials

i. Alerted: ____

ii. Assistance Requested: ____

e. Other (Specify): ____

2. Recommended Response Actions: ____

Part L. Other

____

____

____

____

Part M. Recommendations

____

____

____

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CHAPTER 4 TABLE OF CONTENT

CHAPTER 4 TABLE OF CONTENT

CHAPTER 4 TABLE OF CONTENT

SECTION 4.1 GENERAL RULES

SECTION 4.1 GENERAL RULES

1.0 PURPOSE

This section provides terms and definitions commonly used in Weight and Balance.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Pilot-in-command
- Assigned Personnel in Marketing
- Assigned personnel in Flight Control

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.06

5.0 DEFINITION

5.1. TERMS AND DEFINITIONS

5.1.1. MANUFACTURER'S EMPTY WEIGHT

In the weight of structure, power plant, furnishing, systems and other items of equipment that are considered an integral part of a particular aircraft configuration. It is essentially a dry weight, including only those fluids, which are contained in closed system.

SECTION 4.1 GENERAL RULES

5.1.2. DELIVERY EMPTY WEIGHT

This is Manufacturer's empty weight plus standard items.

5.1.3. BASIC EMPTY WEIGHT

It is delivery empty weight plus/minus the standard item deviations.

5.1.4. OPERATIONAL EMPTY WEIGHT

It determines Basic empty weight plus the operational items.

5.1.5. FLEET OPERATIONAL EMPTY WEIGHT

The operational empty weight, which may be used for a group of airplanes of the same model and configuration, provided the operational empty weight of every member does not vary more than plus or minus one half of one percent of the maximum landing weight from the established operational fleet weight.

5.1.6. STANDARD ITEMS

Consist of equipment and system fluids that are not considered an integral part of a particular airplane configuration, but do not normally vary between configurations. Except for the specific variations due to operational requirements of individual air carriers (see miscellaneous fluids table, Operational Empty Weight build up and equipment list for details) the standard items will include; unusable fuel; engine and constant speed drive system oil; chemical toilet fluid; basic emergency equipment, fire extinguishers, pyrotechnics; buffet, bar and gallery structure; electronic equipment required by operator.

5.1.7. STANDARD ITEM DEVIATIONS

Consist of those Standard Items, which the operator chooses to add to or remove from the Delivery Empty Weight.

SECTION 4.1 GENERAL RULES

5.1.8. OPERATIONAL ITEMS

a) Consist of personnel, equipment and supplies that are necessary for a particular operation and are not included in the Basic Empty Weight. These items will include:

Flight and Cabin Crew and their baggage; manuals and navigational equipment; engine tank oil; food and beverage and related service equipment, washing and drinking water; emergency equipment, life rafts, life vests, cargo handling system and related equipment.

5.1.9. PAYLOAD

Payload consists of passengers, passenger baggage and/or cargo.

5.1.10. USEFUL LOAD

This consists of Payload, usable fuel and engine injection fluid.

5.1.11. MAXIMUM TAXI WEIGHT

The maximum weight authorized for ground manoeuvres by the applicable government regulations and includes taxi run up fuel.

5.1.12. MAXIMUM TAKE OFF WEIGHT

Is the maximum weight authorized at take off brake release by the applicable government regulations and excludes taxi and run-up fuel.

5.1.13. MAXIMUM ZERO FUEL WEIGHT

a) The maximum zero fuel weight is maximum airplane weight minus the usable fuel, engine injection fluid and other consumable propulsion agents. It may include usable fuel in specified tanks when carried in lieu of payload.

b) The addition of usable and consumable items to Maximum Zero fuel weight must be in accordance with the applicable government regulations so that the airplane structure and airworthiness requirements are not exceeded.

SECTION 4.1 GENERAL RULES

5.1.14. ACTUAL ZERO FUEL WEIGHT

is the airplane Operation Empty Weight plus payload, and must never exceed the Maximum Zero Fuel Weight.

6.0 RESPONSIBILITY

N/A

7.0 PROCEDURE

N/A

* * * * *

SECTION 4.1 GENERAL RULES

SECTION 4.2 LOAD SHEET COMPLETIONS

SECTION 4.2 LOAD SHEET COMPLETIONS

1.0. PURPOSE

The purpose of this load sheet completion procedure is to establish guidelines to be followed during load sheet completion.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Pilot-in-command
- Assigned Personnel in Marketing
- Assigned personnel in Flight Control

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.06

5.0 DEFINITION

The load sheet is the standard format recommended by IATA providing a method of determining total load and load distribution on board the airplane, a method of recording allowable weight and centre of gravity limits and a determination that certification limits are not exceeded, and the basic information for the completion of a load message for the Marketing Department’s use.

6.0 RESPONSIBILITY

6.1. Responsibility for completion of the form is as follows:

SECTION 4.2 LOAD SHEET COMPLETIONS

a) At company stations the form shall normally be completed by the Marketing Department or its agent and submitted to the Captain for his review and approval immediately prior to final closeout of the flight.

b) The form shall be completed by the Flight Control for training, test, and Ferry flights, and shall be submitted to the Captain for his review and approval.

c) At station providing neither Flight Control nor Marketing Department Services, the First Officer is responsible for its completion.

7.0 PROCEDURE

7.1. APPROVAL

7.1.1. The approval signature of the Captain, prior to each flight, indicates that he is satisfied and accepts responsibility to ensure that:

a) The airplane is properly loaded

b) The cargo is properly placed & secured

c) The zero fuel weight, maximum allowable weight, and centre gravity of the airplane are within certificate limits.

7.2. FORM COMPLETION

7.2.1. The load Sheet will be completed in duplicate. The original will be given to the Captain for inclusion in the trip envelope. The Marketing Department, its agent, or Flight Control will retain the copy, as appropriate.

7.2.2. A number of blank forms are available in the ship's kit and Flight Control stations. Use black ink when completing the form. Erasure or overwriting of the figures is not permitted. Space is provided on the form for corrections. The entries in the shaded portion of the form are used to complete a load message. At the stations not providing commercial or flight control services, the copy shall be given to the handling agent with a request that he retain it for a period of 30 days.

7.2.3. Flight crew shall accept a loadsheet (or prepare if required) with accurate aircraft weight/mass balance calculations for each flight, and the PIC shall ensure that the loadsheet content is satisfactory prior to each flight.

SECTION 4.2 LOAD SHEET COMPLETIONS

7.3. LAST MINUTE CHANGE (LMC)

7.3.1 The maximum last minute change value that can be added or removed from the aircraft without re-producing new load sheet and trim sheet is as follows. If these limits are exceeded, then the actual zero fuel weight, actual takeoff weight and the actual landing weight must be changed.

7.3.2. The PIC is responsible for preparing or accepting (LMC) to the load sheet to include the maximum allowable difference between planned and actual weights.

S/N	Aircraft type	Maximum Allowed LMC Load in:			Total LMC in Absolute Value(kg)
		Dead Load(kg)	No. of PAX	Takeoff Fuel(kg)	
1	A350-900/1000	500	5	700	700
2	B737-700	200	3	500	500
3	B737-800/8MAX	270*	5*	700	700
4	B737-800F	0	0	700	700
5	B757-200	500	5	1000	1000
6	B767-300/ B772/ B773/ B787	500	5	1000	1000
7	B767F/B777F	0	0	1000	1000
8	Q400	25**	2***	500	500
	*. Do not load "Oa pax compartment" and "1F Cargo compartment" simultaneously.				
	**. FWD or AFT#1 or AFT#2				
	***. Do not load LMC PAX in sections "Oa, Ob, Oc and Oh"				
	***. Do not load LMC PAX in sections Od and FWD cargo simultaneously.				
	***. Do not load LMC PAX in sections Og and AFT cargo simultaneously.				

SECTION 4.2 LOAD SHEET COMPLETIONS

7.3.3. For LMC ± 500kg, only LMC section on load sheet to be completed.

7.3.4. For LMC ± 501kg up to max permissible limit, complete LMC section on load sheet and amend AZFW, ATOW & ALW

7.3.5. For LMC above permissible limits, new computer load sheet required if manual load sheet, amend load sheet figures and issue new trim sheet.

7.3.6. If the weight change affects the trim by more than 4 index units, then a new manual trim sheet must be produced, or a new EDP load and trim sheet.

SECTION 4.2 LOAD SHEET COMPLETIONS

ETHIOPIAN AIRLINES
L O A D   S H E E T   3   CHECKED   APPROVED
EDNO   1   2
ALL WEIGHTS IN KILOGRAMS

4 FROM/TO	5 FLIGHT	6 A/C REG	7 VERSION	8 CREW	9 DATE	1 TIME
ADD BZV	ET831/05	ETAMF	C24S221	2/8	05DEC07	1354

LOAD IN COMPARTMENTS 1   WEIGHT   DISTRIBUTION
20150   1/4595   2/4029   3/5960
4/3750   1
PASSENGER/CABIN BAG   15984   5/1816
153/54/1/1   TTL 209
18/190   SOC 0/0

TOTAL TRAFFIC LOAD   36134   1
DRY OPERATING WEIGHT   91551

1

ZERO FUEL WEIGHT ACTUAL	127685	MAX	133809
TAKE OFF FUEL	30100		
TAKE OFF WEIGHT ACTUAL	157785	MAX	158000 L
TRIP FUEL	19500		
LANDING WEIGHT ACTUAL	138285	MAX	145149
TAXI OUT FUEL	300		

1

LAST MINUTE CHANGES

1

BALANCE AND SEATING CONDITIONS DEST SPEC CL/CPT+/- WEIGHT

BI	86.00	DOI	86.97
LIZFW	91.46	MACZFW	23.42
LITOW	93.46	MACTOW	24.45
LILAW	89.33	MACLOW	22.66
ANU	3.89		

CABIN CLASS
C 18
Y 190
CABIN AREA   1

SECTION 4.2 LOAD SHEET COMPLETIONS

0A30
0B93
0C85
UNDERLOAD BEFORE LMC 215 1
CAPTAINS INFORMATION/NOTES LMC TOTAL +/-
BW 88212 BI 1

PANTRY CODE AMFG /2333

2
LDM
ET831/05.ETAMF.C24S221.2/8
-BZV.85/25/1/0.T12178.1/3055.2/4029.3/1540.4/2210.5/1344.PAX/10/101.PAD/0/0
-FIH.68/29/0/1.T7972.1/1540.3/4420.4/1540.5/472.PAX/8/89.PAD/0/0
SI
DOW 91551
DOI 86.97

CPM 2
ET831/05.ETAMF.11P21P22P
-11P/P1PET/BZV/1715/C0
-13L/LD2ET/FIH/770/B0-13R/LD2ET/FIH/770/B0
-14L/LD2ET/BZV/670/B0-14R/LD2ET/BZV/670/B0
-21P/PIPET/BZV/1915/B0
-22P/P1PET/BZV/2114/CO
-31L/LD2ET/FIH/770/BO-31R/LD2ET/FIH/770/B0
-32L/LD2ET/FIH/770/B0-32R/LD2ET/FIH/770/B0
-41L/LD2ET/BZV/770/B0-41R/LD2ET/BZV/770/B0
-42L/LD2ET/FIH/770/B0-42R/N
-43L/LD2ET/FIH/770/B0-43R/LD2ET/BZV/670/B0
-51/FIH/472/B/M/BZV/150/C.VRI
-52/BZV/1194/B/M.VR0
SI

2
SLS
Et831/05.ETAMF.C24S221.ADD.05DEC07
-BZV.CC/10/0/0.YY/75/25//1.GG/0/0.B/6700.C/3749.M/894
-FIH.CC/7/1/0.YY/61/28/0.GG/1/1.B/7100.M/172

SECTION 4.2 LOAD SHEET COMPLETIONS

FIELD	EXPLANATION
1	Name of load sheet agent (signature not mandatory)
2	Signature of Captain
3	Edition number of load sheet
4	Flight routing
5	Flight number and date (UTC)
6	Aircraft registration
7	Aircraft version
8	Crew: cockpit/cabin
9	Date of load sheet print
10	Time of load sheet print
11	Total weight of dead load and weight of load per compartment
12	Total number/weight of passengers calculated in accordance with MD standard weights adult, children and infants. Number of passengers per class, Seats occupied by cargo, number of seats blocked.
13	Weight calculation
14	Maximum aircraft weights “L”: limiting weight “ADJ”: Weight adjusted
15	Balance conditions DOI: Dry Operating Index LIZFW: Loaded Index at Zero Fuel Weight LILAW: Loaded Index at Landing Weight LITOW: Loaded Index at Take of Weight MACZFW: MAC (%) at Zero Weight

SECTION 4.2 LOAD SHEET COMPLETIONS

	MACTOW: MAC (%) at Take of Weight STAB TO: Stabilizer Setting for take off
16	Number of passengers per cabin section
17	Column for LMC hand amendments
18	Under load based on calculation acc. to item 13
19	Captain information before LMC
20	Load Message (LDM)
21	Container and Pallet Message (CPM)
22	Special Instruction (SI)

* * * * *

SECTION 4.2 LOAD SHEET COMPLETIONS

SECTION 4.3 STANDARD WEIGHTS

SECTION 4.3 STANDARD WEIGHTS

1.0. PURPOSE

The purpose of this section is to establish guidelines on how to deal with the standard weights applied in aviation industry.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Personnel under flight operations

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.06

4.1 There shall be a system approved by the ECAA for obtaining, maintaining and distributing to appropriate personnel current information regarding the mass and balance of each aircraft operated. (ECARAS 9.3.1.16).

4.2 No Ethiopian aircraft shall takeoff without ensuring that the maximum allowable mass for a flight does not exceed the maximum allowable takeoff or landing mass, or any applicable en route performance or landing distance limitations considering the: (ECARAS 8.7.1.4)

a) Condition of the takeoff and landing areas to be used

b) Gradient of runway to be used

c) Pressure altitude

d) Ambient temperature

e) Current and forecast winds

f) Any known conditions which may adversely affect the aircraft performance

SECTION 4.3 STANDARD WEIGHTS

5.0 DEFINITION

5.1 **Basic weight**: The weight of the fully equipped airplane ready for flight. Unless otherwise specified this includes:

a) Normal passenger service equipment such as washing and drinking water, toilet fluids, magazines, blankets, pillows, and lavatory supplies.

b) Normal quantities of oil, hydraulic fluid, water, and other fluids.

c) All items considered standard for the defined operating configuration such as galley over pack, dry supplies, and flyaway kits.

5.2 **Dry operating weight**: The basic weight of the airplane plus the flight deck and cabin crew on assigned duty, exclusive crew baggage.

5.3 **Operating Weight**: Is dry operating weight plus takeoff fuel.

5.4 **Zero Fuel Weight**: is dry operating weight plus payload.

5.5 **Takeoff Fuel**: The required fuel on board the airplane at brake release for takeoff.

5.6 **Takeoff Weight**: The sum of zero fuel weight plus takeoff fuel.

5.7 **Trip Fuel**: The required fuel for operation from origin to destination.

5.8 **Landing Weight**: Is takeoff weight minus trip fuel. (Includes zero fuel weight plus reserve fuel)

5.9 **Traffic Load**: The total weight of passengers and their baggage plus cargo.

5.10 **Payload Fuel**: Fuel required in some instances for weight and balance control but not required as part of takeoff fuel or trip fuel.

5.11 **Payload**: Traffic load plus fuel carried as payload.

6.0 RESPONSIBILITY

N/A

SECTION 4.3 STANDARD WEIGHTS

7.0 PROCEDURE

7.1. STANDARD WEIGHTS

7.1.1. PASSENGER WEIGHT

a) Average passenger weight may be used to compute passenger loads except when special groups such as athletic teams, orchestras, and children's group are carried. In such cases actual passenger weight shall be used. When special passenger groups are carried but form only a part of the total load, the actual weights of the special group shall be used for the balance of the passenger load. In such instances a notation shall be entered on the load manifests identifying the group and indicating the number of persons in the group.

b) Actual passenger weight shall include the weight of minor articles carried on board by the passenger. Weights may be determined by the use of scales or by asking each passenger his weight and estimating the weight of carry-on baggage. In the case of charter flights in which actual weights are provided, such actual weights shall be used to compute passenger load.

	STANDARD	NON STANDARD
Average adult	77	85
Adult male	77	85
Adult female	77	85
Child	35	
Infant	10	

REMARK:

1. 7Kgs cabin baggage weight in included
2. Non standard weights are applicable on the following routes ADD-LOS, LOS-ADD, BKK-ADD, CAN-ADD, DXB-ADD, and HKG-ADD

7.1.2. CREW WEIGHTS

a) The following average weights may be used for crewmembers:

I. Cockpit crew - 107kgs

II. Cabin crew – 99kgs

SECTION 4.3 STANDARD WEIGHTS

7.1.3. BAGGAGE WEIGHTS

a) Actual baggage weights shall normally be used for all flight operations. Average weights shall only be used when baggage weight in facilities are not available.

b) The following average baggage weights may be used in lieu of actual weights:

I. For each piece of checked baggage use and average of 30 kilos for First Class Passenger, 20 kilos for Economy Class.

II. Crew baggage shall be an average of 15 kilos for both flight deck crew & cabin attendants.

NOTE: - Actual baggage weight must be used when special passenger groups are carried.

7.1.4. CARGO WEIGHT

Cargo weight shall be the actual weighed value, including dunning, of equipment required for storage.

7.1.5. PANTRY CODES

Pantry codes consist of standard galley loads for various airplane configurations and/or meal service selections. To be furnished by Marketing.

7.1.6. AIRPLANE WEIGHT AND BALANCE DOCUMENTS

The airplane weight and balance records are maintained by Maintenance, Repair & Overhaul. The basic weight and operating index for each specific airplane is available from Flight Control (Dispatch).

* * * * *

SECTION 5.1 TRAINING PROGRAM

SECTION 5.1 TRAINING PROGRAM

SECTION 5.1 TRAINING PROGRAMS

Table of Contents

SECTION 5.1 TRAINING PROGRAM

SECTION 5.1 TRAINING PROGRAM

SECTION 5.1 TRAINING PROGRAM

SECTION 5.1 TRAINING PROGRAMS

1.0. PURPOSE

The purpose of this training program procedure is to establish guidelines to prescribe the requirements applicable to the company for establishing and maintaining a training program for crewmembers, flight operation officers, and other operations personnel and for the approval and use of training devices in the conduct of the program.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight Crewmembers
- Instructors
- Examiners

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.05

4.1. Training syllabi and checking programs for all operations personnel assigned to operational duties in connection with the preparation and/or conduct of a flight shall be developed to meet the respective requirements of the Authority. No crewmember shall be assigned as a required crew member unless that person meets the training and currency requirements established by the Authority for that respective position.

4.2. The training syllabi shall include training programs for Flight Crew.

4.3. Ethiopian shall ensure that all operations personnel are properly instructed in their duties and responsibilities and the relationship of such duties to the operation as a whole.

4.4. Ethiopian shall have a training program manual approved by ECAA (the Authority) containing the general training, checking, and record keeping policies.

4.5. Detailed Flight Operations personnel records shall be retained to show that all requirements of the training course have been met as approved by the Authority. These records shall be kept for a minimum period of two years after completion of the training. (ECARAS 3.2.15)

4.6. Ethiopian shall have approval of the Authority prior to using a training curriculum for the purpose of qualifying a crewmember, or person performing operational control functions, for duties in commercial air transport.

4.7. Ethiopian shall submit to the Authority any revision to an approved training program and shall receive written approval from the Authority before that revision can be used.

4.8. All flight crew shall be trained and evaluated as required in accordance with the training and examination requirements.

4.9. All flight crew shall undergo appropriate training and examination before operating their previous equipment type once training is completed on a new equipment type.

4.10. Flight crews qualified in different types of airplanes shall keep simulator and system currency of each type in accordance to the state regulation and accomplish the minimum takeoff and landings.

4.11. Instructors, Examiners and flight crew shall use only those documents authorized and published by Flight Training & Standards Department for training and examination purposes.

4.12. All flight crew are trained and examined according to published standards approved by the state.

4.13. Transition training covering each airplane type operated by the company shall be provided for pilots, flight instructors & examiners. Each employee in these classifications who has not previously performed the same duties for the company with respect to specific type of airplane shall be required to accomplish initial training.

4.14. Differences training shall be provided for flight crewmembers for each variation of a particular type of an airplane, which the company operates, for which differences training is required.

4.15. No simulated engine failures or any other emergencies are permitted on actual flights.

4.16. TRAINING AND SERVICE AGREEMENT

4.16.1 Training and service agreements should be signed by candidate trainees before the commencement of the training.

4.16.2 All flight crew will be bonded for a period of four (4) years for senior Captains (A350/B787/777/767/757) and three (3) years for all other flight crewmembers.

4.16.3 Flight crew are required to undertake in writing, not to terminate employment while Ethiopian wishes to retain their service, or the employment is terminated due to their fault, per the Collective Bargaining Agreement, the Labor Law or the Management Policy and Procedure Manual, the employee shall immediately pay the training cost in full amount regardless the number of months of their actual service following the completion of the training.

4.16.4 If a flight crew is assigned to serve the company after successful completion of training and enrolled on different A/C training before the completion of the service period for the earlier training, the commitment period will accumulate, and the crew is supposed to serve the airline for a duration of time amounting to the commitment periods of the trainings.

5.0. DEFINITION

N/A

6.0. RESPONSIBILITY

6.1. Director Flight Training & Standards is responsible for the implementation of this policy and procedure.

6.2. Manager Recurrent & Transition Training shall be responsible to provide a training and procedures manual, approved and accepted by the Authority, for the use and guidance of personnel concerned, including detailed course outlines. This manual shall be amended as necessary to keep the information contained therein up to date. (ECARAS 3.2.18)

6.3. Director Flying and Training is responsible for implementation of Transition Training outlined in this section.

6.4. Director Flight Training and Standards is responsible for implementation of the command upgrading training program.

6.5. The Director Flight Training & Standards is primary responsible to ensure compliance with the appropriate flight crew training.

6.6. The chief pilot is responsible for making sure that each line pilot is not only highly qualified but also that he operates in a safe, efficient and standardized way.

6.7. The instructor pilot (flight and /or simulator) is responsible for developing the student's ability to perform the tasks of a line pilot in a safe, efficient and standard manner.

6.8. The PIC is responsible for ensuring Aerodrome Qualification compliance as outlined in Chapter 2.3 of this manual subject to Category A, B & C Aerodromes.

7.0. PROCEDURE

7.1. INTRODUCTION

7.1.1 The company shall ensure that any training and examination program is approved by the state and published in the Flight Operations Training Program Manual or equivalent document.

7.1.2 Provide Ground, Flight Training and where applicable Examination programs approved by ECAA, which require that all Instructors, Examiners and flight crew (whether employed or subcontracted), are trained for their assigned tasks.

7.1.3 Ground and Flight Training programs shall include qualification:

SECTION 5.1 TRAINING PROGRAM

a) Initial
b) Recurrent(continuing)
c) Transition (conversion)
d) Requalification
e) Upgrade to PIC
f) Familiarization
g) Differences or other specialized trainings.

7.1.4 Ensure that all instructors, examiners, training facilities, devices, equipment and course materials:

a) Have the approval of ECAA and/or the company
b) Meet the required qualification and performance standards
c) Be reviewed periodically to ensure compliance with approved standards

7.1.5 Require that all Instructors, Examiners and flight crew, whether employed or subcontracted, are qualified and standardized for their assigned tasks and are certified by the company and approved by the state as required.

7.1.6 Ensure that all training aids, devices and course materials reasonably reflect the configuration of the fleets for which the respective training and check is being conducted.

7.1.7 Have sufficient instructors, examiners and support personnel to provide the training and check programs.

7.1.8 7.1.8 Training subjects (other than recurrent training) that are applicable to more than one airplane crew member position and that have been satisfactorily completed in connection with prior training for another airplane or another crew member position need not be repeated during subsequent training.

7.1.9 A person who progresses successfully through flight training, is recommended by his instructor or examiner, and successfully completes the appropriate flight check for an examiner need not complete the programmed hours of flight training for the particular airplane.

7.1.10 The company shall prepare and keep current written training program curriculum for a flight dispatcher and flight crewmember requirements for each airplane type. The curriculum must include ground and flight training required by this chapter.

7.1.11 Each training program curriculum must include:

a) A list of principal ground training subjects, including emergency training subjects that are provided.

b) A list of all training devices, mock-ups, system trainers, procedure trainers, or other training aids that the company will use.

c) Detailed descriptions or pictorial displays of the approved normal, abnormal, and emergency manoeuvres, procedures, and function that will be performed during each flight training phase or flight check, indicating those manoeuvres, procedures, and functions that are to be performed during the inflight portions of flight training and flight checks.

d) All list of airplane simulators or other training devices approved for particular manoeuvres, procedures, or functions.

e) The programmed hours of training that will be applied to each phase of training.

7.1.12 In addition to the training described in this section, each training program must also provide ground and flight training, instruction, and practice as necessary to ensure that each crew member and flight dispatcher:

a) Remains adequately trained and proficient with respect to the type of operation in which he serves.

b) Qualifies in new equipment, facilities, procedures, and techniques including modifications to airplanes.

7.1.13 Ensure that flight crews are trained and examined, as applicable, in accordance with the time schedule outlined in this manual.

7.2. TRAINING AND OTHER REQUIREMENTS

An MPL or CPL graduate of Ethiopian Pilot Training School or other institution of equivalent standard holding Commercial Pilot Licence (CPL) may be accepted. All flight crew shall complete the aircraft systems course, simulator training, and line flying under supervision (assisted flights) prior to being assigned to operational duties.

7.2.1. FIRST OFFICER TRAINING REQUIREMENTS

a) Q-400(CPL)

- System /performance Classroom 120 hour
- Full type rating Simulator training long 16 sessions or shortened 9 sessions depending on crew experience
- 3 Observation flights
- Aircraft base training 1 hour (Minimum six landing)

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- Line training: Line training: A minimum of 40 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF
- A line check

b) Q-400(MPL)

- Completion of all MPL stages in accordance with the approved MPL curriculum including Aircraft System /performance course 120 hour
- Final Type Rating Simulator Training (Stage 9) 11 Full Flight Simulator (FFS) sessions (shortened Type Rating).
- 3 observation Flights
- Aircraft base Training 1 hour (Minimum six landing)
- Line training: A minimum of 40 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF.
- A line check

c) B737 (CPL)

- Type rating CBT 120 hours
- Simulator type rating training shortened course 17 sessions or for long course 23 sessions depending on crew experience
- Aircraft base training 1 hour (Minimum six landing)
- 3 Observation flights
- Line training: A minimum of 40 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF.
- A Line check

d) B737 (MPL)

- Completion of all MPL stages in accordance with the approved MPL curriculum including Aircraft System /performance course 120 hour
- Final Type Rating Simulator Training (Stage 9): 12 Full Flight Simulator (FFS) sessions (shortened Type Rating format).
- 3 observation Flights
- Aircraft base Training 1 hour (Minimum six landing)
- Line training: A minimum of 40 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF.
- A Line check

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e) B767

- Type rating CBT 120 hours
- Simulator training (short course 14 and long course 18 sessions)
- Aircraft base training 1 hour (Minimum six landing) for no jet experience.
- 3 Observation flights
- Line training: A minimum of 24 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF.
- A Line check.

f) B777

- Type rating CBT 84 hours
- Simulator training 13 sessions
- 3 observation Flights
- Line training: A minimum of 24 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF
- A Line check
- B777 Difference (ground CBT 84 hours, simulator 5 sessions, 6 sectors of line training)

g) B787

- Type rating CBT 84 hours
- Simulator training (short course 14 and long course 18 sessions)
- Aircraft base training 1 hour (Minimum six landing) for no jet experience.
- 3 Observation flights
- Line training: A minimum of 24 sectors (assisted flights) of which at least 2 sectors as PM and at least half of the training sectors as a PF.
- A Line check
- B787 Difference (ground CBT 84 hours, simulator 6 sessions, 6 sectors of line training)

h) A350

- Type rating CBT 84 hours
- Simulator training 18 sessions
- 3 Observation flights

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- A minimum of 26 assisted line training sectors shall be completed, including at least 2 sectors performed in the Pilot Monitoring (PM) role, and at least half of the training sectors as a PF.
- A minimum of 2 sectors conducted on the A350-1000 variant
- A Line check

7.2.2. **CAPTAIN LINE TRAINING REQUIREMENTS**

a) Q400

- Command upgrade training
- System /performance Classroom 120 hours
- Simulator training 14 sessions
- Aircraft base training 1 hour (Minimum six landing)
- 1 Observation flight
- For F/O upgraded directly from wide body: Route Training at least 100 hours with at least two instructors followed by final check, of which at least 2 as PM and at least half of the training sectors as a PF.
- For SF/O: Route Training at least 75 hours (minimum 30sectors) with at least two instructors followed by final check ride, of which at least 2 sectors as PM and at least half of the training sectors as a PF.
- A Line check

b) B737

- Command upgrade training
- Type rating CBT 120 hours
- Simulator training (short course 18 and long course 24 sessions)
- Aircraft training 1 hour
- 1 Observation flight
- For F/O upgraded directly from wide body: Route Training at least 150 hours with at least two instructors followed by final check ride and company check ride, of which at least 2 as PM and at least half of the training sectors as a PF.
- For SF/O: Route Training at least 100 hours (minimum 30sectors) with at least two instructors followed by final check ride, of which at least 2 as PM and at least half of the training sectors as a PF.
- A Line check

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c) B767

- Command training
- Type rating CBT 120 hours
- Simulator training short course 14 and long course 18 sessions
- 1 Observation flight
- Line Training 18 sectors followed by final check ride, of which at least 1 as PM and at least half of the training sectors as a PF.
- A Line check

d) B777

- Command training
- Type rating CBT 84 hours
- Simulator training 13 sessions
- 1 Observation flight
- Line training: At least 18 sectors, of which 2 at least as PM and at least half of the training sectors as a PF
- A Line Check
- B777 Difference (ground CBT 84 hours, simulator 5 sessions, 6 sectors of line training)

e) B787

- Command training
- Type rating CBT 84 hours
- Simulator training (short course 11 and long course 15 sessions)
- 1 Observation flight
- Line training: At least 18 sectors, of which at least 1 as PM and at least half of the training sectors as a PF
- A Line check
- B787 Difference (ground CBT 84 hours, simulator 4 sessions, 6 sectors of line training)

f) A350

- Command training

- Type rating CBT 84 hours
- Simulator training 15 sessions
- 1 Observation flight
- A minimum of 20 sectors line training, including at least 2 sectors as Pilot Monitoring (PM) and at least half of the training sectors as a PF
- A minimum of 2 sectors conducted on the A350-1000 variant
- A Line check

7.3. INITIAL AND FINAL APPROVAL OF TRAINING PROGRAM

7.3.1. To obtain initial and final approval of a training program, the company must submit to the Ethiopian Civil Aviation Authority:

a) An outline of the proposed program or revision, including an outline of the proposed or revised curriculum, that provides enough information for a preliminary evaluation thereof.

b) Additional, relevant, information as may be requested by the Civil Aviation Authority. If the proposed training program or revision complies with this chapter, the Civil Aviation Authority will grant initial approval, after which the company may conduct the training in accordance with that program. The Civil Aviation Authority then evaluates the effectiveness of the training program and advises the company of deficiencies, if any, which must be corrected.

7.4. GROUND TRAINING

7.4.1. The training and evaluation program for pilots and dispatchers shall include ground instruction applicable to each airplane type operated by the company. Ethiopian airlines flight operations training uses distance learning and/or distance evaluation in the flight crew training and qualification program. The distance learning and/or evaluation is monitored by ground instructors and is approved or accepted by ECAA. The distance learning/evaluation adopted by the training department will have monitoring, recording and evaluation of results of successful and unsuccessful flight crew evaluations. The below ground training shall be included in the initial, transition, conversion, upgrade to PIC and as applicable on differences and recurrent training. The trainings will be specific to aircraft type as required:

a) Flight crew members will receive training in all aspects of aircraft performance. Such training shall include:

I. Weight/mass and balance;

II. Takeoff, climb, cruise, approach and landing performance;

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III. Obstacle clearance;

IV. Fuel planning;

V. Diversion planning;

VI. Effect of inoperative or missing components (MEL/CDL);

VII. Engine-out drift down

b) Flight crew members shall complete training and an evaluation in aircraft systems and limitations, to include a demonstration of competence in the operation of aircraft systems. Major airplane systems such as flight control, electrical, hydraulic system etc. and principles of normal, non-normal, and emergency operation and appropriate procedures:

- Aircraft Operating limitations
- Fuel consumption and cruise control
- Flight planning including EFB
- Normal, Non-normal, and emergency procedures
- The Approved Airplane Flight Manual

c) The company's flight control (dispatch) or flight release procedures.

d) Meteorology: at least sufficient to ensure a practical knowledge of weather phenomena, including thunderstorms and high altitude weather situations.

e) Air traffic control systems, procedures, and phraseologies.

f) Normal and emergency communication procedures.

g) Visual cues prior to and during descent below DH or MDA

h) Other instructions as necessary to ensure competence.

i) For each airplane type;

- A general description
- Performance characteristics
- Engines
- Major components

j) Procedures in adverse weather situations to include:

- Anti-ice/de-ice policies and procedures;

- Contaminated runway operations;
- Thunderstorm avoidance;
- Hot and cold weather operations;
- Operations near volcanic ash;
- Wind shear avoidance and recovery (predictive and actual).

k) Crew Resource Management (CRM/TEM).

l) Specialized trainings:

- Performance-Based Navigation (PBN) (included on initial and recurrent training on every 12 month)
- Performance-Based Communication and Navigation Surveillance System (PBCS)
- Reduced Vertical Separation Minima (RVSM)
- Minimum Navigation Performance Specifications (MNPS/NAT HLA)
- ETOPS/EDTO
- Volcanic ash
- DGR (Dangerous Goods)
- Thunderstorm Avoidance
- Windshear
- TCAS/ACAS
- High altitude airport training
- Unusual attitude/upset recovery (UPRT)
- SMS
- CFIT & Terrain avoidance
- Cold and hot weather operations (aeroplane De-icing/Anti-icing, contaminated taxiways & runways, ground procedures), Winter operations

7.5. FLIGHT TRAINING

7.5.1. Each flight training program for pilots must include the following and the below training are included on initial, transition/conversion, Upgrade to PIC, re-qualification and as applicable during recurrent training (based on the recurrent matrix).

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a) Training and evaluation to demonstrate pilot competence in the applicable manoeuvre and procedures set forth in Flight Operations Training Program Manual. Flight crew members shall complete training in normal and non-normal procedures and maneuvers specific to aircraft type.

i. Pilot Monitoring (PM) Pilot Flying (PF) and other flight crew division of duties (task sharing);
ii. Positive transfer of aircraft control
iii. Consistent checklist philosophy;
iv. Emphasis on a prioritization of tasks (e.g. "aviate, navigate, communicate");
v. Proper use of all levels of flight automation

b) Each initial and recurrent flight training syllabus shall include a type-specific Line-Oriented Flight Training (LOFT) session conducted in a full flight simulator. The LOFT session shall consist of a realistic, uninterrupted line environment scenario with defined CRM and non-technical skill objectives. These competencies shall be observed, evaluated, and debriefed in accordance with CRM principles.

c) No simulated engine failure and non-normal maneuvers are allowed during aircraft training flights. The manoeuvre and procedures that cannot be safely performed in an airplane are trained and evaluated (as applicable) in a state approved type specific simulator, an appropriate training device, static airplane or any other means that ensures pilot competence and includes:

i. Windshear avoidance and recovery,
ii. GPWS knowledge & conduct of associated procedures, response to GPWS alerts and warnings and the avoidance of controlled flight into terrain (CFIT),
iii. TCAS/ACAS training includes duties and responsibilities related to TCAS/ACAS alerting equipment complete training and an evaluation that includes a demonstration of competence in maneuvers and procedures for the proper response to TCAS/ACAS alerts.
iv. Upset prevention and recovery training (UPRT),
v. Engine failure,
vi. Abnormal engine starts,
vii. Rejected take-off,
viii. Emergency Evacuation
ix. Flight crew incapacitation training and evaluation that includes a demonstration of competence in duties and procedures related to flight crew incapacitation.

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d) Procedures, simulated aircraft, weather and environmental conditions are standardized and appropriate for the conduct of the training and evaluation.

e) Each pilot must successfully complete training and practice maneuvers within the limited number of repetitions set forth in Flight Operations Training program Manual that can be performed in an airplane simulator. In the case that this cannot be achieved to a competency standard within the specified repetitions, remedial training may be requested by the instructor. In which case a review and approval from the Training Manager must be obtained.

f) A training and, when applicable, an evaluation, to check the standard piloting technique and ability of the Captain or First Officer to execute both normal and non-normal procedures and the maneuvers (to include, as a minimum, rejected takeoff, emergency evacuation, engine failure) as set forth in Flight Operation Training Manual that can be performed in an airplane or simulator. For tolerance margin of flight training and examination, refer to the training program of each aircraft.

g) The training evaluation syllabus shall include the maneuvers & malfunctions that might be presented during the evaluations. The detailed chronological sequence of the manoeuvre or malfunctions shall not be provided prior to the respective manoeuvre or malfunction being administered. However, this is not intended to preclude the crews from knowing the city pairs to be flown or the general maneuver requirements prior to the examination.

h) Following initial or transition training, a pilot must successfully complete a proficiency check and demonstrate the skill and knowledge level adequate for operating the aircraft to the standard set in the training syllabus or better before starting line & route training and check flights as applicable. This check must be administered by an Examiner.

i) CRM skills training and evaluation such as crew communication, conflict resolution, threat and error management, decision making etc...

7.6. INITIAL NEW-HIRE TRAINING

7.6.1. Initial training shall be provided for pilots, instructors, examiners and flight dispatchers for each airplane type operated by the company. The initial training shall include the following:

a) Basic indoctrination

b) Carriage and handling of restricted articles

c) Crew Resource Management (CRM)

d) General emergency

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7.7. FLIGHT CREW INITIAL TRAINING

7.7.1. Pilots' initial training, as defined earlier in this chapter, is accomplished with the use of the curricula divided into following training phases:

a) Classroom instruction

b) Cockpit procedures training

c) Simulator training

d) Airplane training

e) Route check (assisted flight in the case of F/O)

f) Final Line check

7.7.2. A detailed course content of the above training is given in the Flight Operation Training Program Manual.

7.8. NEWLY HIRED FLIGHT CREWMEMBERS & DISPATCHERS BASIC INDOCTRINATION TRAINING

7.8.1 Basic indoctrination ground training provided for all newly hired crew members and flight Dispatchers on in initial, transition/conversion Training:

a) Duties and responsibilities of crewmembers or Flight Dispatchers, as applicable

b) Relevant Ethiopian Civil Aviation Regulation

c) Relevant content of the Flight Operations Manual (FOM)

d) Air Operator Certificate (authorized operation).

7.8.2. Basic indoctrination training may be excluded from curriculum if the crewmember or flight dispatcher has previously successfully completed this training in the company program. The detail course content of which is given in Flight Operations Training Program Manual.

7.9. INITIAL COURSE FOR FLIGHT CREWMEMBERS

7.9.1 Crewmembers who serve in operations above 25,000 ft must receive instructions in the following:

a) Respiration

b) Hypoxia

c) Duration of consciousness supplemental oxygen at altitude

d) Gas expansion

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e) Gas bubble formation

7.9.2. Physical phenomena indications of decompression.

7.10. CARRIAGE AND HANDLING OF RESTRICTED ARTICLES

7.10.1 An initial and recurrent training program (every 24 months) in the procedures relevant to the transport of dangerous goods shall be provided to each pilot, or any other individual responsible for the carriage and handling of restricted articles. This training shall include up to date instruction in:

a) Regulated commodities and classification
b) Certificate requirement
c) Packaging, marking and labelling
d) Deviation authority
e) Containers for liquids
f) Quantity limitations
g) Special requirements for radioactive materials
h) Notification to Captain
i) Damage to dangerous goods
j) Reporting certain dangerous articles incidents
k) Magnetized materials
l) Cargo location
m) Transportation of dangerous goods by air
n) Special requirements of poisons
o) Storage, handling and loading procedures
p) Emergency response guide.

7.11. CREW MEMBER GENERAL EMERGENCY TRAINING

7.11.1 Crewmember emergency training, as set forth in this section, shall be provided to each required crewmember with respect to specific aircraft type, model, and configuration, and the training is included on initial, transition(conversion), differences, and recurrent training.

7.11.2 Emergency training is accomplished in accordance with the curricula given in Flight Operations Training Program Manual. The training shall ensure flight crew members complete practical training exercises in the use of emergency and safety equipment required to be on board the aircraft. These curricula are divided into the following training phases and is specific to aircraft type.

a) Classroom instruction

b) Aircraft/ mock-ups training

7.12. JOINT EMERGENCY AND SAFETY EQUIPMENT TRAINING

The joint practical emergency and safety training addresses emergency evacuation and coordination among flight crew members and cabin crew members required for the safety of operations. Emergency and safety equipment training shall emphasise the importance of effective coordination and communication between flight crew and cabin crew members.

7.13. ANNUAL JOINT EMERGENCY AND SAFETY EQUIPMENT TRAINING

7.13.1 Emergency training must provide:

a) Instruction in emergency assignments and procedures, including coordination among crewmembers.

b) Individual instruction in the location, function and operation of emergency equipment including:

  I. Equipment in ditching & emergency evacuation.

  II. First aid equipment & its proper use.

  III. Portable fire extinguishers, with emphasis on type of extinguisher to be used on different classes of fires.

c) Instruction in the handling of emergency situations including:

  I. Rapid decompression

  II. Fire in flight or on the ground

  III. Ditching and other abnormal situations

  IV. Illness, injury, or other abnormal situations Involving passengers or crew members

  V. Hijacking and other unusual situations

NOTE: Joint Crew Resource Management (JCRM) training should be conducted between cockpit crew, cabin crew and dispatchers annually.

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7.14. TRIENNIAL JOINT EMERGENCY AND SECURITY TRAINING

7.14.1 The purpose of this training is to enhance on board coordination & mutual understanding of the human factors involved in addressing emergency situations & security threats. This training is included on initial, transition(conversion), differences, and recurrent training. Each crewmember must perform an emergency drill, utilizing the proper equipment and procedures, unless the Civil Aviation Authority finds that, with respect to a particular drill, the crewmember can be adequately trained by demonstration and classroom instruction in the following areas.

a) Ditching

b) Emergency evacuation

c) Fire extinguishing and smoke control

d) Operation and use of emergency exists, including deployment and use of evacuation chutes.

e) Use of crew and passenger oxygen

f) Removal of life rafts from airplane, inflation of life vests, uses of the lifelines, and boarding of passengers and crew.

g) Donning and inflation of life vests and the use of other individual flotation devices.

7.15. SECURITY TRAINING

7.15.1 Flight Operations shall establish, maintain and conduct approved training programs which enable its personnel to take an appropriate action to prevent acts of unlawful interference. Following an act of interference on board an aircraft the PIC or, in his absence, the AOC holder shall submit, without delay, a report of such an act to the designated local authority and the Authority in the State of the operator. (ECARAS 9.4.1.3)

7.15.2 As a minimum, the security training program shall include: (ECARAS 9.4.1.3)

a) Determination of the seriousness of any occurrence.

b) Crew communication and coordination;

c) Policy and procedures associated with flight deck access;

d) Appropriate self-defence responses;

e) Use of non-lethal protective devices assigned to crew members for use as authorized by the State;

f) Understanding the behaviour of terrorists so as to facilitate the ability to cope with hijacker behaviour and passenger responses;

g) Situational training exercises regarding various threat conditions;

h) Flight deck procedures to protect the aircraft;

i) Aircraft search procedures;

j) Guidance on least-risk bomb locations.

k) Post-flight concerns for the crew

7.16. CREW RESOURCE MANAGEMENT (CRM)

7.16.1 During initial, transition/conversion training each flight crew and dispatchers must undergo CRM training and evaluation including threat and error management using facilitators that are trained in human performance and human factors principles. As part of recurrent training completed annually, all elements of the initial CRM course shall be conducted over a three-year cycle. At the end of classroom training, all personnel taking the training will be evaluated using questioners or exam given by the facilitator.

7.16.2 The recurrent CRM training requirement does not apply until a person has completed the applicable initial CRM training.

7.16.3 Joint Crew Resource Management (JCRM) training should be conducted between cockpit crew, cabin crew and flight dispatchers annually. The purpose of this training is to enhance crew coordination, ensuring a mutual understanding of the human factors involved in joint operational control and achieving common learning objectives. The training objective is set out by the flight operations training and operational control management personnel.

NOTE: Recurrent CRM training for flight crewmembers may be accomplished during an approved simulator line operational flight training (LOFT) session.

7.17. SPECIALIZED TRAININGS

7.17.1 This Category of training is intended for flight crew who operates an aircraft equipped with a specific type of new equipment (TCAS, RVSM, MNPS, NAT/HLA, BRNAV/RNP, GPS, ETOPS/EDTO, CFIT, PBN, SMS, DGR, AUPRT, LVO, Volcanic Ash, Adverse weather)

7.17.2 This training must be completed during initial, transition, upgrade to PIC, requalification and recurrent ground and/or flight trainings. The required Specialized Trainings are listed under FOM 5.1(7.4.1).

7.17.3 The training is either a self-study of guidance materials, which will be distributed by the training department periodically or CBT.

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7.18. OTHER TRAININGS

All flight crew shall receive the necessary training to support the introduction of new aircraft type, new policies, procedures, rules/instructions, system and fleet modification/upgrade as recommended by the Flight Operations Review Committee.

7.19. RECURRENT TRAINING

7.19.1 Recurrent training for pilot, as defined earlier in this chapter, is conducted with the use of the curricula divided into the following training bases:

a) Classroom instruction

b) Cockpit procedures training

c) Simulator training

d) Flight training

7.19.2 As a minimum, the ground training portion of each recurrent training program for pilots must include:

a) Flight crew members shall complete training & evaluation in aircraft systems and limitations in every recurrent training, the whole aircraft systems & limitations shall be completed in 36 months to determine the state of the pilot's knowledge with respect to the airplane and the duties of his position.

b) Instructions in the subjects required for initial ground training as appropriate, including emergency training.

c) Biannual questionnaires must be completed by all flight crew members and returned to the training department within one month.

d) Yearly aircraft performance training to include:

I. Takeoff, climb, cruise, approach and landing performance;

II. Weight and balance calculation;

III. Obstacle clearance;

IV. Fuel planning;

V. Diversion planning;

VI. MEL/CDL; and

VII. Engine out drift down.

7.19.3 Each recurrent training program for pilots shall include simulator training in the procedures and manoeuvre prescribed in the Flight Operations Training Program Manual. This training program is two sessions of simulator (proficiency training and proficiency check) and/or aircraft training. The recurrent training shall ensure flight crew members complete training in CRM skills, which may be accomplished as part of simulator proficiency training or line training, as applicable.

7.19.4 Detail course content of the above training is given in Flight Operations Training Program Manual.

7.20. TRANSITION TRAINING

7.20.1 Pilot's transition training is accomplished with the use of the curricula divided into the following phases:

a) Classroom instruction

b) Cockpit procedures training

c) Simulator training

d) Airplane training

e) Route check (assisted flight in the case of F/O)

f) Supervised solo (not required for F/O)

7.20.2 Detailed course content of the above training is given in Flight Operations Training Program Manual.

7.21. FLIGHT CREW TRANSITION GROUND TRAINING

7.21.1 Each transition training program for pilots shall cover each airplane type operated by the company and shall include:

a) Aircraft systems

b) Performance characteristics

c) Engines

d) Major components

e) Principles of normal, abnormal, and emergency operation; and appropriate procedures and limitations.

f) Procedures for avoiding severe weather situations and for operating in or near thunderstorms (including best penetration altitudes), turbulent air

g) (including clear air turbulence), icing, hail and other potentially hazardous meteorological conditions.

h) Operating limitations.

i) Fuel consumption and cruise control.

j) Flight planning

k) Specialized training

l) The Approved Airplane Flight Manual.

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7.22. TRANSITION SIMULATOR AND ROUTE TRAINING

7.22.1 Each transition training program for pilots must include:

a) Each flight crew must successfully complete the simulator Training and practice all of the manoeuvres and procedures set forth in Flight Operation Training Manual that can be performed in an approved simulator.

b) A check in the simulator to the level of proficiency of a captain or first officer as applicable, in those manoeuvres and procedures set forth in Flight Operation Training Manual that can be performed in the simulator.

c) The pilot shall complete the minimum required hours or sectors of line flight under supervision.

d) The final line check is conducted by an examiner, following a successful evaluation check the pilot will be released for line operations.

7.23. ZERO FLIGHT TIME TRAINING

7.23.1 ZFTT will only be approved for type rating training for flight crew of multi-pilot airplanes who meet the minimum flying experience specified for the level of flight simulator to be used on the course, as follows:

a) Flight simulator qualified to Level C. Flight crew undertaking ZFTT shall have completed not less than 1500 hours flight time or 250 route sectors on a relevant airplane type.

b) Flight simulator qualified to Level D- flight crew undertaking ZFTT shall have not less than 500 hours flight time or 100 route sectors on a relevant type.

c) A relevant type of airplane is a turbo-jet, transport category airplane with a MTOW of not less than 10 tons or an approved passenger seating configuration for not less than 20 passengers.

d) Each ZFTT qualification/training course is customized by an instructor administering the course as per the guidance indicated on the syllabus as necessary to address pilot experience, flight crew position and simulator level.

7.23.2 The simulator is specifically certificated and approved for ZFTT by ECAA.

7.23.3 The company conducts all ZFT training in an appropriately approved full Flight Simulator. Before any training sessions are conducted the MEL must be consulted to check that any Simulator defects are allowable for the type of training to be conducted.

7.23.4 Trainees must complete at least six take-offs and landings in a flight simulator, qualified and approved in a minimum of level C simulator, not later than 21 days after the completion of the skill test. This simulator session shall be conducted by the instructor. If these take-offs and landings have not been performed within the 21 days, the operator shall provide refresher training acceptable to the Authority.

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7.23.5 Trainee must be assessed on their ability to land the aircraft safely during the simulator training that the trainee has satisfactorily completed this requirement and is ready for line training in the actual aircraft and it must be recorded by the examiner conducting the training session.

7.23.6 Following completion of a ZFTT program, a pilot must:

a) Commence Line flying under Supervision as soon as possible within 21 days after completing of the skill test. If Line Flying under supervision has not been commenced within the 21 days, the operator shall provide appropriate training acceptable to the authority.

b) Conduct the first four take-offs and landings of the Line Flying under the supervision of the instructor occupying a pilot's seat. Under normal circumstances, at least 3 of these sectors should be conducted with the trainee acting as PF. A final demonstration of competency is completed in an aircraft during actual line operations under the supervision of an evaluator, instructor. This is intended as a final demonstration of takeoff and landing competency during actual line operations.

7.23.7 Pilots not meeting the experience requirements to quality for ZFTT shall be required to complete base training in the actual airplane, (one hr. A/C training comprising of touch and go exercise will be conducted before being released for line training/ check)

NOTE: Touch and go exercise shall be on day light only & have the weather conditions as follows: cloud shall be at least 300m (1000ft) vertically and 1500m (5000ft) horizontally, a minimum visibility of 5km, a cross wind of less than 15kts and no wind shear report is required.

7.24. INTERRUPTIONS IN TRAINING

The following refresher training shall be conducted in the event of interruptions occurring between a skill test and ZFTT, or between a Skill Test and the commencement of Line Flying under Supervision:

TABLE 1

Interruption Length	Refresher Training Requirements
Up to 21 days	None
21 days to 60 days	Repeat ZFTT Training Session
61 days to 12 months	Refresher Training (inc ZFTT session)
More than 12 months	Repeat Type Rating

7.25. UPGRADE TO PIC

7.25.1 Command training and evaluation is accomplished by pilot flight crew members as part of ground, simulator/aircraft or line training as stipulated on the FOTPM.

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7.25.2 Upgrade is essentially offered on the basis of seniority, but also taking into account company requirements, age, professional quality, moral standards and general behavior.

7.25.3 upon completion of the minimum time as jet co-pilot with satisfactory performance, and again subject to company requirements, a F/O may become eligible for upgrade to Captain training in order of his seniority as F/O. Screening occurs prior to a pilot being assigned duties as PIC and includes:

- Training records review;
- Management recommendations and/or review board;
- Training department recommendations and/or review board;
- Verification of minimum experience acceptable to the Authority;
- Successful completion of Command suitability interview

7.25.4 Captains are encouraged to fill out F/O's progress report forms whenever they deem appropriate. In addition, the Training & Recurrent Manager/Respective Chief Pilots may assign an instructor pilot to evaluate the readiness of the F/O for an upgrade depending on his/her seniority. If the evaluation indicates satisfactory progress or if candidate's training file shows satisfactory performance, the F/O will be given written and/or oral examinations in all of the required subjects

7.25.5 For direct entry pilots refer to FOM1.9 (7.1)

7.25.6 All pilots progressing from First Officer to Captain will undertake ground school, designed to emphasize the change in duties and responsibilities between a First Officer and Captain. At the end of the course, the candidate should have a firm understanding of the administrative duties and responsibilities of an Ethiopian Captain.

7.25.7 The F/O, once he joins the company, will be required to improve his knowledge of aircraft systems, emergency procedures, instrument approach procedures and radio procedures.

7.25.8 First Officer upgrade training is accomplished with the use of the curricula divided into the following phases:

a) Classroom instruction or CBT

b) Cockpit procedure training

c) Simulator training

d) Airplane training

e) Route check

f) Supervised solo

7.25.9 For detailed procedures refer to Flight Operations Training Program Manual.

7.25.10 The training will cover all phases of the person's past performance with the airline. A pilot who has a history of poor ground school grades, poor attendance, sloppy appearance, lackadaisical manner in keeping personal records (e.g. licenses, manuals, passports, vaccination cards etc.) may at the option of the Flight Operation Division, be passed over.

7.25.11 Seniority will determine the order in which an F/O becomes eligible for consideration and entry into the training.

7.25.12 The trainees will be given ground school in all subjects required for his upgrading.

7.25.13 In the event that a trainee fails to qualify for Captain upgrade, the review committee chaired by Dir. Flight Training & Standards will determine whether the trainee will be given additional training in an effort to qualify him as a Captain or if he is to be restored to the position of F/O provided there is a vacancy and his previous performance in the above positions was satisfactory. For more information refer FOM 5.1 (7.30).

7.26. DIFFERENCE TRAINING

7.26.1 Flight crewmembers shall complete an appropriate difference training & evaluation course whenever:

a) Operating another variant of an airplane of the same type; or

b) A change of equivalent and/or procedures occurs on type of variants currently operated.

7.26.2 The training must consist of at least the following as applicable to their assigned duties and responsibilities.

a) Ground instruction of 16 hours in each appropriate subject or part thereof required.

b) The required simulator training as described in FOTPM.

c) Line training shall include three take-offs and landings.

7.26.3 Differences training for all variations of particular type airplane may be included in the initial, transition, upgrade or recurrent trainings.

7.27. OPERATION ON MORE THAN ONE VARIANT OR TYPE

7.27.1 Airplane within an approved grouping:

SECTION 5.1 TRAINING PROGRAM

There is no specific minimum requirement for operation on more than one type or variant within an approved grouping once qualified on any one type or variant. The transition training however should cover all differences in flight and handling characteristics, procedures, etc. as specified by the manufacturer. Sufficient number of take-off and landings shall also be conducted in the simulator in each type or variant to proficiency. Actual aircraft training need only be given in and one of the types or variants.

7.27.2 Airplanes not within the same approved Grouping:

It is not Ethiopian's policy to cross utilize flight crew in line operation on more than one type or variant not within the same approved grouping except for instruction purposes. Instructors so dual qualified will be required to meet all qualification and recency requirements for each type. Team Leader Crew Scheduling will ensure that no dual qualified instructor is scheduled to fly either type or variant without meeting all requirements and recencies and that no one is scheduled to fly more than one type in any one duty period.

7.28. ROUTE AREA AND AIRPORT QUALIFICATION TRAINING & REQUIREMENTS

7.28.1 Line training and evaluation will be completed as part of initial, command upgrade, transition/conversion, and differences training consisting of line training and evaluations, approved, or accepted by the ECAA, which ensures flight crew members are qualified to operate in areas, on routes or route segments and into the airports to be used in operations for the Operator. All flight crew members before being scheduled as PIC or FO must demonstrate an adequate and thorough knowledge of:

a) The route to be flown, and the aerodromes, which are to be used.

This shall include knowledge of:

I. Terrain and minimum safe altitudes;
II. Seasonal meteorological conditions;
III. Meteorological, communication and air traffic facilities, services and procedures;
IV. Search and rescue services for the areas over which the aircraft will be flown;
V. Navigational facilities and procedures, including any long-range navigation procedures associated with the route along which the flight is to take place;
VI. Procedures applicable to flight paths over heavily populated areas and areas of high air traffic density;

SECTION 5.1 TRAINING PROGRAM

VII. Airport obstructions, physical layout, lighting, approach aids and arrival, departure, holding and instrument approach procedures and applicable operating minima

b) The procedures applicable to flight paths over heavily populated areas and areas of high traffic density, obstructions, physical layout, approach aids and arrival, departure, holding and instrument approach procedures, and applicable airport operating weather minima.

c) All flight crew members should fly at least for six months on type, as applicable dual rated (B777 and B787) and they shall undergo line training and check on NAT HLA routes before assigned to NAT HLA routes.

NOTE: The requirements for (a) can be satisfied during an annual line check by all Captains demonstrating an adequate knowledge and proficiency in the use of company manuals, briefing information and Jeppesen documents and charts. Sub-section (b) above relates to arrival, departure, holding and instrument approach procedures that are demonstrated and satisfied during bi-annual proficiency checks.

7.29. INSTRUCTORS AND EXAMINERS TRAINING

7.29.1. All instructors, evaluators and line check airmen shall undergo an initial training program, in addition to specific requirements stated below. The training shall include the following:

a. The fundamentals of teaching and evaluation;
b. Lesson plan management;
c. Briefing and debriefing;
d. Human performance issues;
e. Company policies and procedures;
f. Simulator serviceability and training in simulator operation;
g. Dangers associated with simulating system failures in flight;
h. Teamwork, Presentation and Coaching.
i. Techniques of applied instruction and facilitation;
j. Training program development;
k. Assessment of trainees' performance
l. Use of training aids, including flight simulation training devices as appropriate;
m. The simulated or actual weather and environmental conditions necessary to conduct each simulator or aircraft training/evaluation session to be administered.
n. Hazards involved in simulating system failures and malfunctions in the aircraft; and
o. Principles of threat and error management.

SECTION 5.1 TRAINING PROGRAM

7.29.2. Instructors & Examiners perform duties, functions and responsibilities stipulated on this manual. Instructors and evaluators are selected in a manner consistent with the overall objectives of the training program. To achieve this aim, selection criteria and processes includes Confirmation that a minimum level of experience has been attained; A review of the training records of potential selectees; recommendations from Flight Operations management and/or the training department.

7.29.3. The applicable Civil Aviation Authority regulations, company policy, and procedures.

7.29.4. Proper evaluation of performance including the detection of improper and insufficient training and personal characteristics that could adversely affect safety.

7.29.5. The approved methods, procedures and limitations for performing the required normal, abnormal and emergency procedures in the airplane.

7.29.6. Enough flight training and practice in conducting flight checks in the required normal, abnormal, and emergency manoeuvres to ensure his ability to conduct the flight training required by this chapter. This training and practice must be accomplished from both the left and right pilot's seats in the simulator.

NOTE: Refer to the Flight Operation Training Program Manual for more information.

7.30. INSTRUCTORS AND EXAMINERS TRAINING

7.30.1. In addition to the requirements stated above (7.29) instructors and examiners will take the following:

a) A formal observation program that permits supervised practical instruction and observation of experienced instructors administering the course and syllabus lessons.

b) Aircraft jump seat observation program or equivalent, for non-line qualified instructors, to provide familiarity with current and type-related line operations.

c) The observation of the candidate during supervised practical instruction or examination.

7.31. FLIGHT INSTRUCTORS AND EXAMINERS TRAINING

7.31.1. In addition to the requirements stated above (7.29) flight instructors and examiners will take aircraft specific training on the following:

a) An initial seat-specific (right or left seat, as applicable) qualification program for instructors, evaluators, line check airmen and any other pilots, so designated by management, who perform duties from either seat.

b) The appropriate safety measures to be taken for emergency situation that are likely to develop in training. This training must be accomplished from both the left and right pilot's seats.

7.32. EXAMINERS TRAINING

7.32.1. In addition to the requirements stated above (7.29, 7.30 & 7.31) examiners will take the following:

a) The appropriate methods, procedures and techniques for conducting the required checks.

b) The appropriate corrective action in the case of unsatisfactory checks.

7.33. INSTRUCTORS AND EXAMINERS RECURRENT QUALIFICATION

7.33.1. Recurrent qualification process for Instructors and Examiners includes:

a) Standardization meetings as per FOM 1.4 (7.3);

b) Supervised training or examination sessions (simulator or aircraft) by an approved individual (12 months);

c) A minimum of two training or examinations events per six months.

d) A seat-specific (right or left, as applicable) recurrent program for Instructors, Examiners and Line Check Airmen who perform duties from either seat (performed in approved type specific full flight simulator once in every 12 months);

NOTE: - The simulator administrator is responsible for the continuing qualification and recording of a seat specific training taken by all instructors and/or examiners.

e) A jump seat observation flight for non-line qualified instructors and examiners.

7.34. INSTRUCTOR & EXAMINERS TRANSITION TRAINING

7.34.1. Each transition training program for flight or simulator instructors shall include sufficient in flight and/or simulator training and practice in conducting flight checks in the required normal, abnormal and emergency maneuvers to ensure competence to conduct the flight training required by this part. This training and practice must be from both the left and right pilot's seats.

SECTION 5.1 TRAINING PROGRAM

7.34.2. Instruction must include the appropriate safety measures to be taken for emergency situation that are likely to develop in training. This training must be accomplished from both the left and right pilot's seats.

7.35. ASYMMETRIC FLIGHT TRAINING

All asymmetric training will be conducted in a Full Flight Simulator. Pilots are no longer required to demonstrate their ability to control a simulated engine failure in the aircraft.

7.36. FLIGHT CREW TRAINING AND EXAMINATION REQUIREMENTS

All flight crew members shall be trained and evaluated, as applicable, in accordance with requirements specified below.

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SECTION 5.1 TRAINING PROGRAM

TABLE 2

TRAINING & EXAMINATION	GROUND	SIMULATOR OR FLIGHT	LINE
A/C type performance (TE)	I		
A/C type systems and limitations (TE)	3		
A/C type/different type/variants qualification (TE)	I	1	I
Adverse Weather (TE) [Cold weather operations, De-/anti-icing policies, Contaminated runway operations, Thunderstorm avoidance]	3		
Aircraft upset prevention and recovery (T)	3		
Command Training (T)	I	I	I
Common/English Language Proficiency (E)	P		
Company Indoctrination (T)	I		
Crew Resource Management(TE)	3	1	1
Dangerous goods (TE)	2		
Emergency Evacuation	I	1	
ETOPS/EDTO (TE)	I		I
GPWS (CFIT)(TE)	3	3	
Incapacitation (TE)	3	3	
Joint flight & Cabin crew Emergency & practical training on onboard emergency and safety equipment	1		
Joint flight & cabin crew emergency situations (including evacuation and coordination), safety & security threats Practical Training	3		
Joint Flight crew, cabin crew & Dispatchers CRM Training	1		
LOFT		1	
Low Visibility Operations(TE)	I	1	
Normal, Non normal procedures /manoeuvres to include as minimum rejected take off, engine failure (TE)	½**	½**	
Operations Specification/Requirements (AOC)(E)	1*	1*	1*
PBCS(T), MNPS/NAT HLA (T), LRN	I	I	I
PBN(TE)	1	1	
RVSM and RNP/RNAV (TE)	I		
Seat specific qualification (Right seat) (TE)	I	1	
Security(T)	3		
SMS	2		
Special Airport and Route Qualification (TE)	1****	1****	1****
TCAS/ACAS (TE)	3	3	
Unlawful Interference (hijack) (T)	3		
Volcanic Ash (TE)	I	3	
Wind Shear Avoidance and Recovery (TE)	3	3	

SECTION 5.1 TRAINING PROGRAM

Legend

½ training /evaluations shall be satisfactorily completed every 6 months.

1. Shall be satisfactorily completed during initial training and once every 12 months.
2. Shall be satisfactorily completed during initial training and every 24 months.
3. Shall be satisfactorily completed during initial training and once every 36 months.

I Shall be completed during initial type training.

E Evaluation only

P Those demonstrating language proficiency at the Operational Level (Level 4) should be evaluated at least once every three years. Those demonstrating language proficiency at the Extended Level (Level 5) should be evaluated at least once every six years. If the proficiency score is (level 6) he/she doesn't need further evaluation.

T Training only

TE Training and evaluation

Note: Initial refers to initial type rating, transition/Conversion

* Training specific to aircraft operated under the AOC, conducted on appropriate training devices. The training shall include Approaches authorized by the ECAA; Takeoff, approach, and landing visibility and ceiling minima; Allowances for inoperative ground equipment as per MEL; Wind limitations (crosswind, tailwind, headwind) in accordance with aircraft manuals

** First officers' evaluation shall be satisfactorily completed every 12 months.

*** 1st refresher is conducted 18 months following the initial.

**** Unless the PIC has observed or has actually made an approach and landing (or take off), at the aerodrome concerned or special route within in the previous twelve (12) months.

****

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

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1.0. PURPOSE

The purpose of this progression cycle and qualification procedure is to establish guidelines for Progression cycle and qualification requirements.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight crewmembers
- Flight Instructors
- Simulator Instructors
- Examiners
- Dir flight Training and standards

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.05

4.1 It is Ethiopian policy not to assign a captain or First officer as a minimum operating crew unless the requirements stated herein are met.

4.2. Each pilot shall show the aeronautical training and experience used to meet the requirements for a license or rating or recency of experience by a reliable record, Pilot Logbook. (ECARAS 8.4.1.8)

4.3 Documents and records of the following shall be kept in a manner acceptable to The Authority:

4.3.1 Training and/or experience used to meet the requirements for a license, rating, endorsement and/or authorization.

4.3.2 The experience required to show the maintaining of recency of aeronautical experience according to the requirements of ECAA.

4.4 Flight instructor shall not serve as an instructor unless he/she has completed the curricula approved by the Authority.

4.5 It is Ethiopian policy that in the event a student fails to satisfactorily complete an evaluation specified in the curriculum of a required training program, namely

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

Classroom, Cockpit procedure trainer, Simulator and Airplane, a retake of the examination and requalification is necessary.

5.0. DEFINITION

- **Flight Simulation Training Device**. Also known as synthetic flight trainer. Any apparatus in which flight conditions are simulated on the ground:
- **Flight Simulator.** Provides an accurate representation of the flight deck of a particular aircraft type to the extent that the mechanical, electrical, electronic, etc. aircraft systems control functions, the normal environment of flight crewmembers, and the performance and flight characteristics of that type of aircraft are realistically simulated.
- **Instrument Flight Time**. Time during which a pilot is piloting an aircraft solely by reference to instruments and without external reference points.
- **Instrument Ground Time**. Time during which a pilot is practicing, on the ground, simulated instrument flight in a flight simulation training device approved by the Authority.
- **Instrument Time**. Instrument flight time or instrument ground time.
- **Licensing Authority**. The Authority is responsible for the licensing of personnel.
- **Medical Certificate**. The evidence issued by the Authority that the license holder meets specific requirements of medical fitness. It is issued following an evaluation by the Licensing Authority of the report submitted by the designated medical examiner who conducted the examination of the applicant for the license

6.0. RESPONSIBILITY

6.1. Director Flight Training and Standards is responsible for license, qualification and competency requirements stated hereunder.

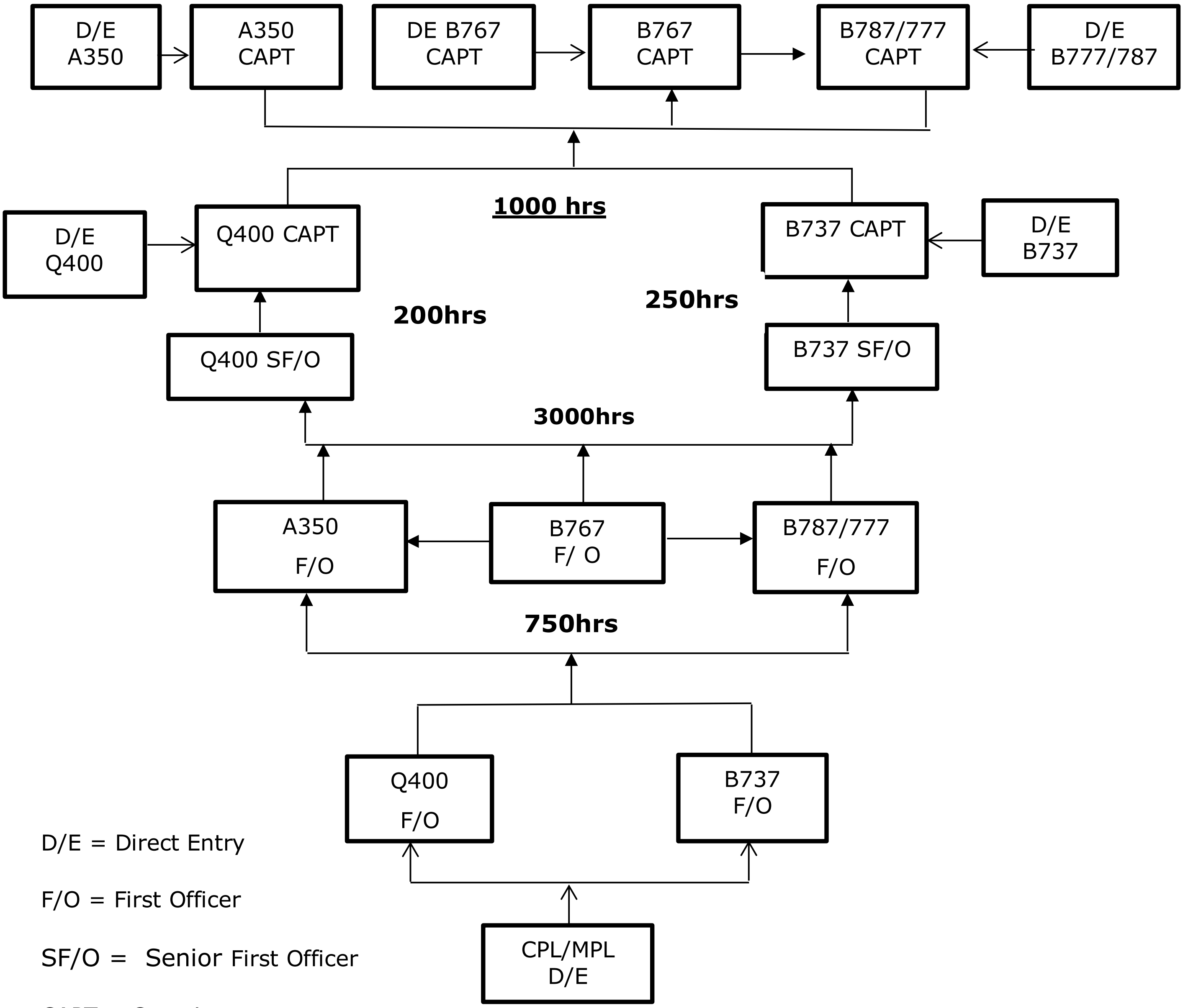
6.2. The Director Flight Training and Standards shall have overall responsibility for ensuring satisfactory integration of flying training synthetic flight training and theoretical knowledge instruction and for supervising the progress of individual trainees and trainers. (ECARAS3.3.5.3)

6.3. Director Flight Training and Standards is responsible to make sure the proper implementation of the policy and procedure in this section.

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.0. PROCEDURE

7.1. PROGRESSION CYCLE



D/E = Direct Entry

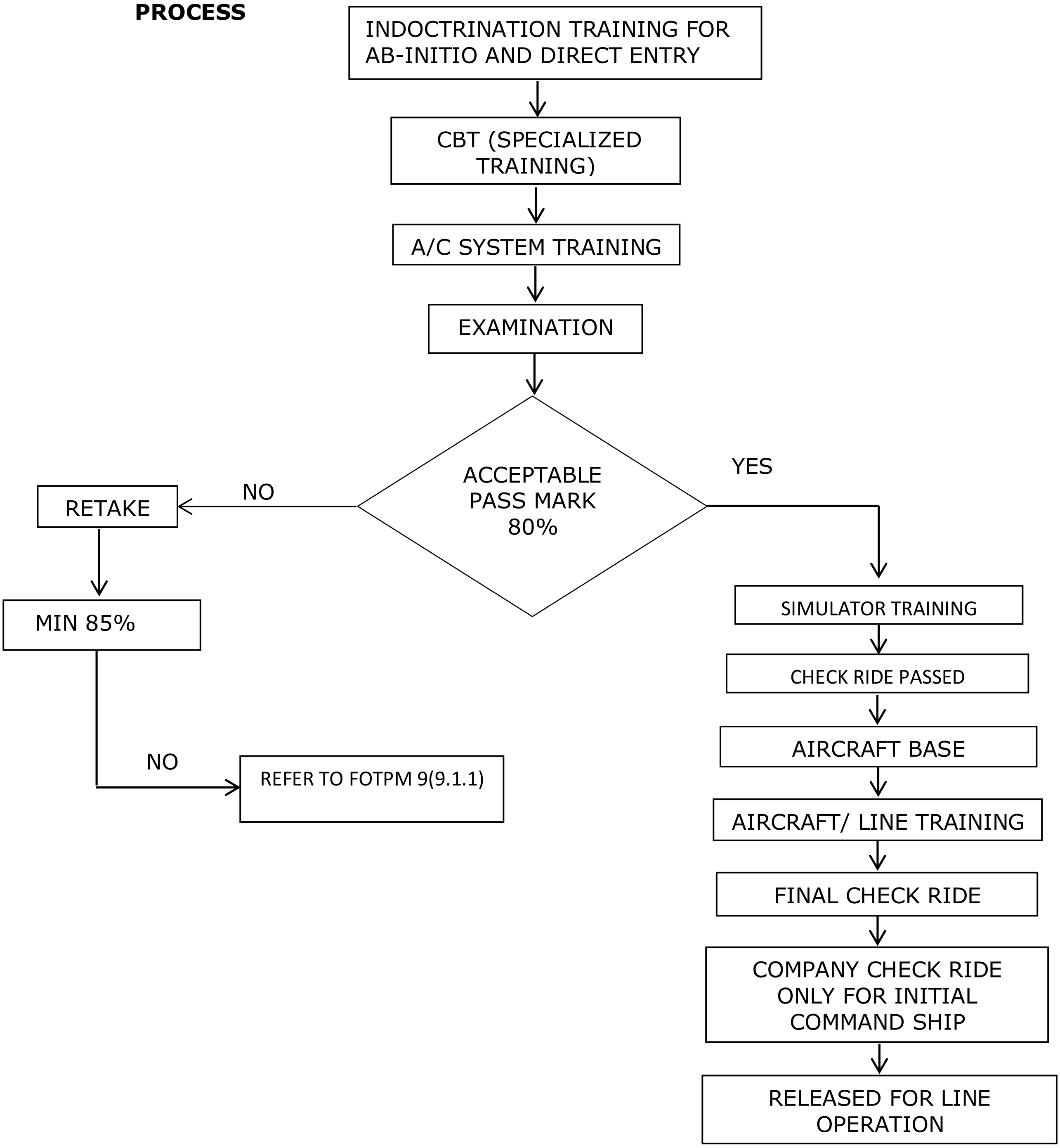
F/O = First Officer

SF/O = Senior First Officer

CAPT = Captain

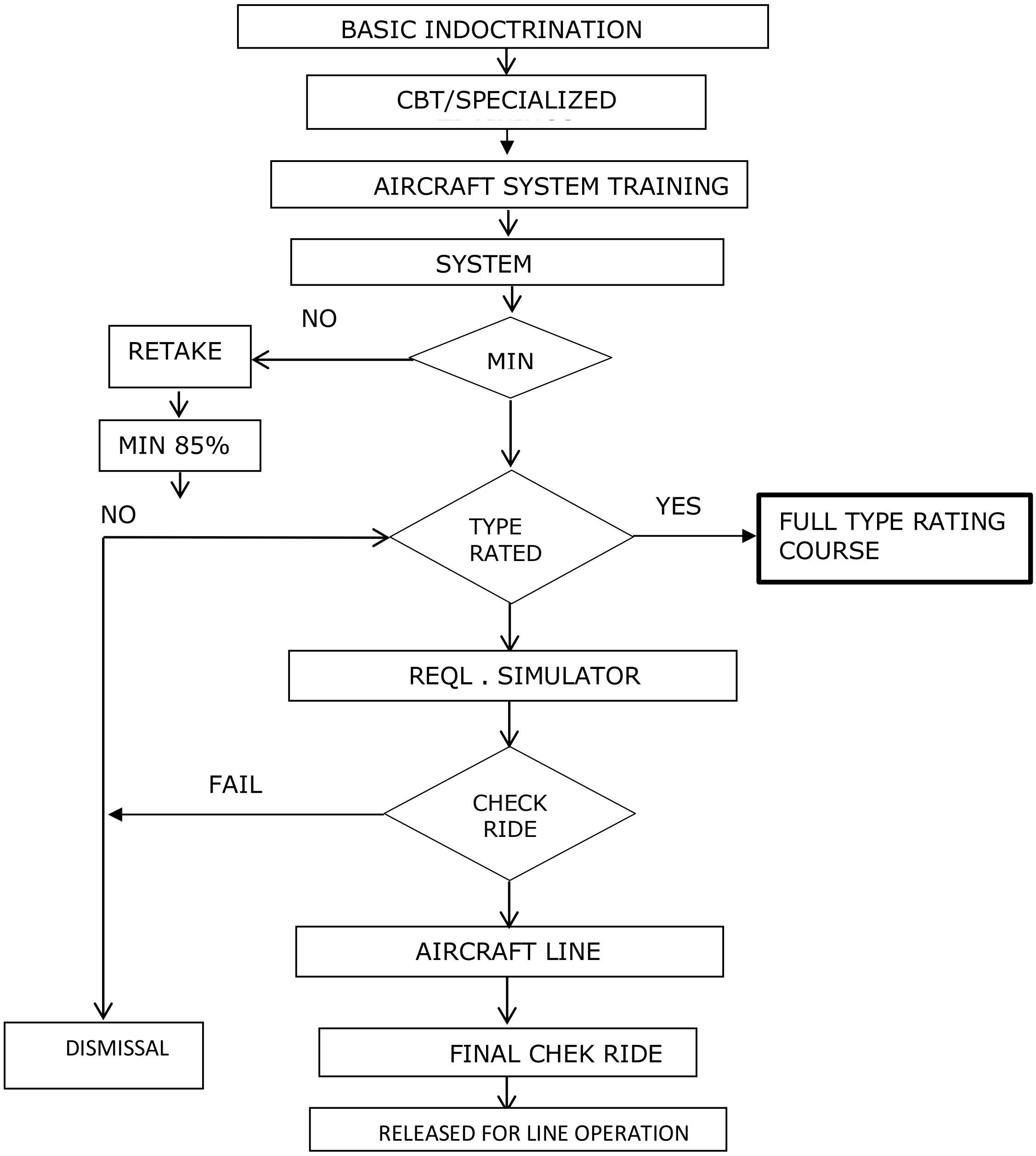
SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.2. AB-INITIO ENTRY & NORMAL FLIGHT CREW QUALIFICATION CRITERIA & PROCESS



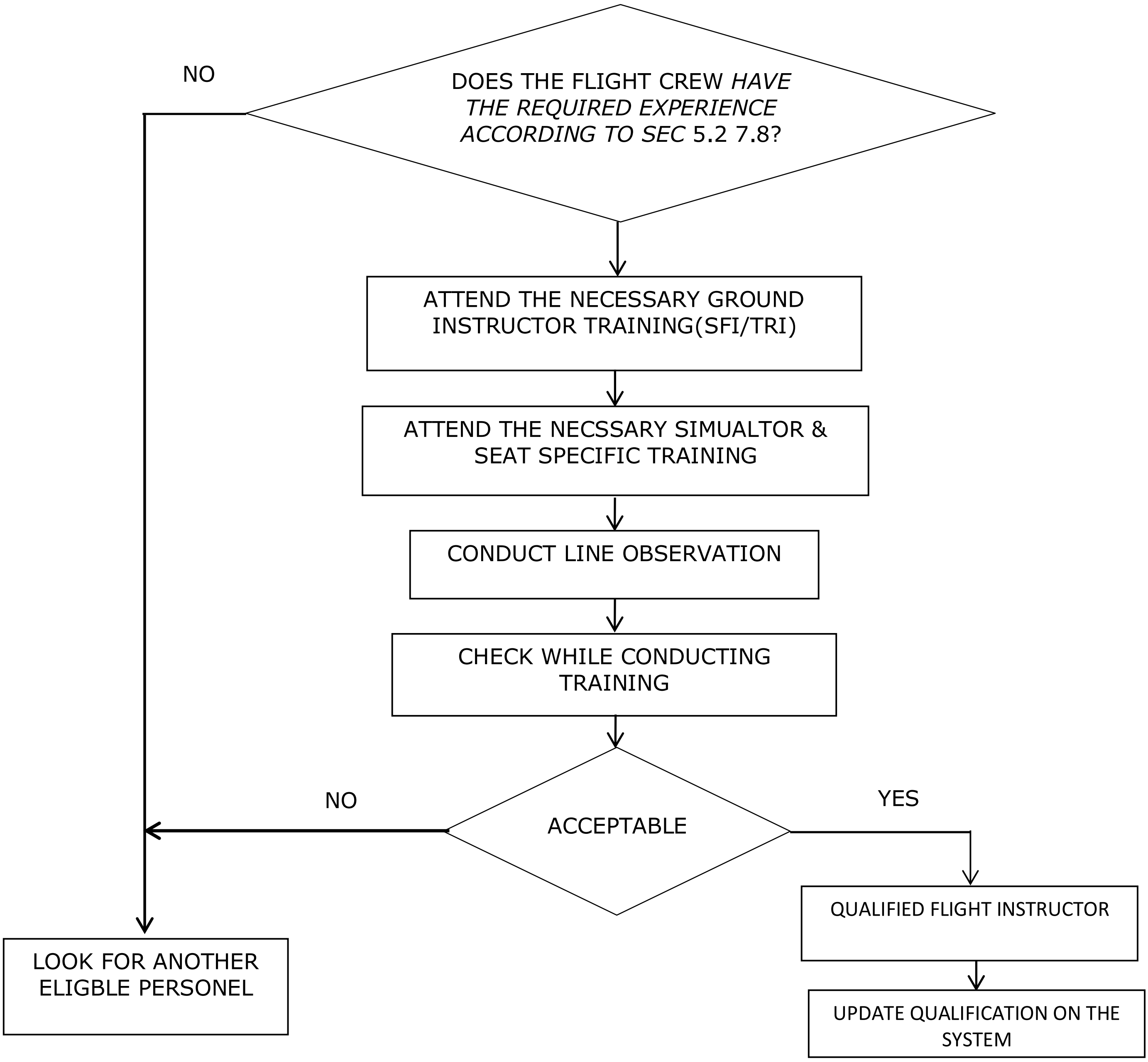
SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.3. DIRECT ENTRY PILOTS QUALIFICATION PROCESS



SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

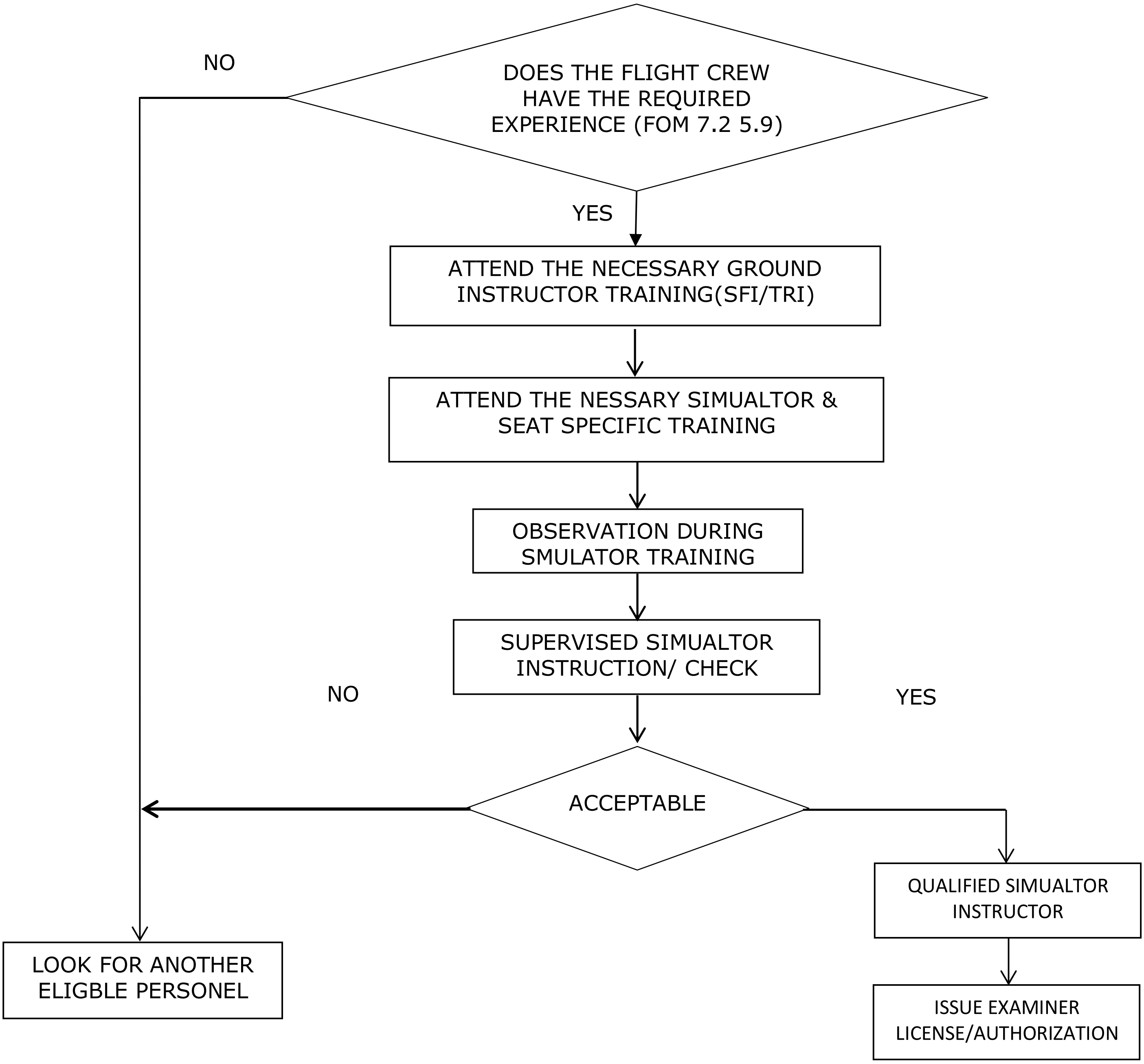
7.4. FLIGHT INSTRUCTOR/EXAMINER QUALIFICATION CRITERIA AND PROCESS



- To be flight examiner he has to serve for 6 months as a type instructor.

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.5. SIMUALTOR INSTRUCTOR/EXAMINER QUALIFICATION CRITERIA AND PROCESS



* To be simulator examiner at least 50 hours of flight instruction as a simulator instructor on the applicable type.

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.6. PILOT-IN-COMMAND AND FIRST OFFICER QUALIFICATIONS

7.6.1. Ethiopian shall not assign a pilot to act as a minimum operating crew of an aircraft unless:

a) He holds an appropriate valid license endorsed for that type of aircraft;

b) He holds a current instrument rating valid for that class of aircraft for IFR operations; he or she has, within the past 6 calendar months, logged at least 6 hours of instrument flight time including at least 3 hours in flight in the category of aircraft and completed at least 6 instrument approaches. (ECARAS8.4.1.10)

c) Within the preceding 90 days that the pilot has operated the flight controls, on the same type or variant of aircraft, three take-offs and landings, or has otherwise demonstrated competence in a flight simulator approved for the purpose. Pilots must notify crew scheduling if they have not made at least three take offs and landings in the specified period;

d) That crew member has completed the appropriate initial or recurrent training phase of the training program appropriate to the type of operation in which the crew member is to serve.

e) Passed an instrument competency check or an instrument proficiency check conducted by the examiner.

NOTE: The instrument competency check shall consist of the same theory and practical requirements of a Proficiency Check for the category and class of license held. The instructor conducting the check shall sign the form and the training department shall retain a record.

f) An annual line check administered by an appropriate flight instructor examiner (captains only)

g) He holds valid medical certificate.

h) Pilot-in-command who has completed an instrument competency check with an authorized representative of the Authority retains currency for IFR operations for 6 calendar-months following that check. (ECARAS 8.4.1.10)

i) First Officer who has completed an instrument competency check with an authorized representative of the Authority retains currency for IFR operations for 6 calendar-months following that check.

j) For detailed training requirements for Flight Crew whose recency has lapsed, refer to the Flight Operations Training Program Manual.

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

k) Specific qualification in accordance of this chapter must also be considered before assigning a flight crewmember for flight duty.

7.6.2. For B777 and B787 at least one recurrent training on each type is required within 12 months.

7.7. INSTRUCTORS AND EXAMINERS

All instructors and examiners are selected based on the standardized processes, meet the required qualification and performance standards, and have the required certification(s)/ approval(s) from the ECAA. They will be evaluated annually to ensure compliance with required qualification and performance standards.

Instructors are responsible for the delivery of training, including imparting knowledge, developing competencies, and preparing trainees for assessment. Examiners, on the other hand, are independent of the instructional process and are tasked solely with evaluating trainee performance to determine if established proficiency standards have been met.

This segregation ensures objectivity, eliminates potential conflicts of interest, and upholds the integrity of the evaluation process. A functional separation is maintained between instructional and examination duties, even when the same individual performs both roles. While some instructors are also designated as examiners, their examining responsibilities are carried out independently and under the authority of the Manager Training Standards, to ensure impartiality and objectivity. This dual-role policy is supported by clear procedures and oversight mechanisms to prevent conflicts of interest and to ensure compliance with ECAA and other regulatory requirements.

Note: Examiners may perform both instructor and examiner duties, in accordance with their assigned tasks, and will carry out only the duties specifically assigned to them.

The below are instructor's responsibilities:

7.7.1. Flight instructors are foundational to the effectiveness and safety of the flight training program. They are responsible for developing competent, confident, and safety-conscious pilots.

7.7.1.1 The instructor is responsible for delivering approved training to both company and customer pilots, using standardized instructional techniques and company manuals.

7.7.1.2 **Instructional Duties:**

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

- Deliver ground, flight, and simulator training in accordance with approved syllabi.
- Demonstrate proper techniques and procedures for all phases of flight.
- Apply and teach company-approved procedures, training manuals, checklists, and standard operating procedures (SOPs).
- Develop and instill sound airmanship, discipline, and CRM principles.
- Ensure all training is conducted in English, maintaining fluency and clarity in communication.

7.7.1.3 **Record Keeping and Administration**

- Accurately maintain and update training records and progress reports for each trainee.
- Provide feedback and performance evaluations to trainees and training management.
- Recommend remedial training where necessary.

7.7.1.4 **Operational Responsibilities**

- When supervising flights, the instructor shall act as the pilot-in-command (PIC) and assume full responsibility for the flight.
- Ensure the trainee captain is seated at the controls during all critical phases: taxi, take-off, approach, and landing.
- Simulated emergency procedures affecting flight characteristics shall not be conducted when passengers are onboard.
- May override any operational decision of the trainee captain or assume command during any safety-critical situation.
- When operating with captains not fully qualified on the aircraft type, the instructor shall occupy a control seat (left or right).

7.7.1.5 **Professional Conduct**

- Maintain high professional standards in conduct, appearance, and interaction with trainees.
- Participate in regular instructor standardization and training workshops.
- Comply with company policies and Civil Aviation Authority guidelines.

7.7.2 **Examiner Roles**

Examiners serve as objective evaluators of pilot proficiency, ensuring training standards and regulatory requirements are met. While some examiners may also act as instructors, their evaluating role is functionally independent.

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.7.2.1 **Evaluation Duties**

- Conduct initial, recurrent, and proficiency flight checks on company and customer pilots.
- Use company-approved evaluation criteria, procedures, and checklists aligned with SOPs and Civil Aviation Authority standards.
- Ensure all checks are fair, objective, and consistent with the documented standards.

7.7.2.2 **Certification and Oversight**

- Must be certified, qualified, and approved by the ECAA or designated examiner.
- Examiner qualification is contingent on satisfactory performance during oversight by the ECAA or designated authority.
- Must demonstrate competence in evaluation, instruction, and English language proficiency.

7.7.2.3 **Standardization and Compliance**

- Report to the Manager, Training Standards and operate under guidance to ensure consistent application of evaluation procedures.
- Participate in examiner standardization sessions and regulatory oversight audits.
- Maintain current knowledge of applicable regulations, operational procedures, and best practices.

7.7.2.4 **Training and Qualification**

- Must meet the same initial and recurrent training requirements as those prescribed for the pilot-in-command (PIC) being evaluated.
- Maintain flight proficiency and current operational experience on the aircraft types being examined.

7.7.2.**5 Conduct and Integrity**

- Maintain the highest levels of professional integrity and impartiality.
- Avoid any conflict of interest during evaluations.
- Provide constructive feedback and recommendations post-check

7.8. SELECTION OF FLIGHT INSTRUCTORS

7.8.1 Flight crewmembers selected for instruction must be thoroughly screened in accordance with the following process:

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

a) He should satisfy the PIC qualification listed above and shall be certified by the company.

b) A pilot who has served for six months or 500 hours as pilot-in-command on type can be selected for instructor pilot.

c) Review of training records to ensure eligibility.

d) Review of personal file and individual performance.

7.8.2 A record of this process shall be kept on the candidates Training File.

7.9. SELECTION OF SIMULATOR INSTRUCTORS

7.9.1. Selection of simulator instructors shall be made in accordance with the following process:

a) Review of training records to ensure eligibility.

b) Review of personal file and individual performance.

c) Hold or have held ATPL.

d) Have a minimum of 500 hours total flight time as a pilot-in-command.

e) Have completed the proficiency check on the applicable airplane type in a FFS within 12 months preceding the application.

f) Have at least third-class medical certificate.

7.9.2. A record of this process shall be kept on the candidates Training File.

7.10. SELECTION OF FLIGHT EXAMINER

7.10.1. Selection of flight examiner shall be made in accordance with the following process:

a) Review of training records to ensure eligibility.

b) Review of personal file and individual performance.

c) Hold a current ATPL.

d) Has served for 6 months as an instructor.

e) Must be a highly qualified senior instructor on the type

f) Has to have a good leadership quality

g) Influence & negotiation

h) Shall be certified by the company

7.10.2. A record of this process shall be kept on the candidates Training File.

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

7.11. SELECTION OF SIMULATOR EXAMINER

7.11.1. Selection of simulator examiner shall be made in accordance with the following process:

a) Review of training records to ensure eligibility.

b) Review of personal file and individual performance.

c) Hold or have held an ATPL.

d) Have a minimum of 500 hours total flight time as a pilot-in-command.

e) Have completed at least 50 hours of flight instruction as a simulator instructor on the applicable type.

7.12. SELECTION OF INSTRUCTOR AND EXAMINER TRAINER

7.12.1. Selection of instructor and examiner trainer shall be made in accordance with the following process:

a) Be an examiner.

b) At least one year total experience as an examiner in Ethiopia.

7.13. SELECTION OF GROUND INSTRUCTORS

7.13.1. Selection of ground instructors shall be made in accordance with the following process:

a) Possess good administrative and presentation skills (using both electronic and conventional media).

b) Be capable of independently producing lesson plans and training material for ground instruction.

c) Have successfully completed an approval instructor course and completed additional supervised training in the conduct of technical classroom instruction for pilot trainees.

7.14. SELECTION OF CREW RESOURCE MANAGEMENT INSTRUCTOR (CRMI)

7.14.1 A CRMI shall have the following qualifications:

a) Adequate knowledge of human performance and limitations (HPL), gained by:
   I. Having obtained a CPL/ATPL with HPL theory, or
   II. Having followed a theoretical HPL course covering the whole HPL syllabus

b) A language proficiency of a minimum of ICAO Operational Level 4

c) Having completed the initial CRM course

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d) Having completed 40 hours of CRM Trainer Training

7.14.2 Selection of a CRMI shall be made in accordance with the following process:

a) Review of training records to ensure eligibility.

b) Review personal files and individual performance

c) Presentation (Any topic given by the panel to see his/her presentation and facilitation skill)

d) Coaching

I. Attends observation CRM classes
II. Facilitates CRM class under the supervision of a CRMI
III. Upon positive feedback from the CRMI, will be assigned as CRMI

7.14.3 A record of this process shall be kept on the candidates Training File.

7.14.4 A CRM trainer should be assessed by a CRM trainer examiner (CRMTE) when conducting their first CRM training course.

7.14.5 A CRMT assessment is valid for a period of 3 years. For currency of the 3-year validity period, the CRM trainer should

a) conduct at least 2 CRM training events in any 12-month period

b) be assessed within the last 12 months of the 3-year validity period by a CRM trainer examiner.

c) receive a CRM trainer refresher training within the 3-year validity period

7.15. FLIGHT INSTRUCTOR DUTIES

7.15.1. He/she shall conduct the following:

a) Base training

b) Route/ Line check

c) Familiarization flights along special routes and airfields designated as CAT C aerodromes.

7.16. FLIGHT EXAMINER DUTIES

7.16.1. He/she shall conduct the following:

a) Base training

b) Route/ Line check

c) Final route check

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d) Familiarization flights along special routes and airfields designated as CAT C aerodromes.

NOTE: No simulated engine failures or any other emergencies are permitted on actual flights.

7.17. SIMULATOR INSTRUCTOR DUTIES

7.17.1. He/she shall conduct the following:

a) Transition, recurrent and update training.

b) Annual/Final Line Checks.

c) LOFT Training.

7.18. FLIGHT SIMULATOR EXAMINER DUTIES

7.18.1. He/she shall conduct the following:

a) Transition, recurrent and update training.

b) Annual/Final Line Checks.

c) LOFT Training.

d) Simulator proficiency or competency training.

7.19. GROUND INSTRUCTORS DUTIES

Ground Instructors are authorized to conduct relevant ground training for transition and recurrent training.

7.20. CREW RESOURCE MANAGEMENT INSTRUCTOR (CRMI) DUTIES

7.20.1. He/she shall conduct the following:

a) Conduct CRM and other training duties.

b) Develop CRM course materials.

7.21. CURRENCY FOR FLIGHT INSTRUCTORS AND EXAMINERS

7.21.1 In order to maintain recency of experience flight instructors shall conduct at least two training events on an aircraft during the preceding 6 months period.

7.21.2 In order to maintain recency of experience flight examiners shall conduct at least two training/checking events on an aircraft during the preceding 6 months period.

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7.21.3 Should the recency requirements not be met, a successful requalification check shall be completed prior to undertaking further training/examining duties.

7.21.4 If no training/checking events have been conducted during the preceding 24 months, the complete qualification process shall be undertaken.

7.22. CURRENCY FOR SIMULATOR INSTRUCTORS AND EXAMINERS

7.22.1 In order to maintain recency of experience simulator instructors shall conduct at least two training events on a simulator during the preceding 6 months period.

7.22.2 In order to maintain recency simulator examiners shall conduct at least two training/checking events on a simulator during the preceding 6 months period.

7.22.3 Should the recency requirements not be met in one year, a successful requalification check shall be completed prior to undertaking further training/examining duties.

7.22.4 If no training/checking events have been conducted during the preceding 24 months, the complete qualification process shall be undertaken.

7.22.5 A non-line qualified simulator instructor will be scheduled for observation flights on the flight deck, in uniform. Two flights per year are required.

Note: Examiners (both flight and simulator) Standardization Program will ensure all designated and ECAA delegated examiners apply consistent, operator-specific evaluation standards. This program includes initial and recurrent standardization tailored to the operator's aircraft types, SOPs, and training policies. All examiners undergo annual reviews and standardization sessions conducted by the Manager Training Standards or standards captains to uphold assessment integrity and alignment with ECAA and other regulatory body requirements.

7.23. CERTIFICATION OF CREWMEMBERS, INSTRUCTORS OR EXAMINERS

The instructor or examiner involved in the training program responsible for a particular ground training subject, segment of flight training, course of training, flight check, or competence check shall certify as to the proficiency and knowledge of the crewmember, instructors, or examiners concerned upon completion of that training or check. That certification shall be made a part of the trainee’s record. When such certification is made by a computer entry a signature is not required, however the certifying instructor or examiner must be identified with that entry.

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7.24. VALIDITY OF EXAMINERS AUTHORIZATION

ECAA authorizes examiners. The authorization is valid for one year. Authorization is renewed by application of Director Flight Training & Standards to ECAA prior to expiry of the authorization.

7.25. QUALITY CONTROL OF INSTRUCTORS/EXAMINERS & TRAINING COURSES

Director Flight Training & Standards shall seek feedback on the course content and Examiner/Instructor performance at the completion of training. This will be achieved by completion of the feedback form by the trainees.

7.25.1. If no training/checking events have been conducted during the preceding 24 months, the complete qualification process shall be undertaken.

7.26. COURSE CONTENT/STRUCTURE

Feedback shall be forwarded to the Training Review Committee for review and incorporation as appropriate.

7.27. POOR PERFORMANCE OF AN INSTRUCTOR/EXAMINER

7.27.1. The following procedure will be followed:

a) A detailed set of documents will be prepared in which a clear description of the poor performance of the Examiner/instructor is contained. A de-briefing of the examiner/instructor will take place. The debriefing will be conducted by Director Flight Training & Standards.

b) Formal notification in writing from Director Flight Training & Standards will precede all action of this nature.

c) Any relevant documentation will go on the training file of the respective examiner/instructor but not his personal file.

7.28. PROFICIENCY CHECKS

7.28.1. Captains & First Officers undergo two (2) proficiency checks every 12 months to demonstrate competence in carrying out normal, non-normal and emergency procedures and handling.

7.28.2. Proficiency check for captains is valid for 6 months in addition to the month of issue. If issued within the final 2 calendar months of validity of a previous proficiency check, the period of validity extended from the date of issue until 6 calendar months from the expiry date of the previous proficiency check.

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7.29. LINE CHECKS

7.29.1. Each PIC undergoes a line check where a pilot is required to operate as pilot flying and pilot monitoring, he should be checked on one sector as pilot flying and on another sector as pilot monitoring to demonstrate his competence in carrying out normal line operations.

7.29.2. The check is conducted by captains nominated by the operator. The period of a line check is 12 calendar months.

NOTE: for more information on line check curricula refer to FOTPM.

7.30. MEDICAL LICENSE ISSUE AND RENEWAL

7.30.1 All flight crew shall hold a valid ECAA class one medical certificate.

7.30.2 Simulator instructors must have third class medical certificate.

7.30.3 The period of validity for a medical certificate shall be six months for all pilots aged 40 or older and one year for pilots younger than 40 years.

7.31. ADMINISTRATION

7.31.1 Upon completion of training, Dir. Flight Training & Standards will forward notice of qualification changes and/or renewal to:

a) ECAA

b) Dir. Flying Operations

c) Mgr. HR & Finance – FOPS

d) Crew Scheduling

e) Team Leader Reward & Retention

f) Team Leader Payroll

g) Training File

7.31.2 All results of successful and unsuccessful evaluations shall be monitored, recorded and evaluated by Dir. Flight Training & Standards Department for the purpose of continuous improvement of the training and check program.

7.32. TRAINING PLAN

A training plan will be prepared annually for each calendar year establishing the number of pilots to be upgraded during the period, in accordance with pilot readiness and company needs. Pilots will have an opportunity to pass flight check portions of each upgrading examinations. The company may at its discretion, offer a second training opportunity to a pilot within 6 – 12 months of his unsuccessful effort, based upon the candidate's past training records and the prospect of success.

7.33. COURSE FAILURE DURING INITIAL HIRE OR TRANSITION TRAINING TO A HIGHER EQUIPMENT

7.33.1 Trainee pilots will be expected to achieve the required standards within the specified transition course described in the Flight Operations Training Program Manual. Those who do not achieve these standards within the specified training program will be subject to a review by the Training Review Committee.

7.33.2 If further training is recommended, any subsequent failure to achieve the required standard within the prescribed training may be handled as per below:

a) Pilots who fail transition training will be offered, if possible, a return to their previous type provided the position reserved for unsuccessful trainees haven't been exhausted & he meets all the requirements for that position. Where this is not possible, e.g. type phase out or upon initial hire the employment of the pilot will be terminated.

b) After successful return on the type & if the fleet phases out thereafter, the employment of the pilot will be terminated if Flight Operations doesn't consider him for another type training.

c) If his training performance isn't to the standard & warrants termination from flying duty the company may consider him to other non-flying duties if a vacancy is available.

d) Pilots who successfully return to their previous type will not be offered a second chance of transition training for at least 6 months and Flight Operations has positive & enough feedback on the improvement of his performance from line & simulator instructors. Subject to terms and conditions of this policy, a pilot who is unsuccessful on a second attempt at a transition course may be allowed to return to his previous fleet, subject to a review by the Training Review Committee. The pilot may be offered another chance of transition training if Flight Operations deems it necessary.

7.34. COMMAND UPGRADE FAILURES:

Command upgrade failure will be handled according to FOM 5.2(7.29.2, a,b,c). At the discretion of the Training Review Committee, after a period of 12 months from an upgrade failure, a First Officer may be considered for a second upgrade attempt. In determining the suitability of a First Officer for a second attempt, the Training Review

SECTION 5.2 PROGRESSION CYCLE AND QUALIFICATION REQUIREMENTS

Committee will pay particular attention to training and standard performance. The Committee expects to see a high level of competency demonstrated through proficiency checks and line operations. If he fails for the second time Flight Operations may offer him another chance if deems necessary.

7.35. PROFICIENCY CHECK FAILURES

Pilots who fail their Proficiency check will repeat the PT & PC with the approval of Dir. Flight Training & Standards. If the pilot fails on the second attempt he will be referred to the Training review committee. The training review committee may recommend a remedial training based on the instructor feedback. After the completion of the additional remedial training he will take the PC. Any subsequent failure or if remedial training isn't recommended it will be dealt as specified in FOM 5.2(7.29) above.

7.36. SEGREGATION OF TRAINING AND CHECKING

7.36.1 Under Ethiopian training structure, qualified instructors shall conduct training. Examining other than ground training exams shall be conducted by designated examiners only.

7.36.2 All examinations (check rides) in conjunction with simulator, aircraft and/or line training shall not be conducted by the same person who conducted the training with the exception of recurrent training.

7.37. INAPPROPRIATE INTERFERENCE

To ensure objectivity is maintained in the training and evaluation program instructors, evaluators and line check airmen shall perform their assigned activities without inappropriate interference from management and/or external organizations including the regulators.

7.38. EXAMINATIONS

7.38.1. All questions for examinations shall comply with the following:

a) Be derived from a recognized database.

b) Be relevant to the instruction given or self-study required.

c) Be focused towards the roles and responsibilities of the person being examined.

d) Be chosen to elicit levels of knowledge and understanding.

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7.39. ACCEPTABLE RESULTS

7.39.1. Passing grade for ground training exams is 80% and trainees are allowed only one retake at this time passing grade will be 85%.

7.39.2. The minimum standard for all flight/simulator check is 3 and satisfactory(s) for all evaluations/checks.

7.39.3. For tolerance margin of flight training and examination, refer to the training program of each aircraft on FOTPM.

7.40. RETAKE

7.40.1. In the event a student fails to satisfactorily complete an evaluation specified in the curriculum of a required training program, the following actions must be taken. These actions are applicable for the following training phases.

a) Classroom

b) Cockpit procedure trainer

c) Simulator

d) Airplane

7.40.2. The student shall be notified as soon as practicable, of the results of the unsatisfactory evaluation by the instructor/Examiner who administered the evaluation. At that time, it shall be determined whether any extraordinary factors contributed to the student's poor performance. Extraordinary factors include student illness, death in the family, and the like.

7.40.3. The responsible Instructor/ Examiner shall submit to the Director Flight Training & Standards the student's training record and any other information or recommendations considered pertinent to the student's performance.

7.40.4. The Director Flight Training & Standards shall review the above information and shall either allow or refuse the student to repeat the evaluation within five working days.

7.40.5. In the event that the student fails to pass evaluation a second time or is not allowed to retake the evaluation, this information, together with the student's record and the Instructor/ Examiner recommendations, shall be presented to the Director Flight Training & Standards.

7.40.6. Director Flight Training & Standards maintains a structured process to identify and support flight crew members struggling to meet the required training standards. Upon identification—through instructor reports, simulator performance, or evaluation trends—an individualized recovery training plan is developed. This plan includes targeted instruction, focused simulator sessions, mentoring, and regular progress reviews. The program is overseen by the Manager Training Standards, with all activities documented. To promote a just culture that supports recovery, ensuring that struggling crew members receive timely assistance while maintaining safety and training integrity.

7.40.7. Director Flight Training & Standards will review all the information available regarding the student who failed to satisfactorily complete an evaluation. Within five working days, he shall notify the VP Flight Operations of the problem and present the student's training records for final review.

7.41. FLIGHT CREW SENIORITY

7.41.1. New entrants from Ethiopian Aviation academy will be having seniority per their ranking as provided from Manager MPL & CADET Training Department.

7.41.2. Pilots who failed transition training will lose seniority automatically and will be dealt with as per FOM5.2 (7.29).

7.41.3. Pilots who failed assessment interview, who couldn't succeed in his/her Simulator or line trainings during command upgrade training will lose their seniority. Second chance for assessment interview will be given after three months if the office believes that the candidate has improved on his previous weakness. Failure on simulator or line training will be dealt with FOM 5.2(7.30)

7.41.4. Pilot who refuse to sit for upgrade assessment interview needs to present his/her justification in writing and he/she will have another chance after request for readiness in writing and his/her case will be evaluated by the office for reconsideration and he will lose his seniority.

7.41.5. Pilots who refuse to take transition training will lose seniority.

7.41.6. 7.41.6. Penalty based demotion should be treated in such a way that the seniority of the pilot will be revoked during his/her lower equipment operation due to the demotion. After the pilot is reinstated to his previous position, his seniority will be on top of the pilots who have less experience from his previous on type experience.

NOTE: If the demotion takes more than six months the pilot is required to accumulate at least 1000 hours on type after being reinstated before being considered for promotion to higher equipment.

7.41.7. Disciplinary measures will not have an impact on pilots' seniority.

7.42. REQUALIFICATION

7.42.1 Re-qualification training shall be provided to restore a previously qualified crewmember to a qualified and current status.

7.42.2 It is the responsibility of each pilot to ensure that his/her recency of experience requirement is fulfilled. If a pilot has not had this requirement satisfied, he should report to his respective chief pilot and/or scheduling ahead of time, so the appropriate

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measures can be taken. Normal control for this requirement will be fulfilled when both Scheduling and pilots track recency of experience by at least one landing per 30 days or a calendar month.

The below table is detailed out Flight Crew Requalification Training Requirements Based on Flight Interruption Period.

Flight Interruption Period	**Ground Training**	**Simulator Training**	**Line Training**
Up to and including 60 days	Not Required	Not Required	Not Required
61 days to 90 days	Not Required	Not Required	2 Sectors (1 Sector Line Training and 1 Sector Line Check)
91 days to 120 days	Current Recurrent Training System course	One Recurrent Training	One Line Check
121 days to 180 days	Current Recurrent Training System course	One Training Session Recurrent Check Recurrent Training	2 Sectors (1 Sector Line Training and 1 Sector Line Check)
181 days to 12 months	Current Recurrent Training System course	Two Training Sessions Recurrent Check Recurrent Training	4 Sectors (3 Sector Line Training and 1 Line Check)
After 12 months to 18 months	Current Recurrent Training System course	Two Training Sessions Recurrent Check Recurrent Training	6 Sectors (5 Sector Line Training and 1 Line Check)
After 18 months to 24 months	All Aircraft Training System course	3 Training Sessions Recurrent Check Recurrent Training	8 Sectors (7 Sector Line Training and 1 Line Check)
After 24 to 48 months	At VP Flight Ops or Director Flight Training discretion: Requalification training with 10 Line Training Sectors and a Line Check or Full Type Rating Course		
After 48 months	Full Type Rating Course	Full Type Rating Course	Full Line Training and Line Check

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SECTION 5.3 LOW VISIBILITY TRAINING CURRICULA

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SECTION 5.3 LOW VISIBILITY TRAINING CURRICULA

1.0. PURPOSE

The purpose of this training curricula procedure is to provide guidelines to be used during the layout of the Training Curricula.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight crewmembers
- Instructors
- Examiners
- Crew Resource Management Instructors

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.05

5.0. DEFINITION

N/A

6.0. RESPONSIBILITY

6.1. The Director Flight Training & Standards is responsible for the compliance with the training requirements outlined in this policy and procedures.

SECTION 5.3 LOW VISIBILITY TRAINING CURRICULA

7.0. PROCEDURES

7.1. LOW VISIBILITY OPERATIONS TRAINING AND QUALIFICATION

7.1.1. GENERAL

a) Pursuant to Ethiopian Civil Aviation Authority approval for All Weather Operations, Ethiopian is required to provide Initial, Transition and Recurrent Training and evaluation to its crews. All weather operation(AWO) training includes the general aspects of Ethiopian AWO, as contained in the Flight Operations Manual, all weather operations, and the aircraft specific aspects as found in the appropriate FCOM and FCTM. The guidelines in this chapter represent the minimum requirements for compliance with Ethiopian Civil Aviation authority. Design and construction of the actual courses are expected to exceed these requirements with the focus on developing safe, capable and knowledgeable pilots for All Weather Operations.

b) All Flight Crew shall be given an initial All Weather Operations training course. Recurrent training is required on an annual basis. This training program is type specific and is included in the regular type qualification, transition (conversion), Upgrade to PIC, re-qualification, and recurrent training.

c) The training program is divided into 4 main parts:

I. Ground Course;

II. Simulator Training;

III. Line Flying Under Supervision; And

IV. Recurrent Training

7.2. GROUND COURSE

7.2.1 Flight Crews will familiarize themselves with the contents of the FCOMs, Flight Operations Manual & other related manuals describing All Weather Operations.

7.2.2 Low visibility CBT course providing a theoretical review of all relevant All Weather Operations aspects.

7.2.3 The following topics shall be covered during the CBT course:

a) The characteristics and limitations of the ILS; and

b) The characteristics of the visual aids;

c) The characteristics of fog;

d) The operational capabilities and limitations of the particular airborne systems;

e) The effects of precipitation, ice accretion, low level wind shear and turbulence

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f) The effect of specific airplane malfunctions;

g) The use and limitations of RVR assessment systems;

h) The principles of obstacle clearance requirements;

i) Recognition of and action to be taken in the event of failure of ground equipment;

j) The procedures and precautions to be followed with regard to surface movement during operations when the RVR is 400 m or less and any additional procedures required for take-off in conditions of 150 m.

k) The significance of decision heights based upon radio altimeters and the effect of terrain profile in the approach area on radio altimeter readings and on the automatic approach/landing systems;

l) The importance and significance of Alert Height if applicable and the action in the event of any failure above and below the Alert Height;

m) The qualification requirements for pilots to obtain and retain approval to conduct Low Visibility Take-offs and Category II or III operations; and

n) The importance of correct seating and eye position.

7.3. SIMULATOR TRAINING FOR LOW VISIBILITY

7.3.1 Flight Crews will be scheduled for one session of All Weather Operations simulator training covering normal & non-normal operation.

a) The following topics & exercises shall be covered during this session;

b) Checks of satisfactory functioning of equipment, both on the ground and in flight;

c) Effect on minima caused by changes in the status of ground installations;

d) Monitoring of automatic flight control systems and auto-land status annunciators with emphasis on the action to be taken in the event of failures of such systems;

e) Actions to be taken in the event of failures such as engines, electrical systems, hydraulics or flight control systems;

f) The effect of known unserviceability's and use of minimum equipment lists;

g) Operating limitations resulting from airworthiness certification;

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h) Guidance on the visual cues required at decision height together with information on maximum deviation allowed from glide path or localizer; and

i) The importance and significance of Decision/Alert Height if applicable and the action in the event of any failure above and below the Decision/Alert Height.

7.3.2 Training shall be divided into phases covering normal operation with no airplane or equipment failures but including all weather conditions which may be encountered and detailed scenarios of airplane and equipment failure which could affect Category II or III operations. If the airplane system involves the use of hybrid or other special systems (such as head up displays or enhanced vision equipment) then flight crew members must practice the use of these systems in normal and abnormal modes during the Flight Simulator phase of training.

7.3.3 Incapacitation procedures appropriate to Low Visibility Take-offs and Category II and III operations shall be practiced.

7.3.4 Initial Category II and III training shall include at least the following exercises:

a) Approach using the appropriate flight guidance, autopilots and control systems installed in the airplane, to the appropriate decision height and to include transition to visual flight and landing;

b) Approach with all engines operating using the appropriate flight guidance systems, autopilots and control systems installed in the airplane down to the appropriate decision height followed by missed approach;

c) Where appropriate, approaches utilizing automatic flight systems to provide automatic flare, landing and roll-out; and

d) Normal operation of the applicable system both with and without acquisition of visual cues at decision height.

7.3.5 Subsequent phases of training must include at least:

a) Approaches with engine failure at various stages on the approach;

b) Approaches with critical equipment failures (e.g. electrical systems, auto flight systems, ground and/or airborne ILS/MLS systems and status monitors);

c) Approaches where failures of auto-flight equipment at low level require either;

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I. Reversion to manual flight to control flare, landing and roll out or missed approach; or

II. Reversion to manual flight or a downgraded automatic mode to control missed approaches from, at or below decision height including those which may result in a touchdown on the runway;

d) Failures of the systems which will result in excessive localizer and/or glideslope deviation, both above and below decision height, in the minimum visual conditions authorized for the operation. In addition, a continuation to a manual landing must be practiced if a head-up display forms a downgraded mode of the automatic system or the head-up display forms the only flare mode; and

e) Failures and procedures specific to airplane type or variant.

7.3.6 The training program must provide practice in handling faults which require a reversion to higher minima.

7.3.7 The training program must include the handling of the airplane when, during a fail passive Category III approach, the fault causes the autopilot to disconnect at or below decision height when the last reported RVR is 300m or less.

7.3.8 Where take-offs are conducted in RVR of 400m and below, training must be established to cover systems failures and engine failure resulting in continued as well as rejected take-offs.

7.3.9 All direct entry pilots regardless of previous low visibility experience must complete the low visibility simulator training during the Conversion Course.

7.4. LINE FLYING UNDER SUPERVISION

7.4.1 The Flight Crew shall conduct at least one Category II/III practice auto land in Category I weather condition or better during his line flying under supervision (route check) phase of training for the completion of the training and qualification process.

7.4.2 Refer to All Weather Operation Chapter 2.19 for flight crew experience requirements.

7.5. RECURRENT TRAINING FOR LOW VISIBILITY

7.5.1 Flight Crews will be scheduled for All Weather Operations simulator recurrent training in concurrent to the Operator Proficiency Check covering normal & non-normal operation in All Weather Operations.

a) A minimum of three approaches shall be conducted in the simulator;

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b) One missed approach shall also be flown during the conduct of the Operator Proficiency Check.

c) The recurrent training shall include;

I. Go-around below alert height during a CAT IIIB approach;

II. Downgrading of approach category due to equipment (ground or airborne).

d) A minimum of 5 CAT II/III approaches, actual or practice shall be done between simulator checks and the Low Visibility Approach Record Card (LARC) must be signed by the Instructor/Examiner during the Proficiency Check. At least one such approach shall be carried out every 60 days.

e) At least one LVTO to the lowest applicable minima shall be flown during the conduct of proficiency check.

f) Pilot incapacitation.

7.5.2 As a principle, no person may serve nor, may any AOC holder use a person as a flight crew member unless within the preceding 12 calendar-months that person has completed the recurrent ground and flight training curricula approved by the Authority. (ECARAS8.10.1.33)

7.6. LOW VISIBILITY TRAINING AND CHECKING

7.6.1. INITIAL TRAINING AND CHECKING

a) Preflight

I. Normal procedures

II. Takeoff Briefing

- FMC
- SID
- EMERGENCY
- Low visibility requirement

b) Engine Start

Normal procedures

c) Taxi

I. Normal procedures

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II. Taxi in low visibility

d) Takeoff

I. Normal procedures

II. Practice RTO

- Engine failure before V1
- Lost visual cue before 100 knots
- Pilot incapacitation before 100 knots

e) Approach and landing

I. Normal procedures

II. All weather operations briefing

III. Demonstrate visual segment in CAT I/II/III A and B

IV. Autoland and reset for another takeoff

f) Approach 1 RVR 75m

Normal CAT III B approach and landing

g) Approach 2 RVR 100m

I. RVR reduces above approach ban RVR 50m

II. Go around due visibility below approved minima

h) Approach 3 RVR 200m

I. A/T fail above approach ban

- Downgrade to CAT III A

II. RVR reduces below approach ban RVR 125/120/100

III. Continue approach to new downgraded minima

i) Approach 4 RVR 250m

I. Engine fails above approach ban (landing flaps)

II. RVR reduces above approach ban

III. Continue approach to new downgraded minima

IV. LAND 2 ASA below approach ban

j) Approach 5 RVR 250m

I. A/T fails below approach ban

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II. RVR reduces below approach ban RVR 175/175/adv

III. Continue approach to new downgraded minima

IV. Go around at 50 feet RA

k) Approach 6 RVR 175m

I. HYD SYS or ELEC SYS failure above approach ban

II. NO AUTOLAND at 50 feet RA

l) Approach 7 RVR 250m

I. RVR reduces below approach ban RVR

II. NO FLARE

m) Approach 8 RVR 125m

Pilot incapacitation below approach ban

n) After landing

I. Normal procedures

II. Low visibility taxi – Call runway vacated

o) Shutdown

Normal procedures

p) Secure

Normal procedures

Repeat Approach 1 to 9 if non-normal crew (e.g. 2 Captains or 2 FOs)

7.6.2 RECURRENT CHECK

a) Preflight

I. Normal procedures

II. Takeoff Briefing

- FMC
- SID
- EMERGENCY
- Low visibility requirement

b) Engine Start

Normal procedures

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c) Taxi

I. Normal procedures

II. Taxi in low visibility

d) Takeoff

I. Normal procedures

II. Practice RTO

- Engine failure before V1
- Lost visual cue before 100 knots
- Pilot incapacitation before 100 knots

III. Reposition – Normal takeoff

e) Climb

Normal procedures

f) Descent and Approach

Preparation and briefing for Low Visibility Approach

g) Approach 1 RVR 75m

Normal CAT III B approach and landing

h) Approach 2 RVR 100m

I. RVR reduces above approach ban RVR 50m

II. Go around due visibility below approved minima

i) Approach 3 RVR 200m

I. A/T fails above approach ban

- Downgrade to CAT III A

II. RVR reduces below approach ban RVR 125/120/100

III. Continue approach to new downgraded minima

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SECTION 5.4 TRAINING DEVICE

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SECTION 5.4 TRAINING DEVICE

SECTION 5.4 TRAINING DEVICE

1.0. PURPOSE

The purpose of this difference training procedure is to establish guidelines for compliance with the difference trainings.

2.0. REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0. PERSONS AFFECTED

- Flight crewmembers

4.0. POLICY

Refer to Management Policy Manual chapter 9 section 9.05

5.0. DEFINITION

N/A

6.0. RESPONSIBILITY

Director Flying and Training is responsible for compliance with the policy and procedure outlined in this section.

7.0. PROCEDURE

7.1. COMPUTER BASED TRAINING

7.1.1 Computer Based Tutorial system (CBT) may be used in lieu of or in conjunction with conventional classroom instruction in the company's approved training program provided that:

a) The CBT course conforms to the subjects and programmed hours specified in the classroom curriculum for the same training

b) The use of CBT is specifically noted in such a classroom curriculum.

c) Both the student evaluation and records requirements of the classroom's instruction curricula are complied with.

7.1.2 Each computer based tutorial system (CBT) used in lieu of or in conjunction with conventional classroom instruction must:

a) Be specifically approved by the Civil Aviation Authority in terms of the presentation and response system.

b) Be specifically approved by the Civil Aviation Authority for the particular subject and personnel duty involved.

7.2 COCKPIT PROCEDURE TRAINER DEVICE

7.2.1 cockpit procedure trainer may be used in company training programs provided that:

a) Be specifically approved by the Civil Aviation Authority for the type airplane and, if applicable, the particular variation within type for which the training or check is being conducted.

b) Be specifically approved by the Civil Aviation Authority for the particular procedures and crew member function involved.

c) Conform to the airplane being simulated in functional and other characteristics required for approval by the Civil Aviation Authority.

d) The cockpit procedure trainer course conforms to the subjects and programmed hours specified in the applicable curriculum.

e) The use of the cockpit procedure trainer is specifically noted in the applicable curriculum.

7.2.2 The cockpit procedure trainer is designed to provide reinforcement to instrument training as required in response to reports to the Manager recurrent & transition training on line flights and training record comments by instructor pilots or examiners training programs. It should be scheduled for any pilot whose performance indicates an instrument deficiency in any area.

7.2.3 If necessary, the company may substitute any cockpit procedure trainer whose curriculum is approved by the Civil Aviation Authority.

7.3 COCKPIT PROCEDURE TRAINER USE

7.3.1 Cockpit procedures trainers may be used in the company's approved training program provided that:

a) The cockpit procedures trainer course conforms to the subjects and programmed hours specified in the ground training curriculum.

b) The use of a cockpit procedures trainer is specifically noted in that ground training curriculum.

SECTION 5.4 TRAINING DEVICE

c) Both the student evaluation and records requirements and the instruction curricula are complied with.

For detailed procedures refer to Flight Operations Training Program

7.4 FLIGHT SIMULATOR

7.4.1 A flight simulator may be used in lieu of the cockpit procedure trainer provided that:

a) Be specifically approved by the ECAA for the type airplane and, if applicable, the particular variation within type for which the training or check is being conducted.

b) That the simulator being used is the same as the aircraft for which training is being conducted.

c) Be specifically approved by the ECAA for the particular manoeuvre, procedure, or crewmember function involved.

d) Conform to the airplane being simulated in performance, functional and other characteristics required for approval by the ECAA.

e) Be given a daily functional pre-flight check before being used.

f) Have a daily discrepancy log kept, with each discrepancy entered in the log at the end of each training or check flight. (this task is accomplished by simulator technicians)

7.4.2 If necessary, the company may substitute an airplane simulator whose curriculum is approved by the ECAA.

7.5 AIRPLANE SIMULATOR TRAINING

7.5.1 Training course utilizing airplane simulator and other training devices may be included in the company's approved training program as provided in this section.

7.5.2 A course of training in an airplane simulator may be included for utilizing proficiency checks as provided under this Chapter, if that course:

a) Provides training in at least the procedures and manoeuvre set forth in the Flight Operations Training Program Manual.

b) Is given by an instructor who meets the applicable requirements of this Chapter.

7.5.3 A qualified examiner must certify the satisfactory completion of course or training.

SECTION 5.4 TRAINING DEVICE

7.6 MINIMUM EQUIPMENT LIST

7.6.1 Before any training sessions are conducted, the simulator MEL and Aircraft MEL must be consulted to check that any simulator/Aircraft defects are allowable for the type of training to be conducted.

7.6.2 For aircraft training the standard MEL allowances apply provided that the particular training to be conducted is not affected by the defect. (E.g. if the training is to conduct a VOR approach the VORs must be functional). Refer to FOTPM APPENDIX C for more information.

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SECTION 5.4 TRAINING DEVICE

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SECTION 5.5 TRAINING REVIEW BOARD

SECTION 5.5 TRAINING REVIEW BOARD

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SECTION 5.5 TRAINING REVIEW BOARD

SECTION 5.5 TRAINING REVIEW BOARD

1.0 PURPOSE

The Purpose of Training Review Board (TRB) is to address training system reviews and make improvement or changes based on feed back from the training assessment system, trend analysis and audit activities. It also addresses training related issues such as unsatisfactory progress during training course as well as complaints from trainees and/or Instructors/Examiners.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- VP Flight Operations;
- Dir. Flight Training & Standards;
- Dir. Flying Operations;
- Chief Pilots;
- Mgr. Boeing Fleet Training;
- Mgr. Airbus and Q400 Training;
- Mgr. MPL and Cadet Training
- Affected Instructor/Examiner and affected Trainee.

4.0 POLICY

4.1 The Training Review Board (TRB) shall ensure that the following standards have been met at all times:

4.1.1 Flight crew are trained and checked to operate the airplanes safely and efficiently under normal and non-normal conditions.

4.1.2 All training materials are current, training devices and facility meet the requirement of ECAA and other authorities as applicable.

SECTION 5.5 TRAINING REVIEW BOARD

4.1.3 All trainees are trained equally and their concerns and complaints are given due consideration.

4.1.4 Standard Operating Procedure and Policy are known and adhered to by all trainees and Instructors/Examiners.

5.0 DEFINITION

Grievance: shall mean a written complaint made for personal relief in a matter of concern and/or dissatisfaction relating to the flight training.

Trainee: shall mean a flight crew member who is undergoing ground or flight training in Flight Operations.

Instructor/Examiner: shall mean a person who is assigned by the Company and/or authorized by ECAA to conduct ground/flight training and check.

6.0 RESPONSIBILITY

6.1 Dir. Flight Training & Standards is primarily responsible for procedures in this section.

7.0 PROCEDURE

7.1 ENHANCING THE TRAINING PROGRAM (TRAINING REVIEW COMMITTEE)

7.1.1 The Training Review Committee meets every six months to review trends and feedback from the outputs of the training and assessment system.

7.1.2 The Committee represents fleet, training, audit and safety interests.

7.1.3 The Committee addresses the impact of regulatory, procedural, technical, fleet and schedule change as well as international best practice upon future training requirements.

7.1.4 The Committee has the important function of providing feedback to be used by Mgr. Recurrent & Transition Training to adjust the training to ensure that safety margins are maintained.

7.1.5 The committee will be comprised of the following personnel or their delegates;

- Dir. Flight Training & Standards (Chairperson)
- Dir. Flying Operations
- Mgr. Flight Operations Safety
- Mgr. Boeing Fleet Training

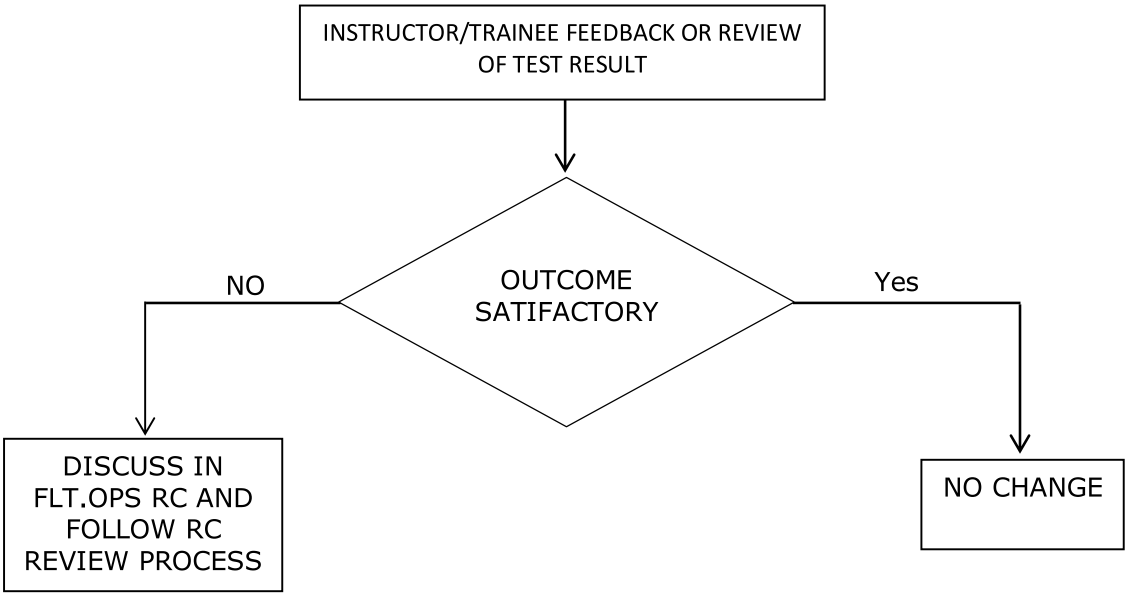
SECTION 5.5 TRAINING REVIEW BOARD

- Mgr. Airbus and Q400 Training Standards
- Chief Pilots

The following personnel may be invited as necessary;

- Head Pilot Training
- Mgr. MPL and Cadet Training
- Head Cabin Crew Training
- ET-ALPA officer

7.2 TREND ANALYSIS & TRAINING PROGRAM IMPROVEMENT



7.3 PERFORMANCE AND GRIEVANCE REVIEW

7.3.1 The Performance review committee meets as required to review pilot performance issues and/or grievance requiring resolution.

7.3.2 From flight training perspective, these issues include, but not limited to;

a) Failure of any check event or unacceptable progress that may result in the down grade of the pilot or termination of employment

b) Any complaint or issue occurred during training or check

SECTION 5.5 TRAINING REVIEW BOARD

7.3.3 The Committee will review all available records including trainees Training file and if required, interviews the concerned Instructor/Examiner and the trainee, it also uses recommendations from Flight Training, Medical Services (if required), and other appropriate inputs when deciding on the action to be taken. Available actions include, but not limited to;

- Demotion
- Return to training or recheck
- Type and duration of remedial training
- Change of Instructor/ Examiner
- Termination of employment (in exceptional circumstances, it will be handled in accordance with the company policy)

7.3.4 Any trainee performance related issue must be reported by the Instructor/Examiner immediately after the event to Dir. Flight Training & Standards or Mgr. Recurrent & Transition Training. In case of any complaint or issue occurred during training or check, it should be reported by the affected person in writing with supportive reasons in 7 days from the training event. No report is accepted if it is reported later than the above period.

7.3.5 Dir. Flight Training & Standards or Mgr. Recurrent & Transition Training will inform the trainee in writing on next course of action in 15 days based on the conclusion of the Committee.

a) The Committee meeting minutes, the report from the affected person and other relevant documents will be kept in a dedicated file in the Flight Training office.

b) The written report by the affected person, the final conclusion report of the Committee and the letter from Dir. Flight Training or Mgr. Transition & Recurrent Training will be placed in the person's personal and/or Training file.

SECTION 5.5 TRAINING REVIEW BOARD

7.3.6 The committee will be comprised of the following personnel or their designates;

- Dir. Flight Training & Standards (Chairperson)
- Mgr. Flight Ops HR (Secretary)
- Mgr. Flight Operations Safety
- Representative from senior line pilot (Local or Expat as appropriate)
- Concerned Chief Pilot

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SECTION 5.6 FLIGHT DISPATCHERS

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SECTION 5.6 FLIGHT DISPATCHERS

SECTION 5.6 FLIGHT DISPATCHERS

1.0 PURPOSE

The purpose of this training program procedure is to establish guidelines to prescribe the requirements applicable to the company for establishing and maintaining a training program for flight dispatchers.

2.0 REVISION HISTORY

Date	Revision No.	Change	Reference Section
30-MAR-2026	25B	Complete Revision in Seven Steps	Whole Section

3.0 PERSONS AFFECTED

- Dispatchers
- Instructors

4.0 POLICY

Refer to Management Policy Manual chapter 9 section 9.05

4.1 The training syllabi shall include training programs for dispatchers.

4.2 All dispatchers are trained and examined according to published standards approved by the state.

4.3 Transition training covering each airplane type operated by the company shall be provided for flight dispatchers. Each employee in these classifications who has not previously performed the same duties for the company with respect to specific type of airplane shall be required to accomplish initial training.

4.4 Differences training shall be provided for flight dispatchers for each variation of a particular type of an airplane, which the company operates, for which differences training is required.

SECTION 5.6 FLIGHT DISPATCHERS

5.0 DEFINITIONS

Flight operations officer/flight dispatcher. A person designated by the operator to engage in the control and supervision of flight operations, whether licensed or not, suitably qualified in accordance with Annex 1, who supports, briefs and/or assists the pilot-in-command in the safe conduct of the flight.

6.0 RESPONSIBILITY

6.1. Manger Flight Dispatch & FMC is responsible for the implementation of this policy and procedure.

7.0 PROCEDURE

7.1 FLIGHT DISPATCHERS INITIAL TRAINING

7.1.1 The Flight Dispatcher's initial training is accomplished with the use of the curricula divided into the following training phases:

a) Classroom instruction

b) Facilities familiarization

c) Route qualification flights

d) Initial flight assignments

7.1.2 A detailed course content of the above training is given if the Flight Operations Training Program Manual.

7.2 GROUND TRAINING FOR FLIGHT DISPATCHER

7.2.1 Each initial training program for flight dispatcher shall include:

a) Use of communications system including the characteristics of these systems and the appropriate normal and emergency procedures.

b) Meteorology, including various types of meteorological information and forecasts, interpretation of weather data (including forecasting of en-route and terminal temperatures and other weather conditions, frontal systems, and wind conditions), and use of actual and prognostic weather charts for various altitudes.

c) The NOTAM system.

d) Navigational aids and publications.

e) Joint responsibilities of pilot and flight dispatcher.

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f) Characteristics of appropriate airports.

g) Prevailing weather phenomena and available sources of weather information.

h) Air traffic control and instrument approach procedures.

i) A general description of the airplane emphasizing operating and performance characteristics, navigation equipment, approach, communication equipment, emergency equipment and procedures, and other subjects having a bearing on the Flight Dispatchers duties and responsibilities.

j) Flight operation procedures including procedures for avoiding severe weather situations and for operating in or near thunderstorms (including best penetration altitudes), turbulent air (including clear air turbulence), icing, hail and other potentially hazardous meteorological conditions.

k) Weight and balance computations.

l) Basic airplane performance dispatch requirements and procedures.

m) Flight planning including track selection, flight time analysis, and fuel requirements.

n) Emergency procedures. These must be emphasized, including the alerting of proper governmental, company, and private agencies during emergencies to give maximum help to airplanes in distress.

o) Crew Resource Management (CRM/TEM) training and evaluation.

7.2.2 Initial ground training for a flight dispatcher must include a competence check given by an appropriate supervisor or ground instructor that demonstrates knowledge and ability in the subjects specified above.

7.2.3 Initial ground training for a flight dispatcher must consist of 305 hours of ground instructions in the subjects stated above.

7.3 FACILITIES FAMILIARIZATION FOR FLIGHT DISPATCHER

7.3.1 Each initial training program for a flight dispatcher shall include a tour of the facilities noted below.

a) Air route traffic control centre.

b) Airport traffic control tower.

c) Station operations.

d) Weather bureau office.

e) Airplane maintenance.

SECTION 5.6 FLIGHT DISPATCHERS

7.4 INITIAL FLIGHT ASSIGNMENT FOR FLIGHT DISPATCHER

Each initial training program for a flight dispatcher shall include five hours flight, over any route or segments, in the cockpit observer's seat in the same airplane type for which he/she is being trained.

7.5 FLIGHT DISPATCHERS RECURRENT GROUND TRAINING

7.5.1 As a minimum, the ground training of each recurrent training program for flight dispatchers must include:

a) A quiz or other review to determine the state of the dispatcher's knowledge with respect to the airplane and the duties of his position.

b) Instructions in the subjects required for initial ground training as appropriate, including emergency training.

c) A competence check given by an appropriate supervisor or ground instructor that demonstrates knowledge and ability with respect to initial ground training subjects.

7.5.2 Recurrent ground training for flight dispatchers shall consist of 20 hours of instructions.

NOTE: Joint Crew Resource Management (JCRM) training should be conducted between cockpit crew, cabin crew and flight dispatchers annually.

7.6 FLIGHT DISPATCHERS TRANSITION GROUND TRAINING

7.6.1 Each transition training program for flight dispatchers shall include:

a) A general description of the airplane emphasizing operating and performance characteristics, navigation equipment, approach, communication equipment, emergency equipment and procedures, and other subjects having a bearing on the flight dispatcher's duties and responsibilities.

b) Weight and balance computations.

c) Basic airplane performance dispatch requirements and procedures.

d) Flight planning including track selection, flight time analysis, and fuel requirements.

SECTION 5.6 FLIGHT DISPATCHERS

e) Emergency procedures. These must be emphasized, including the alerting of proper governmental, company, and private agencies during emergencies to give maximum help to airplanes in distress.

7.6.2 Transition ground training for flight dispatchers must include a competence check given by an appropriate supervisor or ground instructor that demonstrates knowledge and ability in the subjects specified above.

7.6.3 Transition ground training for a flight dispatcher must be 305 hours of instruction.

7.7 FLIGHT DISPATCHER ROUTE QUALIFICATION FLIGHT

Each transition training program for a flight dispatcher shall include five hours flight, in the cockpit observer's seat in the same flight area as the routes to be controlled.

7.8 QUALIFICATION FOR FLIGHT DISPATCHER

7.8.1 A flight dispatcher shall have a route familiarization flight in the preceding 12 months in the cockpit observer's seat in the same flight area as the routes to be controlled.

7.8.2 Each recurrent training program must ensure that each flight dispatcher is adequately trained and currently proficient with respect to the type, including differences if applicable, and crewmember position involved in every 24 months.

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